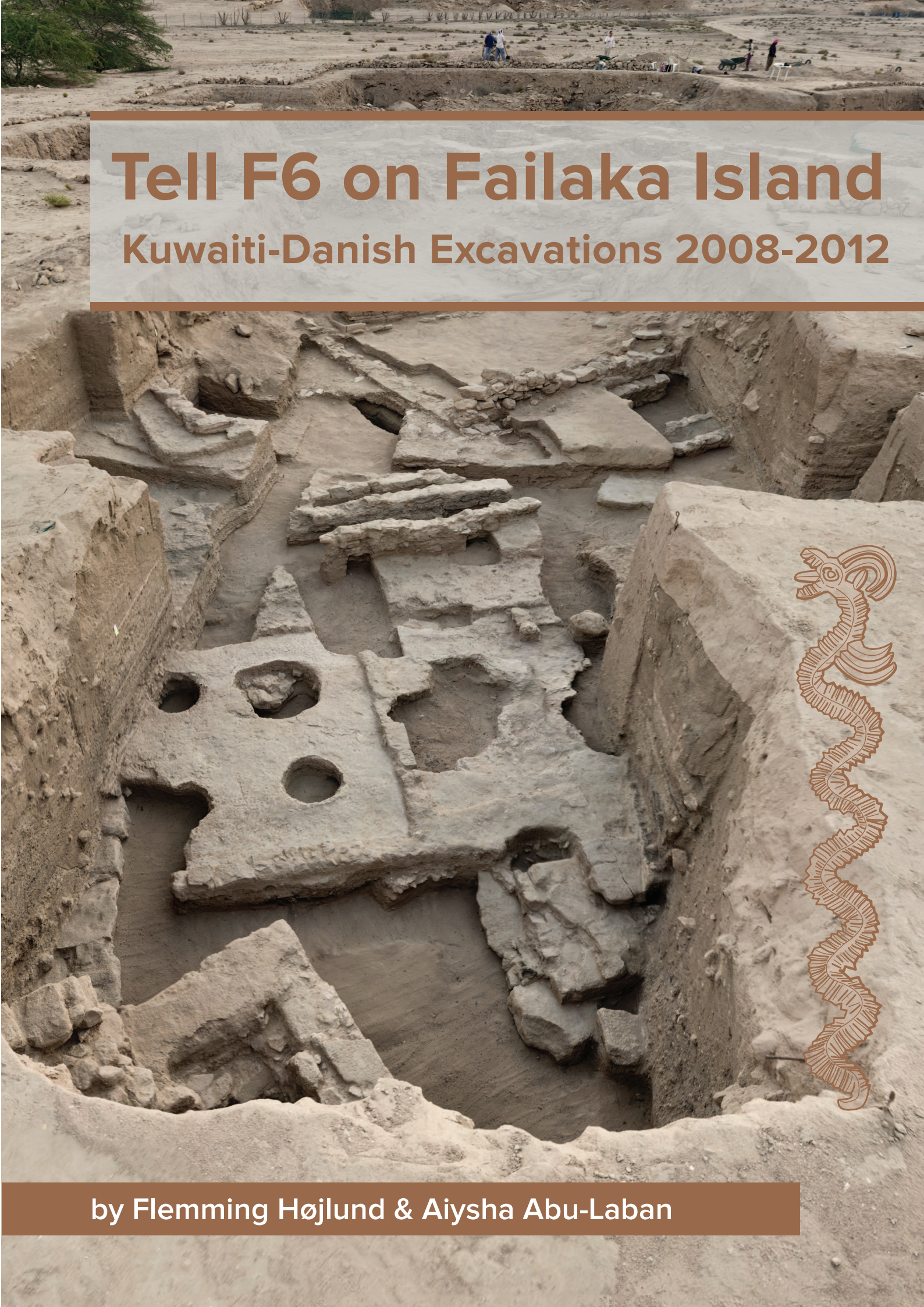




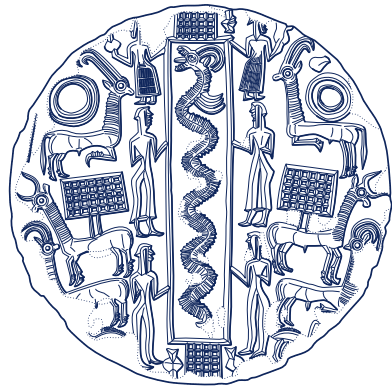
Tell F6 on Failaka Island

Kuwaiti-Danish Excavations 2008-2012



by Flemming Højlund & Aiysha Abu-Laban

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with contributions by Ann Andersson, Jesper Eidem, Jan Heinemeier,
Anna Hilton, Jonathan M. Kenoyer, Holly Pittman, Gunnar Sperveslage,
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National Council for Culture, Arts and Letters, Kuwait

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Fig. 1. Aerial photo of Tell F6 with left "Palace" and right Temple and excavation trenches anno 2009. In the background Zor village.

1. Introduction

by Flemming Højlund

In the 3rd-2nd millennium BC, the Arabian Gulf was an important waterway linking the large cities and states of Mesopotamia and Iran with Southeast Arabia and the Indus Valley (fig. 2). Trade along the *Lower Sea*, as the Gulf was named, demanded supporting stations, and civilizations grew and declined with the fortunes of this trade. In the 1950s, Danish investigations along the southern shores of the Gulf discovered remains of two of these ancient civilizations, *Dilmun* and *Makkan* (Bibby 1969). Survey and excavation in Kuwait beginning in 1958 revealed that the upper Gulf was an important part of Dilmun in the early 2nd millennium. Both before, during and after the flourishing of the Dilmun civilization the island of Failaka located off the Kuwait mainland formed an important role in several Near Eastern trade networks with Mesopotamia as the dominant partner.

On the southwest corner of Failaka, four tells, *Tell F3*, *Tell F4*, *Tell F5* and *Tell F6*, were investigated in 1958-1963 by the Danish expedition (Højlund 2008) (fig. 3). Two of these tells, *F3* and *F6*, dated to the 2nd millennium BC. *Tell F3* contained mostly small *domestic houses*, but in *Tell F6* a 400 m² large building named the “*Palace*” was uncovered (Kjærum & Højlund 2013) (fig. 1). In the 1980s, French archaeologists continued excavation in *Tell F6* and found a *temple* just 15 m east of the “*Palace*” (Calvet & Pic 2008).

Unfortunately, neither the “*Palace*” nor the *Temple* could be dated with any great precision. The construction of the “*Palace*” was placed either in Period 2 or 3A in the Failaka chronology, i.e. a period covering several hundred years in the early 2nd millennium BC (Højlund 1987 p. 146). Evidence for dating the *Temple* was inconclusive pointing to both the end of the 3rd and the beginning of the 2nd millennium BC (Calvet & Pic 1990 p. 104. Pic 2008 p. 52. Højlund 2012 p. 165-167).

In order to clarify the dating of these two grand buildings, to explore their function and relationship and to study the significance of *Tell F6* through history in general *The Kuwaiti-Danish Archaeological Mission to*

Failaka was established in 2008, and a five-year program of excavations was agreed upon.

The initiative to form the Kuwaiti-Danish Mission came from Museum Director Mr Shehab A.H. Shehab on the occasion of the 50-year jubilee of the first archaeological excavation in Kuwait (Højlund 2008). *The Kuwait National Council for Culture, Arts and Letters* has supported our work most generously through its Secretary Generals, Mr Bader Al-Rifai and Mr Ali Hussein Al Youha. Important support has also been received from *Moesgaard Museum* through its Director Mr Jan Skamby Madsen and Head of the Department of Conservation and Science, Dr Peter H. Mikkelsen.

The first season in 2008 lasted from October the 31st to November the 29th. The team consisted of Shehab A.H. Shehab, Khaled Al-Ali, Khalid Salem and Talal Al-Saie from the Kuwaiti side and Lars Krants Larsen, Camilla Bjarnø, Peter Bye Jensen, Mohammed Bander, Anna Hilton, Walid Al-Sadeqi and Flemming Højlund from the Danish side, plus 15 Egyptian labourers, some of who had previous excavation experience from Failaka (Højlund & Krants Larsen 2009).

The 2009 season lasted from October the 14th to December the 16th. The team consisted of Shehab A.H. Shehab, Khaled Al-Ali, Khalid Salem, Talal Al-Saie, Nahar Al-Khaledi and Yves Guichard from the Kuwaiti side and Lars Rønje, Aiysha Abu-Laban, Nilas Hammelev Møller, Maria Reese Næsborg, Ann Andersson, Hélène David-Cuny and Flemming Højlund from the Danish side, plus 20 Egyptian labourers (Højlund 2010).

The 2010 season lasted from October the 16th to December the 8th. The team consisted of Shehab A.H. Shehab, Khaled Al-Ali, Talal Al-Saie, Hamid Al-Mutairi, Johanne Azpeitia, Hélène David-Cuny and Thomas Sagory from the Kuwaiti side and Aiysha Abu-Laban, Aicha Mellah, Jonatan Rose Andersen, Phine Wiborg, Helle Strehle, Sarah Morriss and Flemming Højlund from the Danish side, plus 15 Egyptian labourers (Højlund & Abu-Laban 2011).

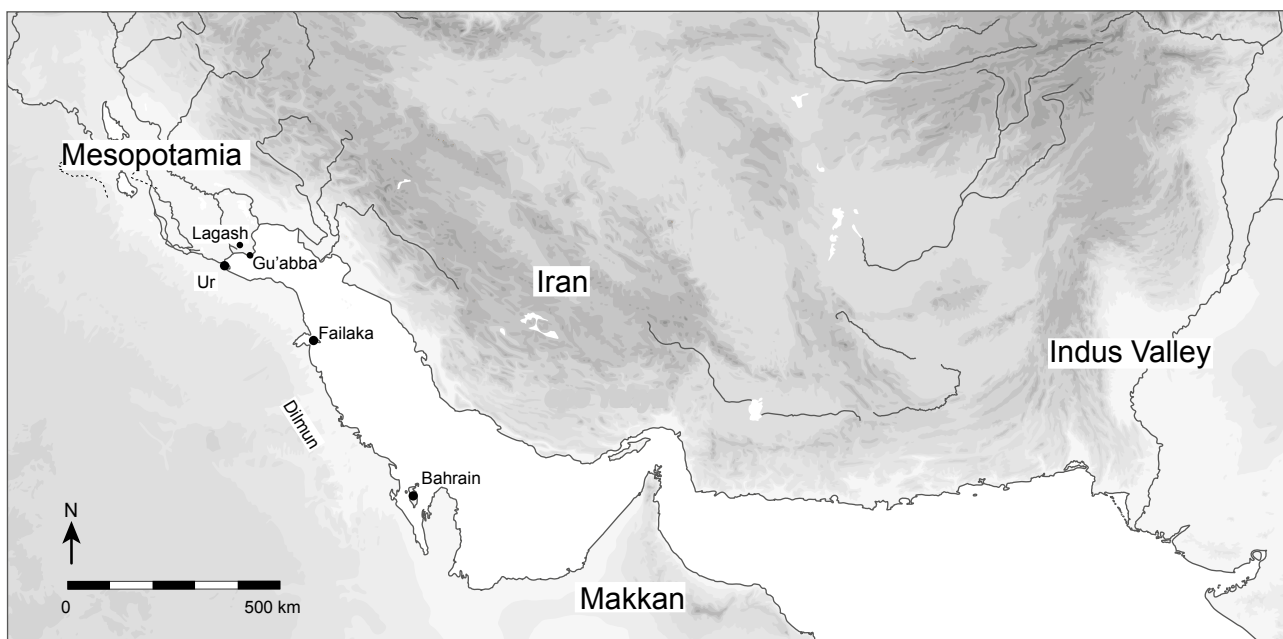


Fig. 2. Map of the ancient world around Failaka Island. The coastal region of Mesopotamia according to Steinkeller (2013 fig. 3).

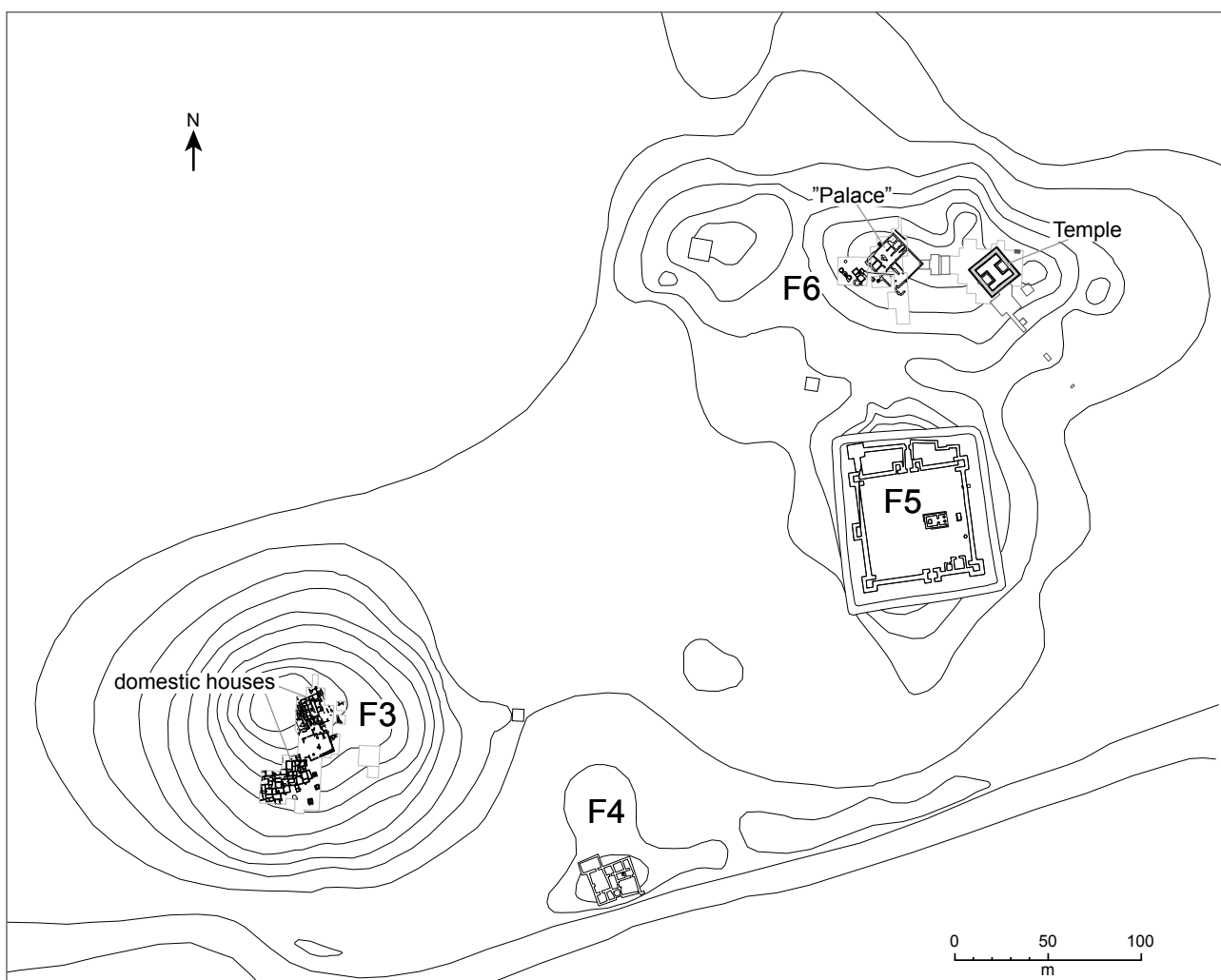


Fig. 3. Map of the archaeological area on the SW corner of Failaka Island with Tells F3, F4, F5, and F6.

The 2011 season lasted from October the 16th to December the 7th. The team consisted of Shehab A.H. Shehab, Khaled Al-Ali, Talal Al-Saie, Hamid Al-Mutairi and Hélène David-Cuny from the Kuwaiti side and Aiysha Abu-Laban, Anna Hilton, Ann Andersson, Casper Skaaning Andersen, Phine Wiborg, Jonatan Rose Andersen, Sarah Morriss, Steffen T. Laursen and Flemming Højlund from the Danish side, plus 15 Egyptian labourers (Højlund & Abu-Laban 2012).

The 2012 season lasted from October the 17th to December the 6th. The team consisted of Shehab A.H. Shehab, Talal Al-Saie, Hamid Al-Mutairi, Khaled Al-Ali, Talal Abdullah Al-Shammar and Hélène David-Cuny from the Kuwaiti side and Aiysha Abu-Laban, Anna Hilton, Ann Andersson, Jonatan Rose Andersen, Phine Wiborg, Christoffer Pelle Hagelquist, Sarah Morriss, Hilary McDonald,

Benjamin Ducke, Paulina Suchowska, Helle Strehle and Flemming Højlund from the Danish side, plus 20 Egyptian labourers (Højlund & Abu-Laban 2013).

The excavation team was quartered in the *Failaka Research Station* located immediately to the east of the ancient tells, and this proved to be a pleasant and efficient place of working thanks to the outstanding helpfulness, efficiency and generosity of our Kuwait colleagues and hosts. We are also grateful to our French colleagues and neighbours, Mathilde Gelin, Olivier Callot and J.-F. Salles and their teams and to the Polish team headed by Piotr Bielinski and Magdalena Zurek who were excavating the Hellenistic Tell F5 and the early Christian settlement at Al Qusur.

The manuscript has benefitted from the constructive scrutiny of Rob Carter and Steffen T. Laursen.

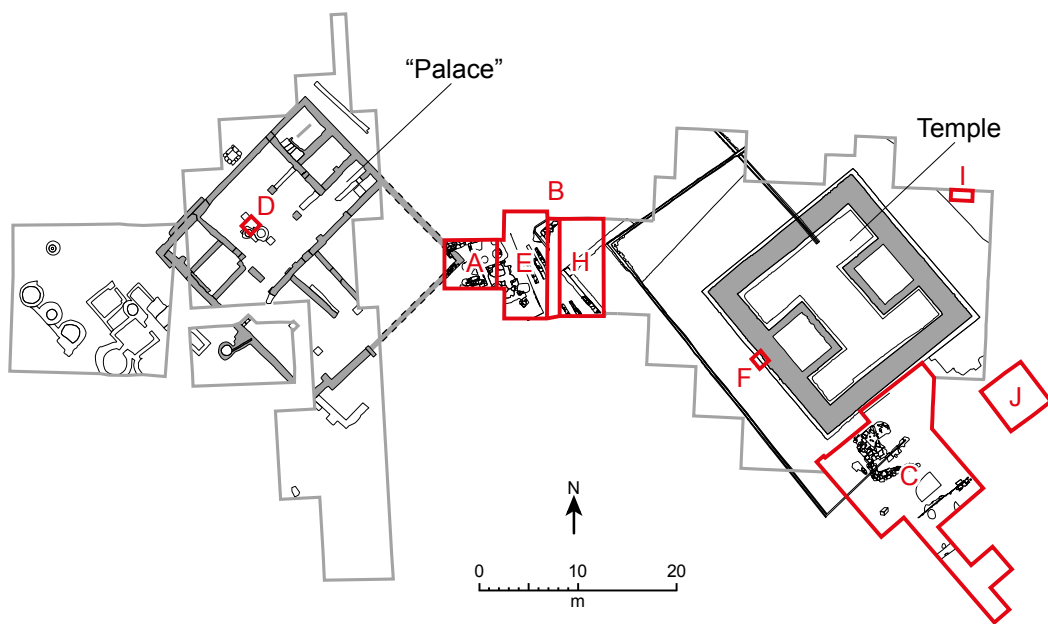


Fig. 4. Trenches A-J in Tell F6 excavated 2008-2012.

G1

G2

2. Stratigraphy and architecture

The 2008-2012 excavations in Tell F6 were carried out in ten different trenches: A, B, C, D, E, F, G, H, I and J, situated to the east of the “Palace” and around the Temple (fig. 4). For Trenches C and J with sub-trenches and Sections 1-16 marked, see fig. 8. For Trenches A, E, B and H with Sections 17-23 marked, see fig. 80. Trench B was a very narrow trench, just cutting down and cleaning the west face of the previous French excavation in order to establish a new section, Section 23.

The layers excavated in these trenches have been referred to seven stratigraphic phases that cover a period of around 1000 years from the late 3rd to the late 2nd millennium BC, followed by an eighth phase that encompasses everything post-dating Phase 7 (fig. 5). The evidence for dating these phases, including radiocarbon dates, and their relationship to the Failaka pottery Periods 1, 2A, 2B, 3A, 3B, 4A and 4B are discussed on p. 93ff and 239ff.

A stratigraphic synthesis of the first seven phases is shown in fig. 6, and they will be described in detail in the following sections. At some point after Phase 7 and before the Hellenistic period Tell F6 became

the object of a systematic search for building stone from the ancient buildings through a network of deep plundering pits.

The present publication deals with the area east of the “Palace” including the Temple. The occupation *within* the “Palace” building has previously been separated into three phases, Early I, Late I and II, following a pre-Palace period (figs. 5-6) (Kjærøum & Højlund 2013 p. 103). The Pre-Palace period corresponds to Phases 2-3 of the present publication, Early Phase I corresponds to Phase 4, Late Phase I corresponds to Phase 6, and Phase II corresponds to Phase 7.

The vertical levelling fixed point (Bench Mark) for the excavation was an iron rod firmly embedded in the soil situated close to Trench A. It had level 4.51 m above sea level (a.s.l.) according to the French excavators of the Temple in Tell F6 (Calvet & Pic 1990 p. 104) and was used in order to facilitate comparison with data from their excavation. The 0-point in the Danish 1960-63 excavations in Tell F6 (Kjærøum & Højlund 2013 p. 99, fig. 132) corresponds approximately to level 3.21 m a.s.l. in the French system. The horizontal measurements were arbitrarily defined for each trench.

Fig. 6. Synthesis of architectural and stratigraphical phases in Tell F6.

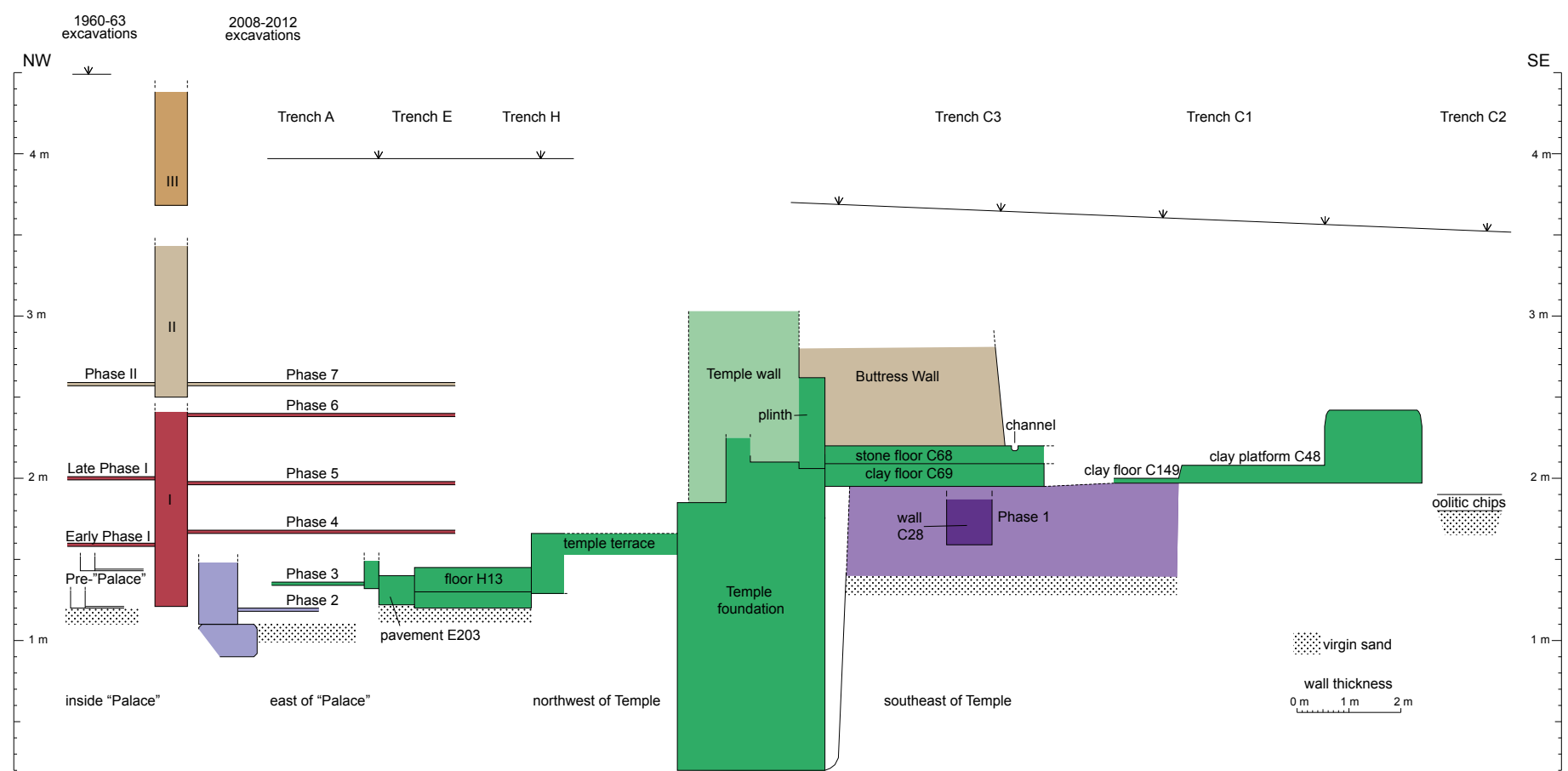




Fig. 7. Trenches C1, C3, CW, CE and CES with the temple south corner in the background and the Mesopotamian wall (C28) overlaid by the temple terrace floor (C68). Looking N, scale unit 50 cm (2010).

2.1. Phase 1. The Mesopotamian Ur III settlement

by Flemming Højlund

Everywhere in the excavated parts of Tell F6 virgin soil was identified as a fine, yellowish-brown sand with small seashells. It was named beach sand, a term that indicates its marine origin. The earliest remains of human presence on Failaka, Phase 1, have been found on the beach sand in the southeast part of Tell F6. Here two walls joining in a corner – a house or a courtyard – were embedded in extensive layers coloured grey by ash (fig. 9). The layers contained pottery, cylinder seals, animal bones, hearths, and a stone-lined pit possibly containing bitumen. The material culture is distinctly Mesopotamian with a few imports from Makkan (UAE-Oman) and Meluhha (Pakistan-India), and it belongs to the Ur III period. The radiocarbon dated samples of Phase 1 fall in the range 2200-2000 cal BC (figs. 5-6).

In the southeast part of Tell F6 a layer, more than half meter thick, of very homogeneous settlement remains was encountered in Trenches C and J (figs. 4 and 7-8) covering an area of at least 22 m from southwest to northeast and at least 12 m from northwest to southeast. The finds clearly became scarcer in the south part of Trench C1. Towards the north, the continuation of this layer was truncated by the construction of the later Temple, but it was visible again in a small sondage, in Trench I to the northeast of the Temple, making the north-south extension c. 40 m (fig. 4).

The occupation layer sealed the natural sterile sand containing many small seashells (*beach sand*). The transition from sterile sand to settlement layer took place around level 1.40 m a.s.l., but cultural finds, especially animal bones, continued to be found occasionally down to c. 1.30 m a.s.l. and in Trench CEE even down to c. 1.10 m a.s.l.

The occupation layer built up to the level of c. 1.95 m a.s.l., i.e. a thickness of 55-65 cm. Originally the layer may have been thicker since the upper part may have been removed by terracing during the construction of the later Temple. In many places, the layer was also partly or completely disturbed by pits dug in search of building stone. Most of these pits seem to post-date Phase 7 and pre-date the Hellenistic period and relate to an extensive plundering of the Temple (p. 59).

Trenches C1-4

In the centre of the excavated area, in Trench C1, two stonewalls meeting in a corner (C28) were uncovered. These may have formed the wall of a house or more likely a courtyard (figs. 7-10). The bases of the walls were laid at a level of c. 1.60 m a.s.l., and they were preserved to a height of c. 40 cm, i.e. up to c. 1.90-2.00 m

a.s.l. The east-west aligned wall was c. 0.8 to 1 m wide and continued to a length of 5 m into Trench CE, but its eastern end may have been robbed away in Trench CEE because of stone plundering.

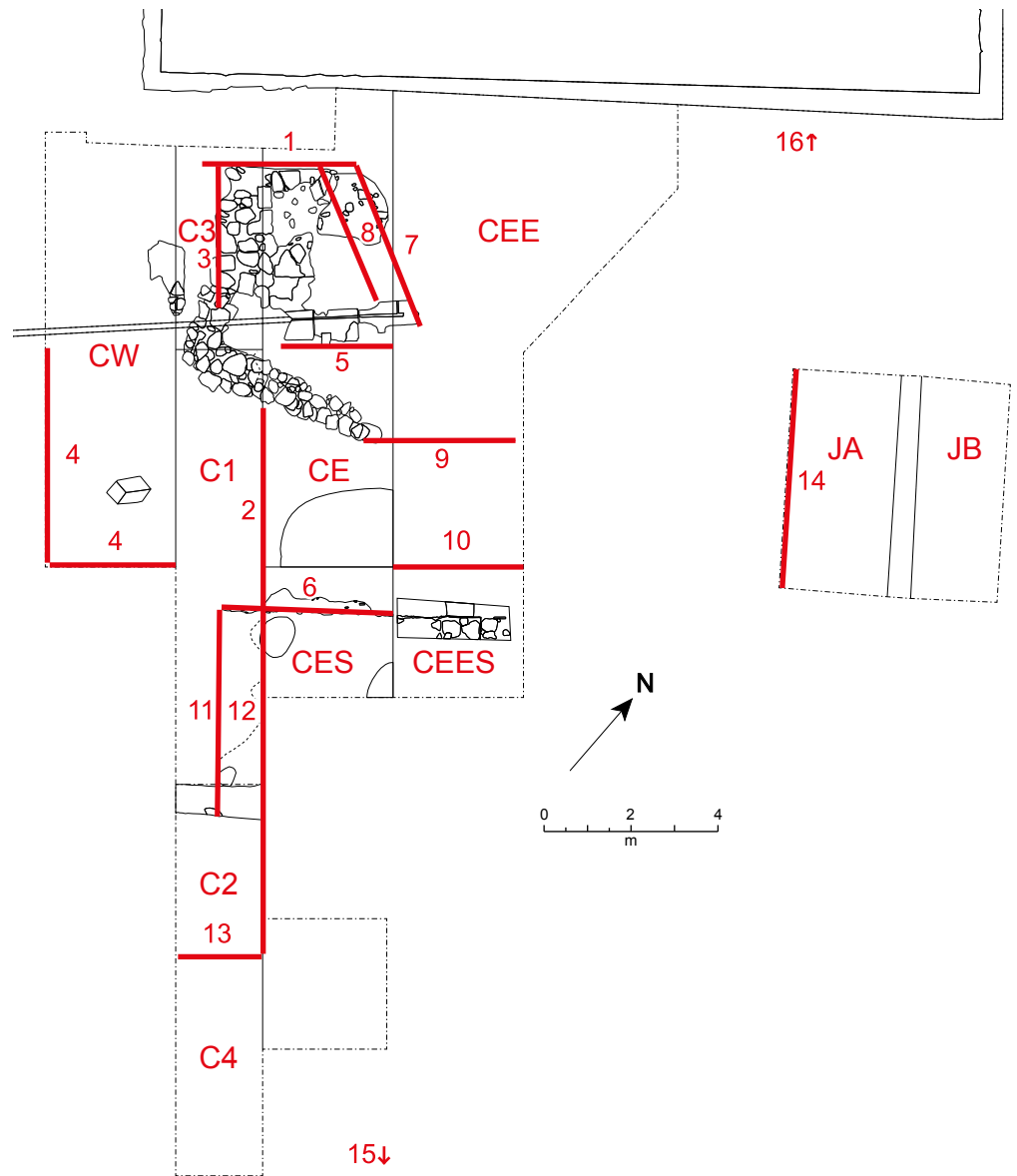
The north-south going wall measured c. 2 m in length with a width of between 0.7-1 m. Towards the north it was covered by a c. 3 m long stretch of a later temple terrace floor (C68) which prevented its excavation. North of this floor, it re-appeared with two sizeable stones (fig. 11) with basal levels at c. 1.60 m a.s.l., giving the wall a total length of 5 m. The continuation of the wall towards the north was truncated by the foundation cut for the later Temple and by an archaeological sondage (Calvet & Pic 1990 plan I, fig. 7).

Wall C28 was constructed using large rounded beach-rock stones, a material available on the beaches of Failaka and traditionally quarried from there (fig. 12). Only sand was observed between the wall stones, but it would have been impossible to construct a wall of such rounded stones without some kind of mortar. Originally, they would probably have been set in clay, but left exposed to rain and sun for a few years, the clay would have disintegrated.¹

Inside the house/courtyard in Trench C1/3 were layers of greyish-brown loose sand containing large amounts of animal bone and pottery datable to the Mesopotamian Ur III period (cf. p. 93, figs. 131-273).

In Trench C1, south of wall C28 and deposited up against it, was a layer of greyish-black beach sand (C22) (fig. 13). The layer was thickest close to the wall and in the east part of the trench. Towards the south it became loose and light grey, and below the clay platform (C48, see below), c. 5 m from the wall, it was very faint. This dark layer (C22) contained good quantities of Mesopotamian Ur III pottery (fig. 14) (cf. p. 93).

Fig. 8. Plan of Trenches C and J with drawn Sections 1-16 marked.



The layer also contained many oyster shells and large quantities of bitumen (cf. p. 210). The bitumen was found in large fragments – some with imprints of plaited palm-leaves, a few with more complicated constructions involving strings – as well as in smaller fragments and in completely disintegrated bituminous powder. This bituminous material may have contributed, together with charcoal and ash, to the dark colour of the layer.

Below the greyish-black sand was loose light grey sand containing numerous animal bones, bitumen pieces and some pottery sherds. A small pit, possibly a posthole, with a diameter of 16 cm extended 40 cm deep into the natural beach sand (fig. 15). South of wall C28, a 1 x 1 m area of the dark sand layer with

plenty of pottery and bitumen was noted to be very hard and also below this layer the sand was very compact and hard. No explanation was found for this phenomenon.

The north leg of wall C28 was covered by a dark grey sand layer, c. 1.80-2.00 m a.s.l., with frequent Ur III pottery sherds (fig. 16), perhaps evidence that the settlement continued after abandonment of the house.

Trench CW

In Trench CW, west of Trenches C1 and C3, large parts of the Mesopotamian Ur III layer were completely removed by stone plundering (figs. 17-18). The disturbance was filled with fist to head sized beach-rock



Fig. 9. Plan of Mesopotamian wall (C28) and shallow pit (C125-126) in Trench C.



Fig. 10. The Mesopotamian wall with corner, overlaid by the temple terrace floor, looking E, scale unit 50 cm (2010).

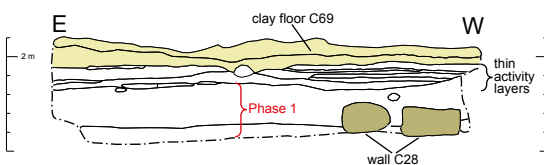


Fig. 11. Section 1 through temple terrace clay floor (C69), activity layers related to temple construction and Mesopotamian settlement layers and wall (C28). In Trench C3, looking SE.

stones and fragments of oolitic limestone in a matrix of compact clay (C143), probably the plundered remains of the later Temple, which was constructed from oolitic limestone ashlar (see below).

In the south of Trench CW, the stone plundering layer ended around c. 1.90 m a.s.l. and below that, a sequence of sandy layers appeared (C224/225/226) which contained bitumen lumps, copper fragments, animal bones, beads and Mesopotamian Ur III potsherds. Layers C224 and C226 consisted of whitish-grey, loose sand with many seashells, with C226 being more compact. Layer C225 was whitish beige,

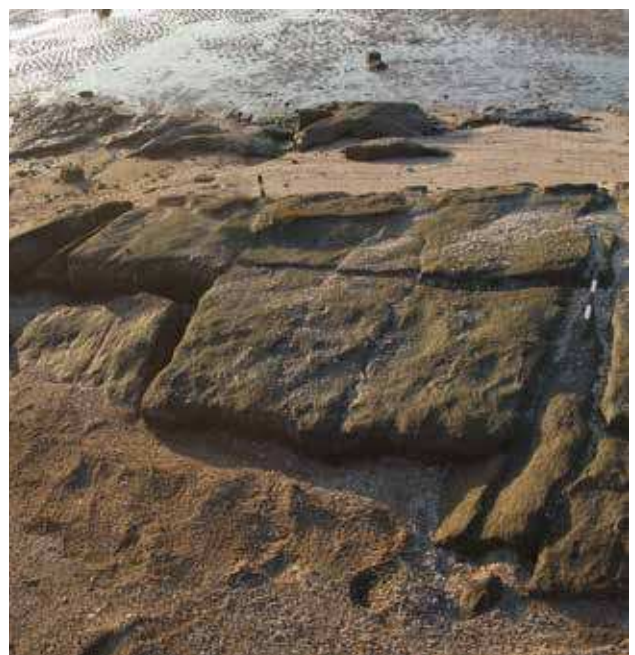


Fig. 12. Beach-rock formation (farush) with extensive traces of quarrying. Located by Massih (2013) along the beach of Failaka opposite Tell F3, scale unit 10 cm.

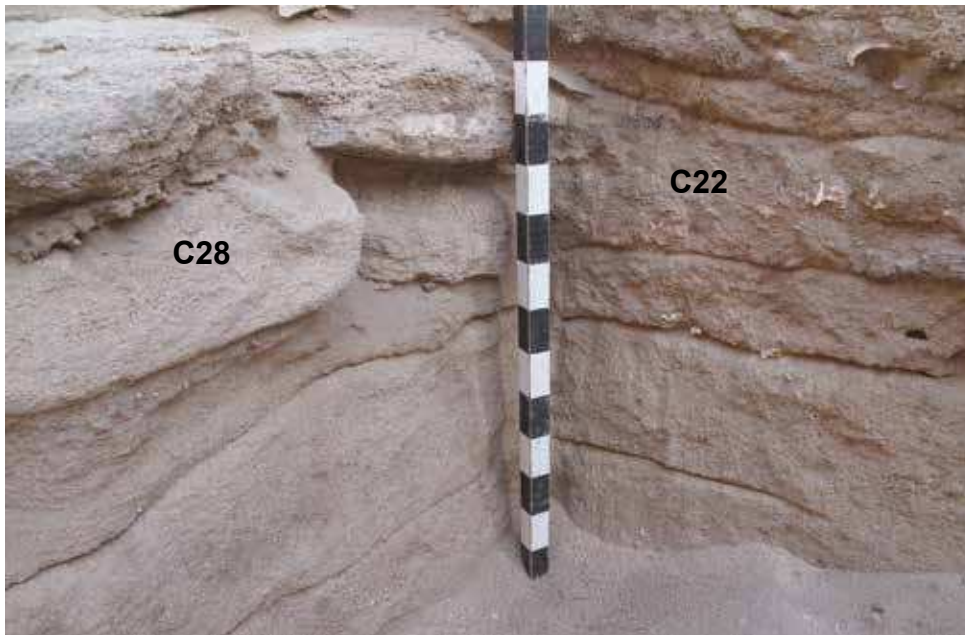


Fig. 13. To the right greyish settlement layer (C22) up against the Mesopotamian wall C28 (left). Trench C1, looking N, scale unit 5 cm (2009).



Fig. 14. Settlement layer (C22) with broken pottery bowl, assembled in fig. 172. To the left Mesopotamian wall (C28). Trench C1, from above, scale unit 10 cm (2009).

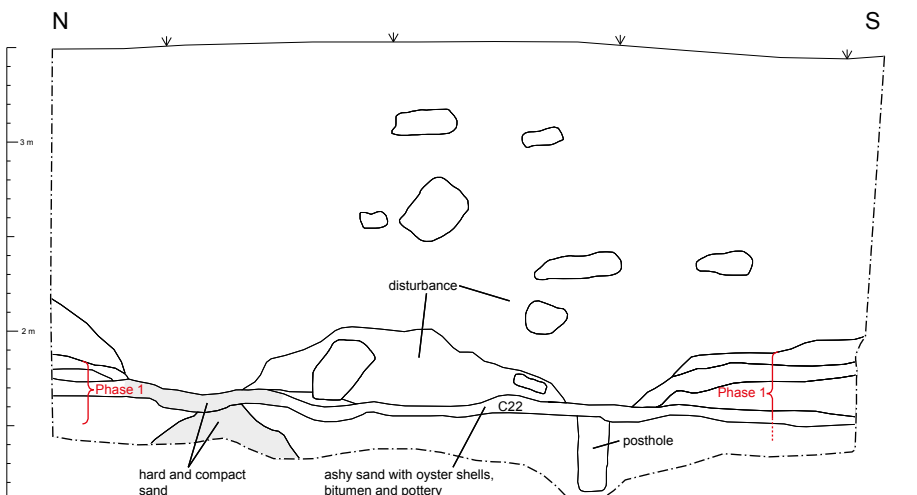


Fig. 15. Section 2, Mesopotamian settlement layer south of wall C28. Trench C1, looking NE.

Fig. 16. Section 3, Mesopotamian wall C28 and settlement layers overlaid by temple terrace stone floor (C68), clay floor (C69) and activity layers related to temple construction. Trench C3, looking NE.

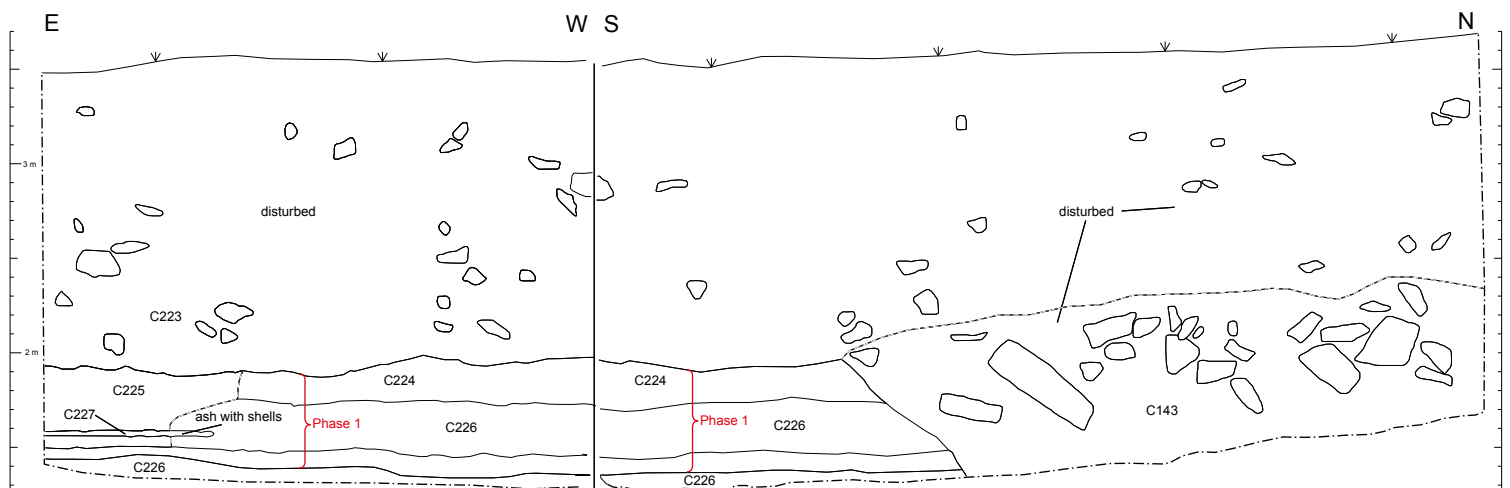
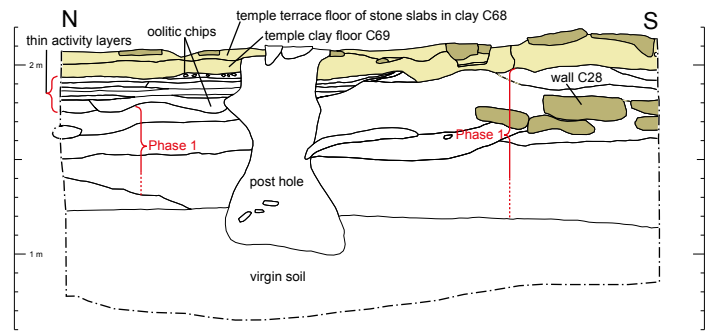


Fig. 17. Section 4, Mesopotamian settlement layers below stone plundering disturbance, cf. fig. 18. Trench CW, looking SE and SW.



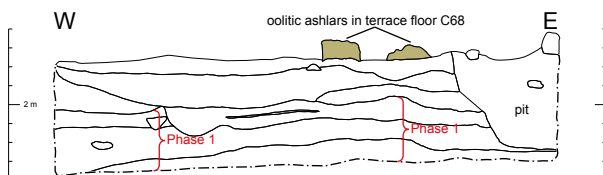
Fig. 18. Southeast and southwest baulk of Trench CW, cf. fig. 17. Mesopotamian settlement layers below stone plundering disturbance (2012).

loose sand, with only few seashells. The majority of finds came from layer C225, i.e. the southeast part of Trench CW, whereas finds were fewer in C224 in the southwest part.

In the south section of Trench CW a 5 cm thick and 1 m wide, dark coloured layer (C227) with burnt sea shells and oyster shells (apparently untouched by fire) appeared at level 1.60 m a.s.l. Southwest of this layer were only few finds in the sandy layers. Below the dark layer, the beach sand continued to contain animal and fish bones, bitumen, beads, and Ur III ceramics until level 1.35 m a.s.l. after which no further finds were encountered.

Trenches CE-CES

The house/courtyard wall (C28) that was first encountered in Trench C continued to the east into Trench CE. On the internal north side of the wall a loose greyish-brown sand with several thin, hard, horizontal layers of clay, separated by thin layers of ash (figs. 19-20) was encountered. The lowest of these activity layers were at 1.64 m a.s.l. and built up to a level of c. 2.00 m a.s.l. with each containing numerous animal bones, bitumen and copper fragments as well as Mesopotamian Ur III pottery. The colour was less dark grey than the layers south of the wall.



South of the wall was a thin layer of loose, dark brownish soil producing a large concentration of Ur III pottery, bones from sheep/goat and fish in addition to oyster shells. Part of a worked shell ladle (fig. 1122) related to the Indus culture was also found in this deposit. Below this layer was loose beach sand at a level of 1.59 m a.s.l.

The loose, dark brownish soil with the same contents of Mesopotamian ceramics, oyster shells and bones, continued in the south of Trench CE where it infilled a shallow pit (C125-126) (figs. 9 and 21). The pit was approximately 0.33 m deep, 1.3 m north to south and 2.3 m east to west. This pit truncated the loose beach sand, which was encountered at c. 1.20-1.24 m a.s.l. and was devoid of any traces of human activity. The pit did not continue into Trench CEE.

The pit contained a Mesopotamian cylinder seal (fig. 752) (1.46 m a.s.l.) together with two fragments of two different round Dilmun stamp seals (figs. 677 and 686), in addition to a complete square Dilmun stamp seal (fig. 747). The pottery was exclusively of Mesopotamian Ur III types and there was no Barbar pottery. Stone plundering disturbance goes very deep in some places in the south part of Trench CE, at least down to 1.35 m a.s.l. and therefore very close to the levels of the pit, but the pit lacked the heterogeneous character of a plundering pit (cf. p. 59).



Fig. 20. Section through temple stone floor (C68) overlaying Mesopotamian settlement layers. Trench C3-C1, looking NW, scale unit 10 cm (2010).



Fig. 21. Mesopotamian wall (C28), left, dark fill in shallow pit (C125-126), right. Trench CE, looking NE, scale unit 50 cm (2010).



Fig. 22. Foreground, section through Mesopotamian settlement layers with ash and shells. Background, plastered front (C8) of temple clay platform (C48). Trench CES, looking SE, scale unit 10 cm (2011).

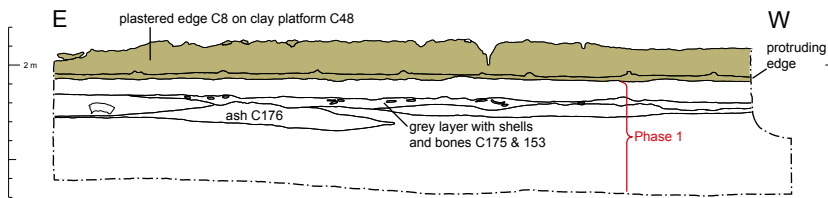


Fig. 23. Section 6, Mesopotamian settlement layers overlaid by plaster front (C8) on temple clay platform (C48), cf. fig. 24. Trench C1 and CES, looking SE.



Fig. 24. Section through Mesopotamian settlement layers below plaster front (C8) on temple clay platform (C48), cf. fig. 23. Trench C1-CES, looking SE, scale unit 10 cm (2011).

Further to the south, in Trench CES, was an undisturbed sequence of layers belonging to the Ur III period beginning at c. 1.90 m a.s.l. (figs. 22-24). A 2-5 cm thick layer of oyster shells (C175) was sealed by a dark grey layer (C153) with many burnt as well as unburnt shells and bones. Immediately below that was a compact ash layer, probably a fireplace (C176), c. 10 cm thick and 80 cm wide at a level 1.65-1.75 m a.s.l., containing Ur III potsherds. This in turn sealed several layers containing animal bones, bitumen lumps, beads and Ur III pottery.

Trench CEE

In most of Trench CEE the stone plundering activities truncated the archaeological sequence to a considerable depth and removed much of the Mesopotamian Ur III settlement layers. However, immediately east

of, and below the temple terrace floors (C68-69), i.e. within the house/courtyard wall C28, a narrow area was undisturbed. Two sections were cut through this sequence.

In the first section was a grey, approximately 10 cm thick layer containing Mesopotamian Ur III sherds (figs. 25-26). The layer sloped slightly from south to north, from c. 1.80 to c. 1.60 m a.s.l. Below this was an ellipsoid, gray layer in level c. 1.60 m a.s.l., containing bitumen lumps, many animal bones, beads and more Ur III sherds. Of special interest was a calcite stone vessel (fig. 771) found at a level 1.66 m a.s.l. Both these grey layers were positioned above the earliest floor level of the house/courtyard.

A sherd of another calcite vessel (fig. 774) came from level 1.38 m a.s.l., i.e. below the first floor. Four more fragments of the same vessel were found elsewhere.

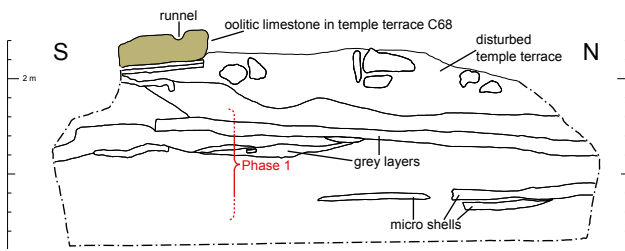


Fig. 25. Section 7, Mesopotamian settlement layers overlaid by oolitic limestone with runnel in temple stone floor (C68), cf. fig. 26. Trench CE and CEE, looking SW.



Fig. 26. Section through temple stone floor (C68) and underlying Mesopotamian settlement layers, cf. fig. 25. Trench CE and CEE, looking SW, scale unit 10 cm (2011).

The other section also featured two grey layers, the upper one was 6-8 cm thick, and the lower one, lighter grey in colour, was 8-20 cm thick (figs. 27-28). Both these layers contained Mesopotamian Ur III sherds and at least the upper one lay above the first floor level of the house/courtyard.

There were no traces, at all, of a continuation of wall C28 in Trench CEE. On the contrary, there was clear evidence that most of Trench CEE was dug through by stone plunderers well into the levels characterised by Mesopotamian Ur III pottery. It must therefore be assumed that the continuation of the wall was removed by that activity. The severe disturbance in Trench CEE may be due to it being located immediately in front of the presumed entrance to the later Temple, an area that would very likely have been architecturally elab-

orate and therefore an area of intense plundering. The disturbance of Trench CEE went as deep as c. 1.50-1.60 m a.s.l. and has thus removed all possible evidence of activity layers associated with this phase of the architecture.

An east-west section was cut through Trench CEE and into Trench CE, parallel with wall C28 (fig. 29). It revealed in Trench CE, below the wall, a dark layer with ash, charcoal and oyster shells (probably a fire-place) in level c. 1.45-1.58 m a.s.l.

In Trench CEE settlement remains were encountered at similar deep levels. At a level of c. 1.50-1.10 m a.s.l. was a concentration of pottery and a large amount of animal bones, primarily sheep/goat, a few bovine and very few fish. A greenish-black Mesopotamian cylinder seal was also found in this location (fig.

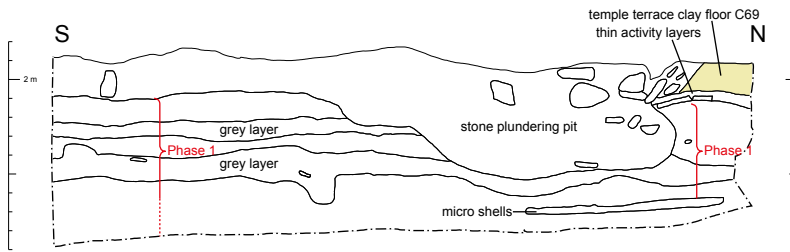


Fig. 27. Section 8, through Mesopotamian settlement layers overlaid by temple terrace clay floor (C69) and thin, activity layers relating to the construction of the temple, cf. fig. 28. Trench CE and CEE, looking SW.



Fig. 28. Section through, right, temple terrace clay floor (C69), thin activity layers relating to the construction of the temple and Mesopotamian settlement layers, cf. fig. 27. Trench CE and CEE, looking SW (2011).

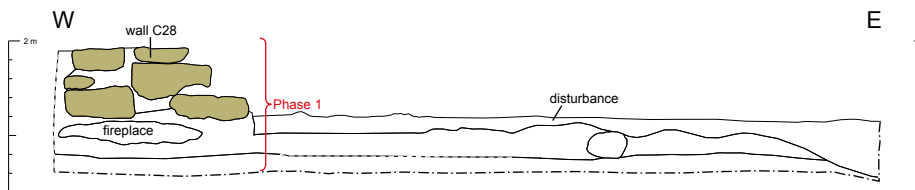


Fig. 29. Section 9, through Mesopotamian wall (C28) and settlement layers. Trench CE and CEE, looking NW.

753) at level 1.18 m a.s.l. There is thus clear evidence for substantial Mesopotamian activities before the wall was built.

A small concentration of beach rock stones were noted at level 1.5-1.2 m a.s.l. c. 1.5 m east of wall C28 with one of the stones covered with traces of melted bitumen (fig. 30). Nearby and in the same level were two upright stones, c. 15 x 25 cm, with five fist-sized beach-rock stones, some fire-cracked. The fill was slightly darkened between and around the feature. Three dark circular spots with diameters of c. 26-29 cm were noted in the light sand at c. level 1.10 m a.s.l. and tentatively interpreted as postholes.

The south baulk of Trench CEE revealed the complete obliteration of every settlement trace down to level 1.65 m a.s.l. where horizontal levels with Mesopotamian Ur III pottery appeared (figs. 31-32).

North part of Trenches CE/CEE

In the north part of Trenches CE and CEE, west of the presumed temple entrance (cf. p. 70), a 1 x 1.5 m sondage was excavated to expose the south face of the temple foundation wall as well as the deep trench in which this wall was constructed (see below p. 76 and figs. 100-102).

It was clear that the building trench cut through the Mesopotamian Ur III layers which originally continued towards the north. The fill in the trench (C208, C210, C218) had the same contents as the undisturbed Mesopotamian levels, i.e. Ur III pottery and a large number of animal bones, primarily goat/sheep, a few bovine and very few fish. Besides, a Mesopotamian cylinder seal (fig. 754), a Dilmun stamp seal (fig. 698), and a Dilmun cylinder seal (fig. 749) was found here. These objects were found with quantities of Mesopotamian Ur III pottery and not a single piece of Barbar pottery.



Fig. 30. Mesopotamian settlement layer exposed with simple stone settings and possible postholes. Trench CEE, looking SW, scale unit 10 cm (2011).

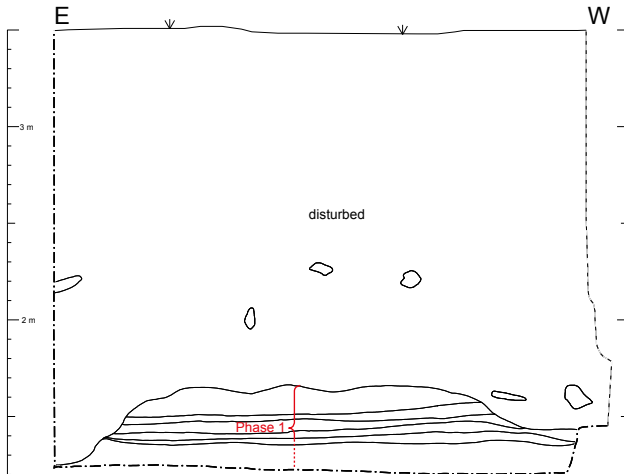


Fig. 31. Section 10, through Mesopotamian settlement layers, cf. fig. 32. Trench CEE, looking SE.

Trench J

Northeast of Trench CEE by a distance of 6 m was Trench J; this was excavated in order to test if the Mesopotamian Ur III level continued in this direction. Again there was considerable disturbance caused by stone plundering activity reaching down in the south of Trench J-A to c. 1.00 m a.s.l. (figs. 33-34) and in the south of Trench J-B even down to 0.67 m a.s.l.

There was evidence of Mesopotamian Ur III occupation across Trenches J-A and J-B. In all sections, a grey layer containing oyster shells and burnt seashells occupied between 1.70 and 1.45 m a.s.l. In the north of both trenches, the grey layer consisted definitely of hearths (J7/14), i.e. greyish ash with a consistency like powder containing lots of both burned and unburned shells and bones, bitumen, and Ur III potsherds.

Below the hearths and the dark grey layer were layers of sand with more Ur III potsherds, bitumen, a few copper fragments, and many animal bones. The finds terminated at c. 1.30 m a.s.l.

In the northeast baulk of Trench J-B, a stone-set pit (J16) was dug into the sand from the dark grey layer (J21) at level 1.52 m a.s.l. (figs. 35-36). The pit had its base at level 0.90 m a.s.l. It was built like a box of almost vertical beach-rock slabs set at right angles to each other. Three of the slabs were excavated and documented, and a fourth removed before the feature was identified. The three slabs were c. 40×40 cm and c. 10 cm thick. The volume of the pit was estimated to be c. 54 litres.

The fill of the pit was a blackish powdery material (J18), possibly bitumen (cf. Van de Velde below p. 214), mixed with some sand and small charcoal pieces and



Fig. 32. Section through Mesopotamian settlement layers, cf. fig. 31. Trench CEE, looking SE, scale unit 10 cm (2011).

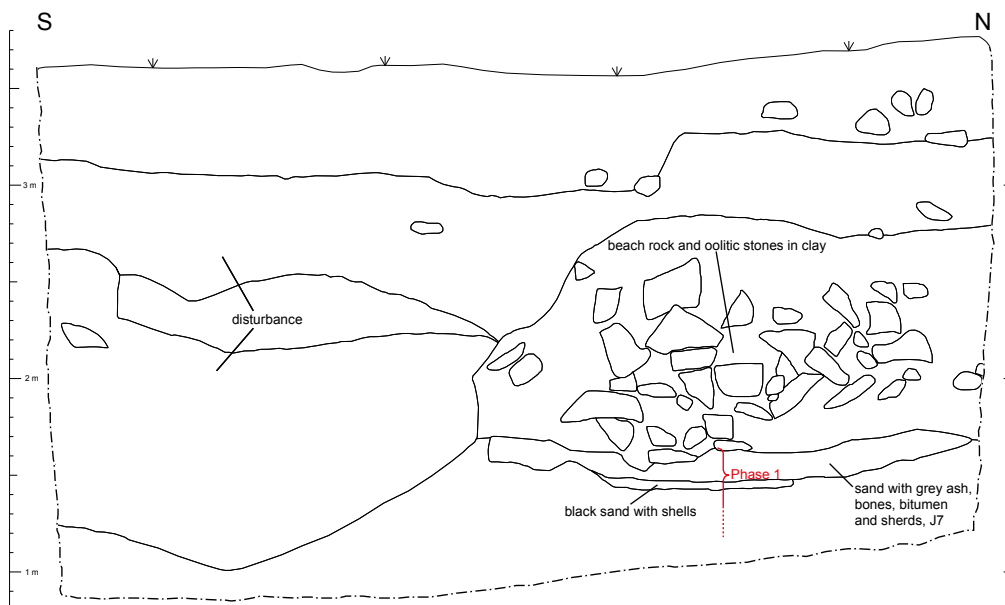


Fig. 33. Section 14 through Mesopotamian settlement layers overlaid by deep stone plundering, cf. fig. 34. Trench J-A, looking SW.



Fig. 34. Section through Mesopotamian settlement layers overlaid by deep stone plundering, cf. fig. 33. Trench J-A, looking SW (2012).

Fig. 35. Section through Mesopotamian settlement layers with stone-set pit (J16). Trench J-B, looking N (2012).



Fig. 36. Section through Mesopotamian settlement layers with stone-set pit (J16). Trench J-B, looking NE, scale unit 10 cm (2012).



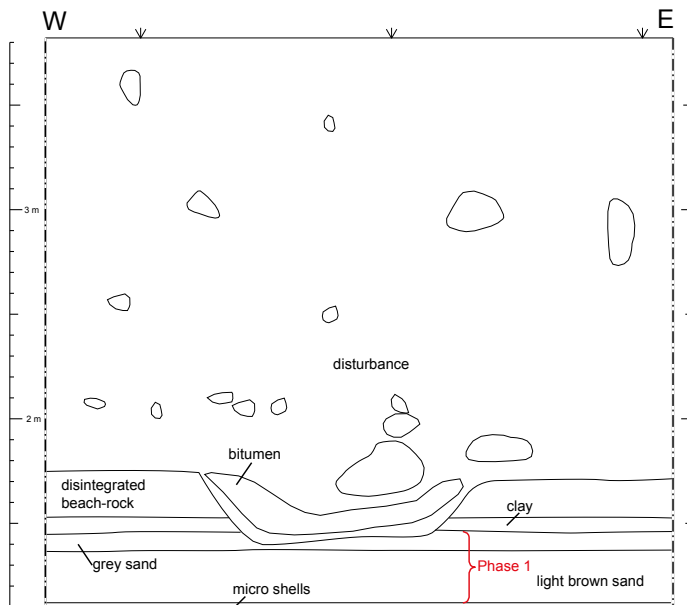


Fig. 37. Section 16, through Mesopotamian settlement layer overlaid by deep stone plundering. Trench I, looking N.

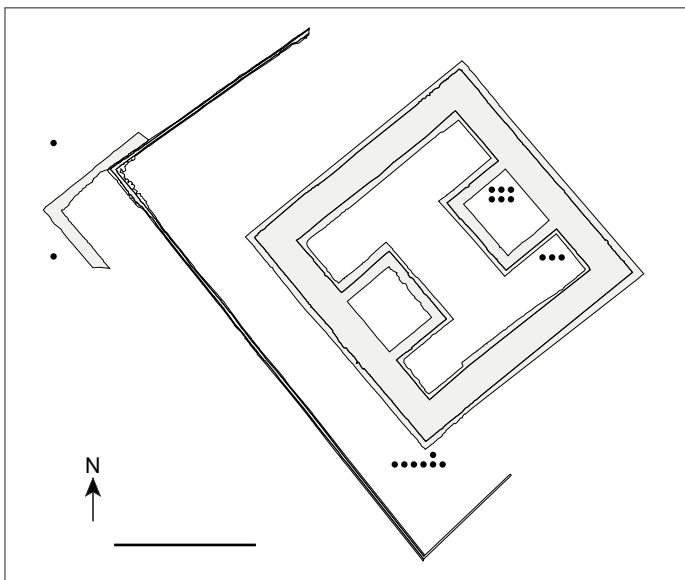


Fig. 38. Distribution of published Ur III potsherds from the French excavations of the Temple (Pic 2008).

several Ur III sherds. The inner sides of the beach-rock slabs were fire cracked, so high temperatures had been reached inside the pit, perhaps to heat up bitumen.

Trench I

A trial trench was excavated to the northeast of the Temple and revealed a grey sandy layer on top of a light-brown sand layer; both contained Mesopotamian Ur III pottery (fig. 37).

Mapping of the Ur III pottery published in the French excavation report (Pic 2008 figs. 15-24) indicates that this settlement originally extended over much of the

area later covered by the Dilmun Temple (fig. 38). The largest number of Ur III sherds was found in the deep trench at the south corner of the Temple, just north of our Trench C. Another centre of gravity for the Ur III sherds is the east part of the Temple.

West of the Temple, in Trenches A, E and H, and in the previous 1960-1963 excavations of the “Palace” (fig. 4), no Ur III sherds were found and in this location the remains from Phase 2 were built directly on the beach sand (see below).



Fig. 39. Trenches A, E and H, looking E. In foreground, SE corner of "Palace" of Phase 4, left, wall A219 of Phase 2. Centre, rooms 1-2, pavement E203, and oven E138/E143 of Phase 3. Scale unit 5 cm (2011).

2.2. Phase 2. The thick-walled buildings in Periods 1-2

by Aiysha Abu-Laban & Flemming Højlund

50 m northwest of the Mesopotamian Ur III settlement of the late 3rd millennium, two substantial buildings were erected directly upon the virgin beach sand in Trench A (figs. 39-40). Between and to the east of these buildings were a number of large hearths. Though heavily disturbed, the few finds recovered in relation to these structures leave no doubt that they belong to the Dilmun civilization. The bulk of the radiocarbon dates from Phase 2 lie in the interval around 1870 ±40 cal BC (figs. 5-6).

The following structures can be assigned to Phase 2: two walls and one piece of walling (A219, A191 and E230), four hearths (A224, E264, E263 and E216) and three floors (A222, A228 and A333) (figs. 39-40). All these structures were built on virgin beach sand, which survived to around the level of 1.00 m a.s.l. None of the structures could be fully excavated as they were partly covered by remains from the following building phases.

Wall A219 ran for more than 3 m in a south-west-northeast direction along the north baulk of Trench A (figs. 41-42). Towards the west, it was built over by the later "Palace" and towards the east, it disappeared into the baulk, and there may have been a corner here, where the wall turned towards the north-west, as it did not continue into Trench E.

The wall was 75 cm thick and constructed of local beach rock. Up to three courses were preserved. Its foundation was made of boulders measuring c. 50×70 cm in size. The two upper courses were built of roughly two rows of stones, c. 30×50 cm, up to 15 cm thick, with 10×15 cm stones set in between.

Wall A219 was torn down before the construction of a thin wall A204 belonging to the following Phase 3, as wall A204 rested directly on top of wall A219 (see below).

In the south of Trench A were the remains of a second substantial, compact wall in local beach rock, measuring 1.4 m north-south and 1.85 m east-west (A191) (fig. 43). The wall was torn down before the construction of the thin-walled Phase 3 building, which covered it towards the east (A213). Towards the west it was probably also damaged by a stone plundering pit (A37, cf. fig. 72).

The wall was preserved to a height of three courses with the foundation consisting of two large, up to 30 cm thick, irregular-shaped boulders, of which one was c. 45×110 cm, and the other c. 65×65 cm. These foundation stones extended 30-40 cm beyond the second

course. The upper two courses were built of c. 40×60 cm, rectangular or square stone slabs. The edge of the wall along the north side consisted of a row of c. 10-15 cm rectangular to square stones. An ashlar was found lying loose to the west of this walling.

It is unclear whether A191 formed the corner of a building, a platform or some other kind of walling, but the sheer size of it suggested that it was part of a monumental building of the same scale as wall A219.

A number of hearths shaped like shallow troughs, located between and to the east of the two thick walls described above were found in the same stratigraphic position: These hearths were also built directly on the natural beach sand and were sealed by Phase 3 architecture.

Between walls A219 and A191 was a sizeable hearth (A224) apparently built into a gap in a plaster floor (fig. 44). It was rectangular, 1.1 m wide and at least 1.3 m long, but it continued towards the east below the thin wall A204 from Phase 3, and it seemed also to continue towards the west below the wall of the later "Palace". In this direction it was, however, less well preserved, having probably been disturbed during the construction of the "Palace". The hearth was built as a 5-6 cm thick layer of clay in a depression in the sterile sand, and its outer edges were marked by a c. 10 cm thick, slightly raised rim of clay. Its outline was marked by a dark brown/reddish, burnt organic material.

The content of the hearth was a dark-brown mixture of ash and charcoal, with lumps of bitumen, plaster and clay, besides sherds of large storage jars, and copper fragments, but there was no clear indication of its specific function.

On both sides of the hearth, it was abutted by a 2-3 cm thin white plaster floor (A222), which was probably damaged during the erection of the "Palace".

Below the southern part of floor A222 was another more solid floor (A228) of plaster with small stones.

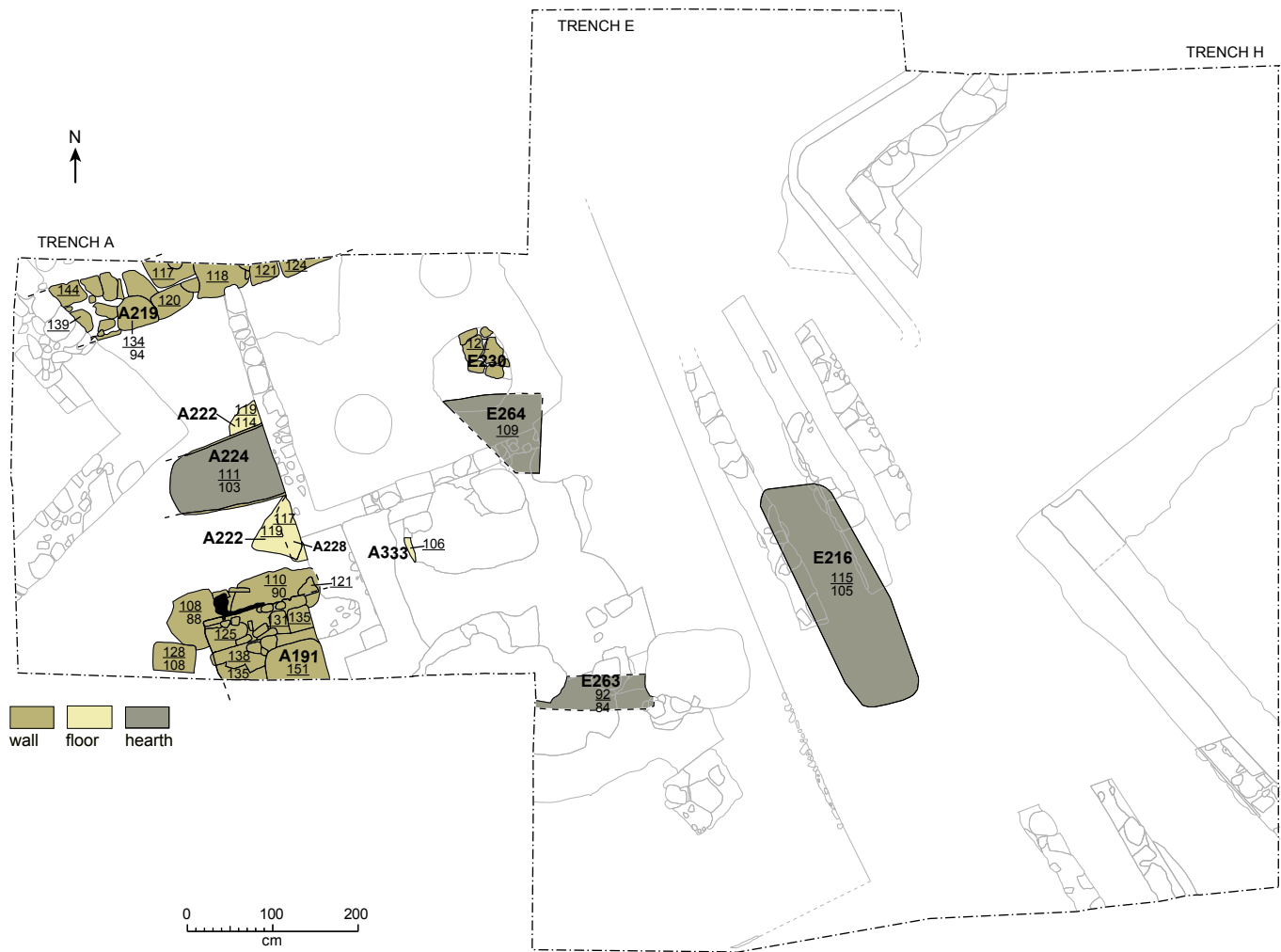


Fig. 40. Plan of structural remains in Trenches A, E and H dating from Phase 2.

A fragment of this floor was preserved close to the north side of wall A191, and in all likelihood the later floor A222 also reached up to wall A191 and perhaps also to A219.

A little further east a floor (A333) of coarse white plaster was identified below stone plundering pit A85. It was almost on the same level as floors A222 and A228 and therefore they are probably connected.

A piece of walling (E230) appeared in the side of a pit (E180) cut through floor A203 belonging to the following phase. The walling was built of mostly irregular beach-rock stones, 15 x 20 cm in size. The construction was covered with sand on to which floor A203 was laid. The nature and extent of E230 is unclear, but some stones noted 1.5 m further east, also below floor A203, may be a continuation of E230.

Close to E230 was a hearth (E264), again situated below floor A203. Only a little of this hearth was visible in pit E180 and from a section of the southeast corner of floor A203 (fig. 45). Hearth E264 seemed to

be oriented east-west with a width of almost 1 m and a length of at least c. 1.2 m.

Approximately 2.4 m further south was another hearth (E263) laid on virgin beach sand and covered by floor E196 and stone feature E262 of Phase 3. Only a small part of the hearth was excavated. It was oriented in an east-west direction with a length of c. 1.4 m. Its west end was destroyed by a stone plundering pit (E58). Its width was uncertain, but no more than 80 cm.

Further to the east a fourth hearth E216 was discovered at the same level and in the same sandy context as the above-mentioned hearths (figs. 40, 47, 50 and 88). E216 was estimated to have been c. 90 cm wide and 2.90 m long. Its content was a mix of greyish and white powder-like ash. It is remarkable that directly above this hearth an oven, E138/E143, was built in Phase 3, with the same orientation as the hearth (i.e. northwest-southeast) (fig. 46) (see below).

As mentioned above, the features of Phase 2 were constructed directly on virgin soil (A227) (figs. 81 and



Fig. 41. North baulk in Trench A, cf. fig. 81, with left, "Palace" wall; back-ground, wall A219 from Phase 2; centre, wall A204 and floor A203 from Phase 3; and above wall A104 from Phase 7. Looking N (2009).



Fig. 42. From above, right, "Palace" wall A40, middle, wall A219 from Phase 2 and left, wall A204 and plaster floor A203 from Phase 3. Looking S, scale unit 5 cm (2009).

84), which appeared around level 1.00 m a.s.l. as light yellowish-brown sand with small shells (beach sand), completely sterile. In a sounding in the south-west corner of Trench A this layer was followed down to 0.75 m a.s.l., and it continued to be devoid of evidence of human occupation. The layers A225 and A226, lying above A227, contained pottery, animal bones, copper fragments and bitumen (fig. 84).

Bitumen was abundant in some of these early levels (1.30-1.00 m a.s.l.), up to layer A188, especially in the south and south-west of Trench A (fig. 84), covering an area of about 10 m² (A215, A220, A221 and A226). It occurred in lumps up to 70x30 cm, often with a lot of stone and plaster fragments and some lumps of green clay, which also characterised the first heavy floor east of the "Palace", A180 (see below). The largest bitumen

Fig. 43. Wall A191 from Phase 2. Trench A, looking S, scale unit 5 cm (2009).



Fig. 44. From above, hearth A224 and plaster floor A222 from Phase 2. Left, wall A204 from Phase 3, and right, "Palace" wall A125 of Phase 4. Trench A, looking S, scale unit 10 cm (2009).



pieces were found close to wall A191, but it is not clear if bitumen was used in the construction of the wall. Perhaps the thin bitumen layers are remains from coating the roofs of the two great buildings (A191 and A219).

In a section below wall A125 of the later "Palace", thin layers of bitumen were observed in the sand of A226 around level 1.00 m a.s.l. and continued 1.50 m away from the wall. The same was observed elsewhere below the "Palace" in layers A238 and A239.

The thick-walled building remains from Phase 2 and the layers containing bitumen were greatly disturbed during the later construction of the thin-walled buildings of Phase 3 and again during the construction of the "Palace" of Phase 4. This resulted in amorphous layers of stone and bitumen covering walls A191, floor A222 and hearth A224 from Phase 2, as well as wall A213 and parts of walls A190 and A204 and floor A217 from Phase 3.



Fig. 45. Hearth E264 from Phase 2, indicated by arrows, sectioned below wall A190 and adjacent floors A203 and E236 from Phase 3. In background, "Palace" corner. Trench E, looking W (2012).

Assigning finds like pottery, stamp seals, beads, and animal bones with certainty to the phase of the thick-walled buildings was difficult as almost all accumulations of floor layers and accompanying occupation deposits were obliterated by the demolition of the buildings and the following rebuilding activities. Only the few finds found in the hearths and directly on the floors or directly below them as well as in the layers between the walls (E225 in hearth E264, E231 below E264; E246 fill in E263; fill in A224) can be referred to this phase with certainty.

Below the north-west part of floor A203 and below the north end of wall A204 was a sandy soil with some pottery, fish bones and copper artefacts (A235, A336, A338), probably belonging to Phase 2, but used to level the area before the construction of Phase 3.

Evidence of Phase 2 was limited to Trenches A and E, whereas there were no signs of this phase in Trench H or further east.

The limited amount of pottery found in Phase 2 can be referred to Failaka Periods 1-2 (p. 95, figs. 274-296). No Mesopotamian Ur III pottery was found in Trenches A, E, and H.

Despite the lack of direct stratigraphic relationship between the Phase 2 and Phase 1 deposits with both lying directly on beach sand, the very clear difference in the pottery of the two phases, coupled with the radiocarbon dates confirm that the Phase 2 occupation is later than Phase 1. The two phases may be separated by more than a hundred years.

2.3. Phase 3. The thin-walled building in Periods 1-2

by Aiysha Abu-Laban & Flemming Højlund

After the thick-walled buildings were demolished, the area was occupied by a thin-walled building with several small rooms fronted to the east by a pavement and a sacrificial fire installation (figs. 39 and 46). The evidence indicates that these buildings were constructed at the same time as the first terrace around the Temple. The associated material culture belongs to the Dilmun civilization. The radiocarbon dates tightly constrain Phase 3 to 1800 ±30 cal BC (figs. 5-6).

The thin-walled building

A building with at least three rooms, oriented north-west-southeast, was uncovered in Trenches A and E (figs. 39 and 46). The walls were badly preserved as they had been torn down to make space for the subsequent construction of the “Palace”. In some areas the stones had been completely removed, leaving only the clayey mortar and never more than one course. The walls were c. 25 cm thick (A204, A190 and A213) and built of one, occasionally two rows of beach-rock stones set in a very hard clayey mortar mixed with small stones and oolitic limestone chips. The largest stones were typically rectangular, up to 20×30 cm, whereas the smaller stones were squarer. There was evidence of plaster coating on the vertical faces of the walls. The oolitic limestone chips probably originated from the cutting of the ashlar used to build the Temple.

Room 1 was delimited by a southwest wall (A204) and a southeast wall (A190). The northeast wall had been removed, and the northwest wall must lie to the north of Trench A. Approximately 9 m² of the room was uncovered.

Wall A204 was 3.05 m long, and its north end rested upon wall A219 from Phase 2, as described above. Only a few stones of wall A190 were preserved and it was c. 2.70 m long and completely damaged in its east end by stone plundering pit E17. There were no stones preserved in the south end of A204 and in the west end of A190, i.e. where the two walls joined. The original presence of a wall was, however, indicated by the fact that the plaster floor inside the room continued a couple of centimetres up as if it lipped up against the vertical face of a wall.

The floor of room 1 (A203) stretched 3.25 m east-west and at least 2.7 m north-south. It was up to 5 cm thick and very solid with a basal layer of plaster with a high density of spherical shaped gravel, coated with a fine white plaster. The floor was slightly higher towards

the east (level c. 1.40 m a.s.l.) than the west (c. 1.30 m a.s.l.) and continued into the north baulk of Trench A.

Three circular pits (A210, E170 and E180) truncated the floor. A possible fourth pit in the north-west corner of the room was damaged by stone plundering pit A26. The pits were filled with brown soil mixed with fragments of plaster and stone, including oolitic limestone chips, charcoal and animal bones.

In the northeast corner room of the “Palace”, i.e. inside the building, the floor was broken by a similar pit found to contain a large storage vessel sunk below the floor level (Kjærum & Højlund 2013 p. 123, fig. 171). Pit A210 may also have originally contained such a vessel, but pits E180 and E170 were too shallow with depths of c. 20 cm and 10 cm respectively.

East of the plaster floor of room 1 was a heavily disturbed area where the east wall of the room probably once stood and was fronted by a pavement (E203). The wall may have been removed in connection with the making of floor A180 of the next Phase 4 as well as by stone plundering pits in the area.

Room 2 was bordered to the north by wall A190, described above, and to the west by wall A213. Wall A213 was c. 1.75 m long, but badly preserved. Only one complete stone and small fragments of stones were left intact. The existence of the wall was indicated by the clayey foundation for its first course of stones, a material identical to that found in both walls A190 and A204.

The east part of room 2 was destroyed by the same stone plundering pit, E17, that damaged room 1, but the room was probably delimited by an east wall in continuation of the east wall of room 1. East of this position, the pavement continued to front the house.

Approximately 6.3 m south of the north wall of room 2 the outer edge of the pavement turned sharply to the west suggesting the position of a south wall in

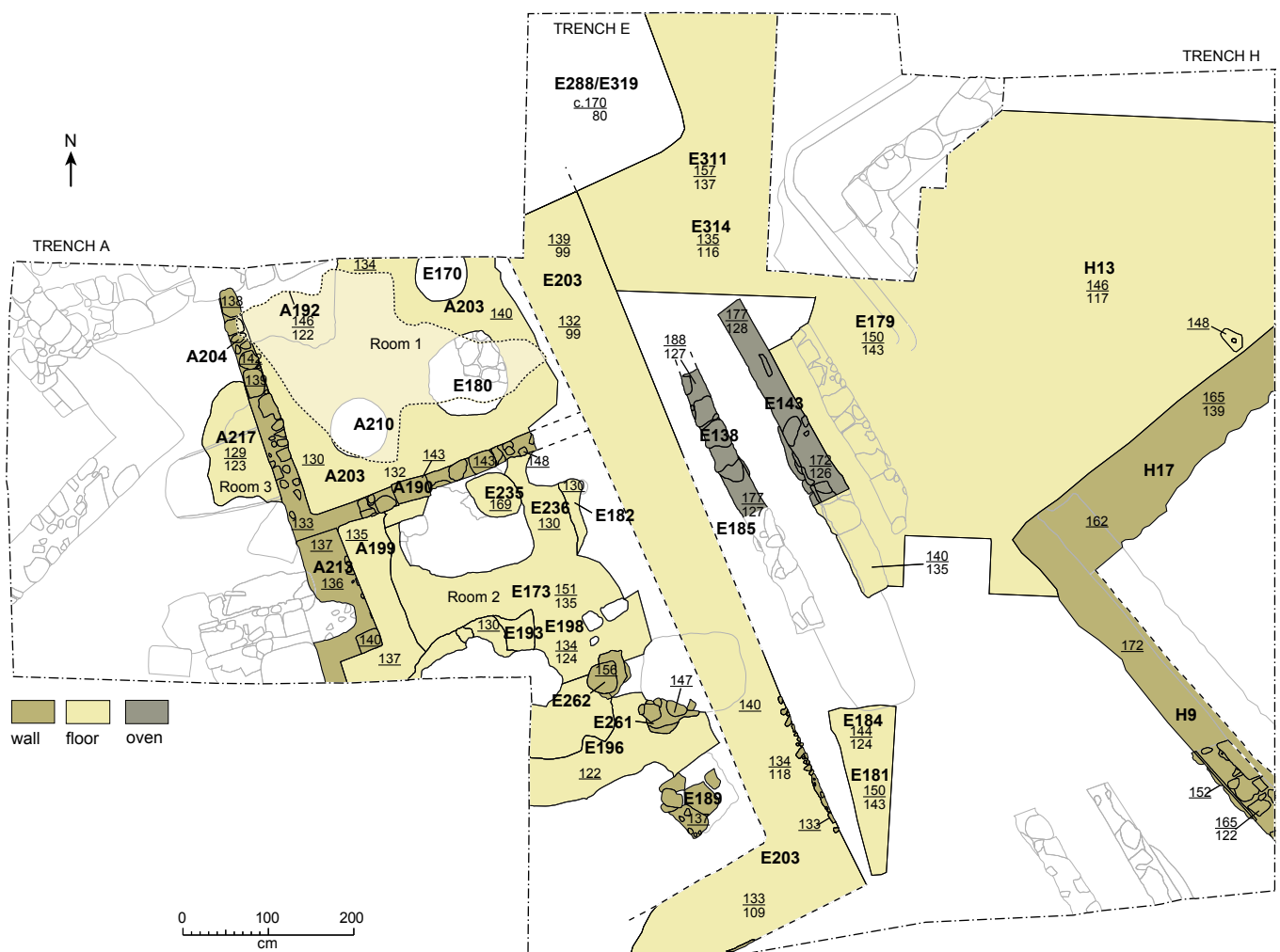


Fig. 46. Plan of structural remains in Trenches A, E and H dating from Phase 3.

the house bordered by the pavement. This would have given the room an internal area of c. 17 m² but the possibility that the room was divided in two cannot be discounted.

The earliest floor in the room (E236) was made of a white, gravelly plaster, similar to the foundation of floor A203. Floor E236 was heavily damaged by stone plundering pit A85 but could be followed 3 m south from wall A190. The eastern edge of the floor (E182) was made of a fine-grained solid plaster. Wall A190 was actually built on the plaster floor of E236.² Towards the east, the floor rested on a compact brown soil, whereas in the north it covered hearth E264 of Phase 2. Towards the south, a floor of a less solid white plaster (E193) was identified. Not much of this floor was exposed, and part of it was damaged by stone plundering pit E58. The level of these floors (1.30 m a.s.l.) suggests that they belong to the earliest phase of the room.

Several floors, E198 and E173 (1.51-1.35 m a.s.l.), were built upon E236, separated by occupational build-up layers E188, E246 and E194. The floor resurfacing layers were made of clayey plaster and were not as solid and compact as the primary surface E236. These floors extended at least half a meter further south and were limited in the east by pavement E203 and stone features E261 and E262. The southern extent of the floors (i.e. E196, E198 and E173) was unclear because the southwest part of Trench E was heavily damaged by a huge clay pit and stone plundering pits which went as deep as 1.25 m a.s.l. (cf. below p. 59). Remains of a damaged plaster floor just east of wall A213 (A199, 1.35 m a.s.l.) was on the same level as E198.

The wall remains in the south end of Trench E (E262, E261 and E189) were much damaged and therefore difficult to reconstruct. Stone feature E262 consisted of irregularly shaped, large beach-rock stones (up to 60 x 30 cm) laid in clay, and floors E173 and E198

Fig. 47. Southern stretch of pavement E203, its eastern edge against the sand. Behind pavement two wallings, E189 and E261. Right, south end of oven E138/E143 with hearth E216 below. Trench E, looking W, scale unit 10 cm (2011).



belonging to Room 2 abutted both this stone feature and the following.

South of E262 was another section of walling, E261, of which only the southern limit was well-preserved (fig. 47). E261 and E262 may originally have been part of the same construction, disturbed by stone plundering (E53).

Wall E189 was built of square stone blocks at least 20 cm thick and had an almost square outline (fig. 47). Only its western limit was well preserved, and it may have extended further eastwards and been part of the east wall of the room, though it was more sturdily built than normal for walls in this house. The distance between walling E261 and E189 and their placing at the probable east wall of room 2 suggested a possible entrance to the room. The width of such a doorway (c. 42 cm) was, however, a little less than normal for the settlement in Tell F3, where doorways varied between 50 and 60 cm (Kjærum & Højlund 2013 p. 23).

West of Room 1 a white plaster floor (A217) abutting wall A204 indicated the presence of a third room. The floor was 2-3 cm thick (1.29-1.23 m a.s.l.) and stretched north-south for 1.5 m and east-west for 80 cm. The western end of the floor was damaged, probably in connection with the construction of the "Palace". It covered the hearth A224 and two floors A222 and A228 from the previous phase.

In the north of Trench A was a series of brownish, rather compact sandy layers (A206, A218 and A223), covering a surface of about 5 m² and characterised by a high concentration of fish bones, pottery and copper fragments. The layers were all together 20-40 cm thick (1.50-0.95 m a.s.l.). They covered part of wall A204 and were interpreted as settlement remains from Phase 3, which were levelled down before constructing the "Palace" walls A40 and A125 and floor A180 of Phase 4.

Occupation layers on the floors of the thin-walled building were preserved to a level of 1.51 m a.s.l., but a higher level of build-up may have been removed by later construction. Finds consisted of potsherds, stamp seals, stone vessel fragments, beads and animal bones. The pottery found in Phase 3 can be referred to Failaka Periods 1-2 (cf. 95, figs. 297-380).

The pavement

Immediately east and south of the thin-walled building was a c. 40 cm thick pavement (E203) (fig. 47). It was very disturbed in the area where it appears in the main east-west section (fig. 81), but at other places (e.g. fig. 88), its upper level was measured to 1.30-1.40 m a.s.l. and its bottom level to c. 1.00 m a.s.l. The pavement had a width of 0.8-1 m and extended c. 8.8 m north-west-southeast. The south part of the pavement could

be followed almost 3 m in an east-west direction. The pavement was constructed directly upon natural sand and consisted of beach-rock stones, oolitic limestone chips, lumps of clay and fragments of plaster, which originated from demolished architecture, everything bonded together by a compact, light brown clay. As far as can be seen, the damaged character of the east part of the thin-walled building taken into account, the pavement must have been built up against the east wall of this building.

The border of the pavement against the sand was a c. 12 cm high, vertical, relatively smooth edge made of larger, up to 40x30 cm stones set in hard clay and coated with clay (fig. 48). This edge could be followed with intervals along the east as well as the south side of the pavement. Towards the north the pavement was cut by a pit (E288/E319) (see below), so its original termination in this direction is unknown. In all likelihood, the pavement was built in order to create a solid surface upon the loose beach sand outside the thin-walled building. Part of the pavement and the east wall of the thin-walled building were probably levelled in connection with the construction of the "Palace" floor A180 in Phase 4.

The northeast side of the pavement (E203) met floor E314 that extended from the temple terrace, in a way that suggested contemporaneity (see below p. 73-74).

Further south an oven (E138/E143) was built between the pavement and the floors outside the temple terrace (see below). Between the pavement and the first oven wall, E138, was an up to c. 40 cm wide space filled with loose yellowish brown sand with a few stones (E185). In the area east of the south part of the pavement were a few traces of surface (E184 and E181).



Fig. 48. East edge of pavement E203 against the sand. A broken pottery vessel (fig. 314) was found up against this edge. Trench E, looking W, scale unit 10 cm (2011).

The oven

East of the pavement was a large oven built of two parallel walls (E138 and E143) separated by a distance of 60 cm. This was constructed parallel to the thin-walled building (figs. 39, 46 and 49). The south ends of the two walls were intact but their north ends were disturbed by stone plundering pit E24. Their preserved lengths were 3.35 m (E138) and 3.8 m (E143) respectively, and their original lengths were probably around 4 m. The walls were between 30 and 44 cm thick and mainly built of local beach-rock with a few oolitic limestone chips set in liberal amounts of fine grained, light brown clay. The same material was applied as a thick cover to both the inner and outer wall faces. A few of the beach-rock stones were rectangular, c. 30x20 cm, but most were irregular and half the size.

The heavily burnt stones and clay found inside the fill probably originated from the walls and a possible roof over the oven. The lowermost layer in the oven, situated upon beach sand, consisted of clay with stone inclusions, possibly constructional remains or a floor of the oven.

The clay applied to the inner faces of the walls was fired reddish, and the stones in the walls were fire cracked, including those lying on top of the deposit between the walls, so it is evident that high temperatures were reached between the two walls. A roof or vault may have bridged the walls creating a long and narrow oven. In front of the south entrance and on both sides of the walls were extensive layers of ashes raked out of the oven.

The fill inside the oven consisted of layers of whitish to greyish, sometimes hard, compact, often powdery ash (E157) and some calcined animal bones as well as stones and clay which had also been exposed to high temperatures (fig. 50). Animal bones were scarce in the lower part of the fill and more common in the upper layer where seven, probably male, goat horn cores were found (preliminary identification by Uerpmann 2009, cf. Yeomans below p. 228). It seems that the oven was used repeatedly over a long time, and that bone and perhaps other material over time was burnt into a whitish-grey powdery material, so that only the bones from the last uses of the oven survived relatively intact in the upper levels. Quite possibly this installation was used to destroy sacrificial remains from the nearby Temple (cf. below p. 253-254).

The oven had two construction phases (fig. 49); the earliest was built at a level of 1.27 m a.s.l., with six courses preserved to a height of 56 cm (fig. 46). In this phase, E138 was preserved to a length of 1.80 m, and E143 was 2.55 m long. Later the walls were extended to

Fig. 49. Oven E138/E143 emptied of its ashes, and parallel wall E164. The two phases of wall E138 are visible. Trench E, looking W, scale unit 5 cm (2011).



Fig. 50. Section through south end of oven E138/E143 with contents of ashes and underlying hearth E216, cf. fig. 88. Trench E, looking NW. scale unit 10 cm (2011).



the south from a height of 1.42 m a.s.l. and preserved to a height of 42 cm (fig. 53). The walls of the late phase additions were 1.5 m long (E138) and 1.25 m long (E143). They were built in larger rectangular boulders, the largest measuring c. 15x60 cm.

The loose yellowish brown sand (E185) that filled the gap between the oven and the pavement to the

west gave no stratigraphic clues to their relationship. The late phase of the east wall, E143, however, rested on floor E179 at a level of c. 1.40 m a.s.l. (figs. 50-51 and 88). This floor covered the area to the east and to the north of the oven and was covered with ash. The floor appears in the east-west main section through Trenches A and E (figs. 46, 52 and 81) as the upper



Fig. 51. In background oven wall E143. In foreground, section through the thick floor E179 consisting mostly of plaster fragments, overlain by wall E164. Trench E, looking W, scale unit 10 cm (2012).



Fig. 52. Main section through Trench E showing two floors, E311 and E314, separated by sand layer, looking N, cf. fig. 81, scale unit 10 cm (2012).

phase of a solid floor that linked the thin-walled building and the pavement with the first temple terrace (E311, 1.57-1.37 m a.s.l.) (cf. p. 73-74). The early phase of this floor (E314, 1.35-1.16 m a.s.l.) can probably be related to the early phase of the oven, which had a

basal level of 1.27-1.28 m a.s.l., although a stone plundering pit (E24) destroyed their exact relationship.

Between the late phase walls was a 6 cm thick clayey layer at the bottom of the walls, at least 70 cm long (E227, 1.45-1.39 m a.s.l.). This clayey surface was

also noted in the north part of the oven (E183, 1.48-1.35 m a.s.l.), covering a 10 cm thick layer of ash and clay lumps originating from the early phase of the oven. The ash deposits in the oven reached almost up to the top of its walls (E144, 1.77 m a.s.l.).

It is noteworthy that the "Palace" floor A180 and its extensions belonging to the following Phase 4, covered the pavement E203, but did not extend further east to cover the oven E138/E143. Taking into account the radical and systematic demolition of the thin-walled building below the "Palace" floor, it is significant that the walls of the oven reached 25 cm above the level of this floor. It must be concluded that the oven continued to be in use after the construction of the "Palace" and the accompanying floor, A180, i.e. Phase 4, and even into Phase 5 (see below p. 50-51).

Below oven E138/E143 and with the same orientation was an earlier hearth (E216), described in detail

above (p. 32), stratigraphically belonging to Phase 2 (fig. 40). The hearth was situated between 1.15 and 1.05 m a.s.l., and there was a layer of sand between the hearth and the oven (figs. 50 and 88). E216 extended c. 1.20 m south of the southern limit of E138/E143 (fig. 47). Outside the oven it was covered by sandy soil with a few clay chunks (E208 1.40-1.10 m. a.s.l.).

The hearth had the same shape, size and stratigraphic position as the other hearths described above for Phase 2. It is remarkable that the oven was constructed immediately above this hearth. The same situation possibly occurred in the case of another oven, H4/H5, excavated close by and described below p. 74. It could be evidence of a continuity of function between the two phases.

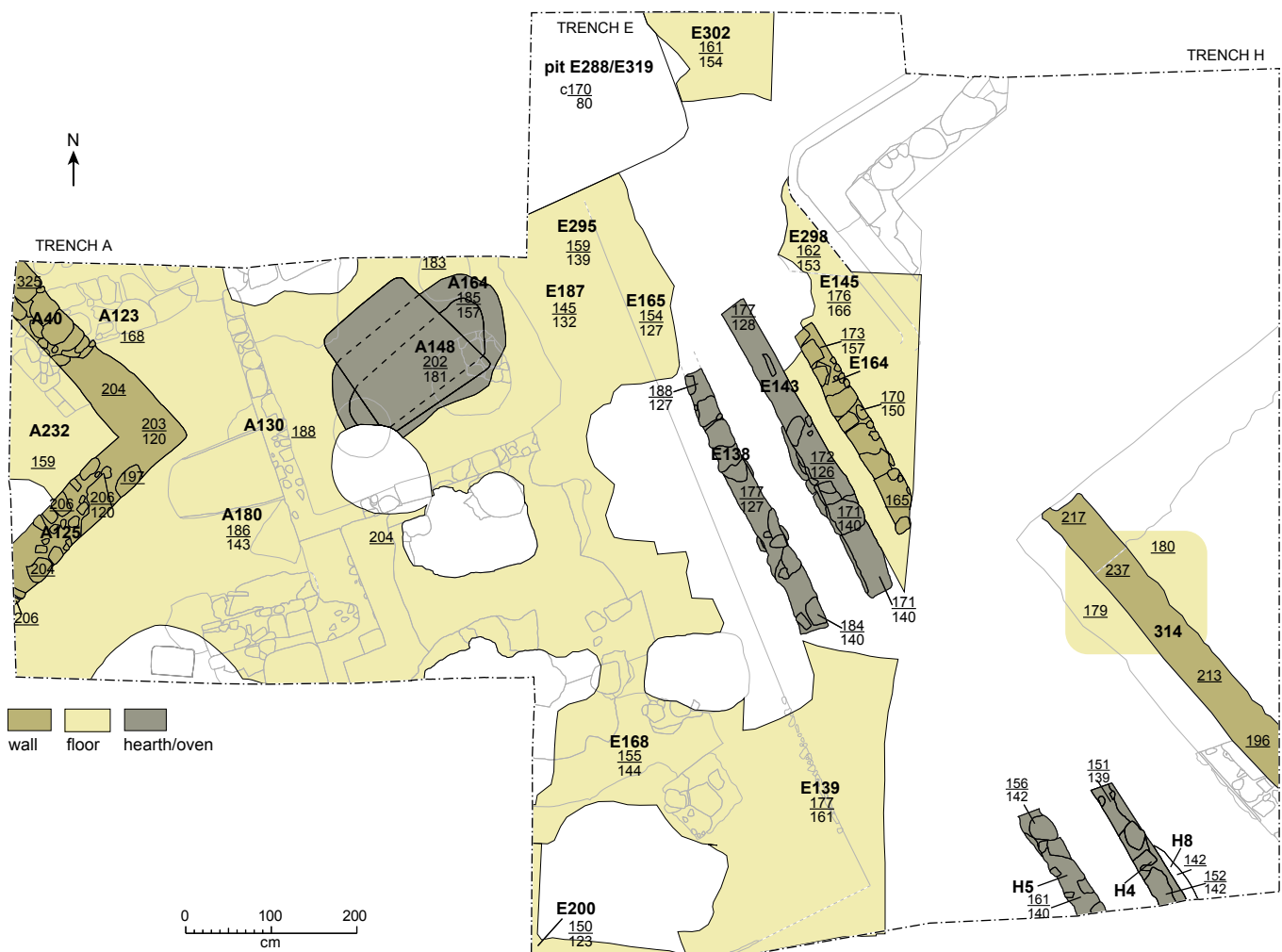


Fig. 53. Plan of structural remains in Trenches A, E and H dating from Phase 4.

2.4.1. Phases 4-5. East of the “Palace” in Periods 1-2

by Aiysha Abu-Laban & Flemming Højlund

After the demolition of the thin-walled building, the area was occupied by a large structure, termed the “Palace” (figs. 53-54). The major part of this building was excavated in 1960-1963 (Kjærsum & Højlund 2013), but the first trench in 2008, Trench A, was positioned to incorporate its still unexcavated south-east corner. Outside the “Palace”, a heavy floor was established and it extended over much of Trenches A and E. On top of this floor a c. 2 m thick series of mostly thin, horizontal deposits – floors as well as refuse layers – accumulated which can be dated to Failaka Periods 1-2, 3B and 4A, during which periods the inner “Palace” was also inhabited. The layers dating to Periods 1-2 are dealt with below. Their material culture belongs to the Dilmun civilization. Phases 4-5 are radiocarbon dated to lie between boundaries of 1800 and 1690 cal BC at 95% probability. Phase 4 has a boundary start of 1771 ±15 cal BC and a boundary end of 1762 ±16 cal BC. Phase 5 has a boundary start of 1751 ±19 cal BC and a boundary end of 1732 ±21 cal BC (figs. 5-6).

The “Palace” corner

The southeast corner of the “Palace” consisted of two walls, A40 and A125 (fig. 53). Wall A40 was oriented northwest-southeast and was exposed in Trench A for 2.8 m (fig. 54). It was preserved to a height of 1.82 m, but was disturbed by one or several stone plundering pits (A64). It was built directly on wall A219 from Phase 2, but in its south end it rested on sand mixed with bitumen and hard-packed clay.

Wall A40 had originally a thickness of 55-65 cm, 65 cm being the typical thickness of the outer “Palace” walls (Kjærsum & Højlund 2013 p. 101), but a shell of beach-rock stones, set in clay, was added to the inner side of the wall, increasing its thickness to c. 80 cm (fig. 55). This later addition rested on the oldest floor inside the building, A232 (fig. 84).

In addition, the outside of wall A40 was at some point strengthened by a buttress-like addition, A178 (fig. 63), consisting of beach-rock stones laid in a hard brown clayey mortar. It was built at the level of c. 1.70 m a.s.l. on a c. 20 cm thick layer of hard packed clay with small stone inclusions.

The upper courses of wall A40 were built of rectangular shaped stones, which measured c. 40×40×20 cm with small stones filling the gaps. The stones were set in a light brown clay mortar with the lower courses more irregular in shape and size.

On the original inner face of wall A40, a plaster-coated area was identified at level 2.15 m a.s.l. The outer face of the wall also had a plaster coating with marks of finger strokes. It was only partially preserved, but at one point, the plaster protruded quite clearly at 1.64 m a.s.l., which corresponded to the level of the first floor outside the “Palace”, A180, where it abutted the wall.

Wall A40 was at right angles to wall A125, which ran in a northeast-southwest direction and was exposed in Trench A for c. 2.90 m (fig. 54). It was preserved to a height of c. 0.86 m, at 2.06 m a.s.l. The bottom level was 1.20 m a.s.l. The wall had a maximum width of 55 cm and was built of two rows of stones with smaller stones between and was preserved up to 6 courses in height. The individual stones were not all visible because of the plaster coating on the outer face, but the three lowest courses, especially towards the corner, were built of large blocks up to 0.4×1 m in size, and they seemed rectangular. Clay was used as mortar and occasionally to coat the outer face of the wall. The plaster coating was c. 0.5 cm thick and marked by finger strokes; it ended in the middle of the lowest course, at level 1.30-1.40 m a.s.l., where the coating curved outwards. The inner face of wall A125 was covered with a different type of white plaster down to 1.46 m a.s.l.

Wall A125 was built on a layer of hard-packed, brown or greenish clay, similar to the material of floor A180 and the deep layer inside the “Palace” walls, A229 (fig. 84). This foundation was up to 15 cm thick and lay directly on a sandy layer mixed with bitumen. A high concentration of fist-sized stones was placed up against the outer side of the wall at its base.

Inside the “Palace”

Only a small part, c. 1.7 m², of the interior of the “Palace” was excavated in Trench A, and much of this space was heavily disturbed by stone plundering pit A64.

Fig. 54. The southeast corner of the “Palace” (A125 and A40) and right, wall 219 from Phase 2. Trench A, looking W (2009).



Fig. 55. Inside the south-east corner of the “Palace” with part of floor A232 exposed. Trench A, from above, looking N, scale unit 10 cm (2012).



The only floor noted in this small sondage was a 2 cm thick, hard, white plaster floor (A232) encountered at level 1.59 m a.s.l. (figs. 55 and 84). This corresponded closely to the level of the first floor noted in the rest of the building during the 1960-1963 excavations

(Kjærøum & Højlund 2013 p. 103). It rested on a c. 50 cm thick layer of generally green clay (A229, 1.44-1.09 m a.s.l.), quite similar to the clayey content of floor A180 outside the “Palace”. As was also the case outside the “Palace”, A229 rested on a compact sandy layer with a

lot of bitumen remains (A239, 1.09 m a.s.l.). There can be no doubt that floor A232 inside the “Palace” was contemporary with floor A180, outside the “Palace”.

Excursus

During excavation in the “Palace” in 1960-63, several horizontal flagstones were noted in the western part of the central room 1 between Pillar NW and Pillar SW. The flagstones were covered by two deep stone-built basins (Kjærum & Højlund 2013 p. 112, figs. 138-143). These two basins were torn down and rebuilt in the 1980's during a restoration of the “Palace”, and it was decided in 2009 to remove them completely in order to investigate the stone flags (Trench D).

Two courses of flags were registered (fig. 56), the first course with upper levels of 1.69-1.72 m a.s.l., and the second with upper levels of 1.83-1.85 m a.s.l., both levels can be referred to Phase I of the “Palace” (Kjærum & Højlund 2013 p. 108-112. The first course corresponds to level -1.49 - -1.52 and the second course to level -1.36 - -1.38 in the levelling system used in 1960-63). The stone flags were well preserved and well dressed in their lower parts, but greatly eroded and decomposed in their upper parts from having been

exposed for almost fifty years. They were of a light greenish stone, finer than the local beach-rock, but otherwise similar.

The flagstones may have escaped the stone plunderers because they were covered by the stone basins, and originally such flagstones may have covered other parts of the floor in room 1 as was the case in rooms 10-11 (Kjærum & Højlund 2013 figs. 187-188 and 193).

The earliest floor outside the “Palace”

In connection with the construction of the “Palace” a heavy floor, A180, up to c. 45 cm thick, was laid down outside and to the east of the “Palace” with the surface at about 1.80-1.60 m a.s.l. (fig. 57). The floor was made of a compact, green or brown clay containing stones in different materials: beach-rock, light-brown to grey sandstone, a whitish cement-like material and oolitic limestone chips and fragments. Many of the stones had plaster adhering to them, indicating that they had been part of earlier architectural constructions, presumably belonging to the previous Phases 2-3. The oolitic limestone chips probably originated from the cutting of the temple ashlars.



Fig. 56. Stone flags between two pillars in the central room of the “Palace”, looking SE, scale unit 5 cm (2009).

Fig. 57. Left, the SE corner of the “Palace”. Outside the “Palace” clay floor A180 exposed. Trench A, looking N, scale unit 5 cm (2009).



Contemporary with floor A180 were floors A123 and A130. Floor A123 was made of white, gravelly plaster, and it abutted part of wall A40. Surface A123 joined with A130, a more fine-grained clayey plaster floor that continued c. 1.70 m eastwards. Floor A123 sloped downwards from its beginning at wall A40 and further eastwards. The use of plaster in the floor close to the “Palace” wall may have been a way of protecting the building from rainwater.

Other floors, more or less similar to A180 and at the same level, covered parts of Trench E. An example of this was floor E168, also made of clay with stone and plaster inclusions and extended for 1.9 m in the south half of Trench E. Another floor, E200, between 10 and 40 cm thick, with the same characteristics as A180 was preserved in the southwest corner of Trench E for less than half a meter as it had been damaged by a stone plundering pit (E65) which went down as deep as level 1.25 m a.s.l.

East of floor A180 in Trench E, floors E165 and E187 were probably also contemporary with A180. In the north part of Trench E floor A180 probably continued as E295 (figs. 85 and 87). Surface E295 was built on a layer, which covered part of the pavement (E203) belonging to Phase 3. Above floor E295 was a compact

layer of brown and green clay mixed with stone and plaster fragments (E282 and E292, 1.65-1.86 m a.s.l.) and similar to a layer found above floor A180 in Trench A.

In the northeast of Trench E were a couple of floors, which could also be assigned to this phase. Floor E302 consisted solely of very small, 2×1 cm, oolitic limestone chips covered by a thin brown clayey layer (figs. 86-87). The floor was damaged in the west by stone plundering pit E90 and perhaps pit E288 (cf. below).

Floor E298 was made of clay. Its western limit was unclear as it was damaged by stone plundering pit E24 (fig. 81). Both floors (E298 and E302) may belong to the earliest “Palace” phase as they immediately post-dated the floors related to the temple terrace (E311) dating to Phase 3. East of floor E298 was the clayey layer E145 that covered wall E164 lying parallel to the oven.

In summary, it can be concluded that a floor contemporary with the establishment of the first phase of the “Palace” extended c. 10 m east-west and c. 11 m north-south. The only area that was not covered by this floor was the area of the oven, E138/E143. All other previous constructions were erased to around a level of 1.40 m a.s.l., but the walls of the oven were preserved to between 1.77 and 1.84 m a.s.l., i.e. above



Fig. 58. Layer of refuse (E137): potsherds, bitumen fragments, shells, and fish bones, from above, Trench E, scale unit 1 cm (2010).

the level of floor A180. This suggests that the oven continued in use after the construction of the “Palace” and its accompanying external surface. Further east, it should be noted that floors of a second temple terrace bordered by a channel have similar levels as floor A180 and related floors and therefore quite possible contemporaneous (see below p. 74-75).

The construction of floor A180 and related floors involved the levelling of the previous thin-walled building to its foundations and preparing the ground surface ready for construction. Apart from the floor level A180, the following makeup layers may be mentioned, A189, A187, A188, A198 and A211 between 1.49-1.27 m a.s.l. They contained a mixture of the same ingredients as floor A180, i.e. clay and stone, as well as remains from the previous settlement levels. The layers were not as compact as A180, and they contained brown, green or bluish-green clay lumps, stone and plaster inclusions as well as brown and orange-brown soil.

Floor A180 and the above-mentioned layers below A180 contained a high density of stone and plaster fragments. The beach-rock stones varied in size from 10 x 10 to 20 x 30 cm and probably originated from previous house walls. There were also many chips of oolitic limestone, 10 x 10 cm in average. The density of stone was highest in the south-west corner of Trench A and along wall A125 (A184 and A186).

A large part of floor A203 and the northern end of wall A204, both belonging to the thin-walled building of Phase 3, were covered with a low heap of white, very hard, fine-grained or more gravelly plaster mixed with rubble, A192 (fig. 46). In some areas, this plaster was mixed with brown or green clay. The thickness of the heap was largest towards the north and diminished in all directions. The most likely explanation of this heap of plaster is that during the construction of the “Palace” the previous floor of the thin-walled building (A203) was used as a workspace to prepare plaster for mortar and floors.

Phase 4. Periods 1-2 layers east of the “Palace”, up to layer E109

Phase 4 is marked by the occupation above floor A180 up to the construction of surface E109/E123. Both of these surfaces could be followed over large parts of Trenches A and E.

Directly upon floor A180 was a layer A179/E158 (1.59-1.53 m a.s.l.) consisting of dark brown and light yellowish brown soil with small chunks of brown or green clay and a variety of stones in beach-rock and a brownish cement-like material, as well as oolitic limestone chips. The layer contained frequent pottery sherds and many fish bones.

Upon layer A179/E158 was a 30 cm thick layer (E137/A175/E279, 1.79-1.50 m a.s.l.). Compared to A179/E158 it was less clayey, rather loose sandy soil with ash patches, charcoal, bitumen fragments and numerous potsherds and fish bones (fig. 58). In several cases, the fish bones occurred as complete articulated skeletons suggesting they had suffered minimal disturbance since deposition (fig. 59). The layer covered the space corresponding to floor A180 (apart from the southwest of Trench A) and E295 (E282/E292, 1.75-1.55 m a.s.l.) and 90 cm further northwards. Layer E147 (1.74-1.65 m a.s.l.) was another layer of refuse, which covered floors E165 and E187.

The main structure in this phase was a *hearth*, A164, situated two meters east of the “Palace” corner and oriented southwest-northeast (figs. 53 and 60). It was constructed in a depression in the above-described layer E137/A175/E279 reaching down to layer A179/

E158. It was 2.20 m long and 1.40 m wide, very regularly shaped in the form of a rectangle with rounded corners except the south-west corner was damaged by stone plundering pit A25. It was built of a layer, up to 15 cm thick, of fine-grained white to slightly greyish plaster on top of a 10 cm thick clay layer that was coloured reddish by heating (A174, 1.67-1.57 m a.s.l.). The hearth had a depth of 20 cm, and the fill inside (A166 and E128, 1.91-1.73 m a.s.l.) consisted of ash, charcoal, lumps of reddish-brown burnt clay, pebble gravel and a few animal bones and pottery sherds.

In the northern half of the trench and near the hearth there were remains of a fine grained plaster floor (A120, 1.79-1.71 m a.s.l.), covered with potsherds, animal bones and copper fragments.

Above floor A120 was a brown sandy soil containing ash patches, charcoal spots, pottery, stone vessel fragments, and animal bones (E130/A172/E131/ A169, 1.91-1.76 m a.s.l.). Covering the hearth A164 and its surroundings and abutting another hearth A148 (see below) were the layers A156, A165, E124, E269 and E275 (2.02-1.83 m a.s.l.) containing ashes, pebble gravel, and some sherds and animal bones.

In the southern half of Trench A was a floor (A170, A171 and A177, 1.88-1.74 m a.s.l.) of hard yellowish-brown clay with plaster fragments and stone inclusions. Another possibly contemporary floor of light brown clay, E139, was situated south of oven E138/E143.



Fig. 59. Layer of refuse (E137): complete fish cranium and fragment of soft-stone bowl (fig. 782), from above, Trench E, scale unit 1 cm (2010).



Fig. 60. Trough-shaped hearth A164, rectangular with round corners. Trench A, looking E, scale unit 10 cm (2009).



Fig. 61. Trench A, in upper right corner shallow hearth A148, square with rounded corners, looking N, scale unit 5 cm (2009).

Fig. 62. Square pit (E319/E288), cut down in virgin soil. Trench E, looking S, scale unit 10 cm (2012).



Approximately 20 cm above hearth A164 was another very regular square shallow hearth, A148 with rounded corners measuring 1.20×1.20 m (figs. 53 and 61). It was made of clay and seemed to have been built on a quite substantial floor A122, made of white plaster mixed with coarse sand and gravel at a level of 2.04 m a.s.l. Near the hearth the floor had been affected by the heat emanating from the feature. The outer edge of the hearth was dark brown and ashy, and the inside was light yellow-beige. It was filled with loose, dark brown and beige clayish soil with ash and charcoal and animal bones mostly from fish.

The floors (A151, A124, A153, A160 and A167) above floor A122 were made of clay and rather fragmentary, sometimes disturbed by stone plundering pits. Between the floors were refuse layers (A149/A156, A152 and A154) with pottery, animal bones, copper fragments and charcoal.

The oven, E138/E143, which is described above under Phase 3, continued to be in use in Phase 4 (fig. 53). Located 35 cm east of the oven and parallel to it was a wall, E164, 2.70 m long, c. 25 cm wide and with only one course preserved to a height of c. 20 cm (figs. 49, 51 and 53). Clay was used more sparingly than in the oven walls, and there were traces of white plaster on its faces. The wall was built on a layer of brown soil (E149) that covered the late floor (E179, 1.50 m a.s.l.) outside the temple terrace. The function of this wall was

unresolved, but it could have been part of an otherwise completely plundered building extending to the east.

In the northwest corner of Trench E a square pit (E319/E288) was cut down from level 1.70-1.80 m a.s.l., into virgin sand to a depth of 0.80 m a.s.l. (figs. 53 and 62). It destroyed the northern extension of the pavement E203, part of both floors outside the temple terrace (E311 and E314) belonging to the previous Phase 3, as well as floor A180 (E295) belonging to the phase described here. Only the southeast part of the pit was excavated, measuring 2.15 m northwest-southeast, the rest lying in the unexcavated area west and north of Trench E.

The lower part of the pit, E319, had a high content of clay, and towards its middle, the clay had a whitish colour, which in combination with a lot of ashes indicated that the area had been exposed to heat (E317 and E320). Conversely, in the east part of the pit, i.e. closer to its periphery, the clay had not been exposed to heat. Its exact function is unknown.

The upper part of the pit, E288, level c. 1.80 to 1.30 m a.s.l., contained loose brown soil with a lot of small clay lumps, small plaster fragments and ashy lenses, as well as animal bones and pottery. There were a few oolitic limestone chips, but generally, stones were of beach-rock, fist sized or smaller.

The pottery found in Phase 4 can be referred to Failaka Periods 1-2 (p. 95, figs. 381-500).

Phase 5. Periods 1-2 layers east of the “Palace”, from layer E109

This phase is marked by the construction of floor E109 (1.92-2.18 m a.s.l.), and ends with the construction of floor E92, both surfaces extending over large parts of Trenches A and E. E109 was a white c. 4-7 cm thick plaster floor (fig. 63). It came within 60 cm of the “Palace” and could be observed in the north and east sections of Trench A (figs. 81-83). It may originally have reached all the way to the “Palace” walls, however, this area was damaged by stone plundering. It covered large parts of Trench A and E where it was excavated as A146, E123, E112, E121 and E274/E277 (figs. 85-87). In the south part of Trench E it was destroyed by a huge clay pit (see p. 58). Towards the west, i.e. in Trench A, it was less solid and more clayey. In both floors E274/E277 (2.00-1.94 m a.s.l.) and E123 (2.13-1.97 m a.s.l.) the plaster was coarse and gravelly.

In Trench A and the baulk between Trench A and E and especially towards the north, the floor was covered with ash and fish bones (A146/E106, 2.20-2.06 m a.s.l.) (figs. 82-83). On top of these deposits, a clayey floor

containing plaster fragments (E103/E104, 2.31-2.15 m a.s.l.) was constructed covering more or less the same area as floor E109. This floor was slightly fragmented and covered with sandy, ashy layers with many animal bones, seashells, and potsherds (E87, E89, E99, E100, A138, A139, A142, A144, 2.45-2.21 m a.s.l.).

The oven E138/E143 was still accessible and in use in Phase 5 as indicated by two radiocarbon dates from its upper ash layers (cf. p. 250). The upper part of the oven was not preserved.

The layers that covered the oven E138/E143 (E105, E110, E134, 2.33-1.64 m a.s.l.) consisted of brown soil mixed with crumbled plaster, pebble gravel and some clay chunks. The amount of finds such as potsherds, animal bones and copper fragments was quite low. These layers were interspersed with thin layers of sandy soil (E102, E114 and E119, 2.15-1.87 m a.s.l.).

The pottery found in Phase 5 can be referred to Failaka Periods 1-2 (p. 95, figs. 501-534).

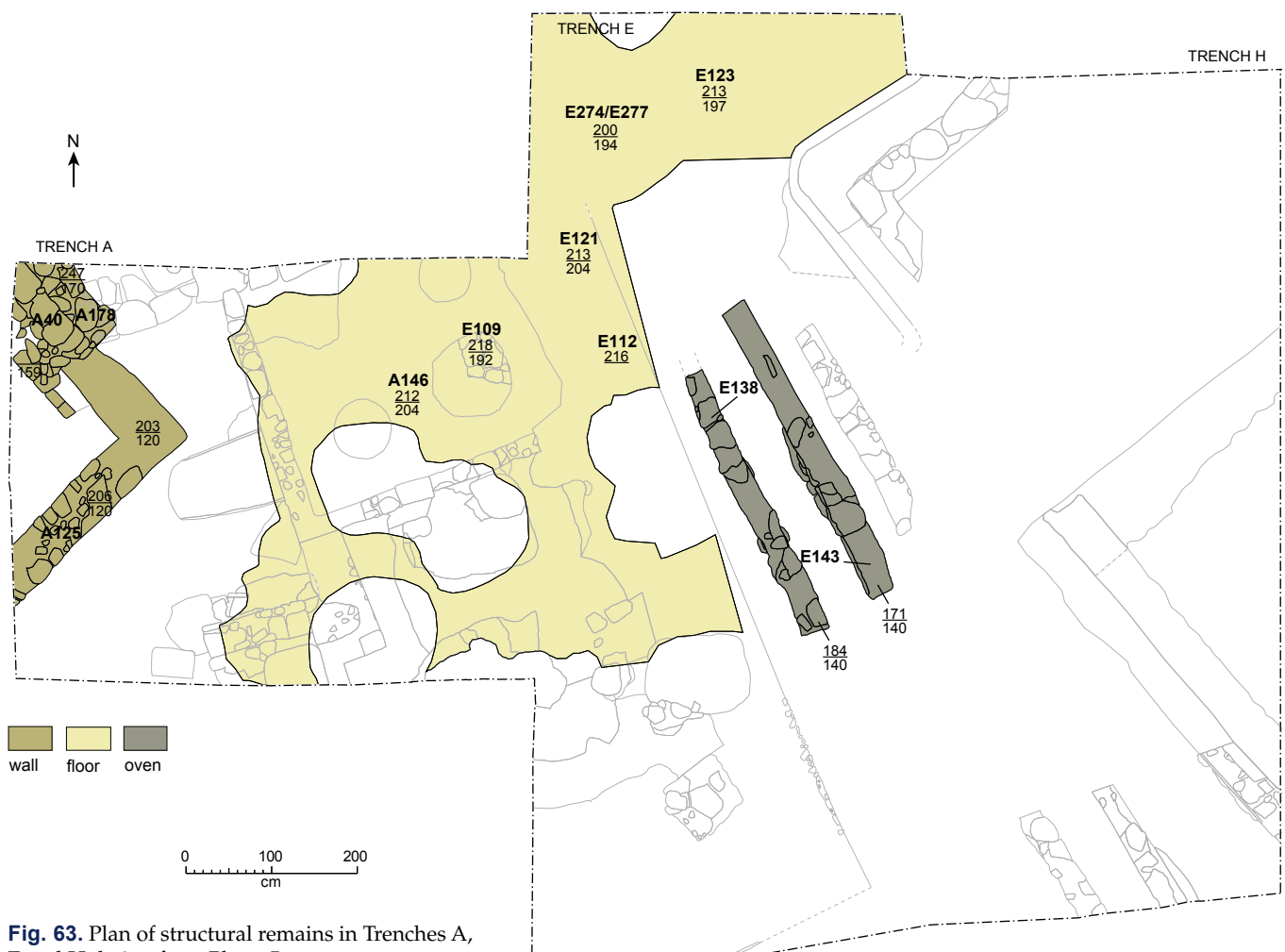


Fig. 63. Plan of structural remains in Trenches A, E and H dating from Phase 5.

2.4.2. Phase 6. East of the “Palace” in Period 3B

by Aiysha Abu-Laban and Flemming Højlund

From the excavations reported here no finds can be attributed to Failaka Period 3A (fig. 5), and from the old excavations inside and outside the “Palace” only few finds could be dated to this period. However, in Period 3B the “Palace” was reoccupied. One or two buildings with substantial floors were added in the north of Trench E, and several narrow channels, probably for conducting water, were cut into the floors (figs. 64-65). The material culture combines Dilmun and Mesopotamian elements (figs. 5-6).

In this period a rather heavy wall, B71, situated in the northeast of Trench E, dominated the area east of the “Palace” (figs. 64-65). It formed a corner with one wall running northeast-southwest and another running northwest-southeast. The orientation of the wall conformed to the general orientation of the “Palace” and other buildings in the area. The uncovered part of the

wall was 2.30 m long, and its width was 50 cm. Large stones measuring up to 50 x 40 cm were used to construct the wall, laid in clay with smaller stones in between. The wall was severely plundered, and only one course was preserved with a base level of 2.20 m a.s.l. and a top level of 2.45 m a.s.l. The external southwestern face of the wall was covered with a thin plaster layer.

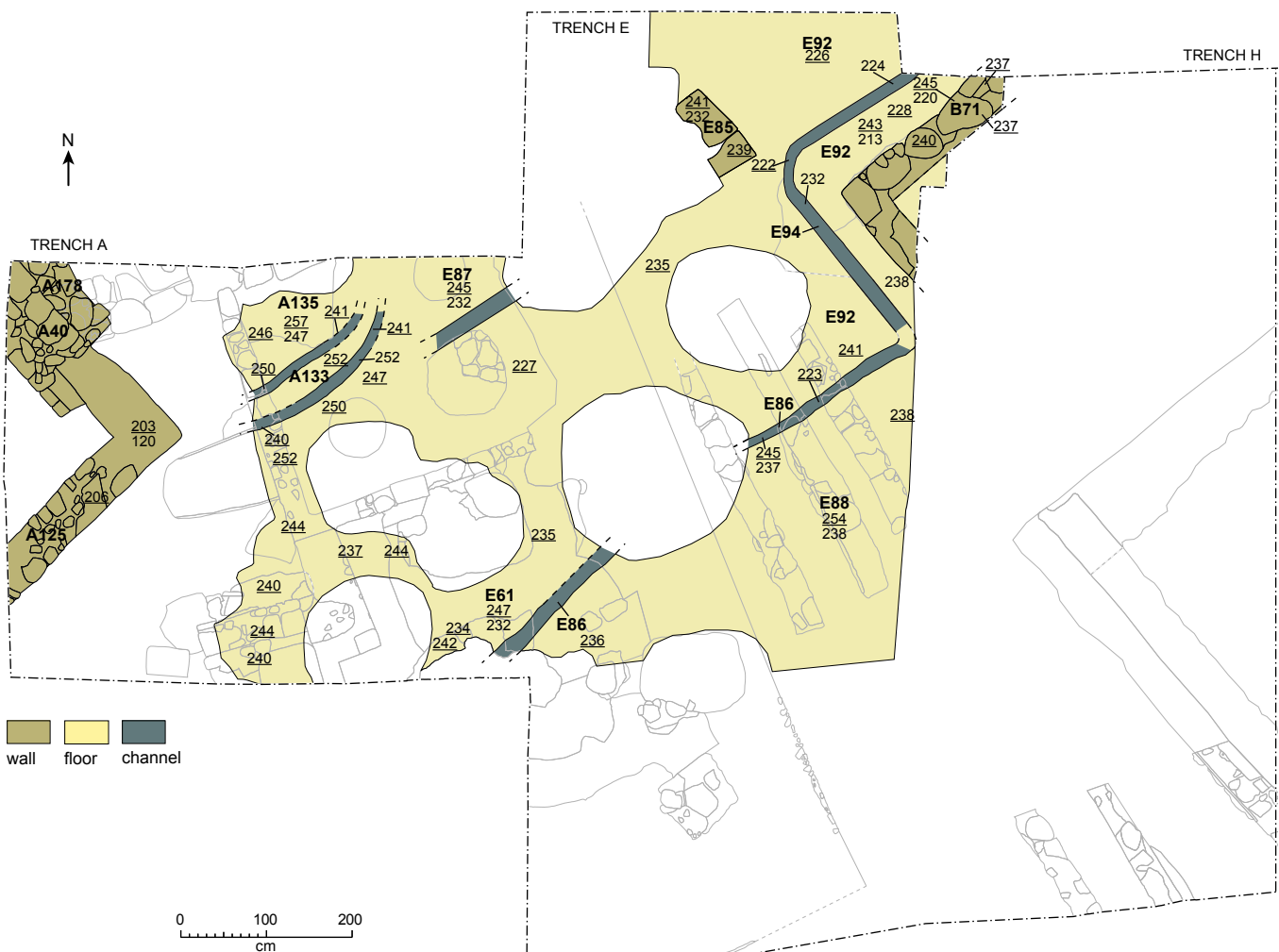


Fig. 64. Plan of structural remains in Trenches A, E and H dating from Phase 6.



Fig. 65. Foreground wall corner B71 with channel E92 of Phase 6. Background wall E85. Trench E, looking W, scale unit 50 cm (2010).

The wall was constructed upon a foundation of gravelly plaster (E92). In Trench B's section 23, directly below the wall the foundation had a thickness of c. 25 cm (2.20-1.95 m a.s.l.) (fig. 89), and the same thickness was measured along the north baulk of Trench H.

1.1 m west of wall B71 were the plundered remains of a contemporary wall, E85 (2.41-2.32 m a.s.l.) with only two ashlar preserved (figs. 64-65).

The foundation layer of wall B71 (E92) consisting of compact, white, coarse plaster mixed with gravel (2.43-2.13 m a.s.l.) extended, with a thickness of 10-20 cm, to the northern limit of Trench E and from there several meters southwards (figs. 86-87). A narrow, 12-14 cm wide and 10 cm deep channel (E94) was cut down into this floor that had an upper level of 2.32 m a.s.l. (figs. 64-65). The bottom of the channel measured around 2.21-2.24 m a.s.l. The channel followed the direction of wall B71, i.e. it turned the corner in a perpendicular angle.

South of E94 another channel (E86) traversed the entire Trench E over 6 m (figs. 64-66). It was 10-25 cm wide and 8 cm deep and dug into a plaster floor coated with clay, E88 (2.54-2.38 m a.s.l.), and further towards the west it was dug into a 10 cm thick, coarse, some-

what gravelly plaster floor, E61 (2.47-2.32 m a.s.l.), less compact than E92 and laid upon E100. The area of possible connection of the two channels was disturbed by stone plundering pit B47, but as they had the same dimensions, were constructed at the same level, and partly in floors of the same material, they must have been part of the same system.

In the northeast of Trench A was a plaster floor A135 (2.57-2.47 m a.s.l.) (figs. 81-83) and further south a related floor A131/A134 (2.55-2.49 m a.s.l.), both quite brittle and fragmented. Two parallel narrow channels (A133, 2.52-2.40 m a.s.l.), running in a northeast-southwest direction were dug into floor A135 (figs. 64 and 67). The channels were between 10-18 cm wide and not more than 10 cm deep. Their western continuation was disturbed by a stone plundering pit, and they could not be followed towards the east.

A possible channel cut in E87/E89 was observed in the baulk between Trench A and E (fig. 64).

The channels may have been intended to drain away waste fluids or transport water, perhaps into a pit to be mixed with clay for construction. However, they were not preserved so well that it could be seen which direction the water would have been running.

Fig. 66. Floor E92 traversed by channel E86 and walls E85 and B71 partly exposed. Trench E, looking N, scale unit 50 cm (2010).



Fig. 67. Two channels, A133, cut in floor A135. Trench A, looking S, scale unit 5 cm (2009).



The north half of Trench E with walls B71 and E85 and their accompanying floors E92, E88 and E61 were covered by layers (E35, E77, E78, E81 and E83, 2.67-2.39 m a.s.l.) consisting of rather loose, brown soil with a little gravel, a few small plaster fragments and a few animal bones and artefacts made of copper and stone. The potsherds in these layers were extraordinarily small and heavily worn and belong to Failaka Period 3B (p. 96, figs. 535-605).

The floor A135 contained a good deal of bird bones that were otherwise rare. The concentration of bird bones continued in the same area down to layer A139 (2.42-2.34 m a.s.l.).

In the east end of Trench A (E76, 2.62-2.41 m a.s.l.) were layers consisting of loose, brown soil with a few small plaster fragments and gravel. The quantity of finds such as animal bones and artefacts made of copper and stone was low.

The floor level inside the building, Late Phase I, is in this period c. -1.20 in the levelling system used in 1960-63 (Kjærø & Højlund 2013 p. 108), which corresponds to c. 2.00 m a.s.l. in the system used here, i.e. 30-40 cm deeper than outside the building.

2.4.3. Phase 7. East of the “Palace” in Period 4A

by Aiysha Abu-Laban & Flemming Højlund

In Period 4A, the “Palace” continued to be in use after a major reconstruction. Outside the building, there was a substantial accumulation of floors and floor deposits. A large wall parallel to the “Palace” may belong to a new neighbouring building (fig. 68). Besides this, the only structure encountered was a large clay treatment pit. The material culture seems purely Mesopotamian, devoid of any Dilmun remnants (figs. 5-6).

A substantial, up to 7 cm thick, white plaster floor was laid at the start of this phase (fig. 68). In Trench A this floor, A35, was noted 60 cm east of the “Palace” at a level of between 2.68 and 2.54 m a.s.l. (figs. 69, 73 and 81-83). It continued c. 8 m towards the east into Trench E where it was excavated as E40 (2.65 m a.s.l.) (fig. 70) and appeared in Section 23 through Trench B as layers B51

and B53 (figs. 89-90). The floor was constructed directly on layers dating to Period 3B, and a few potsherds of Period 3B date, animal bones and copper fragments were found embedded into the floor plaster.

Above floor A35/E40 was a series of floors and floor deposits with a total thickness of c. 60 cm (2.65-3.25 m a.s.l.). Most of the layers deposited on floor A35/E40



Fig. 68. Plan of structural remains in Trenches A, E and H dating from Phase 7.



Fig. 69. Left, plaster floor A35 of Phase 7, right “Palace” wall A40. Trench A, looking S, scale unit 50 cm (2009).



Fig. 70. Background, plaster floor E40 of Phase 7, foreground clay pit emptied. Trench E, looking N, scale unit 50 cm (2010).

were clayey and compact. In Trench A layers A32 and A24 (2.80-2.68 m a.s.l.) and in Trench E layers E23, E27, E28, E36, and E37 (2.86-2.60 m a.s.l.) were excavated. There was generally rather few finds such as pottery, all dating to Period 4A, animal bones, and stone objects. The latest layer with a homogenous pottery inventory of Period 4A was found at a level of 3.30 m a.s.l. (layer E69) in the baulk between Trench A and E.

In the north section of Trench A, the remains of a substantial wall (A104) built of beach-rock stones was encountered at a distance of c. 1.5 m to the northeast of the “Palace” wall (2.96-3.56 m a.s.l.) (figs. 41 and 81). As such it is a likely continuation of a wall uncovered in 1960-1963 along the northeast side of the “Palace” and at the same level and probably part of a building parallel to the “Palace” (Kjærø & Højlund 2013 fig. 156).

In the south of Trench A an almost circular fireplace with a rim of burnt clay measuring 60 cm in diameter (A31/A34) (2.58 m a.s.l.) was constructed on floor A32, and there was another fireplace A33 (2.52-2.39 m a.s.l.) in floor A32, cut partly down into lower floor A35.

In the north of Trench E to where floor A35/E40 did not extend, there were a number of contemporary, compact clayey layers (E14, E21, E29 and E34, 2.84-2.56 m a.s.l.) (figs. 85-87). A large heap of plaster, E22, partly extending below the floor (figs. 68, 70 and 85), situated in the northwest corner of Trench E did not seem to serve any architectural function and could just have been a stockpile.

The clay pit

The south part of Trench E was taken up by a great pit, at least 1.5 m deep and extending c. 6 m east-west and at least 7 m north-south (figs. 68 and 70). Section 23 provided a 6.25 m north-south section through the pit with a clear termination towards the north, whereas the southern edge lay beyond the excavated area (figs. 89-90). Towards the west, the limit of the pit was found in the west part of Trench E, and its eastern border could be seen in the south baulk of Trench H and in a French parallel section (Calvet & Pic 1986 fig. 5 lower section right).

The pit contained a number of thick horizontal clay layers but none of them filled the entire pit. In the section of Trench B (fig. 89) the deepest clay layer in the pit was identified as layer B33 (c. 1.70-1.80 m a.s.l.), followed by B31 (c. 1.80-1.90 m a.s.l.), both light grey, very compact clay. Further layers were noted in the section as compact clay, B30, B28, B25, B23, B21, B18, B17, B16 and B14. A similar depth (1.75 m a.s.l.) to the pit was measured in the south section of Trench H.

During excavation of the south part of Trench E several layers of very hard, light brown clay were uncovered: E18 (c. 2.79-2.71 m a.s.l.), E30 (c. 2.60 m a.s.l.), E32 (c. 2.55 m a.s.l.), E41 (c. 2.50 m a.s.l.), E43 (c. 2.46 m a.s.l.) and E44 (c. 2.24 m a.s.l.). A substantial layer of hard chunky clay (E49) was found at level of c. 1.96 m a.s.l. and below E49 were two even more hard clay layers (E59 and E60) at a level of c. 1.89 m a.s.l. All of these clay levels had very uneven surfaces, and none extended for more than one and a half meter in the horizontal dimension.

Between the clay layers and beside them were thick layers of loose sand, E26 (2.68-2.58 m a.s.l.), E38 (2.67-1.96 m a.s.l.), and E46 (2.12-1.89 m a.s.l.). In the section through Trench B (fig. 89), these sand layers, B24, B26-27, B29-30 and B32, could be seen to have been deposited in thin horizontal layers, sometimes interrupted by thin layers of fine clay, dry and broken up in small fragments, as often seen today on Failaka after drying up of temporary water ponds (fig. 71), i.e. probably deposited as a natural process.

From the sandy layers a good quantity of pottery was retrieved, consistently dating to Failaka Period 4A, as well as pieces of copper, animal bones, and some fragments of soft stone jars. No traces of fireplaces were noted. The abovementioned finds did not cluster immediately upon the clay layers but were evenly distributed in the sand. The majority of the finds came from the sand layers E38 and E46.

The uneven surface of the clay levels, their limited extension in the horizontal level, the even distribution



Fig. 71. Temporary water pond on Failaka after drying up with a thin layer of fine clay, dry and broken up (2013).

of the finds in the sand and the lack of concentration of finds on the surface of the clay levels, all argue against an interpretation of the clay levels as actual activity zones with any long-term exposure. The pit was rather repeatedly used for the preparation of clay in large quantities during several events of construction work, each separated from the other by a period of time.

A clay preparation pit would be dug close to where the clay was to be used, probably either the "Palace" or the Temple. Both buildings were either re-built (the "Palace", Kjærsum & Højlund 2013 p. 99-108) or enlarged (the Buttress Wall around the Temple, cf. below p. 91) in Period 4A, and both these operations would have required substantial amounts of clay.

The clay pit was disturbed by several stone plundering pits that were dug even deeper than the clay preparation pit (fig. 70).

Not only does the pottery found in the clay pit date to Failaka Period 4A, the same date applies to the pottery from the layers that accumulated upon floor A35 and that were cut by the pit (p. 96, figs. 606-667).

The uppermost layers covering Trench E, dating to Period 4A (E69 and E72, 3.30-3.08 m a.s.l.) (fig. 82), resemble normal settlement layers consisting of greyish brown, loose sandy soil with potsherds, animal bones and copper fragments. Moreover, there were a couple of ash pits, e.g. E71 in the baulk between Trenches A and E and A36 in Trench A.

A possible second clay pit of equally great dimensions was noticed in Trench J-B (cf. p. 92).

2.4.4. Phase 8. The Post-4A Period

by Aiysha Abu-Laban & Flemming Højlund

The upper part of Tell F6, from the Period 4A layers to the present surface, consisted of a homogeneous deposit. No pottery datable to Failaka Period 4B, as had been encountered frequently on Tell F3, was found, and no distinctive layers or structures could be separated that could for example be related to the late walls that were uncovered near the surface of the tell in 1962-63 (Phase III, cf. Kjærums & Højlund 2013 p. 151). From somewhere in this level a great many pits had been dug, presumably in search of building stones (fig. 72). Directly above the "Palace" (and the Temple) these plundering pits merged into a single cohesive zone of disturbance, whereas in the peripheries of the buildings where the plundering had been less extensive, the individual plundering pits could be distinguished. A number of ashlar in a fine beach rock belong to this Post-4A Period and could possibly relate to Phase III of the "Palace" (figs. 5-6).

The upper layers

The surface of the tell in Trenches A and E lay around 4 m a.s.l., and the upper first half to one meter consisted of a homogenous, light brownish sandy soil with a mixture of 2nd millennium BC pottery, and a few Hellenistic sherds.

The stone plundering pits

Trenches A and E were pock-marked with many, more or less round pits, c. 1.5 m in diameter, becoming narrower the deeper they went and reaching c. 2.30-2.50 m below the present surface of the tell (fig. 72). They generally contained rather loose, dark grey sand and a mixed assemblage of sherds from all periods represented in Trenches A and E, i.e. Periods 1-4A. No Hellenistic sherds were found in the stone plundering pits. The pits were dug through levels dating to Period 4A and therefore post-date this period.

Similar pits were identified during the earlier Danish and French excavations in Tell F6 and were probably dug in order to search for building stones (Calvet & Pic 1986 p. 14-16. Kjærums & Højlund 2013 p. 103).

One pit, A26, visible in the north section of Trench A, reached down to wall A219 from Phase 2 (figs. 41 and 81). In the west part of Trench A, above the "Palace", several pits which merged into one big disturbance, named A64, removed parts of the "Palace" walls (figs. 54 and 84). In the east section of Trench A, a large pit, A85, went as deep as c. 1.20 m a.s.l. (figs. 82-83). In the south section of Trench A, a huge pit, A37, went down to a level of 1.25 m a.s.l. In the middle of Trench A, a large pit, A25, turned out to consist of two separate pits that had merged.

In Trench E several round, deep pits were regularly spaced across the trench, E90 (figs. 86-87), E75, E24, B47, E17, E53, and E58 (fig. 66). The south-west corner of Trench E was especially disturbed, by a deep pit (E65) which reached a level of c. 1.25 m a.s.l. Some pits did not reach that deep, e.g. E75 only reached level 2.11 m a.s.l.

Stratigraphically, it was not possible to decide based on the section through Trench B whether pit B47 dates before or after the large clay pit (figs. 89-90). B47 was first observed in the horizontal plane at a level of c. 2.53 m a.s.l.

Fine beach-rock ashlar

Eight ashlar cut in a hard, light grey, fine beach-rock, gravely with many shells, were found in Trench A, lying scattered in stone plundering pits, left there by the stone plunderers, perhaps covered during collapse of their pits (fig. 73).

At least ten ashlar cut in the same fine beach-rock were found during the excavation of the "Palace" in 1960-63. They occurred in Trenches B2, A1, C1 and SS 0-9, but without any clear context or dating, apart from the fact that they could not be shown to be part of the "Palace" (Kjærums & Højlund 2013 figs. 131, 191-192, 202).

These ashlar were well cut, with flat, even faces, right angles and regular corners, though some were damaged. The largest was 24x72.5x90 cm and the smallest 28.5x35x58 cm (fig. 74). Some had cuttings on their sides to accommodate the neighbouring stones in a wall or in gate constructions (fig. 75). Two were shaped like beams with lengths of c. 150 cm and cross sections of c. 40 cm and with deep depressions cut in one end, possibly parts of a gate (fig. 74 nos. 17-18, and figs. 76-77).



Fig. 72. Plan of stone plundering pits in Trenches A and E.

The fact that such ashlar were only found in Trench A and in the 1960–63 excavations west of Trench A and not in Trench E may indicate that they belonged to a building with the approximate same location as the “Palace”.

In 1962/63 a heavy wall of several meters’ length built of well-cut ashlar, up to 90 x 50 cm in size, was uncovered in the uppermost levels of Tell F6 (Kjærsum & Højlund 2013 p. 103, 151, fig. 136). The exact type of stone was not described, and no cultural material was related to the wall; the corresponding layers may have been deflated off the surface of the mound. Perhaps the ashlar found scattered in the excavated parts of Tell F6 belonged to this building, which at the moment can only be given a post-Period 4A dating.

Fine beach-rock chips

In Trench C2/4, 20 m south of the Temple, a thin layer of beach-rock chips was found (figs. 107 and 112). The chips were in the same hard, fine beach-rock, light

grey, gravelly with many shells (fig. 78), as the ashlar described above and they indicate that some cutting of ashlar of this material had taken place there. A 3 m extension of the trench was made towards the east in the hope of finding datable pottery with the chipped stone layer.

The layer, situated between level 2.50 and 2.60 m a.s.l., produced 136 stone chips varying from 2.3 x 2.1 x 0.8 cm to 13.5 x 10 x 3.5 cm (fig. 79). It is likely that the beach-rock chips were residue from the cutting of the beach-rock ashlar mentioned above, and that these ashlar belonged to a building that once crowned Tell F6.

Unfortunately, the layer of fine beach-rock chips only contained a few undatable pottery sherds. However, it superimposed by 67 cm a layer of oolitic limestone chips that are interpreted as remains from the cutting of the ashlar for the Dilmun Temple (see below), so at the moment only a dating later than the Temple can be ascribed to this layer.

Fig. 73. Fine beach-rock ashlars lying loose in stone plundering pit A64; right plaster floor A35. Trench A, looking N, scale unit 10 cm (2008).



Fig. 74. Dimensions in cm of ashlars in fine beach-rock from the 1960-63 and the 2008-2012 excavations in Tell F6. For damaged ashlars the minimum dimension is given.

2008-2012				
Nº	L	H	Th	
1	>54	>50	35	
2	>63	>55	44	
3	76	61	49	
4	>82	>77	30	
5	82	67	27	
6	>71	>60	43	
7	>66	62	27	
8	58	35	29	
1960-1963				
Nº	L	H	Th	
9	>81	43	22	
10	>90	47	20	
11	?	?	26	
12	90	73	24	
13	55	51	20	
14	54	50	34	
15	165	35	27	
16	122	44	37	
17	>152	44	43	
18	>163	37	35	



Fig. 75. Ashlar in fine beach-rock found in topsoil of Trench A in 2008, cf. fig. 74 no. 3.

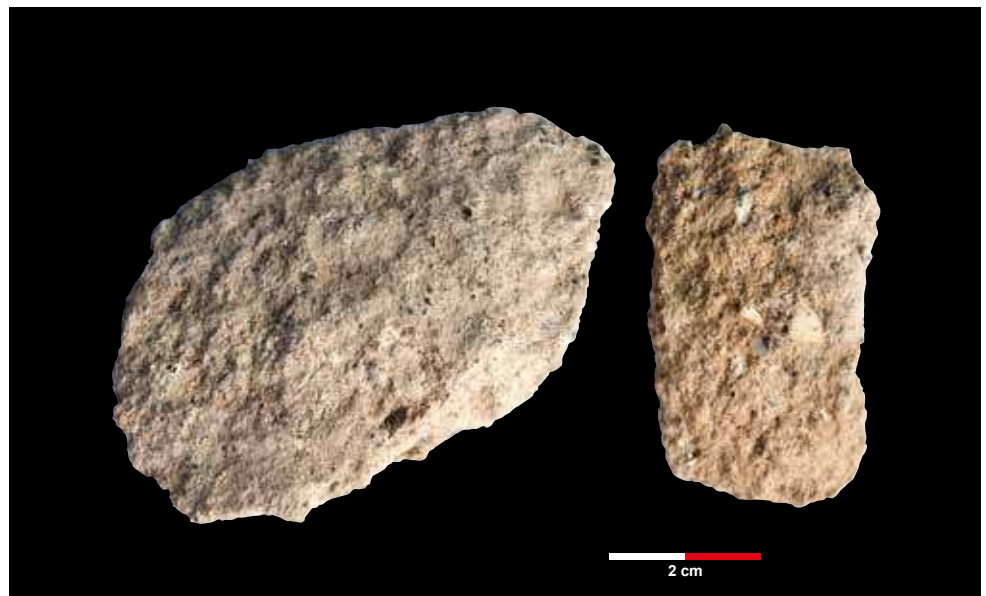


Fig. 76. Ashlar in fine beach-rock excavated in the “Palace” in 1960-1963, cf. fig. 74 no. 17.



Fig. 77. Ashlar in fine beach-rock excavated in the “Palace” in 1960-1963, cf. fig. 74 no. 18.

Fig. 78. Chips of fine beach-rock found in level C102 in Trench C2/4.



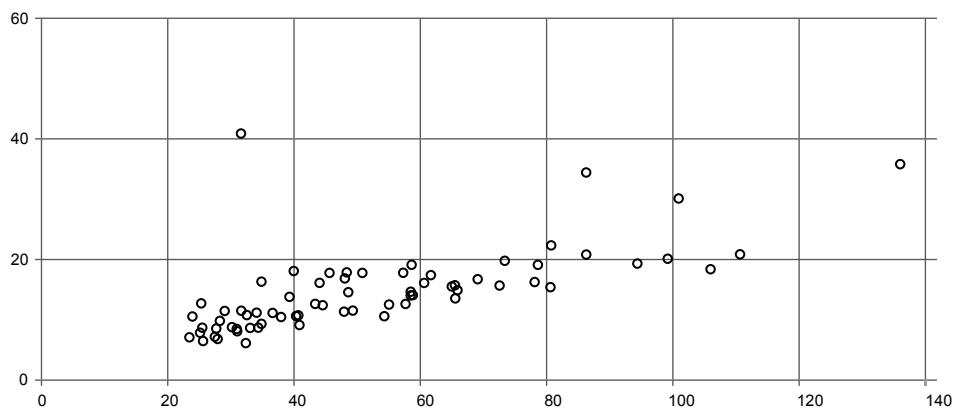


Fig. 79. Length and thickness in mm of chips in fine beach-rock from levels C63 and C102 in Trenches C2/4 and eastern extension.

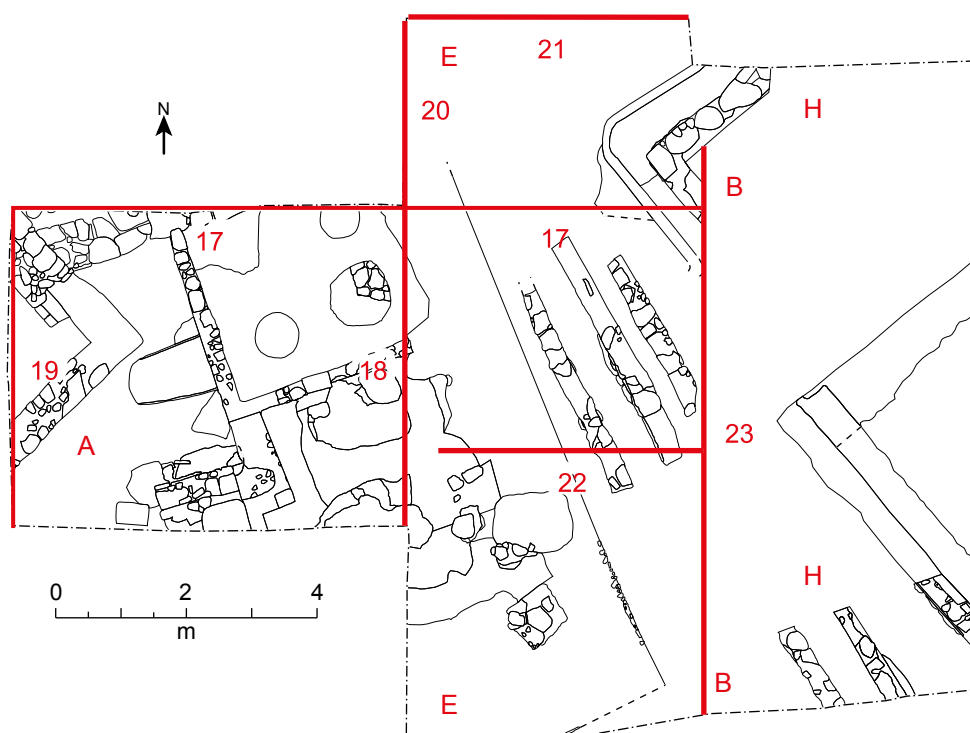
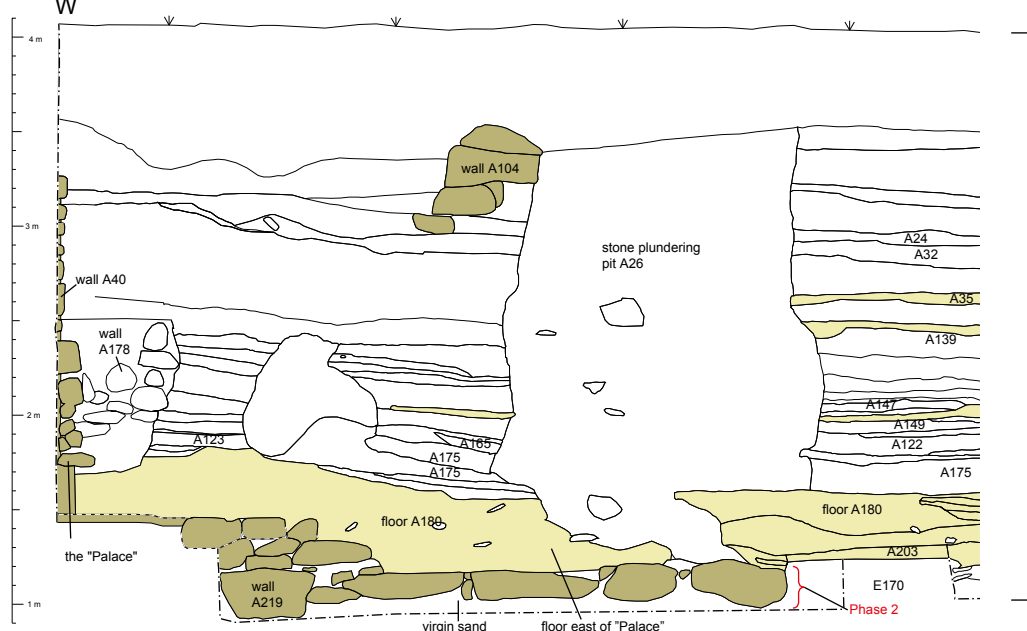
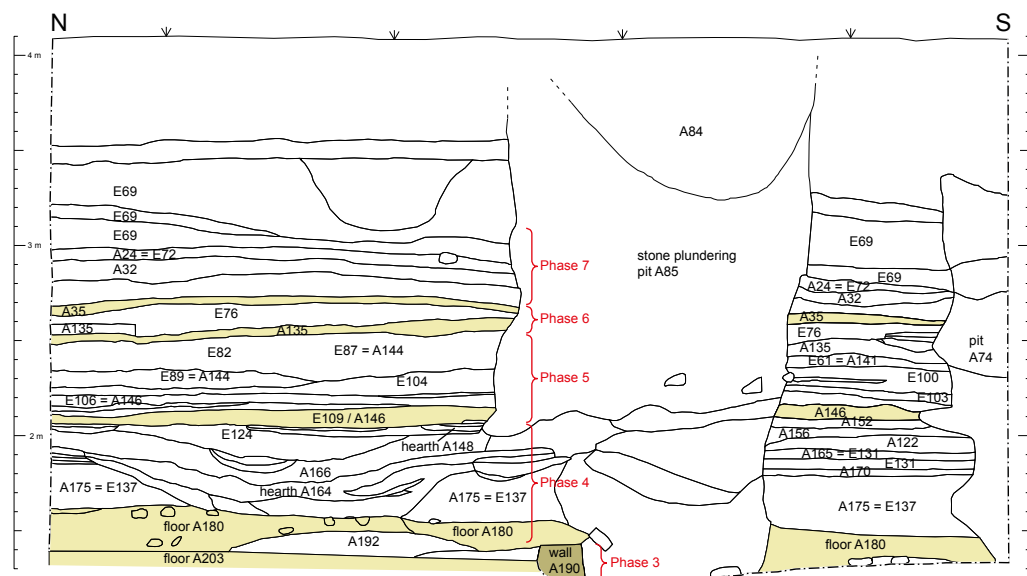


Fig. 80. Plan of Trenches A, B, E and H with drawn Sections 17-23 marked.

W



N



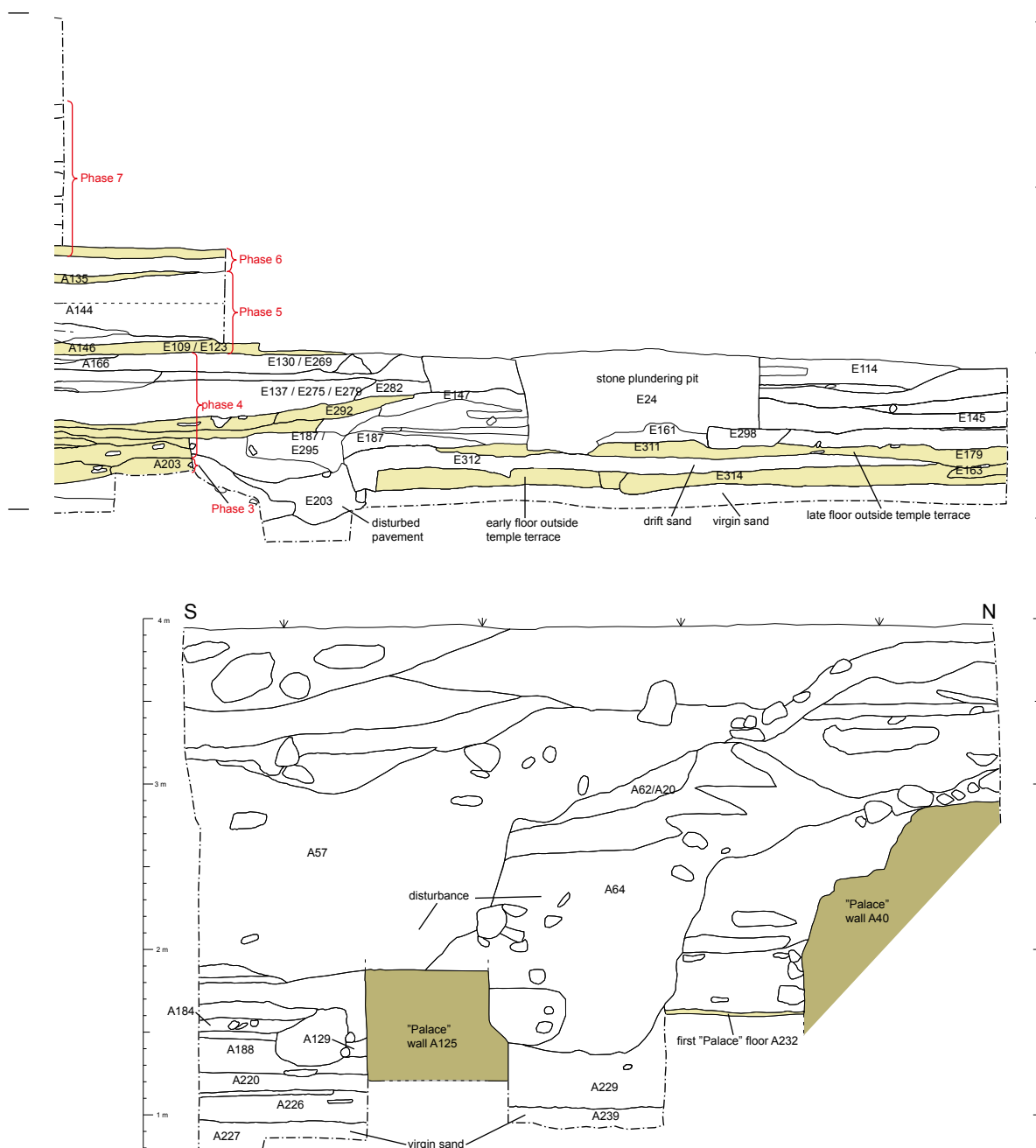


Fig. 84. Section 19 through walls A125 and A40 and floor A232 of the “Palace”, disturbed by stone plundering. Trench A, looking W.

Fig. 85. Section 20 through floor E109 (Phase 5) and pit E288/E319. Trench E, looking W, cf. fig. 87.

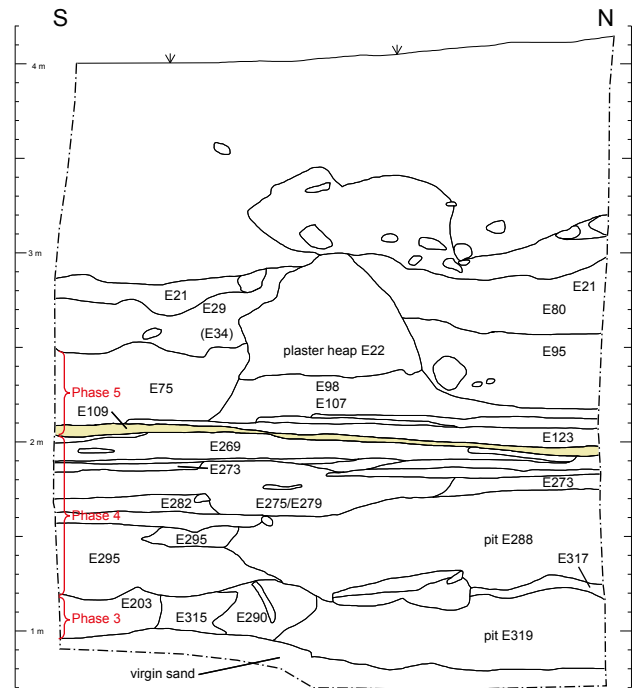


Fig. 86. Section 21 through floors E92 (Phase 6), E274 (Phase 5) and pit E288/E319 (Phase 4), and floors E311 and E314 (Phase 3). Trench E, looking N, cf. fig. 87.

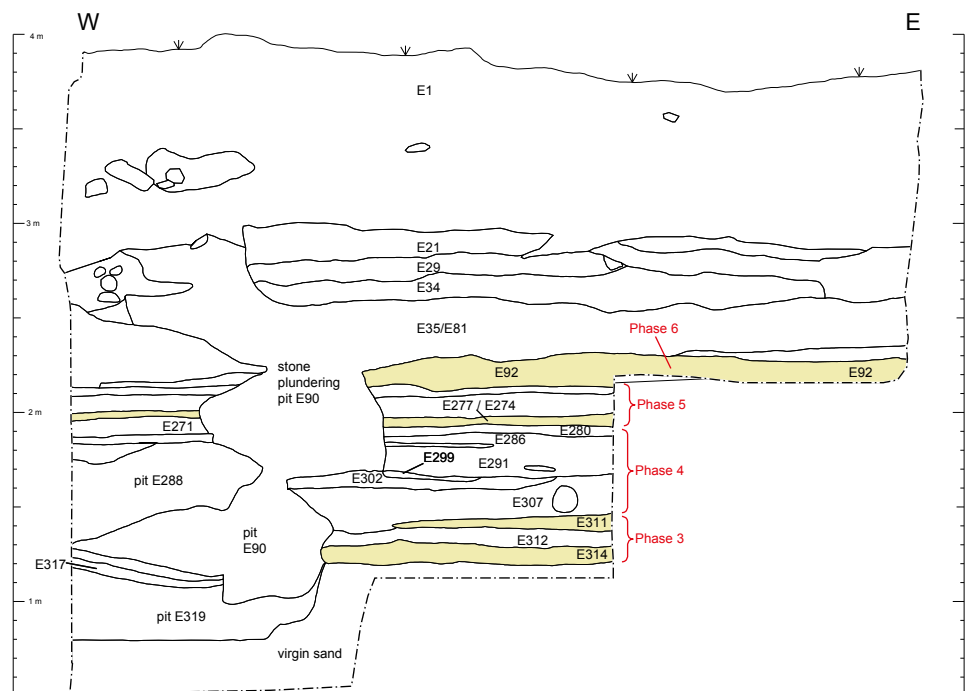




Fig. 87. North part of Trench E after excavation. Left, Section 20, middle, Section 21, right wall B72 and centre bottom, pit E288/E319, looking N, scale unit 10 cm (2012).



Fig. 88. Section 22 through south end of oven E138/E143 with underlying hearth E216, and pavement E203 left, cf. fig. 50. Trench E, looking N.

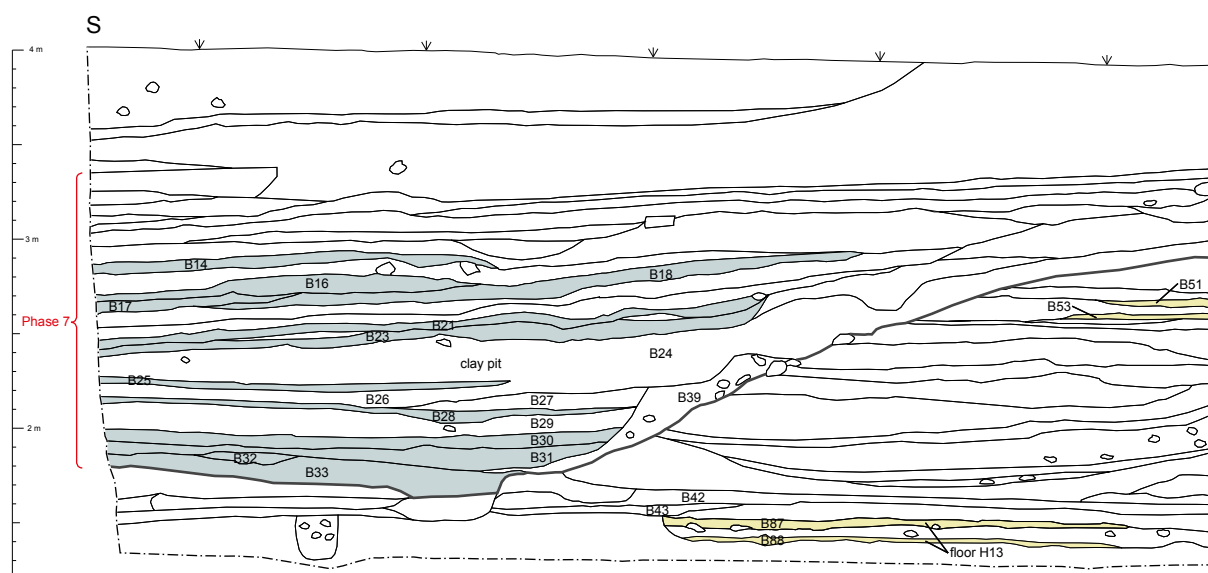
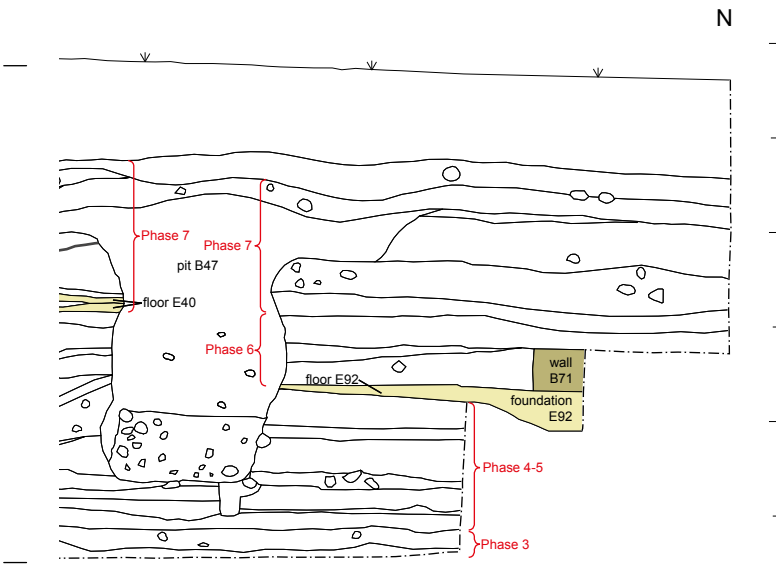


Fig. 89. Section 23 through clay pit with clay layers marked. Besides floor E40 from Phase 7, floor E92 with wall B71 from Phase 6, and floors B87-B88 (H13) from Phase 3, cf. fig. 90. Trench B, looking W.



Fig. 90. Section through settlement layers from white plaster floor E40 to virgin soil, left clay pit emptied, cf. fig. 89. Trench B/E, looking SW (2009).



2.5.1. The Dilmun Temple, Northwest

by Flemming Højlund

In the southeast of Tell F6, a temple with many similarities to the Barbar temple in Bahrain was constructed in the early 2nd millennium BC. To the northwest the Temple was bordered by a low terrace (fig. 46). Outside this terrace was a well-built floor relating the Temple to the thin-walled building of Phase 3, described above. The first terrace was later replaced by a second, bordered by a channel, which seems to be contemporary with the erection of the "Palace" dating to Phase 4 (figs. 5-6).

Fifteen meters southeast of the "Palace", the foundations for a substantial building were uncovered in the 1980s by French archaeologists (figs. 1, 3-4). The building was severely plundered of its fine ashlar, but enough was preserved to allow it to be convincingly interpreted as a temple (Calvet & Pic 2008).

The central temple building measured c. 20 x 20 m, i.e. almost the same dimensions as the "Palace", and it was constructed on 2.8 m wide and 1.5 m deep foundations. The walls of the building had been removed, apart from a few fragments of finely cut ashlars that were left *in situ* (fig. 91). They were found embedded in the upper surface of the foundations that were covered with a fine white plaster (fig. 92), and they could be identified as oolitic limestone. The surface colour of the ashlar fragments are now light brownish, but below the surface they are white (fig. 93). Impressions of scores of similar ashlars were found all over the plastered surface of the foundations making it possible to reconstruct the ground plan of the building (Calvet & Pic 1990 p. 104, plan 1). It had an outer wall, c. 2.1

m thick, consisting of two shells of ashlars with an internal cavity filled with rough stones laid in a white mortar. The entrance to the building was in the middle of the southeast side, and inside two internal rooms were demarcated with 70-80 cm thick ashlar walls.

The floor was covered with up to 23 cm thick, c. 0.6-1.1 m large quadrangular flags of oolitic limestone set in a brownish, sandy plaster, of which a number were left *in situ* (fig. 91:5) (Calvet & Pic 1990 plate I: 363, 374; pers. obs. 2008). The surface colour of the flags are now light brownish, but below the surface, they are white. The edges of the flags are very regularly cut with a joint of c. 1 cm.

In terms of dimensions, construction techniques and materials etc., the Failaka Temple shows many close similarities to the Barbar temples whereas the posited parallels to Syrian temple towers (Calvet 2002 p. 266. 2008 p. 15-19) seem doubtful (Højlund 2012 p. 170-171. Kjærum & Højlund 2013).

Because of its severe plundering, there is only little evidence for the various functions that took place

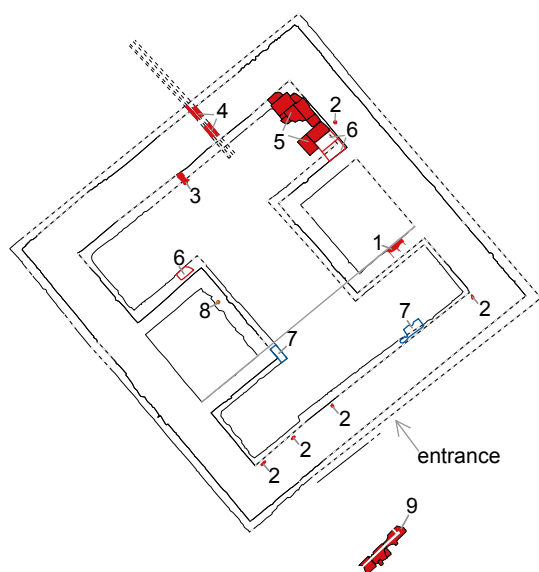


Fig. 91. Plan of the Temple in Tell F6 with 1) fragments of two finely cut oolitic limestone ashlars *in situ*, inner wall of Temple, 2) fragments of oolitic limestone ashlars *in situ*, outer wall of Temple, 3) oolitic limestone block, 31 x 45 cm, 13 cm thick, lying *in situ* within the outer temple wall at approximately the same level as the plastered surface of the foundation wall, 4) oolitic limestone ashlars with cut runnel passing through the outer temple wall, 5) floor flags in oolitic limestone, 6) impression of floor flags in brown plaster, 7) brown plaster, 8) impressions of feet in the plaster surface, 9) oolitic limestone ashlars with cut runnel south of the Temple. Entrance to Temple marked by arrow. (Basic plan after Calvet & Pic 1990 plate 1).



Fig. 92. Fragments of two oolitic lime stone ashlars embedded in the plastered surface of the temple foundation, cf. fig. 91: 1. The surface of the ashlars has turned a light brown, but immediately below the surface, they retain their original white colour. Scale unit 5 cm (2010).



Fig. 93. Close-up of oolitic limestone with a fresh break.

in the Temple. A channel was cut through the outer northwest wall at floor level, continuing to at least 10 m away from the building (Calvet & Pic 1986 fig. 39, plan I: 358). The excavators suggested that the channel may have begun in the centre of the building, carrying water out of the building (Calvet & Pic 1990 p. 107-109).

The way the channel was cut through the wall ashlar, and the way it continued far away from the Temple, is closely paralleled in phase II of the Barbar temple, where several channels were observed leading from central altars and ending far away from the Temple (Andersen & Højlund 2003 fig. 103, plan 3: 3). It was argued that they removed water used to wash and clean the altars after bloody sacrifices (Andersen & Højlund 2003 p. 323-324). A similar function could be proposed for the Failaka channel, and it is probably therefore no coincidence that the channel led from the Temple towards the northwest, as this must have been the rear of the building, with the entrance lying to the southeast.

Below the floor level of the Temple, two basins had been constructed using stone slabs with the gaps carefully sealed with plaster. They may have been intended to hold water for certain rituals, like the well chamber in the Barbar temple, as suggested by the excavators (Calvet & Pic 1990 p. 107, pl. I: 350 and 353).

Northwest of the Temple

Towards the northwest, the Temple was enclosed by a low terrace, uncovered in the 1980s and re-excavated in 2012 (figs. 46, 94-95). A low wall, H17, bordered the terrace on its northwest side for twenty meters (Calvet & Pic 1990 pl. I: 311). The wall was preserved to a height of no more than c. 20 cm and had an average width of 60 cm. It was built of local beach-rock stones that varied in size and shape, the largest stones measured c. 40x30 cm, set in compact clay. The outer face of the wall was regularly built and coated with a thick white plaster. The inner side of the wall was irregular and unplastered, evidently not intended to be seen but to be covered by the sandy fill of a terrace platform. No floor surface of this terrace was preserved (Calvet & Pic 1986 fig. 5).

Towards the southwest the terrace was bordered by wall H9, uncovered in the 1980s (Calvet & Pic 1990 pl. I: 327) and re-excavated in 2011 (figs. 46 and 96). It was preserved up to three courses high (c. 45 cm). The stones were set in clay that also covered parts of the outer face of the wall, from top to bottom, whereas

the inside of the wall was irregular as in the case of wall H17. The upper part of the face of the wall was covered with a c. 2 cm thick, white plaster with traces of finger marks. The bottom of the plaster indicated the original ground surface (1.52 m a.s.l.), and the wall thus had a two-course high foundation, built in loose beach sand. The two walls met in a clearly demarcated corner (Calvet & Pic 1986: fig. 38b).

The plaster coating on the face of H17 continued onto an adjacent floor, H13 (Calvet & Pic 1990 pl. I: 313), that was re-excavated in 2012 (figs. 46, 94-95 and 97). The floor had a level around 1.46 m a.s.l., higher close to wall H17 and sloping slightly downwards towards the north. It was a thick hard clayey floor filled with plaster fragments and other stones re-used from previous constructions. A 50 cm deep sondage, which was cut in front of wall H17 by the French, was re-opened in 2012, and it revealed that the floor was built on loose beach sand (virgin soil).

Close to the terrace wall, the floor, H13, included large flat slabs of fine plaster. The slabs measured c. 35x25 cm in average and were 3-4 cm thick. They were



Fig. 94. The NW corner of the Temple fronted by a low terrace bordered by walls H17 and H9, later replaced by a terrace bordered by a channel (306/307). Trenches E and H, looking E (2012).



Fig. 95. Wall H17 bordering the first temple terrace with related floor H13. To the left, corner of second terrace with channel (306/307). Trench H, looking SE, scale unit 10 cm (2012).

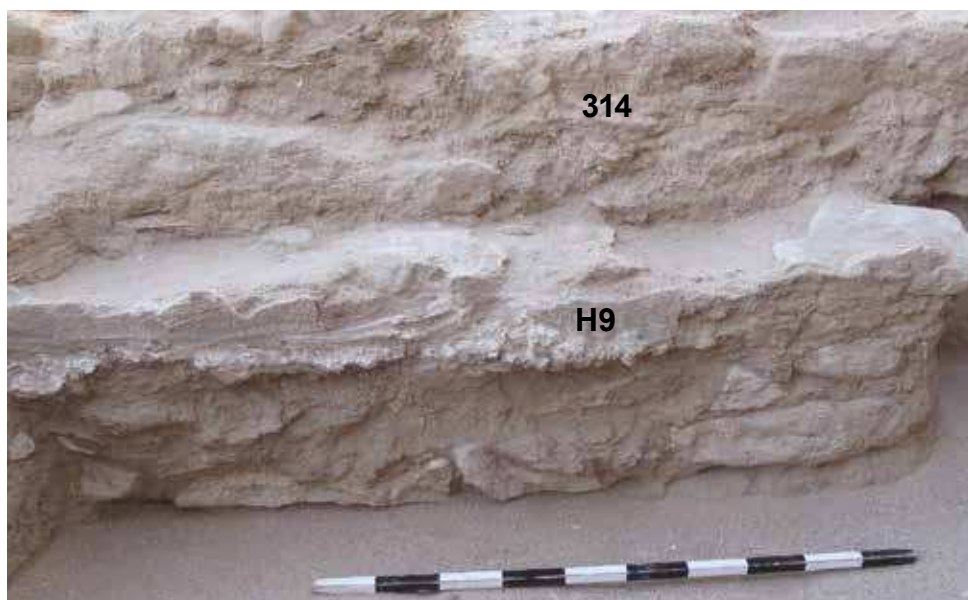


Fig. 96. Wall H9 with foundation and plastered front, bordering the first temple terrace. Behind, wall 314, partly resting on wall H9. Trench H, looking NE, scale unit 10 cm (2011).

undoubtedly broken-up fragments of an older floor re-used in a new context, and in some cases it could be seen that the original plaster floor covered and included a layer of small oolitic chips.

Floor H13 extended at least 6 m north-south and 8 m east-west as it continued into Trench E. In the west section in Trench B it corresponded to floor B88 (1.39 m a.s.l.), which was covered by another floor, B87 (1.51 m a.s.l.) (figs. 89-90). The upper floor was probably removed during the French excavation in Trench H as a layer rich in finds which is sometimes described as a floor (Calvet & Pic 1986 p. 24: 300). In 2012, a similar

layer of ashy occupation debris could be seen to cover floor H13 in the north section of Trench H.

Both floors were uncovered in the north of Trench E (figs. 52 and 81). They both consisted of plaster fragments, local beach-rock stones and oolitic limestone chips bonded together in brown compact clay. Between the earliest floor E314 (1.35-1.16 m a.s.l.) and the latest floor E311/E179 (1.56-1.44 m a.s.l.) was a layer of drift sand with only few finds (E312, 1.50-1.31 m a.s.l.). The earliest floor was laid on loose, sterile beach sand.

Exactly where the east-west main section crosses Trenches A and E (figs. 52 and 81) the pavement (fig.

Fig. 97. Floor H13 lying outside the first temple terrace bordered by walls H17 and H9. Trench H, looking E, scale unit 10 cm (2012).



46: E203) fronting the thin-walled building of Phase 3 was very disturbed. However, in the immediate neighbourhood of the section the pavement met with floor E314, and the top level of the pavement (1.32-1.39 m a.s.l.) corresponded well with the first of the floors outside the temple terrace. It is difficult to avoid the conclusion of contemporaneity between these two structures, i.e. between the thin-walled building of Phase 3 and the first floor outside the first Temple terrace.

South of the main section, the area between the pavement E203 and the temple floors was taken up by the oven E138/E143. The east wall of the oven, E143, in its late phase, was clearly resting on the second floor (E179) (figs. 50 and 88).

Oven H4/H5

1.5 m west of terrace wall H9 and c. 3.50 m south-east of oven E138/E143 was another two-walled oven, H4/H5, first encountered in the 1980s (Calvet & Pic 1990 Plate I: 346) and re-excavated in 2011 (figs. 53 and 98). Oven H4/H5 bore a strong similarity to oven E138/E143 as it consisted of two parallel walls with the same distance to each other (60 cm) and the same orientation. The walls were built of beach-rock stones set in

clay, and they had almost the same width as in E138/E143 being not more than 32 cm wide. Only one course of the walls was preserved, to a height of 25 cm with base constructed at a level of 1.40-1.42 m a.s.l. The west wall was uncovered with a length of 1.46 m, the east wall in a length of 1.60 m and both walls continued into the south baulk of the trench. The inside of the walls were extensively marked by fire, and between the walls was a thick layer of ashes. Below the oven was a thick layer of dark ashes (H8) which could be remains of an earlier hearth upon which oven H4/H5 was built, much like hearth E216 below oven E138/E143 (see p. 42).

The south section of Trench H was too disturbed to throw light on the relationship between temple terrace wall H9 and oven H4/H5, but they seem to have been constructed at around the same level (c. 1.40-1.50 m a.s.l.).

New terrace with channel

At some point, a new terrace was constructed around the Temple. It was uncovered on the southwest and northwest side of the Temple during the French investigations (fig. 105). The terrace was bordered by a nar-



Fig. 98. Oven H4/H5.
Trench H, looking S, scale
unit 10 cm (2011).

row channel integrated into its construction (Calvet & Pic 1990 Pl. I: 306/307), and it was covered by a gravelly plaster floor at a level of 1.70-1.90 m a.s.l. (310).

Wall H9 that bordered the first terrace towards the southwest was later used as foundation for a real wall, i.e. a wall with two faces, wall 314 (figs. 53, 94-96) (Calvet & Pic 1990 Pl. I: 314). Inside this wall a white plaster floor, 312, was noted at a level of 1.80 m a.s.l. The section drawing (Calvet & Pic 1986 fig. 5, p. 17) indicated that it was laid out directly on a sandy layer. West of wall 314 and abutting it was another floor, a fragmented white plaster floor at the same level (1.79 m a.s.l.) (Calvet & Pic 1986 Pl. I: 316, fig. 5, p. 17, fig. 6, p. 21) and covering both wall H9 and oven H4/H5.

The continuation of wall 314 towards the north was destroyed by stone plundering (Calvet & Pic 1986 p. 24). A few meters further north a stone plundering pit (B47) was seen in the section through Trench B (figs.

89-90). However, neither pit nor wall continued into Trench E, so the wall must have turned northeast. The sloping position of the layers south of the pit actually indicates the existence of a wall at this place.

According to the excavators the channel and terrace (306/307) was integrated with the so-called Buttress Wall (Calvet & Pic 1990 p. 109), but there seems to be no compelling arguments for that, on the contrary, see below p. 91. It is more relevant that the floor of the new terrace bordered by a channel has the same level (1.70-1.90 m a.s.l.) as the floors observed on both sides of wall 314 (1.79-1.80 m a.s.l.) and that these levels correspond to the levels of the first floor (1.60-1.80 m a.s.l.) established outside the "Palace" in Phase 4. These floors, the channel and wall 314 seem to be part of a renovation of the Temple that may well have included the first floor outside the "Palace" (1.60-1.80 m a.s.l.) as well as the "Palace" itself.

2.5.2. The Dilmun Temple, Southeast

by Flemming Højlund

The trench in which the foundation wall for the Temple was constructed was seen to cut through the Mesopotamian Ur III layers in Trench C. As was the case northwest of the Temple two consecutive terraces were noted to the southeast (figs. 99, 104). First, a thick floor of clayey plaster followed by a floor made of flat stone slabs and again this second phase was bordered by a channel that must have been connected to the channel previously noted along the west and north side of the Temple. Below the clay and stone terraces was a series of thin clay, sand and plaster layers, probably activity levels from the construction of the Temple (fig. 16). A 4 m wide clay platform was found running parallel to the southeast wall of the Temple at a distance of 11.7 m, probably limiting the terrace and perhaps at the same time creating the foundation for a wall (figs. 99 and 110). Further to the south a wide area was covered by oolitic limestone chips from the cutting of the temple ashlar (figs. 5-6).

The area south of the Dilmun Temple was heavily disturbed by stone plundering. This activity had removed most pre-existing structures from the surface of the tell to a depth of several meters turning it into homogenous soil in texture, hardness and colour. The disturbed layer consisted of light brownish clayish sand, quite loose, with scattered fist to head sized beach-rock stones, lumps of clay and fragments of oolitic limestone, probably from the Temple itself, besides two complete oolitic limestone ashlar.

The trench for the temple foundation

Immediately south of the Temple, west of the presumed entrance a 1 x 1.5 m sondage was made, exposing the southern face of the temple foundation (figs. 99-100). The foundation wall was built from beach rock stones laid in brown clay, which also covered the wall face. The outline of the trench in which the foundation wall was constructed, could be seen in the horizontal level (fig. 101) and in three different sections against the foundation wall (fig. 102). The foundation trench cut through the Mesopotamian Ur III settlement layers (described above p. 15) which originally continued towards the north.

At three different points the base of the foundation wall was measured to 0.3 m, 0.3 m and 0.63 m a.s.l., and the outer limit of the foundation trench was followed up to level 1.4 m a.s.l. where it was disturbed by stone plundering. The French excavators measured the foundation wall's base to 0.3-0.4 m a.s.l. (Calvet & Pic 1990 fig. 10: section 9.9, Pl. II: section 2.2: 337 and 392).

The Temple's southeast terrace

Southeast of the Temple, a stone floor (C68) had escaped the destruction of the stone plunderers (figs. 7, 10, 99 and 104). It was preserved over an area of 5 x 6.5 m in Trenches C1, C3, CE, CEE and CW and was constructed at a level of 2.09-2.24 m a.s.l. The stones were of the local beach rock, mostly rough-surfaced, some revealed marks from cutting. A large block of white plaster with oolitic limestone chips was included in the floor cover.

The stone layer covered a thick layer of hard greenish clayey plaster (C69) encountered at a level of 1.95-2.10 m a.s.l. (figs. 11, 16 and 27), probably a previous floor in its own right. The continuation of such a floor in the same level can be seen 6 m further south in front of the clay platform (fig. 99: C149, see p. 81).

Patches of the stone floor (C68) set on a 17 cm thick clayey plaster containing a few oolitic limestone chips (C69) was preserved on top of the Mesopotamian wall C28 (fig. 103).

Approximately 2.7 m from the Temple the thick clay layer (C69) was cut by a 1 m deep pit with a diameter of 30-60 cm that cut down into the natural sand (fig. 16). Perhaps this was intended to hold a large temple standard, cf. the common stamp seal motive composed around a central standard topped with moon and sun symbols (Kjærum 1983 nos. 108-138. Abu-Laban above figs. 698, 710, 727). It is unclear if the posthole also breached the stone floor.

The second water channel

Along the southeast edge of the stone floor and an integral part of it was a 2.5 m long runnel cut into large oolitic limestone blocks (figs. 25-26, 99 and 104-106).

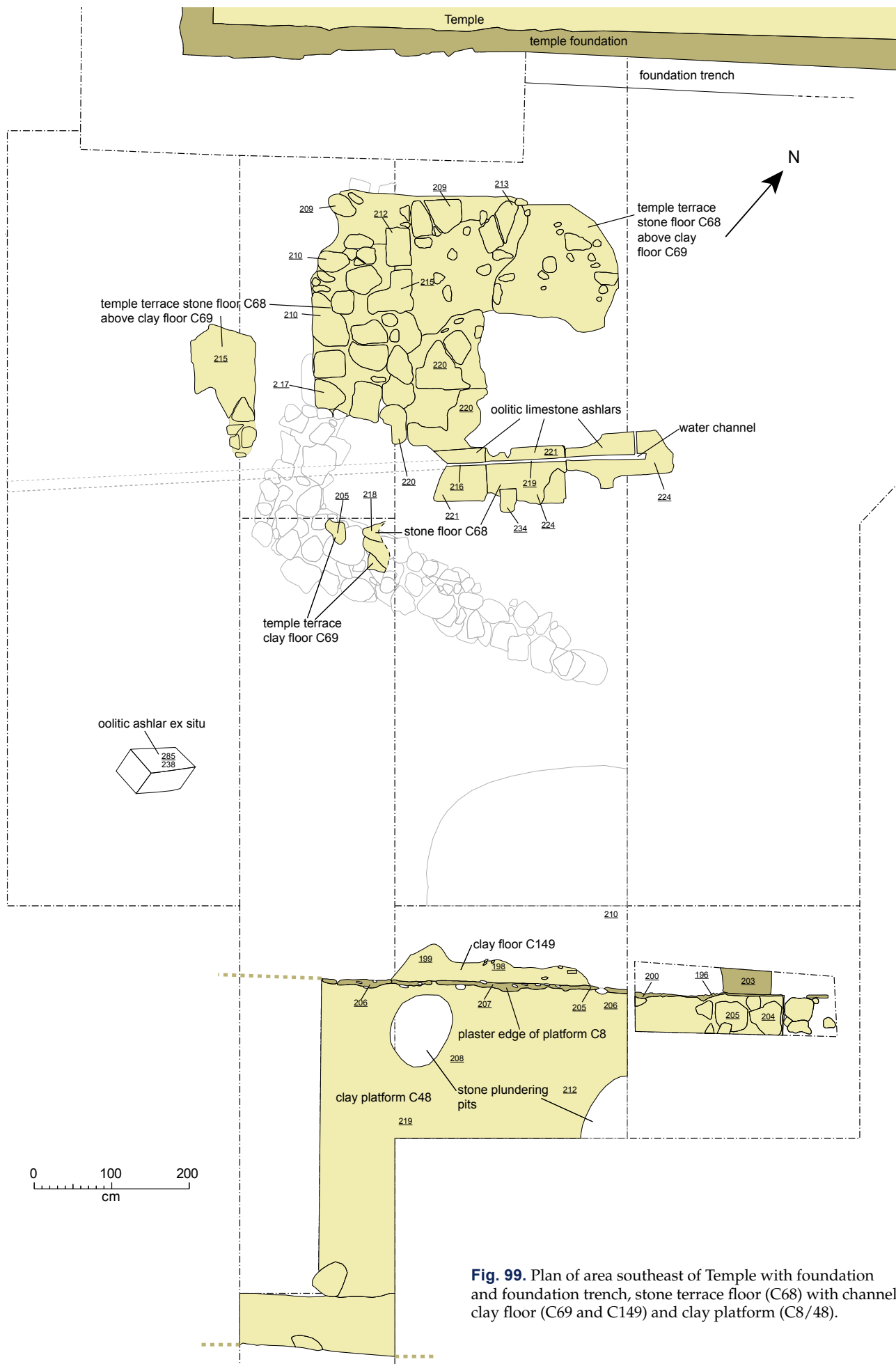


Fig. 99. Plan of area southeast of Temple with foundation and foundation trench, stone terrace floor (C68) with channel, clay floor (C69 and C149) and clay platform (C8/48).

Fig. 100. The SE foundation for the Temple exposed, built of rough beach-rock set in clay and coated with clay. The upper part of the foundation wall was uncovered in the 1980s and hence the clay has washed away. Looking NW, scale unit 10 cm (2011).



Towards the east the runnel bifurcates, one arm leading off towards the northwest, the other continuing towards the northeast (figs. 99 and 104). The blocks were extremely worn and battered, and there were clear traces along the channel of further construction that had been removed, e.g. south of the runnel a broken ashlar was sitting *in situ*, possibly the last remains of an east-west wall delimiting one side of the runnel (fig. 106).

Southwest of the Temple was another channel (fig. 105), excavated by the French mission, running parallel to the temple wall and with the same distance to the wall of c. 5 m as the channel described above (Calvet & Pic 1990 Pl. I: 307). The level of the southeast channel is 2.16-2.19 m a.s.l., i.e. more than 30 cm above the level of the southwest channel which has levels between 1.87 and 1.50 m a.s.l. The channel continued northwest of the Temple ending at a level of 1.45 m a.s.l. There can be no doubt that the channels all fit together in one system designed to transport water from the southeast front of the Temple to its rear side.



Fig. 101. The SE temple foundation with the trench for the foundation exposed. Looking NW, scale unit 10 cm (2011).



Fig. 102. Section in front of the Temple showing the foundation trench cutting through sterile beach sand layers, looking SW, scale unit 10 cm (2011).



Fig. 103. Remains of clay floor C69 and stone floor C68 on top of Mesopotamian wall C28, cf. fig. 99. Trench C1, looking N, scale unit 1 cm (2010).

The same difference in levels characterises the floors related to the channels in the two areas where they have been uncovered, i.e. they are higher southeast of the Temple than to the northwest.

The French excavators interpreted the west channel (307) as contemporary and related to the Buttress Wall (Calvet & Pic 1990 p. 109), but nothing prevents the Buttress Wall from being a much later construction added on top of the terrace and the channel (see p. 91).

The channel leading from the inside of the Temple to its rear, described p. 71 (fig. 105), seems to belong to the original Temple, whereas the channel going around the Temple, described here, must be part of a renovation of the Temple, possibly contemporary with the construction of the “Palace” in Phase 4 (see above

p. 75). The two channels cross each other behind the Temple and could not function at the same time. If these channels are related to altars and specifically the need to clean altars after bloody sacrifices, as argued above p. 71 (cf. Andersen & Højlund 2003 p. 323-324), then we may be permitted to suggest that the altar was at some point moved from the inside of the building to the area in front of the entrance on the southeast side.

The construction activity levels

Below the clay/plaster floor (C69) came a series of thin, 1-4 cm thick layers at a level of c. 1.75-1.95 m a.s.l. (figs. 11, 16, 27-28). This sequence of layers was excavated and sectioned in three different places in the north

Fig. 104. Temple terrace stone floor C68 with (left) channel cut in oolitic limestone blocks. In foreground the floor is disturbed by stone plundering. Trenches C1, C3, CE and CEE, looking SW, scale unit 10 cm (2011).



Fig. 105. Plan of the Temple with two channels, the first (blue) and the second (red), and Buttress Wall (grey).

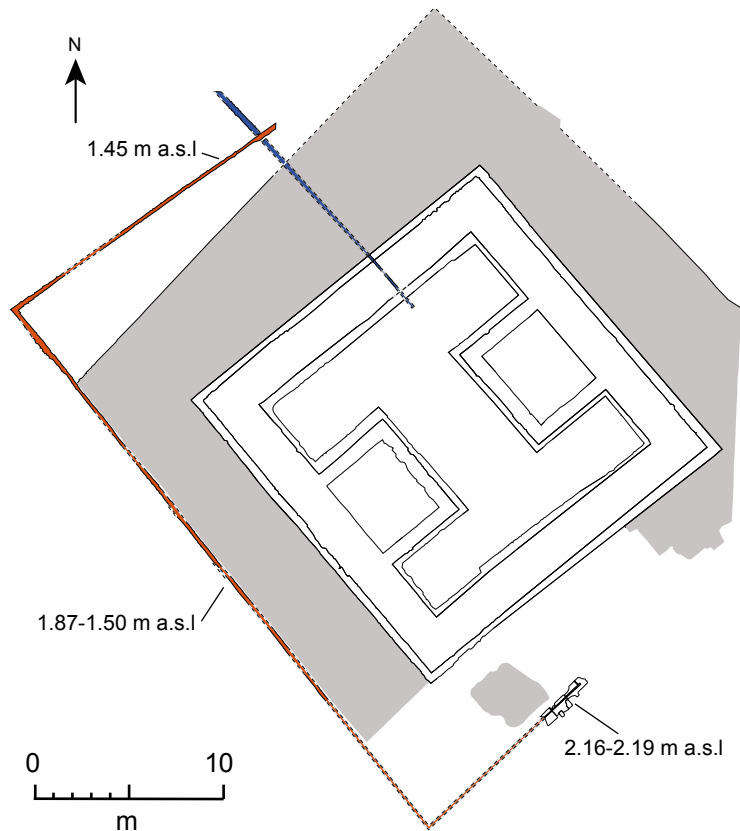




Fig. 106. Temple terrace floor (C68) with oolitic limestone ashlars with cut channel. Trench CE, looking SE, scale unit 10 cm (2010)

of Trenches C3 and CE. The layers consisted of either clay, plaster or sand, with some bitumen and oolitic limestone chips. The slight thickness of the layers and the fact that the sequence of materials was different in each section make it unlikely that they were intentionally constructed floors and rather suggest that they were activity levels formed during the construction of the Temple, consisting of materials used in the construction.

These thin clay, sand and plaster layers extended up to a distance of c. 3.3 m from the ashlar wall of the Temple, i.e. only below the north part of the stone and clay floors (C68-69). Samples of charcoal from two of these activity layers (fig. 11) have been radiocarbon dated (cf. p. 250).

The clay platform

C. 11.7 m south of the Temple a c. 4 m wide platform (C48) was uncovered at level of c. 2.1 m a.s.l., i.e. the same level as the Temple terrace floors (fig. 99). The clay platform was followed for over 7.7 m from Trench C1 to CEES, and it ran parallel to the southeast wall of the Temple. The platform was sectioned in the middle of Trench C1 (fig. 107), and the core of the platform could be seen to consist of sand reinforced at the north and south ends and close to its upper surface by fist sized stones set in hard clay. The surface of the platform was heavily eroded/worn and damaged by stone plundering pits (fig. 99). In the north part of the platform stones continued down to level 1.63 m a.s.l., or alternatively the platform may have been constructed on top of an

earlier wall of flat stones that was, however, not accompanied by any datable cultural material.

The clay platform was limited in the north by a 10 cm thick and up to 20 cm high edge of greyish-white plaster in which oolitic limestone chips were embedded (C8) (figs. 22-24, 99, 108). The chips were c. 5 cm wide and between 5 and 20 cm long. The lowest two centimetres of the plastered front protruded slightly (figs. 23-24). Towards the east, right in front of the part of the Temple's southeast wall where the entrance is likely to have been, several features occurred that are probably related to such an entrance. The plaster edge turned 90 degrees towards the south, the platform was reinforced with large stones and just north of the plaster edge was a square construction of oolitic limestone chips set in plaster (figs. 99 and 109).

The south end of the platform was only uncovered in Trench C1 where it seemed more solid with a thicker white clay layer (C16) reaching up to a level of 2.34 m a.s.l. (fig. 110). In the east section of Trench C1 faint traces of a pit could be seen almost from the surface (fig. 107), cutting down to the south part of the platform. This is probably a plundering pit that removed stones from a wall constructed on the southern, more solid part of the platform.

North of the platform, a horizontal clayey surface (C149) was uncovered in Trench CES, at a slightly lower level, 1.98 m a.s.l., stretching only 0.5 m northwards before it was disturbed by stone plundering activity (figs. 99 and 111). This surface probably originally continued north and went flush with the clayey floor in front of the Temple (C69).

Fig. 107. Amalgamation of Sections 11 and 12, cf. fig. 8, through clay platform (C8 and C48) and layers of oolitic limestone chips and fine beach-rock chips. Trenches C1-2, looking NE.

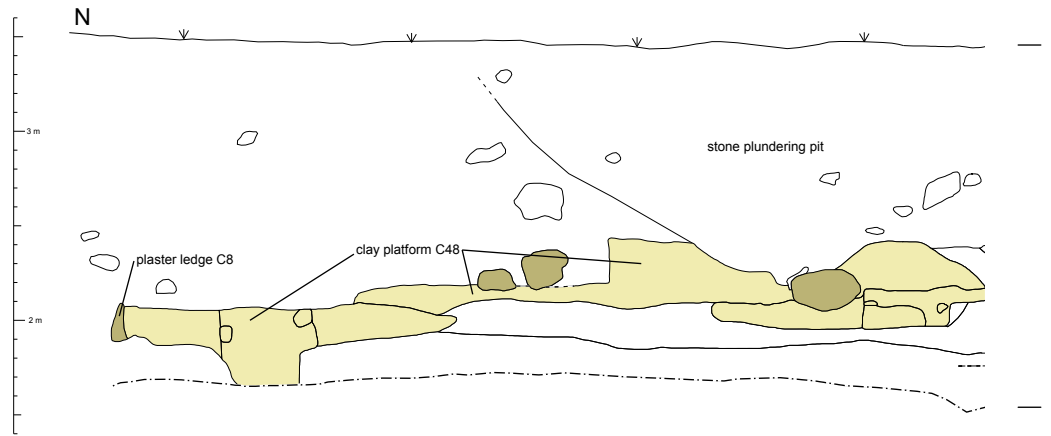


Fig. 108. Plaster edge (C8) of clay platform (C48) with inset oolitic limestone chips. Trench CES, from above, scale unit 10 cm (2011).



Fig. 109. The continuation of the clay platform into Trench CEES, looking NW, scale unit 5 cm (2014)



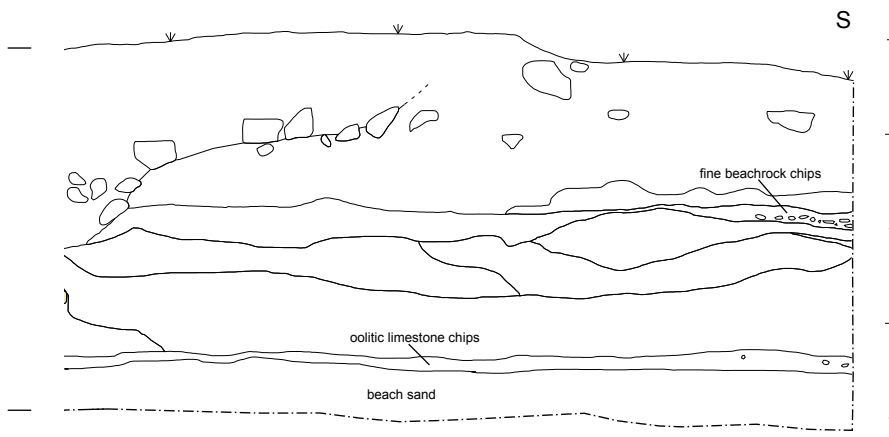


Fig. 110. Clay platform (C8 and C48). Trenches C1-2, looking NE, from above, scale unit 5 cm (2009).

The orientation of the clay platform, the level of its upper surface and the clay floor that extends north relates the platform to the Temple and to the temple terrace floor, most likely as the termination of a terrace and as a foundation for a wall surrounding the Temple, possibly in continuation of wall 314 on its southwest side (fig. 53, p. 75).

A layer of oolitic limestone chips

Between c. 18 and 26 m from the southeast wall of the Temple, a thin layer (C19) of oolitic limestone chips was observed right above virgin soil in Trench C2 at a level of c. 1.75 m a.s.l. (fig. 112). 82 oolitic limestone

chips were collected beside the broken off corner of a finely dressed ashlar (figs. 113-114). The chips vary in size from 13.5 x 9 x 5.5 cm to 1.5 x 2 x 0.7 cm (fig. 115). They derive from the cutting of limestone ashlar, probably for the Temple, and as such indicated the ground level at the time of the temple construction. Similar levels of oolitic limestone chips were located during the excavation of the Barbar temples in Bahrain (Andersen & Højlund 2003 p. 327. Højlund *et al.* 2005 p. 124).

No cultural material was found in the chip layer. It was covered by a c. 50 cm thick layer of white sand, patched with yellowish brown sand, but with minimal horizontal stratification. This layer contained some Barbar sherds and a triangular prismatic stamp seal



Fig. 111. North part of the clay platform (C48) with attached clay floor C149 to the left of plaster edge C8. Trenches C1 and CES, looking NE, scale unit 10 cm (2011)

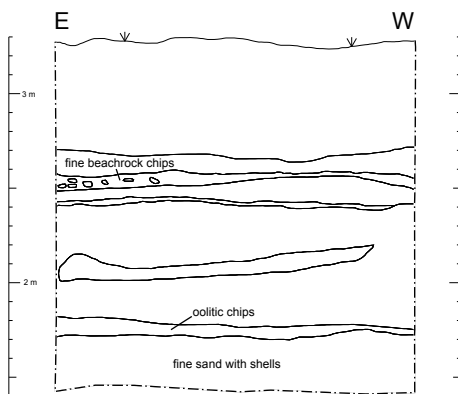


Fig. 112. Section 13 through layers with fine beach-rock chips and oolitic limestone chips. Trench C2, looking SE.

(fig. 748, level 2.05 m a.s.l.). The seal is in Arabian Gulf style (Abu-Laban below p. 141) and belongs to the late 3rd millennium BC and must therefore be out of context, since the white sand layer must be later than



Fig. 113. Oolitic limestone chips from layer C19 in Trench C2.

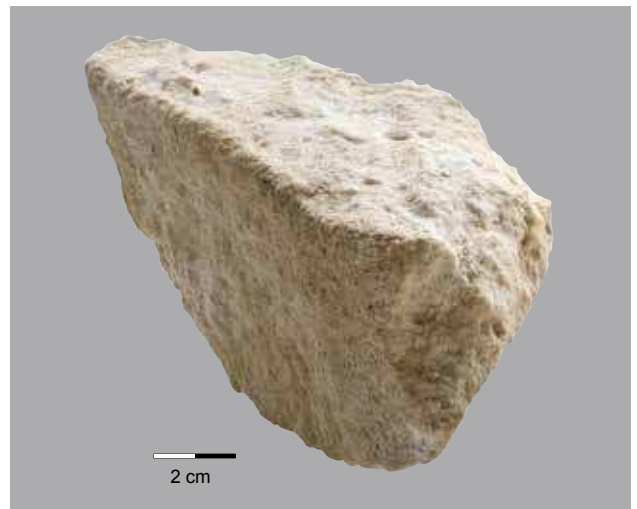


Fig. 114. Well dressed corner fragment of oolitic limestone ashlar from layer C19 in Trench C2.

the construction of the Temple, i.e. early 2nd millennium BC.

Further south, at a distance of 45 m from the Temple a similar, 10 cm thick layer of oolitic limestone chips was noted at a level of 1.31-1.38 m a.s.l. in Trench G1 (figs. 4 and 116-117). The layer sloped a little down towards the south. As many as 1165 chips were collected from this layer, varying from 14 x 10 x 3.3 cm to 0.5 x 0.5 x 0.3 cm in size (fig. 118). The chips from Trench C and G1 resemble each other, although those from G1 include more small chips (fig. 115). The layer seems to have accumulated rapidly as the edges of the oolitic chippings were still sharp. The chip layer rested upon a sequence of thin horizontal layers of fine sand interrupted by layers of fine sand and small shells. The level on which the chips were found corresponds with what must have been the ground level surrounding the Temple at the time of its construction.

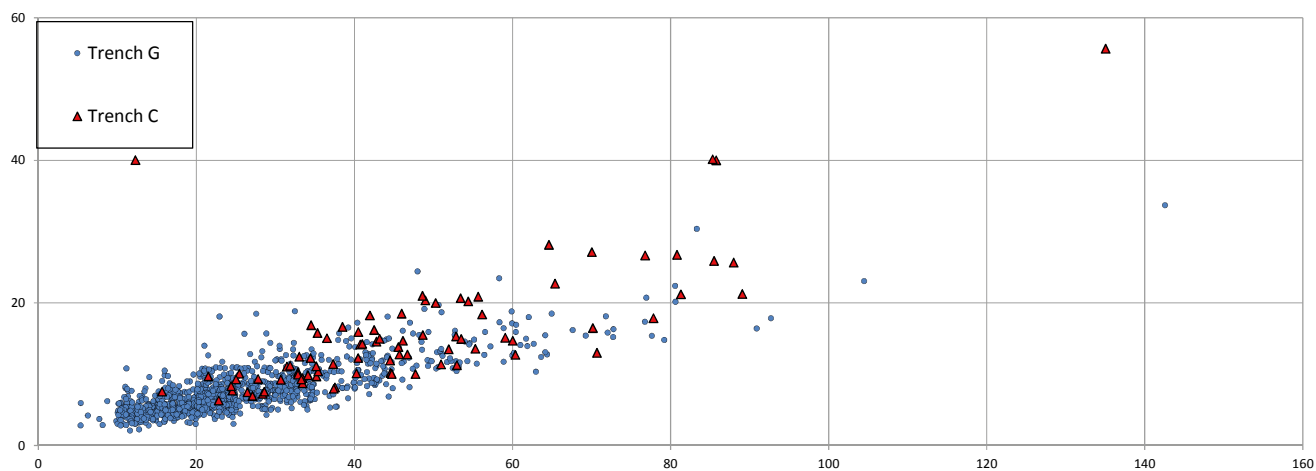


Fig. 115. Diagram of thickness and length in mm of oolitic limestone chips from Trenches C2 and G.

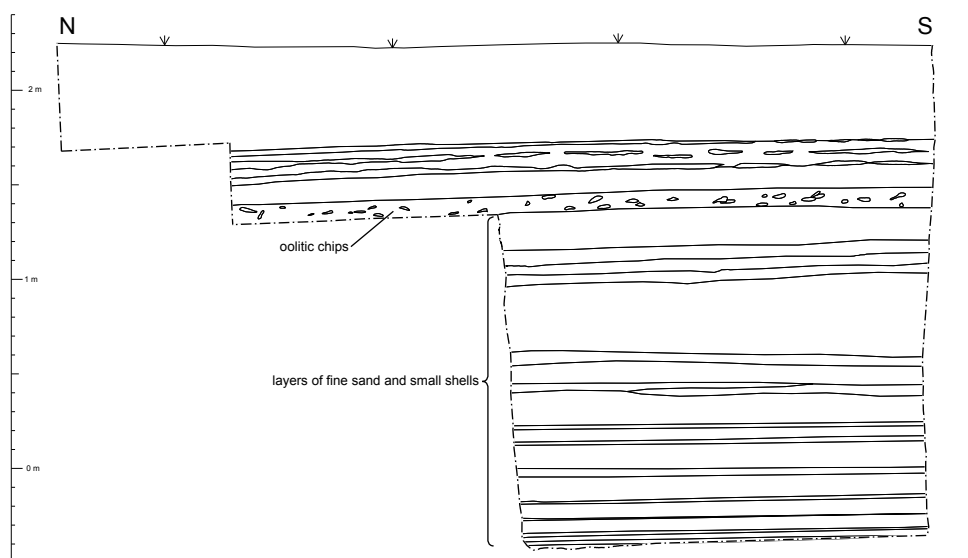


Fig. 116. Section 15 through layer of oolitic limestone chips upon layers of fine sand and small shells. Trench G, looking NE.



Fig. 117. NE baulk of Trench G with layer of oolitic limestone chips (2011).



Fig. 118. Heap of oolitic limestone chips from Trench G. Scale unit 1 cm (2011).

Fig. 119. Oolitic limestone ashlar, 63 cm long, found out of context in Trench CW.



Fig. 120. Oolitic limestone ashlar found out of context in Trench CW. Scale unit 1 cm.

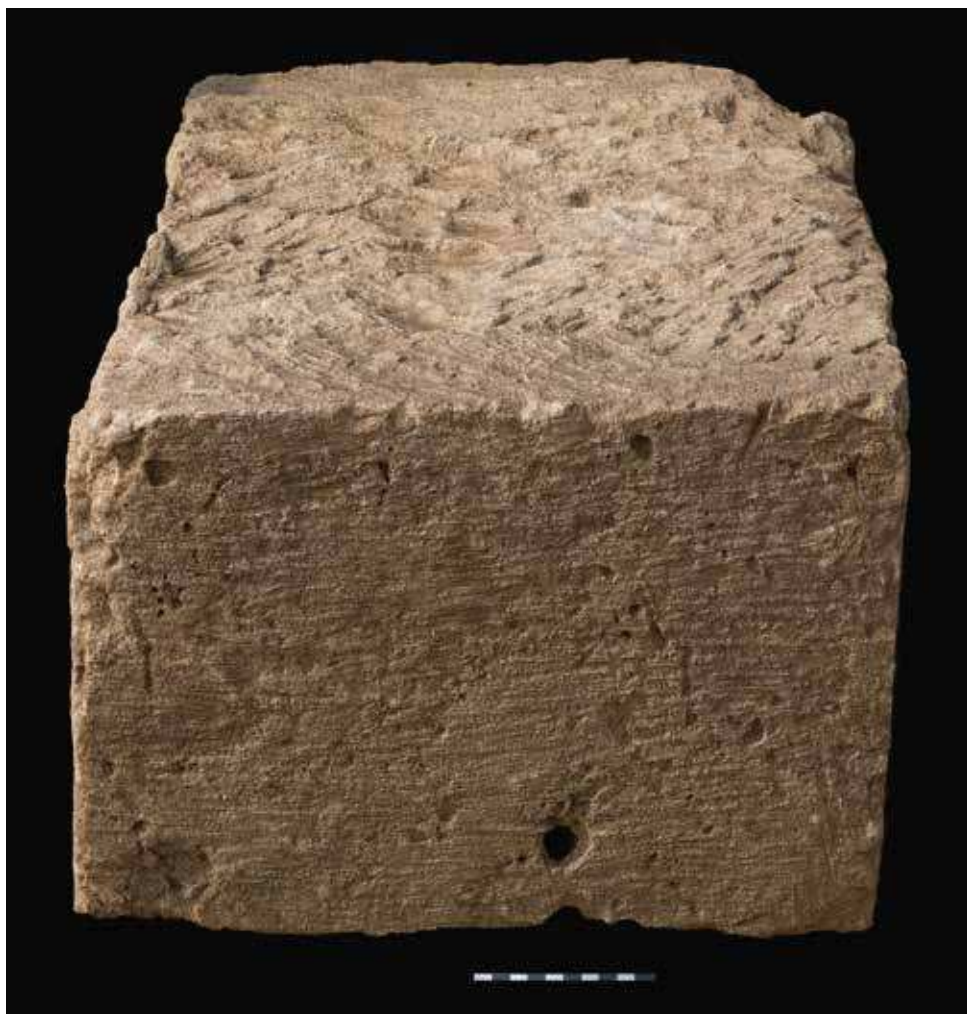




Fig. 121. Oolitic limestone ashlar found out of context in Trench J. Scale unit 1 cm.



Fig. 122. Corner of oolitic limestone ashlar from layer C123 in Trench CEE.

In Trench G2, 17 m further south (fig. 4), no oolitic limestone chips were found.

Oolitic limestone ashlars

A complete oolitic limestone ashlar was found in Trench CW a few meters southwest of the Temple, out of context, lying loose in the upper layer of the tell, in all likelihood stemming from the Temple, being lost by the stone plunderers (figs. 99, 119-120). The dimensions of the ashlar were 32.5-37 x 44.5-39.5 x 63 cm. Its surface was covered by cutting traces made by an adze with a 6 cm wide edge, comparable to the cutting traces seen on the Barbar temple ashlars (Andersen & Højlund 2003 p. 327-328). The most even side, probably the front, measured 32 x 44.5 cm and was covered with parallel lines (fig. 120), possibly from a special technique of evening the side, if not signs of prolonged erosion which have removed the cutting marks. Another side is cut in three different levels to accommodate the neighbouring blocks (fig. 119); in addition, an 11 x 32 cm large and 2.5 cm thick patch of plaster still adhered to this side.

Another complete oolitic limestone ashlar of unique, slightly curved shape was found in Trench J. Its dimensions were 10-24 x 41 x 65 cm (fig. 121).

Other fragments of oolitic limestone ashlars have been found scattered in disturbed areas of Trenches C and J, e.g. a corner of an ashlar with the same evened surface as the above-described (fig. 122), a corner of a 14 cm thick ashlar, and a small, 10 x 19.5 x 6 cm, completely preserved ashlar. Similar small ashlars are a recurrent feature of the ashlar walls of the Barbar temple (Andersen & Højlund 2003 figs. 150, 161, 194, 200, 201).

2.5.3. Erosion of the façade of the Dilmun Temple

by Flemming Højlund

The plinth along the southwest temple façade was severely eroded due to lengthy exposure to the elements, and this erosion damage was subsequently repaired (fig. 123).

The southwest façade of the Temple was furnished with a plinth integrated during the original construction. In 2008, it was noted that, below the upper course of dressed stones, the plinth (Calvet & Pic 1990, Plate I: 319) had a special construction of numerous small chips of oolitic limestone set in a hard greyish plaster (figs. 123-124). When this layer was removed in 2009 (figs. 125-126), several courses of ashlar were revealed, which had been severely damaged by wind and sand erosion over a considerable length of time (fig. 124: 2 and 4). The upper and lower courses (fig. 124: 1 and 4) were of a light greenish, very hard gravelly stone with many shells, whereas the middle courses (fig. 124: 2) were of a different type of stone and comprised of a light greyish, rather fine, gravelly, less hard sandstone. It seems evident that the plaster layer with the oolitic chips was a repair of the eroded ashlars.

Of the 69 chips recovered from the small exposure, 65 were of oolitic limestone and 4 of a different more gravelly variety with shells, a fine beach-rock. All the oolitic chips seem to have been flaked from the same block that had an eroded, rough, golden surface layer, 0.5-1.5 cm thick, below which was a 2 mm thick greenish layer (fig. 127). It therefore does not seem to have been a freshly quarried stone. The heavily eroded surface of the block indicates that it could have been lost in the tidal zone, perhaps during the original construction of the Temple, only to be salvaged from there later.

The severe erosion noted on the rather hard sandstone in the plinth made it evident that had the temple façade not been protected on this front then the much softer oolitic limestone façade would soon have been in a deplorable state. It is therefore likely that the



Fig. 123. The repaired southwest façade of the Temple with chips of oolitic limestone embedded in plaster. To the left, Buttress Wall with remains of plastered surface indicated by arrow, looking NE, scale unit 10 cm (2009).

plinth was constructed to protect the oolitic limestone façade against wind/sand erosion. The construction of the plinth is contemporary with the first phase of the Temple (cf. figs. 124 and 126) and not in itself a later repair and protection of the oolitic limestone façade, and testifies to some previous experience with using oolitic stone in construction.

The repair of the plinth occurred at an indeterminate time. It is perhaps not unlikely that it took place at the same time as the renovation of the temple that entailed the installation of a new terrace bordered by a channel. At any rate, it was completed before the construction of the Buttress Wall (see below) and was completely concealed by this wall (fig. 124: 10).

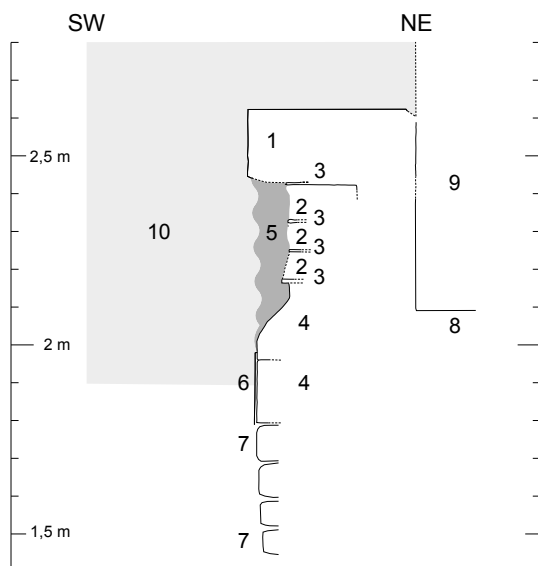


Fig. 124. Section through the southwest façade of the Temple. 1) very hard, gravelly stone with many shells, light greenish. 2) fine, gravelly sandstone, light greyish, less hard. 3) white mortar with glazed-like surface. 4) same as 1. 5) very hard, greyish plaster with oolitic chips, repair of eroded 2 and 4. 6) very hard, greyish plaster. 7) temple foundation wall of rough beach-rock stone in clay. 8) plastered foundation of outer temple wall (Calvet & Pic 1990 plate 1: 320). 9) original position of plundered outer temple wall, evidenced by impressions in 8. 10) Buttress Wall.

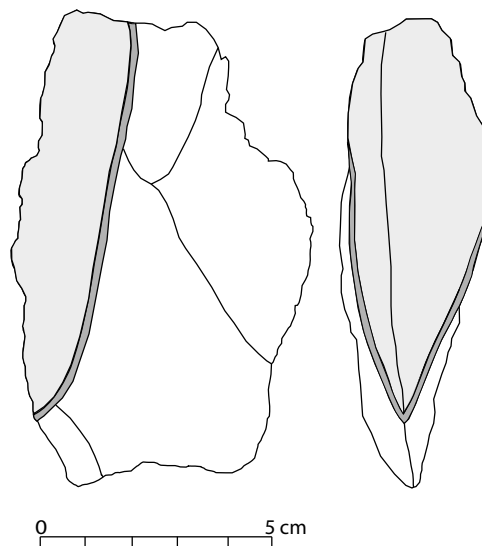


Fig. 125. The eroded façade of the Temple, after removal of repair, looking N, scale unit 5 cm (2009).

Fig. 126. The eroded façade of the Temple, after removal of repair, with foundation exposed, looking NE (2014).



Fig. 127. Chip of oolitic limestone from the repair of the southwest façade of the Temple, flaked from a block with an 0.5-1.5 cm thick, eroded surface layer (grey) above a 2 mm thick greenish layer (dark grey).



2.5.4. The Buttress Wall around the Dilmun Temple

by Flemming Højlund

Sometime after the repair of the erosion of the temple façade, a four meter wide wall was erected around the temple building, perhaps in order to create a large platform (fig. 128). Charcoal mixed into the plaster with which the façade of this building was covered has been radiocarbon dated to 1400 -1300 cal BC at 68% probability, corresponding to Period 4A (figs. 5-6).

The remains of an at least 4 m wide wall was revealed around all sides of the Temple (fig. 105) (Calvet & Pic 1990: p. 109, plate 1: 330, 388, 370, 375). The wall was built sometime after the repair of the temple façade, described above, and it was constructed in rough beach-rock laid in clay, in 1.8-3.5 m wide buttress-like sections up against the ashlar wall (Calvet & Pic 1990, plate 1: 330-332, 339 and 375) (figs. 123, 124: 10 and 128). The outer faces of the buttress sections were plastered (fig. 123). The method of building the wall in buttress-like sections may have been a practical way of organizing the work of several separate work teams. The purpose of the Buttress Wall is uncertain, but it may have been to enlarge the Temple and to create a high platform (Calvet & Pic 1990 p. 116). It is uncertain if the building continued to function as a temple in this phase.

The basal level of the Buttress Wall was c. 2.10 m a.s.l. southeast of the Temple, and c. 1.70 m a.s.l. to the southwest and to the northwest (Calvet & Pic 1990

plan I: 330, fig. 7). Its outer faces were either vertical or oblique, and the wall was preserved up to a height of c. 2 m (level 3.65 m a.s.l.)

The Buttress Wall was severely plundered for its stone material, and along the west end of the southeast side it had almost completely disappeared. In Trenches C and CE all that was left was a c. 20-40 cm thick layer of rough, uncut beach-rock stones packed in hard, compact clay (C59, C112) with top levels at c. 2.21-2.42 m a.s.l. and base levels at 2.09-2.24 m a.s.l. (figs. 129-130), resting directly on the terrace stone floor (C68), described above (p. 76).

The Buttress Wall was referred by the excavators to their level V and dated to Failaka Period 3A, *i.e.* the Old Babylonian period. However, the three pottery rim sherds referred to as evidence of this dating (Pic 2008 fig. 14, nos. 63-65) are not diagnostic of Period 3A or any other period (cf. Højlund 1987 p. 119).

Charcoal embedded in the plastering of the Buttress Wall on the southwest side of the Temple (Calvet



Fig. 128. The Buttress Wall (middle) built up against the plinth at the southwest temple wall (right), looking N, scale unit 5 cm (2010).

Fig. 129. The rough beach-rock stones remaining from the Buttress Wall appearing in Trench CE, looking W, scale unit 50 cm (2010)



Fig. 130. In the foreground, a mass of rough beach-rock stones laid in clay, i.e. the remains of the Buttress Wall lying on the stone floor (C68). Behind it, the Mesopotamian wall (C28) and the clay platform (C8/C48). Trenches C3, C1, C2 and C4, looking SE, scale unit 5 cm (2009).



& Pic 1990, plate 1: 330) has on the other hand been radiocarbon dated to c. 1300-1400 BC, corresponding to Failaka Period 4A (cf. p. 250).

A large quantity of clay went into the Buttress Wall and that clay may have been prepared in the deep pit uncovered in Trench E dated to Period 4A (see p. 58) or in another possible clay pit encountered in Trench J.

A second clay pit

A large pit was noticed in the south section of Trench J-B. It appeared at a level of 2.82 m a.s.l. and went as deep as 1.02 m a.s.l. This pit was filled with layers of alternating clay and sand without stones or other material, sloping from east to west. The continuation of the layers towards the west was cut by a vertical stone plundering disturbance that went down to at least 0.67 m a.s.l., but the bottom of which was not reached. The stratification and size of the pit resembles the clay pit in Trench E (see above), and it may have had the same function.

3. Finds

The objects found during the excavation were registered under a number that consists of three parts: the name of the trench (e.g. A), the number of the layer in this trench in which the object was found (e.g. A128) and the number of the object within this trench (e.g. X738), i.e. A128-X738 means object no. 738 found in Trench A in layer A128.

If possible, the objects are referred to one of eight stratigraphic phases, Phases 1-8 (figs. 5-6). Phase 1

dates to the late 3rd millennium BC; Phases 2-5 belong to Failaka Periods 1-2 in the early 2nd millennium BC, Phases 6 and 7 belong to Failaka periods 3B and 4A in the middle to late 2nd millennium BC, and Phase 8 encompasses what is later than Phase 7, including the stone plundering pits. For a discussion of the dating of these phases see below and p. 239. Objects may also be marked as having been found in layers, which were *Disturbed* or where the dating is *Unknown*.

3.1. Pottery

by Flemming Højlund

During excavation all pottery rim and base sherds as well as decorated side sherds were kept, and of these most rim and base sherds and some side sherds are illustrated here (figs. 131-667). Several pottery traditions are represented. In the oldest phase, Phase 1, dating to the end of the 3rd millennium BC (fig. 5) the pottery almost exclusively belongs to the *Mesopotamian* tradition and must either have been made by Mesopotamian potters on the island or, more likely, imported to Failaka from the north. Only a few sherds derive from the southeast, i.e. Bahrain, the Oman peninsula, and the Indus Valley region.

In the following four phases, Phases 2-5, dating to the early 2nd millennium BC (fig. 5), the pottery almost exclusively belongs to the local East Arabian *Barbar* tradition, with only a few sherds imported from Mesopotamia or from the southeast.

In Phase 6, dating to the mid 2nd millennium BC, the Barbar pottery tradition continues to play a role, but is supplemented by a great deal of pottery in a Mesopotamian tradition.

In Phase 7, dating to the late 2nd millennium BC, all pottery follows the Mesopotamian tradition and there seems to be no remnants of a Barbar pottery tradition.

Throughout this section reference will be made to

the Barbar pottery types (Types 1-53) and the Mesopotamian types (Types 54-100) defined on the basis of material from the 1958-1963 excavations in Tell F3 and Tell F6 (Højlund 1987). Furthermore, reference will be made to the Barbar types (Types B1-76), the Mesopotamian types (Types M1-M27), the Umm an-Nar types (Types U1-12) and the Eastern types (Types E1-10) defined on the basis of material from Excavation 520 at Qala'at al-Bahrain (Højlund & Andersen 1994). With respect to the description of wares reference is given to the typology consisting of ware types A, B, C, D, E, F, G and H that was developed on the basis of material from the 1958-1963 excavations in Tell F3 and F6 (Højlund 1987 p. 103-106).

Phase 1

Mesopotamian pottery. The pottery assemblage from the first phase is very homogenous (figs. 131-273). The ware is hard fired, tempered with a little fine sand and light brownish, yellowish or greenish. Only some of the larger vessels and some ring bases are tempered with chaff. Body sherds are occasionally decorated with horizontal or wavy multiple incised lines or horizontal ridges on the shoulder.

Identical shapes are well-known from Mesopotamia, and several rims have close parallels in period Ur III contexts, although some of them may also continue into the early Isin-Larsa period: elongated “droopy” rims (figs. 131-149) (Armstrong & Gasche 2014 pl. 109); large everted multi-ribbed rims (figs. 152-153) (Type M4) (McCown & Haines 1967 pl. 87: 14 and pl. 148: 2; Armstrong & Gasche 2014 pl. 136); medium and small sized multiple-ribbed rims (figs. 154-159) (McCown & Haines 1967 pl. 84:3-4; Invernizzi *et al.* 2002-03 pl. 9: 20; Boehmer, Pedde & Salje 1995 Pl. 16: b and Pl. 17: Grab 34e; Armstrong & Gasche 2014 pl. 72-73); vessels with a concave rim/neck and a sharp shoulder ridge (figs. 161-167) (McCown & Haines 1967 Pl. 85: 2; Boehmer, Pedde & Salje 1995 Pl. 17: Grab 34a; Armstrong & Gasche 2014 pl. 70); bowls with concave rim, high carination and string-cut base (Type M15) (figs. 168-171 and 182-188) (Armstrong & Gasche 2014 pl. 43; De Meyer 1984 pl. 8:5 and pl. 10: 2), and bowls with straight side, rounded rim and string-cut base (Type M13) (figs. 173-180 and 182-188) (Armstrong & Gasche 2014 pl. 11). The evidence thus points strongly to a dating of this pottery assemblage to the Ur III period.

Additional types include simple, everted rims (figs. 189-213), large vats with sharp ribs on the side and thick rims with ridges with transversal incisions (Type M17) (figs. 214-217). The shoulder of storage vessels may be decorated with ridges (Types M22-24) (figs. 223-227) or groups of incised horizontal or wavy lines (Type M26) (figs. 228-237). A few other incised decorations were found (figs. 238-240). Large bottom-less cylinders occur (figs. 241-243), a shape that becomes very common in the Barbar tradition (B44-48). Most storage vessels must have had round bases, as ring-bases are few.

The scarcity of parallels with the Mesopotamian pottery found in Periods Ia-b and IIa at Qala’at al-Bahrain is surprising since this has also been referred to the Ur III period, although probably beginning already in the late Akkadian period (Højlund & Andersen 1994 p. 168). Though a few shapes occur both on Failaka and at Qala’at al-Bahrain, e.g. the large multiple-ribbed rim type (Højlund & Andersen 1994 fig. 248), the bowl with straight side, rounded rim and string-cut base (Type M13) and the bowl with concave rim, high carination and string-cut base (Type M15), it is surprising that the most common type on Failaka, the elongated “droopy” rim, only occurs once at Qala’at al-Bahrain (Højlund & Andersen 1994 fig. 253, from an undatable context), and that three of the most common Mesopotamian types found at Qala’at al-Bahrain, a rounded, slightly thickened rim on a vertical or splaying neck (Type M1),

a small, everted rim (Type M2) and a large everted rim (Type M3) hardly occur on Failaka.

This might perhaps be explained by the different nature and function of the two sites: Failaka was a specialised trade settlement inhabited in this period by people from Mesopotamia producing all the pottery they needed. Conversely, Qala’at al-Bahrain was the main city of the Dilmunites, and in Periods Ia and Ib the Dilmunites relied on importing storage vessels and fine wares from Mesopotamia and the Oman peninsula. The only pottery they produced was cooking vessels. In the following Period IIa the Dilmunites became self-sufficient in pottery. They began production of storage vessels and fine ware, which brought about a sharp decline in import of pottery from Mesopotamia and Oman (Højlund & Andersen 1994 p. 129-177). The Mesopotamian pottery found at Qala’at al-Bahrain in Period Ia-Ib was therefore a deliberate selection on the part of the Dilmunites of the complete Mesopotamian Ur III pottery inventory, and this might possibly explain the noted differences.

Another possibility is that the differences are due to the chronology of the occupation sequences. The Mesopotamian pottery found at Qala’at al-Bahrain may date slightly different from the Mesopotamian pottery found on Failaka. A tentative suggestion might be that the Mesopotamian pottery from Qala’at al-Bahrain Periods Ia-Ib dates to the Late Akkadian-Early Ur III period and the Failaka assemblage belongs to the mid to late part of the Ur III period.

A cylindrical shape perforated before firing and probably open in both ends is quite common in the Failaka material (figs. 244-256). The diameter is the same as found at Qala’at al-Bahrain, normally 12-18 cm (Type M20). It is probably just a coincidence that this type has not been published from Mesopotamian Ur III sites.

A half-moon shaped impression on the shoulder of a storage vessel (fig. 257) has a Mesopotamian Ur III parallel (Armstrong & Gasche 2014 pl. 50:8).

During the French excavations in the Temple area, similar Mesopotamian Ur III pottery was recovered (cf. Pic 2008 figs. 15-24).

A small number of sherds in *grey ware* has incised and stamped decoration sometimes inlaid with a white material (figs. 258-266). Similar sherds were found in the previous French excavations in Tell F6 (Pic 2008 fig. 29). Of special interest is a hanging vessel with a ribbed rim and pierced shoulder (fig. 258). The shape of this rim has close similarities to a Mesopotamian Ur III jar (Armstrong & Gasche 2014 pl. 73:

3-4), whereas the pierced holes, the decoration and the grey ware combine these sherds with a group of incised and stamped grey-ware hanging vessels from Mesopotamia and Elam that date to the late Ur III–Early Isin-Larsa period (Armstrong & Gasche 2014 pl. 75). Two grey-ware sherds with stamped and incised decoration from disturbed contexts are included here as well (figs. 265-266).

Barbar pottery. No rims or other diagnostic sherds in the Barbar tradition have been found, just one side sherd in what looks like Barbar ware, i.e. red sand-tempered with white dots.

Umm an-Nar pottery. A single rim (Type U8), a base (Type U10) and two wavy-ridged side-sherds (Type U9) (figs. 267-270) in a red fine sand-tempered, hard-fired, dense ware belong to a shape well-known from Qala'at al-Bahrain where it has been referred to the Umm an-Nar tradition (Højlund & Andersen 1994 figs. 310-322).

SE Iran - SW Pakistan. A straight-sided beaker in grey ware with incised herringbone decoration in vertical panels below horizontal lines (fig. 271) may belong in this period, but comes from an undatable context. It resembles Incised Grey Ware from SE Iran or SW Pakistan (cf. During Caspers 1970 fig. 46:3).

Indus Valley pottery. A shoulder sherd in a ware changing between light grey, light brown and light red, with no sand, decorated in black on a red slip (fig. 272) resembles pottery from the Indus Valley or surrounding areas. Another, 1 cm thick, decorated shoulder sherd probably also derives from the East; it is decorated in plum red on a cream-slipped red ware tempered with a coarse black material, but was found in a disturbed area (fig. 273).

Phases 2-5

Barbar pottery. Almost all pottery found in the four phases, 2-5, belong in the Barbar tradition (figs. 274-534). The internal similarity between the pottery of these phases is so great that they must belong to a rather short time span of the Dilmun culture.

The triangular rims resemble the variety of such shapes found in Qala'at al-Bahrain Periods IIb-c (figs. 274-275, 297-310, 381-420, and 501-515) (Højlund & Andersen 1994 figs. 611-712). The same applies to the hole-mouth rims (figs. 284-286, 335-343, 450-462, 529) with an important addition. Phases 3 and 4 both

feature the lip-shaped, up-turned hole-mouth rim of Type B19 (figs. 344-346, 463-468), one of the diagnostic shapes of Qala'at al-Bahrain Period IIc. The number of sherds that can safely be referred to Phases 2 and 5 is so small that the absence of Type B19 rims in these two phases may be purely accidental, and at any rate, the remaining sherds from Phases 2 and 5 have a close similarity to the inventory of Phases 3-4.

The absence of plates with upper chamfered rim face (Types B31 and B67), another diagnostic rim for Qala'at al-Bahrain Period IIc, is of no chronological relevance since no plates at all occur in the assemblages dealt with here. An open bowl (figs. 347-348, 475-479) has parallels at Qala'at al-Bahrain in Period IIc (B24).

In general, ware Types A and B dominate over Type C in all four phases.

Wheel-made and painted pottery, including rippled sherds (Type 47) (figs. 279-282, 315-334, 435-448, 522-528) are present in all four phases, and in phase 5 several wheel-made thin triangular rims (Type 2) (figs. 516-520) occur. Two side-sherds, one in fine, hard, red ware with a few white halos and one in fine, hard, red ware of Type H, both decorated with hatched triangles in black-on-red (figs. 329-330), resemble Type B74, another Qala'at al-Bahrain Period IIc-diagnostic type. A broken vase in a red, fine A or H ware, hand-made (fig. 314), was found in the sand up against the east edge of the pavement E203 in front of the thin-walled building of Phase 3 (figs. 46-48).

Flat plates with a simple rim (Type 43) (figs. 292, 362-364, 492), probably bake-plates, occur in these phases. In Bahrain, this type is first found in the IIIrd Barbar Temple (Højlund & Andersen 2003 figs. 579-583) and in Period "IIF" at Qala'at al-Bahrain. They are probably imitations of a Mesopotamian shape that appears in the late Isin-Larsa period (Armstrong & Gasche 2014 pl. 1).

A unique vessel in A ware (fig. 422) was found broken in many pieces to the north of, but in a slightly lower level than hearth A164, a Phase 4 context (fig. 53). Part of the globular body was decorated with horizontal ridges, as is normal on necked storage vessels in the Barbar tradition, but in this case, the area covered by ridges was circular and clearly separated from the rest of the body that was smooth and unridged. The way this vessel breaks the traditional rules of Barbar pottery production makes it unique.

Mesopotamian pottery. Not a single piece of pottery diagnostic of the previous Ur III phase was found in Phases 2-5. Conversely, there are a few sherds in

light-brownish, light-greenish or light-yellowish, fine sand- or straw-tempered ware belonging to shapes that are commonly found in Mesopotamia in the Isin-Larsa period: a storage vessel with a splayed elongated rim (Type 54A) (figs. 373-374) (Armstrong & Gasche 2014 pl. 127-128), a tall cylindrical bottle (figs. 377-380, 500) (Armstrong & Gasche 2014 pl. 71), a multiple-grooved vertical rim (fig. 295) (Woolley 1934 pl. 265, type 225; Armstrong & Gasche 2014 pl. 76), and a platter with horizontal thickened rim (fig. 376) (Armstrong & Gasche 2014 pl. 9).

Eastern sherds. Some of the painted/slipped sherds could perhaps be imports from the East, e.g. a 1.9 cm thick side-sherd with a polished surface in a fine, dense ware with some mica. A fine pink ware sherd, wheel-made, no clear slip, decorated in black-on-red with a row of people holding hands (fig. 283), from a disturbed area, derives from the same vessel as a sherd from the old excavations in Tell F6, found in a Period 1 context (Højlund 1987 fig. 478).

A correlation of the pottery periods defined on the basis of the 1958-1963 excavations in Tells F3 and F6 and the phases presented in this study takes as its point of departure the presence of hole-mouth rims of Type B19 in Periods 1, 2A and 2B as previously noted (Højlund & Andersen 1994 p. 173), as well as in the two middle of the four phases studied here, Phases 3-4. The absence in Period 1 (located only in Tell F6) of additional types diagnostic of Qala'at al-Bahrain Period IIc, e.g. Types B28, B31 and B67, which are present in Periods 2A and 2B in Tell F3, is paralleled in the four phases examined here, and may be accidental or perhaps in some way related to the difference in function between Tell F3 (domestic architecture) and Tell F6 (public buildings).

The variety of thick triangular rims in Phases 2-4 corresponds with that of Periods 1 and 2A, whereas a few examples of Type 1G, the tall smoothed variant, in Phase 5 resembles the significant increase of this variant in Period 2B, as do the rather many wheel-made thin triangular rims (Type 2).

The pottery assemblage of Phases 2-5 has a great resemblance to the pottery found in the settlement of Al-Khidr on the northwest corner of Failaka (Benediková ed. 2010 figs. 161-180), as well as material from Pottery period 3 distinguished at Saar in Bahrain (Carter 2005 p. 276-277).

It should be noted that not a single sherd diagnostic of Failaka Period 3A (Højlund 1987 p. 119-121) was found during the 2008-2012 excavations.

Phase 6

A small part of the pottery from this phase belongs in the Barbar tradition (figs. 535-557). Especially noteworthy is the simple flared rim (Type 5) with horizontal scored lines on the shoulder (Type 48) (figs. 543-544), which is commonly found in Period 3A and 3B assemblages in Tell F3 and Tell F6 (Højlund 1987 figs. 519 and 534). The ware belongs to Types A, B and C.

The major part of the pottery falls into the Mesopotamian tradition that is well known from Period 3B assemblages in Tell F3 and Tell F6 (figs. 558-605) (cf. Højlund 1987 fig. 538-561). Prominent are storage vessels especially with a triple-ribbed rim (Type 56) (figs. 562-568), bowls with slightly drawn in to vertical rim without rim decoration (Type 67A) (figs. 579-585) or the multiple-grooved variant (Type 67C) (figs. 586-596), beakers with a knob foot (Type 78) (figs. 577-578) and a thin, splayed rim (Types 61-62) (figs. 571-576). A single storage vessel rim of Type 57D with red paint on both inside and outside (fig. 570) should be mentioned. The ware types are predominantly Type E with a few Type G.

Phase 6 is contemporary with Failaka Period 3B and with Period IIIa at Qala'at al-Bahrain (Højlund & Andersen 1997 figs. 129-158) that show the same combination of late Barbar pottery supplemented by a very characteristic Mesopotamian pottery assemblage.

The lack of parallels in South Mesopotamia to this obviously Mesopotamian tradition pottery has been discussed elsewhere (Højlund 1987 p. 158-160. 1989b. 2007a p. 14-15) and will be dealt with again below p. 256. In this connection a few similarities between Failaka Period 3B and pottery from the House of Ur-Utu at Tell ed-Der dated late in the 17th century are of interest: Bowls with incurving rim are uncommon in the Old Babylonian period, but diagnostic for Failaka Period 3B, especially with multiple-grooved rim (figs. 586-596), and two such bowls were found in the house of Ur-Utu (Gasche 1989 pl. 25: 46 and pl. 28:2). In the same house were a number of beakers with very constricted bases (Gasche 1989 pl. 33: 6 and 35: 5 and 17) which are typologically close to the knob foot (Type 78) (figs. 577-578), another type diagnostic of Failaka Period 3B.³ It could thus be suggested that the – at the moment – unknown pottery following after that of the House of Ur-Utu and dating to the 16th century would have a close similarity to that of Phase 6.

Phase 7

The pottery of this phase belongs in the Mesopotamian tradition and consists predominantly of storage

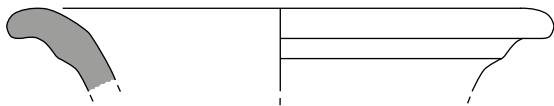


Fig. 131. 2:5.

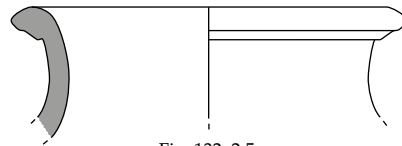


Fig. 132. 2:5.

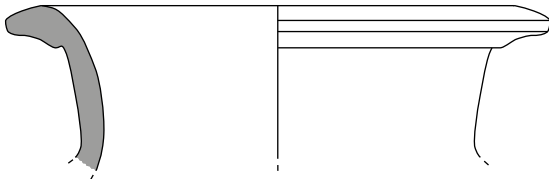


Fig. 133. 2:5.

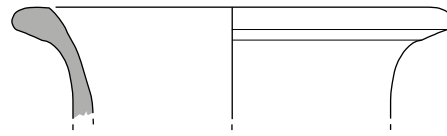


Fig. 134. 2:5.

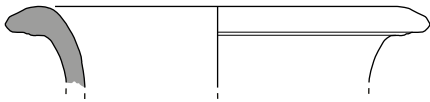


Fig. 135. 2:5.

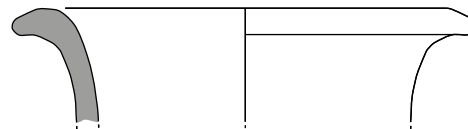


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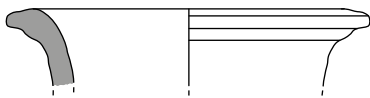


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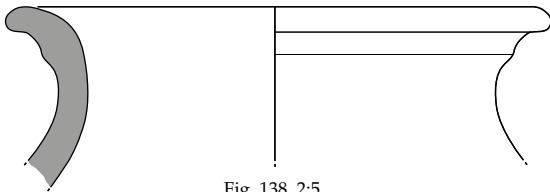


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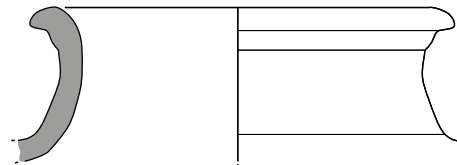


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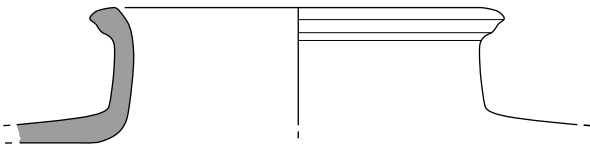


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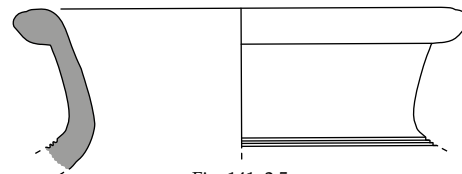


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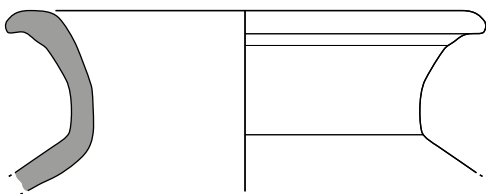


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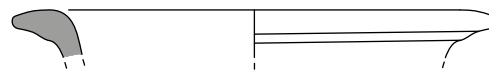


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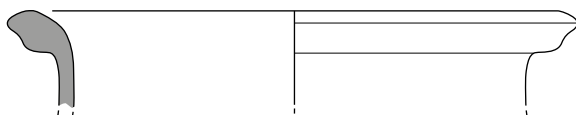


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Fig. 146. 2:5.

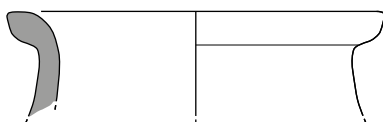


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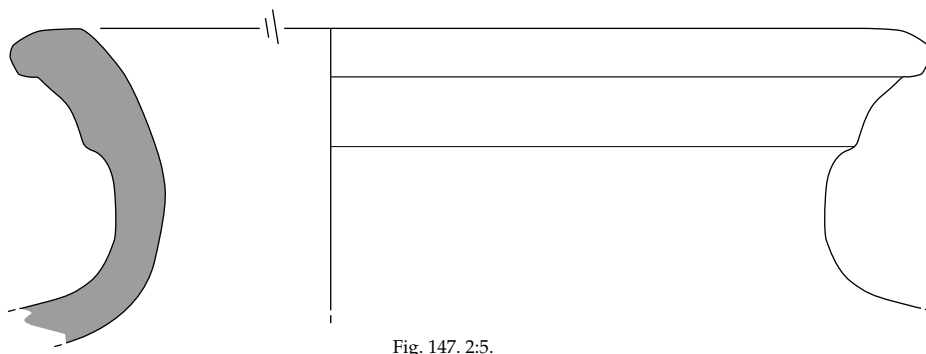


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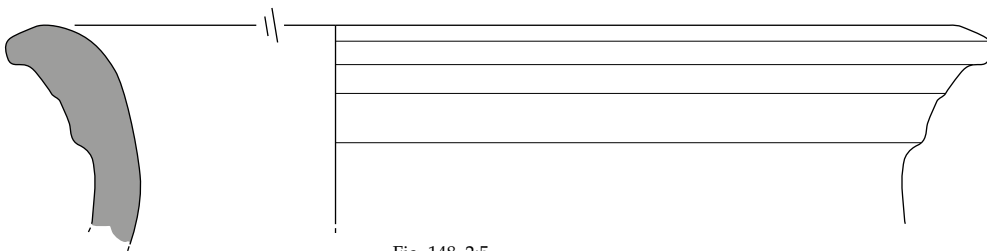


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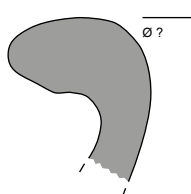


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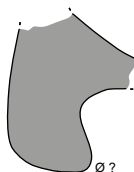


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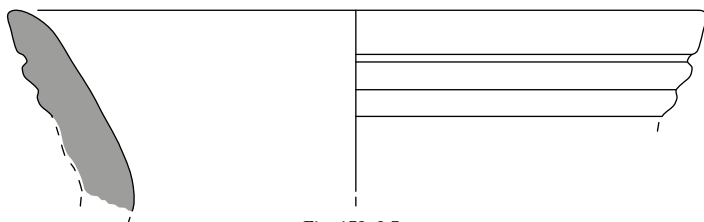


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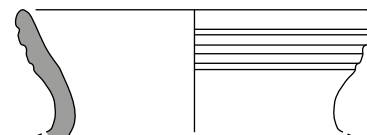


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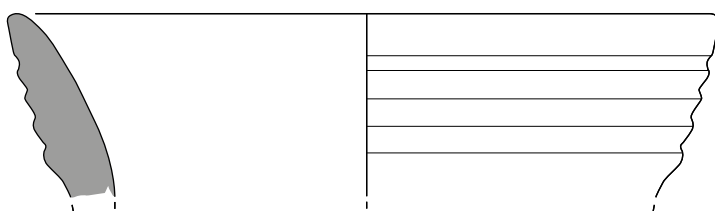


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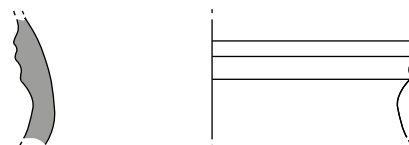


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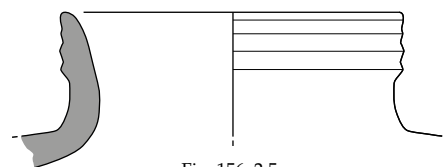


Fig. 156. 2:5.



Fig. 160. 2:5.



Fig. 159. 2:5.



Fig. 158. 2:5.

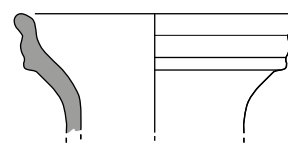


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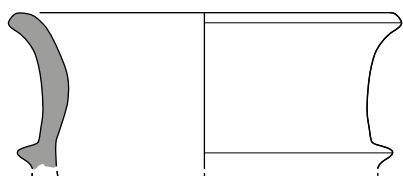


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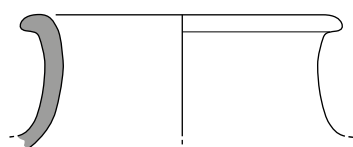


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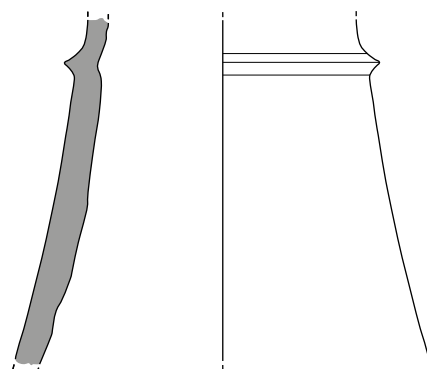


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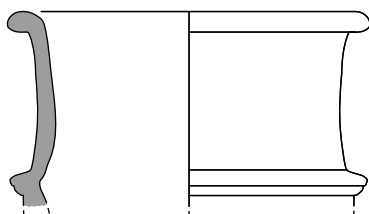


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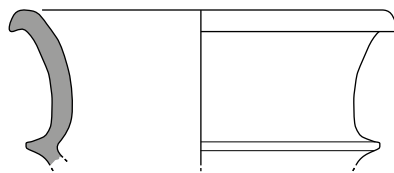


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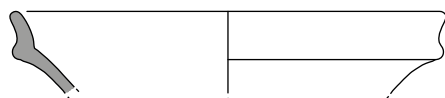


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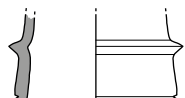


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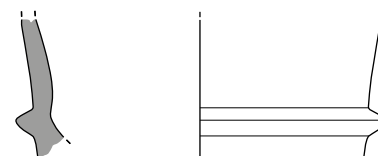


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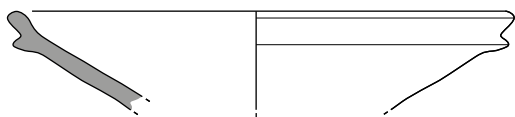


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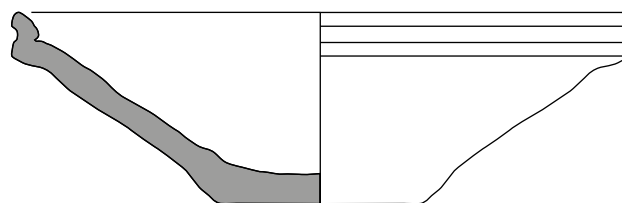


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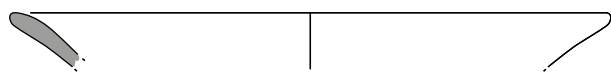


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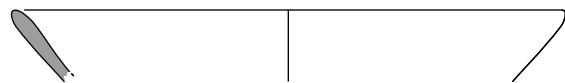


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Fig. 175. 2:5.



Fig. 176. 2:5.



Fig. 177. 2:5.

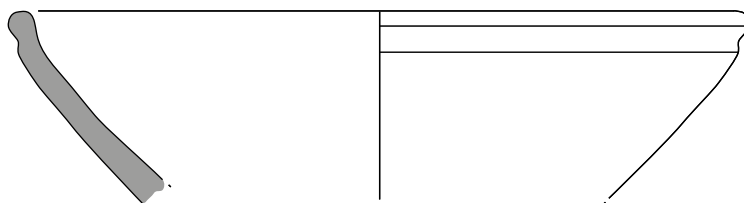


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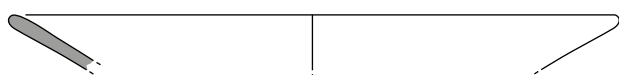


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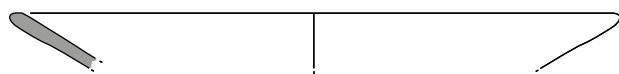


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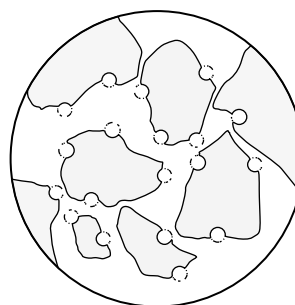


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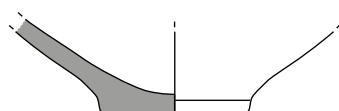
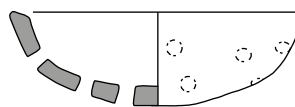


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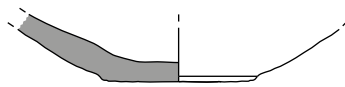


Fig. 185. 2:5.



Fig. 188. 2:5.



Fig. 183. 2:5.

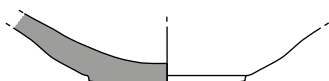


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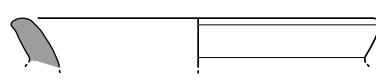


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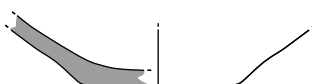


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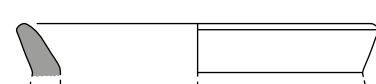


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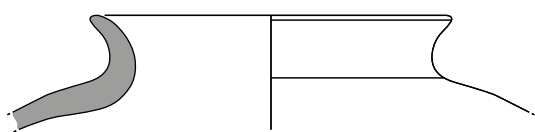


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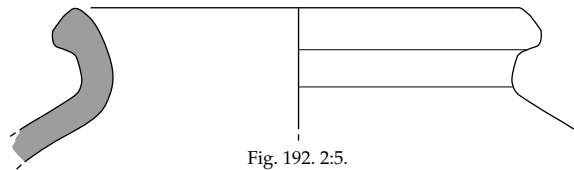


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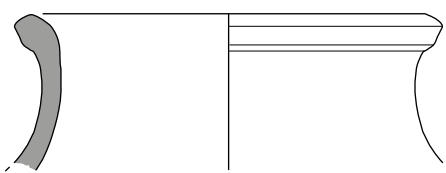


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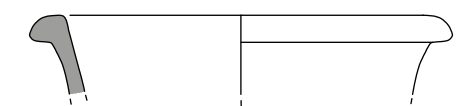


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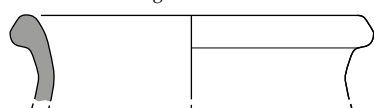


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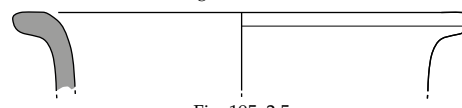


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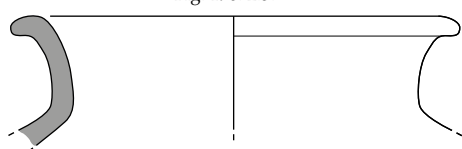


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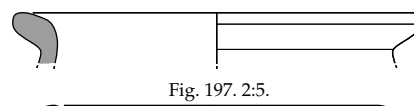


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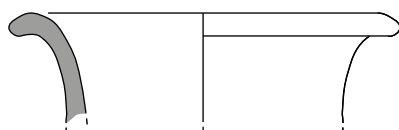


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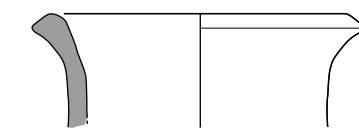
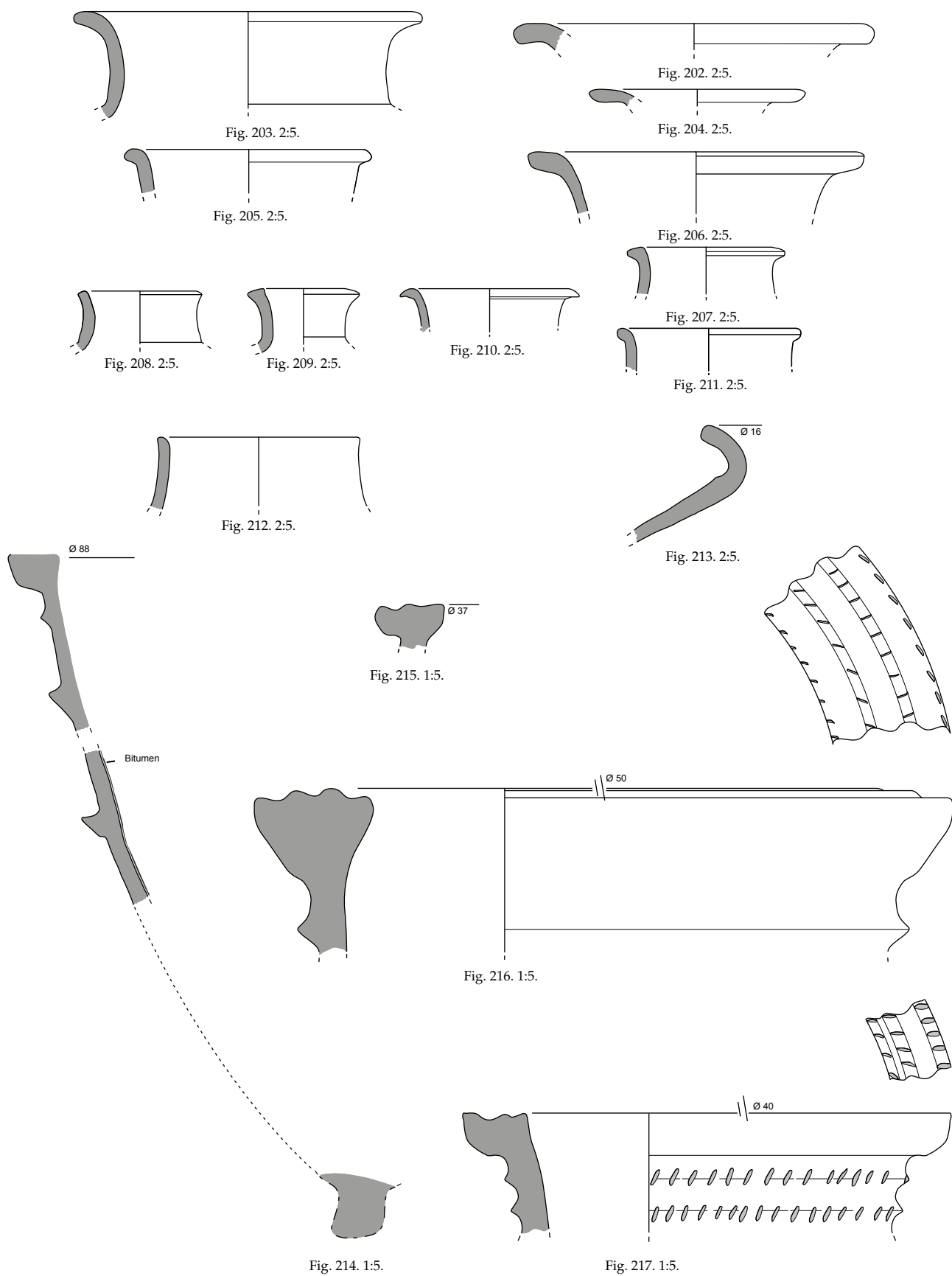


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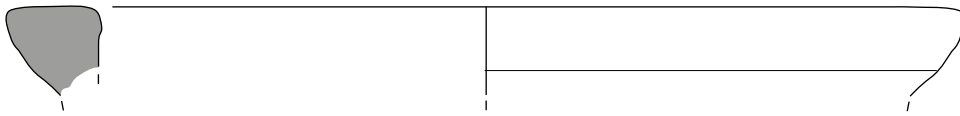


Fig. 218. 2:5.



Fig. 219. 2:5.

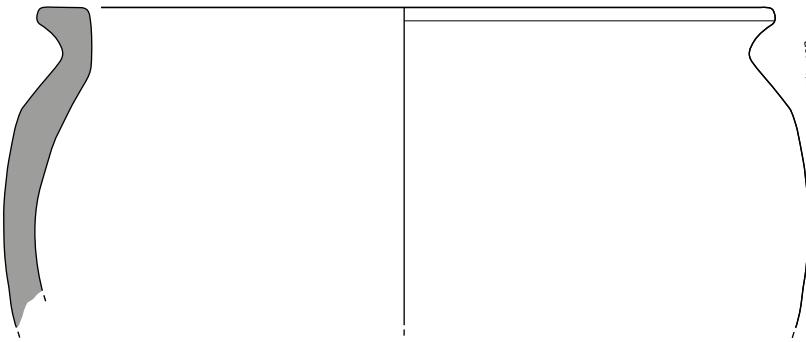


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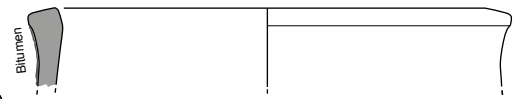


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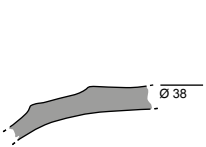


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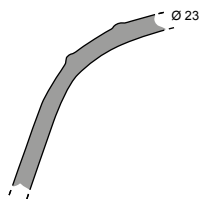


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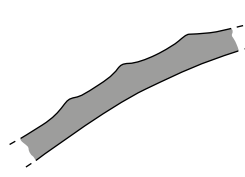


Fig. 225. 2:5.



Fig. 226. 2:5.

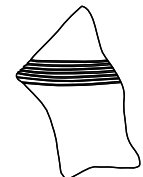


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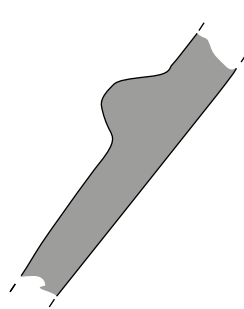


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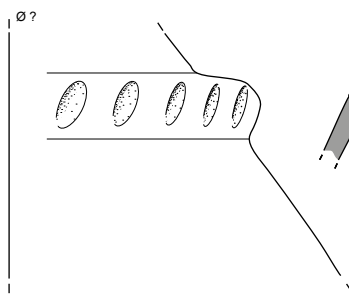


Fig. 228. 2:5.



Fig. 231. 2:5.



Fig. 232. 2:5.

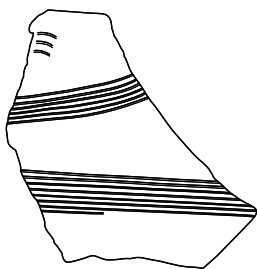


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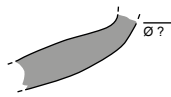


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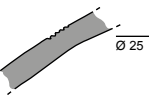


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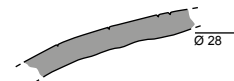


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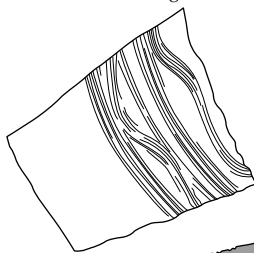


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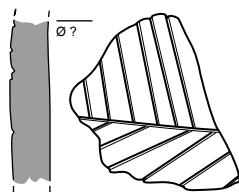


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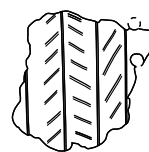


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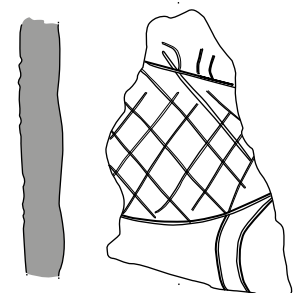


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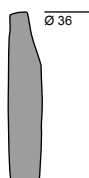


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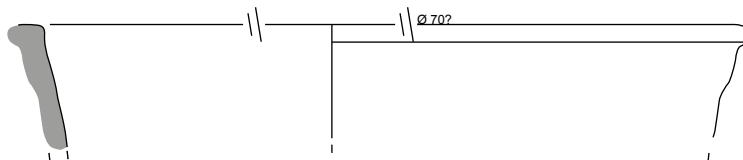


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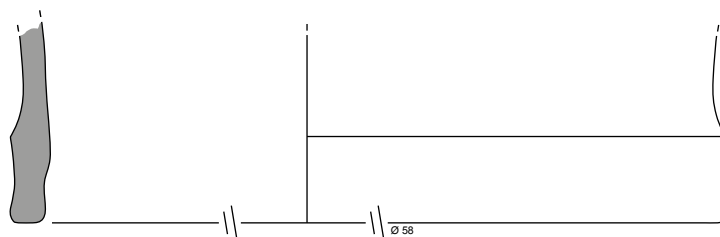
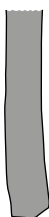


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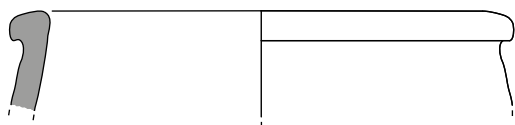


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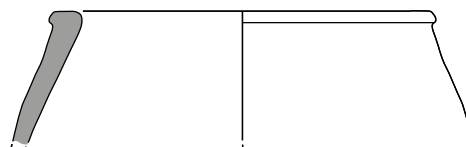


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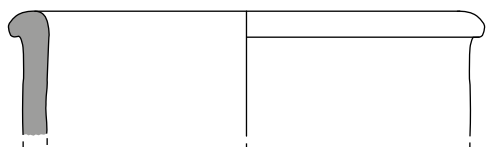


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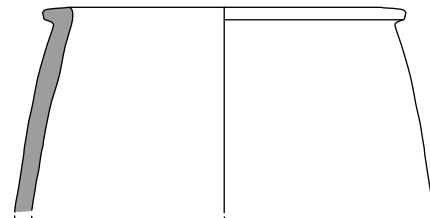


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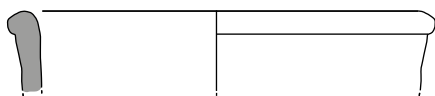


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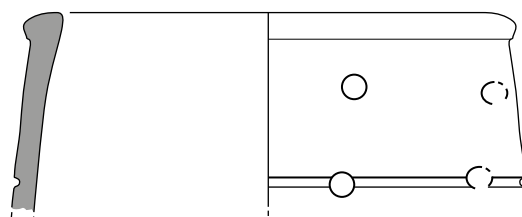


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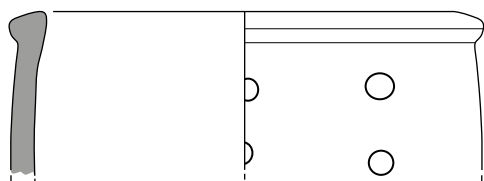


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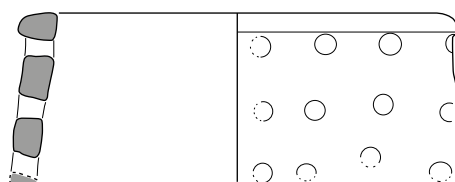


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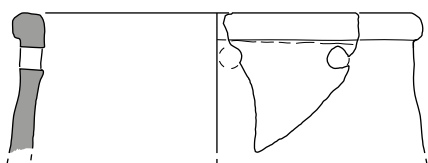


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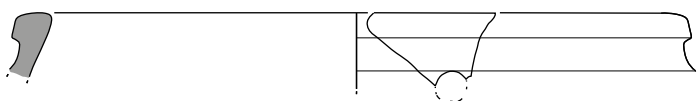


Fig. 253. 2:5.



Fig. 254. 2:5.

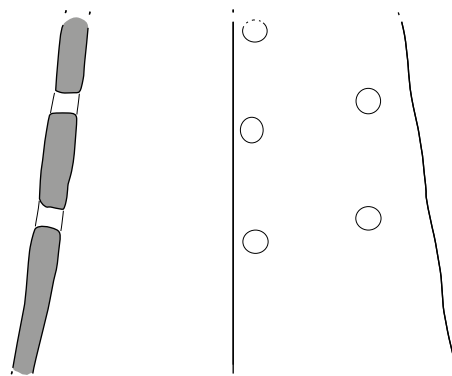


Fig. 255. 2:5.

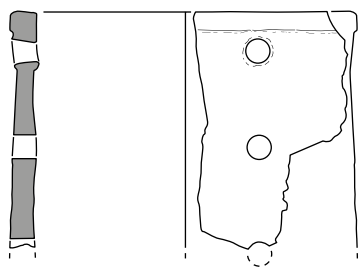


Fig. 256. 2:5.

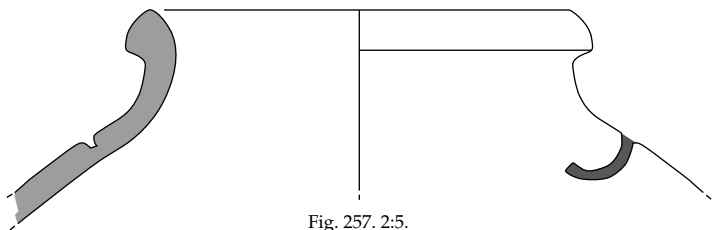


Fig. 257. 2:5.

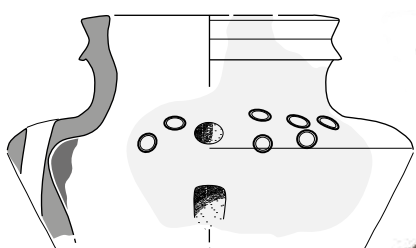


Fig. 258. 1:2.

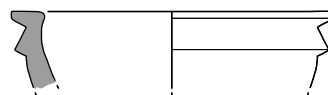


Fig. 259. 1:2.

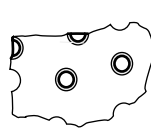
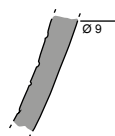


Fig. 260. 1:2.

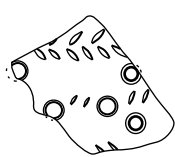


Fig. 263. 1:2.

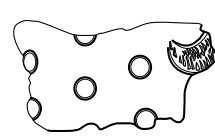
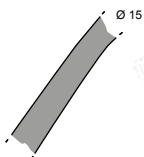


Fig. 261. 1:2.

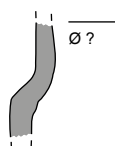


Fig. 264. 1:2.

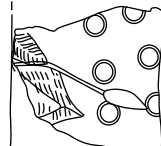
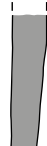


Fig. 262. 1:2.

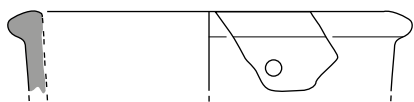


Fig. 265. 1:2.

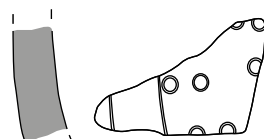


Fig. 266. 1:2.

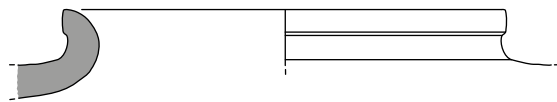


Fig. 267. 1:4.

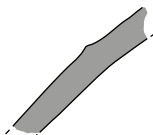


Fig. 268. 1:4.

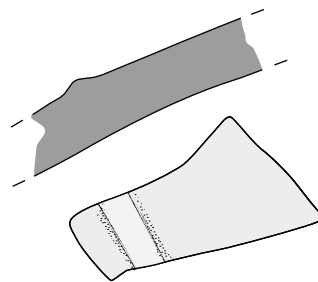
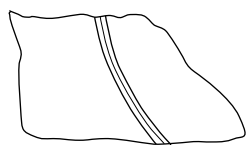


Fig. 269. 1:4.



Fig. 270. 1:4.



Fig. 271. 1:2.

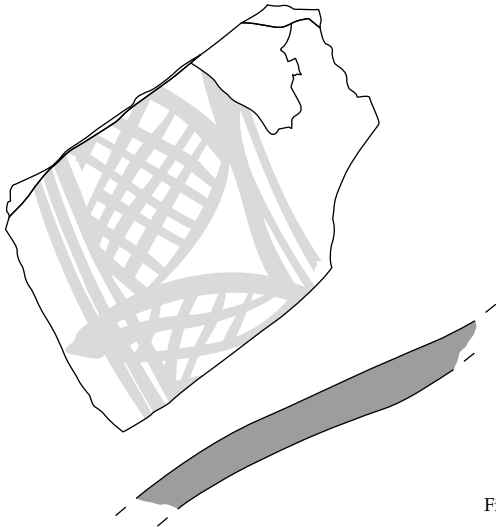


Fig. 272. 1:2.

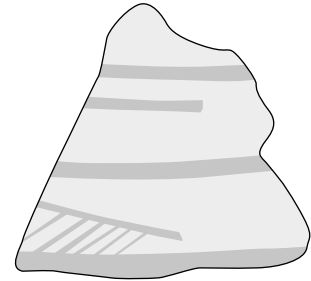


Fig. 273. 1:2.

Fig. 131-273. Pottery from Phase 1.

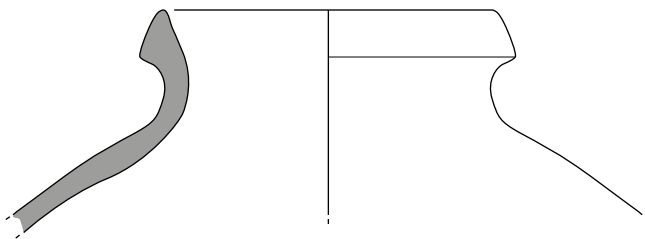


Fig. 274. 2:5.



Fig. 275. 2:5.

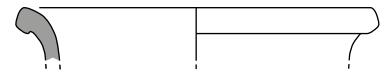


Fig. 277. 2:5.



Fig. 276. 2:5.

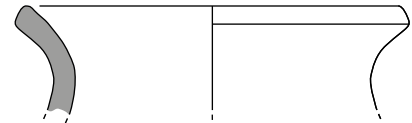


Fig. 278. 2:5.

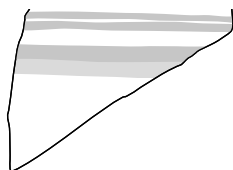


Fig. 279. 2:5.

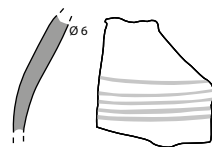


Fig. 280. 2:5.

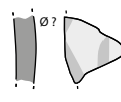


Fig. 281. 2:5.

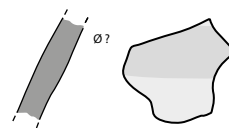


Fig. 282. 2:5.



Fig. 283. 2:5.

Fig. 274-296. Pottery from Phase 2.

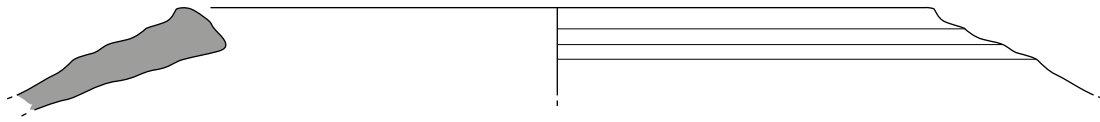


Fig. 284. 2:5.

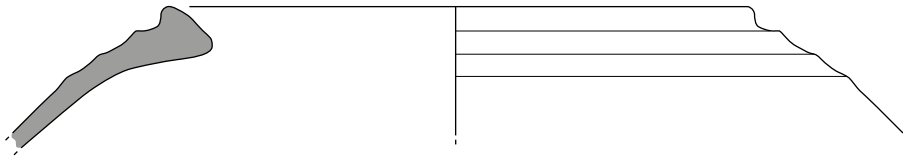


Fig. 285. 2:5.

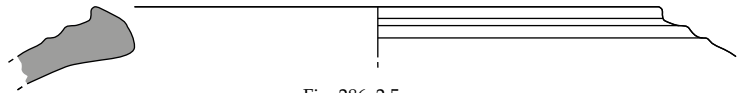


Fig. 286. 2:5.

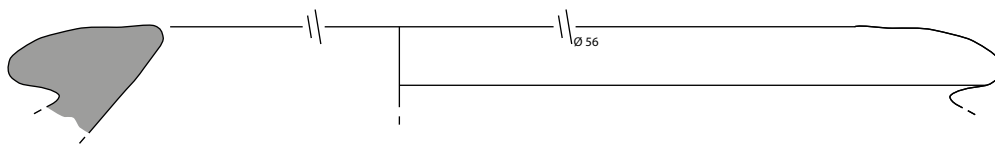


Fig. 287. 2:5.

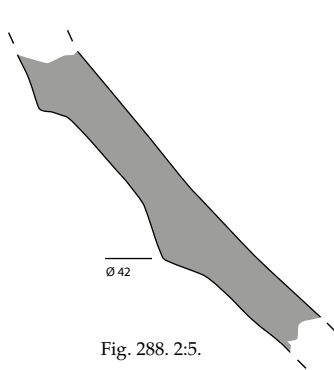


Fig. 288. 2:5.

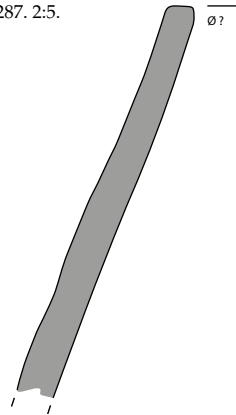


Fig. 289. 2:5.

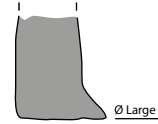


Fig. 290. 2:5.

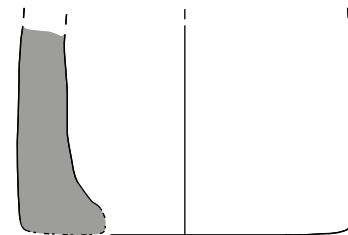


Fig. 291. 2:5.

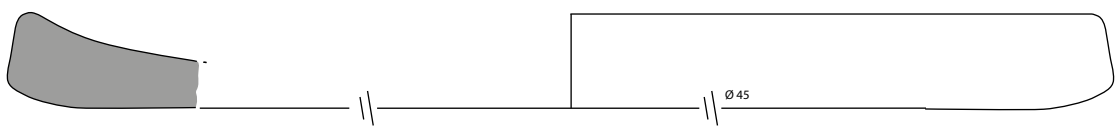


Fig. 292. 2:5.



Fig. 293. 2:5.

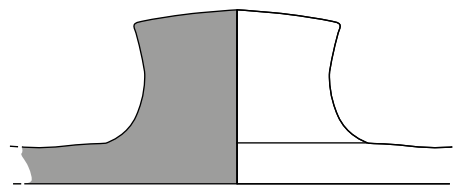


Fig. 294. 2:5.

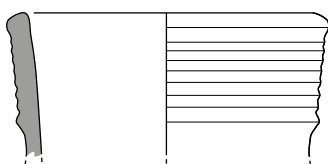


Fig. 295. 2:5.

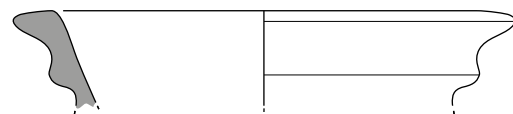
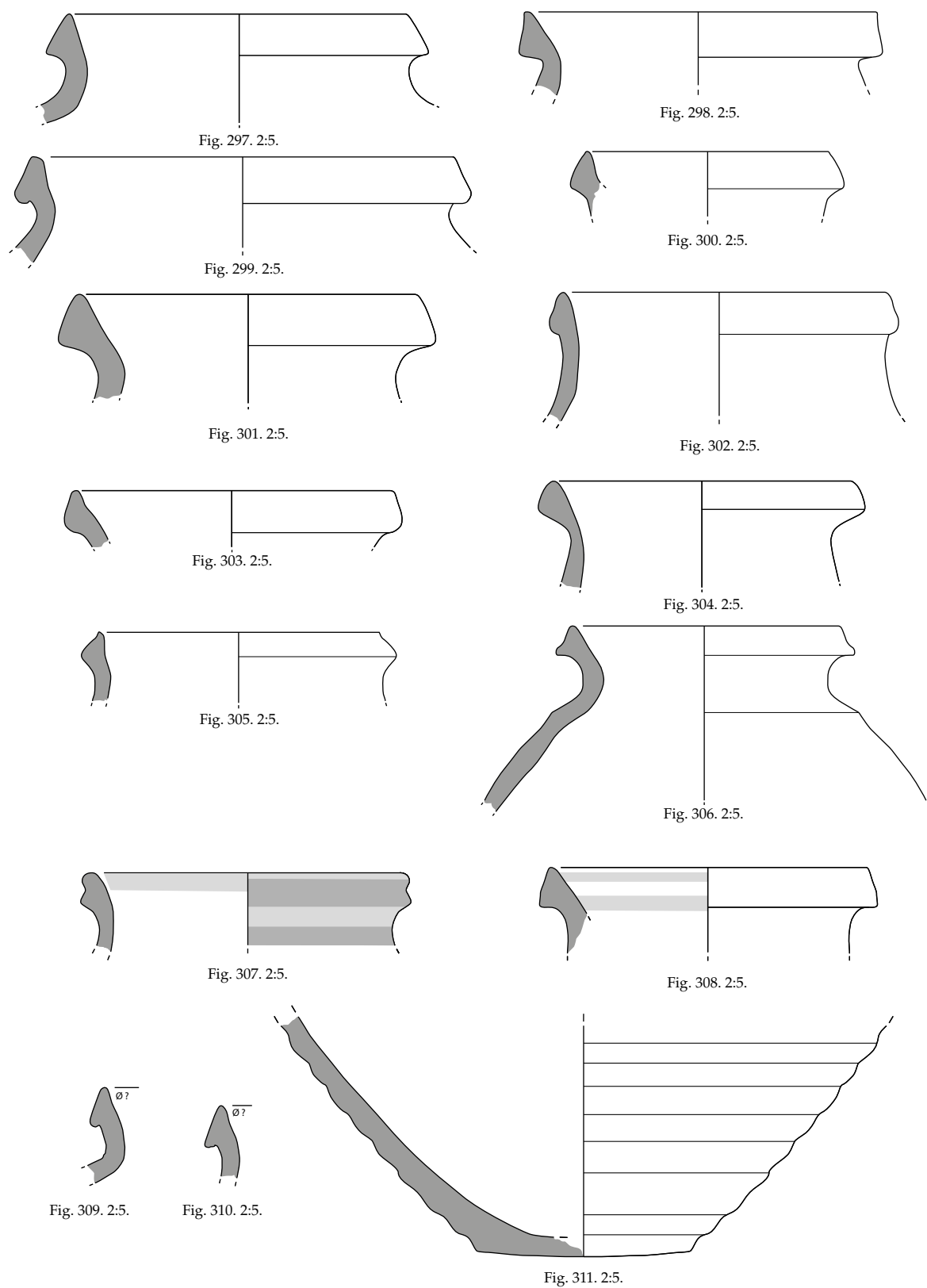
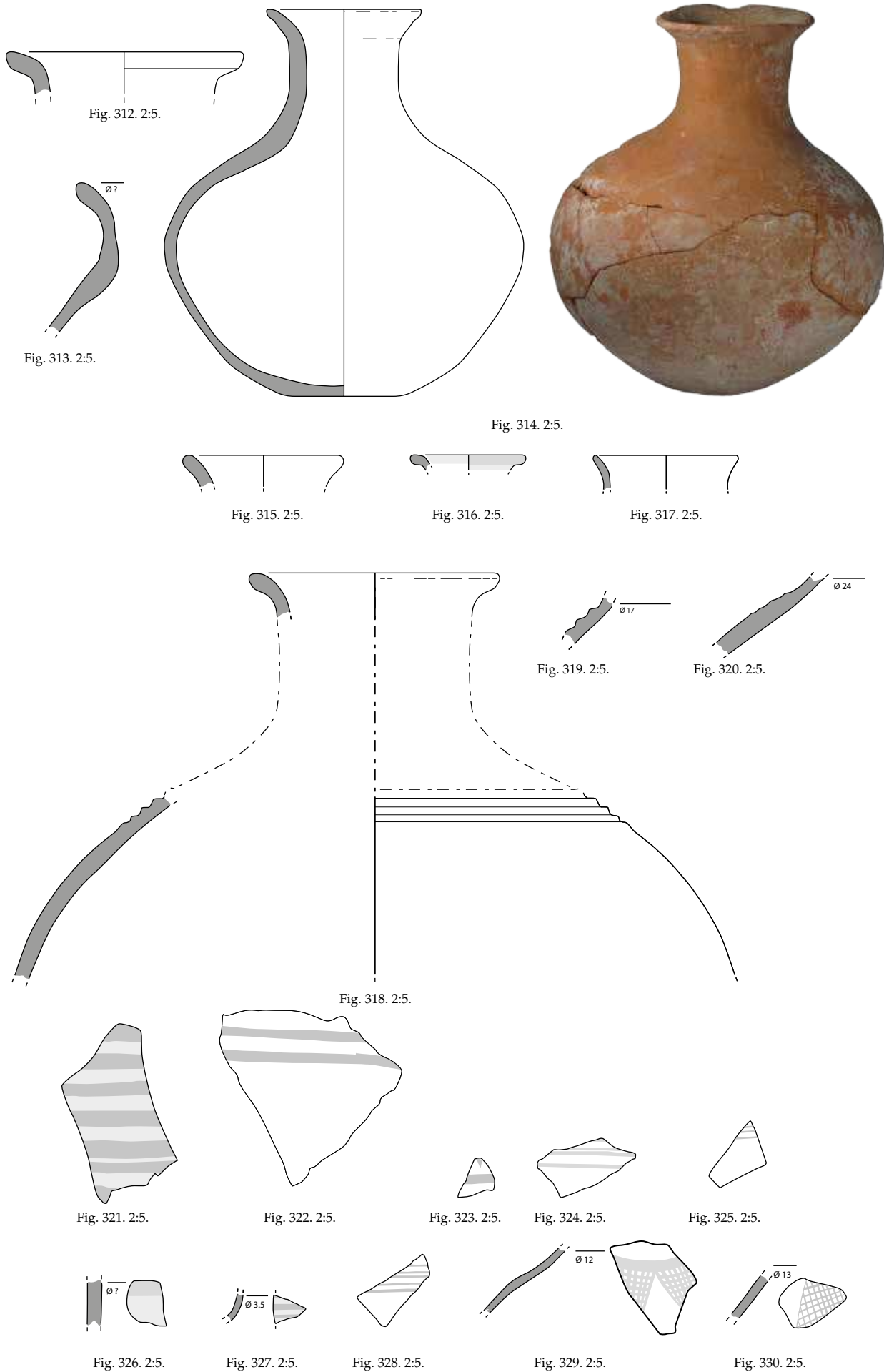


Fig. 296. 2:5.





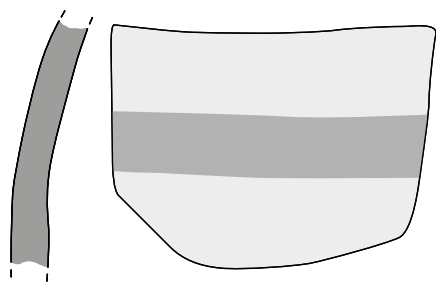


Fig. 331. 2:5.

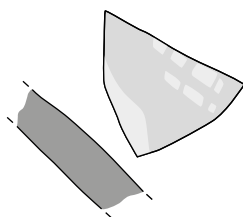


Fig. 332. 2:5.



Fig. 333. 2:5.



Fig. 334. 2:5.



Fig. 335. 2:5.

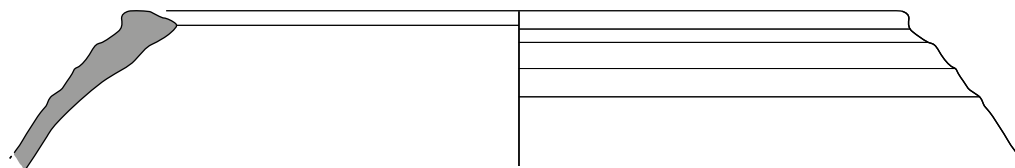


Fig. 336. 2:5.

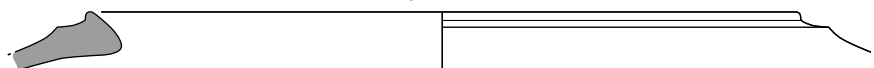


Fig. 337. 2:5.

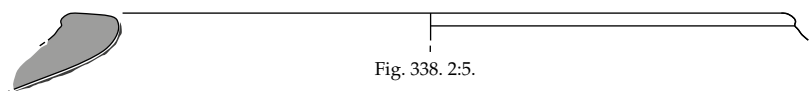


Fig. 338. 2:5.

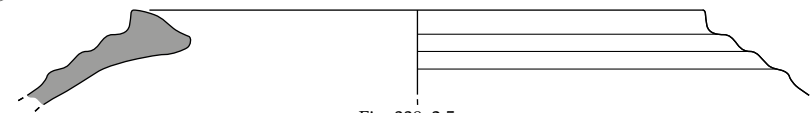


Fig. 339. 2:5.

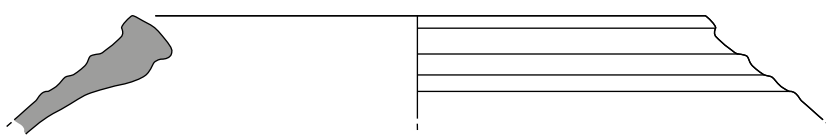


Fig. 340. 2:5.

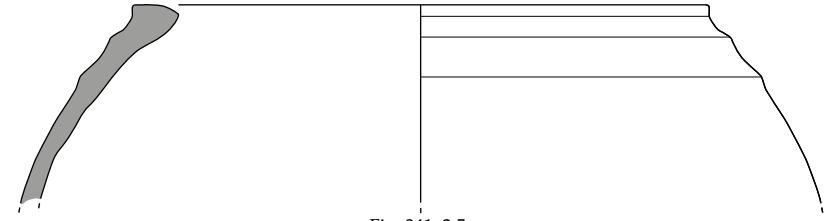


Fig. 341. 2:5.



Fig. 342. 2:5.



Fig. 343. 2:5.

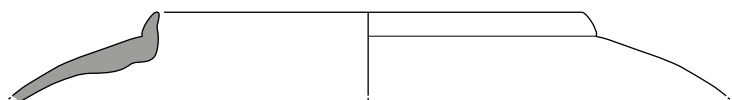


Fig. 344. 2:5.



Fig. 345. 2:5.

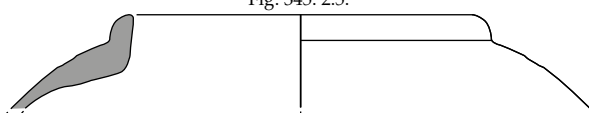


Fig. 346. 2:5.

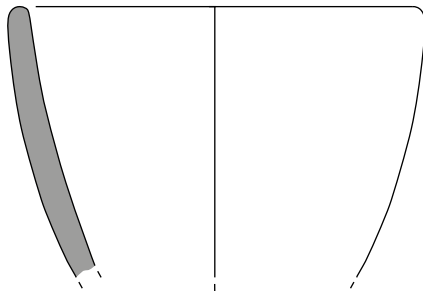


Fig. 347. 2:5.

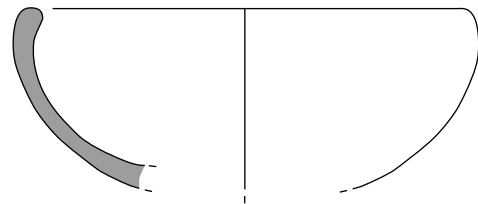


Fig. 348. 2:5.

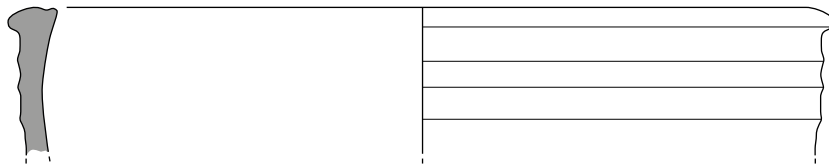


Fig. 349. 2:5.

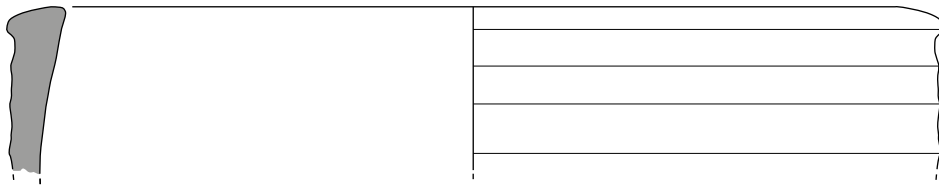


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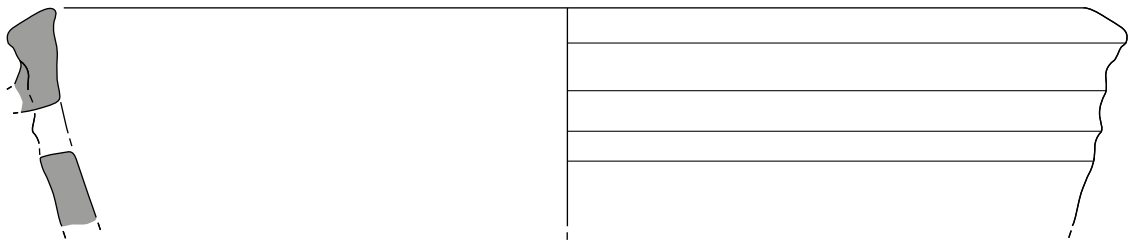


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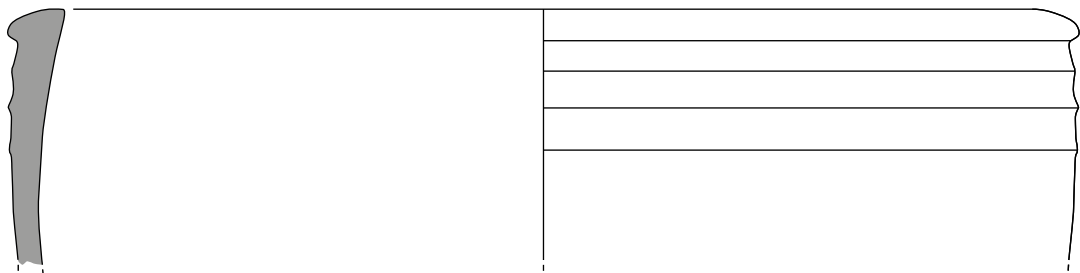


Fig. 352. 2:5.



Fig. 353. 2:5.

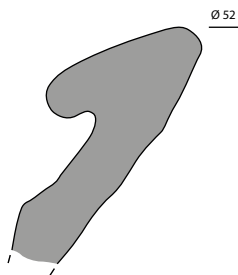


Fig. 354. 2:5.

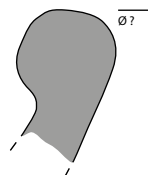


Fig. 355. 2:5.

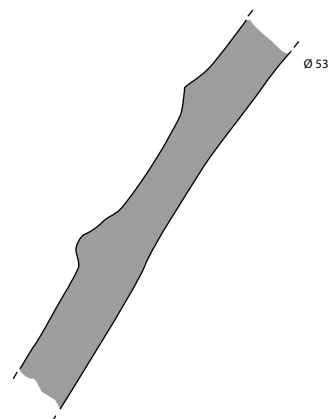


Fig. 356. 2:5.

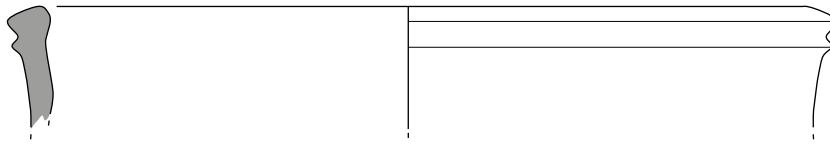


Fig. 357. 2:5.

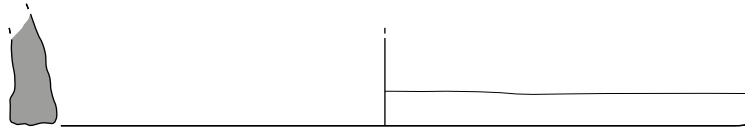


Fig. 358. 2:5.



Fig. 359. 2:5.



Fig. 360. 2:5.

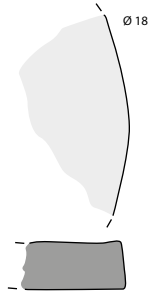


Fig. 361. 2:5.



Fig. 362. 2:5.

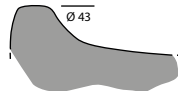


Fig. 363. 2:5.



Fig. 364. 2:5.

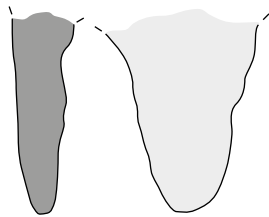


Fig. 365. 2:5.

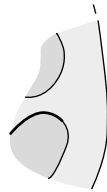


Fig. 366. 2:5.

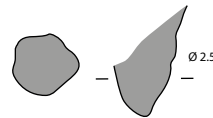


Fig. 367. 2:5.



Fig. 368. 2:5.

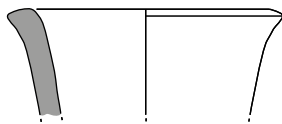


Fig. 369. 2:5.

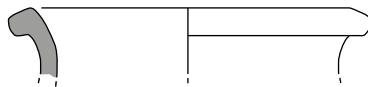


Fig. 370. 2:5.

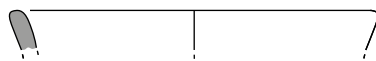


Fig. 371. 2:5.

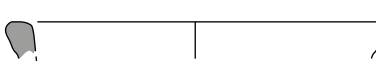


Fig. 372. 2:5.

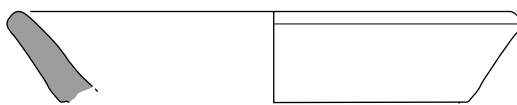


Fig. 373. 2:5.

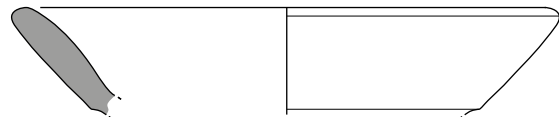


Fig. 374. 2:5.

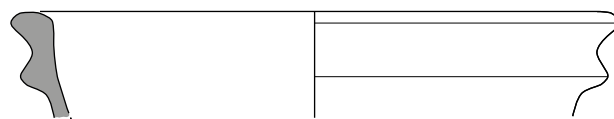


Fig. 375. 2:5.

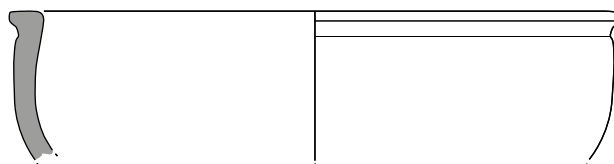


Fig. 376. 2:5.

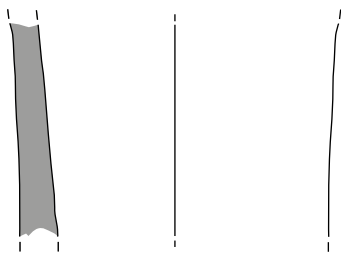


Fig. 377. 2:5.

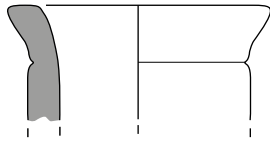


Fig. 378. 2:5.

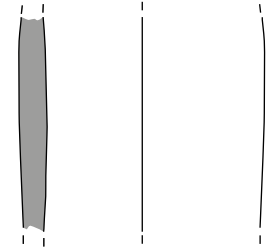


Fig. 379. 2:5.

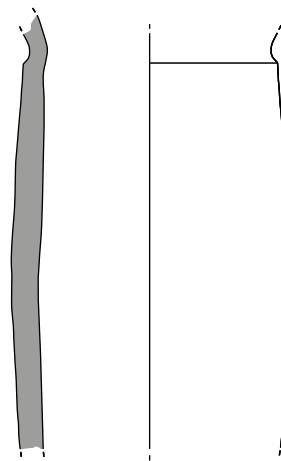


Fig. 380. 2:5.

Fig. 297-380. Pottery from Phase 3.

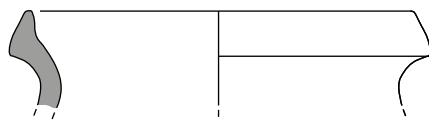


Fig. 381. 2:5.

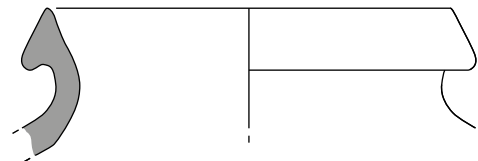


Fig. 382. 2:5.

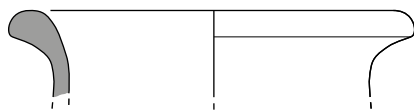


Fig. 383. 2:5.

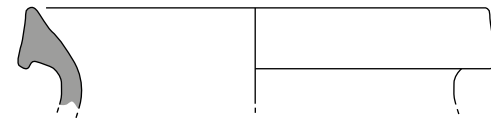


Fig. 384. 2:5.

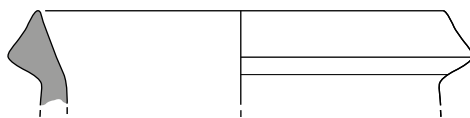


Fig. 385. 2:5.

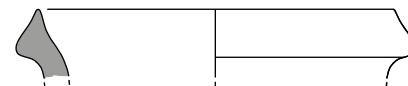


Fig. 386. 2:5.

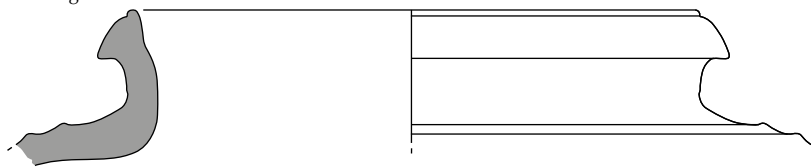


Fig. 387. 2:5.

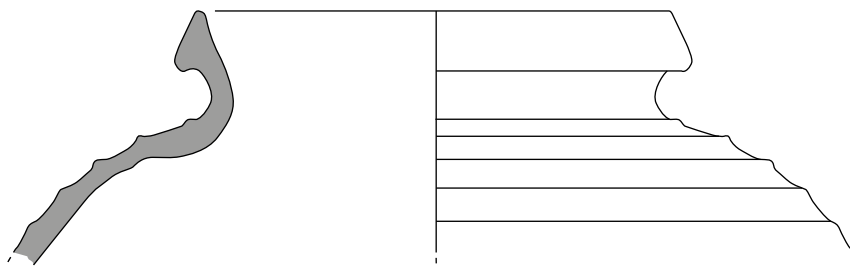


Fig. 388. 2:5.

Fig. 381-500. Pottery from Phase 4.

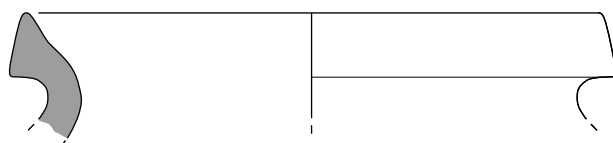


Fig. 389. 2:5.

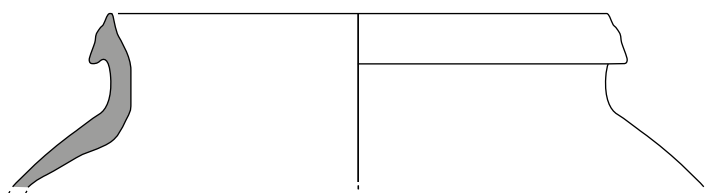


Fig. 390. 2:5.

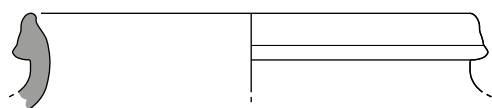


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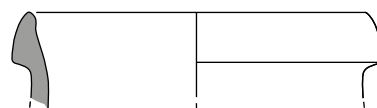


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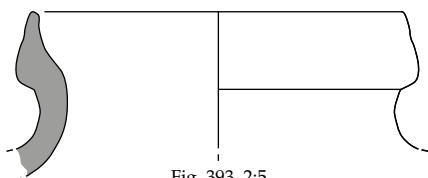


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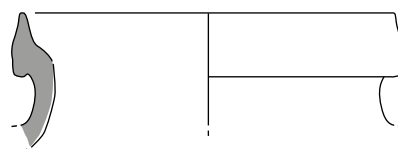


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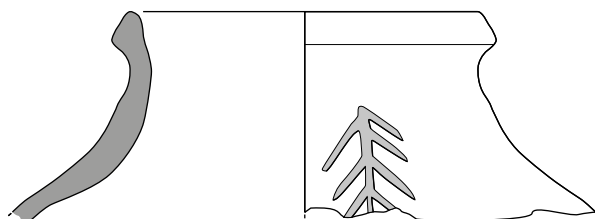


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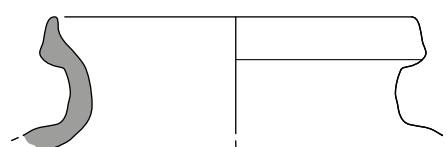


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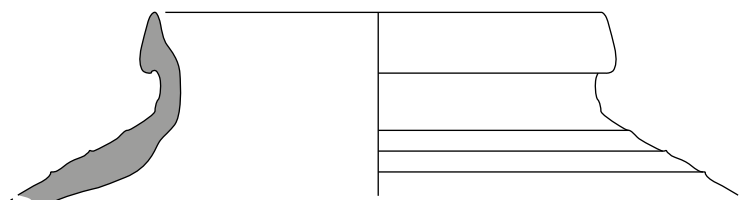


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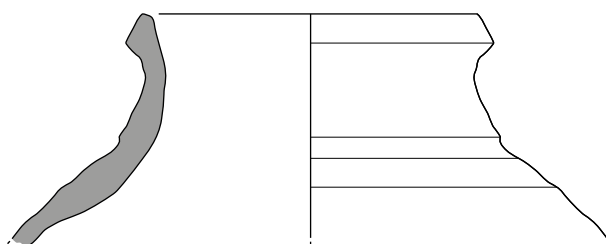


Fig. 398. 2:5.

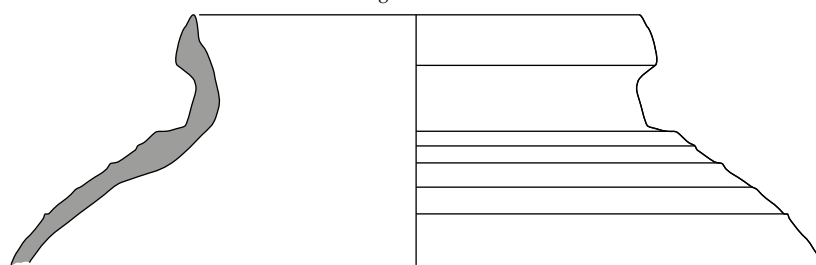
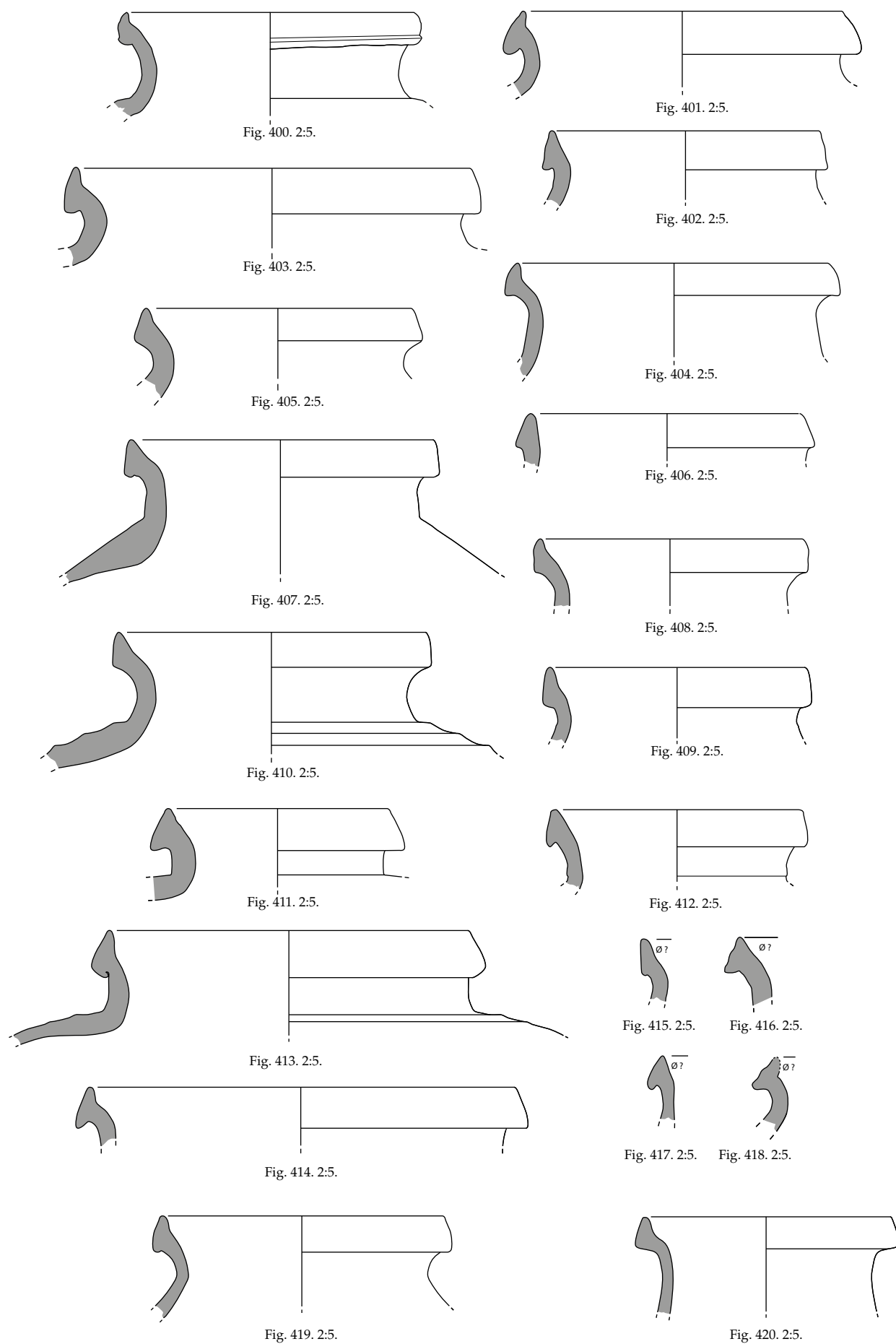


Fig. 399. 2:5.



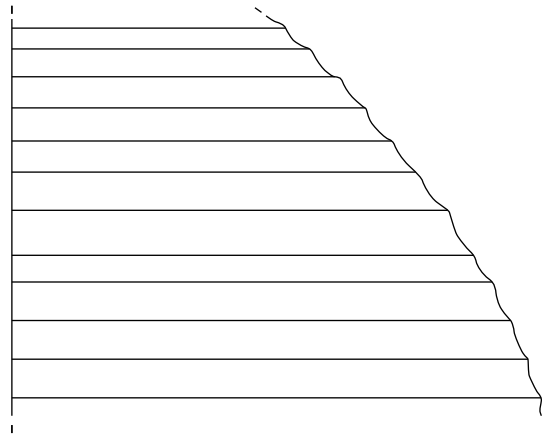
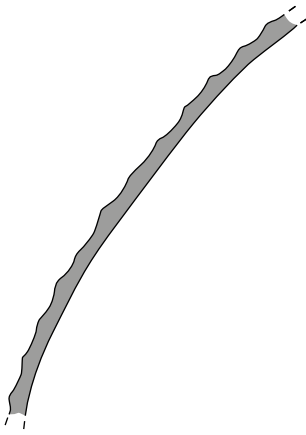


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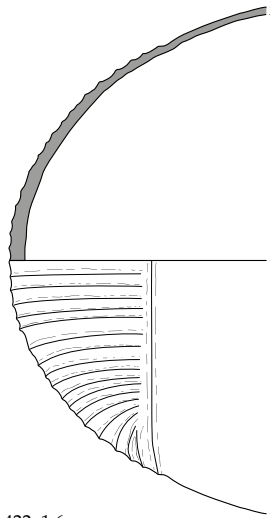
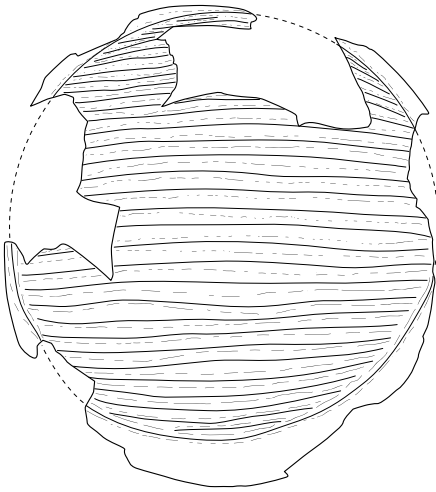


Fig. 422. 1:6.

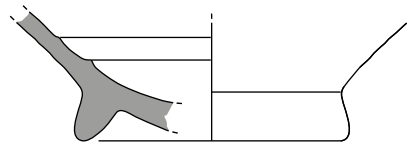
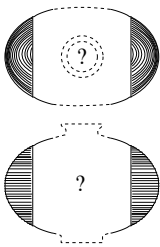


Fig. 423. 2:5.

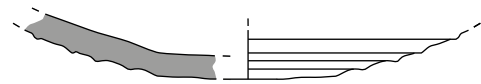


Fig. 424. 2:5.

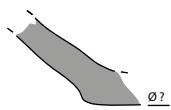


Fig. 425. 2:5.

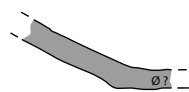


Fig. 426. 2:5.

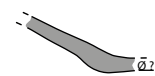


Fig. 427. 2:5.

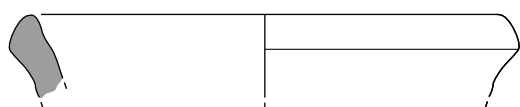


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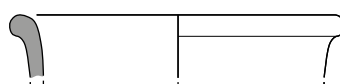


Fig. 429. 2:5.



Fig. 430. 2:5.

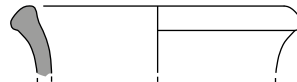


Fig. 431. 2:5.

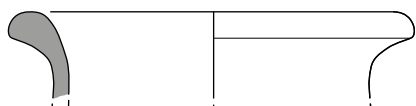


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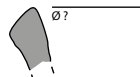


Fig. 433. 2:5.



Fig. 434. 2:5.

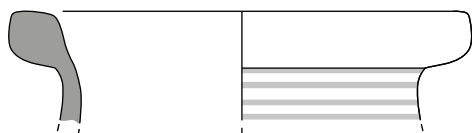


Fig. 435. 2:5.



Fig. 436. 2:5.

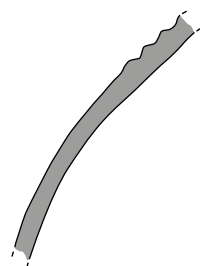


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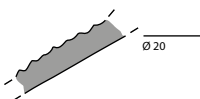
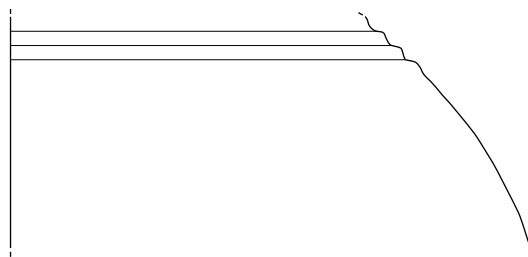


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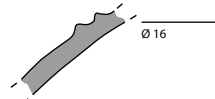


Fig. 439. 2:5.



Fig. 440. 2:5.



Fig. 441. 2:5.

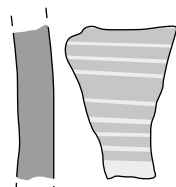


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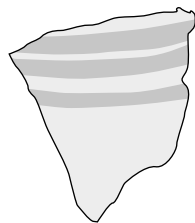


Fig. 443. 2:5.



Fig. 444. 2:5.

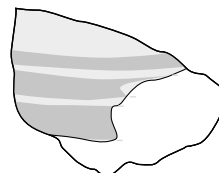


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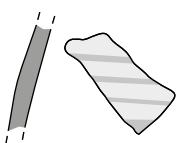


Fig. 446. 2:5.

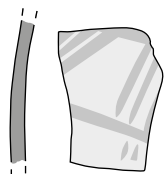


Fig. 447. 2:5.



Fig. 448. 2:5.

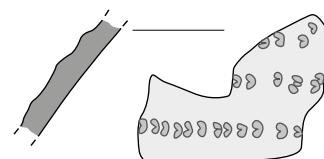
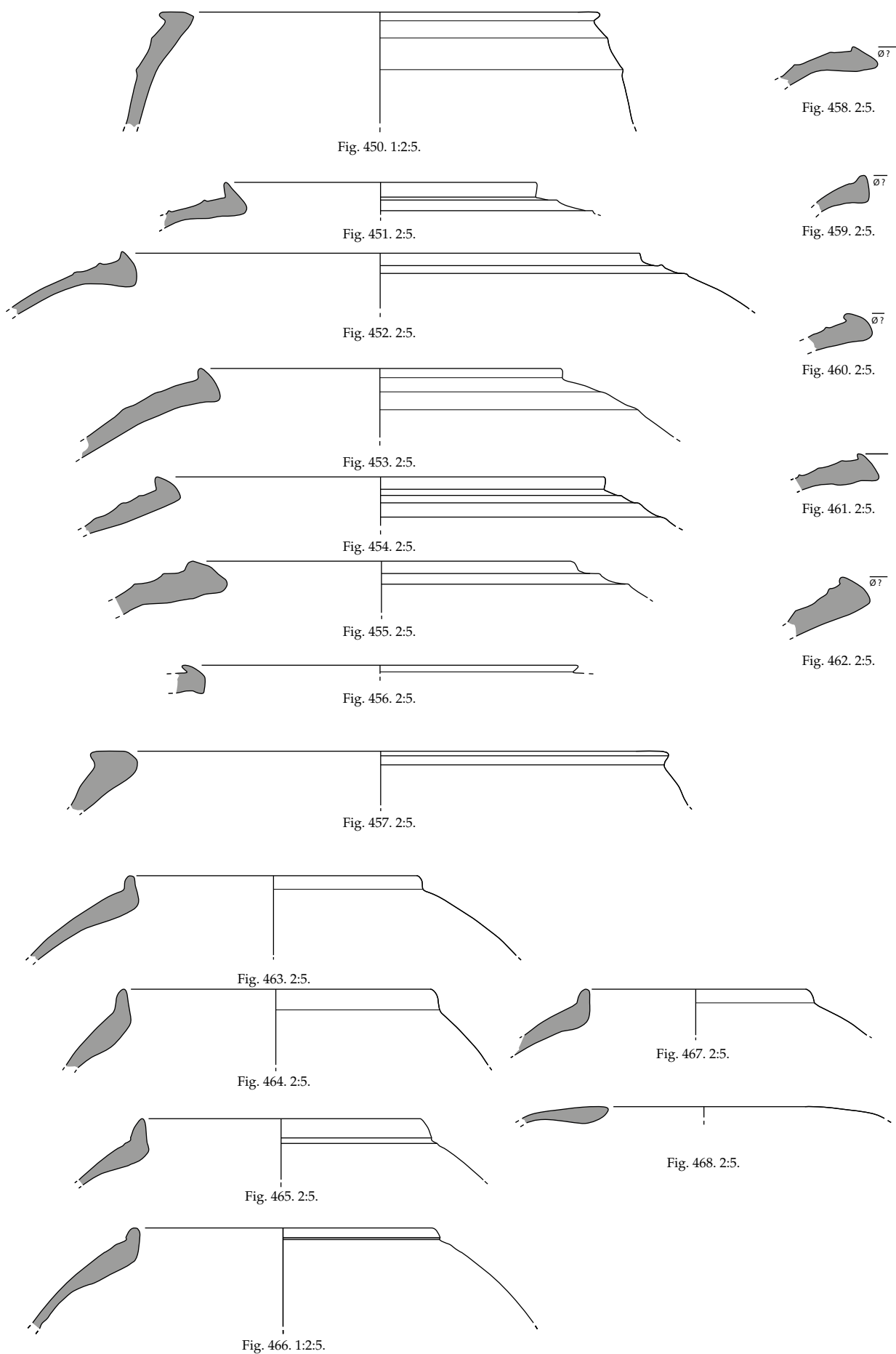


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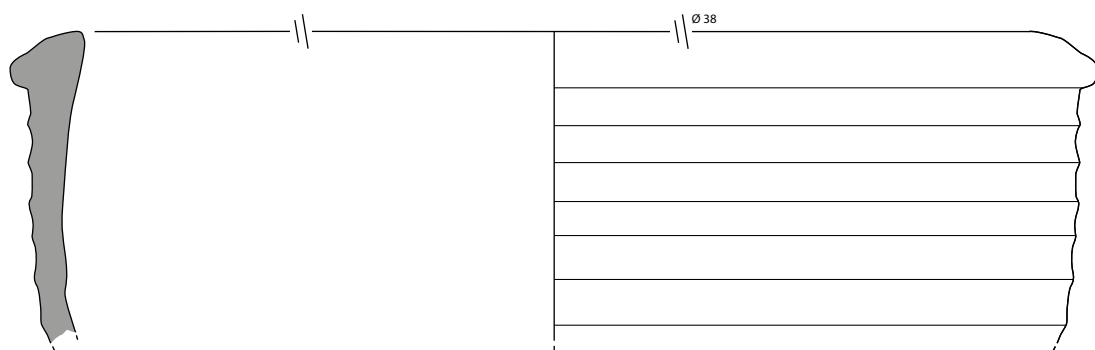


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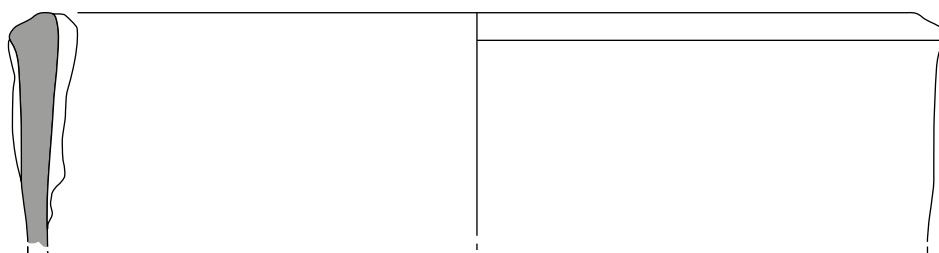


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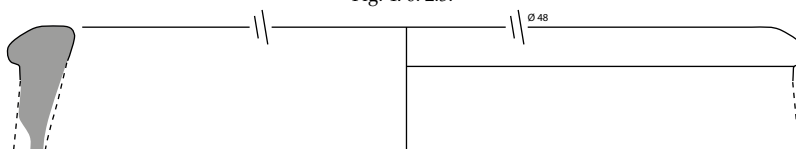


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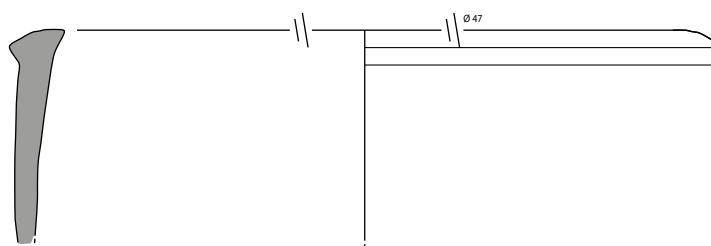


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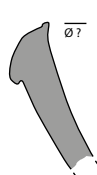


Fig. 473. 2:5.

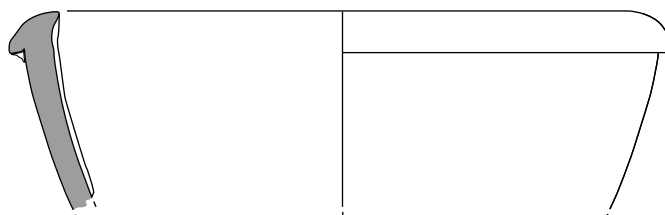


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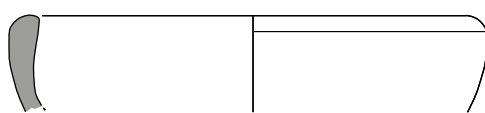


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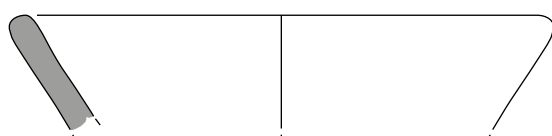


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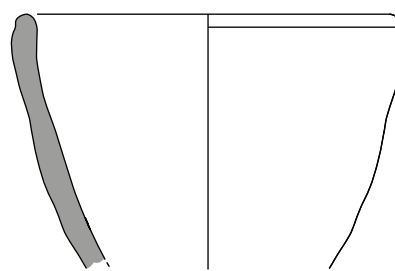


Fig. 476. 2:5.



Fig. 478. 2:5.

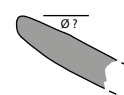


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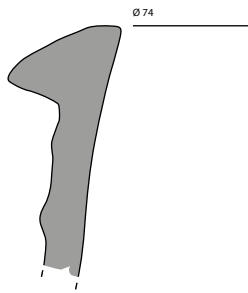


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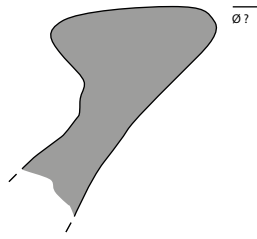


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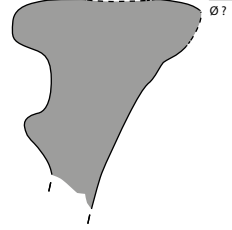


Fig. 482. 2:5.

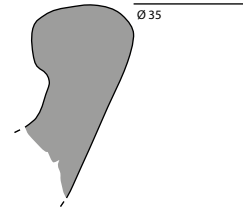


Fig. 483. 2:5.

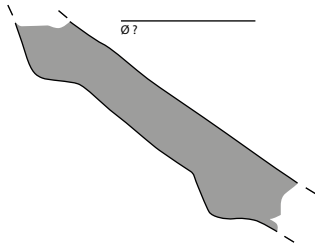


Fig. 484. 2:5.

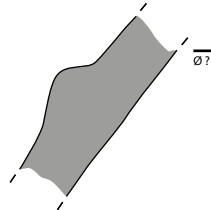


Fig. 485. 2:5.

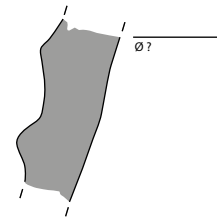


Fig. 486. 2:5.

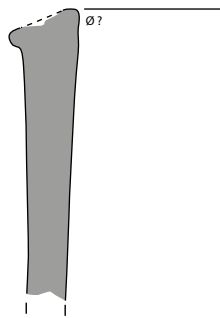


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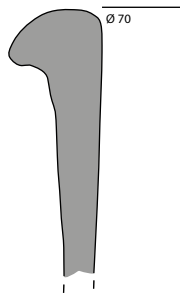


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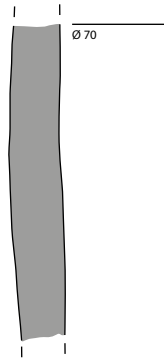


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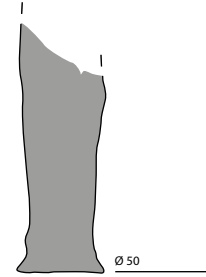


Fig. 490. 2:5.



Fig. 491. 2:5.

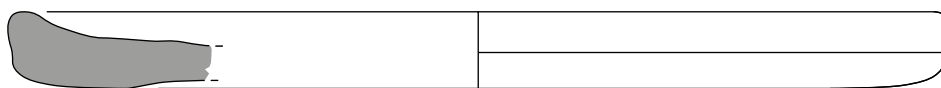


Fig. 492. 2:5.

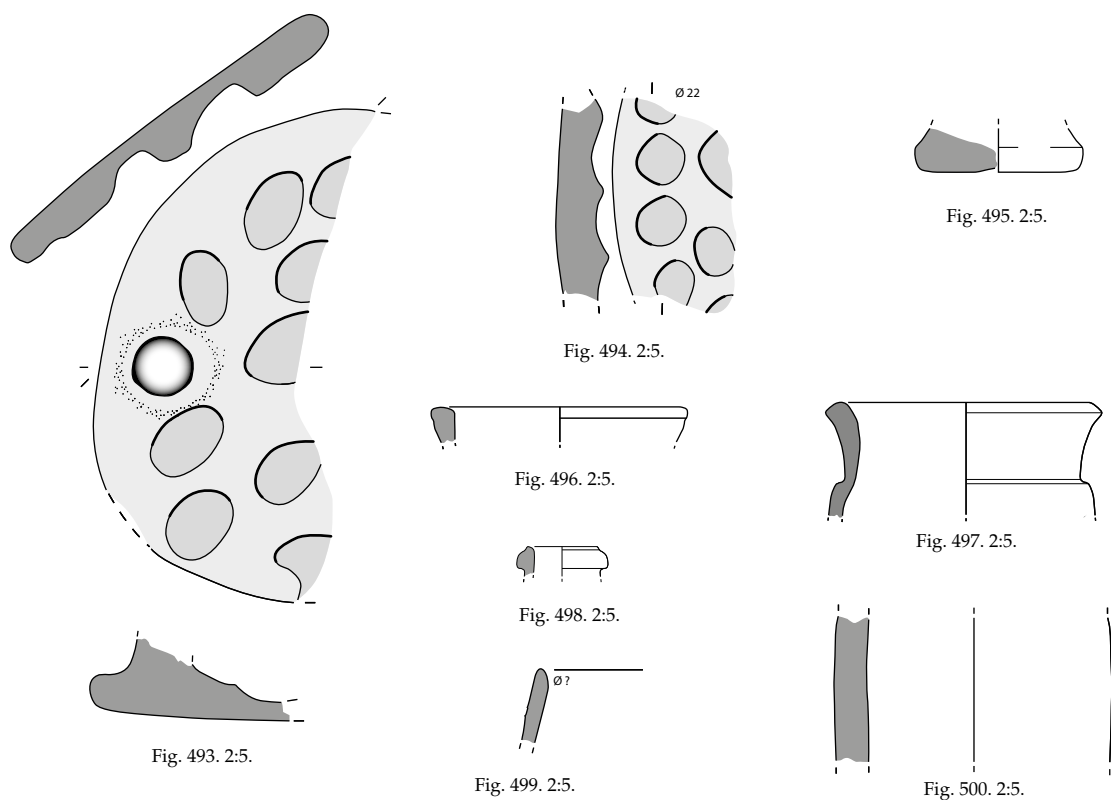


Fig. 381-500. Pottery from Phase 4.

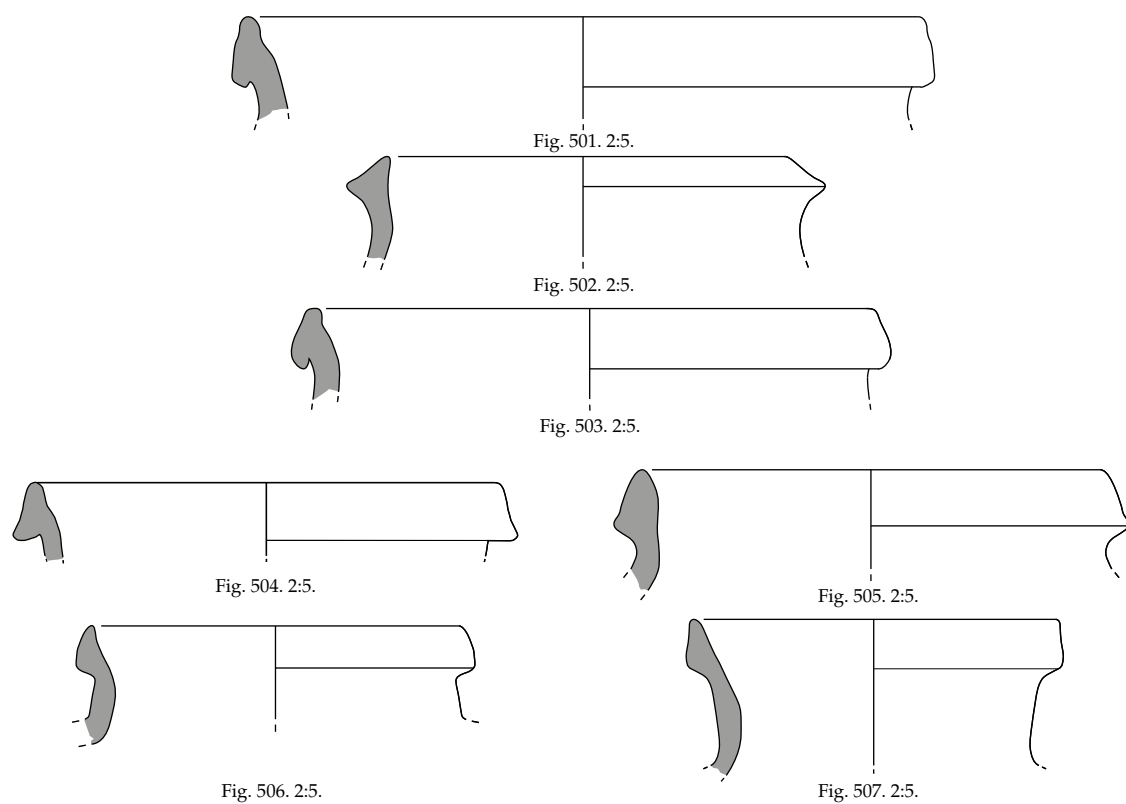


Fig. 501-534. Pottery from Phase 5.

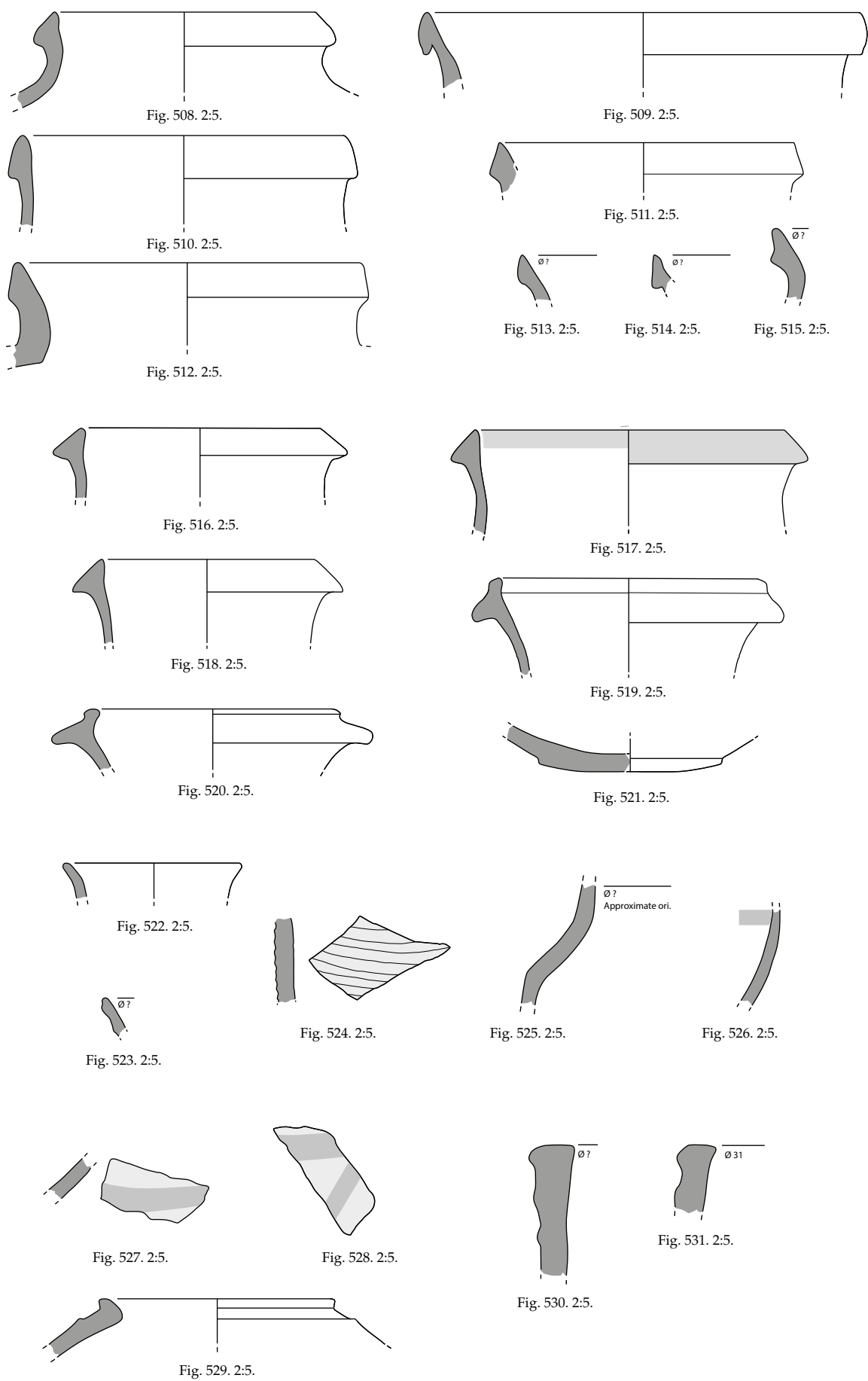


Fig. 501-534. Pottery from Phase 5.

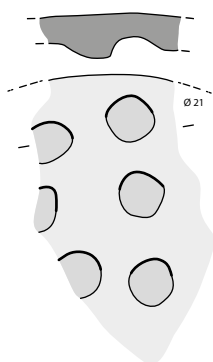


Fig. 532. 2:5.

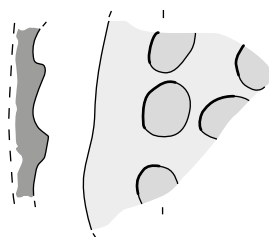


Fig. 533. 2:5.

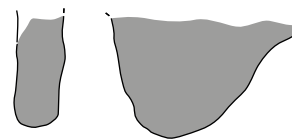


Fig. 534. 2:5.

Fig. 501-534. Pottery from Phase 5.



Fig. 535. 2:5.

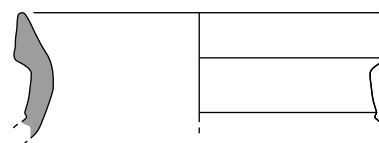


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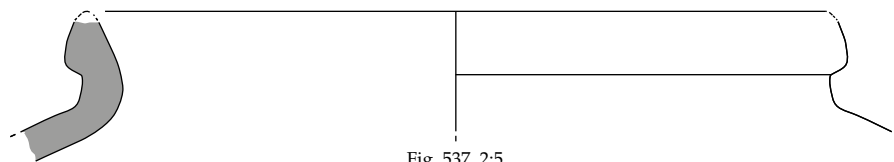


Fig. 537. 2:5.

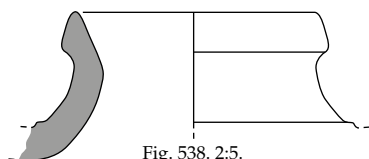


Fig. 538. 2:5.

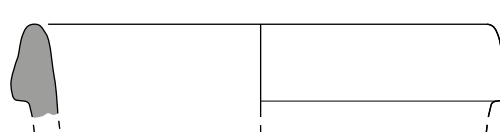


Fig. 539. 2:5.

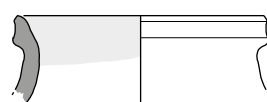


Fig. 540. 2:5.

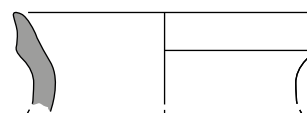


Fig. 541. 2:5.



Fig. 542. 2:5.

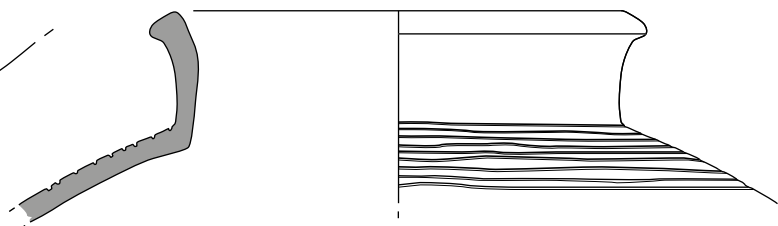


Fig. 543. 2:5.

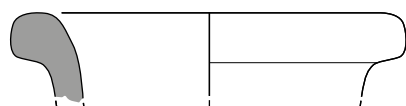


Fig. 545. 2:5.

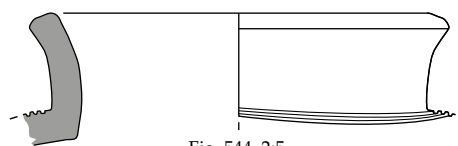


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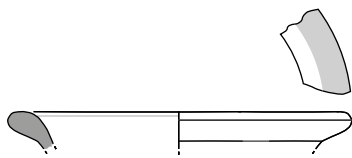


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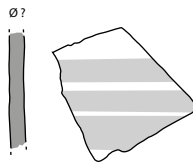


Fig. 547. 2:5.

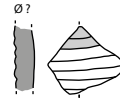


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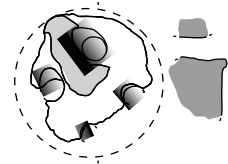


Fig. 549. 2:5.



Fig. 550. 2:5.



Fig. 551. 2:5.



Fig. 552. 2:5.



Fig. 553. 2:5.

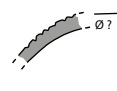


Fig. 554. 2:5.

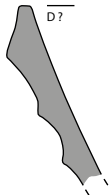


Fig. 555. 2:5.

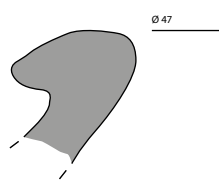


Fig. 556. 2:5.

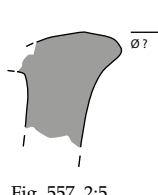


Fig. 557. 2:5.

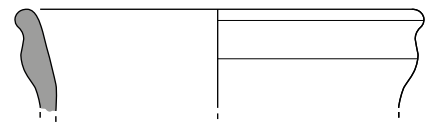


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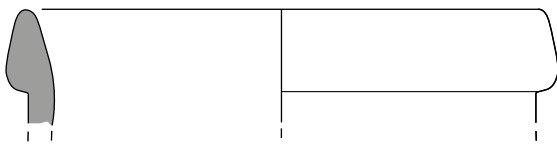


Fig. 559. 2:5.

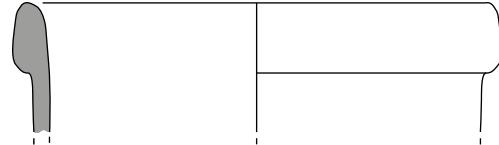


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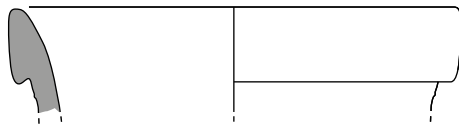


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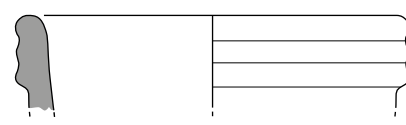


Fig. 562. 2:5.

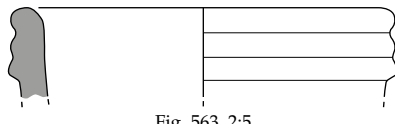


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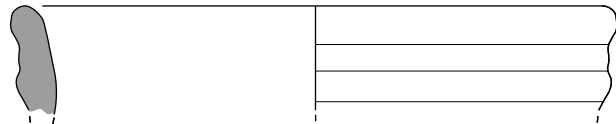


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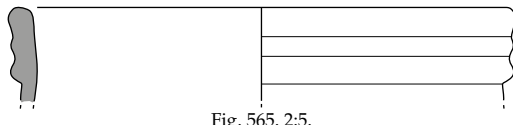


Fig. 565. 2:5.

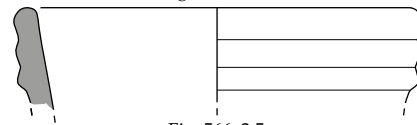


Fig. 566. 2:5.

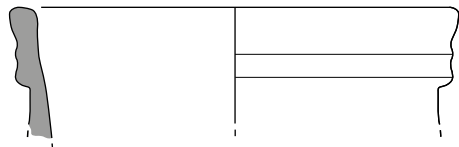


Fig. 567. 2:5.

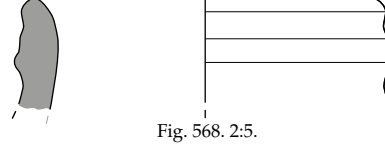


Fig. 568. 2:5.

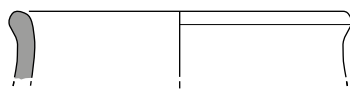


Fig. 569. 2:5.

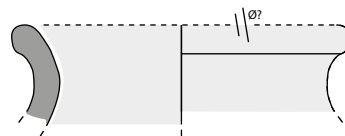


Fig. 570. 2:5.

Fig. 535-605. Pottery from Phase 6.

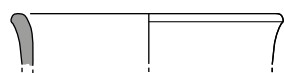


Fig. 571. 2:5.



Fig. 572. 2:5.

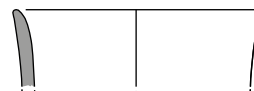


Fig. 573. 2:5.

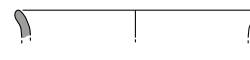


Fig. 574. 2:5.



Fig. 575. 2:5.



Fig. 576. 2:5.

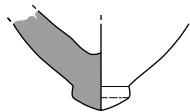


Fig. 577. 2:5.

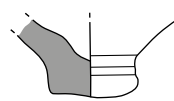


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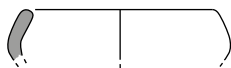


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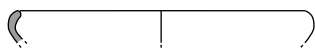


Fig. 580. 2:5.



Fig. 581. 2:5.



Fig. 582. 2:5.

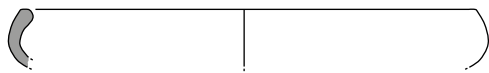


Fig. 583. 2:5.

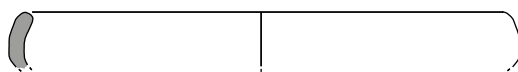


Fig. 584. 2:5.

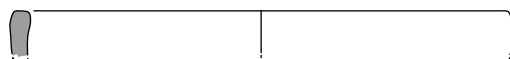


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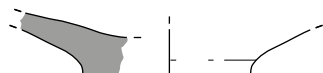


Fig. 597. 2:5.

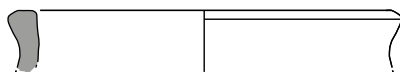


Fig. 599. 2:5.



Fig. 600. 2:5.

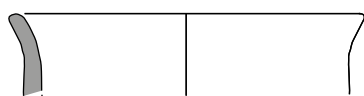


Fig. 601. 2:5.



Fig. 586. 2:5.

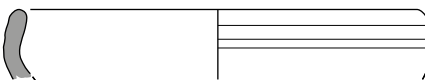


Fig. 587. 2:5.



Fig. 588. 2:5.

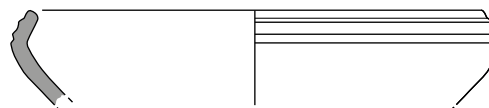


Fig. 589. 2:5.



Fig. 590. 2:5.



Fig. 591. 2:5.

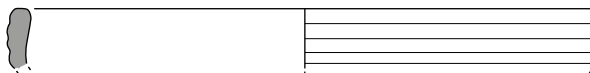


Fig. 592. 2:5.

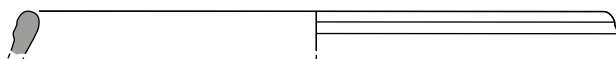


Fig. 593. 2:5.



Fig. 594. 2:5.

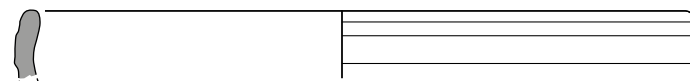


Fig. 595. 2:5.

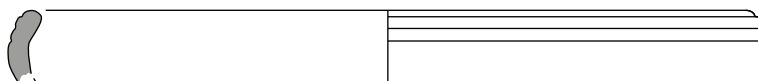


Fig. 596. 2:5.

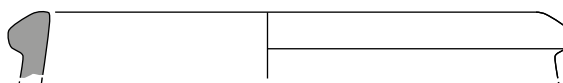


Fig. 598. 2:5.

Fig. 535-605. Pottery from Phase 6.

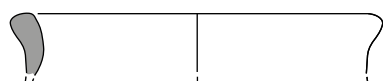


Fig. 602. 2:5.

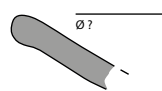


Fig. 603. 2:5.



Fig. 604. 2:5.



Fig. 605. 2:5.

Fig. 535-605. Pottery from Phase 6.

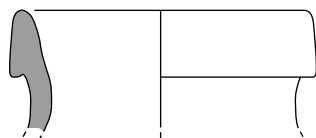


Fig. 606. 2:5.



Fig. 607. 2:5.

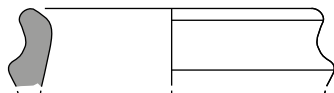


Fig. 608. 2:5.

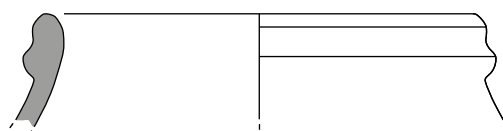


Fig. 609. 2:5.

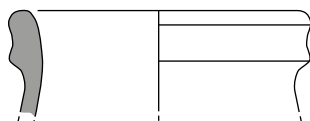


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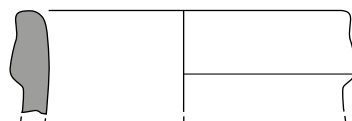


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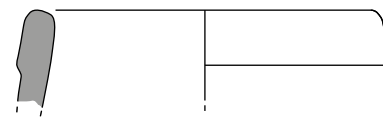


Fig. 612. 2:5.

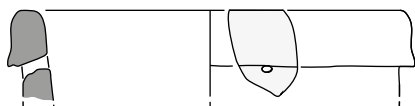


Fig. 613. 2:5.

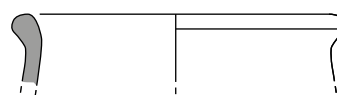


Fig. 614. 2:5.

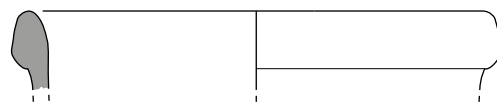


Fig. 615. 2:5.

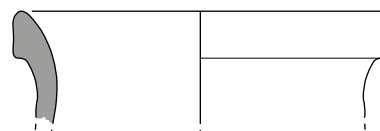


Fig. 616. 2:5.

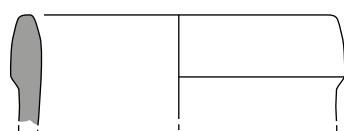


Fig. 617. 2:5.

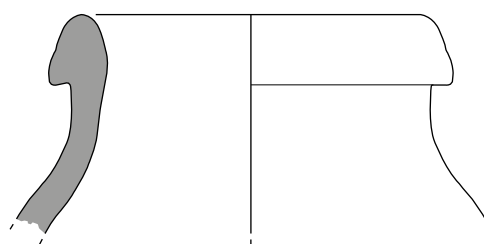


Fig. 618. 2:5.

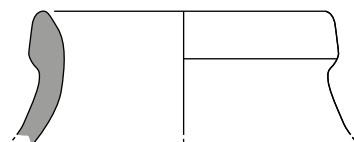


Fig. 619. 2:5.



Fig. 621. 2:5.



Fig. 620. 2:5.



Fig. 622. 2:5.

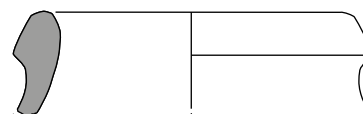


Fig. 623. 2:5.

Fig. 606-667. Pottery from Phase 7.

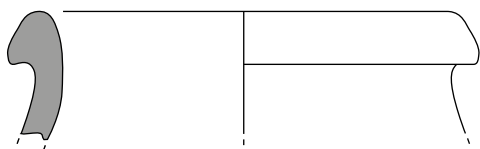


Fig. 624. 2:5.

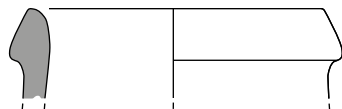


Fig. 626. 2:5.



Fig. 628. 2:5.

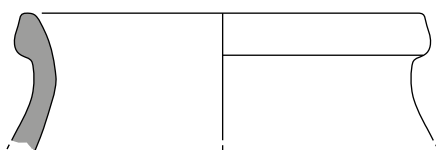


Fig. 630. 2:5.

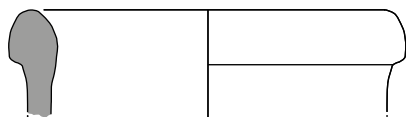


Fig. 633. 2:5.

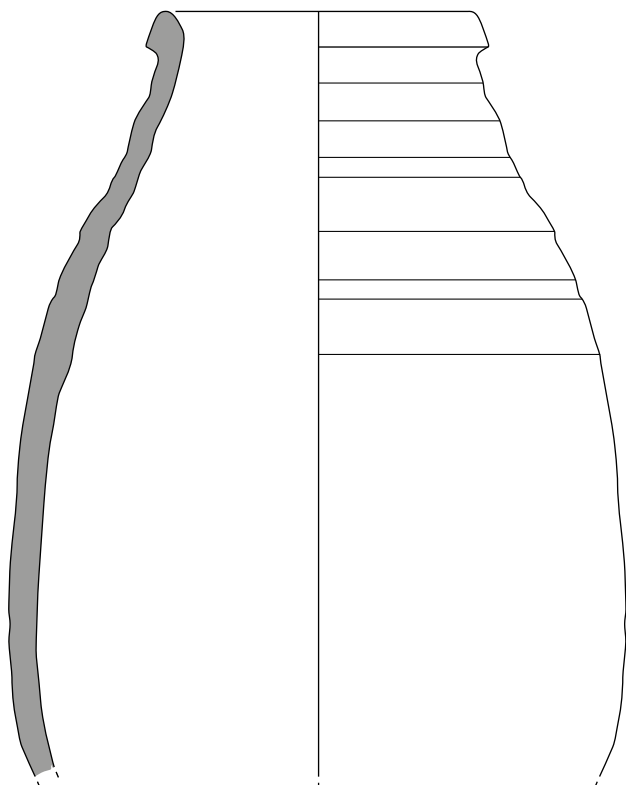


Fig. 634. 2:5.

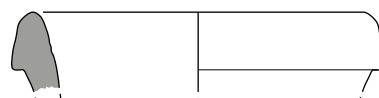


Fig. 625. 2:5.

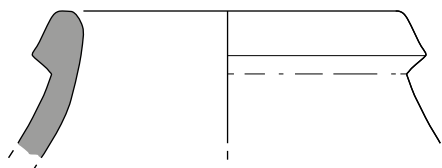


Fig. 627. 2:5.

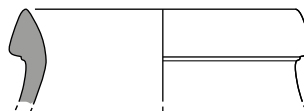


Fig. 629. 2:5.

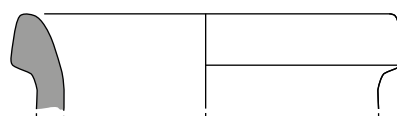


Fig. 631. 2:5.

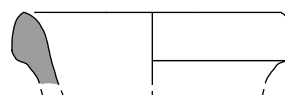


Fig. 632. 2:5.

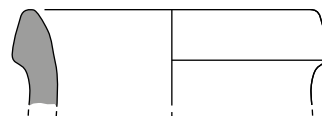


Fig. 635. 2:5.

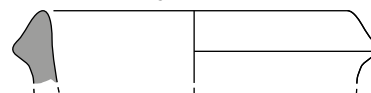


Fig. 636. 2:5.



Fig. 637. 2:5.

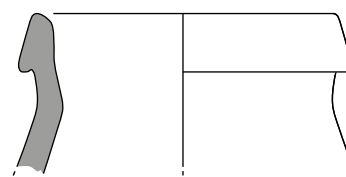


Fig. 638. 2:5.

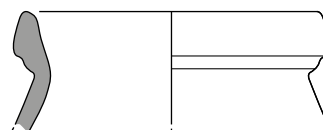


Fig. 639. 2:5.

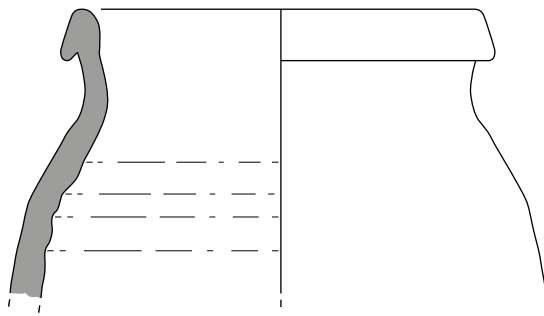


Fig. 640. 2:5.



Fig. 641. 2:5.

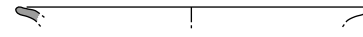


Fig. 642. 2:5.

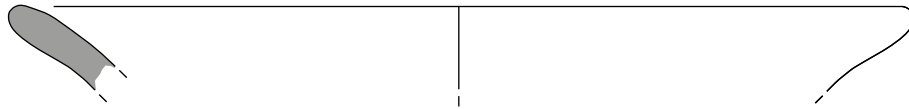


Fig. 643. 2:5.



Fig. 644. 2:5.



Fig. 645. 2:5.



Fig. 646. 2:5.

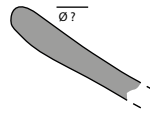


Fig. 647. 2:5.

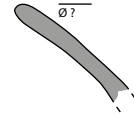


Fig. 648. 2:5.

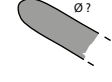


Fig. 649. 2:5.

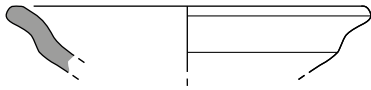


Fig. 650. 2:5.

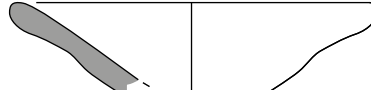


Fig. 651. 2:5.

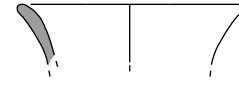


Fig. 652. 2:5.

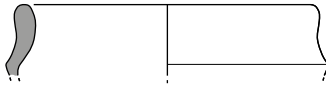


Fig. 653. 2:5.

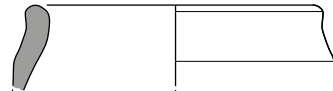


Fig. 654. 2:5.

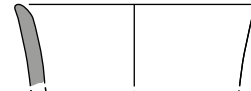


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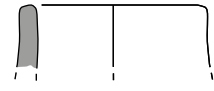


Fig. 656. 2:5.

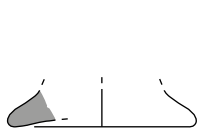


Fig. 657. 2:5.

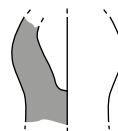


Fig. 658. 2:5.

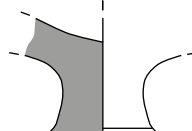


Fig. 659. 2:5.

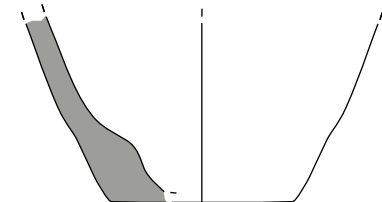


Fig. 660. 2:5.

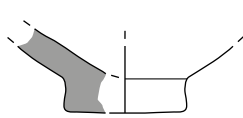


Fig. 661. 2:5.

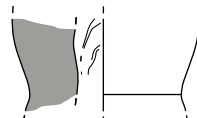


Fig. 662. 2:5.

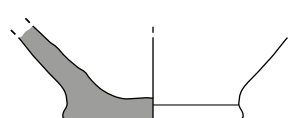


Fig. 663. 2:5.

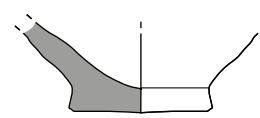


Fig. 664. 2:5.

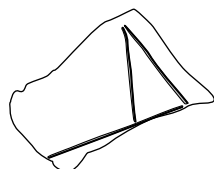


Fig. 665. 2:5.



Fig. 666. 2:5.

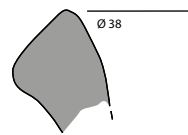


Fig. 667. 2:5.

vessels, a few rims of each of the following types, 54B (figs. 606-607), 55A (figs. 608-609), 55B (figs. 610-611), 57A (figs. 612-613), 57D (fig. 614), and the great majority of Type 57B (figs. 615-640). Bowls normally have a simple, plain rim (Type 68) (figs. 641-649) with only a few wavy-sided examples (Type 69) (figs. 650-651). Beakers with thin rims (Type 62 and 61A?) (figs. 652-656) are present with either hollow (Type 82) or massive feet/bodies (figs. 657-664). A few side sherds have incised signs (figs. 665-666) (cf. Højlund 1987 fig. 716). The ware is of either Type E (sometimes with some straw), Type F or Type G.

A single 1.7 cm thick body sherd may be an import from the East. The ware is very hard fired, light brown, extremely fine, with no visible sand, some mica, and the surface eroded.

The Phase 7 pottery resembles Period 4A assemblages found in previous excavation at Tell F3 and F6 (Højlund 1987 figs. 562-598) as well as Period IIIb assemblages at Qala'at al-Bahrain (Højlund & Ander-

sen 1997 figs. 159-283), and it has many parallels to late 2nd millennium Kassite pottery from Mesopotamia (see references in Højlund 1987; Højlund & Andersen 1994 and 1997; Armstrong & Gasche 2014).

Phase 8

With the exception of one faience sherd none of the pottery types diagnostic of Failaka Period 4B (Højlund 1987 p. 127-128) were found in the 2008-2012 excavations at Tell F6. The exception is a fragment of a small bowl in white faience from a Post Period 4A context (E24-X412). The shape is known from Period 4B in Tell F3 (Højlund 1987 fig. 423). No types diagnostic of Qala'at al-Bahrain Period IIIc were found either (Højlund & Andersen 1994 p. 185-197). The latest pottery identified is a small number of Hellenistic sherds of types well known from the neighbouring Tell F5 (Hannestad 1983).

3.2. Dilmun stamp and cylinder seals

by Aiysha Abu-Laban

A total of 91 seals were found in Tell F6 in the 2008-2012 excavations. This chapter deals with 75 of these seals which were locally produced within the Arabian Gulf region and which roughly date from the late 3rd to the mid 2nd millennium BC. They are described in detail in the catalogue below. The remaining seals, fifteen Mesopotamian cylinder seals and one Egyptian scarab, are dealt with elsewhere (see below Pittman p. 157 and Sperveslage p. 155).

The majority of the seals (N=67) have a circular shaped disc with a domed back/boss (fig. 668), four seals have a disc shape, without a boss, and one seal is cuboid with a prism-shaped dorsal side. The boss usually has three incised parallel lines and two circles with a central dot on each side and corresponds thus to the Dilmun type (cf. Kjærøum 1980 p. 45). A perforation for suspension runs perpendicular to the three lines. Motifs are engraved on the obverse side for the bossed seals, and on one or both faces for the discoid seals, and they are all in *intaglio*.

These 72 seals belong to one of the following styles: Style IA, Style IB, Style II or Style III (fig. 669). These four styles have been defined and described on the basis of seals uncovered in the 1950s and 1960s in Tells F3, F5 and F6 on Failaka and at Qala'at al-Bahrain and the Barbar temples in Bahrain (Kjærøum 1980, 1983, 1994 and 2003), and they will be followed here with the addition that Style IA has been divided into Style IAa and Style IAb (Abu-Laban in prep.). The following brief description of Style IAa and Style IAb will focus on the imagery, and other aspects will be dealt with further below.

Style IAa seals have only few, in average three to five motifs, which occupy the field. Motifs are generally wider and more deeply gouged than in Style IAb. The human figures have semi-circular heads and the face and torso are shown in profile. The eye is commonly rendered and consists of a triangle, the nose also has a triangular shape and is rather protruding, the lips consist of two lines, and the beard or chin is pointed (e.g. fig. 679).

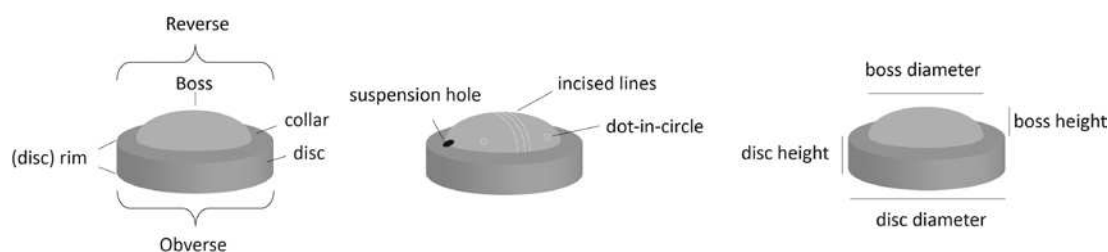


Fig. 668. Terminology.

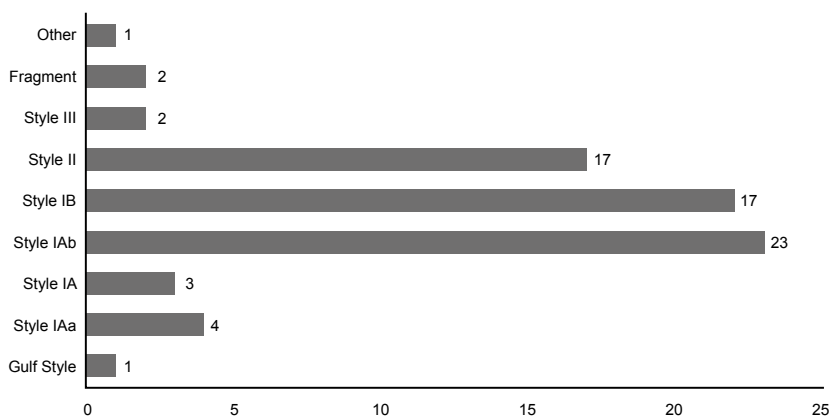


Fig. 669. Frequency of styles (N=75).

Style IAb comprises a wider range of images and themes, e.g. bull-men, hybrid monsters, and architectural symbols such as shrines and temple facades. The field is generally filled with more than ten motifs, which are rendered in a smaller scale to accommodate the space. Several actions such as drinking together and holding animals are combined in the same field. There is also more emphasis on conveying a hierarchical difference between garbed and nude humans. Human figures differ from *Style IAa* in that the head is less accentuated, and the eye is not rendered; the chin or beard is either shown pointed as in *Style IAa* or engraved down to the neck. The nose, lips and beard/chin are less protruding, and in a few examples the nose and lips consist of horizontal lines of equal length. Human chests are narrower, and animals such as ungulates have slimmer bodies. There is more variation in the position of ungulates which may, besides standing, be crouched, rampant, jumping or crossed (e.g. fig. 691).

The remaining three seals belong to rare types: a triangular prismatic seal decorated in Arabian Gulf style (Kjærøum 1994 p. 341), a cylinder seal decorated in *Style IAa*, and an imitation of an Egyptian scarab.

Size and shape

In the catalogue below, the seals have been ordered according to the style of their obverse decoration. There are slight differences between seals of different styles with respect also to size and shape for the seals which have a circular disc and a boss. Thus the seals of *Style IAa* have a diameter of 23.3-30 mm (median: 25.3 mm) and a height of 10.2-14.2 mm (median: 10.4 mm) (fig. 670). There are among the *Style IAb* seals two seals of extraordinary size resulting in a larger variation as the diameter ranges between 20.3-52.8 mm (median: 26.4 mm), and the height is 9.3-18 mm (median: 11.4 mm). *Style IB* seals are smaller than the *Style IAa* and *IAb* seals, as they have a diameter of 16-26.5 mm (median: 23 mm) and a height of 8.3-13.7 mm (median: 10.8 mm). *Style II* seals are even smaller with a diameter of 14.6-23.4 mm (median: 21.4 mm)

Style	Disc diameter	Median	Seal height	Median	Collar width	Median
IAa	23.3-30	25.3	10.2-14.2	10.4	3.8-6.7	5.1
IAb	20.3-52.8	26.4	9.3-18.0	11.4	2.2-8.5	4.7
IB	16.0-26.5	23.0	8.3-13.7	10.8	1.3-5.2	3.3
II	14.6-23.4	21.4	6.9-12.9	9.7	0.7-3.2	1.9
III	20.6-23.2		10.0-10.2		1.2	1.2

Fig. 670. Dimensions in mm according to style.

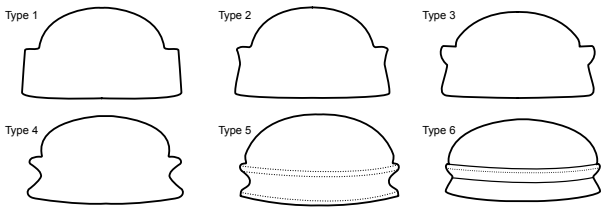


Fig. 671. Disc profile types.

Style/ Disc profile type	1	2	3	4	5	6
Style IAa	1	2	0	0	0	0
Style IAb	5	14	3	0	0	0
Style IB	2	1	0	0	0	0
Style IB	0	0	5	16	0	0
Style II	0	0	1	4	11	1
Style III	0	0	0	0	2	0

Fig. 672. Distribution of disc profile types according to style (N=68).

and a height of 6.9-12.9 mm (median: 9.7 mm). The two *Style III* seals have a diameter of 20.6-23.2 mm and a height of 10-10.2 mm.

The shape of the circular disc, the disc profile, also varies according to style (figs. 671-672). *Styles IAa* and *IAb* normally have a concave disc profile (Type 2), but a couple are flat or slightly concave (Type 1). Type 3, concave with an upper flattened or rounded rim edge, is found in three seals of *Style IAb*, but are more common in *Style IB*. Seals with a marked concavity with an upper and lower rounded rim edge (Type 4) are most common in *Style IB*. Within *Style II* the disc is low and the disc edges are narrower and sometimes flattened instead of rounded (Type 5), and in a few examples the concavity of the disc is so narrow that it seems rather grooved (V-shaped) (Type 6). Both seals of *Style III* have a disc shape of Type 5.

The proportions of the disc in relation to the boss also vary according to style (fig. 673). Kjærøum has already demonstrated this difference between Arabian Gulf/Gulf Type seals and Dilmun seals (i.e. *Style IA* seals) from Bahrain (1994). The relative disc height, i.e. the proportion of the disc in relation to the total

Style	Relative disc height (%)	Median (%)	Relative boss diameter (%)	Median (%)
IAa	57-68	59	72-77	74
IAb	44-68	57	68-83	77
IB	44-67	54	71-88	78
II	35-61	45.4	81-95	87
III	39	39	94	94

Fig. 673. Relative disc height and boss diameter according to style.

height of the seal, is highest among style IAa seals with the disc height forming 57-68% of the total height (median: 59%). Style IAb and IB seals have slightly lower discs as they constitute 44-68% and 44-67% respectively of the total height (median: 57% and 54%). Style II and Style III have discs comprising less than half of the total height, 35-61% and 39% respectively (median 45.4% and 39%).

Moreover, there is also a difference in the proportion of the disc diameter in relation to the boss diameter (fig. 673). The relative boss diameter, i.e. the boss diameter in relation to the disc diameter, is lowest among the style IAa seals as the boss forms 72-77% of the disc diameter (median: 74%) and thus the seal has a wide collar (median: 5.1 mm) (fig. 670). The relative boss diameter among Style IAb and IB seals is slightly higher, 68-83% and 71-88% respectively (median: 77% and 78%) with a narrower collar width (median: 4.7 mm and 3.3 mm) and still higher among Style II and Style III seals, 81-95% and 94% respectively with a quite narrow collar (median: 1.9 mm and 1.2 mm).

Averse decoration

The standard decoration of the boss consisting of three parallel lines with two dot-in-circles on each side is found on all seals in styles IAa, IAb (though the seal in fig. 697 has four lines) and IB. Three Style II seals lack this decoration: Fig. 727 has only two lines, whereas figs. 731 and 735 have neither lines nor dot-in-circles. None of the Style III seals has lines or dot-in-circles.

Iconography

The few Style IAa seals show humans, both garbed and nude, holding animals, a deity with two garbed humans holding a horizontal line, and one seal shows animals only. Style IAb seals contain a higher variety of themes, mainly the capturing or controlling of animals or hybrid monsters which can be tied by their legs or held by their horns. Hybrid serpent monsters are always placed in the center of the field and flanked by humans. Banquet scenes are also common and show garbed humans and a deity drinking from a jar through a straw, garbed humans sitting opposite an offering table or holding an animal while drinking. One seal shows an erotic scene, and one has humans holding hands. In a few cases bullmen and monkeys are engaged in the scene, and one seal shows rampant animals in a circular composition.

Style IB shows a higher frequency of nude humans, monkeys and bullmen mainly in association with ani-

mals, especially ungulates, which they either hold or stand next to. A few show them holding standards with astrals mounted on top. There is one seal which shows humans in a boat, and one showing seated garbed humans at a banquet scene.

Style II seals are dominated by prayer scenes where nude humans, bullmen with the assistance of monkeys either hold or have their arms directed towards astral symbols (the rosette motif is common) which are often mounted on a staff or standard.

The only Style III seal shows a prayer scene where garbed humans have their arms towards astral symbols held by a bird.

The iconography of the cylinder seal, the triangular prismatic seal and the scaraboid seal is discussed in the catalogue below.

Material

With respect to stone type there are slight differences between the seal styles. The majority of the seals (N=68) are made of soft stone, which has been white glazed and heat treated. The stone is normally fine grained and homogenous with a foliated structure (lamellar fracture) and contains small black inclusions (porphyroblasts), sometimes also red inclusions, but these are less common in seals of Style II than IA. The stone can vary in colour and in its level of opacity/translucency as described below; it corresponds to David-Cuny's Types B and X on the seals from Al-Khidr (David-Cuny & Azpeitia 2012 p. 22-23) and has some resemblance to Hilton's Type 4 on the soft stone vessels from Failaka (Hilton 2014 p. 12-13 and pers.com.).

Most of the seals are of a green coloured soft stone (N=46). Two-thirds of them (N=32) are dark grayish green with a bluish hue and they are semi-translucent. This colour is most common among Style IAa and IAb seals. Eight seals are of a green semi-translucent stone (with brownish hue) which predominates among Style IB seals. Seven seals are of a light green to slightly whitish opaque and these are mostly Style II seals. The rest are grey (N=3), bluish grey (N=5), brown (N=1) or black (N=1). For ten seals it was impossible to determine the core colour of the stone.

The difference in colour does not necessarily imply that the stone was quarried from different deposits. The colour and the degree of opacity/translucency of soft stone are easily manipulated e.g. by heating (Law 2011 p. 258).

Two Style II seals (figs. 731 and 735) are most likely of a hard stone, possibly chalcidony. Their surfaces are rather chalky, i.e. they rub off and have an exception-

ally homogenous white colour, which can be observed in overheated objects made of carnelian. Unlike seals made of softstone, there is no evidence of cracklines.

Three Style II seals are made of marine mollusk shells of gastropod species (figs. 736, 732, 741).⁴ Seals made of sea shell are only known in Style II.

Both Style III seals (figs. 743 and 744) are made of a hard, semi-translucent pink carnelian. Their pale surface and interior suggest that they have both been heat treated (pers.com. Ann Anderson 2014).

The triangular prismatic seal in Arabian Gulf style (fig. 748) is made of a green translucent soft stone, and the scaraboid seal (fig. 750) is made of a whitish soft stone, not known from other Dilmun seals.

Context

The majority of seals (81%, N=59) were found in Trenches A and E, i.e. the area between the southeast corner of the “Palace” and the northwest corner of the Temple (fig. 4). No seals were found in Trench H during the 2008-2012 excavations, but previous French excavations in the same area produced 11 seals (Beyer 1986 p. 89 ff.). The rest of the stamp seals (N=13), the prismatic seal and the cylinder seal were found south of the Temple in Trench C and will be dealt with below. Only twelve seals, including the scaraboid seal fig. 750, were found in disturbed or unknown context, e.g. from topsoil, stone plundering pits or on the surface of the tell (six seals from Trenches A and E, and six seals from Trench C).

The excavated area between the “Palace” and the Temple covers 71.5 m². A total of 54 out of 60 seals were found here in undisturbed layers. The highest concentration of seals (N=50) was retrieved from the layers dated to Failaka Periods 1-2 (fig. 5) of which more than half (N=30) predate the “Palace” building (from c. 1.74 m a.s.l. and below). The seals recovered from Phase 2 (N=8) were found adjacent to walls A219 and A191 and in sandy layers (figs. 40 and 674). The seals from Phase 3 (N=22) were found in deposits west of the thin-walled building, and in floor fill in the southeastern part of the building (fig. 46). A high concentration of seals were found in and just above the pavement fill (E203), especially its southern part, and a couple of seals were found in floors and deposits east and south of the oven E138/E143.

Most seals from Phase 4 were found in the fill of floor A180 (N=9) (fig. 53), and seven seals came from activity layers above floor A180, concentrated in the northeastern end and they predate hearth A164. Only four seals belong to Phase 5, and they come from fill covering oven E138/E143 (figs. 53 and 63).

BC	Style/Phases	Style IAa	Style IA	Style IAb	Style IB	Style II	Style III
1350	Phase 8		1	6	1	1	1
1400	Phase 7				2		1
1500	Phase 6					1?	
1600	Phase 5			1	1	2	
1750	Phase 4	2		6	4	4	
1800	Phase 3	1	1	5	7	8	
1900	Phase 2			1	5	2	
2100	Phase 1	1	1	4	1		

Fig. 674. Distribution of seal styles according to phases among 2008-2012 seals (N=72).

From Phase 6, i.e. Period 3B, only one seal fragment possibly of Style II, was retrieved (fig. 64). From Phase 7 two seals in Style IB and one in Style III came from the clay pit dating to Period 4A (fig. 68).

The number of seals is somewhat higher in Phases 2-3, pre-dating the “Palace”, than in Phases 4-5 contemporary with the “Palace”, especially when taking into account that the seals found in the fill of floor A180 could very well derive from Phase 3. Floor A180 was a very substantial floor, up to 45 cm in thickness, laid out contemporary with the construction of the “Palace” and consisting partly of debris from the previous phases.

A similar pattern may be interpreted from previous excavations. A total of 39 seals were recovered by the Danish excavations (1960-1963) in layers below the first floor of the “Palace”, whereas 35 seals were uncovered from the layers above. When taken into consideration that the area excavated below the first “Palace” floor was much more limited than the area excavated above the first “Palace” floor, it must be concluded that seals are significantly more numerous in layers pre-dating the construction of the “Palace”.

Moreover, from the French excavations of the Temple, 11 seals were found in a layer (Calvet & Pic 1986 p. 24) on floor H13 outside the temple terrace walls H17 and H9 (fig. 46) (cf. above p.), a context that can be referred to Phase 3.

There may be several reasons for the decline in seal frequency. One explanation is that it could be related to a reorganization of space northwest of the Temple prompted by the construction of the “Palace”. The activities involving frequent seal use and seal loss may have been moved to another location. It is, however, in general unclear why the seals ended up in the loca-

tions where they were found, and it is not certain that seals were simply lost where they were used.

The majority of seals (79%) were in good condition but five seals were severely worn and eleven seals were broken. The possibility that they were thrown away because they were worn out can therefore be discounted. A fair number of seals (N=25) were actually found in construction layers, e.g. floors, consisting of debris from older buildings, indicating the possibility that seals were hidden away in wall cracks and wall niches and accidentally re-deposited during demolition of houses (Højlund 2016).

Dating the seal styles

Among the seals excavated in 2008-2012, Style IA dominates with 40% (N=30), Style IAa (5%, N=4) being much less common than Style IAb (31%, N=23) (fig. 669). There are almost as many Style IB seals (30%, N=22) as Style IAb, and slightly less Style II seals (23%, N=17). Only two seals belong to Style III (3%).

This frequency of styles is similar to that found amongst seals from previous excavations on Failaka (i.e. from the Danish excavations in 1958-1963 (Kjærøum 1983), the American excavations in 1973-1974 (Howard-Carter 1984), the French excavations in 1984-1988 (Beyer 1986 and Pic 1990), and the Slovak excavations in 2004-2008 (David-Cuny & Azpeitia 2012). In this corpus of 512 seals there is also a predominance of Style IAb and Style IB (27% and 30% respectively). Style II seals on the other hand are fewer in number (13%). Style IAa forms 11%, followed by Style III (4%). The Gulf Type and the Proto-Dilmun Style together represent only 1% of the seals (Abu-Laban in prep.).

Dating the seal Styles IA, IB, II and III on Failaka has been hampered by the lack of secure stratigraphic context information retrieved during the previous excavations. Nevertheless, Kjærøum presented some thoughts on the chronology of the seal styles (1980 p. 46. 1994 p. 349-350). According to him seals of Style IA were alone in the deepest level of Tells F3 and F6. Seals of Style IB were found in the following level together with Style IA, and Style II seals first appeared in the next level and were dated not earlier than the middle of the 18th century. There is no mention of the stratigraphic context of Style III seals, but on the basis of cuneiform inscriptions they were dated to the late Old Babylonian or Early Kassite period (Kjærøum 1994 p. 348).

The extreme scarcity in Bahrain of seals of Style IB and especially Style II was considered by Kjærøum to be "hardly due to different style developments within

Dilmun's provinces, but must be chronologically and economically determined", i.e. caused by a decline of occupation in Bahrain after the collapse of the Dilmun civilization *versus* a prolonged occupation on Failaka (1994 p. 348). This tentative scheme for the evolution of seal styles was presented in a chronological table (Kjærøum 1994 fig. 1758), though Kjærøum's text clearly accorded with some indeterminate overlap between the styles.

The 2008-2012 excavations produced precise stratigraphic context information for the majority of the retrieved seals, and this has enabled a revised dating of the seal styles, especially Styles IB and II. Styles IB and II seals are found in contexts dating to Failaka Periods 1-2; in all four stratigraphic Phases 2-5 (figs. 5 and 674) they were found together in the same frequency and always together with seals in Style IA, more often in the Style IAb variety (N=13) than in Style IAa (N=3).

A similar picture of coexistence of seal styles IA, IB and II emerges from the site of Al-Khidr (David-Cuny & Azpeitia 2012). Four stratigraphic phases belonging to the Dilmun period were identified at Al-Khidr. How the pottery and other finds uncovered from the site relate to these four phases has not been published, however the majority of the pottery assemblage can be dated to Failaka period 2A-2B (Benediková 2010 p. 184, figs. 161-180), i.e. contemporary with Phases 2-5 in the 2008-2012 excavations in Tell F6. The number of Style IAa seals is very low (4%), compared to Style IAb (11%) and IB (28%), and the frequency of Style II seals is very high (42%) (fig. 675).

It must be concluded that Style IB and Style II were local Failaka seal styles that were developed in the final stage of Style IA, simultaneously with the full establishment of Failaka as a trading entrepot, instead of styles belonging to a period after the collapse of the Dilmun civilisation.

New evidence for the dating of Style III was not provided by this excavation.

Style	Quantity
Style IAa	3
Style IAb	8
Style IA	4
Style IB	20
Style II	30
Style II-III	4
Fragmented	2

Fig. 675. Frequency of seal styles at Al-Khidr (N=71).

Trench C

The excavated area south of the Temple covers 144.5 m², however the seals from undisturbed contexts were found within a few meters of the exposed architectural structures, i.e. the Mesopotamian wall (C28) dating to Phase 1 and the terrace belonging to the later Temple (C68-69). The Dilmun cylinder seal (fig. 749) was found with an Ur III Mesopotamian cylinder seal (fig. 754) in temple foundation fill south of the temple wall (fig. 99). Five stamp seals (figs. 705, 677, 747, 686 and 692) were found with three Mesopotamian cylinder seals (figs. 752, 756 and 753) in the sandy layers dating to the Ur III period. One style IB seal (fig. 704) was found in the clay platform (fig. 99: C48), i.e. a much later context closely related to the Dilmun temple.

The seals found in undisturbed context are of both Style IAa (N=1), IAb (N=4) and IA (N=1) and were found in Ur III layers with one exception. A Style IB seal (fig. 705) was recovered south of wall C28, however at a higher level (1.71 m a.s.l.) than the other seals. This group of seals seem typologically earlier than the seals found in Trenches A, E and H northwest of the Temple. It prompts the question of how to explain their presence in an Ur III level. Do they belong in this level or are they intrusions? The period between the end of the Ur III period and the beginning of the Dilmun occupation on Failaka is not particularly well illuminated in the excavation presented here, and this question is perhaps better left for future investigations. The only seal in Arabian Gulf style, the triangular prismatic seal, was found in an undated context.

Catalogue

The catalogue includes the following: *The object number* consisting of the trench and layer number followed by an X-number. *Context*: name of trench and layer number, stratigraphic phase and the height above sea level (m a.s.l.). *Material*: assessing the material is based on visual examination through an 18 mm/30X loupe. For the seals made of stone, the description includes the colour of the core if visible. *Treatment*: unless otherwise stated the standard treatment of the stone seals included heating and (white) glazing. *Condition*: a description of the extent of breakage and degree of wear. *Size*: in mm, in parenthesis if estimated. *Abbreviations* (fig. 668): *Disc H*: the height of the disc. *Boss H*: the height of the boss. *Boss D*: the diameter of the boss. *Disc D*: the diameter of the disc. *Suspension hole D*: the diameter of the suspension hole. *Disc profile*: the cross-section of the disc edge divided into six types (fig. 671): Type 1. Straight to

slightly concave. Type 2. Concave. Type 3. Concave with an upper flattened or rounded rim edge. Type 4. Deep concave with an upper and lower rounded rim edge. Type 5. Deep concave, low disc with an upper and lower rounded or flattened rim edge and Type 6. Deep, grooved (v-shaped), low disc, with an upper and lower rounded or flattened rim edge. *Reverse*: unless otherwise stated the standard decoration on the boss consists of three parallel lines engraved across the boss, and four circles with a dot in the center placed symmetrically with two on either side of the lines. The diameter of the circles is also provided. *Obverse*: The description of the image on the obverse is oriented according to the impression.

Photographs of the reverse, obverse, side and impression were taken by H. David-Cuny.

Seals with a circular disc and a boss, in Style IA

No. C4-X9 (fig. 676). *Context*: Trench C, layer 9, Phase 8. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: One quarter of the seal is preserved, joined with a seal (KM819) from previous excavations in Tell F3. *Size*: Disc H 6.4 mm, Boss D 21.4 mm, Disc D 23.8 mm, Suspension hole D 3.5 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 2 mm. *Obverse*: Four ungulate protomes joined in the central field.

No. C125-X317 (fig. 677). *Context*: Trench C, layer C125, Phase 1, 1.56-1.53 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Small part of disc and boss is preserved, slight evidence of wear. *Size*: Boss D (22.3 mm), Disc D (24.3 mm), Suspension hole not preserved. *Disc profile*: Type 1. *Reverse*: 3 lines, 2 dot-in-circles preserved, D 2.9 mm. *Obverse*: Not preserved.

No. E188-X864 (fig. 678). *Context*: Trench E, layer E188, Phase 3, 1.46 m a.s.l. *Material*: Soft stone, dark grayish green (opaque). *Treatment*: Standard. *Condition*: Complete, wear especially on the obverse. *Size*: Disc H 5.5 mm, Boss H 3.9 mm, Boss D 17 mm, Disc D 23.5 mm, Suspension hole D 2.7 mm. *Disc profile*: Type 1. *Reverse*: Standard, D 2.6 mm. *Obverse*: In the central field a palm tree flanked by two ungulates with their back to the tree. A motif below the tree. A star below the pizzle of the ungulate to the right.

Seals with a circular disc and a boss, in Style IAa

No. E153-X675 (fig. 679). *Context*: Trench E, layer E153, Phase 4, 1.74-1.59 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 8.1 mm, Boss H 6.1 mm, Boss D 21.7 mm, Disc D 30 mm, Suspension hole D 4.2 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 2.4 mm. *Obverse*: In the central field a human with a flounced garment holds with one hand the horn of a recumbent ungulate and with the other hand a jar which is also held by a nude human standing behind the garbed human. In front of the garbed human a crescent staff. To the left a possible monkey holds the neck of the ungulate. Below the arm of the garbed human a star, and behind the ungulate a crescent.

No. E205-X907 (fig. 680). *Context*: Trench E, layer E205, Phase 3, 1.49 m a.s.l. *Material*: Soft stone, grayish green (bluish hue).

Treatment: Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 6 mm, Boss H 4.2 mm, Boss D 19 mm, Disc D 25.5 mm, Suspension hole D 3.1 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 3.4 mm. *Obverse:* In the central field, a human, wearing a horned head dress and a flounced garment, holds a horizontal line flanked by two humans. The human to the right wearing a plain garment holds a staff with the other hand. The human to the left wearing a flounced garment holds with one hand a staff and the other hand is raised towards the horned human. An oval shaped motif, possibly a jar, is attached into the horizontal line between the deity and the human with the flounced garment.

No. E289-X1266 (fig. 681). *Context:* Trench E, layer E289, Phase 4, 1.56 m a.s.l. *Material:* Soft stone, dark grey (opaque). *Treatment:* Standard. *Condition:* Part of boss and obverse is broken, slight evidence of wear. *Size:* Disc H 6 mm, Boss H 4.4 mm, Boss D 18 mm, Disc D 23.3 mm, Suspension hole D 3.8 mm. *Disc profile:* Type 1. *Reverse:* Standard, 3 dot-in-circles preserved, D 3.5 mm. *Obverse:* A cross divides the field into four quadrants. In two of the quadrants a crab, in a third quadrant a possible monkey holding a staff. Remains of a dot-in-circle in the lower part of the fourth quadrant.

Seals with a circular disc and a boss, in Style IAB

No. E1-X8 (fig. 682). *Context:* Trench E, layer E1, Phase 8. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, minor evidence of wear on the obverse. *Size:* Disc H 7 mm, Boss H 4.6 mm, Boss D 19.1 mm, Disc D 24.8 mm, Suspension hole D 3.7 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 2.1 mm. *Obverse:* In the top field, two crossed ungulates, the one to the right has its forelegs raised towards a lentoid motif. To the right a nude human holds the neck of a bull standing below the rampant ruminant. Between the horns of the bull a jar. Below the bull an unidentified motif. A nude human in a running (or kneeling) position holds the leg of one of the crossed ungulates. Behind and above the nude human are two unidentified figures.

No. A3-X26 (fig. 683). *Context:* Trench A, layer A3, Phase 8. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Part of the boss and disc is broken off, minor evidence of wear. *Size:* Disc H 6.9 mm, Boss H 5.2 mm, Boss D 21.5 mm, Disc D 28.4 mm, Suspension hole D 3.3 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 1.7 mm. *Obverse:* In the upper right field a winged animal facing an unidentified motif. Below, two ungulates joined from their backs and from their snouts. To the left a nude human holds the neck of one of the ungulates. To the right an ungulate has its head turned towards the joined ungulates. Below the head of the ungulate a star.

No. A-X193 (fig. 684). *Context:* Trench A, layer A, Phase 8. *Material:* Soft stone, dark grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, wear on the obverse and slightly on the boss. *Size:* Disc H 8.7 mm, Boss H 6.7 mm, Boss D 26.5 mm, Disc D 34.6 mm, Suspension hole D 3.7 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 3.4 mm. *Obverse:* In the central field, two ungulates are crossing each other, and above their heads a crescent. The ungulates are flanked by two humans wearing flounced garments. The human to the left holds a tube which leads from the mouth and down to a jar. The human to the right holds with one hand the back of the ungulate to the right and with the other hand a tube leading down to a jar. Between the human and the ungulate, a star. In the bottom a radiating sun or star motif.

No. E75-X321 (fig. 685). *Context:* Trench E, layer E75, Phase 8. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Part of the disc is broken off, worn especially on the obverse. *Size:* Disc H 6.6 mm, Boss H 3.9 mm, Boss D 19 mm, Disc D 27.7 mm, Suspension hole D 3.4 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 1.5 mm. *Obverse:* In the central lower field a square podium with net-pattern, on which two protomes are mounted. Between the protomes a palm frond. To the left a human with a flounced garment sits on a trapezoid shaped stool with horizontal lines and holds with one hand the horn of the protome to the left. To the right is another seated human with a flounced garment.

No. C125-X336 (fig. 686). *Context:* Trench C, layer C125, Phase 1, 1.56-1.53 m a.s.l. *Material:* Soft stone, light yellow (opaque). *Treatment:* Standard. *Condition:* Small part of the disc and a minor part of the boss is preserved, slight evidence of wear. *Size:* Disc H 7.2 mm, Boss D (20.5 mm), Disc D (24.3 mm), Suspension hole D 2.9 mm. *Disc profile:* Type 2. *Reverse:* 3 lines, dot-in-circle not preserved. *Obverse:* An ungulate stands in the lower field facing left. Below its neck a palm frond and behind its tail a crescent. Parts of two other ungulates, one behind the standing ungulate and one above.

No. A162-X387 (fig. 687). *Context:* Trench A, layer A162, Phase 8. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 6.1 mm, Boss H 5.8 mm, Boss D 18.5 mm, Disc D 23.6 mm, Suspension hole D 3.1 mm. *Disc profile:* Type 2. *Reverse:* Standard, D 2.5 mm. *Obverse:* In the central field a bull-man and a human with a flounced garment crossing each other. They have one of their arms raised holding a crescent. The human holds a scorpion and the bull-man holds an ungulate. Between the head of the bull-man and the ungulate is a jar. At the bottom a fowl.

No. A180-X484 (fig. 688). *Context:* Trench A, layer A180, Phase 4, 1.70 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* A small part of disc is broken off, slight evidence of wear. *Size:* Disc H 6.5 mm, Boss H 5.8 mm, Boss D 19.7 mm, Disc D 25 mm, Suspension hole D 3.8 mm. *Disc profile:* Type 1. *Reverse:* Standard, D 1.7 mm. *Obverse:* The seal can be oriented in two directions. In the central field a male holds with one hand the leg of a female to the right and the neck of a rampant ungulate with other hand to the right. The male is in a phallic position and stands on a square podium with net-pattern. The female has her legs stretched and her arms extended along the legs. Above each of the female's arms, two fowls with spread wings facing the female. Behind the ungulate a foot motif.

No. C146-X462 (fig. 689). *Context:* Trench C, layer C146, Disturbed. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 5.4 mm, Boss H 3.9 mm, Boss D 19.3 mm, Disc D 23.7 mm, Suspension hole D 3.5 mm. *Disc profile:* Type 1. *Reverse:* Standard, D 1.3 mm. *Obverse:* In the right field a human with a flounced garment holds a scorpion and with the other hand a rectangular shaped stand with horizontal lines. To its left is an ungulate with its back to the staff and looking towards its back. The hind legs are tethered with a rope. Above its back and below the head, an oval shaped motif with convex ends.

No. E134-X629 (fig. 690). *Context:* Trench E, layer E134, Phase 5, 1.64 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of

wear. *Size*: Disc H 7 mm, Boss H 4.2 mm, Boss D 16.9 mm, Disc D 20.3 mm, Suspension hole D 3.2 mm. *Disc profile*: Type 1. *Reverse*: Standard, D 2.1 mm. *Obverse*: In the central field, a winged monster with horns, hairy back and tail. To the right, a nude human with one leg raised and bent holds the monster by its neck. To the right a square podium with net-pattern, above it a crescent and below it a jar.

No. E147-X660 (fig. 691). *Context*: Trench E, layer E134, Phase 4, 1.68 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 7.6 mm, Boss H 9 mm, Boss D 40.2 mm, Disc D 48 mm, Suspension hole D 3.8 mm. *Disc profile*: Type 3. *Reverse*: Standard, D 2 mm. *Obverse*: In the top field from right to left a fowl, a trident shaped leaf, a jar, a nine-leafed rosette, and a jar. To the left is a human with a flounced garment and horned helmet. The deity sits on a trapezoid shaped stool with horizontal lines and chevron pattern and holds a tube leading down to a jar. A nude human, in front of the deity, holds with one hand the same tube. Below the deity a seated human with a flounced garment holds the same tube. Behind the human a palm frond. In the centre two bulls facing the humans. The bulls have a square podium with net-pattern above their backs. Between the horns of each bull a jar. Behind the bull in the top centre is a human with a flounced garment holding with one hand the podium and with the other a crescent motif with a star on top. To the right a nude human in a running (kneeling) position holds the same crescent motif. Another nude human holds the square motif with net-pattern above the bull. The human holds with the other hand a crescent motif. Behind the nude human a rampant ungulate and behind it a foot motif. Behind the bull in the lower field a palm frond.

No. C186-X694 (fig. 692). *Context*: Trench C, layer C186, Phase 1, 1.63 m a.s.l. *Material*: Soft stone, light yellow to white. *Treatment*: Standard. *Condition*: Only a small fragment of the disc and boss is preserved, slight evidence of wear. *Size*: Disc H 7.2 mm, Boss D (20.5 mm), Disc D (26 mm), Suspension hole D 2.8 mm. *Disc profile*: Type 2. *Reverse*: lines not preserved, 1 dot-in-circle preserved, D 1.8 mm. *Obverse*: Not preserved.

No. E141-X724 (fig. 693). *Context*: Trench E, layer E141, Phase 3, 1.34 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, minor evidence of wear on boss. *Size*: Disc H 5.2, Boss H 4.8, Boss D 18.7 mm, Disc D 24.7 mm, Suspension hole D 3.6 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 2.9 mm. *Obverse*: In the central field a circle with an inner cross and inside each quadrant is a smaller quadrant motif. The outer circle and the cross have incised lines. At the bottom a standing bull and in front of it a staff with horizontal lines (ball staff). Behind the bull is a recumbent bull and in front of it a palm frond. Above the recumbent bull is a courant feline, above the standing bull a rampant feline and below it a crescent motif.

No. E165-X728 (fig. 694). *Context*: Trench E, layer E165, Phase 4, 1.39 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 6.3 mm, Boss H 5.1 mm, Boss D 19.7 mm, Disc D 23.7 mm, Suspension hole D 2.8 mm. *Disc profile*: Type 1. *Reverse*: Standard, D 1.8 mm. *Obverse*: In the left field a nude human with bent knees holds with one hand a tube in a jar and with the other hand a palm-tree mounted on top of a triangular shaped motif with horizontal lines in the top and net-pattern below. To the right an ungulate facing the palm-tree.

No. E171-X735 (fig. 695). *Context*: Trench E, layer E171, Phase 3, 1.48 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, large flakes on the obverse, slight evidence of wear. *Size*: Disc H 8 mm, Boss H 5.8 mm, Boss D 21.8 mm, Disc D 26.4 mm, Suspension hole D 2.9 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 1.7 mm. *Obverse*: In the centre a circle with hatched lines. Attached above the circle a V-shaped motif of hatched lines. Between the two legs of the V is a horizontal hatched line to which a five leafed rosette is attached. This motif is mirrored in the lower field below the circle flanked by two humans with flounced garments holding on to the central circle. Between the human to the right and the circle is a jar.

No. A221-X737 (fig. 696). *Context*: Trench A, layer A221, Phase 2, 1.13 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 6.7 mm, Boss H 5.8 mm, Boss D 19.6 mm, Disc D 25.5 mm, Suspension hole D 3.4 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 2.1 mm. *Obverse*: Three humans with flounced garments hold each other's hands. The human to the left and the one to the right are holding a gate motif with the other hand. At the top a crescent.

No. A218-X738 (fig. 697). *Context*: Trench A, layer A218, Phase 3, 1.12 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, flakes around the obverse edges, slight evidence of wear. *Size*: Disc H 7.9 mm, Boss H 10.1 mm, Boss D 44.2 mm, Disc D 52.8 mm, Suspension hole D 4.9 mm. *Disc profile*: Type 3. *Reverse*: 4 lines, 4 dot-in-circles, D 2 mm. *Obverse*: The central field is dominated by a horned serpent monster facing left. The serpent is placed within a rectangular frame which has on both ends a square podium with net-pattern. On each side of the podia is a jar. The upper part of the rectangular frame (shrine) is flanked by two humans with flounced garments facing the opposite direction. They hold the top corner of the frame with one hand and a jar with the other. Below them are two humans with plain garments facing and holding with one hand the frame. Below them are two nude humans holding the frame. Three ungulates stand in the left and the right. The ungulates in the top have their front legs bent and are facing the humans. Above their backs a sun motif with two concentric circles. Above the sun motif to the right a jar. A possible jar is above the sun motif to the left. Behind the ungulate to the left a vertical line. Below the ungulates two bulls with their backs to the humans. A square podium with net-pattern is mounted to their backs. Below them, two ungulates with their backs to the humans and having their head towards the humans. Below the ungulates are two fowls with their backs to the humans.

No. C208-X754 (fig. 698). *Context*: Trench C, layer C208, temple foundation trench, Phase 1, 1.06-1.01 m a.s.l. *Material*: Soft stone, brown (surface). *Treatment*: Standard, transparent glazing. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 6 mm, Boss H 4.7 mm, Boss D 17.1 mm, Disc D 23.5 mm, Suspension hole D 3 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 2.2 mm. *Obverse*: In the left field a bull-man holds a crescent standard with a rosette on top. Behind the bull-man a jar. To the right of the standard a cross motif. In the far right an ungulate with its head turned towards its back. Above the head of the ungulate a jar. At the bottom a bull.

No. A180-X824 (fig. 699). *Context*: Trench A, layer A180, Phase 4, 1.47 m a.s.l. *Material*: Soft stone. *Treatment*: Glazed, overheated. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 7.1

mm, Boss H 3.4 mm, Boss D 18.6 mm, Disc D 23 mm, Suspension hole D 2.7 mm. *Disc profile*: Type 1. *Reverse*: Standard, D 2.1 mm. *Obverse*: In the central field a nude human sitting on a rectangular stool. The human holds the neck of an ungulate and with the other hand a gate motif. Behind it a possible palm frond.

No. E202-X905 (fig. 700). *Context*: Trench E, layer E202, Phase 3, 1.26 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 6.6 mm, Boss H 4.7 mm, Boss D 19.2 mm, Disc D 25.1 mm, Suspension hole D 3.3 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 1.8 mm. *Obverse*: In the central field a palm-tree with a pointed end. To its left a nude human sits on a triangular shaped stool. The human holds a tube leading from the mouth down to a jar. Between the tree and the human's head a jar. To the right an ungulate with its head turned towards its back. Between the leaflets and the head of the ungulate a triangular motif.

No. E143-X1036 (fig. 701). *Context*: Trench E, layer E143, Phase 3, 1.66 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 5.9 mm, Boss H 4.3 mm, Boss D 18.8 mm, Disc D 24.6 mm, Suspension hole D 3 mm. *Disc profile*: Type 3. *Reverse*: Standard, D 1.6 mm. *Obverse*: In the central field a rectangular podium with net-pattern. Above the podium a human visible from head to torso holding with each hand the neck of two ungulates. The ungulates are in a seated position with their backs to the human, only one of their hind legs is visible. At the base of the podium are two joined ungulates, only their head, neck and one front leg visible. Below the joined ungulates, a four legged animal flanked by a palm frond. To the right a monkey holds the neck of one of the joined ungulates. Above the monkey a palm frond and a star. To the left a crescent and below it a palm frond and a crescent.

No. E315-X1353 (fig. 702). *Context*: Trench E, layer E315, Phase 4. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, minor flake on boss and obverse, slight evidence of wear. *Size*: Disc H 7.3 mm, Boss H 4.1 mm, Boss D 19 mm, Disc D 25 mm, Suspension hole D 3.1 mm. *Disc profile*: Type 2. *Reverse*: Standard, D 1.9 mm. *Obverse*: Two humans with flounced garments sitting on a stool and drinking from a tube in a jar. The human to the right sits on a two-legged, square stool with horizontal lines. The backrest is rectangular with horizontal lines. The human holds with one hand the head of a rampant ungulate. Between them is a triangular motif. Behind the rampant ungulate a rectangle with horizontal lines and below it a fowl. The human to the left sits on a square stool with horizontal lines. Below the stool is a triangle with horizontal lines. At the bottom a bull with a lowered head and below its neck a palm frond.

Discoid seal in Style IAB

No. E291-X1265 (fig. 703). *Context*: Trench E, layer E291, Phase 4, 1.76 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 9.2, Disc D 25.2 mm, Suspension hole D 3.4 mm. *Disc profile*: Type 2. *Face 1*: In the central field two hybrid serpent monsters joined by their heads, a crescent motif is placed on each side of the joined head. To the left a bull-man holds the serpent. A nude human holds the back of the bull-man. Behind the bull-man's head a jar. Between the bull-man and the serpent a palm frond. To the right of the serpent monster a recumbent ungulate with its head turned towards its back. A nude human holds the ungulate's horn. On each side of

the human a jar. Below the ungulate a fowl facing the serpent. The field is bordered by a circle with denticulated lines. *Face 2*: In the top central field a crescent and below a square motif with two inner square motifs with net-pattern. On each side of the niche a palm frond. The niche is flanked by two humans with flounced garments, sitting on a square-shaped stool with horizontal lines and a backrest. The humans hold the lower edge of the facade. A table with hoofed legs is situated between the humans. On the table a monkey sitting and holding the neck of a fowl. Behind the monkey another fowl facing the monkey. On each side of the table column a square motif with net-pattern and a third one below the table. To the right of the table a fowl.

Seals with a circular disc and a boss, in Style IB

No. C23-X73 (fig. 704). *Context*: Trench C, layer C23, temple clay platform, 1.97 m a.s.l. *Material*: Soft stone, bluish grey. *Treatment*: Standard. *Condition*: Complete, worn on obverse. *Size*: Disc H 5.6 mm, Boss H 4.1 mm, Boss D 16.8 mm, Disc D 20.7 mm, Suspension hole D 2.6 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 1.7 mm. *Obverse*: In the bottom field an ungulate and below its neck a crescent shaped motif. Behind the ungulate to the right a nude human holds a palm frond. At the top a triangular shaped motif with net-pattern and below it a fowl.

No. C32-X84 (fig. 705). *Context*: Trench C, layer C32, Phase 1, 1.71 m a.s.l. *Material*: Soft stone, green (semi-opaque). *Treatment*: Standard. *Condition*: Complete, minor flakes on the reverse, slight evidence of wear. *Size*: Disc H 5.3 mm, Boss H 3.9 mm, Boss D 16.5 mm, Disc D 21 mm, Suspension hole D 3.2 mm. *Disc profile*: Type 3. *Reverse*: Standard, D 1.7 mm. *Obverse*: In the central field a motif consisting of two concentric circles. Three recumbent ungulates surround the motif. Above the back of two of the ungulates a jar and above the back of the third ungulate a circle.

No. E59-X240 (fig. 706). *Context*: Trench E, layer E59, Phase 7, 1.89 m a.s.l. *Material*: Soft stone, grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, minor flakes on the disc, wear on the obverse and reverse. *Size*: Disc H 5.3 mm, Boss H 6.7 mm, Boss D 19 mm, Disc D 24.1 mm, Suspension hole D 2.7 mm. *Disc profile*: Type 3. *Reverse*: Standard, D 1.8 mm. *Obverse*: In the central field a nude human holds with one hand a human with a plain garment and with the other an ungulate protome. Behind the protome an arrow shaped motif. The field is bordered by a circle with denticulated lines.

No. C128-X354 (fig. 707). *Context*: Trench C, layer C128, Phase 8. *Material*: Soft stone, green. *Treatment*: Standard. *Condition*: Small part of disc and boss is preserved, slight evidence of wear. *Size*: Disc H 7.1 mm, Boss D (19.7 mm), Disc D (23.7 mm), Suspension hole D not preserved. *Disc profile*: Type 4. *Reverse*: lines not preserved, 1 dot-in-circle preserved, D 1.9 mm. *Obverse*: Not preserved.

No. E134-X561 (fig. 708). *Context*: Trench E, layer E134, Phase 5, 1.90 m a.s.l. *Material*: Soft stone, bluish grey (opaque). *Treatment*: Standard. *Condition*: Complete, heavily worn. *Size*: Disc H 4 mm, Boss H (4.6 mm), Boss D 15.4 mm, Disc D (17 mm), Suspension hole D 2.8 mm. *Disc profile*: worn. *Reverse*: lines not preserved, 3 dot-in-circles preserved, D 2 mm. *Obverse*: In the bottom field a bull and below its head an ungulate head. Above the bull is an ungulate and above that a square podium with net-pattern. Behind the ungulates to the left a lentoid motif with horizontal lines.

No. E137-X587 (fig. 709). *Context:* Trench E, layer E137, Phase 4, 1.77 m a.s.l. *Material:* Soft stone, green, brownish hue (translucent). *Treatment:* Standard. *Condition:* Complete, some wear on boss and obverse. *Size:* Disc H 4.9 mm, Boss H 4.1 mm, Boss D 17.1 mm, Disc D 20 mm, Suspension hole D 2.8 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 1.7 mm. *Obverse:* In the bottom field a bull with the tail of a serpent attached to its back. To the right a nude human holds with one hand one leg of a monkey with its arms raised. With the other hand the human holds a fish. Below it another fish facing downwards. At the top a trident shaped motif. At the bottom a scorpion facing downwards.

No. E149-X661 (fig. 710). *Context:* Trench E, layer E149, Phase 3, 1.65 m a.s.l. *Material:* Soft stone. *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 5.2 mm, Boss H 5.1 mm, Boss D 17 mm, Disc D 21.6 mm, Suspension hole D 3 mm. *Disc profile:* Type 3. *Reverse:* Standard, D 1.8 mm. *Obverse:* In the left field a bull has a star in a crescent staff mounted to its back. Both above and below the ungulate's head is a crescent. Behind the bull, to the right, a nude human holds a star in a crescent staff. At the top a crescent.

No. E142-X664 (fig. 711). *Context:* Trench E, layer E142, Phase 3, 1.74 m a.s.l. *Material:* Soft stone, green, brownish hue (semi-translucent). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 6 mm, Boss H 6.1 mm, Boss D 22 mm, Disc D 26.5 mm, Suspension hole D 2.8 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 2.7 mm. *Obverse:* In the top central field a star in a crescent staff which is bent held by a nude human standing to the right. Behind the nude human is a monkey with arms raised, and behind the monkey a crescent. Below the monkey a volute standard on a square podium. To its right a crescent and to its left a star. Between the nude human and the monkey a possible stool. Below the nude human and the crescent staff an ungulate and in front of its legs a hook shaped motif. Above its horns a serpent like motif and in front of it a volute staff. Below the ungulate a bull standing on top of a horizontal line, possibly a podium. In front of the bull to the right a nude human with one hand reaching towards the neck of the bull. Above the nude human a possible fish.

No. E171-X734 (fig. 712). *Context:* Trench E, layer E171, Phase 3, 1.52 m a.s.l. *Material:* Soft stone, light grayish green to white (opaque). *Treatment:* Standard. *Condition:* Complete, flake on the obverse, slight evidence of wear. *Size:* Disc H 5.6 mm, Boss H 3.4 mm, Boss D 11.4 mm, Disc D 16 mm, Suspension hole D 3 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 1.7 mm. *Obverse:* In the central field a standard on to which an unidentified motif, possibly a fowl, is mounted. The standard is flanked by two ungulates looking backwards. At the top right a crescent.

No. A221-X736 (fig. 713). *Context:* Trench A, layer A221, Phase 2, 1.18 m a.s.l. *Material:* Soft stone. *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 5 mm, Boss H 3.2 mm, Boss D 15 mm, Disc D 18.3 mm, Suspension hole D 2.3 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 1.9 mm. *Obverse:* In the lower central field two ungulates joined from their backs. The ungulates stand on a square podium with a net-pattern and on each upper end is a crescent motif attached. Above the joined ungulates is an ungulate also with the head lowered. Above the ungulate is a fish and in the left field a monkey with raised arms.

No. E171-X738 (fig. 714). *Context:* Trench E, layer E171, Phase 3, 1.45 m a.s.l. *Material:* Soft stone, green (translucent). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 4.5 mm, Boss H 3.8 mm, Boss D 16.6 mm, Disc D 21.9

mm, Suspension hole D 2.7 mm. *Disc profile:* Type 3. *Reverse:* Standard, D 1.6 mm. *Obverse:* In the central field a rectangular motif with net-pattern mounted on a staff, attached in its lower end to a square podium with net-pattern. At the top centre a bucranium which has a crescent on each side. Two nude humans hold the rectangular motif. Behind the human to the left a serpent.

No. A221-X766 (fig. 715). *Context:* Trench A, layer A221, Phase 2, 1.09 m a.s.l. *Material:* Soft stone, green with brownish hue (translucent). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 6 mm, Boss H 5.1 mm, Boss D 20 mm, Disc D 24 mm, Suspension hole D 2.8 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 2.5 mm. *Obverse:* In the central field two humans crossing each other with a crescent mounted on top of their heads. In both crescents is a star. Between their heads a palm frond. They hold each an ungulate. In the lower field is a staff with a dot-in-circle flanked by two dot-in-circles.

No. A209-X767 (fig. 716). *Context:* Trench A, layer A209, Phase 3, 1.12 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 5.4 mm, Boss H 5.1 mm, Boss D 17.9 mm, Disc D 21.9 mm, Suspension hole D 2.8 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 2.4 mm. *Obverse:* The seal can be oriented in two directions. At the bottom a bull with a hook-shaped motif mounted between its horns. In front of the bull a scorpion facing downwards. Behind the bull an oval shaped motif. Above the back of the bull a monkey holds a nude human. Below them a crescent.

No. A226-X790 (fig. 717). *Context:* Trench A, layer A226, Phase 2, 1.02 m a.s.l. *Material:* Soft stone, light grayish green to white (opaque). *Treatment:* Standard. *Condition:* Complete, heavily worn. *Size:* Disc H 7.6 mm, Boss H 3.8 mm, Boss D 20.4 mm, Disc D 24.9 mm, Suspension hole D 3.2 mm. *Disc profile:* possibly Type 4. *Reverse:* Standard, D 2 mm. *Obverse:* In the bottom field a boat with a mast and a possible sail which is flanked by two nude humans holding the sail. Behind the humans a crescent. In the upper field a scorpion facing right.

No. E173-X827 (fig. 718). *Context:* Trench E, layer E173, Phase 3, 1.49 m a.s.l. *Material:* Soft stone, dark grey (opaque). *Treatment:* Standard. *Condition:* Fragmented, slight evidence of wear. *Size:* Disc H 5.7 mm, Boss D 18 mm, Disc D 22 mm. Suspension hole not preserved. *Disc profile:* Type 4. *Reverse:* 3 lines, dot-in-circles not preserved. *Obverse:* In the top central field a horizontal line with a palm frond on each end. Between the palm fronds a staff with a dot-in-circle. To the right a human holds the tail of a bull.

No. E187-X865 (fig. 719). *Context:* Trench E, layer E187, Phase 4, 1.35 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 6.4 mm, Boss H 5.7 mm, Boss D 20 mm, Disc D 25.6 mm, Suspension hole D 3 mm. *Disc profile:* Type 4. *Reverse:* Standard, D 2.2 mm. *Obverse:* In the right field a bull-man holds a sun consisting of two concentric circles, the outer circle has incised lines. To the left of the bull-man an ungulate and below it a bull with a head of an ungulate attached to its neck. Above the bull a palm frond which is bent. At the top a scorpion facing left and to its left a crescent.

No. A180-X898 (fig. 720). *Context:* Trench A, layer A180, Phase 4, 1.35 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, evidence of wear. *Size:* Disc H 5.7 mm, Boss H 5.1 mm, Boss D 18.6 mm, Disc D 21.3 mm, Suspension hole D 2.8 mm. *Disc profile:* Type 4. *Reverse:*

Standard, D 1.9 mm. *Obverse*: In the right upper field a star in a crescent. At the centre a horned nude human holding with one hand the horns of a bull. A monkey-like creature holds the other horn of the bull. The bull has its back to an undefined motif consisting of a horizontal line and two vertical lines. The human holds with the other hand, to the right, a staff. On the top of the staff is an ungulate head attached and below are three fish attached from their mouth.

No. A235-X905 (fig. 721). *Context*: Trench A, layer A235, Phase 2, 1.14 m a.s.l. *Material*: Soft stone, dark grey (opaque). *Treatment*: Standard. *Condition*: Complete, worn especially around the disc edges, possible polishing of disc edges. *Size*: Disc H 6.3 mm, Boss H 6.7 mm, Boss D 22.3 mm, Disc D 25.1 mm, Suspension hole d 3.2 mm. *Disc profile*: possibly Type 4. *Reverse*: Standard, D 2.5 mm. *Obverse*: In the central field two bulls with a joined head and neck. Two monkeys each hold the snouts of the joined animals. Below them are two bulls with their backs to each other. A palm frond near the neck of the bull to the right. Above the back of the joined bull to the right is a square.

No. E279-X1186 (fig. 722). *Context*: Trench E, layer E279, Phase 4, 0.76 m a.s.l. *Material*: Soft stone, green. *Treatment*: Standard. *Condition*: Complete, a small part of boss is broken, slight evidence of wear. *Size*: Disc H 6.7 mm, Boss H 5.3 mm, Boss D 19.7 mm, Disc D 25.4 mm, Suspension hole D 3 mm. *Disc profile*: Type 3. *Reverse*: Standard, D 2 mm. *Obverse*: In the central field a bull-man flanked by two ungulates with their heads turned backwards. The bull-man facing right has one arm raised towards the ungulate in the right. The bull-man and the ungulates are standing on a rectangular podium with horizontal lines. On each side of the podium a four leafed rosette and at the top on each side of the bull-man's head a star motif.

No. E285-X1215 (fig. 723). *Context*: Trench E, layer E285, Phase 7, 1.84 m a.s.l. *Material*: Soft stone, dark brown (translucent). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 7.9 mm, Boss H 5.8 mm, Boss D 18.8 mm, Disc D 24.2 mm, Suspension hole D 3.5 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 1.4 mm. *Obverse*: In the top field two ungulates sitting with their back to each other on top of a square podium with net-pattern. Only the head, neck and the two front legs are visible. The podium and ungulates are flanked by two seated humans wearing garments with horizontal pattern. The humans hold with one hand the legs of the ungulates. Below the podium two jars placed on a horizontal line, possibly a podium.

No. E179-X1216 (fig. 724). *Context*: Trench E, layer E179, Phase 3, 1.48 m a.s.l. *Material*: Soft stone, light grayish green, brownish hue (semi-opaque). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 5.9 mm, Boss H 4.9 mm, Boss D 16.7 mm, Disc D 20.3 mm, Suspension hole D 3 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 1.8 mm. *Obverse*: In the right field a nude human holds the leg of a monkey. The monkey holds one of the horns of a hybrid animal which has an ungulate head and the body of a fowl. The animal stands above a bull facing left. Below the neck of the bull a possible crescent motif. At the top left a serpent facing downwards.

Discoid seal in Style IB

No. E246-X1031 (fig. 725). *Context*: Trench E, layer E246, Phase 2, 1.13 m a.s.l. *Material*: Soft stone, greyish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, evidence of wear, a hole in the central part of face 2. *Size*: Disc H 7 mm, Disc D 23 mm, Suspension hole D 3.8 mm. *Disc profile*: Type 4. *Face 1*: In the right field a bull and below its head a crescent motif.

Above it an ungulate in a jumping position. Between the two ungulates is another crescent motif. To the right behind the ungulate a serpent-like creature which is entwined. Between the ungulate and the serpent a wavy line, possibly also a serpent. *Face 2*: In the central field an outturned volute standard. To its left an ungulate looking towards its back. Above the head of the ungulate a trident motif. To the right of the standard a fish attached to a staff held by a nude human holding another staff with the other hand.

Seals with a circular disc and a boss, in Style II

No. A165-X361 (fig. 726). *Context*: Trench A, layer A165, Phase 4, 1.82 m a.s.l. *Material*: Soft stone, black (translucent). *Treatment*: Standard. *Condition*: Complete, part of the surface of the obverse is damaged. *Size*: Disc H 4 mm, Boss H 6 mm, Boss D 20.7 mm, Disc D 22.8 mm, Suspension hole D 2.7 mm. *Disc profile*: Type 6. *Reverse*: Standard, D 1.3 mm. *Obverse*: In the central field a rosette with pointed shaped leafs. Around the rosette four ball staffs with a rosette in a crescent. Between the rosettes in a crescent are out-turned volutes in a ball staff.

No. C139-X450 (fig. 727). *Context*: Trench C, layer C139, Phase 8. *Material*: Soft stone, light grayish green to white (opaque). *Treatment*: Standard. *Condition*: Large part of the boss and minor parts of the disc are broken off, slight evidence of wear. *Size*: Disc H 3.9 mm, Boss H 6.1 mm, Boss D 20.3 mm, Disc D 22.2 mm, Suspension hole D 2.5 mm. *Disc profile*: Type 5. *Reverse*: 2 lines, 4 dot-in-circles, D 1.2 mm. *Obverse*: In the top central field a standard with a radiating sun flanked by two bull-men with both arms raised towards the standard. Behind each bull-man a rosette and below the standard another rosette. Below the bull-men are two ungulates with their heads directed towards their back. Behind each of the ungulate a rosette and at the bottom another rosette.

No. E134-X583 (fig. 728). *Context*: Trench E, layer E134, Phase 5, 1.75 m a.s.l. *Material*: Soft stone, light grayish green to white (opaque). *Treatment*: Standard. *Condition*: Complete, minor part of disc is broken off, slight evidence of wear. *Size*: Disc H 3.4 mm, Boss H 6.3 mm, Boss D 22.7 mm, Disc D 24.8 mm, Suspension hole D 2.1 mm. *Disc profile*: Type 5. *Reverse*: Standard, D 1.1 mm. *Obverse*: In the top central field a rosette and below it a recumbent ungulate with the head directed backwards. The ungulate is flanked by two bull-men. The bull-man to the left has both arms raised towards the rosette, and behind him a nude human with both arms raised. The bull-man to the right holds the neck of the ungulate and behind him a palm frond. At the bottom a nude human with his arms directed towards two bulls on each side. The bulls face each other and have their front legs placed each on a square shaped podium with horizontal lines.

No. A209-X679 (fig. 729). *Context*: Trench A, layer A209, Phase 3. *Material*: Soft stone, green (semi-opaque). *Treatment*: Standard. *Condition*: Part of the boss and disc is broken off, slight evidence of wear. *Size*: Disc H 4.7 mm, Boss H 6.4 mm, Boss d 18.6 mm, Disc d 22.6 mm, Suspension hole d 2.5 mm. *Disc profile*: Type 3. *Reverse*: 3 lines, 3 dot-in-circles preserved, D 1.6 mm. *Obverse*: In the top field a rosette and below it two crossed bull-men. The bull-man to the left holds a palm frond and the bull-man to the right holds the tail of an ungulate. Below the ungulate a possible monkey holding the tail of one of the bull-men. To the left an ungulate and below it a fowl with spread wings. At the bottom a square podium with net-pattern.

No. E171-X732 (fig. 730). *Context*: Trench E, layer E171, Phase 3, 1.49 m a.s.l. *Material*: Soft stone, bluish grey (opaque). *Treatment*: Standard. *Condition*: A small part of the boss and disc is broken. Slight evidence of wear. *Size*: Disc H 4.3 mm, Boss H 4.4 mm, Boss D 18.8 mm, Disc D 21.4 mm, Suspension hole D 2.2 mm. *Disc profile*: Type 5. *Reverse*: Standard, D 1.3 mm. *Obverse*: In the central field a rosette flanked by two bull-men. The bull-men hold an ungulate protome raised above the rosette. Behind each bull-man a palm frond. The bull-men are flanked by two nude humans. The nude human to the right holds the nape curl of one of the bull-men and the nude human to the left has both arms raised. Below the rosette a fowl and at the bottom a bull with lowered head.

No. A218-X735 (fig. 731). *Context*: Trench A, layer A218, Phase 3, 1.16 m a.s.l. *Material*: Stone, white surface. *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 4.6 mm, Boss H 7.7 mm, Boss D 23.4 mm, Disc D 24.4 mm, Suspension hole D 3.1 mm. *Disc profile*: Type 5. *Reverse*: none. *Obverse*: In the central field a ball standard with a rosette in a crescent. The standard is placed on a rectangular podium with vertical lines. Two bull-men have their arms raised towards the standard. Behind each bull-man a palm frond. Below the arms of the bull-man to the left are two jars.

No. A180-X826 (fig. 732). *Context*: Trench A, layer A221, Phase 4, 1.66 m a.s.l. *Material*: Marine mollusk shell (possibly *Lambis truncate* (sebae) sp.). *Condition*: Complete, minor breaks, worn around the seal. *Size*: Disc H 3 mm, Boss H 4.6 mm, Boss D 21 mm, Disc D 22.5 mm, Suspension hole D 2.3 mm. *Disc profile*: worn. *Reverse*: none. *Obverse*: In the central field a star in a crescent standard placed on a square podium with horizontal lines. The standard is flanked by two bull-men with their arms raised towards the standard. The bull-men are flanked by two monkeys. The arm gestures of the monkeys are unclear.

No. A180-X828 (fig. 733). *Context*: Trench A, layer A180, Phase 4, 1.38 m a.s.l. *Material*: Soft stone, light grayish green to white (opaque). *Treatment*: Standard. *Condition*: Complete, minor breaks, slight evidence of wear. *Size*: Disc H 4 mm, Boss H 4.8 mm, Boss D 16.1 mm, Disc D 18.4 mm, Suspension hole D 2.6 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 0.8 mm. *Obverse*: In the top central field a bucranium and below it a square podium with net-pattern. The bucranium and the podium are flanked by two rampant ungulates. Below the podium are two ungulates facing each other and between them a palm frond.

No. A338-X870 (fig. 734). *Context*: Trench A, layer A338, Phase 2, 0.79 m a.s.l. *Material*: Soft stone, light grayish green to white (opaque). *Treatment*: Standard. *Condition*: Complete, top of the boss is broken off. Slight evidence of wear. *Size*: Disc H 4.2 mm, Boss H 3.7 mm, Boss D 16.6 mm, Disc D 19.3 mm, Suspension hole D 2.7 mm. *Disc profile*: Type 5. *Reverse*: Standard, D 1.2 mm. *Obverse*: The field is divided into four quadrants by a cross motif which consists of two rectangular lines with hatched pattern. In each quadrant a rosette. The field is bordered by a circle with denticulated lines.

No. E198-X889 (fig. 735). *Context*: Trench E, layer E198, Phase 3, 1.29 m a.s.l. *Material*: Stone, white surface. *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 5.9 mm, Boss H 7 mm, Boss D 21.6 mm, Disc D 23.4 mm, Suspension hole D 3.1 mm. *Disc profile*: Type 5. *Reverse*: none. *Obverse*: In the central field a star in a circle. Four crescent standards are attached to the circle and above each crescent is a dot. Between each standard is an ungulate looking towards its back.

No. E198-X890 (fig. 736). *Context*: Trench E, layer E198, Phase 3, 1.34-1.18 m a.s.l. *Material*: Marine mollusk shell (possibly species *Charonia tritonis*), the side with the bulging rim shows a former *varix*, and the spiral sculpture is still visible. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 4.3 mm, Boss H 4.1 mm, Boss D 18.6 mm, Disc D 21.7 mm, Suspension hole D 2.5 mm. *Disc profile*: Type 5. *Reverse*: 3 lines. *Obverse*: In the central field a ball standard on to which a rosette in a crescent is mounted. Two bull-men on each side hold the standard. Behind each bull-man is a rosette.

No. E199-X906 (fig. 737). *Context*: Trench E, layer E199, Phase 3, 1.29 m a.s.l. *Material*: Soft stone. *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 3.1 mm, Boss H 3.7 mm, Boss D 12 mm, Disc D 14.6 mm, Suspension hole D 2.6 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 1.1 mm. *Obverse*: In the central field a dot-in-circle motif to which four ungulate protomes are attached.

No. E225-X963 (fig. 738). *Context*: Trench E, layer E225, Phase 2, 1.16 m a.s.l. *Material*: Soft stone. *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 4.4 mm, Boss H 6.6 mm, Boss D 21.3 mm, Disc D 24.1 mm, Suspension hole D 3 mm. *Disc profile*: Type 5. *Reverse*: Standard, D 2 mm. *Obverse*: In the central field, inside a circle, a rosette with four trapezoid-shaped leaves. Four ungulate protomes are attached to the circle. In the field between the protomes an ungulate and a star in a crescent are placed side one another.

No. E202-X1053 (fig. 739). *Context*: Trench E, layer E202, Phase 3, 1.04 m a.s.l. *Material*: Soft stone. *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 4.1 mm, Boss H 4.6 mm, Boss D 14.9 mm, Disc D 17 mm, Suspension hole D 1.6 mm. *Disc profile*: Type 5. *Reverse*: Standard, D 1 mm. *Obverse*: In the top field two crescents and below them a rosette. Two bull-men have their arms raised towards the rosette. Behind the bull-man to the left a palm frond. At the bottom an ungulate with its head turned backwards.

No. E270-X1166 (fig. 740). *Context*: Trench E, layer E270, Phase 5, 1.93 m a.s.l. *Material*: Soft stone, green (semi-translucent). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 5.6 mm, Boss H 5.7 mm, Boss D 18.5 mm, Disc D 21.7 mm, Suspension hole D 2.9 mm. *Disc profile*: Type 4. *Reverse*: Standard, D 1.8 mm. *Obverse*: In the right field a bull-man holds the tail of an ungulate. Above the ungulate at the top centre a rosette and to the left a crescent. Below the ungulate a bull and in front of the bull, at the bottom to the left, a monkey facing the bull.

Discoid seal in Style II

No. A188-X661 (fig. 741). *Context*: Trench A, layer A188, Phase 3, 1.27 m a.s.l. *Material*: Marine mollusk shell, species cannot be identified. *Treatment*: None. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 5.5 mm, Disc D 19.9 mm, Suspension hole D 2.9 mm. *Disc profile*: Type 5. *Face 1*: An eight pointed leaved rosette dominates the field. Between each leaf is a dot motif. *Face 2*: Undecorated.

No. A229-X897 (fig. 742). *Context*: Trench A, layer A229, Phase 4, 0.89 m a.s.l. *Material*: Soft stone, light bluish grey (opaque). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 6 mm, Disc D 20.8 mm, Suspension hole D 2.3 mm. *Disc profile*: Type 5.

Face 1: In the top central field a rosette, on each side a pair of nude humans holding a protome with double snouts and

horns. The protomes are held above a podium with vertical lines. Behind the nude human in the far right is a serpent facing downwards. Behind the nude human in the far left a palm frond. Below the rosette three parallel wavy lines which are attached to a dot-in-circle motif situated on top of a possible jar placed on top of a rectangular podium with vertical lines. Two monkeys have their arms towards the jar. Below the podium a serpent facing left. Behind the monkey to the left a rosette.

Face 2: In the central field a rosette in a circle. Around the circle are four ungulates and a nude human. The nude human holds with one hand a hook shaped object and with the other hand an unidentified object. Between two of the ungulates a rectangular motif with vertical lines. Between the two larger ungulates is a serpent.

Seals with a circular disc and a boss, in Style III

No. E51-X209 (fig. 743). *Context:* Trench E, layer E51, Phase 7, 2.07 m a.s.l. *Material:* Chalcedony, carnelian (pink). *Treatment:* Heated. *Condition:* Complete, minor evidence of wear. *Size:* Disc H 3.9 mm, Boss H 6.1 mm, Boss D 21.9 mm, Disc D 23.2 mm, Suspension hole D 4 mm. *Disc profile:* Type 5. *Reverse:* None. Copper string in situ in suspension hole. *Obverse:* In the central field a bird facing right with both wings elevated. Above the bird a star in a crescent. The bird stands on top of a rectangular podium. The bird is flanked by two humans wearing garments with zig-zag pattern, the one to the right possibly wearing a horned head dress and the other a pigtail. They have both hands raised towards the star in a crescent. The seal is bordered by two circles with a chevron pattern.

No. C130-X372 (fig. 744). *Context:* Trench C, layer C130, Phase 8. *Material:* Chalcedony, carnelian (pink). *Treatment:* Heated. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 4 mm, Boss H 6.2 mm, Boss D 19.5 mm, Disc D 20.6 mm, Suspension hole D 2.5 mm. *Disc profile:* Type 5. *Reverse:* none. *Obverse:* Blank.

Seal fragments

No. C52-X117 (fig. 745). *Context:* Trench C, layer C52, Phase 8. *Morphology:* Circular disc with boss. *Disc profile:* not preserved. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* A small fragment of the seal is preserved. *Size:* Too fragmented. *Reverse:* 3 lines, dot-in-circles not preserved. *Obverse:* Not preserved.

No. E92-X479 (fig. 746). *Context:* Trench E, layer E92, Phase 6, 2.43-2.13 m a.s.l. *Morphology:* Circular disc (possibly with boss). *Disc profile:* too fragmented. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* A small fragment of the disc and obverse is preserved. *Size:* Disc D 26.7 mm, Suspension hole D not preserved. *Reverse:* not preserved. *Obverse:* In the left field an ungulate looking towards its back. Below the ungulate a possible jar and to the left a possible palm frond. *Style:* Possibly Style II.

Cuboid seal with a prism shaped back, in Style IAB

No. C126-X328 (fig. 747). *Context:* Trench C, layer C126, Phase 1, 1.21 m a.s.l. *Material:* Soft stone, grayish green (bluish hue). *Treatment:* Standard. *Condition:* Complete, slight evidence of wear. *Size:* Disc H 5.7 mm, Boss H 4.3 mm, Length: 16.5 mm, Width: 13.3 mm, Suspension hole D 2.9 mm. *Disc profile:* Type 1. *Reverse:* Standard, D 2.4 mm. *Obverse:* Dominating the field is a recumbent ungulate looking towards its back. A monkey with outstretched legs holds the horns of the ungulate. Behind the monkey a crescent motif.

Triangular prismatic seal in Arabian Gulf Style

No. C66-X142 (fig. 748). *Context:* Trench C, layer C66. Between c. 18 and 26 m from the southeast wall of the Temple a thin layer (C19) of oolitic limestone chips was observed right above natural soil in Trench C2 at level c. 1.75 m a.s.l. (fig. 112). Above this layer, which indicates the ground level at the time of the temple construction, was a c. 50 cm thick layer of white sand with patches of yellowish brown sand, but with only minimal horizontal stratification. This layer contained some Barbar sherds and a triangular prismatic stamp seal found at level 2.05 m a.s.l. The seal is in Arabian Gulf style, i.e. it belongs to the late 3rd millennium BC and must therefore be intrusive in this context since the white sand layer must be later than the construction of the temple, i.e. 2nd millennium BC. *Material:* Soft stone, dark green (translucent). *Treatment:* Standard. *Condition:* Complete, extensive wear around the edges. *Size:* Length 25.2 mm, Cross section 13.5 x 13 x 13 mm, Suspension hole D 4.2 mm.

Face 1: Nude human with circular head, the semi-trapezoid chest is facing forwards, and the lower body shown in profile with one leg raised and bent. One hand holding an arrow pointing downwards, the other hand raised and bent at the elbow. Because of wear it is unclear if he is holding something. The area above the head is heavily worn but two lines running from the human's head suggest horns. Resembles humans on Gulf Type seals, e.g. from Al-Hajjar (Kjærums 1999 p. 120, fig. 148) and Madinat Hamad (ibid p. 116, fig. 133). The leg posture, one knee raised, is not known from other Gulf Type seals but occurs in later Dilmun seals (cf. fig. 690).

Face 2: Oriented in the opposite direction. A bird with the head in profile facing left and with the body shown front-on with elevated wings, legs, toes and a triangular tail. Below the bird a foot.

Face 3: Oriented in the same direction as Face 1. In the lower field an ungulate with two long horns, possibly an antelope, standing on a horizontal line. The ungulate has its snout on an arrow-shaped motif pointing upwards. Above, facing right, a scorpion with a circular body, a narrow tail and two pincers; the mouth marked by two protruding lines.

Discussion: Parallels to the human figure are found in two sealings from Harappa and in particular a seal from Mohenjo Daro (Vats 1940 Pl. XCIII, figs. 305 and 310). They have a triangular prismatic shape and are made of clay and faience. Fig. 305, which is quite worn, shows on all three faces a figure which has an upper body of a human and legs of a bull. Sealing no. 310 has on one face a human with bull horns and perhaps a tail. This motif has a strong resemblance to a rectangular shaped seal from Mohenjo Daro (Marshall 1931 Pl. CXI fig. 356). The figure is standing with one arm raised and besides the bull horns, the ear and a tail suggest a bull-man rather than a deity or hero.

The foot on face 2 is common in Gulf Type seals, e.g. from Qala'at al-Bahrain (Kjærums 1994 p. 325, fig. 1731). The frontally facing bird is seen on a Gulf Type seal from the Karzakkan Cemetery in Bahrain but without the legs depicted (Laursen 2010 p. 128, fig. 111). The motif is known also from contemporary Mesopotamian cylinder seals, e.g. from Al-Maqsha (Crawford 1999 p. 91, fig. 80).

The scorpion on face 3 is strikingly similar to a Gulf Type seal from Qala'at al-Bahrain (Kjærums 1994 p. 325, fig. 1731). The arrow-shaped motif below the snout of the ungulate may have parallels on Indus seals (e.g. Kenoyer 2010 p. 50, fig. 12c).

The close parallels to Gulf Type seals in both engraving techniques and images make it quite certain that the triangular prismatic seal was locally produced in the same area. The integration of foreign iconographic elements, typically from the Indus, but also from Mesopotamia, is also evidenced among Gulf Type seals. It can thus be safely dated to the end of the 3rd millennium BC as Gulf Type seals (Laursen 2010).



Fig. 676. 1:1.



Fig. 677. 1:1.



Fig. 678. 1:1.



Fig. 679. 1:1.



Fig. 680. 1:1.



Fig. 681. 1:1.



Fig. 682. 1:1.



Fig. 683. 1:1.



Fig. 684. 1:1.



Fig. 685. 1:1.



Fig. 686. 1:1.



Fig. 687. 1:1.



Fig. 688. 1:1.



Fig. 689. 1:1.



Fig. 690. 1:1.



Fig. 691. 1:1.



Fig. 692. 1:1.



Fig. 693. 1:1.



Fig. 694. 1:1.



Fig. 695. 1:1.



Fig. 696. 1:1.



Fig. 697. 1:1.



Fig. 698. 1:1.



Fig. 699. 1:1.



Fig. 700. 1:1.



Fig. 701. 1:1.



Fig. 702. 1:1.



Fig. 703. 1:1.



Fig. 704. 1:1.



Fig. 705. 1:1.



Fig. 706. 1:1.



Fig. 707. 1:1.



Fig. 708. 1:1.



Fig. 709. 1:1.



Fig. 710. 1:1.



Fig. 711. 1:1.



Fig. 712. 1:1.



Fig. 713. 1:1.



Fig. 714. 1:1.



Fig. 715. 1:1.



Fig. 716. 1:1.



Fig. 717. 1:1.



Fig. 718. 1:1.



Fig. 719. 1:1.



Fig. 720. 1:1.



Fig. 721. 1:1.



Fig. 722. 1:1.



Fig. 723. 1:1.



Fig. 724. 1:1.



Fig. 725. 1:1.



Fig. 726. 1:1.



Fig. 727. 1:1.



Fig. 728. 1:1.



Fig. 729. 1:1.



Fig. 730. 1:1.



Fig. 731. 1:1.



Fig. 732. 1:1.



Fig. 733. 1:1.



Fig. 734. 1:1.



Fig. 735. 1:1.



Fig. 736. 1:1.



Fig. 737. 1:1.



Fig. 738. 1:1.



Fig. 739. 1:1.



Fig. 740. 1:1.



Fig. 741. 1:1.



Fig. 742. 1:1.



Fig. 743. 1:1.



Fig. 744. 1:1.

Fig. 745. 1:1.

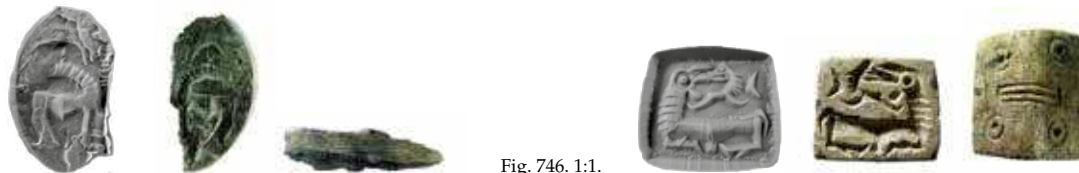


Fig. 746. 1:1.

Fig. 747. 1:1.



Fig. 748. 1:1.



Fig. 749. 1:1.

Fig. 750. 1:1.

Within the Arabian Gulf region two other seals and a seal blank of triangular prismatic shape have been found from May-sar 1 in Oman (Weisberger 1980 p. 85-86), from Al-Hajjar (Dur-ing Caspers 1983 p. 662) and from Qala'at al-Bahrain (Kjærum 1994 p. 339-340).

Cylinder seal in Style IAa

No. C218-X789 (fig. 749). *Context*: Trench C, layer C218, temple foundation trench, Phase 1, 0.96 m a.s.l. *Material*: Soft stone, dark grayish green (bluish hue). *Treatment*: Standard. *Condition*: Complete, slight evidence of wear. *Size*: H 22.6 mm, D 13.5 mm, Suspension hole D 3.7 mm.

Field: A human with a flounced garment holds with one hand one of the legs of a feline (to the left) and with the other hand a rod and ring (to the right) directed towards a deity wearing a horned headdress and a flounced garment sitting on a two-legged stool and raising a cup in one hand. Behind the deity is another human motif with a flounced garment holding the stool. Behind the feline a nude human holding with a raised arm an arrow stabbed into the feline's neck and with the other raised arm a serpent. A jar is behind and in front of the deity. The top and the bottom fields are bordered by a horizontal line.

Discussion: This seal is locally produced and decorated in Style IAa. The presentation scene and certain iconographic elements imitate Mesopotamian cylinder seals. The rod-and-ring symbol is not known from other Dilmun seals and is usually held by the enthroned (or standing) deity or ruler in Mesopotamian glyptic (e.g. Collon 1986 Pl XI, fig. 105). A couple of Dilmun stamp seals from Bahrain show a seated deity or a ruler holding a cup (e.g. Al-Sindi 1999 no. 24).

Four other cylinder seals in Dilmun style are known; one came from the previous excavations in Tell F6 (Kjærum 1983 No. 373), one from the Saar Settlement in Bahrain (Crawford 2001 No. 4741:11) and two from Susa in Iran (Peyronel 2001-2003

figs. 10-11). They are all in Style IAa with the exception of one of the Susa cylinders which is closer to the Proto-Dilmun Style with humans having more 'bulky' bodies and round heads.

Scaraboid seal

No. E247-X1032 (fig. 750). *Context*: Trench E, layer E247, Unknown. *Morphology*: Oval with prism back, and two deep grooved parallel lines running across on each side of the disc. *Material*: Soft stone, white to green in colour. *Treatment*: Heated and glazed. *Condition*: Complete, slight evidence of wear. *Size*: Disc H 4.7 mm, Boss H 3.4 mm, Length: 17.4 mm, Max width: 10.5 mm, Suspension hole D 2.5 mm. *Reverse*: Two dot-in-circles, D 2.7 mm, in the upper field, below each of the circles are two parallel lines and below them a vertical line with incised lines along it. *Obverse*: At both narrow ends are dot-in-circles, between them are two motifs which could be a schematic rendering of ungulates.

Discussion: A seal with similar features was previously found in Tell F3 also in an undated context (Kjærum 1983 No. 367). It has a deep groove around the disc and both faces are engraved with ungulates. Other parallels are found in Oman such as two seals from the 2nd millennium site Bib Bid (DA28522 and D20784) (pers. comm. H. David-Cuny 2014). DA28522 has the same decoration on the back as the seal described here and on the obverse are two possible animals shown perpendicular to each other. The engravings on the reverse of seal DA20784 is very similar to that on the obverse of the Tell F3 seal described here and the disc also has two grooved lines on the side. None of these seals can be dated precisely but they may have been influenced by scarabs imported from the Levant probably in the first half of the 2nd millennium, an example of which was found in Trench CW (see Sperveslage below p. 155. Kjærum 1983 Nos. 315-316).

3.3. A scarab

by Gunnar Sperveslage

During the 2008-2012 excavations in Tell F6 a scarab and a scaraboid shaped seal were discovered. The latter (fig. 750) originates from an unknown context and is discussed by Abu-Laban along with the Dilmun stamp seals (see above p. 154). The scarab C223-X807 (fig. 751), found in season 2012 will be discussed in this chapter.

The scarab was unearthed in Tell F6 in the south-western corner of Trench CW in layer C223 at a level of 2.32-1.85 m a.s.l. Contextual data concerning chronology and meaning are not available as the layer was disturbed.

The scarab is made of steatite and pierced longitudinally. The piece is broken along the drilling, so only half of the scarab is preserved. Its preserved length is 17 mm, the preserved width is 9 mm and the actual thickness 8 mm. The surface is much eroded and the original dark bluish-green glaze is almost completely gone with visible remains on the back and inside the incisions on the reverse and the side.

On the back of the scarab only a single line is visible that separates the prothorax from the elytra, not permitting any stylistic definition. The left side is preserved very well. The scarab is placed on a rather high base; the legs of the beetle are carefully incised. The foreleg and the mid leg are plain, the hind leg is notched.

Due to the fragmentary character of the scarab, only part of the decoration on the reverse is preserved. The reverse is framed by a single line and originally had three vertical columns of hieroglyphic signs. The top was presumably decorated by two antithetic Udjat-eyes, as the spiral on top of the right column should belong to an Udjat. It is followed by *k3*-arms ("spirit, essence"), a *w3h*-sign ("to endure") and an *nh*-sign ("life, to live"). The middle column begins with a *nfr*-sign ("good, perfect"), followed by a beetle (*hpr*, "manifestation, essence") and is closed by a *h*-sign ("appearance"). The hieroglyphic signs in the middle column might have been inspired by Egyptian royal names, which on Egyptian scarabs are often placed in the centre and often bear the *hpr*-sign, like *Mn-hpr-Rw*, the name of Thutmose III. The signs on this piece, however, do not correspond to any Egyptian royal name. On scarabs of the Middle Kingdom as well as on Canaanite Middle Kingdom imitations pseudo-

royal names are frequently attested (Hornung and Staehelin 1976 p. 51). The left column is completely lost, but might have shown an antithetic design in analogy to the right column.

Although each hieroglyphic sign has a phonetic value and a specific meaning, the decoration of the scarab does not read as a text. The signs are symbols for good luck and health, functioning as protecting and apotropaic elements for the owner of the scarab (Keel 1995 p. 167-169; Hornung and Staehelin 1976 p. 168-171). Style and arrangement of the signs point to a non-Egyptian origin of the scarab, as similar patterns are well attested on Canaanite adaptations of Egyptian scarabs during the Second Intermediate Period (Dynasties 13-15) (Ben-Tor 2007). Paired Udjat-eyes and antithetic columns are common on Canaanite scarabs, which are inspired by late Middle Kingdom pieces from Egypt (Ben-Tor 2007 p. 131, Pl. 79). Several pieces similar to C223-X807 from Tell F6 on Failaka have been found at Tell el-Ajul and other Canaanite sites in contexts dated to 1800-1500 BC.⁵ Furthermore the high base and the style of the legs with a notched hind leg are characteristic for Dynasties 12-15 (Keel 1995 p. 52-57). These observations provide a date for C223-X807 to 1800-1500 BC or MB IIB, respectively.

A dating of the scarab to the first half of the 2nd millennium BC does not necessarily imply that it was transferred to Failaka in this period. There are a few occurrences in the Levant and on the Arabian Peninsula, where scarabs have been recovered from contexts much later than their date of production (Keel 1995 p. 52-57). These pieces might either be interpreted as heirlooms or as survivals (Keel 1995 p. 262-263). On Failaka, however, another scarab and scaraboid seal have previously been found during the 1960-1963 excavations in Tell F6 (Kjærø 1983 p. 128-129, Nos.



Fig. 751. 2:1.

315–316; Sperveslage 2014 p. 139–141, Nos. 104–105). The scarab was found inside the 2nd millennium “Palace” but in a rather disturbed context, and the scaraboid seal is without any specific find information. Both are made of glazed steatite and, according to the decoration on the reverse, seem to be local productions. The scaraboid (Kjærum 1983 p. 128–129, No. 316; Sperveslage 2014 p. 139–141, No. 105) shows a pastoral scene, which often appears on Dilmun stamp seals (cf. Kjærum 1983 Nos. 234–252.). The reverse of the scarab is decorated with three incised animals, either Antilopidae or Caprinae (Kjærum 1983 p. 128–129, No. 315; Sperveslage 2014 p. 139–141, No. 104). Similar patterns again are attested on several Dilmunite seals (Kjærum 1983 Nos. 317, 320, 322). The shape of both seals in turn is clearly of Egyptian origin.

Scarab C223-X807 from Tell F6 was most probably produced in a Canaanite workshop and hence transferred to Failaka through Mesopotamian intermediaries or by direct exchange between Canaan and Dilmun. It does not indicate direct contact between Egypt and Dilmun. In the 2nd millennium BC, Egyptian scarabs were widespread in the Ancient Near East, particularly in the Levant, and were distributed along networks of trade and communication. It is, however, remarkable that as early as the 2nd millennium BC scarabs were imported to and imitated in Dilmun. The scarabs influenced the Dilmunite glyptic, as a new oval shaped scaraboid stamp seal was introduced among the common round shaped ones.⁶

3.4. Mesopotamian cylinder seals

by Holly Pittman

Fifteen cylinder seals or fragments are recorded from the excavations in Tell F6 undertaken by the Danish archaeological team between 2008 and 2012. They include four complete and two fragmentary seals of Ur III date, one of late Old Babylonian date, four from faience in Common Mitannian style, one seal of shell that is perhaps Kassite in date, two severely damaged chalcedony seals and one blank cylinder of banded agate. Most were found in primary or secondary contexts while others were found on the surface or in highly disturbed contexts. When enough of the imagery remains, they can all be identified as imports from Mesopotamia.

Ur III cylinders

Among these cylinder seals, the most significant are those of Ur III date (figs. 752-757). The four complete exemplars carry standard presentation scene (Collon 1982 p. 145-46) of a worshipper lead by a goddess toward an enthroned goddess. All of the seals, including the two fragments, carry inscriptions that record personal name and occasionally patrimony. One inscription identifies its owner as a *dubsar* (scribe), son of a *nu-banda* (foreman). In addition to the seals presented here, six additional Ur III seals, including four complete cylinders and two fragments (Potts 2010 nos. 7, 8, 9, 103, 104, 105) are published from earlier seasons on Failaka. Together these dozen cylinders indicate that there was a substantial Mesopotamian administrative presence among the earliest settlers on the island of Failaka. One cylinder, most recently published by H       David-Cuny (2012 p. 1), belonged to "Eshgina, the man of great boats, son of Ur Abba". This title, which first appears in the Early Dynastic period, identifies the owner as a high ranking mariner merchant, higher in rank than the seal of the scribe published here.

It is significant that all of the standard presentation scenes from Failaka to date are toward a goddess as the enthroned figure. In an exhaustive study of the Ur III seals from Umma, Rudolf Mayr (1997) argues that, although there are exceptions, as a rule the identity of the enthroned figure can be correlated with the rank of the *dubsar* owner, the lowest holding a contest scene seal, the next a presentation scene to a goddess, the next to an enthroned god, the next highest to a divine

king, and in the seals of the highest ranking officials the owner is shown in audience with the king (Mayr 1997, Asher-Greve and Westenholz 2013 p. 201). The fact that all of the Ur III seals from Failaka depict the presentation scene toward a goddess indicates that all of the officials were of a lower middle rank in the administrative hierarchy.

The identity of the goddess in Ur III presentation scenes is rarely indicated either by attributes or through inscription. Two of the four complete seals presented here associate the enthroned goddess with a waterfowl or Anserini. On fig. 754 the goddess is shown enthroned, placing her feet on waterfowl while on fig. 753 a waterfowl faces the goddess in the field. The waterfowl is the most common attribute associated with goddesses on Ur III seals, and it appears in other compositions as well. An exhaustive consideration of Neo-Sumerian goddesses associated with birds by Asher-Greve argues that the identity of the goddesses on Ur III seals was not exclusive, and that there existed multiple iconographies for divine entities. Regarding the association of Anserini and goddesses she concludes that "the flexible symbolism derived from the habits of Anserini is well-suited for goddesses with multifaceted character of multiple functions" (Asher-Greve and Westenholz 2013 p. 219).

Such a high concentration of Ur III seals is unknown elsewhere in the Persian Gulf, either in Bahrain, the Emirates or Oman. Together they confirm a late third millennium date for the earliest period of occupation on the island of Failaka, Phase 1 (cf. fig. 5), also suggested by ceramics and radiocarbon dating (see below Heinemeier & H       p. 243). Their presence bears witness to the fact that the Neo-Sumerian "sea-faring merchants of Ur," so well-known for the later Isin-Larsa period through Oppenheim's classic study (1954), grasped the importance of the newly emerging island landmass for their commercial enterprise. To reconstruct from the textual record, following the collapse of Akkad, the ancient city of Ur regained its prominence and became the center of political power. Located directly on the mouth of the Gulf, Ur III merchants identified a convenient foothold in the emerging dry land of Failaka, allowing them to directly engage in trade with Magan, bypassing the intermediary port of trade at Dilmun/Bahrain.

Outside of Failaka, only two examples of Ur III seals have been found in the region of the Persian Gulf. One from Thaj in Saudi Arabia, very worn, is of uncertain date (Potts 2010 p. 35). The other, from grave 27/a at Al Maqsha in Bahrain, deserves further discussion in the context of the seals from Failaka (André-Salvini 1999 p. 91). It is a large well carved cylinder of hematite which originally carried a standard presentation scene toward a god denoting the owner's middle high rank in the hierarchy. The original presentation scene included an inscription (now erased) as long as three lines, traces of which are visible in the modern impression. By the time this cylinder reached its final destination in Bahrain, it had been recut twice. During the first recutting, the original inscription was erased and was replaced by a heraldic device consisting of a spread-winged bird of prey above crossed rampant lions. That this device replaced the original inscription is apparent by the presence of a faint but clearly present line to its left. This line should not be understood as a staff held by the left facing lion (André-Salvini 1999 p. 91) but rather as a trace of the original inscription; in addition faint traces of wedges can be seen near the bird of prey. At the same time as the heraldic device was added, the filling motifs of the sun-disk in moon-crescent on staff, and the tall vase were also added. This is obvious from the fact that all of these design elements are made of thin weak lines not shared by the original carving of the goddess and the enthroned god. The second recutting involved the removal of the worshipper figure who was in turn replaced by the existing three-line inscription which identifies the seal's final owner as "Kaga, the man of great boats, son of Abbagula". Given the series of recuttings, it seems most likely that Kaga was not functioning during the rule of the Ur III kings, but rather belonged to the merchant mariners of the post Ur III period.

The dating of this seal in the post Ur III period, is relevant for our discussion of the Ur III seals from Failaka. It highlights the importance of Failaka for the marine commerce of the Ur III period to the exclusion of the earlier and later port of trade at Dilmun/Bahrain. This supports the conclusion inferred from the fact that Dilmun is rarely mentioned in the Ur III economic documents that the traders colonized Failaka and organized their commerce directly with Magan, by passing the island of Bahrain. Dilmun as the island of Bahrain becomes dominant again only after the collapse of the Neo-Sumerian dynasty, when the merchants working from Ur under the Kings of Isin and Larsa depended on Dilmun as their primary

port of trade. Ur and Guappa remained the central ports for Mesopotamia receiving huge quantities of copper, timber, precious stones, and other exotica arriving from the resource rich lands to the east until the political unrest following the death of Hammurapi when the trade from the Gulf comes to an end, replaced by copper and timber from the northwest and tin from the northeast. Failaka becomes important again to Mesopotamian traders after the middle of the 2nd millennium as testified through the large numbers of Mitanni and Kassite style seals.

Late Old Babylonian cylinder

Four Old Babylonian seals are known from previous excavations on Failaka which can be situated in the period before the end of the reign of Hammurapi (Potts 2010 p. 24-5). Apparently no imported seals of the Old Babylonian period are known from Bahrain although Potts attributes three seals, which must be local productions, to the Old Babylonian style (2010). The example presented here (fig. 758) is the only seal of late Old Babylonian period so far documented from the Gulf. Its date contemporary with middle years of Samsuiluna (c. 1730 BC) to the last old Babylonian ruler Samsuditana (c. 1600 BC) is evident from its highly schematic style of elongated figures formed through flat carving combined with the extensive and unmitigated use of the drilling. The date of these seals is confirmed through their impressions on dated tablets (Buchanan 1981 p. 366).

Common Mitanni style cylinder seals

Twenty-three seals of the Mitanni style have been found in earlier excavations on Failaka (Potts 2010) while another dozen found in Bahrain have been published (Potts 2010). All are of the Common Style, made of faience/sintered quartz. Seals of this type are known in large numbers in Mesopotamia and western Iran and are used infrequently in administrative contexts impressed on tablets or clay sealings (Kjærø 1997). The four examples described here (figs. 759-762) are typical of the style and were certainly imported.

Kassite? cylinder

Many imported seals of Kassite date have been found in the excavations at Failaka reflecting the fact that it was most probably under the control of the Kassite kings during the middle centuries of the 2nd mil-

lennium BC (Kjærøum 1983, Potts 2010). The highly worn cylinder published here (fig. 765) is tentatively associated with a Kassite date because of the striding winged quadruped in the upper field as well as the two register composition showing rows of quadrupeds moving in opposite directions. However, its worn condition and the lack of close comparisons make it difficult to be certain.

Catalogue

UR III cylinder seals

No. C125-X314 (fig. 752). From layer C125 in Trench CE, a Phase 1 context, corresponding to the Ur III period (fig. 5). Soft stone, steatite. Standard presentation scene with cuneiform inscription that has been recut (cf. Eidem p. 162). The carving on the seal is worn. From left: a bald male worshipper wears plain robe with fringe and raises his hand in prayer; a goddess wearing a flounced garment and a horned headgear having a single pair of horns grasps his wrist and approaches and raises her hand in greeting to an enthroned goddess wearing a flounced garment, extending her hand in greeting and seated on a support which has been erased. Between the two goddesses is a moon crescent above a libation vessel. The four-line inscription has been recut replacing a shorter one traces of which remain visible in the modern impression. The last line of the inscription has been arranged to accommodate the back of the seated goddess. The inscription identifies the owner as *dubsar*, a scribe, son of a *nu-banda* (an official).

No. C198-X714 (fig. 753). From layer C198 in Trench CEE, a Phase 1 context. Soft stone, very fine grained, chlorite. Standard presentation scene with a cuneiform inscription (cf. Eidem p. 162). The surface of the seal is not worn and there are no signs of recutting. The style of carving is linear with no sign of modelling. From the left: a bald worshipper wearing a long robe without obvious fringe raises his right hand in prayer; a goddess wearing a flounced garment grasps his wrist, raises her hand in greeting and approaches a goddess wearing a flounced garment, gesturing in greeting and enthroned on a temple façade. Between the goddesses is a waterfowl in profile facing right toward the enthroned goddess; a two-line cuneiform inscription.

No. C208-X755 (fig. 754). From Layer C208 in Trench CEE, the temple foundation trench, a Phase 1 context. Soft stone, chlorite. Standard presentation scene with cuneiform inscription (cf. Eidem p. 162). Surface of the stone is chipped but not worn. The carving is schematic with no indication of recutting. From the left: a bald worshipper wearing a long robe with fringe raises his hand in prayer; a goddess wearing a fringed robe arranged over one shoulder grasps his wrist, raises her hand in greeting, and approaches a goddess wearing a flounced garment extending her hand in greeting and seated on the back of a water fowl while resting her feet on the back of another water fowl. Between the two goddesses is a crescent moon. Behind the goddess is a two-line cuneiform inscription, poorly carved: Two Sumerian names, PN son of PN.

No. KM 8198 (fig. 755). Surface find from the area of the "Palace" in Tell F6 (fig. 4). Soft stone, fine grained, chlorite. Standard presentation scene with a cuneiform inscription (cf. Eidem p. 162). Seal surface is not worn, but is extensively chipped. The

style of carving is modelled and its quality is high. There are no signs of recutting. From the left: a bald worshipper wearing a long robe with fringed ends raises his hand in prayer; a goddess wearing a plain robe over one shoulder grasps his wrist, raises her hand in greeting, and approaches a goddess wearing a flounced garment, a headgear with one range of horns, extending her hand in greeting and enthroned on a temple façade throne. Between the two goddesses is a winged sun disk? (broken away) above a crescent on a standard; below a scorpion above a stand. Behind the goddess is a two-line cuneiform inscription.

No. C117-X289 (fig. 756). From layer C117 in Trench CE, a Phase 1 context. Banded white translucent/white opaque quartz. Fragment of cylinder preserving several partial rows of illegible cuneiform signs.

No. JA2-X12 (fig. 757). From layer J2 in Trench JA, a disturbed context. Banded agate. Fragment of a cylinder seal preserving a partial line of cuneiform inscription (cf. Eidem p. 163).

Late Old Babylonian cylinder seal

No. KM8325 (fig. 758). Found northeast of Temple in Tell F6 in spoils from the 1980s excavations. Banded quartz. Late Old Babylonian. Samsuiluna to Samsiditana. A presentation scene. The white stone is heavily eroded. The schematic carving based on flat planes and small drillings is typical of the post-Hammurapi period. From the left: Two worshippers wearing long robes and separated by a vertical line of eight small drillings face an enthroned figure wearing a long flounced robe and perhaps extending one hand in greeting.

Mitannian style cylinder seals

No. E20-X79 (fig. 759). From layer E20 in Trench E, a Phase 7 context corresponding to the late Kassite period (fig. 5). Beige faience cylinder broken in two pieces. From left, a partially preserved and heavily damaged human figure seated facing right and raising a hand in greeting; a male figure in *kneelauf* position with right hand raised and left hand at waist gestures toward a schematic flowering tree constructed of vertical, horizontal and diagonal lines that end in fruit denoted by drill marks. A single line defines top and bottom.

No. E69-X252 (fig. 760). From layer E69 in Trench E, a Phase 7 context. Cylinder seal of greenish faience, very worn. From the left: a horned quadruped stands facing right toward a cervid with head turned back. Above an incised border. Two vertical lines enclose marks meant to imitate cuneiform signs.

No. C139-X433 (fig. 761). From layer C139 in Trench CES, a disturbed context. Small fragment of greenish faience cylinder preserves the two front legs of confronted quadrupeds standing on a register line.

No. C139-X443 (fig. 762). From layer C139 in Trench CES, a disturbed context. Fragment of a greenish faience cylinder seal preserves illegible drilled marks.

Chalcedony cylinder seals

No. E16-X67 (fig. 763). From layer E16 in Trench E, a Phase 7 context. Chalcedony. Fragments of highly eroded cylinder seal with no discernable design.

No. C104-X211 (fig. 764). From layer C104 in Trench CE, a disturbed context. Chalcedony. A complete but highly eroded cylinder seal with no discernable design.



Fig. 752. 1:1.



Fig. 753. 1:1.



Fig. 754. 1:1.



Fig. 755. 1:1.

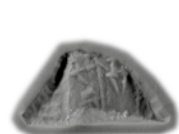


Fig. 756. 1:1.



Fig. 757. 1:1.



Fig. 758. 1:1.



Fig. 761. 1:1.



Fig. 759. 1:1.



Fig. 762. 1:1.



Fig. 760. 1:1.



Fig. 763. 1:1.



Fig. 764. 1:1.



Fig. 765. 1:1.



Fig. 766. 1:1.

Kassite? cylinder seal

C151-X485 (fig. 765). From layer C151 in Trench CES, a disturbed context. White shell cylinder heavily eroded. Two register composition: Above from left: a winged quadruped striding right toward a second quadruped; below: one or two quadrupeds striding toward the left.

Cylinder seal blank

No. C108-X457 (fig. 766). From layer C108 in Trench CE, a disturbed context. White-black-red veined agate.

3.5. Cuneiform inscriptions

by Jesper Eidem

The recent finds of cuneiform inscriptions from the 2008-2012 excavations in Tell F6 add rather significantly (c. 17%) to the corpus of cuneiform inscriptions known from Failaka, and conveniently collected by Glassner (2008), who lists a total of 53 items with inscriptions. As yet the corpus includes only a few fragmentary tablets, but will surely grow as exploration proceeds.

Seal legends

Six cylinder seals, all of Ur III - early Isin-Larsa date, found in Tell F6, carry cuneiform legends, wholly or partially preserved. The iconography and general cultural context is discussed elsewhere in this volume by Pittman (p. 157). Here it may be noted that three of the legends display anomalies: in No. 1 the inscription was recut, and an extra line was added to accommodate the title of the seal owner's father, not part of a standard legend; in No. 2 two signs are reversed (i.e. in mirror image when impressed); and finally in No. 4 the whole legend is reversed. These features strongly suggest that the legends were added to seals imported/brought from Mesopotamia *ad hoc* and probably locally.

From Mesopotamia we have some tablets with 'models' for seal inscriptions (Feingold 2014 p. 36). Clearly, the stone cutter would have needed not only a model of the inscription, but also a reversed version to guide his concrete work. Presumably the easiest way to obtain this would have been to impress the dry tablet with the model into a piece of moist clay. In some cases, however, the inscription was engraved without such preparation, and the seals produce impressions with the legend reversed. According to Collon (1993 p. 105) such examples are generally "provincial, late ..., or unskilled pieces and modern forgeries also fall into this category". Surely the provincial context may apply in the Failaka cases, not least the curious mixture on seal no. 2.

Given the relative large number of cylinder seals found on Failaka (Potts 2010 lists 66, to which may now be added 15 more!), the paucity of impressions is noteworthy, and would fit other peripheral contexts where the cylinders apparently were little used in sealing protocols, so that one suspects the actual seals may have been more important than their potential impressions.

The legible personal names in the legends are Sumerian, except that of the seal owner in no. 6, which is Akkadian.

No. 1. C125-X314 (fig. 752). Cuneiform legend in 4 lines (- or columns). The last line, which provides the title of the seal owner's father, slightly overlaps the seated goddess, and was clearly not part of the original design.

lú-^dnin-šubur
dub-sar
dumu al-ba-ni-dug₄
nu-bandà

"Lu-Nin-šubur, scribe, son of Albanidug, foreman"

No. 2. C198-X714 (fig. 753). Legend in 2 lines. The signs GÌR in l. 1 and 'EN' in l. 2 are reversed.

gìri-né-ì-sa₆
dumu lú-^dEN+ZU

"Girnesa, son of Lu-Suen"

No. 3. C208-X755 (fig. 754). Legend in 2 lines, poorly engraved.

lú-^dšeš.ki
dumu ka-ka

"Lu-Nanna, son of Kaka"

No. 4. KM 8198 (fig. 755). Legend in 2 lines. The signs are reversed, and badly written. The last signs in both lines are squeezed, and damaged, so that reading of both PNs are uncertain.

hu-bur²-ur-ra
<x> dumu ur-^dlamma?

"Hubur-urra, son of Ur-Lamma"

No. 5. C117-X289 (fig. 756). Fragment of end piece of cylinder with a few signs preserved from the end of two lines of a legend.

No. 6. JA2-X12 (fig. 757). Fragment of seal with first line of a legend preserved.

nu-ur-^d⟨x⟩

"Nur-DN,"

Stone plates

Two fragments of stone plates and one of a bowl (cf. below Hilton p. 164) carry remains of cuneiform inscriptions. In one case the well-known Dilmunite deity Inzag is involved, but otherwise the few signs preserved suggest no secure readings or reconstructions. The type of lapidary signs on the fragments was in use throughout the 2nd and 1st millennia, and thus do not help provide a firmer date for the objects. The notes here reproduce those published in Hilton *et al.* (2012 p. 416).

No. 7. C104-X210 (fig. 811)

an[?] ⟨x⟩ [.....]

This beginning of a line seems to have a vertical orientation, and may involve the name of a deity (an), while the second sign may belong to a lower line.

No. 8. E73-X265 (fig. 794)

[.....] ^din-za-[ak]

[.....] ⟨x⟩ [.....]

No. 9. E110-X471 (fig. 786)

3.6. Stone vessels

By Anna Hilton

A total of 114 fragments of soft and hard stone vessels were found during the 2008-2012 excavations in Tell F6. The sherds were distributed in Trenches A, B, E and H located northwest of the temple and Trenches C and J situated south of the temple (figs. 4 and 767).

A typological analysis of the sherds has referred them to a series of morphological types that were defined through a study of the stone vessel material uncovered during previous Danish excavations on Failaka in 1958-1963 (Hilton 2014 p. 17-41). A few new vessel types have been identified and given the new type names of 14, 72, 73, 74, 80, 81 and 82. Additionally a number of new variations of already existing types, types 4E, 10D, 18Bb and 23C, were defined.

The 114 sherds dealt with in this chapter have, if possible, been referred to as one of eight stone vessel styles that were defined in the abovementioned study (fig. 768). Three of these styles date from the mid to late 3rd millennium: the *Figurative Style*, the *Undecorated Style* and the *Umm an-Nar Style*. From the 2nd millennium come five further stone vessel styles: the *Wadi Suq Style*, the *Late Bronze Age Style*, the *Figurative Failaka Style*, the *Geometric Failaka Style* and the *Plain Failaka Style* (Hilton 2014 p. 51-162).

The typology of stone fabrics that was developed for the 1958-1963 stone vessels comprised of seven soft stone types (Types 1-7) and four harder stone types (Types 8-11) (Hilton 2014 p. 12-16, figs. 4-14) has also been referred to in this study, with the addition of a new type (Type 15), a hard stone with a distinctive red colour with white veins (fig. 769 and 786).

The 114 sherds are presented below in their find contexts, and begin with the earliest stratigraphic

Trench/Phase	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7	Phase 8	Disturbed
Trench A	–	–	–	4	1	2	1	3	–
Trench B	–	–	–	–	–	–	–	–	2
Trench E	–	–	8	10	2	1	14	9	–
Trench H	–	–	–	–	–	–	–	1	–
Trench C/J	12	–	–	–	–	–	–	–	44
Total sherds	12	–	8	14	3	3	15	13	46

Fig. 767. Distribution of 114 stone vessel sherds on trench and phase.

phase, Phase 1, dating to the Mesopotamian Ur III period and continue through Phases 3, 4, 5, 6, and 7, all dating in the 2nd millennium BC, ending with Phase 8 (cf. figs. 5-6). No sherds belonged to Phase 2.

The distribution of sherds of different styles through this sequence of 3rd-2nd millennium BC phases (fig. 768) supports the conclusions of the previous study (Hilton 2014). Most 3rd millennium BC material was found in the Ur III layers (N=11), though some of these sherds were redeposited in later layers as would be expected.

The occurrence of Umm an-Nar sherds in the 2nd millennium BC layers does not necessarily imply that Umm an-Nar vessels continued to be used in the 2nd millennium BC on Failaka, but rather that they arrived on Failaka after having fallen out of use to be reused for other purposes. This is suggested by the earliest findings of two Umm an-Nar sherds in Phase 3, which both are heavily worn and have cutting marks suggesting that they were reshaped and reused.

The later 2nd millennium styles, Wadi Suq, Late Bronze Age, Figurative, Geometric and Plain Failaka styles, occur in Phases 3-8 as expected.

It was noted that from Phase 1, and through all later phases, broken stone vessels were occasionally recarved and used for other purposes, e.g. pendants. This was also the case for the material excavated in 1958-1963 (Hilton 2014 p. 163-166).

Style/ Phase	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7	Phase 8	Disturbed
3 rd Mill. Figurative Style	4	–	–	–	1	–	–	1	1
3 rd Mill. Undecorated Style	7	–	–	2	–	–	–	1	5
3 rd Mill. Umm an-Nar Style	–	–	2	1	–	–	1	2	–
2 nd Mill. Wadi Suq Style	–	–	–	5	–	–	5	–	2
2 nd Mill. Late Bronze Age Style	–	–	–	–	–	–	–	–	1
2 nd Mill. Figurative Failaka Style	–	–	–	–	–	1	1	1	5
2 nd Mill. Geometric Style	–	–	–	–	–	–	–	–	5
2 nd Mill. Plain Style	–	–	2	–	1	1	3	3	5
Undetermined	1	–	4	6	1	1	5	5	22
Total sherds	12	–	8	14	3	3	15	13	46

Fig. 768. Distribution of 114 stone vessel sherds on style and phase.

Stone type/ Phase	Phase 1	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7	Phase 8	Disturbed
Stone type 1	4	–	–	1	1	–	–	1	2
Stone type 2	–	–	3	–	–	1	4	2	13
Stone type 3	–	–	2	5	1	1	3	1	8
Stone type 4	1	–	3	4	–	1	6	5	12
Stone type 5	–	–	–	–	–	–	–	–	–
Stone type 6	–	–	–	1	–	–	–	–	–
Stone type 7	–	–	–	–	–	–	–	–	–
Stone type 8	–	–	–	–	–	–	–	–	–
Stone type 9	6	–	–	–	–	–	–	1	5
Stone type 10	1	–	–	1	–	–	–	–	–
Stone type 11	–	–	–	–	–	–	2	2	4
Stone type 12	–	–	–	–	–	–	–	–	–
Stone type 13	–	–	–	–	–	–	–	–	–
Stone type 15	–	–	–	–	1	–	–	–	–
Undetermined	–	–	–	2	–	–	–	1	2
Total sherds	12	–	8	14	3	3	15	13	46

Fig. 769. Distribution of 114 stone vessel sherds on stone type and phase.

A few of the stone vessel fragments from the 2008–2012 excavations have previously been published, e.g. a group of unique stone plates of which some are inscribed in cuneiform (Hilton *et al.* 2012), and two figurative decorated sherds that join with sherds excavated in 1958–1963 (Hilton 2014 p. 31, 34, 106, 185, figs. 255–263).

Photos and drawings are reproduced at a scale of 1:2. Diameters on drawings are given in cm. D indicates the diameter of a given sherd (if a rim, the diameter refers to the rim diameter, if a base it refers to the base diameter and if a sidesherd the measurement refers to the maximum diameter), L indicate the maximum length of a sherd, W indicate the maximum width and Th indicate its maximum thickness.

Catalogue

Phase 1

From the layers belonging to Phase 1 or the Ur III settlement 12 sherds were found in Trenches C (N=11) and J (N=1) (fig. 5). Nine out of the 12 sherds joined with sherds from contemporary or other layers, as stated below.

The 12 sherds derive from two bowls and three vases with one indeterminate body sherd. All vessels, except possibly the body sherd, are in terms of stone types,

decoration and form typical of the 3rd millennium BC stone vessel repertoire with parallels at Bronze Age sites such as Shahdad, Shahr-I Sokhta, Susa and Tepe Yahya.

The bowls include a base from a chlorite bowl in the 3rd millennium BC Figurative Style (fig. 770) and a conical calcite bowl of the 3rd millennium BC Undecorated Style (fig. 771).

One tall and cylindrical vase of Casanova's *Série IV* (2003 p. 283–288) is represented with two fragments (figs. 772–773). The other vase is a unique elongated cylindrical shape with a small diameter that seems to belong to the 3rd millennium Undecorated Style, but no parallels were found to confirm this impression (fig. 774). A body sherd was carved secondarily into a pendant (fig. 775).

Bowls

No. C117-X273, C117-X471, C117-X472 and C117-X473 (fig. 770). *Context:* Trench C, layer C117, Phase 1. *Type:* Four joining sherds of cylindrical bowl with flat base with semi-angular base edge. Of the original decoration only two horizontal lines are preserved, cf. types 46 and 17A (Hilton 2014 p. 23, 32, 52–54, figs. 27, 28, 30). *Size:* D 160 mm, L 24 mm, W 45 mm, Th 19 mm. *Stone type:* Type 1, greenish soft stone. *Comparanda:* Susa, Iran (Miroschedji 1973 p. 53, figs. 5.7, 5.9, 5.11, 5.13); Shahdad, Iran (Hakemi 1997 p. 694, fig. Ra.1); Tepe Yahya, Iran (Lamberg-Karlovsky 1988 fig. 3.b); al-Rufay'a, Tarut, KSA (Zarins 1978 pl. 66, figs. 50, 109). *Style:* 3rd Millennium Figurative Style.

No. C207-X740 (fig. 771). *Context:* Trench C, layer 207, Phase 1. *Type:* Open plain conical bowl, everted straight sloping rim with pointed lip and flat base with rounded base edge. Type 14. *Size:* D 148 mm, L 53 mm, Th 3–10 mm. *Stone type:* Type 9, white-beige calcite, medium hard stone. *Comparanda:* Shahdad, Iran (Hakemi 1997 p. 606, fig. Fb.7); Shahr-I Sokhta, Iran (Ciarla 1979 tab. 1.a, tab. 2.a); Mundigak, Afghanistan (Potts 1986 Pl. XXVII, fig. a); Ur, Sippar, Mesopotamia (Potts 1989 p. 153, 154, figs. 1, 4). *Style:* 3rd Millennium Undecorated Style.

Vases

No. C78-X164, C83-X173 (fig. 772). *Context:* C83-X173: Trench C, layer C83, Phase 1. C78-X164: Trench C, layer C78, Disturbed. *Type:* Two joining sherds from closed plain vase, vertical rim with externally thickened lip, cf. types 39B and 71B (Hilton 2014 p. 30, 41, 62, figs. 85, 88). Horizontal striations on the inner rim surface may derive from a lid. *Size:* D 120 mm, L 110 mm, W 65 mm, Th 7.5–14 mm. *Stone type:* Type 9, white-beige calcite, medium hard stone. *Comparanda:* Ur, Kish, Nippur, Tell Brak, Tello in Mesopotamia and Susa, Shahdad, Tepe Sialk in Iran along with Mundigak IV, 3 and southern Bactria in Afghanistan (Casanova 2003 p. 283; Potts 1994 p. 228–232, tab. 6.2, fig. 1); Barbar Temple, Bahrain (Casanova 2003 p. 283). *Style:* 3rd Millennium Undecorated Style.

No. J4-X37 (fig. 773). *Context:* Trench J, layer J4, Phase 1. *Type:* Closed plain vase, vertical rim with externally thickened lip, cf. types 39B and 71B (Hilton 2014 p. 30, 41, figs. 85, 88). *Size:* D 160 mm, L 37 mm, W 38 mm, Th 8–12 mm. *Stone type:* Type 10, Grey limestone with whitish fossil corals (*Waagenophyllum*), hard stone. The other type 39B vessels found on Failaka are made of calcite, Type 9. *Comparanda:* Shahdad, Iran (Hakemi

1997 p. 610, fig. Fr.8, p. 694, figs. Ra.2, Ra.3); Shahr-I Sokhta (Ciarla 1979 tab. 1.b and 2.e). *Style*: 3rd Millennium Undecorated Style.

No. C13-X31, C125-X322, C147-X466, C215-X765 and C226-X869 (fig. 774). *Context*: C13-X31, C125-X322, C215-X765 and C226-X869: Trench C, layers C13, C125, C215 and C226, Phase 1. C147-X466: Trench C, layer C147, Disturbed. *Type*: Open plain cylindrical elongated vase, straight rim with rounded lip. Type 82. The internal cavity was hollowed with grinding slots indicated by raised ridges. On the shoulder is a secondary vertical cut. Four of the sherds join; the fifth sherd (C125-X322) does not, although it clearly derives from the same vessel. *Size*: D 58 mm, L c. 164 mm, Th 11-13 mm. *Stone type*: Type 9, non-banded beige calcite, medium hard stone. *Style*: 3rd Millennium Undecorated Style?

Other

No. C136-X407 (fig. 775). *Context*: Trench C, layer C136, Phase 1. *Type*: Plain body sherd reworked into a pendant with a suspension hole. Type 74. *Size*: L 25 mm, W 20 mm, Th 4.5-6 mm. *Stone type*: Type 4, light grey soft stone. *Comparanda*: No specific characteristics, but this stone type was used to produce Umm an-Nar and Wadi Suq vessels on the Oman Peninsula in the late 3rd and 2nd millennium BC. *Style*: ?

Phase 3

Layers from this phase in Trench E contained 8 sherds that can be grouped into three bowls, two jars and three body sherds of which two join. Phase 3 dates to Failaka Periods 1-2 (fig. 5).

They include a bowl (fig. 776) and a jar (fig. 779) of Umm an-Nar type produced and used on the Oman Peninsula in the late 3rd millennium BC (2300-2000 BC). Both vessels are quite worn and have cutting marks from reshaping and reuse.

Two bowls (figs. 777-778) appear to be of Plain Failaka Style produced during the 2nd millennium BC. These bowls have parallels to vessels uncovered by the 1958-1963 expeditions.

A tiny rim fragment stems from either an Umm an-Nar or a Wadi Suq jar (E259-X1054), but too little remains to estimate the overall vessel form. Lastly, three body sherds (E211-X962, E141-X630 and E141-X842) are too small to be assigned to any specific vessel form or style.

Bowls

No. E248-X1046 (fig. 776). *Context*: Trench E, layer E248, Phase 3. *Type*: Closed ovaloid bowl, rim with rounded lip sloping slightly inwards, decorated with dot in single circle and horizontal lines, cf. type 22A (Hilton 2014 p. 25, fig. 106). Lowest part cut secondarily, and a number of thin incised lines on the external surface. *Size*: D 150 mm, L 30 mm, W 42 mm, Th 5.5-6 mm. *Stone type*: Type 4, greyish soft stone. *Comparanda*: Al-Rufay'a, Tarut, KSA (Zarins 1978 pl. 71, no. 252); Saar, Bahrain (Killick & Moon 2005 p. 206, figs. 5.24.a-c). *Style*: Umm an-Nar Style.

No. E162-X753 (fig. 777). *Context*: Trench E, layer E162, Phase 3. *Type*: Open plain ellipsoidal bowl, vertical rim with externally thickened lip and rounded base, cf. small variation of type 1C (Hilton 2014 p. 18, figs. 324-331). *Size*: D 80 mm, L 27 mm, Th 5-9 mm. *Stone type*: Type 2, blue-greenish to greyish soft stone. *Style*: Plain Failaka Style.

No. E141-X615 (fig. 778). *Context*: Trench E, layer E141, Phase 3/Disturbed. *Type*: Open plain ellipsoid bowl, everted straight sloping rim with rounded lip. Type 10D. *Size*: D 340 mm, L 25 mm, W 39 mm, Th 8-12.5 mm. *Stone type*: Type 3, grey soft stone. *Style*: The large diameter points towards the Plain Failaka Style.

Jar

No. E307-X1312 (fig. 779). *Context*: Trench E, layer E307, Phase 3. *Type*: Closed geometrically decorated globular jar, inverted curving wall with pierced lug-handles and flat base with semi-angular base edge, decorated with dot in double circles and horizontal line, cf. types 33Aa, 43A and 61A (Hilton 2014 p. 28, 32, 37, figs. 114-115, 160-166, 168). Chisel marks on internal surface between base and wall. The pierced lug-handles were drilled from two sides, and the holes have a diameter of 4 mm. The jar is worn and the decoration and lug-handles have nearly vanished. Parts of the external surface are reworked; 5-10 mm of the rim has been cut away and polished secondarily. *Size*: D 50 mm, L 57 mm, Th 14 mm. *Stone type*: Type 4, light grey soft stone. *Comparanda*: Shimal, UAE (Häser 1991 p. 224, fig. 2.a); Fariq al-Akhrash, Tarut (Bibby 1973 p. 37, fig. 33, Zarins 1978 pl. 77, figs. 40, 594); Saar settlement, Bahrain (Killick & Moon 2005 p. 208, fig. 5.25.f). *Style*: Umm an-Nar Style.

Phase 4

From the layers of this phase a total of 14 sherds were excavated in trenches A (N=4) and E (N=10). These sherds can be grouped into eight different bowls and two undetermined body sherds. Seven sherds join with other sherds. Phase 4 dates to Failaka Periods 1-2 (fig. 5).

Two bowls belong to the 3rd Millennium Undecorated Style, one of them a bell-shaped chlorite bowl and the other a spherical calcite bowl (fig. 780). A third bowl belongs to the Umm an-Nar repertoire.

Other bowls were produced during the 2nd millennium BC and include two spouted Wadi Suq bowls (figs. 782-783) and a geometric decorated spherical bowl of the Wadi Suq, Geometric Failaka Style or Late Bronze Age repertoire (fig. 781).

A small body sherd (E319-X1350) has a simple decoration of parallel lines with some resemblance to decoration found in the 2nd millennium BC Figurative Failaka Style, but it is not likely that this style occurred as early as Phase 4. As explained elsewhere (Hilton 2014 p. 104-107) the Figurative Failaka Style has close similarities to Style III of the stamp seals, but not a single Style III stamp seal nor a single unequivocal Failaka Figurative Style sherd has been found as early

as Phase 4 though this phase is well furnished with both stamp seals and soft stone vessel fragments.

A unique 'hybrid' bowl base decorated with both figurative and geometric decoration (fig. 784) stems from the same bowl as a rim sherd found in the previous excavations (Hilton 2014 fig. 255).

Lastly, two undecorated body sherds (E147-X711 and E288-X1268) are too small to be placed within a specific style or estimate their overall shape.

Bowls

No. E300-X1280. *Context:* Trench E, layer E300, Phase 4. *Type:* A plain body sherd from bell-shaped bowl, cf. type 5A (Hilton 2014 p. 20, figs. 69-77). *Size:* D c. 200 mm, L 36 mm, W 16 mm, Th 6-8 mm. *Stone type:* Type 1, greenish soft stone with porphyroblast inclusions. *Comparanda:* Shahdad, Cemetery A, Iran (Hakemi 1997 p. 605, figs. Fa.1, Fa.3); Tepe Yahya, Iran (Lamberg-Karlovsky 1970 figs. 23.E, 23.H, 23.O); Ur, Royal tombs, Mesopotamia (Woolley 1934 p. 235, no. 51); Al-Rufay'a, Tarut, KSA (Zarins 1978 pl. 64, figs. 33, 34, pl. 65, fig. 587); Hili North, UAE (David 2011 figs. 231, 232: DLA/m90); Jiroft, Iran (DA 2003 p. 140, figs. 148c, 149c-e). *Style:* 3rd Millennium Undecorated Style.

No. E137-X594 (fig. 780). *Context:* Trench E, layer E137, Phase 4. *Type:* Closed plain spherical bowl, incurving rim with rounded lip. Type 80. *Size:* D 100 mm, L 45 mm, W 4 mm, Th 6-14 mm. *Stone type:* Type 9, white-beige calcite, medium hard stone. *Comparanda:* Shahr-I Sokhta, Iran (Ciarla 1979 tab. 1.b, tab. 2.g); Jiroft, Iran (DA 2003 p. 141, fig. 146a). *Style:* 3rd Millennium Undecorated Style.

No. A163-X366, A163-X369 and A3-X27 (fig. 781). *Context:* A163-X366 and A163-X369: Trench A, layer A163, Phase 4. A3-X27: Trench A, layer A3, Phase 8. *Type:* Three joining sherds from an open spherical bowl, vertical rim with rounded lip, decorated with dot in single circle, horizontal and vertical incised lines, cf. type 9B (Hilton 2014 p. 21, figs. 127-133). *Size:* D 130 mm, L 28 mm, W 70 mm, Th 5 mm. *Stone type:* Type 4, green to light brown soft stone. *Style:* No specific parallels were found. The shape of the bowl is common during the Wadi Suq period, whereas the arrangement and the extensive geometric decoration are more typical in the Geometric Failaka Style or the Late Bronze Age repertoire.

No. A175-X455, E137-X603, E137-X628 and E137-X637 (fig. 782). *Context:* A175-X455, E137-X603, E137-X628, and E137-X637: Trench A, layer A175 and Trench E, layer E137, Phase 4. *Type:* Four joining sherds from an open spherical bowl with open spout, vertical rim with pointed lip, decorated with dot in single circle, horizontal and oblique lines, cf. types 13B, 60E (Hilton 2013: p. 23, 36, figs. 148, 150). The external surface is sooted. *Size:* D 190 mm, L 100 mm, W 202.5 mm, Th 5-15 mm. *Stone type:* Type 3, grey soft stone. *Comparanda:* Shimal, UAE (Velde 2003 p. 107, fig. 5.11); Saar Settlement, Bahrain (Killick & Moon 2005 p. 208, pl. 5.25.d). *Style:* Wadi Suq Style.

No. E124-X522 (fig. 783). *Context:* Trench E, layer E124, Phase 4. *Type:* Round, open spout at bowl rim, three thin grooves incised on the external surface, cf. type 60E (Hilton 2014 p. 36, figs. 148-149). The spout was deliberately cut from the bowl indicating reuse. *Size:* D -, L 13 mm, W 22 mm, Th 5-8 mm. *Stone type:* Type 3, grey soft stone. *Comparanda:* Shimal, UAE (Velde 2003 p. 107,

fig. 5.11); Saar Settlement, Bahrain (Killick & Moon 2005 p. 208, pl. 5.25.d). *Style:* Wadi Suq Style.

No. E308-X1333. *Context:* Trench E, layer E308, Phase 4. *Type:* Open plain ellipsoidal small bowl, everted sloping rim with pointed lip, cf. type 19A (Hilton 2014 p. 24, fig. 100). *Size:* D c. 70 mm, L 21 mm, W 13 mm, Th 2.5-4 mm. *Stone type:* Type 4, grey soft stone. *Comparanda:* Hili North, UAE (David 1996 p. 36, fig. 5.1); Qala'at al-Bahrain, Bahrain (Højlund & Andersen 1994 p. 388, fig. 1916); Al-Rufay'a, Tarut, KSA (Zarins 1978 pl. 64, figs. 77, 90). *Style:* Umm an-Nar Style.

No. A129-X454 (fig. 784). *Context:* Trench A, layer A129, Phase 4. *Type:* Ring base with angular base edge and concentric stepped bands on lower part, decorated with a combination of carved figurative branch stem and incised geometric dot in circle, cf. type 51E (Hilton 2014 p. 34, fig. 255). A rim sherd of type 1F excavated in 1958-63 at Tell F6 (Hilton 2014 p. 19, fig. 255) belongs to the same vessel. *Size:* D 55 mm, L 68 mm, W 60 mm, Th 6.5-10 mm. *Stone type:* Type 6, reddish with black inclusions, soft stone. *Style:* This bowl is unique, in form, stone type, as well as its combination of figurative and geometric decoration.

Other

No. E319-X1350. *Context:* Trench E, layer E319, Phase 4. *Type:* Small body sherd figuratively decorated in low relief with incised parallel lines (cf. Hilton 2014: figs. 213, 234, 249). *Size:* D -, L 19 mm, W 14 mm, Th 7.5 mm. *Stone type:* Type 4, greenish-grey soft stone. *Style:* Undetermined.

Phase 5

Three sherds derive from this phase, two from Trench E; one from a 3rd Millennium Figurative Style bowl (fig. 785), and one inscribed in cuneiform in a distinctive red and white hard stone (fig. 786), which resembles a bowl excavated in the previous excavations and classified as 2nd millennium BC Plain Failaka Style. A small body sherd from Trench A (A144-X294) could not be classified. Phase 5 dates to Failaka Periods 1-2 (fig. 5).

Bowls

No. E270-X1167 (fig. 785). *Context:* Trench E, layer E270, Phase 5. *Type:* A body sherd near the base of an open cylindrical bowl with flat base with semi-angular base edge, decorated in low relief with basketry and thick zig-zag band, cf. types 46 and 17A (Hilton 2014 p. 23, 32, 52-54, figs. 27-30). *Size:* D 120 mm, L 17 mm, W 25 mm, Th 6-9 mm. *Stone type:* Type 1, greenish soft stone. *Comparanda:* Susa, Iran (Miroshedji 1973 p. 53, figs. 5.9, 5.12); Tepe Yahya, Iran (Kohl 2001 p. 213, fig. 9.3); Al-Rufay'a, Tarut (Zarins 1978 pl. 70, fig. 303); Jiroft, Iran (Pittman 2003 p. 85). *Style:* 3rd Millennium Figurative Style.

No. E110-X471 (fig. 786). *Context:* Trench E, layer E110, Phase 5. *Type:* Closed plain ellipsoidal bowl, incurved rim with flattened lip, cf. type 25B (Hilton 2014 p. 26, fig. 355). Inscribed in cuneiform (Eidem above p. 163). A comparable, inscribed vessel in soft stone was found in previous excavations (Hilton 2014 p. 48, fig. 354). *Size:* D 200 mm, L 21 mm, W 36 mm, Th 7-10 mm. *Stone type:* Type 15, red and white coloured hard stone. *Style:* Plain Failaka Style.

Phase 6

From this phase derive three body sherds; from Trench A two bowl sherds that include a geometric decorated sherd and a Figurative Failaka Style sherd (figs. 788 and 789) and from Trench E a bowl in Plain Failaka Style (fig. 787). All three sherds belong to 2nd millennium stone vessel styles. Phase 6 dates to Failaka period 3B (fig. 5).

It should be noted that fig. 789 is the oldest securely dated example of the Figurative Failaka Style found on Failaka, and that its occurrence in a period 3B context supports the dating of the Figurative Failaka Style and stamp seal Style III that has previously been suggested (Hilton 2014 p. 169-170. Højlund 2007a p. 14-15).

Bowl

No. E81-X339 (fig. 787). *Context:* Trench E, layer E81, Phase 6. *Type:* Closed plain ellipsoid bowl, inverted sloping rim with externally thickened lip, cf. type 28C (Hilton 2014 p. 27, figs. 332-333). *Size:* D 200 mm, L 27.5 mm, W 32 mm, Th 5-6 mm. *Stone type:* Type 3, light grey soft stone. *Style:* Plain Failaka Style.

Other

No. A132-X234 (fig. 788). *Context:* Trench A, layer A132, Phase 6. *Type:* Sherd from the lower part of a large bowl decorated with horizontal lines and dot in double circles. Type 73. Two mending holes with diameters between 5-3.5 mm. *Size:* D 200 mm, L 24 mm, W 38 mm, Th 7.5-9 mm. *Stone type:* Type 4, light grey soft stone. *Style:* Too little remains of the shape to refer it to either the Wadi Suq or the Geometric Failaka Style.

No. A141-X285 (fig. 789). *Context:* Trench A, layer A141, Phase 6. *Type:* Body sherd from bowl decorated in low relief, possibly an animal, likely type 67B (Hilton 2014 p. 40, figs. 249-252). Based on similarities in carving style, size and stone type it seems that the sherd came from the same bowl as figs. 815-816. *Size:* D 80 mm, L 25 mm, W 23.5 mm, Th 5-7 mm. *Stone type:* Type 2, grey soft stone. *Style:* Figurative Failaka Style.

Phase 7

From this phase 14 sherds were found in Trench E and 1 sherd in Trench A. Three of these sherds join with other sherds. Phase 7 dates to Failaka period 4A (fig. 5).

The sherds belong to a 3rd millennium Umm an-Nar vase (fig. 793), two 2nd millennium BC Wadi Suq jars (figs. 791-792), a 2nd millennium BC Plain Failaka Style bowl (fig. 790) and two plates of which one is inscribed in cuneiform (figs. 794-795).

Additionally, eight body sherds including two Wadi Suq sherds (figs. 797), a Figurative Failaka Style sherd (fig. 798) and a plain sherd recarved into a pendant (fig. 796). The remaining four body sherds were too small to be determined in respect to form and style (A32-X163, E12-X57, E12-X61 and E20-X75).

Bowl

No. E20-X76 (fig. 790). *Context:* Trench E, layer E20, Phase 7. *Type:* Closed plain ellipsoidal bowl, incurved rim with externally thickened lip, cf. type 20A (Hilton 2014 p. 25, figs. 356-360). *Size:* D 115 mm, L 26 mm, W 24.5 mm, Th 7-12 mm. *Stone type:* Type 2, greenish-grey soft stone. *Style:* Plain Failaka Style.

Jars

No. E12-X62 (fig. 791). *Context:* Trench E, layer E12, Phase 7. *Type:* Closed conical jar, inverted sloping rim with rounded lip, possibly type 33Ab (Hilton 2014 p. 28, figs. 173-178). It is difficult to establish whether or not this jar originally was ornamented with geometric decoration as it is heavily worn. *Size:* D 70 mm, L 28 mm, W 24 mm, Th 5-10 mm. *Stone type:* Type 4, light grey soft stone. *Comparanda:* Shimal, UAE (Velde 2003 p. 107, figs. 5.1, 5.2, 5.6). *Style:* Wadi Suq Style.

No. E12-X63 (fig. 792). *Context:* Trench E, layer E12, Phase 7. *Type:* Plain round base with rounded base edge from jar or bowl, cf. types 41A or 41B (Hilton 2014 p. 30-31, figs. 118-120, 169-172). *Size:* D 90 mm, L 43 mm, W 29 mm, Th 7-11 mm. *Stone type:* Type 3, light grey soft stone. *Comparanda:* Shimal, UAE (Velde 2003 p. 107, fig. 5.1); Saar settlement, Bahrain (Killick & Moon 2005 p. 208, figs. 5.25.c, 5.25.e). *Style:* Wadi Suq Style.

Vase

No. E12-X59, E8-X25 and E24-X149 (fig. 793). *Context:* E12-X59: Trench E, layer E12, Phase 7. E8-X25 and E24-X149: Trench E, layers E8 and E24, Phase 8. *Type:* Three joining sherds from an open decorated cylindrical vase with the entire surface covered in horizontal lines; an open version of type 32A (Hilton 2014 p. 28, figs. 108-109). Type 81. *Size:* D 85 mm, L 50 mm, W 60 mm, Th 2-4 mm. *Stone type:* Type 4, light grey soft stone. *Comparanda:* Hili North, UAE (David 1996 p. 36, fig. 5.14; 2011 fig. 227: DLA/m10, DLA/m7); Tarut, Al-Rufay'a, KSA (Zarins 1978 pl. 66, no. 115). *Style:* Umm an-Nar Style.

Plates

No. E73-X265 (fig. 794). *Context:* Trench E, layer E73, Phase 7. *Type:* Open plate with straight side and internally thickened lip, inscribed in cuneiform (Eidem above p. 163), cf. type 18Ab (Hilton 2014 p. 24, fig. 402. Hilton *et al.* 2013 p. 414-417, fig. 3). *Size:* D ?, L 25 mm, W 31 mm, Th 4 mm. *Stone type:* Type 11, grey-greenish hard stone. *Comparanda:* Ur, Mesopotamia (British Museum: BM 118555). *Style:* Plain Failaka Style.

No. E23-X88, E24-X157, C196-X706, E4-X14 (fig. 795). *Context:* E23-X88: Trench E, layer E23, Phase 7. E24-X157 and E4-X14: Trench E, layers E24 and E4, Phase 8. C196-X706: Trench C, layer C196, Disturbed. *Type:* Open plate with straight side and internally thickened lip, cf. type 18Ab (Hilton 2014 p. 24, fig. 402. Hilton *et al.* 2013 p. 414-416). *Size:* D 360 mm, L 27 mm, W 17 mm, Th 33 mm. *Stone type:* Type 11, grey-greenish hard stone. *Comments:* All four sherds seem to derive from the same plate based on similar shape and stone type, however just two sherds joined (E23-X88 and C196-X706). *Comparanda:* Ur, Mesopotamia (British Museum: BM 118555). *Style:* Plain Failaka Style.

Other

No. E38-X134 (fig. 796). *Context:* Trench E, layer E38, Phase 7. *Type:* Plain body sherd. Type 74, secondarily shaped into a pendant with a pierced hole. *Size:* D 60 mm, L 32 mm, W 24 mm, Th 3 mm. *Stone type:* Type 4, grey soft stone. *Style:* Plain Failaka Style?

No. E18-X80. *Context:* Trench E, layer E18, Phase 7. *Type:* Body sherd decorated with horizontal lines and dot in single circles. Type 73. *Size:* D 200 mm, L 18 mm, W 33.5 mm, Th 6-8 mm. *Stone type:* Type 2, grey-greenish soft stone. *Style:* Wadi Suq Style?

No. E23-X89, E23-X95 (fig. 797). *Context:* Trench E, layer E23, Phase 7. *Type:* Two joining body sherds decorated with dot in single circle and oblique lines. Type 73. *Size:* D 110 mm, L 29 mm, W 32 mm, Th 7.5 mm. *Stone type:* Type 2, greenish-grey soft stone. *Style:* Wadi Suq Style.

No. E42-X181 (fig. 798). *Context:* Trench E, layer E42, Phase 7. *Type:* Body sherd with figurative decoration in low relief, showing lower part of animal legs or more likely legs of a bull-man facing a pole or staff, cf. type 65B (Hilton 2014 p. 39, fig. 234). *Size:* D 200 mm, L 38 mm, W 53 mm, Th 6.5-10 mm. *Stone type:* Type 4, light grey soft stone. *Style:* Figurative Failaka Style.

Phase 8

Thirteen sherds came from Phase 8 layers and they derive from Trenches A (N=3), E (N=9) and H (N=1). Seven of the sherds join with other sherds of which five have already been presented.

In Phase 8 layers a 3rd Millennium BC Figurative Style bowl with a characteristic hut-motif was found (fig. 799) along with a 2nd millennium BC Plain Failaka Style bowl (fig. 800), a geometric decorated jar from either the late part of the Wadi Suq period or the Late Bronze Age repertoire (fig. 801) and a Figurative Failaka Style body sherd (fig. 802).

Furthermore, a calcite body sherd likely coming from a 3rd millennium BC Undecorated Style vessel (E68-X248) and three body sherds (A20-X97, E9-X30 and E17-X414) too small in size to reveal their form and style.

Bowls

No. H14-X13 (fig. 799). *Context:* Trench H, layer H14, Phase 8. *Type:* Body sherd from cylindrical bowl decorated with hut motif and bevelled squares in low relief, cf. type 63A (Hilton 2014 p. 38, figs. 38-50). *Size:* D 120 mm, L 37 mm, W 31 mm, Th 5-7 mm. *Stone type:* Type 1, green soft stone. *Comparanda:* Jiroft, Iran (Madjidzadeh 2003 p. 48-49, figs. 67, 73; Vallat 2003 p. 91, figs. 70a, 71b); Tepe Yahya, Iran (Lamberg-Karlovsky 1988 p. 74-75, fig. 1.c, f; 1970 pl. 23j, fig. 21i, pl. 23j, fig. 21g); Tarut, KSA (Zarins 1978 pl. 66, nos. 109, 273, 568). *Style:* 3rd Millennium Figurative Style.

No. E8-X26 (fig. 800). *Context:* Trench E, layer E8, Phase 8. *Type:* Closed plain ellipsoidal bowl, incurving rim with externally thickened lip, cf. type 20A (Hilton 2014 p. 25, figs. 356-360). *Size:* D 200 mm, L 9 mm, W 32 mm, Th 9 mm. *Stone type:* Type 2, greenish-grey soft stone. *Style:* Plain Failaka Style.

Jar

No. E7-X24, B-X4, B-X60 (fig. 801). *Context:* E7-X24: Trench E, layer E7, Phase 8. B-X4: Trench B, Disturbed. B-X60: Trench B, Disturbed. *Type:* Flat base with angular base edge that appears to stem from a conical jar, decorated with oblique and hori-

zontal lines, cf. types 33D and 43A (Hilton 2014 p. 28, 32, figs. 183-184, 200-201). None of the sherds join, but they appear to derive from the same jar based on similar stone type, decoration, shape and size. *Size:* D 300 mm, L c. 60 mm, W c. 47 mm, Th 12-43 mm. *Stone type:* Type 2, green soft stone. *Comparanda:* Jebel al-Buhais, UAE (Jasim 2012 p. 67, 84, 94, 108, 223-224, figs. 81, 103.3, 103.5, 115.4, 134.12, 268.1-3, 269.1). *Style:* Late Wadi Suq or Late Bronze Age Style.

Jar or Vase

No. A30-X147 (fig. 802). *Context:* Trench A, layer A30, Phase 8. *Type:* Body sherd from a closed vase or jar, decorated in low relief with panels of bulls and humans worshipping deity separated by hatched band, cf. type 42B (Hilton 2014 p. 31, figs. 256-257, 262). Previously published with five joining sherds from the same vessel excavated in 1958-1963 in Tell F6 (Hilton 2014 p. 31, 106, 185, figs. 256-263). *Size:* D c. 120 mm, L 53.5 mm, W 61 mm, Th 8-11 mm. *Stone type:* Type 4, light beige soft stone. *Comparanda:* No stone vessel parallels, but a small terracotta head (KM1492) and two Style III stamp seals (Kjærum 1983 nos. 345, 347) have similarities with the deity. *Style:* Figurative Failaka Style.

Disturbed contexts

Fortysix sherds were uncovered in disturbed contexts. They came from Trench B (N=2), Trench C (N=42) and Trench J (N=2). Seven sherds join with sherds presented above. Twentytwo sherds are too small or uncharacteristic to be determined in terms of style.

The remaining 24 sherds can be grouped in different styles. They include a 3rd millennium BC Figurative Style vase (fig. 810) and 3rd millennium BC Undecorated Style sherds, of which one sherd had been semi-worked into a pendant (fig. 822).

There are also a number of sherds of Wadi Suq (fig. 813) and Late Bronze Age styles (fig. 821). Most of the identifiable sherds belong to Figurative Failaka Style bowls and sherds (figs. 803, 815-818, 820), Geometric Failaka Style bowls and sherds (figs. 804-806, 819) and Plain Failaka Style bowls, a plate and sherds (figs. 807-808, 811 and 814).

Bowls

No. C5-X41 (fig. 803). *Context:* Trench C, layer C5, Disturbed. *Type:* Open ellipsoid bowl, everted sloping rim with externally thickened lip with centred indentation, with indications of figurative decoration in low relief. Type 4E, i.e. a figurative version of types 4A and 4B (Hilton 2014 p. 19-20, figs. 304, 336-339). *Size:* D c. 200 mm, L 42 mm, W 34 mm, Th 5.5-11 mm. *Stone type:* Type 3, light grey soft stone. *Style:* Figurative Failaka Style.

No. C152-X597 (fig. 804). *Context:* Trench C, layer C152, Disturbed. *Type:* Open spherical to ovaloid bowl, vertical rim with externally thickened lip, decorated with horizontal lines, cf. type 1E (Hilton 2014 p. 19, figs. 289-292). *Size:* D 130 mm, L 8 mm, W 18.5 mm, Th 4-5 mm. *Stone type:* Type 4, light grey soft stone. *Comparanda:* Tell F6, Failaka (Calvet & Pic 1986 p. 64-65, no. 104). *Style:* Geometric Failaka Style.

No. C139-X451 (fig. 805). *Context:* Trench C, layer C139, Disturbed. *Type:* Closed ovaloid to ellipsoid bowl, with rim slightly sloping inwards with rounded or flattened rim, decorated with dot in double circles and horizontal lines. Two to three indentations on external surface close to rim. Type 23C, a restricted version of type 11B and decorated version of 23A (Hilton 2014 p. 22, 26, figs. 309-312, 345-348). *Size:* D 150 mm, L 21 mm, W 19 mm, Th 4.5-5 mm. *Stone type:* Type 4, beige-grey soft stone. *Comparanda:* Ur, Mesopotamia (British Museum, BM 124426) (Reade & Searight 2001 p. 161-162, no. 7). *Style:* Geometric Failaka Style.

No. J13-X72 (fig. 806). *Context:* Trench J, layer J13, Disturbed. *Type:* Closed geometrically decorated ovaloid to ellipsoid bowl, with rim slightly sloping inwards with rounded or flattened rim, decorated with dot in double circles and horizontal lines. Two to three indentations on external surface close to rim. Type 23C, a restricted version of type 11B and decorated version of 23A (Hilton 2014 p. 22, 26, figs. 309-312, 345-348). *Size:* D 260 mm, L 26 mm, W 12 mm, Th 4-6 mm. *Stone type:* Type 4, grey soft stone. *Comparanda:* Ur, Mesopotamia (British Museum, BM 124426) (Reade & Searight 2001 p. 161-162, no. 7). *Style:* Geometric Failaka Style.

No. C132-X384 (fig. 807). *Context:* Trench C, layer C132, Disturbed. *Type:* Closed plain ellipsoidal bowl, incurved rim with externally thickened lip, cf. type 20A (Hilton 2014 p. 25, figs. 356-360). Fine horizontal striations on external and internal surfaces suggest finishing on a lathe. *Size:* D 120 mm, L 8 mm, W 12 mm, Th 10 mm. *Stone type:* Type 4, dark soft stone. *Style:* Plain Failaka Style.

No. C139-X435 (fig. 808). *Context:* Trench C, layer C139, Disturbed. *Type:* Open plain ellipsoidal bowl, straight sloping rim with externally thickened lip, cf. type 1C (Hilton 2014 p. 18, figs. 324-331). *Size:* D 200 mm, L 15.5 mm, W 32 mm, Th 6-7.5 mm. *Stone type:* Type 2, grey soft stone. *Style:* Plain Failaka Style.

No. C108-X593 (fig. 809). *Context:* Trench C, layer C108, Disturbed. *Type:* Closed plain cylindrical bowl, rim sloping slightly inwards with rounded rim, cf. type 24B (Hilton 2014 p. 26, figs. 422-423). *Size:* D 230 mm, L 375 mm, W 28.5 mm, Th 4-6 mm. *Stone type:* Type 4, grey soft stone. *Style:* Miscellaneous.

Vase

No. C108-X235 (fig. 810). *Context:* Trench C, layer C108, Disturbed. *Type:* Closed conical vase, outwardly sloping rim with out-curved rounded lip, with hatched decoration in low relief, possibly basketry, cf. type 37 (Hilton 2014 p. 29, fig. 55). *Size:* D 110 mm, L 16 mm, W 26 mm, Th 4-5 mm. *Stone type:* Type 1, green soft stone. *Comparanda:* Susa, Iran (Miroshedji 1973 p. 53, figs. 5.1, 5.3); Jiroft, Iran (Madjidzadeh 2003 p. 35, fig. 60). *Style:* 3rd millennium Figurative Style.

Plate

No. C104-X210 (fig. 811). *Context:* Trench C, layer C104, Disturbed. *Type:* Open plate with concave side and internally thickened lip, inscribed in cuneiform (cf. Eidem above p. 163), cf. type 18B (Hilton 2014 p. 24, figs. 403-405). Previously published together with similar plates made from hard and soft stones (Hilton *et al.* 2013 p. 412-420, fig. 2). *Size:* D c. 340 mm, L 20 mm, W 27 mm, Th 11 mm. *Stone type:* Type 11, grey-greenish hard stone. *Comparanda:* Ur, Mesopotamia (British Museum: BM 118555). *Style:* Plain Failaka Style.

Lid

No. C177-X603 (fig. 812). *Context:* Trench C, layer C177, Disturbed. *Type:* Knob of stemmed handle from circular lid, cf. type 62B (Hilton 2014 p. 37, figs. 189-190, 193). *Size:* D 13 mm, L 11 mm, W 13 mm, Th –. *Stone type:* Type 4, grey soft stone. *Comparanda:* Shimal, UAE (David 1996 p. 40, figs. 6.19); Saar Settlement, Bahrain (Killick & Moon 2005 p. 211, figs. 5.26.e-f). *Style:* Umm an-Nar or Wadi Suq Style.

Other

No. C128-X347 (fig. 813). *Context:* Trench C, layer C128, Disturbed. *Type:* Flat base with rounded base edge, decorated with horizontal lines and dot in single circles, cf. type 42A (Hilton 2014 p. 31, figs. 181-182). *Size:* D 140 mm, L 48 mm, W 27 mm, Th 5.5-8 mm. *Stone type:* Type 2, greenish to greyish soft stone. *Comparanda:* Shimal, UAE (De Cardi 1988 p. 67, fig. 12.1); Shimal, UAE (Vogt & Franke-Vogt (eds.) 1987, figs. 25.7-10). *Style:* Wadi Suq Style.

No. C196-X791 (fig. 814). *Context:* Trench C, layer C196, Disturbed. *Type:* Ring base with rounded base edge, cf. type 54B (Hilton 2014 p. 35, figs. 376-377). *Size:* D 100 mm, L 52 mm, W 32 mm, Th 5-13 mm. *Stone type:* Type 2, grey soft stone. *Style:* Plain Failaka Style.

No. C3-X3 (fig. 815). *Context:* Trench C, layer C3, Disturbed. *Type:* Body sherd decorated in low relief displaying a human face with long hair; one shoulder is showing and the arm is bent at the elbow with the hand placed near the face, possibly in a gesture of worship. Type 72. Based on carving style, size and stone type the sherd appears to belong to the same vessel as figs. 789 and 816. *Size:* D 110 mm, L 26 mm, W 19.5 mm, Th 4.5-5.5 mm. *Stone type:* Type ?, brownish-grey soft stone. *Style:* Figurative Failaka Style.

No. C5-X69 (fig. 816). *Context:* Trench C, layer C5, Disturbed. *Type:* Body sherd decorated in low relief with part of human or animal, possibly the head of a bird (vulture with crooked neck?). Type 72. Based on carving style, size and stone type the sherd appears to belong to the same vessel as figs. 789 and 815. *Size:* D c. 100 mm, L 11 mm, W 18 mm, Th 3-5.5 mm. *Stone type:* Type ?, brownish-grey soft stone. *Style:* Figurative Failaka Style.

No. C5-X72 (fig. 817). *Context:* Trench C, layer C5, Disturbed. *Type:* Body sherd decorated in low relief with caprid and possibly part of a crescent moon, cf. type 67B (Hilton 2014 p. 40, fig. 251). *Size:* D 175 mm, L 35 mm, W 30 mm, Th 16-19 mm. *Stone type:* Type 4, light grey soft stone. *Style:* Figurative Failaka Style.

No. C177-X609 (fig. 818). *Context:* Trench C, layer C177, disturbed. *Type:* Body sherd decorated in low relief with ungulate lying with folded legs above hatched band, cf. types 65B and 67B (Hilton 2014 p. 39-40, figs. 232, 251). *Size:* D c. 220 mm, L 28 mm, W 32 mm, Th 7.5-10 mm. *Stone type:* Type 2, grey soft stone. *Style:* Figurative Failaka Style.

No. C128-X338 (fig. 819). *Context:* Trench C, layer C128, Disturbed. *Type:* Body sherd decorated in low relief with bottom part of date-palm tree. Type 72. *Size:* D –, L 39.5 mm, W 18.5 mm, Th 4.5-6 mm. *Stone type:* Type ?, greenish-grey soft stone. *Style:* The sherd may belong to the 3rd millennium Figurative Style or perhaps more likely the Figurative Failaka Style.

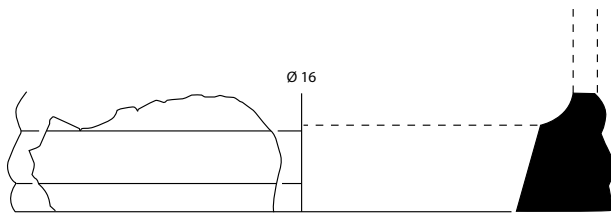


Fig. 770. 1:2.

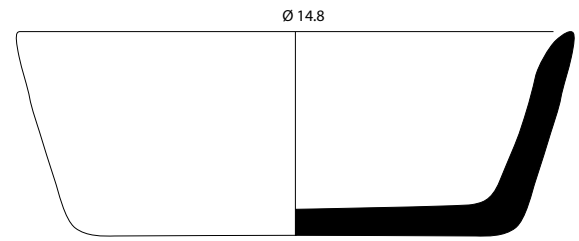


Fig. 771. 1:2.

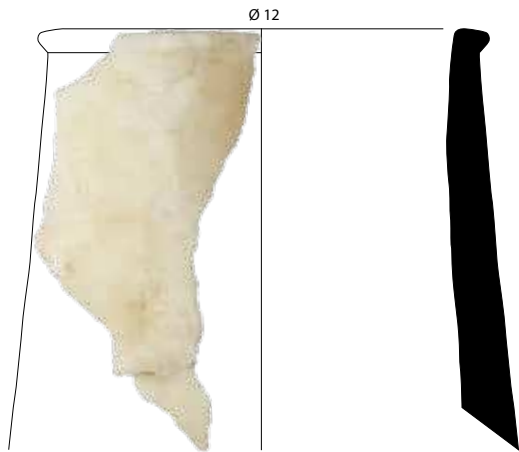


Fig. 772. 1:2.

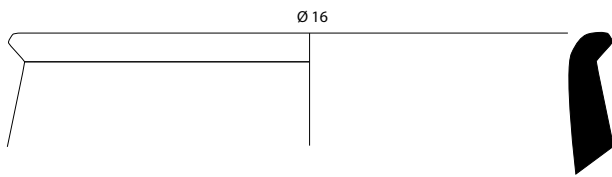


Fig. 773. 1:2.

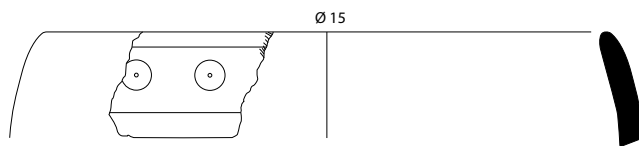


Fig. 776. 1:2.



Fig. 775. 1:2.

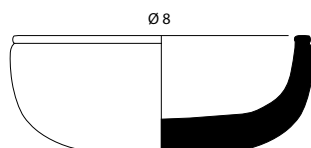
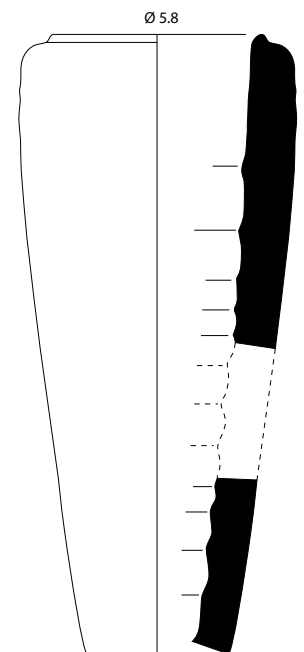


Fig. 777. 1:2.



Fig. 778. 1:2.



Fig. 774. 1:2.

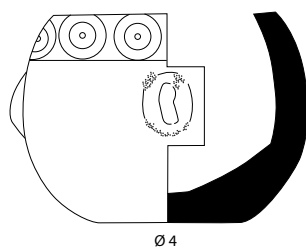


Fig. 779. 1:2.



Fig. 780. 1:2.

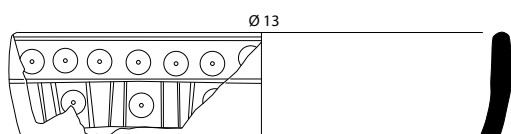


Fig. 781. 1:2.

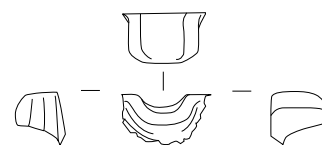


Fig. 783. 1:2.

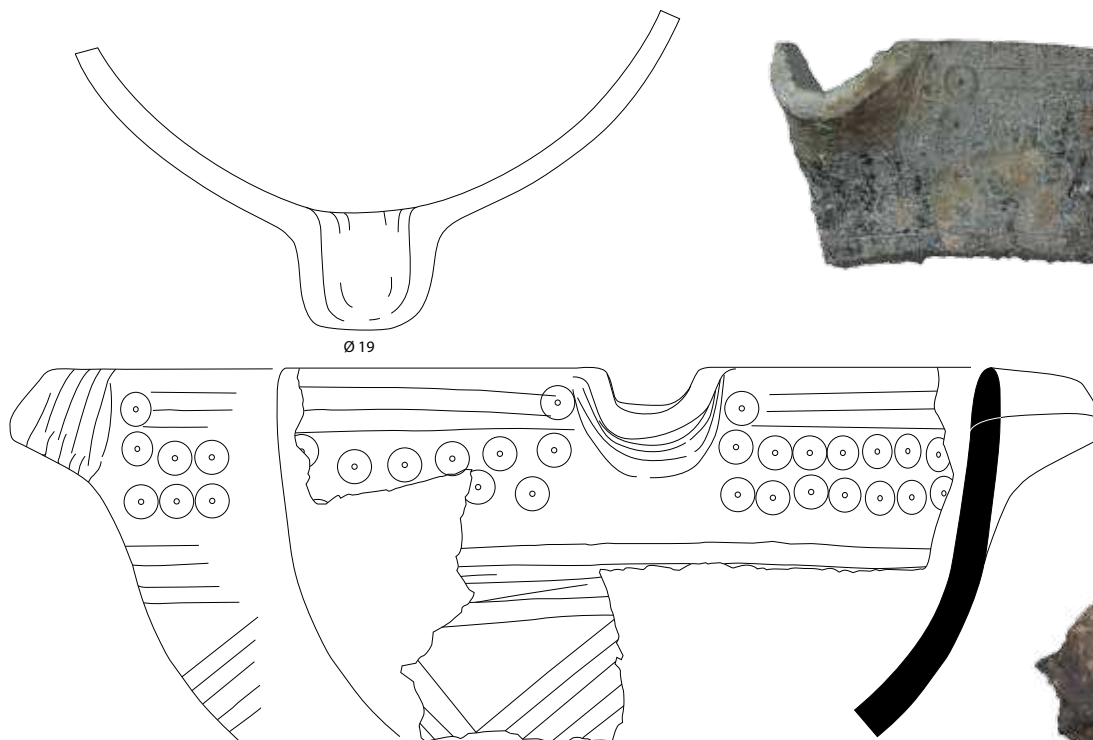


Fig. 782. 1:2.

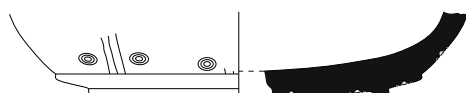


Fig. 784. 1:2.



Fig. 785. 1:2.

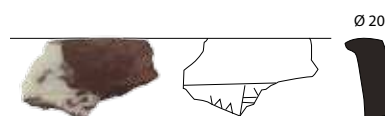


Fig. 786. 1:2.



Fig. 787. 1:2.

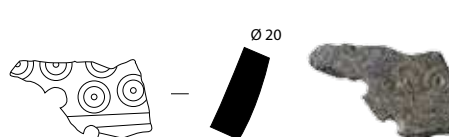


Fig. 788. 1:2.



Fig. 789. 1:2.

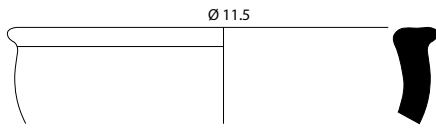


Fig. 790. 1:2.

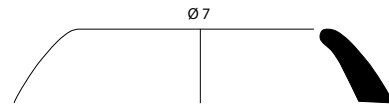


Fig. 791. 1:2.

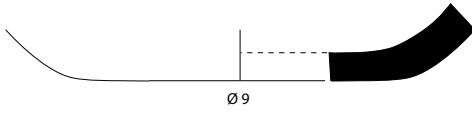


Fig. 792. 1:2.

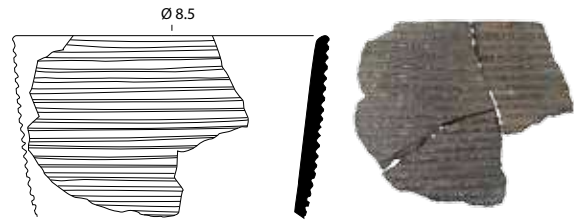


Fig. 793. 1:2.

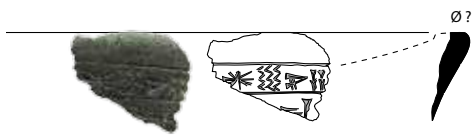


Fig. 794. 1:2.

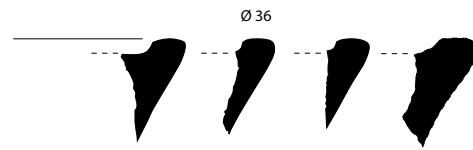


Fig. 795. 1:2.



Fig. 796. 1:2.



Fig. 797. 1:2.

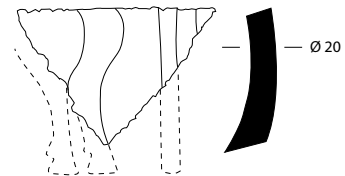


Fig. 798. 1:2.

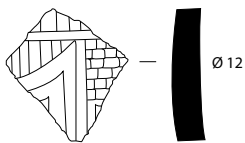


Fig. 799. 1:2.

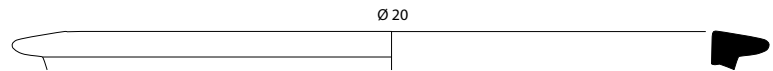


Fig. 800. 1:2.

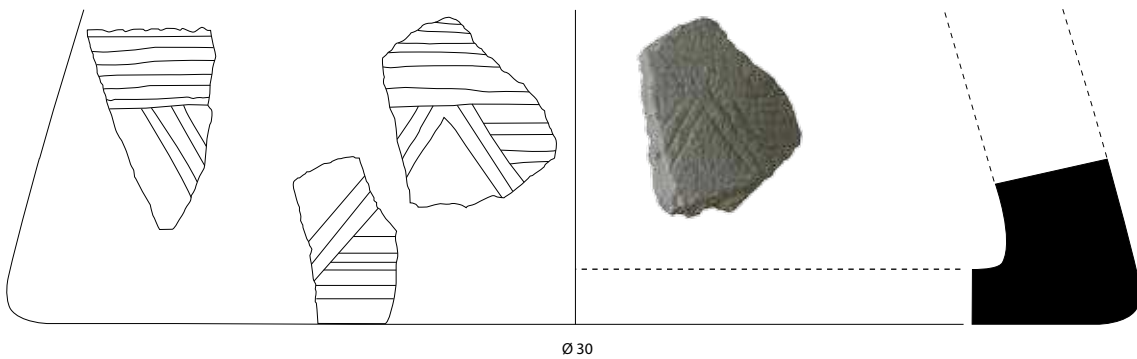


Fig. 801. 1:2.

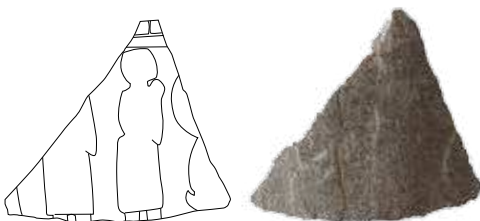


Fig. 802. 1:2.

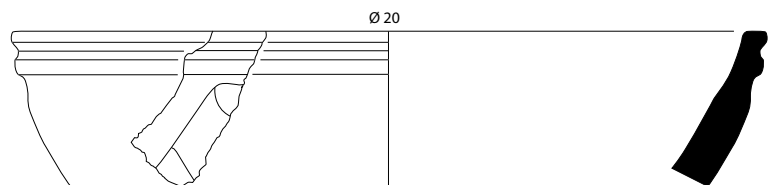


Fig. 803. 1:2.

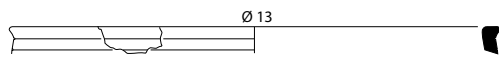


Fig. 804. 1:2.

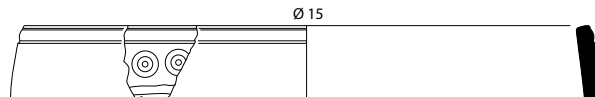


Fig. 805. 1:2.

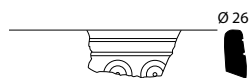


Fig. 806. 1:2.

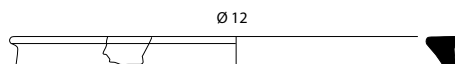


Fig. 807. 1:2.

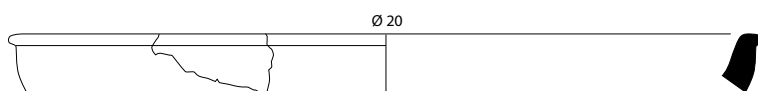


Fig. 808. 1:2.

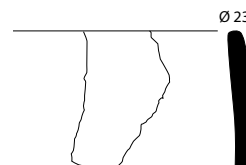


Fig. 809. 1:2.

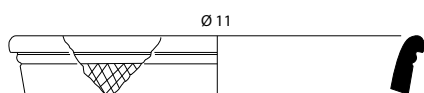


Fig. 810. 1:2.

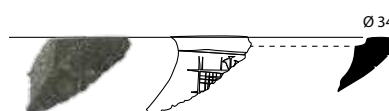


Fig. 811. 1:2.

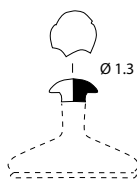


Fig. 812. 1:2.

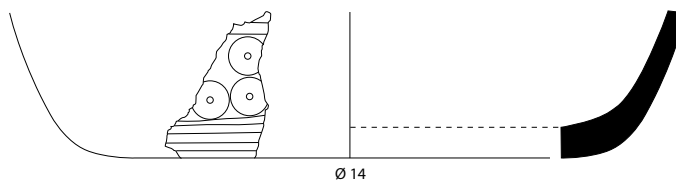


Fig. 813. 1:2.

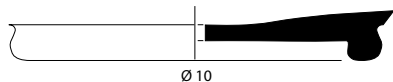


Fig. 814. 1:2.



Fig. 815. 1:2.

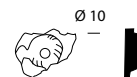


Fig. 816. 1:2.

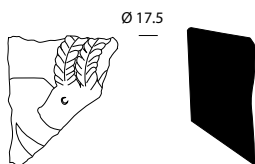


Fig. 817. 1:2.



Fig. 818. 1:2.

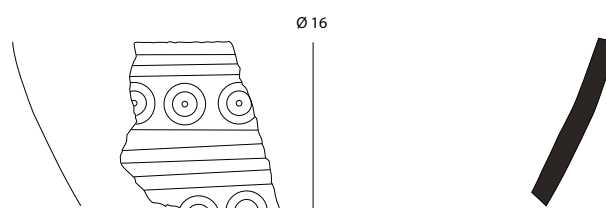


Fig. 820. 1:2.

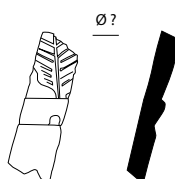


Fig. 819. 1:2.



Fig. 821. 1:2.



Fig. 822. 1:2.

No. C108-X594, C108-X595 (fig. 820). *Context:* Trench C, layer C108, Disturbed. *Type:* Two joining body sherds, possibly from an open spherical bowl, decorated with horizontal lines and dot in double circles, types 1B or 6A (Hilton 2014 p. 18, 20, figs. 286, 288, 300). *Size:* D 160 mm, L 48 mm, W 40 mm, Th 4-5.5 mm. *Stone type:* Type 3, light grey soft stone. *Style:* Geometric Failaka Style.

No. C129-X367 (fig. 821). *Context:* Trench C, layer C129, Disturbed. *Type:* Body sherd from a conical jar, decorated with dot in single circle, horizontal and vertical lines, cf. type 70 (Hilton 2014 p. 40, figs. 204-207). *Size:* D 160 mm, L 23 mm, W 28

mm, Th 5-6.5 mm. *Stone type:* Type 2, greenish-grey soft stone. *Comparanda:* Qusais, Qattara, Shimal, UAE (Velde 2003 p. 110, figs. 6.9, 6.10, 6.11). *Style:* Late Bronze Age Style.

No. CW-X484 (fig. 822). *Context:* Trench CW, Disturbed. *Type:* Plain body sherd apparently from a cylindrical vase, cf. type 71B (Hilton 2014 p. 41, fig. 88). The sherd is secondarily worked on all four edges, shaped into a semi-finished pendant. *Size:* D 140 mm, L 41 mm, W 26 mm, Th 5-10 mm. *Stone type:* Type 9, white-beige calcite, medium hard stone. *Style:* 3rd Millennium Undecorated Style.

3.7.1. Beads

by Ann Andersson

During the 2008-2012 excavations at Tell F6 on Failaka a total of 200 beads of different forms and materials were found. Of these 153 were complete and 47 fragments. When possible the broken beads have been reconstructed in illustration. The beads were recovered from the excavation Trenches A, C, D, E, G, I and J (fig. 4) with the majority from Trench C (N=136, 68%) whilst the other trenches contributed smaller numbers, e.g. Trench E with 29 beads (14.5%) and Trench A with 15 beads (7.5%).⁷

Context

Whenever possible the beads have been referred to one of the eight stratigraphic phases (figs. 5-6). The highest number of dateable beads is assigned to Phase 1, dating to the late 3rd millennium BC (62 beads, 31%); the remaining phases (2-8) dating to the 2nd millennium BC have far fewer beads or none at all (fig. 823).

Typological classification

The material has been classified to 27 general form types, based on Beck (1928 Pl. I-III) with some modifications (figs. 824 and 827). Common forms include bicones, barrels and spheres with circular cross sections of different proportions (i.e. disc, short, standard or long) (figs. 825-826). These forms make up 69% of the assemblage. Some form types are closely related to the forms with circular cross sections, e.g. barrels with elliptical cross section and a single bicone with an elliptical cross section. In total these form types form 73.5% (N=147) of the entire assemblage.

Phase/ Trench	Phase 1	Phase 3	Phase 4	Phase 5	Phase 7	Phase 8	Disturbed	Unknown	TOTAL
A	-	-	3	1	1	10	-	-	15
C	53	-	1	-	-	-	82	-	136
D	-	-	-	-	-	-	-	1	1
E	-	4	2	-	14	8	1	-	29
G	-	-	-	-	-	-	-	8	8
I	-	-	-	-	-	-	-	1	1
J	9	-	-	-	-	-	1	-	10
TOTAL	62	4	6	1	15	18	84	10	200
%	31	2	3	0.5	7.5	9	42	5	100

Fig. 823. Beads distributed on phases and trenches.

Perforation techniques

Four of Beck's perforation types (I, II, III and IV) (1928 Pl. IV) have been identified through visual examination and are noted in the catalogue below (figs. 828-829). The drill technology of a selection of the beads has been further analysed by Kenoyer (below p. 199).

A modification has been made to distinguish between the general double cone type (type I) and a specific type of perforation, which is also perforated from two sides and appears as a double cone in cross section. However, the perforation is a pecked drilling made with a stone tipped percussor, which is a different technique than the drillings made with stone and copper drills (Kenoyer 2001 p. 160. 2003 p. 16). This perforation type has here been categorized as type I.1 (a "pecked double cone") (see Kenoyer below p. 199).

Two additional terms are used: "Not preserved" includes perforations of fragmented beads, where the perforation type could not be determined, even though part of the perforation is preserved. The term "Unidentified" includes perforations of beads, where the perforation type could not be identified, even though the perforation is completely preserved.

The type I drilling method ("double cone") is not highly represented in any of the phases. It occurs in Phase 1 and in Disturbed contexts. Type I.1 ("pecked double cone") is likewise poorly represented and almost only in Phase 1. The majority of the beads have been perforated by the type II method ("drilled from both ends"), which is generally connected to barrels and bicones made from mineral materials. The type III drilling method ("single cone") drilled from one end only is more likely to occur with spherical forms, such as oblates, circulars and ellipsoids in mineral materials. The type IV perforation technique ("plain") is either a drilling with parallel walls made in mineral materials or a perforation with parallel walls made without a drill, such as for instance a perforation in beads made from artificial material.

Materials

The materials used in the bead assemblage have been divided into three primary groups; mineral (N=145 beads, 72.5%), organic (N=20, 10%) or artificially produced (N=35, 17.5%) (figs. 830-831).

Cross section/ Profile	I. Circular	II. Elliptical	III. Ovoid	IV. Lenticular	V. Plano-convex	VI. Semicircular	VII. Circle and flat	IX. Square	XIII. Hexagonal	TOTAL
1.a. Spherical	48	-	-	2	1	-	1	-	-	52
1.b. Barrel ¹	48	8	-	4	1	4	6	-	-	71
1.c. Convex cone	5	-	-	-	-	-	-	-	-	5
1.d. Convex truncated cone	2	-	-	-	-	-	-	-	-	2
1.e. Convex bicone	-	-	1	-	-	-	-	-	-	1
1.f. Truncated convex bicone	40	1	-	-	-	1	-	-	-	42
1.g. Pear-shape	-	-	-	-	-	-	-	1	-	1
2.b. Cylinder	7	-	-	-	-	-	-	2	-	9
2.f. Truncated bicone ²	2	-	-	-	-	-	-	-	-	2
4.f.b. Cylinder with two convex ends ³	2	1	-	2	-	-	-	2	1	8
TOTAL ⁴	154	10	1	8	2	5	7	5	1	193
%	77	5	0.5	4	1	2.5	3.5	2.5	0.5	96.5

Fig. 824. Beads distributed on profile and cross section types. Notes:
1 Includes a fluted barrel (C139-X369, fig. 969).
2 Includes a gadrooned truncated bicone (C139-X441, fig. 992).
3 Includes a spiral patterned cylinder with two convex ends (E59-X231, fig. 913).
4 Beads made from complete shells (n=3, 1.5%), a drop pendant (n=1, 0.5%) or the unidentified forms (n=3, 1.5%) are not included in this table.

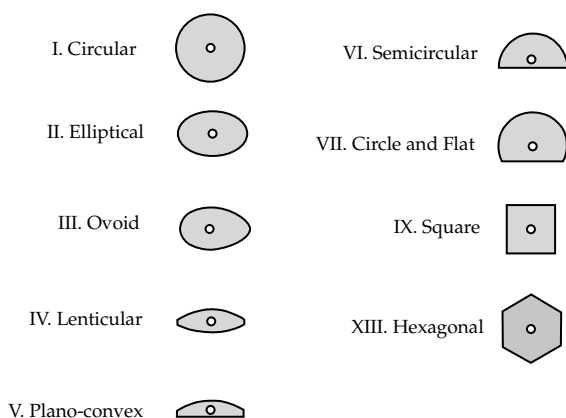


Fig. 825. Cross sections represented in the 2008-2012 Failaka assemblage (modified from Beck 1928: plate I).

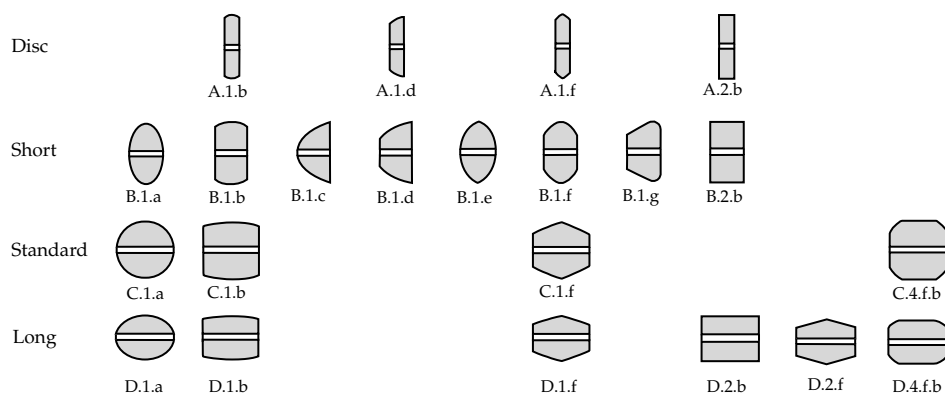


Fig. 826. Profiles represented in the 2008-2012 Failaka bead assemblage (modified after Beck 1928: plate II-III).

Form Type/ Phase	IB.1.a.	IC.1.a.	ID.1.a.	IA.1.b.	IB.1.b.	IC.1.b.	ID.1.b. ¹	IB.1.c.	IA.1.d.	IB.1.d.	IA.1.f.	IB.1.f.	IC.1.f.	ID.1.f.	IA.2.b.	IB.2.b.	ID.2.b.	IB.2.f. ²	ID.2.f.	ID.4.f.b. ³	II.C.1.b.	II.D.1.b.	II.D.1.f.	II.D.4.f.b.
Phase 1	10	2	-	3	6	1	12	1	-	1	3	10	-	3	1	1	-	-	-	1	1	-	-	1
Phase 3	-	-	-	-	1	-	-	-	-	-	-	1	1	1	-	-	-	-	-	-	-	-	-	-
Phase 4	1	-	-	-	-	-	2	-	-	-	-	-	-	-	1	-	1	-	-	-	-	-	-	-
Phase 5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-
Phase 7	4	2	1	-	-	-	2	-	-	-	-	-	-	-	-	-	-	-	-	1	-	1	-	-
Phase 8	4	-	-	-	1	-	4	1	-	-	-	1	-	-	-	-	-	-	-	-	-	1	-	-
Disturbed	19	2	1	-	-	2	12	3	1	-	1	7	3	4	-	2	1	1	-	-	-	5	1	-
Unknown	2	-	-	-	-	-	2	-	-	-	-	-	4	1	-	-	-	-	-	-	-	-	-	-
Total	40	6	2	3	8	3	34	5	1	1	4	19	8	9	2	3	2	1	1	2	1	7	1	1

Cont. below

Form Type/ Phase	III.B.1.e.	IV.C.1.a. ⁴	IV.C.1.b.	IV.D.1.b.	IV.B.4.f.b.	IV.C.4.f.b.	V.C.1.a. ⁵	V.D.1.b.	VI.B.1.b.	VI.D.1.b.	VI.C.1.f.	VII.C.1.a.	VII.D.1.b.	IX.B.1.g.	IX.D.2.b.	IX.D.4.f.b.	XIII.B.2.b.	XXI.B.2.a.	XXVII.A.I.	Unid.	Total
Phase 1	-	-	-	-	1	-	-	-	-	-	-	-	2	1	1	-	-	-	-	-	62
Phase 3	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	4
Phase 4	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	6
Phase 5	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1
Phase 7	-	-	-	1	-	-	-	-	-	-	-	1	-	-	-	-	-	-	1	1	15
Phase 8	-	-	-	-	-	-	-	-	-	-	-	-	3	-	-	1	-	-	1	1	18
Disturbed	1	2	2	1	-	1	-	1	1	3	1	-	1	-	1	1	1	1	-	1	84
Unknown	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	10
Total	1	2	2	2	1	1	1	1	1	3	1	1	6	1	2	2	1	1	3	3	200

Fig. 827. Beads Form types distributed on phases. Notes: **1** Includes one fluted bead (C139-X369, fig. 969). **2** Includes one gadrooned bead (C139-X441, fig. 992). **3** Includes one bead with spiral pattern (E59-X231, fig. 913). **4** Includes an eye bead (X8-J2, fig. 1000). **5** Includes an eye bead (X7-I1, fig. 1027).

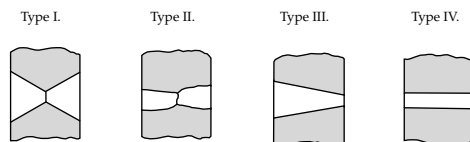


Fig. 828. Bead perforations represented in the Failaka bead assemblage (modified after Beck 1928: plate IV).

Type/ Phase	Type I	Type I.1	Type II	Type III	Type IV	Not pres.	Unid.	TOTAL
Phase 1	6	5	18	15	14	1	3	62
Phase 3	-	-	4	-	-	-	-	4
Phase 4	-	-	3	1	1	-	1	6
Phase 5	-	-	1	-	-	-	-	1
Phase 7	-	-	5	2	6	-	2	15
Phase 8	-	-	10	4	-	2	2	18
Disturbed	2	1	24	20	19	14	4	84
Unknown	-	-	7	2	-	1	-	10
TOTAL	8	6	72	44	40	18	12	200
%	4	3	36	22	20	9	6	100

Fig. 829. Bead perforation types distributed on all phases.

Mineral materials

Agate

Agate is a silicate (like chalcedony and quartz) characterized by curved bands or zones of differing colour with a vitreous to waxy luster (Bishop *et al.* 2005 p. 132). Agate is a relatively frequently occurring stone material (N= 15, 75%); one bead is identified as an agate eye bead (fig. 1000), a bead cut to resemble an eye by using the natural banding of the stone.

Chalcedony variants

Chalcedony encompasses carnelian, chrysoprase, jasper and a group simply labelled chalcedony, representing 103 beads (51.5%). Most varieties of chalcedony but not all are uniformly coloured and “... *commonly has a banded structure, which is not always obvious to the naked eye.*” (Bishop *et al.* 2005 p. 132).

Carnelian is a red to reddish brown variant of chalcedony (Bishop *et al.* 2005 p. 130) and the most commonly used material for beads at Failaka (N=94, 47%). However, much of the carnelian at Failaka appears discoloured and the surfaces are an opaque white colour with some degree of orange or red hue reflecting

Material	Form/ Material	IB.1.a.	IC.1.a.	ID.1.a.	IA.1.b.	IB.1.b.	IC.1.b.	ID.1.b.	IB.1.c.	IA.1.d.	IB.1.d.	IA.1.f.	IB.1.f.	IC.1.f.	ID.1.f.	IA.2.b	IB.2.b.	ID.2.b.	IB.2.f.	ID.2.f.	ID.4.f.b.	IC.1.b.	ID.1.b.	
Mineral	Agate	-	-	-	-	-	-	5	-	-	-	-	-	-	1	-	1	-	-	-	-	-	2	
	Chalcedony	19	4	1	-	3	2	20	2	-	-	1	19	7	6	-	-	1	-	1	-	-	2	
	Carnelian	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Chrysoprase	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Jasper	-	-	-	-	1	-	1	-	-	1	-	-	-	-	-	-	-	-	-	-	-	1	
	Chalcedony	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Lapis Lazuli	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Limestone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Porphyry	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-
	Quartz	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	
	Milky quartz	3	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Rock crystal	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Smoky quartz	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	
	Quartz	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Turquoise	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
Unid. stone	-	-	-	-	-	-	1	-	-	-	-	-	-	-	2	-	-	-	-	-	-	-	1	
Organic	Bitumen	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Bone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Ostrich eggshell	-	-	-	3	2	-	-	1	1	-	2	-	-	-	-	-	-	-	-	-	-	-	
	Shell	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Conus Ebreaus	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Cypraea	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	2	1	-	-	-	-	-	
Unid. shell	-	-	-	-	-	-	-	2	-	-	-	-	-	1	-	-	-	-	-	-	-	-		
Artificial	Heated steatite	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Faience	14	-	-	-	1	1	2	-	-	-	1	-	-	-	-	-	-	1	-	2	-	-	
	Glass	2	1	1	-	1	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	
	Gold	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	
	Paste	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	Unid. Art. Mat.	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	
	TOTAL	40	6	2	3	8	3	34	5	1	1	4	19	8	9	2	3	2	1	1	2	1	7	

Material	Form/ Material	Cont. below																						
		IID.1.f.	IID.4fb.	IIIB.1.e.	IVC.1.a.	IVC.1.b.	IVD.1.b.	IVB.4fb.	IVC.4fb.	VC.1.a.	VD.1.b.	VIB.1.b.	VID.1.b.	VIC.1.f.	VII.C.1.a.	VIED.1.b.	IXB.1.g.	IXD.2.b.	IXD.4fb.	XIIIB.2.b.	XXIIB.2.a.	XXVIIA.I.	Unid.	Total
Mineral	Agate	-	-	-	1	-	1	1	1	-	-	-	-	-	-	2	-	-	-	-	-	-	-	15
	Chalcedony																							
	Carnelian	1	-	-	-	1	-	-	-	-	-	-	-	-	1	-	-	-	1	1	-	-	1	94
	Chrysoprase	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1
	Jasper	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	4
	Chalcedony	-	-		-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	4
	Lapis Lazuli	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2
	Limestone	-	-		-	-	-	-	-	-	-	-	2	-	-	-	-	-	-	-	-	-	-	2
	Porphyry	-	-		-	-	-	-	-	-	-			-	-	-	-	-	-	-	-	-	-	1
	Quartz																							
Organic	Milky quartz	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	2
	Rock crystal	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	5
	Smoky quartz	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	1
	Quartz	-	-	-	-	-	-	-	-	-				-	-	-	-	-	-	-	-	-	-	1
	Turquoise	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2
	Unid. stone	-	1	-	-	1	-	-	-	-	1	-	1	-	-	2	-	1	-	-	-	-	-	11
	Bitumen	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	1
	Bone	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	-	-	-	-	1
	eggshell	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	9
	Ostrich																							
Artificial	Shell																							
	Conus Ebreaus	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	-	1
	Cypraea	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	2	-	2
	Unid. shell	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	6
	Heated steatite	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1
	Faience	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	22
	Glass	-	-	-	-	-	-	-	-	1	-	-	-	-	-	-	-	-	-	-	-	-	2	9
Gold	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	
Paste	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	
Unid. Art. Mat.	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	1	
TOTAL	1	1	1	2	2	2	1	1	1	1	1	3	1	1	6	1	2	2	1	1	3	3	200	

Material	Phase/ Material	Phase 1	Phase 3	Phase 4	Phase 5	Phase 7	Phase 8	Disturbed	Unknown	TOTAL
Mineral	Agate	5	-	-	-	-	2	7	1	15
	Chalcedony	26	4	4	1	4	7	41	7	94
	Carnelian	-	-	-	-	-	1	-	-	1
	Chrysoprase	-	-	-	-	2	-	2	-	4
	Jasper	4	-	-	-	-	-	-	-	4
	Chalcedony	1	-	-	-	-	-	1	-	2
	Lapis Lazuli	-	-	-	-	-	-	2	-	2
	Limestone	-	-	-	-	-	-	-	-	1
	Porphyry	-	-	-	-	-	1	-	-	1
	Quartz	-	-	-	-	-	-	2	-	2
	Milky quartz	4	-	-	-	-	-	-	1	5
	Rock crystal	1	-	-	-	-	-	-	-	1
	Smoky quartz	1	-	-	-	-	-	-	-	1
	Quartz	-	-	-	-	-	-	-	-	1
Organic	Turquoise	-	-	-	-	-	1	1	-	2
	Unid. stone	1	-	-	-	-	2	8	-	11
	Bitumen	-	-	-	-	-	-	1	-	1
	Bone	1	-	-	-	-	-	-	-	1
	Ostrich eggshell	7	-	-	-	-	-	2	-	9
	Shell	-	-	1	-	-	-	-	-	1
	Conus Ebreas	-	-	-	-	1	1	-	-	2
Artificial	Cypreaea	1	-	1	-	-	1	3	-	6
	Unid. shell	-	-	-	-	-	-	-	-	-
	Heated steatite	-	-	-	-	-	-	1	-	1
	Faience	9	-	-	-	3	-	10	-	22
	Glass	-	-	-	-	4	2	2	1	9
	Gold	-	-	-	-	-	-	1	-	1
	Paste	-	-	-	-	1	-	-	-	1
TOTAL	Unid. Art. Mat.	1	-	-	-	-	-	-	-	1
	TOTAL	62	4	6	1	15	18	84	10	200

Fig. 831. Beads materials distributed on phases.

the original colour of the stone material. Sometimes this presents itself as a thin translucent “film” on part of the bead surface, but more commonly in the Failaka assemblage the bead is completely discoloured. Brunet ascribes such discolouration of carnelian beads to a process of overheating the stone material (2009 p. 63).⁸ Overheating of the stone is likely to have happened during the manufacture of the beads as carnelian was heated in order to bring out the red colour of the material (Brunet 2009 p. 63). However, since there are fragmented beads with the fracture surfaces similarly discoloured in the Tell F6 2008-2012 assemblage, at least some of the beads must have been discoloured after or during breakage (for instance figs. 878, 936, 943 and 996). The beads from the Saar burial complex (Bahrain), from sites like Shimal and Dhayah (Ras al-Khaimah) and from the Jebel al-Buhais necropolis (Sharjah) are similarly discoloured (Brunet 2009 p. 63; Jasim 2012 fig. 9; Mughal 1983 p. 68). Most of the bead forms (spheres, barrels and bicones of different proportions) in the

2008-2012 Failaka bead assemblage made from carnelian are quite common and can be found at most sites in the Gulf where carnelian beads are recorded.

The origin for carnelian (and carnelian beads) is traditionally ascribed to the Indus region (located in parts of modern Afghanistan, India, Pakistan) since it provided a significant source of carnelian and the region has been known for the expertise of its craftsmen producing beads. This allowed them to manufacture quality products, which were in high demand. Both the craftsmen and their products may have travelled widely along the trade networks (Lankhton 2003 p. 35). However, carnelian can also be found as pebbles in rivers and in minor deposits (Moorey 1994 p. 97). In the Gulf region carnelian sources have been reported from al-Ghail in Ras al-Khaimah (UAE) and from Iran (the Bushire peninsula) (Brunet 2009 p. 60; Law 2011 p. 282; Vogt 1996 p. 112).⁹ An Indus provenience exclusively for the bead raw material or the beads cannot, therefore, be assumed.¹⁰ The source of Bushire is interesting as it is the closest source to Failaka, and Law considers the possibility of it being an “...important source for consumers in ancient Mesopotamia.” (Law 2011 p. 282). However, besides a few blanks present in the 1958-1963 bead assemblage (Andersson in prep.) no firm evidence of bead manufacture has been identified at Failaka and therefore the proximity of the Bushire source may not have been important. As mentioned above, much of the carnelian shares the feature of discolouration with southern sites in Bahrain and Oman, which might suggest a connection.

Chrysoprase. One bead has tentatively been identified as manufactured from chrysoprase. The stone material appears to have a microcrystalline structure and a deep green colour (fig. 933).¹¹ The bead is attributed to Phase 8 and might be of a later date than the Dilmun occupation at Failaka.

Jasper. Four beads are identified as jasper (N=4, 2 %). Jasper is an “...opaque chalcedony and is generally red, but yellow brown, green and grey-blue varieties occur. Jasper is rarely uniformly coloured; the colour is often distributed in spots or bands.” (Bishop et al. 2005 p. 132-133). The jasper beads are found in Phase 7 and in disturbed contexts (figs. 914-915 and figs. 967 and 1016, respectively).

Chalcedony. The microcrystalline stone materials that could not be ascribed to the aforementioned varieties have simply been identified as chalcedony (N=4, 2%). All were found in Phase 1 contexts (figs. 849, 862, 866 and 890).

Lapis lazuli

Only two beads are identified as being made from lapis lazuli (figs. 863 and 1001). Lapis lazuli is a rock with a characteristic azure blue colour (Bishop *et al.* 2005 p. 143). The two beads come from Phase 1 and a disturbed context respectively. Lapis lazuli is a good indicator of the long distance that some of the materials found at Failaka travelled since this material is only found at a few sources in Afghanistan. However, not all of the sources known today are thought to have been exploited in antiquity. Scholars refer to the deposits of the Sar-I Sang in the Badakhshan province of Afghanistan as the most convincing source (Herrmann 1968 p. 28; Moorey 1994 p. 86; Law 2011 p. 531). The scarcity of lapis lazuli at Failaka might reflect the decline in contact and maritime trade between Mesopotamia and the Indus region, which occurred around 1900 BC (Moorey 1994 p. 87; Lankton 2003 p. 39). The lapis lazuli bead from a secure context in Phase 1 can be ascribed to a period before the decline in contact between the two regions. The bead fig. 1001 from a disturbed context has parallels at the Saar burials in Bahrain (Ibrahim 1982 p. 84 and 218, fig. 1; Crawford 1999 p. 97, fig. 97).

Limestone

Two beads are tentatively identified as limestone (N=2). They both come from disturbed contexts (figs. 1008-1009) and are manufactured from an opaque stone in dusty pink and orange hues with slight blurry grey lines.

Porphyry

One bead has been tentatively identified as made out of porphyry. The stone has a characteristic opaque dark red colour with a purple hue and a pattern of closely spaced white to beige spots (fig. 926). The material is unique and was found in a Phase 8 context. Porphyry does not appear to have been widely used for ornaments in the Dilmun cultural sphere as other sites have not reported this material in their bead assemblages. Purple porphyry is reported to originate only in Egypt (at the Mons Porphyrites/ *Gebel Dokhan* in the Eastern Desert), but usually the mining of the material is connected with the Roman period as the only two quarries discovered so far are dated to this period.¹² The use of the stone is attested in Egypt in earlier periods (Predynastic and Early Dynastic periods) but the quarries supplying this material remain unknown (Aston, Harrell and Shaw 2000 p. 48). The bead was found in topsoil and could be of a date later than the Dilmun occupation at Failaka.

Quartz variants

Nine beads have been identified as made from quartz (4.5%), either milky quartz, smoky quartz or rock crystal.

Milky Quartz is white and translucent (Bishop *et al.* 2005 p. 130). Two beads of this material were recovered from disturbed contexts (figs. 997 and 1011).

Smoky Quartz is smoky brown to almost black (Bishop *et al.* 2005 p. 130). Only one example from a Phase 1 context in Trench J has been identified (fig. 889).

Rock Crystal is colourless and transparent (Bishop *et al.* 2005 p. 130). Five beads made from rock crystal have been recovered (figs. 835-837, 891 and 1020; 2.5%); all except one come from Phase 1 layers in Trench C. The fifth bead (fig. 1020) was recovered during investigations of structures in the central room of the "Palace". Sources of rock crystal are known in a number of different regions including India, Iraq (near Eridu), Iran, Turkey and Cyprus (Moorey 1994 p. 95). In the Gulf rock crystal beads have been found in the graves at Saar, Bahrain (Ibrahim 1982 p. 36 and 84, fig. 6) and in the 1958-1963 Tell F3 material from Failaka (Andersson in prep.).

Quartz. The macrocrystalline stone materials that could not be identified as to the more specific preceding varieties have simply been identified as quartz. The stone material has individual crystals that can be seen by the naked eye. Only one bead was simply identified as quartz (fig. 886).

Turquoise

Two beads have been identified as made from turquoise (figs. 921 and 999; N=2). They were both found in disturbed contexts. Sources of turquoise are present in the Sinai Peninsula, in Iran (Nishapur and Damgham mines in northeastern Iran), Afghanistan and the Kyzul Kum desert in Uzbekistan (Moorey 1994 p. 101-103, Law 2011 p. 90).¹³ The rarity of the material at Failaka could indicate that contacts with regions with access to turquoise sources might not have been very strong and perhaps only secondary in nature. The trade in the mineral material might have been outside the sphere of the Gulf trading network and therefore not reaching Failaka (Moorey 1994 p. 103).

Unidentified stone materials

A portion of the stone material remains unidentified (N=11, 5.5% of the assemblage). This material consists of a range of stone types, which have been found in

Phase 1 (N=1), Phase 8 (N=2) and disturbed contexts (N=8). Although made from an unidentified stone material, C223-X810 (fig. 1003) from disturbed contexts has the same form as agate beads found in a Sar el-Jisr burial at Bahrain (Lombard 1999 p. 71, fig. 65; Ibrahim 1982 p. 85 and 218, fig. 3).

Organic materials

Bitumen

One bitumen object interpreted as a bead has been identified (fig. 1007; 0.5%). Although bitumen was a common material used for a range of objects, it does not appear to have been a desired material for ornaments at Failaka. At the Saar settlement (and within the Saar temple) at Bahrain, some bitumen objects have been suggested as large “bitumen beads” (c. 2-2.5 cm in diameter), but these have also alternatively been suggested as floats for fishing nets or spindle whorls (Crawford, Killick and Moon 1997 p. 63, fig. 51; Killick and Moon 2005 p. 181, 193-195, fig. a-h; Moon 2000 p. 65). The examples from Saar are cylindrical or spherical with large perforations, while the example from Failaka is a short barrel with a semi circular cross section and a relatively small perforation (c. 1-1.3 mm). Bitumen beads covered with gold foil have been reported at sites in Mesopotamia, such as Ur and Abu Salabikh (Killick and Moon 2005 p. 193, Postgate and Moon 1982 p. 134, pl. Vc), but in general plain bitumen beads do not appear to have been favoured, and there is no trace of gold foil on the Failaka bitumen bead.

Bone

Only one bead is made from bone (fig. 892; 0.5%). The bone has been worked to a degree that obscures any diagnostic features and it has not been possible to determine the species of animal from which it derived. The bead was found in a Phase 1 context.

Ostrich eggshell

Nine beads are made from ostrich eggshell (figs. 844-846, 850-851, 865, 868, 973-974, 4.5%). They are very homogeneous in form, all of them being thin disc beads with circular cross sections. Only one bead stands out being larger in diameter and having a larger perforation (fig. 868). The majority have been found in Phase 1 layers (N=7). The material was identified by comparison to fragments of ostrich eggshell found at Tell F6 in Trenches A, C and E (cf. figs. 1130-1133), which might indicate the presence of raw materials or of formerly complete ostrich eggs at Failaka.

The ostrich was widespread in many regions in the Near East and Africa, but was probably not present at Failaka unless brought there by the people who settled the island (Moorey 1994 p. 127-128; Lombard 1999 nos. 56-57).¹⁴ It is far more likely that the ostrich eggshell was brought to the island either as raw material for beads, as complete objects later recycled into beads or that the finished beads were brought there. Complete ostrich eggs with their tops cut off and used for vessels were favoured as grave goods in the Dilmun period (Ibrahim 1982 p. 35, plate 44; Højlund 2007; Lombard 1999 p. 70, figs. 56 and 57). Ostrich eggshell beads are not reported from many sites¹⁵ but possible parallels of similar size and shape come from two burials in Qarn al-Harf, Ras al-Khaimah. However, these objects show extensive use wear and may have been sewn onto clothing, rather than strung. None of the ostrich eggshell beads from Failaka show this kind of wear. The Qarn al-Harf beads date to the late 3rd millennium or the early 2nd millennium (pers. comm. Hilton 2015; Hilton 2013 p. 15, figs. 30 and 31).

Shell

Few beads are made from complete shells, which have only been slightly modified, or from worked shell material (N=9, 4.5%). Some of the shell types have been identified (e.g. the *Conus Ebraeus* and *Cypraea*), while some could not be fully identified as the working of the shell has removed most diagnostic features.

Conus Ebraeus. The worked black-and-white cone bead (fig. 902) from Phase 4 has parallels in Bahrain (Killick and Moon 2005 p. 177, fig. 5.7q). At the Saar settlement such shells are considered clear evidence of imported goods (from Oman), as the *Conus Ebraeus* do not occur north of the Musandam Peninsula (Killick and Moon 2005 p. 180). Additional parallels come from grave contexts in Bahrain and these shells are also considered evidence of transit trade (Lombard 1999 p. 71, fig. 64; Mughal 1983 Plate LIV). The example from Failaka is thus an indicator of contacts with regions beyond the Musandam Peninsula.

Cypraea sp. One complete and one fragmented worked cowrie shell (*Cypraea* sp.) were found in Phase 7 (fig. 917) and Phase 8 contexts (fig. 934) (N=2). The cowrie shells were likely ornamental items, strung or sewn onto fabrics in a manner that would display the natural aperture of the shell. The *Cypraea* have been identified as two different species, i.e. *Monetaria caputserpentis* (fig. 934) and *Erosaria turdus* (fig. 917) (pers. comm. Abu-Laban and Mienis 2015).¹⁶ Similarly worked cowrie

shells have not been reported from Al-Khidr, Building G3, or the Tell F6 Temple.¹⁷ At the Saar settlement three worked cowrie shells have been recovered, but the perforations in these shells are smaller than the examples from Failaka (Killick and Moon 2005 p. 180 and fig. 5.7 i-j). One was found at Qala'at al-Bahrain (Højlund & Andersen 1994 p. 414, fig. 2067). Similarly worked cowries are also reported from Wadi Suq period burials in Sharjah (Jasim 2012 figs. 15.14-15 and 237.4). The use of cowries as adornments in the Near East is well documented (Golani 2014).

Unidentified shell

Shell rings and apices. The assemblage contains one shell ring (fig. 883) and three shell apices (figs. 900, 927 and 972) (N=4, 2%). They have lost most of their diagnostic features after being worked and it is not possible to identify the exact species of the shell, although they likely belong to the *Conus* family, which is found naturally at Failaka (pers. comm. Abu-Laban and Mienis 2015). They were found in Phase 1 (fig. 883), Phase 4 (fig. 900), Phase 8 (fig. 927) and disturbed contexts (fig. 972). Shell rings are found at Al-Khidr (Benedikova 2010 p. 115, fig. 107c-f) and at the Saar settlement (Killick and Moon 2005 p. 180, fig. 5.7s),¹⁸ at Qala'at al-Bahrain (Højlund & Andersen 1994 p. 413-414, figs. 2064, 2067 and 2068), and in grave contexts on Bahrain (Ibrahim 1982 pl. 54:9). Worked shell rings are not restricted to the Gulf and can be found in archaeological contexts in regions such as the Aegean, Syria, Cyprus, Mesopotamia and Iran. They could have been worn as parts of belts (Cluzan 2008 p. 327, Frifelt 1991 p. 184).¹⁹ The shell apices may have been used as ornaments, but might also have been by-products of shell ring and/or shell seal manufacture. The apices are found in other settlement contexts at Al-Khidr (Benedikova 2010 p. 115, fig. 107a-b), Saar (Killick and Moon 2005 p. 180, fig. 5.11p and q) and Qala'at al-Bahrain (Højlund & Andersen 1994 p. 414-415, figs. 2065, 2068).

Shell beads in other forms. A small quantity (N=2, 1%) of ornaments consists of 'regular' bead forms cut out of unidentified shell (Beck 1928 p. 29). They have likewise been worked to a degree that has removed diagnostic features of the shell (figs. 984 and 990). They were recovered from Phase 1 and disturbed contexts.

Artificial materials

Heated steatite

One bead has been identified as heated steatite (enstatite).²⁰ It is a long barrel with flutes cut vertically along

the axis and was found in a disturbed context (fig. 969). This is a unique form in the assemblage but it has a parallel from the French excavations of the Tell F6 Temple at Failaka which was a bead described as faience (Calvet and Pic 1986 p. 70-71, fig. 28, no. 131). The bead was attributed to level Va dated to the early centuries of the 2nd millennium (Calvet and Pic 1986 p. 24).

Glass

The glass beads excavated at Failaka are generally poorly preserved due to the saline environment of the soil. This makes it difficult to assess their original appearance. Glass beads are relatively few in number (N=9, 4.5%) and date to Phase 7 (figs. 906, 909-910) and Phase 8 (fig. 922) only. This is as expected as glass technology developed into a widespread phenomenon around 1700-1500 BC (Lankhton 2003 p. 39; Moorey 1994 p. 192). Few parallels can be found for the glass beads but fig. 1027 found in a context of unknown date, is similar to glass eye beads from New Kingdom Egypt (Lankhton 2003 p. 43, fig. 4.5, in particular the central bead).

Faience

A number of faience beads (N=22; 11%) were recovered from Phase 1 (N=9, 4.5%), Phase 7 (N=3, 1.5%) and disturbed contexts (N=10, 5%). Faience is generally poorly preserved. No faience beads were found at Al-Khidr, but a small number were found in the Tell F6 Temple (N=19) (Calvet and Pic 1986 p. 66-71, figs. 115, 120, 131, 134 and 140). A few examples (N=3) were also found at the Saar settlement in Bahrain (Killick and Moon 2005 p. 186, fig. 5.11.l-m), and in period Ib, IIa and IIc contexts (N=6) at Qala'at al-Bahrain (Højlund & Andersen 1994 p. 391, figs. 1944-1948, 1950, 1962). Two types of oblates with circular cross sections (I.B.1.a.) stand out in the 2008-2012 faience material. Four similar beads come from Phase 7 (fig. 905) and from disturbed contexts (figs. 944-946). These are mint green in colour and all have a diameter of approx. 2.30-3.00 mm. A somewhat similar, but slightly larger type (diameter of approx. 3.70-4.60 mm) was found in disturbed contexts (figs. 948-952) and Phase 1 contexts (figs. 839-841 and 852). However, these are whitish beige and greyish black in colour. A unique example of a short truncated gadrooned bicone made of faience (fig. 992 from disturbed contexts) has a parallel in the Tell F6 Temple bead assemblage (Calvet and Pic 1986 p. 70-71, fig. 28, no. 134).²¹

Gold

A single small gold bead was found in a disturbed context (fig. 991). Generally, gold survives well in the

archaeological record, but as a valuable metal, the material is prone to recycling. The small amount of gold recovered from Failaka might reflect that this material was carefully conserved and/or reused. Alternatively, gold may have been a material, which was only present at Failaka in very small amounts. Gold has not been identified at the other sites of Failaka, but a piece of gold sheet was found in disturbed layers of Trench C at Tell F6 (fig. 1135).²²

Paste

One bead (fig. 907) has been classified as a paste bead in a bright turquoise colour. The material is delicate in nature and has a worn surface. The bead was found in a Phase 7 context. A blue paste bead found in the Saar settlement is suggested as paste made of ground lapis lazuli (Killick and Moon 2005 p. 186, fig. 5.11n). The bead from Tell F6 may be of a comparable material. Alternatively, the bead may find parallels in the Tell F6 Temple bead assemblage from the French 1984-1985 excavations. These have a slightly smaller diameter of 3.80 mm (compared to 4.90-5.00 mm of fig. 907) and are described as globular blue faience beads (Calvet and Pic 1986 p. 70-71, fig. 28, no. 131).

Unidentified artificial material

The material of one bead has not been identified (fig. 843) although faience is the most probable material.

Discussion

The number of beads found in excavations in Tell F6 seems surprisingly large. To the 200 beads from the 2008-2012 excavations presented above may be added c. 350 beads found during the 1960-1963 excavations of the "Palace" (Andersson in prep.). Additionally 51 beads were recovered during the 1984-85 French excavations of the Temple in Tell F6 (Calvet and Pic 1986 p. 66-71),²³ as well as ten beads in different materials (agate, carnelian, lapis lazuli, unidentified stone and bronze) from the American excavations at Tell F6 (Howard-Carter 1984). This makes the number of beads (c. 611) found at Tell F6 significantly larger than any other settlement site at Failaka and Bahrain.

Also Tell F3, which was excavated by Danish archaeologists from 1958-1963, produced a significant amount of beads, c. 170 (Andersson in prep.).

For comparison, the site of Al-Khidr at the north coast of Failaka produced just five beads (a spherical bronze bead and four spherical soft stone beads), one pendant, three semi products, two shell rings, one shell apex and two perforated shell ornaments (Benediková 2010 figs. 67f, 90a, 104d and f, 106a-b and c,

and 107a-f). Spherical bronze and soft stone beads are not found in the Tell F6 2008-2012 assemblage.

On Bahrain the Barbar temple excavations only uncovered ten beads and a shell disc (Andersen and Højlund 2003 p. 316-317) and the settlement at Qala'at al-Bahrain (periods Ib-IIIb) dating to about the same period as the occupation at Failaka yielded 30 beads (Højlund & Andersen 1994 p. 391-392, figs. 1941-1966; Højlund & Andersen 1997 p. 36, 73, figs. 95, 96 and 301).

Even though Saar is by far the most extensively uncovered Dilmun settlement, the excavations only produced 104 beads (Killick and Moon 2005 p. 181-186). The quality of the beads also differs from the Failaka assemblage as a large percentage of the Saar beads are made from clay and are interpreted as local products imitating stone beads (N=47, c. 45%) (Killick and Moon 2005 p. 181). Clay beads were not found in the Tell F6 2008-2012 assemblage.

The comparison with other settlement sites on Failaka and in Bahrain makes it difficult to avoid the conclusion that the many beads of semi-precious stone, faience and glass found at Tell F6 and Tell F3 were not primarily intended for local consumption. Failaka was favourably situated to mediate trade going to and from Mesopotamia and down the Gulf, and the many beads found at Tell F3 and F6 may very well have been lost during these activities.

The area of the Ur III settlement layers (Phase 1) in Trenches C and J was slightly larger than that of the settlement layers uncovered in Trenches A and E dating to the following phases (2-7). This fact does not, however, account for the many more beads found in Phase 1 contexts than in the following phases. It appears that activities involving trade/transport in beads were especially prevalent in Phase 1.

Catalogue

The beads presented below have first been divided according to phase and secondly according to the form of the bead. When dealing with fragmented beads an estimate has been made of their original form. Thirdly, the beads have been divided according to material under each category (mineral, organic, artificial) in alphabetical order and fourthly according to the given excavation trench and lastly according to find number.

Beads with any kind of additional decoration (i.e. being gadrooned, fluted, etc.) have been categorized according to their form and not according to decoration, as otherwise suggested by Beck (1928 p. 24) and placed at the end of their respective category with a clear description of their decoration.

A number of abbreviations are used. The prefix “Est.” in front of any of the abbreviations listed below denotes that the dimensions given are estimated on basis of the preserved feature on a fragmented bead. The prefix “Pres.” in front of a measurement denotes that it is the preserved dimensions of a fragmented bead that is given. L: Length. H: Height. W: Width. D: Diameter. E: End measurements. EH: End height. EW: End width. ED: End diameter. PD: Perforation diameter.

Phase 1

No. C136-X398 (fig. 832). *Context:* Trench C, layer C136, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.80; D 7.60-8.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.90-2.50 mm.

No. C187-X659 (fig. 833). *Context:* Trench C, layer C187, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 3.95; D 5.10-5.40; ED 3.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.60-1.90 mm.

No. C187-X661 (fig. 834). *Context:* Trench C, layer C187, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.05; D 7.20-7.30; ED 2.80-4.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III (see Kenoyer below p. 201); PD 1.65-2.15 mm.

No. C35-X90 (fig. 835). *Context:* Trench C, layer C35, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 5.25; D 7.10-7.25; ED 3.00-3.80 mm. *Material:* Rock crystal. *Perforation:* Type III; PD 1.80-2.30 mm.

No. C117-X271 (fig. 836). *Context:* Trench C, layer C117, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* Pres.L 7.45; D 9.40-9.60; ED 3.40 mm. *Material:* Rock crystal. *Perforation:* Type II; PD 2.20-2.45 mm.

No. C124-X310 (fig. 837). *Context:* Trench C, layer C124, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.40; D 7.70-8.10; ED 3.4-4.20 mm. *Material:* Rock crystal. *Perforation:* Type III (See Kenoyer below p. 201); PD 3.40-4.20 mm.

No. C136-X397 (fig. 838). *Context:* Trench C, layer C136, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.70; Pres.D 9.10; Est.ED 6.00 mm. *Material:* Faience. *Perforation:* Type IV; Est.PD 3.50 mm.

No. C153-X578 (fig. 839). *Context:* Trench C, layer C153, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 2.00; D 3.90; ED 2.50 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.90 mm.

No. C180-X611 (fig. 840). *Context:* Trench C, layer C180, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.80; D 3.90; ED 3.60 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.90 mm.

No. C180-X622 (fig. 841). *Context:* Trench C, layer C180, Phase 1. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.70; D 4.00; ED 3.25 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.55 mm.

No. C181-X626 (fig. 842). *Context:* Trench C, layer C181, Phase 1. *Type:* Circular/Circular; I.C.1.a. *Size:* L 8.80; D 9.40-9.50; ED 3.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 201); PD 2.00-2.10 mm.

No. J4-X20 (fig. 843). *Context:* Trench J, layer J4, Phase 1. *Type:* Circular/Circular; I.C.1.a. *Size:* L 8.25; D 8.90; ED 4.80 mm. *Material:* Unidentified artificial material. *Perforation:* Type IV; PD 1.50-1.60 mm.

No. C13-X152 (fig. 844). *Context:* Trench C, layer C13, Phase 1. *Type:* Barrel disc/Circular; I.A.1.b. *Size:* H 1.60; D 5.25; ED 5.15 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.30-1.50 mm.

No. C180-X623 (fig. 845). *Context:* Trench C, layer C180, Phase 1. *Type:* Barrel disc/Circular; I.A.1.b. *Size:* L 1.60; D 5.00; ED 4.85 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.65 mm.

No. C183-X636 (fig. 846). *Context:* Trench C, layer C183, Phase 1. *Type:* Barrel disc/Circular; I.A.1.b. *Size:* L 1.60; D 5.35-5.40; ED 5.30 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.90 mm.

No. C39-X100 (fig. 847). *Context:* Trench C, layer C39, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 6.30; D 10.00; ED 7.70 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; PD 1.00 mm.

No. C218-X790 (fig. 848). *Context:* Trench C, layer C218, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 5.90; D 8.70-8.85; ED 6.40-7.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; PD 1.25-3.95 mm.

No. C125-X323 (fig. 849). *Context:* Trench C, layer C125, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 2.24; D 5.20-5.45; ED 5.20-5.45 mm. *Material:* Chalcedony. *Perforation:* Type I; PD 2.40 mm.

No. C141-X437 (fig. 850). *Context:* Trench C, layer C141, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 1.97; D 5.70; ED 5.70 mm. *Material:* Ostrich eggshell. *Perforation:* Type IV; PD 1.75-1.80 mm.

No. C186-X657 (fig. 851). *Context:* Trench C, layer C186, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 1.90; D 5.50-5.55; ED 5.40 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.25-2.10/2.60 mm.

No. C168-X542 (fig. 852). *Context:* Trench C, layer C168, Phase 1. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 1.80; D 4.00; ED 3.40 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.80 mm.

No. C89-X187. *Context:* Trench C, layer C89, Phase 1. *Type:* Standard barrel/Circular; I.C.1.b. *Size:* Pres.L 3.00; D 3.00; Est.ED 2.00 mm. *Material:* Faience. *Perforation:* Unidentified; PD 0.80 mm.

No. C218-X771 (fig. 853). *Context:* Trench C, layer C218, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 7.60; D 4.40; ED 3.95 mm. *Material:* Agate. *Perforation:* Type III; PD 1.45-1.60 mm.

No. J4-X33 (fig. 854). *Context:* Trench J, layer J4, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* Pres.L 12.90; D 5.00; ED 2.60-3.20 mm. *Material:* Agate. *Perforation:* Type II; PD 1.50 mm.

No. C61-X131 (fig. 855). *Context:* Trench C, layer C61, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 11.30; D 5.20-5.30; ED 4.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.00-2.50 mm.

No. C117-X298 (fig. 856). *Context:* Trench C, layer C117, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 10.70; D 5.80; ED 4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 201); PD 1.90 mm.

No. C136-X393 (fig. 857). *Context:* Trench C, layer C136, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 11.25; D 6.30-6.90; ED 3.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.50-1.60 mm.

No. C187-X660 (fig. 858). *Context:* Trench C, layer C187, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 5.95; D 4.90; ED 4.30 mm. *Material:* Carnelian. *Perforation:* Type II (See Kenoyer below p. 201); PD 1.60-1.90 mm.

No. J11-X57 (fig. 859). *Context:* Trench J, layer J11, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 12.25; D 4.10; ED 3.20-3.50 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.65-1.75 mm.

No. J12-X67 (fig. 860). *Context:* Trench J, layer J12, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 10.00; D 5.80; ED 4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.00-1.80 mm.

No. J12-X83 (fig. 861). *Context:* Trench J, layer J12, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 11.40; D 8.30-8.50; ED 4.80 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.80-1.90 mm.

No. C215-X766 (fig. 862). *Context:* Trench C, layer C215, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.45; D 5.25; ED 3.40-3.60 mm. *Material:* Chalcedony. *Perforation:* Type II; PD 1.45-1.90 mm.

No. C186-X655 (fig. 863). *Context:* Trench C, layer C186, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.45; D 4.90; ED 2.10-3.80 mm. *Material:* Lapis Lazuli. *Perforation:* Type II; PD 0.70-0.75 mm.

No. C135-X391 (fig. 864). *Context:* Trench C, layer C125, Phase 1. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 26.50; D 8.50-8.70; ED 5.30 mm. *Material:* Faience. *Perforation:* Type III; PD 2.30-3.50 mm.

No. C213-X757 (fig. 865). *Context:* Trench C, layer C213, Phase 1. *Type:* Short convex cone/Circular; I.B.1.c. *Size:* L 1.70; D 4.80; ED 4.20-4.80 mm. *Material:* Ostrich eggshell. *Perforation:* Type II; PD 1.50 mm.

No. C213-X756 (fig. 866). *Context:* Trench C, layer C213, Phase 1. *Type:* Short truncated convex cone/Circular; I.B.1.d. *Size:* L 2.50; D 7.20; ED 5.70 mm. *Material:* Chalcedony. *Perforation:* Type II; PD 2.25-2.50 mm.

No. C7-X23 (fig. 867). *Context:* Trench C, layer C7, Phase 1. *Type:* Convex truncated bicone disc/Circular; I.A.1.f. *Size:* L 3.40; D 8.40-9.00; ED 6.70-7.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; PD 0.80-1.00 mm.

No. C198-X713 (fig. 868). *Context:* Trench C, layer C198, Phase 1. *Type:* Truncated convex bicone disc/Circular; I.A.1.f. *Size:* L 1.75; D 15.75-16.30 mm; ED 15.30 mm. *Material:* Ostrich eggshell. *Perforation:* Type III; PD 3.80-3.90 mm.

No. C181-X625 (fig. 869). *Context:* Trench C, layer C181, Phase 1. *Type:* Truncated convex bicone disc/Circular; I.A.1.f. *Size:* L 1.40; D 3.40-4.00; ED 3.00 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.60 mm.

No. C82-X178 (fig. 870). *Context:* Trench C, layer C82, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 3.95; D 5.20-5.40; ED 3-3.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.60-1.80 mm.

No. C136-X396 (fig. 871). *Context:* Trench C, layer C136, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 4.50; D 5.20; ED 3.10 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.40-1.90 mm.

No. C184-X643 (fig. 872). *Context:* Trench C, layer C184, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 3.50; D 4.60-4.75; ED 3.20-3.40 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.15-1.30 mm.

No. C186-X693 (fig. 873). *Context:* Trench C, layer C186, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 4.00; D 7.15; ED 5.15 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 2.10-2.40 mm.

No. C197-X707 (fig. 874). *Context:* Trench C, layer C197, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 2.95; D 6.80-6.90; ED 6.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I; PD 2.00-2.10 mm.

No. C211-X768 (fig. 875). *Context:* Trench C, layer C211, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.20; D 8.60-8.70; ED 6.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; PD 0.80 mm.

No. C198-X712 (fig. 876). *Context:* Trench C, layer C198, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 4.30; D 5.60; ED 3.70 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III (See Kenoyer below p. 201); PD 1.20-2.10 mm.

No. C211-X769 (fig. 877). *Context:* Trench C, layer C211, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 4.55; D 6.75; ED 4.75 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III (See Kenoyer below p. 201); PD 1.45-1.90 mm.

No. C225-X859 (fig. 878). *Context:* Trench C, layer C225, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.00; Est.D 10.80; Est.ED 9.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; Est.PD 3.75-7.60 mm.

No. C225-X867 (fig. 879). *Context:* Trench C, layer C225, Phase 1. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 3.15; D 8.20-8.40; ED 5.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.70 mm.

No. C164-X530 (fig. 880). *Context:* Trench C, layer C164, Phase 1. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 4.95; D 4.15; ED 2.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III (See Kenoyer below p. 201); PD 1.70-2.00 mm.

No. J12-X60 (fig. 881). *Context:* Trench J, layer J12, Phase 1. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 12.50; D 6.30; ED 4.20 mm. *Material:* Agate. *Perforation:* Type II; PD 2.00-2.10 mm.

No. J9-X100 (fig. 882). *Context:* Trench J, layer J9, Phase 1. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 4.90; D 3.40-3.80; ED 3.00-3.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.15-1.20 mm.

No. C19-X55 (fig. 883). *Context:* Trench C, layer C19, Phase 1. *Type:* Cylinder disc/Circular; I.A.2.b. *Size:* H 2.90; D 16.30 mm. *Material:* Shell. *Perforation:* Unidentified; PD 10.10 mm.

No. J9-X84 (fig. 884). *Context:* Trench J, layer J9, Phase 1. *Type:* Short cylinder/Circular; I.B.2.b. *Size:* L 1.80; D 4.80-5.10; ED 4.80-5.10 mm. *Material:* Agate. *Perforation:* Type IV; PD 0.90-1.70 mm.

No. C211-X767 (fig. 885). *Context:* Trench C, layer C211, Phase 1. *Type:* Long cylinder with two convex ends/Circular; I.D.4.f.b. *Size:* L 7.00; D 2.90; ED 1.90 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.30 mm.

No. C180-X621 (fig. 886). *Context:* Trench C, layer C180, Phase 1. *Type:* Standard barrel/Elliptical; II.C.1.b. *Size:* L 9.90; H: 6.00; W: 10.15; E: 5.65 x 8.75 mm. *Material:* Quartz. *Perforation:* Type III; PD 1.60-2.35 mm.

No. C14-X28 (fig. 887). *Context:* Trench C, layer C14, Phase 1. *Type:* Long cylinder with two convex ends/Elliptical; II.D.4.f.b. *Size:* L 10.30; D 4.20-5.00; ED 3.80-4.20 mm. *Material:* Unidentified stone material. *Perforation:* Type IV; PD 1.80-2.00 mm.

No. C186-X654 (fig. 888). *Context:* Trench C, layer C186, Phase 1. *Type:* Short cylinder with two convex ends/Lenticular; IV.B.4.f.b. *Size:* L 9.40; W 12.00; H 4.60; EW 11.20; EH 4.00 mm. *Material:* Agate. *Perforation:* Type III; PD 2.00-2.20 mm.

No. J12-X63 (fig. 889). *Context:* Trench J, layer J12, Phase 1. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* L 25.00; W 13.60; H 11.50; EW 9.90; EH 6.60 mm. *Material:* Smoky quartz. *Perforation:* Type II; PD 2.70-2.80 mm.

No. C117-X291 (fig. 890). *Context:* Trench C, layer C117, Phase 1. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* Pres.W 15.25; Pres.L 11.80 mm. *Material:* Chalcedony. *Perforation:* Not preserved; Est.PD 4.35 mm.

No. C89-X186 (fig. 891). *Context:* Trench C, layer C89, Phase 1. *Type:* Short Pear-shape/Square; IX.B.1.g. *Size:* H 6.40; Pres.W 5.50; Pres.L 5.10 mm. *Material:* Rock crystal. *Perforation:* Unidentified; Est.PD 1.70 mm.

No. C186-X656 (fig. 892). *Context:* Trench C, layer C186, Phase 1. *Type:* Long cylinder/Square; IX.D.2.b. *Size:* L 7.55; D 5.20-5.80; ED 5.10-5.30 mm. *Material:* Bone. *Perforation:* Type II; PD 3.00 mm.

Phase 3

No. E210-X939 (fig. 893). *Context:* Trench E, layer E210, Phase 3. *Type:* Short barrel/Circular; I.B.1.b. *Size:* L 6.00; D 6.70; ED 4.80 mm. *Material:* Carnelian. *Perforation:* Type II (See Kenoyer below p. 206); PD 2.10-2.30 mm.

No. E184-X912 (fig. 894). *Context:* Trench E, layer E184, Phase 3. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.95; D 6.70; ED 3.70-4.00 mm. *Material:* Carnelian. *Perforation:* Type II (See Kenoyer below p. 201); PD 1.60-1.70 mm.

No. E248-X1148 (fig. 895). *Context:* Trench E, layer E248, Phase 3. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 4.20; D 4.15; ED 2.80 mm. *Material:* Carnelian. *Perforation:* Type II (See Kenoyer below p. 206); PD 1.35-1.45 mm.

No. E316-X1351 (fig. 896). *Context:* Trench E, layer E316, Phase 3. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 6.60; Est.D; 6.00; Est.ED 4.90 mm. *Material:* Carnelian. *Perforation:* Type II; Est.PD 1.50-2.00 mm.

Phase 4

No. C113-X256 (fig. 897). *Context:* Trench C, layer C113 = temple terrace stone floor C68, Phase 4? *Type:* Oblate/Circular; I.B.1.a. *Size:* L 7.90; D 9.55; ED 3.20-4.95 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.75-3.55 mm.

No. A172-X400 (fig. 898). *Context:* Trench A, layer A172, Phase 4. *Type:* Long Barrel/Circular; I.D.1.b. *Size:* L 16.50; D 7.70; ED 5.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 206); PD 2.50 mm.

No. E147-X827 (fig. 899). *Context:* Trench E, layer E147, Phase 4. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.10; D 5.85-5.95; ED 4.90 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 206); PD 2.00 mm.

No. A201-X888 (fig. 900). *Context:* Trench A, layer A201, Phase 4. *Type:* Cylinder disc/Circular; I.A.2.b. *Size:* H: 8.70; D 32.40 mm. *Material:* Shell. *Perforation:* Unidentified; PD 7.30-9.00 mm.

No. E200-X1035 (fig. 901). *Context:* Trench E, layer E200, Phase 4. *Type:* Long Cylinder/Circular; I.D.2.b. *Size:* L 15.60; D 3.90; ED 3.80 mm. *Material:* Carnelian. *Perforation:* Type II (See Kenoyer below p. 206); PD 2.20-2.30 mm.

No. A172-X389 (fig. 902). *Context:* Trench A, layer A172, Phase 4. *Type:* Bead consisting of a complete shell; XXVII.A.1. *Size:* L 23.50; D 15.00-15.55 mm. *Material:* Shell (*Conus Ebraeus*). *Perforation:* Unidentified; PD 1.70-2.10 mm.

Phase 5

No. A128-X413 (fig. 903). *Context:* Trench A, layer A128, Phase 5. *Type:* Long truncated bicone/Circular; I.D.2.f. *Size:* L 5.55; D 4.80-5.00; ED 3.15 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.70 mm.

Phase 7

No. A32-X165 (fig. 904). *Context:* Trench E, layer A32, Phase 7. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 8.40; D 9.00-10.00; ED 4.80-5.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 206); PD 2.10-2.20 mm.

No. E11-X38 (fig. 905). *Context:* Trench E, layer E11, Phase 7. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.20; D 3.00; ED 2.30 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.20 mm.

No. E69-X251 (fig. 906). *Context:* Trench E, layer E69, Phase 7. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.30; D 8.90; ED 5.40-5.60 mm. *Material:* Glass. *Perforation:* Type IV; PD 2.60 mm.

No. E32-X128 (fig. 907). *Context:* Trench E, layer E32, Phase 7. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 4.00; D 4.90-5.00; ED 3.30 mm. *Material:* Paste. *Perforation:* Type III; PD 1.80-1.90 mm.

No. E20-X68 (fig. 908). *Context:* Trench E, layer E20, Phase 7. *Type:* Circular/Circular; I.C.1.a. *Size:* L 6.70; D 7.15; ED 2.90-4.20 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III (See Kenoyer below p. 206); PD 1.80-2.50 mm.

No. E34-X294 (fig. 909). *Context:* Trench E, layer E34, Phase 7. *Type:* Circular/Circular; I.C.1.a. *Size:* L 6.80; D 7.15-7.20; ED 3.30 mm. *Material:* Glass. *Perforation:* Type IV; PD 2.20 mm.

No. E25-X83 (fig. 910). *Context:* Trench E, layer E25, Phase 7. *Type:* Ellipsoid/Circular; I.D.1.a. *Size:* Est.L 15.60; Est.D 12.80; Est.ED 6.00-7.00 mm. *Material:* Glass. *Perforation:* Type IV; PD 3.35 mm.

No. E23-X82 (fig. 911). *Context:* Trench E, layer E23, Phase 7. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.85; D 7.30-7.70; ED 4.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II (See Kenoyer below p. 206); PD 1.95 mm.

No. E54-X220 (fig. 912). *Context:* Trench E, layer E54, Phase 7. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 27.00; D 9.80; ED 4.00 mm. *Material:* Faience. *Perforation:* Type IV; PD 2.40 mm.

No. E59-X231 (fig. 913). *Context:* Trench E, layer E59, Phase 7. *Type:* Spiral patterned long cylinder with two convex ends/Circular; I.D.4.f.b. *Size:* Pres.L 17.00; D 6.20; ED 3.70 mm. *Material:* Faience. *Perforation:* Type IV; PD 2.40 mm.

No. E21-X73 (fig. 914). *Context:* Trench E, layer E21, Phase 7. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* L 18.10; W 6.60; H 5.20; ED 3.50-4.00 mm. *Material:* Jasper. *Perforation:* Type II (see Kenoyer below p. 206); PD 1.40-1.60 mm.

No. E74-X280 (fig. 915). *Context:* Trench E, layer E74, Phase 7. *Type:* Long barrel/Lenticular; IV.D.1.b. *Size:* L 21.50; H 4.80; W 13.50; E 3.00 x 7.30 mm. *Material:* Jasper. *Perforation:* Type II (See Kenoyer below p. 206); PD; 1.90-2.00 mm.

No. E214-X940 (fig. 916). *Context:* Trench E, layer E214, Phase 7. *Type:* Standard barrel/Circle and flat; VII.C.1.a. *Size:* L 5.80; D 5.20-5.70; ED 2.70-3.70 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.50-1.60 mm.

No. E74-X279 (fig. 917). *Context:* Trench E, layer E74, Phase 7. *Type:* Bead consisting of a complete shell; XXVII.A.1. *Size:* L 25.50; W 21.30; H 8.00; EW 4.60-5.50 mm. *Material:* Cowrie shell (*Erosaria turdus*). *Perforation:* - .

No. E59-X233. *Context:* Trench E, layer E59, Phase 7. *Type:* Unidentified. *Size:* - *Material:* Glass. *Perforation:* - .

Phase 8

No. A22-X111 (fig. 918). *Context:* Trench A, layer A22, Phase 8. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 7.90; D 9.90-10.10; ED 3.80-4.40 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.40-2.70 mm.

No. A28-X113 (fig. 919). *Context:* Trench A, layer A28, Phase 8. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 7.30; D 8.70-8.85; ED 5.30-6.90 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.80-2.60 mm.

No. E24-X159 (fig. 920). *Context:* Trench E, layer E24, Phase 8. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.00; D 8.50; ED 4.30-4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.80-2.50 mm.

No. A1-X36 (fig. 921). *Context:* Trench A, layer A1, Phase 8. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.90; D 11.10-11.20; ED 5.70-6.90 mm. *Material:* Turquoise. *Perforation:* Type II; PD 2.70-2.80 mm.

No. A10-X69 (fig. 922). *Context:* Trench A, layer A10, Phase 8. *Type:* Short barrel/Circular; I.B.1.b. *Size:* Pres.L 5.00; Est.D 9.00; Est.ED 6.10-7.00 mm. *Material:* Glass. *Perforation:* Not preserved; Est.PD 3.10 mm.

No. E17-X110 (fig. 923). *Context:* Trench E, layer E17, Phase 8. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 9.60; D 7.00; ED 4.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.00 mm.

No. E17-X126 (fig. 924). *Context:* Trench E, layer E17, Phase 8. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.20; D 6.20; ED 5.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.60-1.70 mm.

No. E24-X131 (fig. 925). *Context:* Trench E, layer E24, Phase 8. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 16.60; D 8.30; ED 4.40

mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.30 mm.

No. E1-X2 (fig. 926). *Context:* Trench E, layer E1, Phase 8. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 31.30; D 10.25-10.45; ED 4.60 mm. *Material:* Porphyry. *Perforation:* Type II; PD 2.10-2.25 mm.

No. A1-X21 (fig. 927). *Context:* Trench A, layer A1, Phase 8. *Type:* Short convex cone/Circular; I.B.1.c. *Size:* L 6.90; D 14.80-16.60 mm. *Material:* Shell. *Perforation:* Unidentified; PD 3.60-3.85 mm.

No. A155-X813 (fig. 928). *Context:* Trench A, layer A155, Phase 8. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.74; D 6.40; ED 5.90 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.55 mm.

No. E10-X41 (fig. 929). *Context:* Trench E, layer E10, Phase 8. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* L 12.15; H 5.30; W 6.50; E 3.90 x 4.30 mm. *Material:* Unidentified stone material. *Perforation:* Type III; PD 2.20-2.40 mm.

No. A10-X70 (fig. 930). *Context:* Trench A, layer A10, Phase 8. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* L 21.10; H 11.95; W 12.10; ED 4.00 x 5.55 mm. *Material:* Agate. *Perforation:* Type II; PD 2.60 mm.

No. A28-X128 (fig. 931). *Context:* Trench A, layer A28, Phase 8. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* L 16.65; H 9.60; W 13.90; ED 4.80 x 8.60 mm. *Material:* Agate. *Perforation:* Type II; PD 2.40 mm.

No. E8-X27 (fig. 932). *Context:* Trench E, layer E8, Phase 8. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* L 28.30; H 13.75; W 16.50; E 8.00 x 10.00 mm. *Material:* Unidentified stone material. *Perforation:* Type II; PD 3.60 mm.

No. A6-X45 (fig. 933). *Context:* Trench A, layer A6, Phase 8. *Type:* Long cylinder with two convex ends/Square; IX.D.4.f.b. *Size:* L 11.75; D 5.40-5.80; ED 5.00-5.30 mm. *Material:* Chrysoprase. *Perforation:* Type II; PD 2.80-3.00 mm.

No. A85-X806 (fig. 934). *Context:* Trench A, layer A85, Phase 8. *Type:* Bead consisting of complete shell; XXVII.A.1. *Size:* L 32.00; W 21.80; E 5.53-5.98 mm. *Material:* Cowrie shell (*Monetaria caputserpentis*). *Perforation:* - .

No. E17-X165. *Context:* Trench E, layer E17, Phase 8. *Type:* Unidentified. *Size:* Pres.L 15.00; Pres.W: 9.00 mm. *Material:* Glass. *Perforation:* Not preserved.

Disturbed contexts

No. C59-X154 (fig. 935). *Context:* Trench C, layer C59, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.60; D 9.10; ED 3.90-4.11 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.60-2.70 mm.

No. C76-X155 (fig. 936). *Context:* Trench C, layer C76, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* Pres.L 6.80; Pres.D 12.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; Pres. PD 1.90 mm.

No. C108-X302 (fig. 937). *Context:* Trench C, layer C108, Disturbed. *Type:* Short oblate/Circular; I.B.1.a. *Size:* L 3.93; D 6.50; ED 4.20-5.25 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.75-2.30 mm.

No. C128-X353 (fig. 938). *Context:* Trench C, layer C128, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 3.80; D 4.60-5.30; ED 3.80-4.70 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.20-2.50 mm.

No. C130-X371 (fig. 939). *Context:* Trench C, layer C130, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 5.90; D 7.50-7.60; ED 4.40 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.75-2.65 mm.

No. C132-X383 (fig. 940). *Context:* Trench C, layer C132, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 8.20; D 8.60-9.40; ED 3.70-4.65 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.70-2.50 mm.

No. C152-X499 (fig. 941). *Context:* Trench C, layer C152, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* Pres.L 6.60; Pres.D 8.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Not preserved; Est.PD 2.50 mm.

No. C179-X629 (fig. 942). *Context:* Trench C, layer C179, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 6.00; D 7.60-7.90; ED 3.15 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.50-2.25 mm.

No. C223-X830 (fig. 943). *Context:* Trench C, layer C223, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* Pres.L 2.60; Pres.D 7.70 mm. *Material:* Carnelian (discoloured). *Perforation:* Not preserved; PD 1.85 mm.

No. C5-X45 (fig. 944). *Context:* Trench C, layer C5, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.35; D 3.00 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.00 mm.

No. C15-X46 (fig. 945). *Context:* Trench C, layer C15, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* Est.L 1.25; Est.D 3.00 mm. *Material:* Faience. *Perforation:* Type IV; Est.PD 1.00 mm.

No. C15-X48 (fig. 946). *Context:* Trench C, layer C15, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.30; D 2.50 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.00 mm.

No. C128-X348 (fig. 947). *Context:* Trench C, layer C128, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 2.95; D 3.75; ED 2.90 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.30 mm.

No. C163-X547 (fig. 948). *Context:* Trench C, layer C163, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.60; D 3.70; ED 3.00 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.60 mm.

No. C163-X570 (fig. 949). *Context:* Trench C, layer C163, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.85; D 4.60; ED 3.90 mm. *Material:* Faience. *Perforation:* Type IV; PD 2.00 mm.

No. C179-X608 (fig. 950). *Context:* Trench C, layer C179, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.70; D 4.00; ED 2.90 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.90 mm.

No. C179-X628 (fig. 951). *Context:* Trench C, layer C179, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.75; D 4.15-4.50; ED 3.30 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.90 mm.

No. C163-X752 (fig. 952). *Context:* Trench C, layer C163, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 1.50; D 4.30; ED 3.50 mm. *Material:* Faience. *Perforation:* Type IV; PD 1.80 mm.

No. C221-X796 (fig. 953). *Context:* Trench C, layer C221, Disturbed. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 2.25; D 4.45; ED 2.50 mm. *Material:* Glass. *Perforation:* Type IV; PD 1.25 mm.

No. C78-X159 (fig. 954). *Context:* Trench C, layer C78, Disturbed. *Type:* Circular/Circular; I.C.1.a. *Size:* L 6.90; D 6.95-7.60; ED 3.80-4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 2.40-3.30 mm.

No. C223-X819 (fig. 955). *Context:* Trench C, layer C223, Disturbed. *Type:* Circular/Circular; I.C.1.a. *Size:* L 7.50; D 8.05-8.25; ED 3.00-3.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.60-2.30 mm.

No. C130-X380 (fig. 956). *Context:* Trench C, layer C130, Disturbed. *Type:* Ellipsoid/Circular; I.D.1.a. *Size:* L 12.90; D 9.60-9.80; ED 3.50-5.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.80-2.60 mm.

No. C106-X219 (fig. 957). *Context:* Trench C, layer C106, Disturbed. *Type:* Standard barrel/Circular; I.C.1.b. *Size:* L 7.10; D 7.25; ED 5.95 mm. *Material:* Carnelian. *Perforation:* Type II; PD 2.10-2.20 mm.

No. C179-X642 (fig. 958). *Context:* Trench C, layer C179, Disturbed. *Type:* Standard barrel/Circular; I.C.1.b. *Size:* L 6.70; D 6.00-6.40; ED 5.40-5.85 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.15-2.20 mm.

No. C78-X161 (fig. 959). *Context:* Trench C, layer C78, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 11.25; D 6.95; ED 3.90-4.40 mm. *Material:* Agate. *Perforation:* Type II; PD 2.00 mm.

No. C128-X467 (fig. 960). *Context:* Trench C, layer C128, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 7.15; D 4.00-4.20; ED 3.00-3.20 mm. *Material:* Agate. *Perforation:* Type II; PD 1.50-1.90 mm.

No. C116-X264 (fig. 961). *Context:* Trench C, layer C116, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.00; D 6.65-7.15; ED 4.20-5.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.60-2.40 mm.

No. C108-X303. *Context:* Trench C, layer C108, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* Pres.L 4.70; Pres. W 5.30; Pres.E 3.90 mm. *Material:* Carnelian (discoloured). *Perforation:* Not preserved; Est.PD 1.80 mm.

No. C130-X374 (fig. 962). *Context:* Trench C, layer C130, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 7.40; D 6.00-6.30; ED 5.30-5.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.80 mm.

No. C130-X375 (fig. 963). *Context:* Trench C, layer C130, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 6.20; D 4.30; ED 3.30-3.95 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.60-1.65 mm.

No. C169-X575 (fig. 964). *Context:* Trench C, layer C169, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 13.35; D 6.15; ED 5.00 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.60-1.80 mm.

No. C177-X606 (fig. 965). *Context:* Trench C, layer C177, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 8.25; D 5.00; ED 3.25 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.60 mm.

No. C196-X721 (fig. 966). *Context:* Trench C, layer C196, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 3.40; D 2.80; ED 2.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.10-1.20 mm.

No. C158-X506 (fig. 967). *Context:* Trench C, layer C158, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* Pres.L 5.15; D 5.20-5.80; ED 4.00-4.50 mm. *Material:* Jasper. *Perforation:* Not preserved; PD 2.35-2.50 mm.

No. C128-X344 (fig. 968). *Context:* Trench C, layer C128, Disturbed. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 38.10; D 13.40; ED 6.10-6.30 mm. *Material:* Unidentified stone material. *Perforation:* Type II; PD 2.80-3.30 mm.

No. C139-X369 (fig. 969). *Context:* Trench C, layer C139, Disturbed. *Type:* Fluted long barrel/Circular; I.D.1.f. *Size:* L 10.80; D 6.40-6.60; ED 3.50 mm. *Material:* Heated steatite. *Perforation:* Type IV; PD 2.10 mm.

No. C128-X340 (fig. 970). *Context:* Trench C, layer C128, Disturbed. *Type:* Short convex cone/Circular; I.B.1.c. *Size:* L 3.80; D 5.20-5.50; ED 4.00-5.00 mm. *Material:* Carnelian. *Perforation:* Type III; PD 1.75-2.20 mm.

No. C152-X498 (fig. 971). *Context:* Trench C, layer C152, Disturbed. *Type:* Short convex cone/Circular; I.B.1.c. *Size:* L 3.95; D 4.80-5.00; ED 2.50-3.00/5.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.10-1.75 mm.

No. C115-X265 (fig. 972). *Context:* Trench C, layer C115, Disturbed. *Type:* Short convex cone/Circular; I.B.1.c. *Size:* L 8.80; D 18.90-21.60 mm. *Material:* Shell. *Perforation:* Unidentified; PD 7.60-9.00 mm.

No. C179-X627 (fig. 973). *Context:* Trench C, layer C179, Disturbed. *Type:* Truncated convex cone disc/Circular; I.A.1.d. *Size:* L 1.75; D 5.40-5.50; ED 5.00 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.90 mm.

No. C106-X220 (fig. 974). *Context:* Trench C, layer C106, Disturbed. *Type:* Convex truncated bicone disc/Circular; I.A.1.f. *Size:* L 1.80; D 8.10; ED 6.90-7.20 mm. *Material:* Ostrich eggshell. *Perforation:* Type I; PD 1.95 mm.

No. C108-X269 (fig. 975). *Context:* Trench C, layer C108, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.80; D 7.90-8.10; ED 5.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 2.00-2.10 mm.

No. C177-X607 (fig. 976). *Context:* Trench C, layer C177, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 2.40; D 3.15-3.40; ED 2.00-2.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.20-1.40 mm.

No. C177-X653 (fig. 977). *Context:* Trench C, layer C177, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 4.65; D 5.60-6.25; ED 4.50-4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.65-2.30 mm.

No. C196-X720 (fig. 978). *Context:* Trench C, layer C196, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 2.20; D 3.20; ED 2.40 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.35-1.45 mm.

No. C223-X813 (fig. 979). *Context:* Trench C, layer C223, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f.

Size: L 3.50; D 7.60; ED 5.70-6.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type I.1; PD 1.95 mm.

No. C223-X816 (fig. 980). *Context:* Trench C, layer C223, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 5.10; D 6.00; ED 4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.80 mm.

No. C223-X837 (fig. 981). *Context:* Trench C, layer C223, Disturbed. *Type:* Short truncated convex bicone/Circular; I.B.1.f. *Size:* L 3.30; D 5.00; ED 3.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.30-1.50 mm.

No. C128-X349 (fig. 982). *Context:* Trench C, layer C128, Disturbed. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 7.10; D 7.20-7.50; ED 5.95 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.20 mm.

No. C123-X449 (fig. 983). *Context:* Trench C, layer C123, Disturbed. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 5.45; D 4.80-5.00; ED 3.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.38 mm.

No. C41-X110 (fig. 984). *Context:* Trench C, layer C41, Disturbed. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 8.70; D 7.50-8.20; ED 5.90 mm. *Material:* Shell. *Perforation:* Type II; PD 3.10-3.30 mm.

No. C48-X459 (fig. 985). *Context:* Trench C, layer C48, Disturbed. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 6.90; D 5.00; ED 2.80 mm. *Material:* Carnelian (discoloured). *Perforation:* Unidentified; possibly Type IV; PD 1.20 mm.

No. C196-X730 (fig. 986). *Context:* Trench C, layer C196, Disturbed. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 4.40; D 3.60; ED 2.70 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.25-1.30 mm.

No. C223-X839 (fig. 987). *Context:* Trench C, layer C223, Disturbed. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 7.40; D 4.45; ED 2.70 mm. *Material:* Unidentified stone material. *Perforation:* Unidentified; PD 1.50-1.60 mm.

No. C223-X841 (fig. 988). *Context:* Trench C, layer C223, Disturbed. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 6.10; D 4.40; ED 2.40-2.60 mm. *Material:* Unidentified stone material. *Perforation:* Type III; PD 1.60 mm.

No. E66-X238 (fig. 989). *Context:* Trench E, layer E66, Disturbed. *Type:* Short cylinder/Circular; I.B.2.b. *Size:* Est.L 9.10; Est.D 12.20 mm. *Material:* Glass. *Perforation:* Not preserved.

No. C137-X414 (fig. 990). *Context:* Trench C, layer C137, Disturbed. *Type:* Short cylinder/Circular; I.B.2.b. *Size:* L 2.75; D 2.95-3.10; ED 3.00 mm. *Material:* Shell. *Perforation:* Type IV; PD 1.50 mm.

No. C128-X456 (fig. 991). *Context:* Trench C, layer C128, Disturbed. *Type:* Long cylinder/Circular; I.D.2.b. *Size:* L 3.90; D 3.30; ED 3.20 mm. *Material:* Gold. *Perforation:* Unidentified; PD 2.10 mm.

No. C139-X441 (fig. 992). *Context:* Trench C, layer C139, Disturbed. *Type:* Short gadrooned truncated bicone/Circular; I.B.2.f. *Size:* L 8.20; D 11.90-12.40; ED 4.90 mm. *Material:* Faience. *Perforation:* Type IV; PD 3.20 mm.

No. C130-X379 (fig. 993). *Context:* Trench C, layer C130, Disturbed. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* L 11.45; D 7.00-7.90; ED 4.40-5.10 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 2.20-2.40 mm.

No. C152-X497 (fig. 994). *Context:* Trench C, layer C152, Disturbed. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* L 11.75; D 5.25-5.80; ED 4.15-4.50 mm. *Material:* Agate. *Perforation:* Type II; PD 2.15-2.20 mm.

No. C196-X722 (fig. 995). *Context:* Trench C, layer C196, Disturbed. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* L 7.40; D 4.90-5.60; ED 3.80-4.50 mm. *Material:* Agate. *Perforation:* Type II; PD 1.30-2.40 mm.

No. C177-X600 (fig. 996). *Context:* Trench C, layer C177, Disturbed. *Type:* Long barrel/Elliptical; II.D.1.b. *Size:* Pres.L 4.30; Pres.D 5.30; Pres.ED 4.50 mm. *Material:* Carnelian (discoloured). *Perforation:* Not preserved; Est.PD 1.90 mm.

No. C137-X532 (fig. 997). *Context:* Trench C, layer C137, Disturbed. *Type:* Long Barrel/Elliptical; II.D.1.b. *Size:* Pres.L 3.30; Pres.H 7.60; Pres.W 11.30; E 6.00 x 8.80 mm. *Material:* Milky quartz. *Perforation:* Not preserved; PD 2.55 mm.

No. C5-X43 (fig. 998). *Context:* Trench C, layer C5, Disturbed. *Type:* Long truncated convex bicone/Elliptical; II.D.1.f. *Size:* L 5.25; D 3.60-4.10; ED 2.70 mm. *Material:* Carnelian. *Perforation:* Type IV; PD 1.40 mm.

No. C104-X209 (fig. 999). *Context:* Trench C, layer C104, Disturbed. *Type:* Short convex bicone/Ovoid; III.B.1.e. *Size:* L 7.60; H 2.80; W 3.25; ED 2.95-3.00 mm. *Material:* Turquoise. *Perforation:* Type IV; PD 1.20 mm.

No. J2-X8 (fig. 1000). *Context:* Trench J, layer J2, Disturbed. *Type:* Circular/Lenticular; IV.C.1.a. *Size:* Pres.L 10.00; Pres.H 8.00; Pres.W 9.30 mm. *Material:* Agate. *Perforation:* Not preserved; PD 1.60 mm.

No. C53-X120 (fig. 1001). *Context:* Trench C, layer C53, Disturbed. *Type:* Circular/Lenticular; IV.C.1.a. *Size:* Pres.L 10.90; Pres.H 5.50; Pres.W 9.50 mm. *Material:* Lapis Lazuli. *Perforation:* Not preserved; Est.PD 1.90 mm.

No. C30-X105 (fig. 1002). *Context:* Trench C, layer C30, Disturbed. *Type:* Standard barrel/Lenticular; IV.C.1.b. *Size:* L 8.85; D 8.40; E 2.10 x 4.15 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.60-1.70 mm.

No. C223-X810 (fig. 1003). *Context:* Trench C, layer C223, Disturbed. *Type:* Standard barrel/Lenticular; IV.C.1.b. *Size:* L 21.70; H 9.90; Est.W 21.40; E 5.80 x 11.00 mm. *Material:* Unidentified stone material. *Perforation:* Type II; PD 3.10-3.50 mm.

No. C132-X389 (fig. 1004). *Context:* Trench C, layer C132, Disturbed. *Type:* Long Barrel/Lenticular; IV.D.1.b. *Size:* Pres.L 26.40; Pres.H 11.80; Pres.W 15.50 mm. *Material:* Agate. *Perforation:* Not preserved.

No. C152-X548 (fig. 1005). *Context:* Trench C, layer C152, Disturbed. *Type:* Standard cylinder with two convex ends/Lenticular; IV.C.4.f.b. *Size:* Pres.L 7.90; Pres.H 6.00; Pres.W 7.70; Pres.E 4.30 x 5.70 mm. *Material:* Agate. *Perforation:* Not preserved; Est. PD 1.80 mm.

No. C137-X394 (fig. 1006). *Context:* Trench C, layer C137, Disturbed. *Type:* Long barrel/Plano-convex; V.D.1.b. *Size:* Pres.L 20.00; Pres.H 11.00; Pres.W 11.50; Pres.E 10.50 x 8.60 mm. *Material:* Unidentified stone material. *Perforation:* Not preserved; Est. PD 3.50 mm.

No. C15-X54 (fig. 1007). *Context:* Trench C, layer C15, Disturbed. *Type:* Short Barrel/Semicircular; VI.B.1.b. *Size:* Pres.L 14.30; Pres.H 8.70; W 16.90; Est.ED 5.50 x 7.00 mm. *Material:* Bitumen. *Perforation:* Unidentified; PD 1.00-1.30 mm.

No. C5-X75 (fig. 1008). *Context:* Trench C, layer C5, Disturbed. *Type:* Long barrel/Semicircular; VI.D.1.b. *Size:* L 22.25; H 10.00; W 15.60; E 5.00 x 8.00 mm. *Material:* Limestone. *Perforation:* Type II; PD 2.60-2.70 mm.

No. C104-X213 (fig. 1009). *Context:* Trench C, layer C104, Disturbed. *Type:* Long Barrel/Semicircular; VI.D.1.b. *Size:* L 25.00; H 13.00; Est.W 20.00; E 5.00 x 6.70/9.00 x 12.30 mm. *Material:* Limestone. *Perforation:* Type II; PD 2.10-2.80 mm.

No. C130-X376 (fig. 1010). *Context:* Trench C, layer C130, Disturbed. *Type:* Long barrel/semi circular; VI.D.1.b. *Size:* L 21.20; H 8.30; W 11.50; E 5.00 x 5.10/3.50 x 5.30 mm. *Material:* Unidentified stone material. *Perforation:* Type II; PD 1.60-1.90 mm.

No. C4-X8 (fig. 1011). *Context:* Trench C, layer C4, Disturbed. *Type:* Standard truncated convex bicone/Semicircular; VI.C.1.f. *Size:* L 25.00; H 13.00; Est.W 20.00; E 5.00 x 6.70/9.00 x 12.30 mm. *Material:* Milky Quartz. *Perforation:* Type II; PD 2.60-2.90 mm.

No. C139-X446 (fig. 1012). *Context:* Trench C, layer C139, Disturbed. *Type:* Long barrel/Circle and flat; VII.D.1.b. *Size:* Pres.L 10.80; Pres.H 9.60; Pres.W 8.30; Pres.E 6.50 x 5.40 mm. *Material:* Unidentified stone material. *Perforation:* Not preserved; Est. PD 2.50 mm.

No. C132-X386 (fig. 1013). *Context:* Trench C, layer C132, Disturbed. *Type:* Long cylinder/Square; IX.D.2.b. *Size:* L 19.20; H 6.10; W 6.70; E 4.00 x 4.80/4.30 x 5.30 mm. *Material:* Unidentified stone material. *Perforation:* Type II; PD 2.80-3.00 mm.

No. C5-X39 (fig. 1014). *Context:* Trench C, layer C5, Disturbed. *Type:* Long cylinder with two convex ends/Square; IX.D.4.f.b. *Size:* L 3.40; D 2.80; ED 2.40 mm. *Material:* Carnelian (discoloured). *Perforation:* Type III; PD 1.10-1.30 mm.

No. C223-X831 (fig. 1015). *Context:* Trench C, layer C223, Disturbed. *Type:* Short cylinder/Hexagonal; XIII.B.2.b. *Size:* L 2.60; D 3.00; ED 2.90-3.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Type IV; PD 1.50 mm.

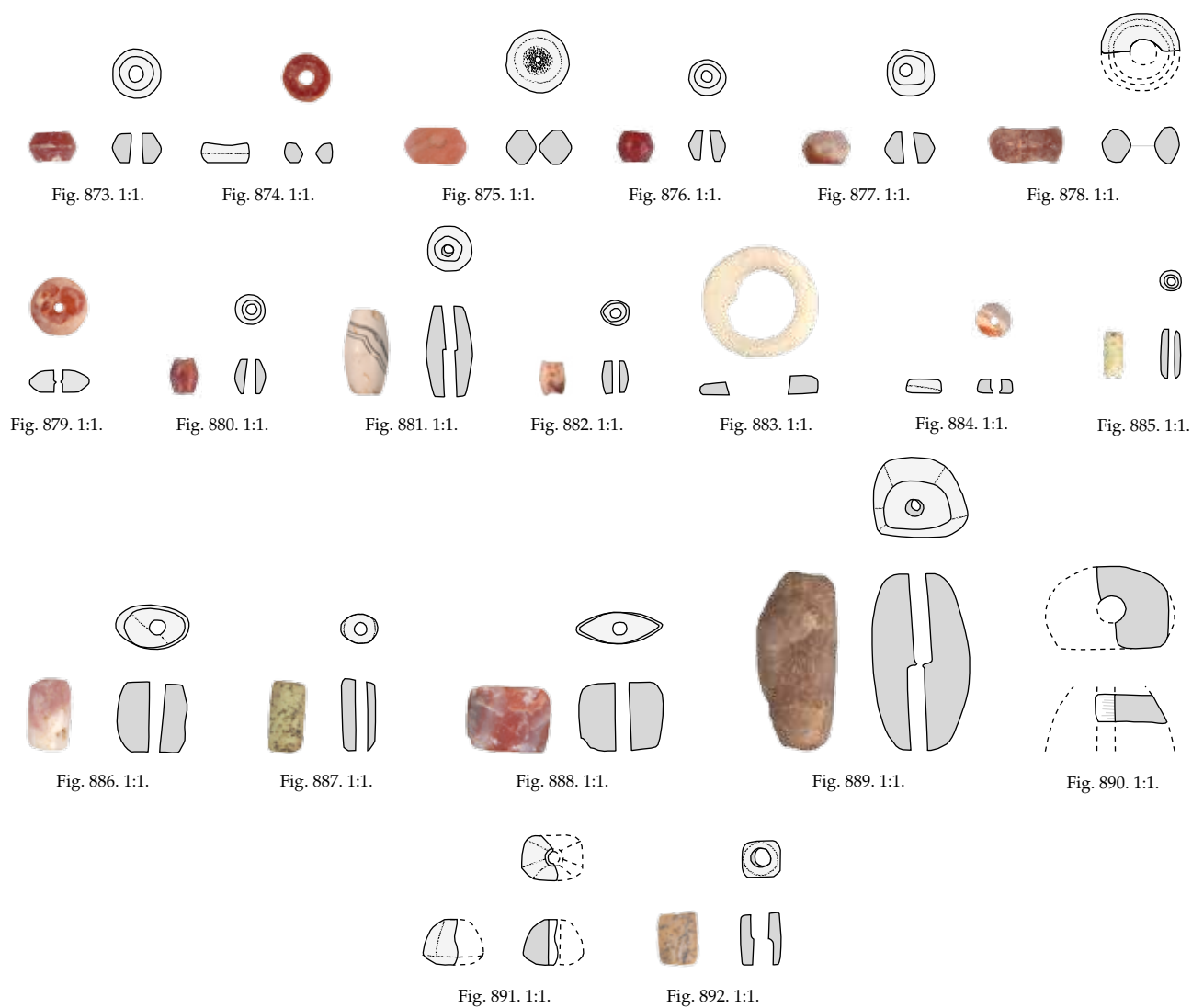
No. C179-X630 (fig. 1016). *Context:* Trench C, layer C179, Disturbed. *Type:* Drop pendant; XXII.B.2.a. *Size:* L 9.70; H 5.90; W 10.40 mm. *Material:* Jasper. *Perforation:* Type III; PD 0.95-1.85 mm.

No. C146-X463 (fig. 1017). *Context:* Trench C, layer C146, Disturbed. *Type:* Unidentified. *Size:* Pres.L 3.70; Pres.D 5.90; Pres.ED 3.40 x 5.00 mm. *Material:* Carnelian (discoloured). *Perforation:* Not preserved; PD 2.10 mm.

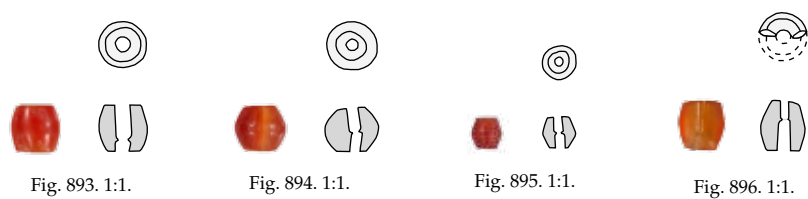
Unknown

No. G(2)1-X11 (fig. 1018). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 3.50; D 5.10-5.30; ED 3.25-4.20 mm. *Material:* Carnelian. *Perforation:* Type III; PD 1.50-1.90 mm.

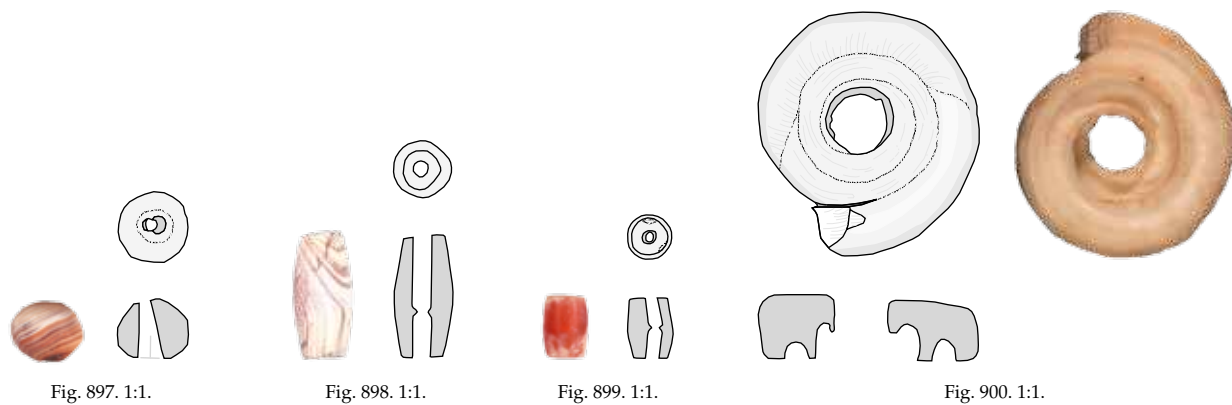




Figs. 832-892. Beads from Phase 1.



Figs. 893-896. Beads from Phase 3.



Figs. 897-902. Beads from Phase 4.

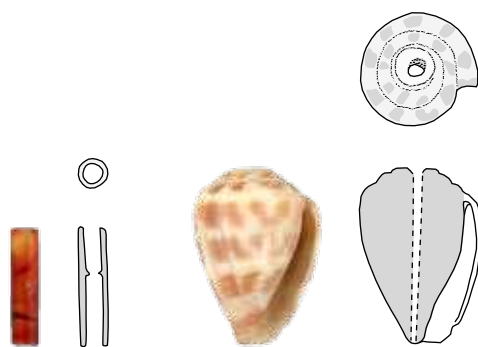


Fig. 901. 1:1.

Fig. 902. 1:1.

Figs. 897-902. Beads from Phase 4.

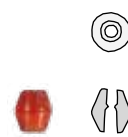


Fig. 903. 1:1.

Figs. 903. Beads from Phase 5.

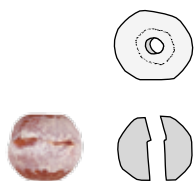


Fig. 904. 1:1.



Fig. 905. 1:1.



Fig. 906. 1:1.



Fig. 907. 1:1.

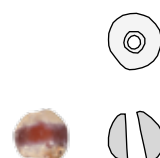


Fig. 908. 1:1.



Fig. 909. 1:1.

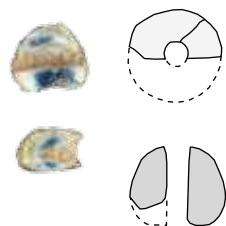


Fig. 910. 1:1.

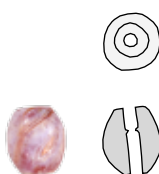


Fig. 911. 1:1.



Fig. 912. 1:1.



Fig. 913. 1:1.



Fig. 914. 1:1.

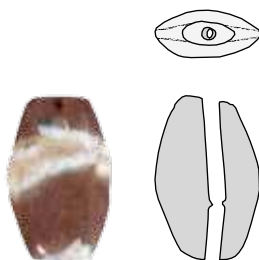


Fig. 915. 1:1.



Fig. 916. 1:1.

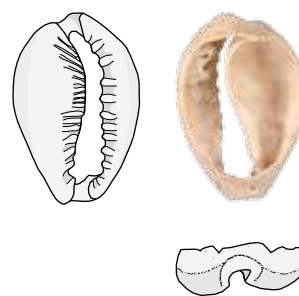


Fig. 917. 1:1.

Figs. 904-917. Beads from Phase 7.

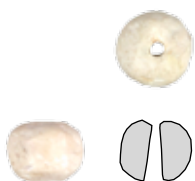


Fig. 918. 1:1.

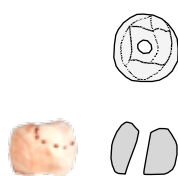


Fig. 919. 1:1.



Fig. 920. 1:1.



Fig. 921. 1:1.

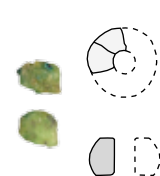


Fig. 922. 1:1.

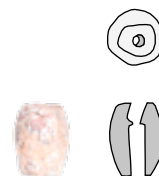


Fig. 923. 1:1.

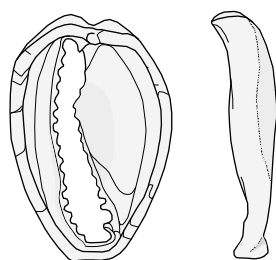
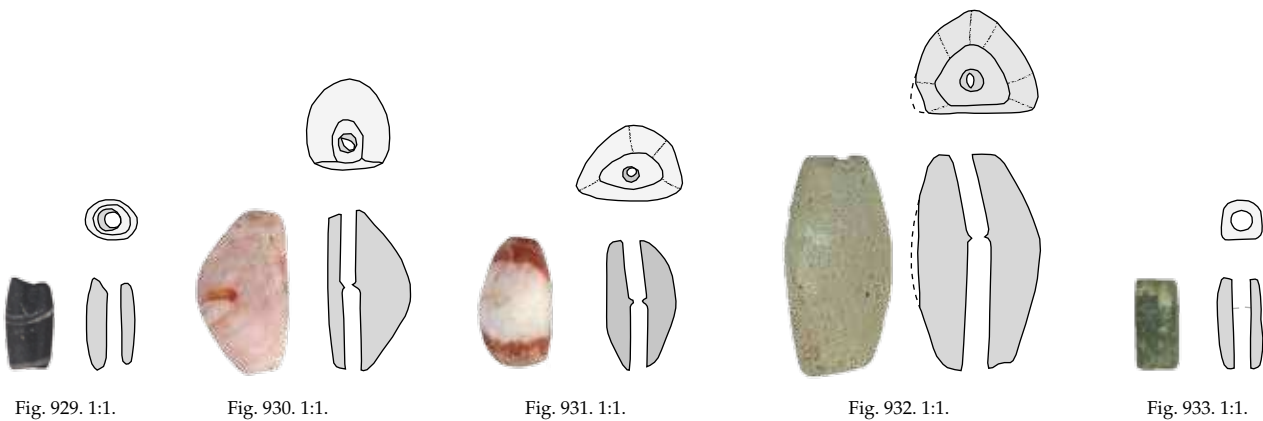
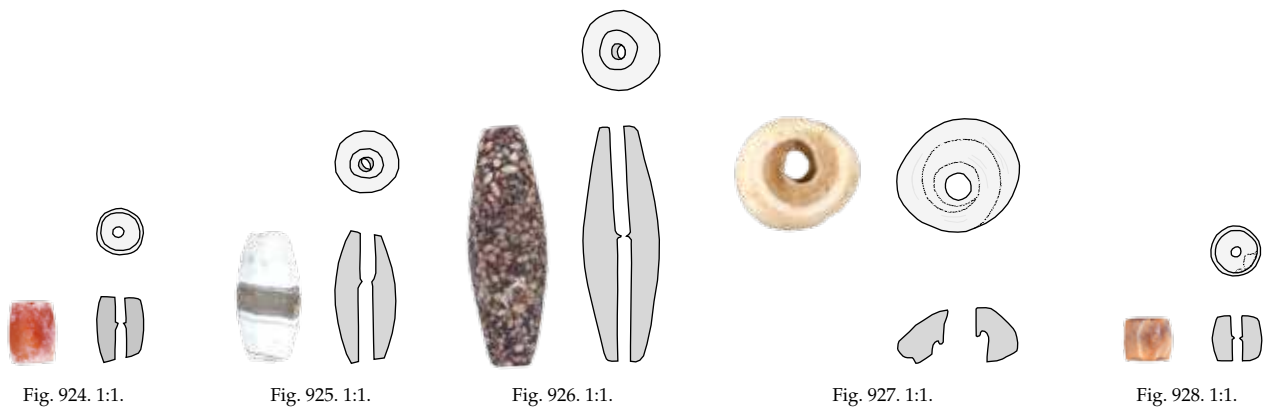
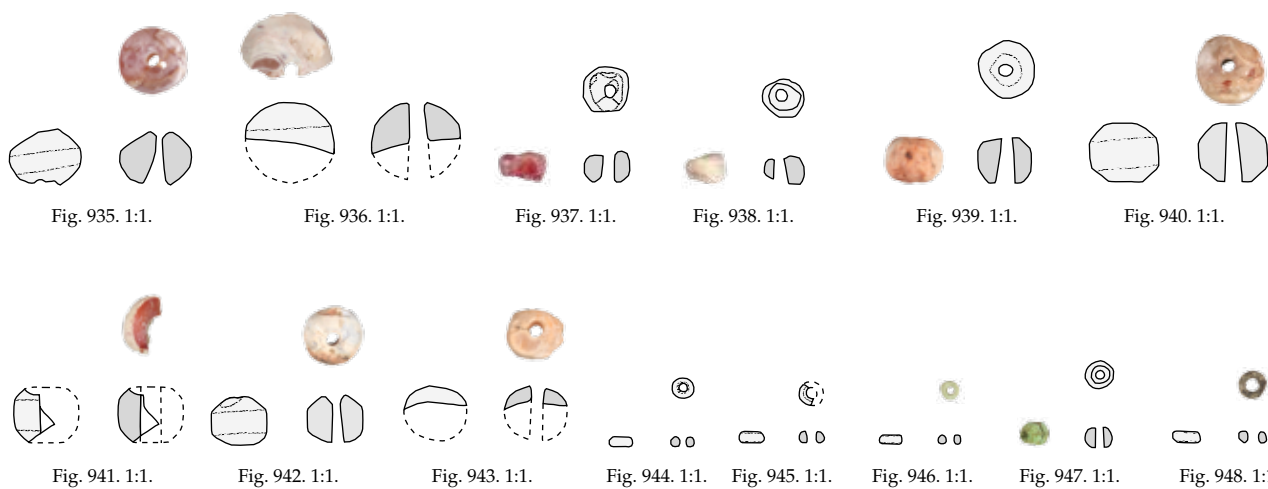
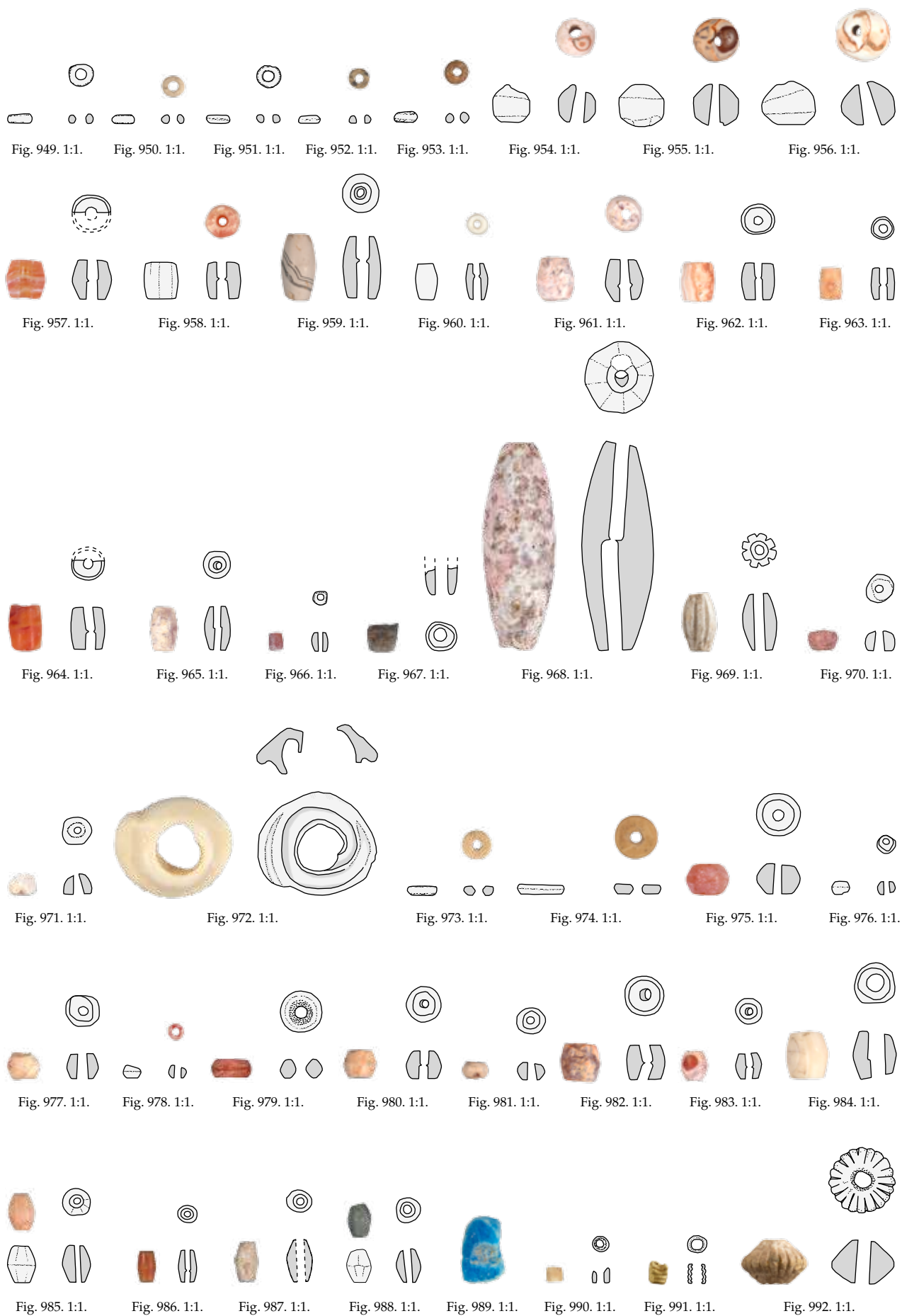


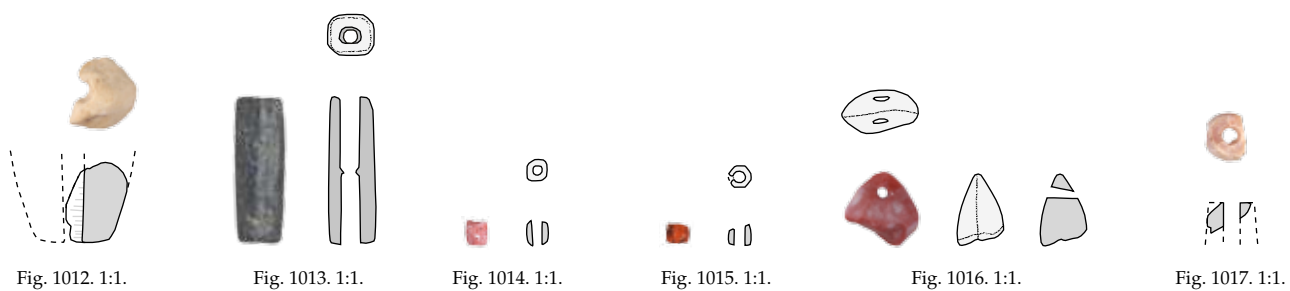
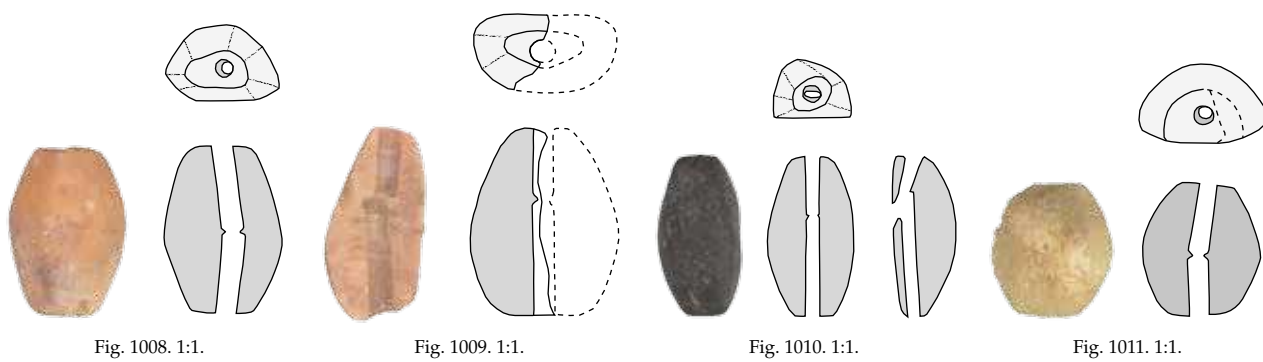
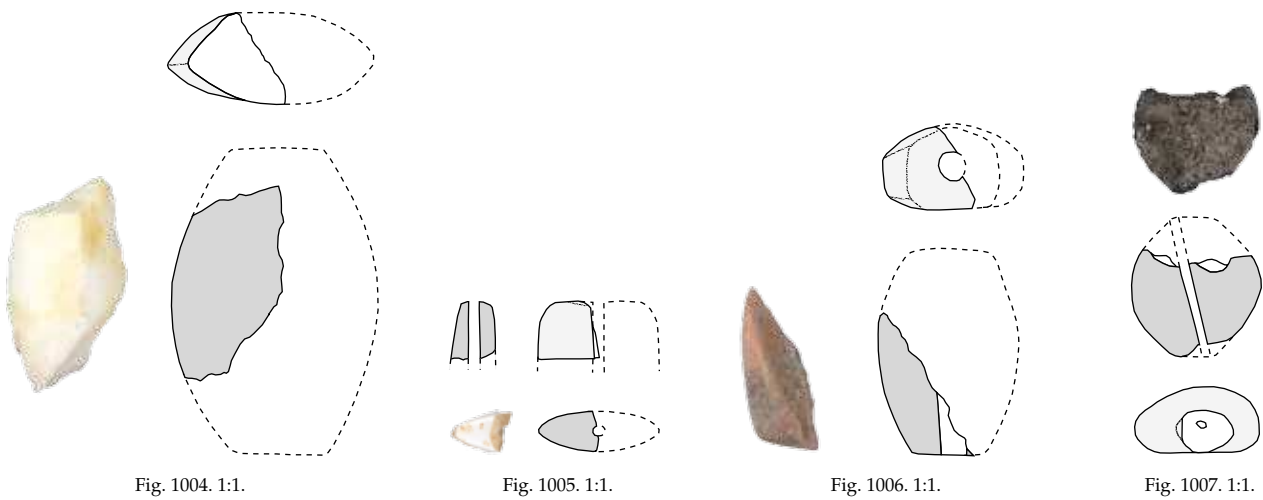
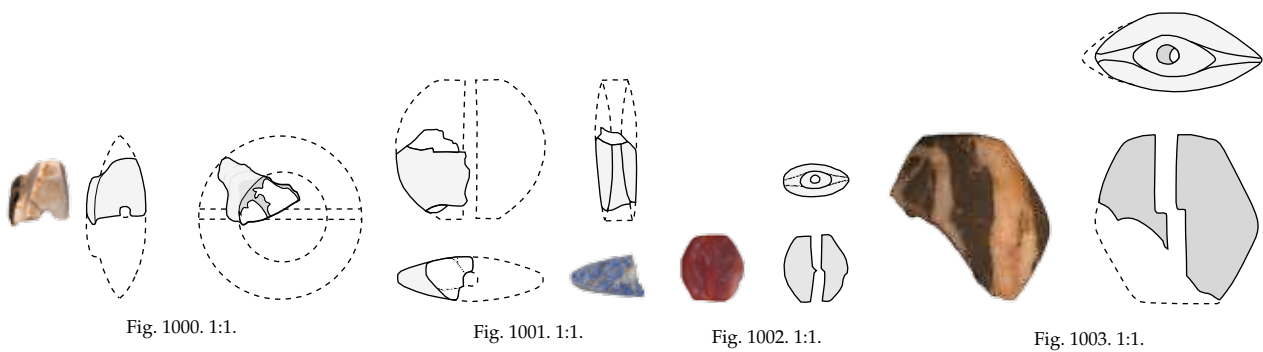
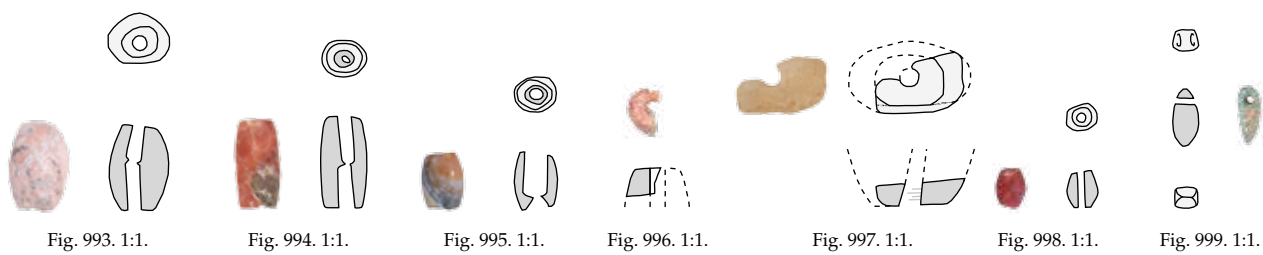
Fig. 934. 1:1.

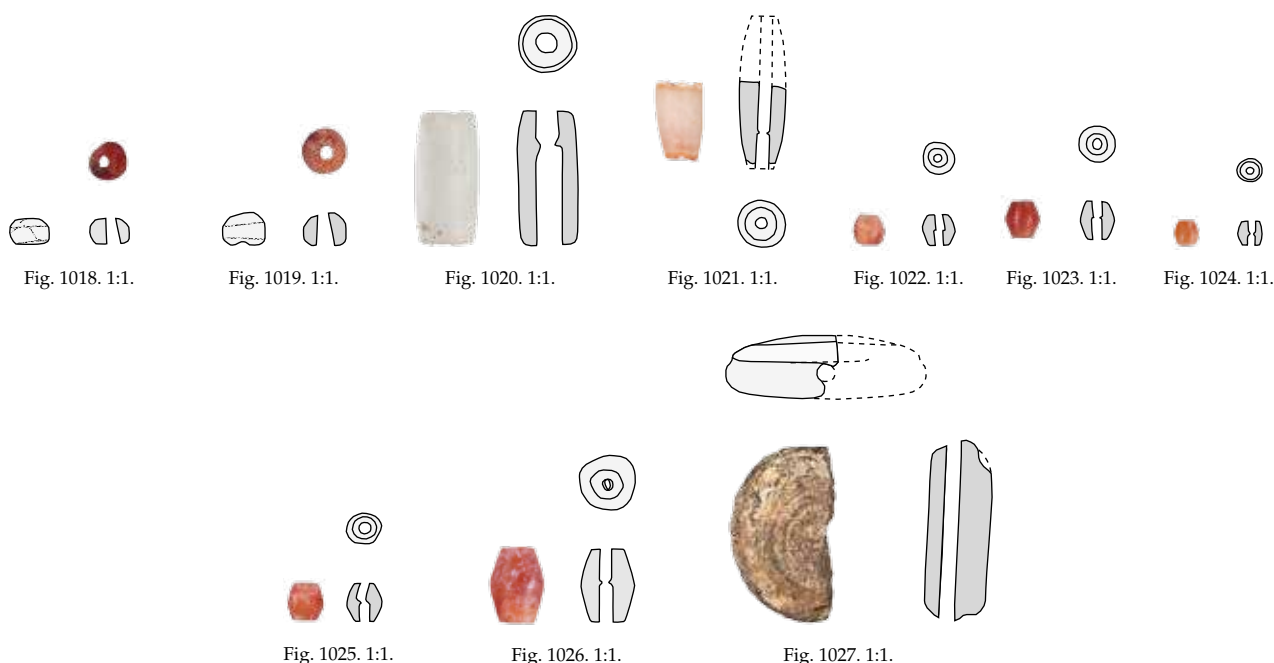
Figs. 918-934. Beads from Phase 8.



Figs. 935-1017. Beads from disturbed contexts.







Figs. 1018-1027. Beads from unknown contexts.

No. G(2)1-X14 (fig. 1019). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Oblate/Circular; I.B.1.a. *Size:* L 4.15; D 5.90; ED 3.25-3.90 mm. *Material:* Carnelian. *Perforation:* Type III; PD 1.50-2.15 mm.

No. D1-X1 (fig. 1020). *Context:* Trench D; Unknown. *Type:* Long barrel/Circular; I.D.1.b. *Size:* L 17.40; D 7.90-8.20; ED 6.40-6.60 mm. *Material:* Rock crystal. *Perforation:* Type II; PD 2.80-3.20 mm.

No. G(1)-X5 (fig. 1021). *Context:* Trench G1, layer G(1)1, Unknown. *Type:* Long barrel/Circular; I.D.1.b. *Size:* Pres.L 10.80; D 6.30; Pres.ED 4.20-4.45 mm. *Material:* Agate. *Perforation:* Type II; PD 1.50-1.60 mm.

No. G(1)-X4 (fig. 1022). *Context:* Trench G1, layer G(1)1, Unknown. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 4.00; D 4.20; ED 2.50-2.80 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.10-1.30 mm.

No. G(2)1-X12 (fig. 1023). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 4.90; D 4.60; ED 3.00 mm. *Material:* Carnelian. *Perforation:* Type II; PD 1.45 mm.

No. G(2)1-X13 (fig. 1024). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 3.25; D 3.25; ED 2.25-2.80 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.00 mm.

No. G(2)1-X17 (fig. 1025). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Standard truncated convex bicone/Circular; I.C.1.f. *Size:* L 4.80; D 4.50-5.20; ED 3.00-3.30 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.30-1.45 mm.

No. G(2)1-X10 (fig. 1026). *Context:* Trench G2, layer G(2)1, Unknown. *Type:* Long truncated convex bicone/Circular; I.D.1.f. *Size:* L 9.90; D 7.00-7.30; ED 4.20-4.60 mm. *Material:* Carnelian (discoloured). *Perforation:* Type II; PD 1.70-1.90 mm.

No. I1-X7 (fig. 1027). *Context:* Trench I, layer I1, Unknown. *Type:* Circular/Plano-convex; V.C.1.a. *Size:* L 23.20; Pres.W 14.10; Pres.H 8.55 mm. *Material:* Glass. *Perforation:* Not preserved; Est.PD 2.70 mm.

3.7.2. Bead Drill Hole SEM Analysis

By Jonathan Mark Kenoyer

The drill hole impressions of stone beads recovered from the Kuwaiti-Danish Archaeological Mission to Failaka have been studied using Scanning Electron Microscopy to determine the nature of drilling and other aspects of bead production. For a general study of the beads recovered during the 2008-2012 excavations in Tell F6 on Failaka see Andersson above p. 176. Information on provenience, general context and chronology (fig. 5), as well as photographs, drawings and some overall measurements were provided to the author by Ann Andersson and Flemming Højlund. The bead drill hole impressions were taken by Conservator Helle Strehle and the SEM study was conducted by the author in the SEM laboratory at the Department of Animal Sciences, University of Wisconsin, Madison, USA.

A table with all of the basic data is provided below (fig. 1049), along with narrative discussion of each bead and its drill hole. The standard bead terminology developed on the basis of the bead type descriptions proposed by Beck (1928) is used along with modifications resulting from comparative bead research in the Indus region and Mesopotamia. Basic bead measurements used are presented in fig. 1028 and drill types are presented in fig. 1029. The drill hole impressions were studied using the Scanning Electron Microscope and photographs of specific parts of the drill impression were taken for documenting the nature of the drilling. The magnification used for the photographs was 300X, 100X, 50X, and 25X or lower. This allows for detailed study of the drilled surface at the higher magnification, and an overview of the drill hole and its general features at lower magnification. Both high

and low magnifications are necessary to determine the nature of the drilling and to interpret the drill type that was used in perforation.

The beads from Failaka appear to have been produced using multiple different technologies, which can be interpreted in various ways. Since there are no sources of carnelian on the island and there is no report of local production, all of the beads were probably brought to the site in finished form. The carnelian used to make the beads could have come from any of the major sources in Egypt (Harrell 2012), Arabia (Vogt and Franke-Vogt 1987, Inizan *et al.* 1992, Guba 2007, Brunet 2009), Anatolia and Iran (Moorey 1985), or the Indus (Kenoyer *et al.* 1991, Law 2011). However, a study of the manufacturing technology, and particularly the drilling of the beads, can provide a general idea about the possible sources of the beads and also their use life. The most obvious interpretation is that the beads made with different shaping and drilling technologies were produced in different workshops that may be separated spatially or chronologically. On the basis of ongoing research conducted on bead manufacture and drilling, it is possible to suggest some of the general areas of production and also some more specific areas (Gwinnett and Gorelick 1981, Gwinnett and Gorelick 1987, Kenoyer and Vidale 1992, Gwinnett and Gorelick 1998, Kenoyer 2003, Ludvik, Kenoyer *et al.* 2015).

There are three basic types of drilling represented in the sample of 19 beads from the current excavation analysed here: stone drills, solid copper drills and tubular copper drills (fig. 1029). Pecked drilling (fig. 1029:1) is, however, present in the larger collection of

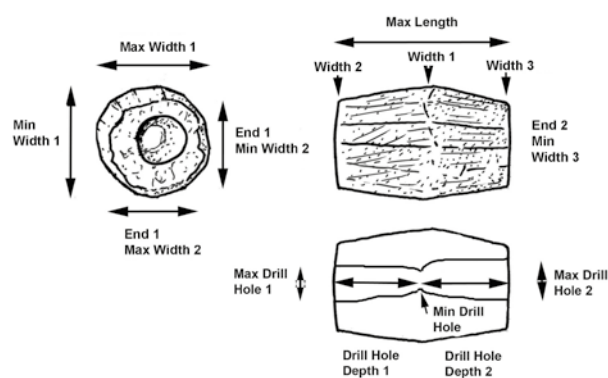


Fig. 1028. Major bead measurements.

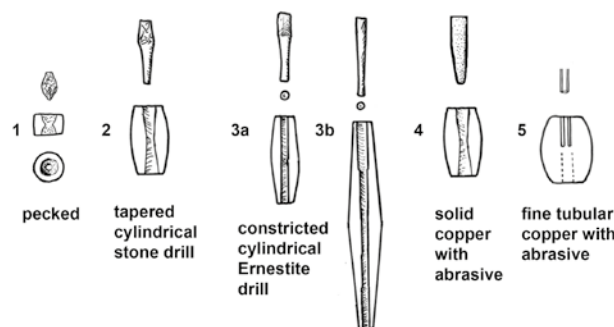


Fig. 1029. Drill Types.

beads analysed by Andersson (see above p. 176) and is relatively common at sites in Mesopotamia, Arabia and the Indus Valley region.

The stone drills can be divided into tapered cylindrical drills and constricted cylindrical drills (Kenoyer and Vidale 1992, Kenoyer 2005). The copper drills were used with some form of abrasive, possibly emery or corundum (Kenoyer ongoing experimental studies). The drilling was sometimes done from two sides of the bead, generally meeting in the center of the bead, but some beads were drilled to a greater depth on one side than the other. The alignment of the two drill holes is sometimes perfectly centered, but in many beads the alignment is off center and some barely connect. In some examples the bead was drilled from one side only and the opposite end popped off due to the pressure of the drilling. In some beads the trajectory of the drill hole is very straight and demonstrates a very stable drilling apparatus, while in other beads the drill hole shows many irregularities that come from lateral movement of the drill. The regular and even drill holes may indicate the use of a frame for holding the drill and the bead would also have been held in a vise. The irregular drilling may represent less stable drilling apparatus such as hand held drilling or an unstable frame. All of these bead drilling techniques and styles are well documented from other sites in the Indus Valley region or Mesopotamia. On the basis of wear in the drill hole or the ends of the beads, it appears that some of the beads were used and curated for some time before passing into the archaeological record. However, some beads have no evidence for string wear or use-wear inside the drill hole or at the ends. This means that they were brought to the site directly from production areas without long use. It is also possible that some beads were strung with large knots on either side that would restrict the movement of the thread and this could be the explanation for the lack of wear on the interior drill hole.

The broader implications of the types of beads and their presence at the site will need to be determined by careful analysis of other artifacts that they were found with and their context. The following section will focus on the types of bead drilling and manufacture represented in this sample and detailed descriptions of each bead.

Stone Drilling: Tapered Cylindrical Drill (fig. 1029:2)

The use of stone drills to perforate agate and carnelian in the ancient world is well attested from Egypt

to China and involves the use of a hard form of jasper or chert that is flaked and retouched to prepare a long tapering drill bit (Kenoyer 2003). In the course of drilling the flake scars are worn away and the shape of the drill becomes a long tapered cylinder. The tip of the drill often develops a slight depression or dimple as the result of drilling pressure. This type of drill would have been hafted on the end of a wooden rod and turned with a bow drill. Drilling by hand turning is a possibility for softer stone beads, but for carnelian and banded agate a bow drill and steady pressure on the drill are required for perforation to proceed efficiently. The bead would have been fixed in some sort of vise and the drill had to be held very steady to avoid snapping the stone bit due to lateral movement in the process of drilling. The degree of lateral movement is often visible in the irregular pattern of drilling striae indicating slight changes in the angle of the drill bit relative to the bead.

Beads that have been drilled with tapered stone drills have tapering drill holes with undulating circular striae that have a very smooth surface. The angle of drilling is usually somewhat irregular and this results in some irregularly aligned striae. The tips of the drills are slightly rounded but in some cases the negative impression of the dimple is visible in the drill impression. Tapered cylindrical stone drills with dimples on the tip have been found at sites in Egypt (Stocks 2003), Mesopotamia (Moorey 1994), Iran (Piperno 1973, Gwinnett and Gorelick 1981) and the Indus region (Barthélemy de Saizieu 2003, Kenoyer 2005). This drilling technology is distinct from the constricted cylindrical drills used exclusively by the Indus drillers.

Stone Drilling: Constricted Cylindrical Drill (fig. 1029:3a)

The constricted cylindrical drill is a very specialized form of drill developed by bead makers of the Indus Civilization using a unique stone raw material that is called *Ernestite* (Kenoyer and Vidale 1992). The actual source of this stone is not known, but it was obtained from somewhere in the southern Indus Valley, either in Gujarat or Balochistan (Law 2011). This type of drill was specially fashioned using a combination of sawing, chipping and grinding to create a drill bit that is wider at the tip with a narrow shaft that then widens at the base where the drill is hafted (Prabhakar, Bisht *et al.* 2012).

Only one of the beads appears to have been drilled with a very distinctive technology using a constricted

cylindrical stone drill that is diagnostic of the Indus civilization. This bead shape also conforms to the shapes found at sites in Gujarat and the Indus Valley as a whole. The bead was discovered in layers of the site dating from what would correspond to the Late Harappan period, circa 1900-1300 BC in the Indus region, when this drilling technique may have been discontinued in the Indus. This suggests that the bead may have been curated or traded at a later time and represents an old bead that has been passed down for many hundreds of years.

Solid Copper Drill (fig. 1029:4) and Tubular Copper Drill (fig. 1029:5)

All of the other beads from the collection appear to have been drilled using solid or tubular copper drills and emery (corundum and other minerals) abrasive. In this part of the ancient world, emery (corundum, Moh's hardness 9) would have been available from Anatolia (e.g. Izmir, Turkey) (Gorelick and Gwinnett 1983) and in the Eastern Mediterranean on the island of Naxos (Heimpel *et al.* 1988). Emery and other varieties of corundum are available in numerous locations in South Asia, with the most well known sources being in Central India (Rewa District, Madhya Pradesh) (Ball and Holland 1898). The coarse drilling of the beads from Failaka is identical to experimental drilling using emery abrasives that has been carried out by the author using micro tubular and solid copper drills on agate, carnelian and jade. This type of drilling is also seen in beads from Troy, Anatolia and Eastern Mediterranean region (Ludvik, Kenoyer *et al.* 2015) and therefore this technology may be more closely related to those regions rather than the Indus. Tubular drilling of beads with micro-tubular drills using a fine abrasive is documented in Late Harappan Period in the Indus, but the abrasive does not appear to be the same coarseness as is seen in the drilling of the beads from Failaka and was possibly quartz rather than emery.

Bead and Drilling Description

C117-X298, Phase 1, c. 2200-2000 BC (figs. 856, 1030a-b). Long barrel, mottled light red-orange carnelian bead. It has a tapered cylindrical inverted biconical drill hole perforated from two sides. Based on the SEM analysis it appears to have been drilled with a tapered cylindrical stone drill. The drilling depth from one side is deeper than the other. It also appears that the drill hole was started on both sides using a slightly larger drill bit. The remainder of the hole was perforated with a smaller diameter stone drill. This use of a larger stone drill, followed by a smaller stone drill is not seen on any other beads that have been studied from this site but is known from sites in Oman that the author has recently studied.

C124-X310, Phase 1, c. 2200-2000 BC (figs. 837, 1031a-b). Short irregular spherical rock quartz bead. It is semi-translucent white in color. It has a long tapered drill hole that is drilled from one side and pops out at the opposite side of the bead. This type of drilling indicates that the bead was placed in a vise only once. Based on the SEM analysis that shows a rough abraded surface with irregular drilling striae, it appears to have been drilled with a tapered solid copper drill using emery abrasive. The drilling surface is smoothed from string wear at the narrow end of the bead hole, which suggests that this bead was used for a very long time before it was lost or buried.

C164-X530, Phase 1, c. 2200-2000 BC (figs. 880, 1032a-b). Long Biconical, red-orange carnelian. It has a long tapered conical drill hole that was drilled from one side and popped out at the opposite end. It appears to have been drilled with a solid copper drill with abrasive. The abraded surface does not show a lot of wear from the string, popped end does not show heavy wear.

C181-X626, Phase 1, c. 2200-2000 BC (figs. 842, 1033a-b). Spherical, banded red-orange carnelian with whitened surface. Long tapered inverted biconical drilling, stepped with collar on one side, drilled from two sides. Tubular copper with emery abrasive, drill holes not aligned, deeper from one side than the other, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill.

C187-X660, Phase 1, c. 2200-2000 BC (figs. 858, 1034a-c). Long Barrel, light pink carnelian. Tapered inverted biconical drilling, from two sides, tapered cylindrical stone drill, drilling depth from one side deeper than the other, tapered stone drill with collar, possibly from chipped drill tip fragment, smaller drill tip at final drilling on both sides. Irregular stepping from lateral movement of the drill during the drilling process, smooth surface characteristic of stone drilling but slight collar is unique and may be from a buildup of drill tip slurry. Does not show heavy wear from string abrasion in the narrow drill hole.

C187-X661, Phase 1, c. 2200-2000 BC (figs. 834, 1035a-c). Short Irregular Spherical, Red-orange carnelian. Long tapered conical drilling, popped out at one end, from one side, solid copper with abrasive, abraded surface does not show a lot of wear from the string, popped end does not show heavy wear.

C198-X712, Phase 1, c. 2200-2000 BC (figs. 876, 1036a-c). Short Biconical, red-orange carnelian. Long tapered cylindrical drilling, popped out at one end, from one side, tapered cylindrical stone drill, polished surface does not have string wear at the edges of the drill hole or where the drill tip popped out, smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling.

C211-X769, Phase 1, c. 2200-2000 BC (figs. 877, 1037a-b). Short Biconical, red-orange carnelian. Long tapered conical drilling, popped out at one end, drilled from one side, solid copper with abrasive, abraded surface does not show a lot of wear from the string, popped end does not show heavy wear.

E184-X912, Phase 3, c. 1800 BC (figs. 894, 1038a-b). Short Biconical, red-orange carnelian. Tapered cylindrical inverted biconical drilling, drilled from two sides, tapered cylindrical stone drill, smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling, drill holes not aligned but do meet in the middle.

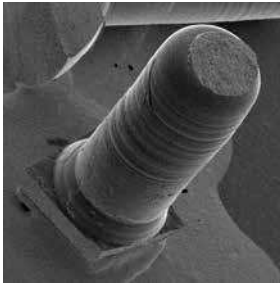


Fig. 1030a. 20x.



Fig. 1030b. 300x.

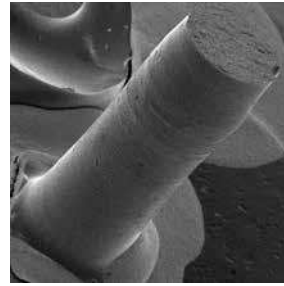


Fig. 1031a. 22x.

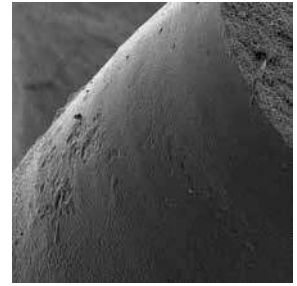


Fig. 1031b. 100x.

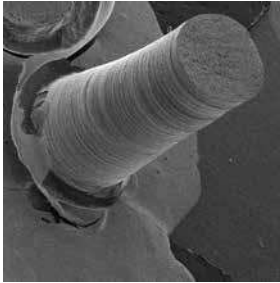


Fig. 1032a. 20x.

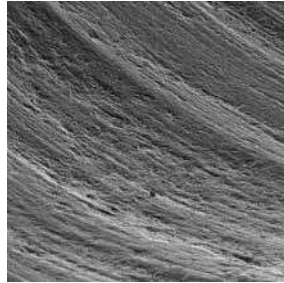


Fig. 1032b. 300x.

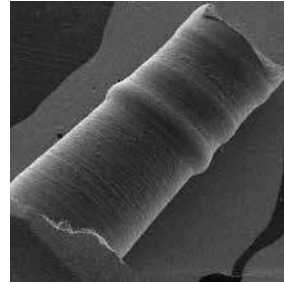


Fig. 1033a. 20x.

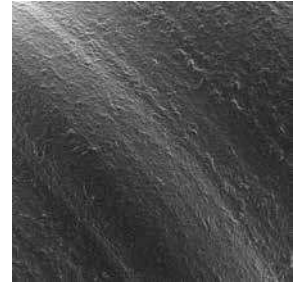


Fig. 1033b. 100x.

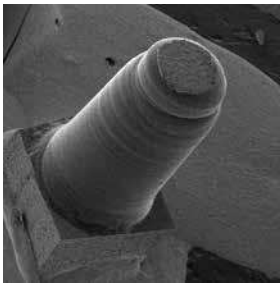


Fig. 1034a. 25x.

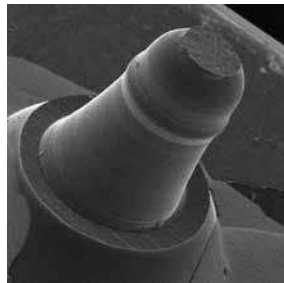


Fig. 1034b. 25x.

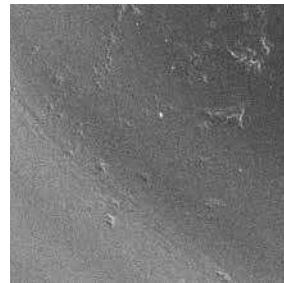


Fig. 1034c. 300x.

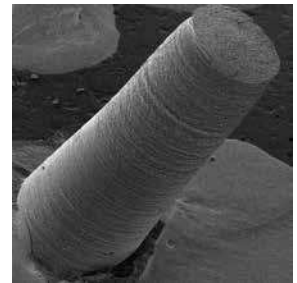


Fig. 1035a. 20x.

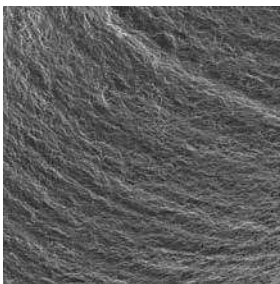


Fig. 1035b. 100x.

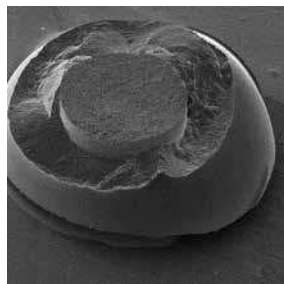


Fig. 1035c. 25x.

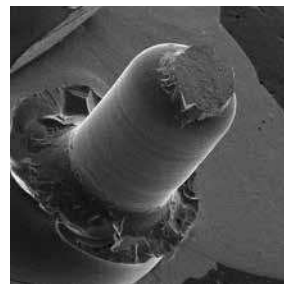


Fig. 1036a. 20x.

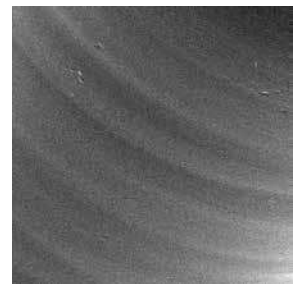


Fig. 1036b. 100x.

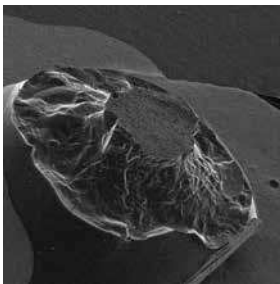


Fig. 1036c. 25x.

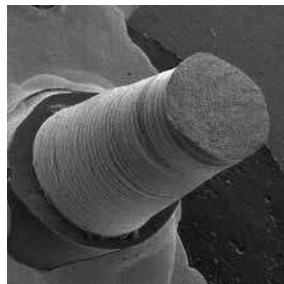


Fig. 1037a. 20x.

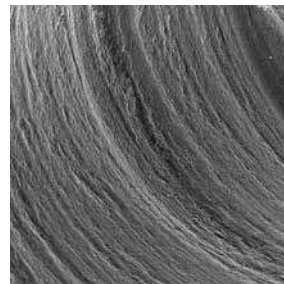


Fig. 1037b. 100x.

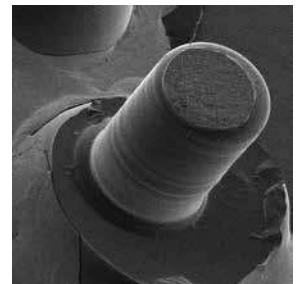


Fig. 1038a. 25x.

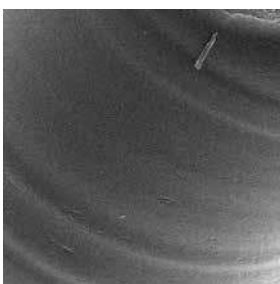


Fig. 1038b. 100x.

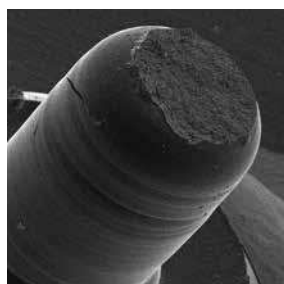


Fig. 1039. 50x.

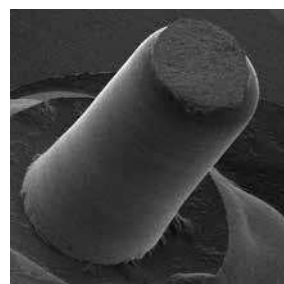


Fig. 1040a. 20x.

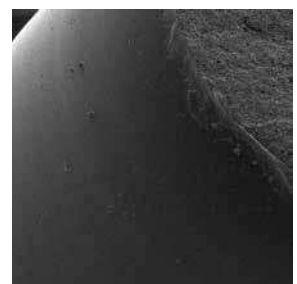


Fig. 1040b. 100x.

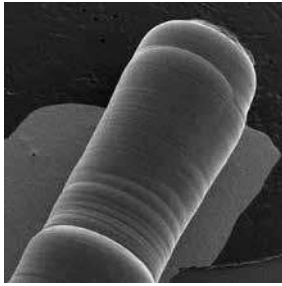


Fig. 1041a. 25x.

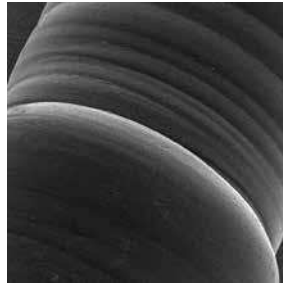


Fig. 1041b. 50x.

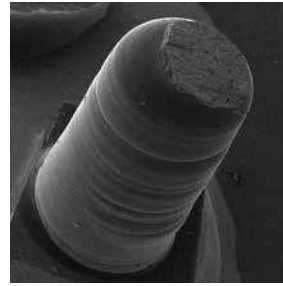


Fig. 1042a. 25x.

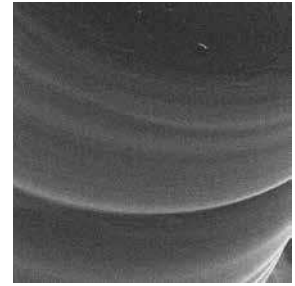


Fig. 1042b. 100x.

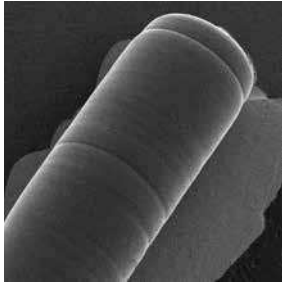


Fig. 1043a. 20x.

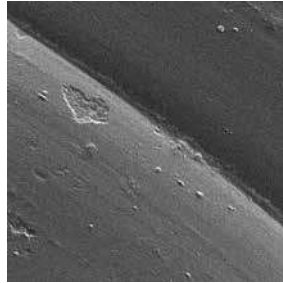


Fig. 1043b. 300x.

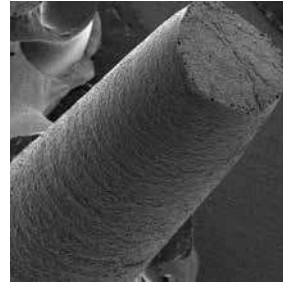


Fig. 1044a. 25x.

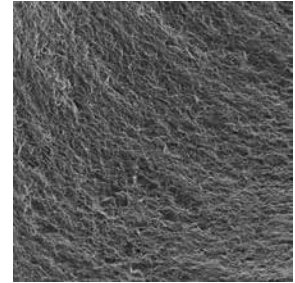


Fig. 1044b. 100x.

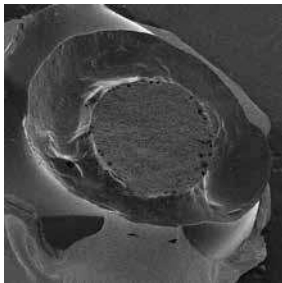


Fig. 1044c. 25x.

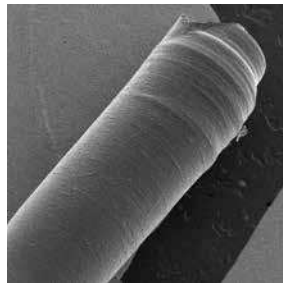


Fig. 1045a. 25x.

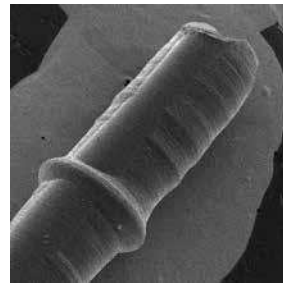


Fig. 1045b. 20x.

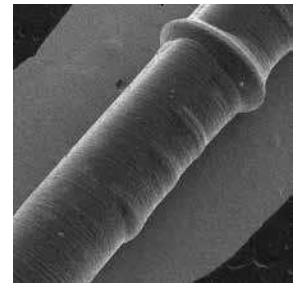


Fig. 1045c. 20x.

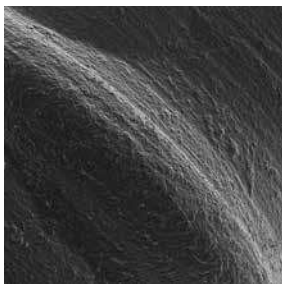


Fig. 1045d. 100x.

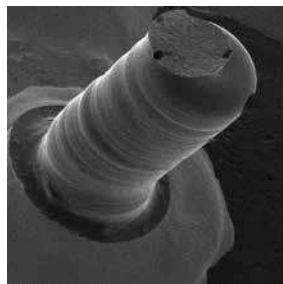


Fig. 1046. 25x.

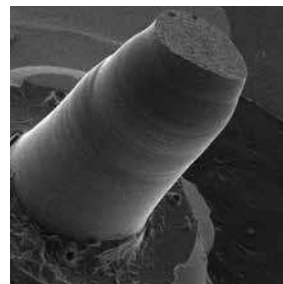


Fig. 1047a. 25x.

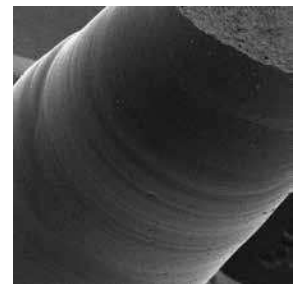


Fig. 1047b. 50x.

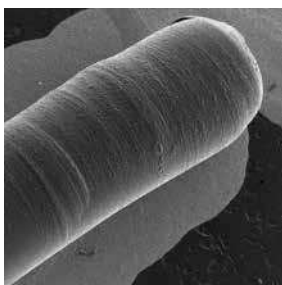


Fig. 1048a. 25x.

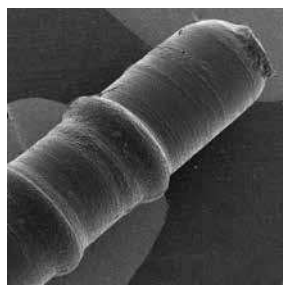


Fig. 1048b. 20x.

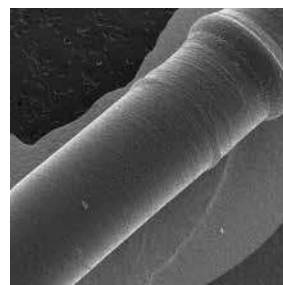


Fig. 1048c. 20x.

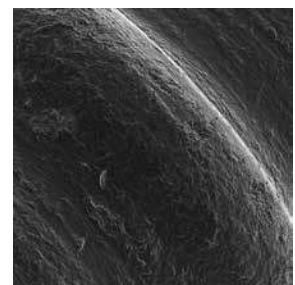


Fig. 1048d. 100x.

Reported	Reported	Reported	Kenoyer	Reported	Kenoyer	Kenoyer	Kenoyer	Kenoyer
Object Number	Context	Bead Shape	Bead Shape	Raw Material	Raw Material	Raw Material Color	Bead Drill hole type	Drilling Directions
A172-X400	Phase 4, c. 1750 BC	Long Barrel/Circular	Long Barrel	Carnelian	Carnelian	banded red-orange, whitened surface	long tapered cylindrical inverted biconical, stepped	from two sides
E200-X1035	Phase 4, c. 1750 BC	Long Cylinder/Circular	Long Cylindrical	Carnelian	Carnelian	banded red-orange	straight cylindrical from both sides, slight stepping	from two sides
E21-X73	Phase 7, c. 1450 BC	Long Barrel/Elliptical	Long irregular elliptical barrel	Jasper	Banded Jasper	Yellow-Brown-Orange-White	straight cylindrical from both sides, with collars on one side	from two sides
E74-X280	Phase 7, c. 1450 BC	Long Barrel/Lenticular	Long Lenticular Barrel	Jasper	Banded Jasper	Yellow-Brown-Orange-White	straight cylindrical from both sides, with collars on one side	from two sides
C117-X298	Phase 1, c. 2200-2000 BC	Long Barrel/Circular	Long Barrel	Carnelian	Carnelian	mottled light red-orange	cylindrical stepped	from two sides
C124-X310	Phase 1, c. 2200-2000 BC	Oblate/Circular	Short Irregular Spherical	Rock Crystal	Rock Quartz	semi translucent white	long tapered conical, popped out at one end	from one side
C164-X530	Phase 1, c. 2200-2000 BC	Long truncated convex bicone/Circular	Long Bicone	Carnelian	Carnelian	red-orange	long tapered conical, popped out at one end	from one side
C181-X626	Phase 1, c. 2200-2000 BC	Circular/Circular	Spherical	Carnelian	Carnelian	banded red-orange, whitened surface	tapered inverted biconical, stepped with collar on one side	from two sides
C187-X660	Phase 1, c. 2200-2000 BC	Long Barrel/Circular	Long Barrel	Carnelian	not determinable from photo	light pink	tapered inverted biconical	from two sides
C187-X661	Phase 1, c. 2200-2000 BC	Oblate/Circular	Short Irregular Spherical	Carnelian	Carnelian	red-orange	long tapered, popped out at one end	from one side
C198-X712	Phase 1, c. 2200-2000 BC	Short truncated convex bicone/Circular	Short Bicone	Carnelian	Carnelian	red-orange	long tapered cylindrical, popped out at one end	from one side
C211-X769	Phase 1, c. 2200-2000 BC	Short truncated convex bicone/Circular	Short Bicone	Carnelian	Carnelian	red-orange	long tapered, popped out at one end	from one side
E184-X912	Phase 3, c. 1800 BC	Short truncated convex bicone/Circular	Short Bicone	Carnelian	Carnelian	red-orange	tapered cylindrical inverted biconical	from two sides
E248-X1148	Phase 3, c. 1800 BC	Standard truncated convex bicone/Circular	Long Bicone	Carnelian	Carnelian	red-orange	tapered cylindrical inverted biconical	from two sides
E147-X827	Phase 4, c. 1750 BC	Long barrel/Circular	Long Barrel	Carnelian	Carnelian	red-orange, whitened surface	long tapered cylindrical inverted biconical, stepped	from two sides
E210-X939	Phase 3, c. 1800 BC	Short barrel/Circular	Short Barrel	Carnelian	Carnelian	red-orange	tapered cylindrical inverted biconical	from two sides
E20-X68	Phase 7, c. 1450 BC	Circular/Circular	Short Irregular Spherical	Carnelian	Carnelian	banded red-orange, whitened surface	long tapered, popped out at one end	from one side
E23-X82	Phase 7, c. 1450 BC	Long barrel/Circular	Long Barrel	Carnelian	Carnelian	banded red-orange	tapered cylindrical inverted biconical	from two sides
A32-X165	Phase 7, c. 1450 BC	Oblate/Circular	Short Spherical	Carnelian	Carnelian	banded red-orange, whitened surface	tapered cylindrical inverted biconical, irregular stepping	from two sides

Fig. 1049. Bead analysis database.

Kenoyer	Kenoyer		Drill Measure- ments	
Drill type	Drill hole comments	Additional Comments	Max Length (mm)	Max width/ dia (mm)
tapered cylindrical stone drill	drilling depth from one side deeper than the other, irregular stepping from lateral movement of the drill during the drilling process, smooth surface characteristic of stone drilling, two sizes of stone drills used in drilling	does not look like Indus drilling	16,50	7,70
constricted cylindrical drill	drilling depth from one side deeper than the other, straight sided, long cylindrical stepping, very smooth surface characteristic of Indus style constricted cylindrical Erenstite stone drill	proably Indus bead, curated	15,60	3,90
Tubular copper with abrasive	drilling depth from one side deeper than the other, redrilled after initial drilling, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill, rough pitted surface distinctive of emery abrasive		18,10	6,60
Tubular copper with abrasive	drill holes not aligned, deeper from one side than the other, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill, rough pitted surface distinctive of emery abrasive		21,50	13,50
constricted cylindrical stone drill	drilling depth from one side deeper than the other, started drilling on both sides with a larger drill bit and then used a smaller drill bit for the rest of the drilling, smooth regular drilling striae, stable drill with cylindrical shape	looks like constricted stone drill, but surface is too undulating and it does not look like Indus drilling	10,70	5,80
solid copper with abrasive	drilling surface is smoothed from string wear at the narrow end of the bead hole, drill hole surface is worn and pitted from post depositional weathering or accretions, but it is still possible to determine the use of abrasive and type of drill		6,40	8,10
solid copper with abrasive	abraded surface does not show a lot of wear from the string, popped end does not show heavy wear, rough pitted surface distinctive of emery abrasive		4,95	4,15
Tubular copper with abrasive	drill holes not aligned, deeper from one side than the other, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill, rough pitted surface distinctive of emery abrasive		8,80	9,50
tapered cylindrical stone drill	drilling depth from one side deeper than the other, tapered stone drill with collar, possibly from chipped drill tip fragment, smaller drill tip at final drilling on both sides, irregular stepping from lateral movement of the drill during the drilling process, smooth surface characteristic of stone drilling but slight collar is unique and may be from a buildup of drill tip slurry, Does not show heavy wear from string abrasion in the narrow drill hole.		5,95	4,90
solid copper with abrasive	abraded surface does not show a lot of wear from the string, popped end does not show heavy wear, rough pitted surface distinctive of emery abrasive		6,05	7,30
tapered cylindrical stone drill	polished surface does not have string wear at the edges of the drill hole or where the drill tip popped out, smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling		4,30	5,60
solid copper with abrasive	abraded surface does not show a lot of wear from the string, popped end does not show heavy wear, rough pitted surface distinctive of emery abrasive		4,55	6,75
tapered cylindrical stone drill	drill holes not aligned but do meet in the middle, smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling, drill holes not aligned but do meet in the middle		5,95	6,70
tapered cylindrical stone drill	smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling, drill holes not aligned but do meet in the middle, two sizes of stone drills used for drilling		4,20	4,15
tapered cylindrical stone drill	drill holes not aligned and slight chipping where the tips of the holes meet, equal drilling from each side, surface smooth, regular drilling striae, stable drill, two sizes of stone drills used for drilling		8,10	5,95
tapered cylindrical stone drill	drilling depth from one side deeper than the other, drill holes are well centered, smooth surface, regular drilling striae, stable drill		6,00	6,70
solid copper with abrasive	abraded surface does not show a lot of wear from the string, popped end does not show heavy wear, rough surface distinctive of emery abrasive		6,70	7,15
tapered cylindrical stone drill	drilling depth from one side deeper than the other, smooth surface, highly irregular drilling striae, undulating ridges, unstable drilling,		8,85	7,70
tapered stone drill	drill holes not aligned, deeper from one side than the other, drilling angles are highly irregular indicating unstable vise or drill angles, traces of redrilling using different size of drill, smooth surface,		8,40	10,00

E248-X1148, Phase 3, c. 1800 BC (figs. 895, 1039). Long Biconical, red-orange carnelian. Long tapered cylindrical inverted biconical drilling, from two sides, tapered cylindrical stone drill, smooth surface characteristic of stone drilling, regular drilling striae show that the drilling apparatus was very stable and very little lateral movement while drilling, drill holes not aligned but do meet in the middle. Two sizes of drills used for drilling.

E210-X939, Phase 3, c. 1800 BC (figs. 893, 1040a-b). Short Barrel, red-orange carnelian. Long tapered cylindrical inverted biconical drilling, drilled from two sides, tapered cylindrical stone drill, smooth surface, regular drilling striae, stable drill, drilling depth from one side deeper than the other, drill holes are well centered.

A172-X400, Phase 4, c. 1750 BC (figs. 898, 1041a-b). Long Barrel, banded red-orange carnelian, whitened surface. Long tapered cylindrical inverted biconical drilling, irregular stepping from lateral movement of the drill during the drilling process, drilled from two sides, tapered cylindrical stone drill, smooth surface characteristic of stone drilling, drilling depth from one side deeper than the other, does not look like Indus drilling. Two sizes of stone drills used in drilling.

E147-X827, Phase 4, c. 1750 BC (figs. 899, 1042a-b). Long Barrel, red-orange carnelian, whitened surface. Long tapered cylindrical inverted biconical, stepped, from two sides, tapered cylindrical stone drill, drill holes not aligned and slight chipping where the tips of the holes meet, equal drilling from each side. Two sizes of drills used for drilling.

E200-X1035, Phase 4, c. 1750 BC (figs. 901, 1043a-b). Long Cylindrical, banded red-orange carnelian. Straight cylindrical from both sides, slight stepping, from two sides, constricted cylindrical drill, drilling depth from one side deeper than the other. The smooth drilling surface and the stepping indicate the use of multiple sizes of Ernestite drills. This appears to be an Indus bead (2600-1900 BC) or made by Indus trained bead makers using Indus carnelian and Indus style Ernestite drills. The bead shape is found in the Indus, but is not that common.

E20-X68, Phase 7, c. 1450 BC (figs. 908, 1044a-c). Short Irregular Spherical, banded red-orange carnelian, whitened surface. Long tapered conical drilling, popped out at one end, drilled from one side, solid copper with abrasive, abraded surface does not show a lot of wear from the string, popped end does not show heavy wear, rough surface distinctive of emery abrasive.

E21-X73, Phase 7, c. 1450 BC (figs. 914, 1045a-d). Long irregular elliptical barrel, Yellow-Brown-Orange-White, banded jasper. Straight cylindrical from both sides, with collars on one side, drilled from two sides, tubular copper drilling with abrasive, drilling depth from one side deeper than the other, redrilled after initial drilling, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill.

E23-X82, Phase 7, c. 1450 BC (fig. 911, 1046). Long Barrel, banded red-orange carnelian. Tapered cylindrical inverted biconical drilling, drilled from two sides, tapered cylindrical stone drill, smooth surface, drilling depth from one side deeper than the other, drill holes not aligned, highly irregular drilling striae, undulating ridges, unstable drilling.

A32-X165, Phase 7, c. 1450 BC (figs. 904, 1047a-b). Short Spherical, banded red-orange carnelian, whitened surface. Tapered cylindrical inverted biconical, irregular stepping, from two sides, tapered stone drill, smooth surface, drill holes not aligned, deeper from one side than the other, drilling angles are highly irregular indicating unstable vise or drill angles, traces of redrilling using different size of drill.

E74-X280, Phase 7, c. 1450 BC (figs. 915, 1048a-d). Long Lenticular Barrel, Yellow-Brown-Orange-White banded jasper. Straight cylindrical drilling from both sides, with collars on one side, drilled from two sides. Tubular copper with abrasive, drill holes not aligned, deeper from one side than the other, the long side has the collared projections, the other side may have been drilled with a solid copper drill or a collapsed tubular drill, rough pitted surface distinctive of emery abrasive.

3.8. Copper

by Flemming Højlund

Only a few fragments of crucibles and moulds were found in the 2008-2012 excavations in Tell F6, and they are all mentioned below. They testify to copper working activities outside the excavated area. The copper-based metal objects, of which a selection is presented below, are heavily corroded, fragile and fragmented and identification of form and function is rarely possible.

Crucibles and moulds

Phase 2

Base part of large crucible (fig. 1050); slightly curved piece of reddish clay, tempered with some sand and a lot of straw, one side of which is heavily vitrified, with a lump of copper attached (E225-X964) (cf. Højlund & Andersen 1994 p. 370-375. Andersen & Højlund 2003 p. 280-282, figs. 748-751).

Phase 3

Three fragments of two or three different moulds; one straight side (fig. 1051) and two corners (fig. 1052), ware heavily tempered with straw and sand, light-brown surface, reddish below surface and a black, non-oxidized, core (A218-X750). An extra 2-3 mm layer of clay has been applied to the outer surface of one of the moulds, probably a repair. The inner bottom of the straight-sided fragment has a grey discolouration extending c. 1 cm up the side as seen also on moulds from Qala'at al-Bahrain and the Barbar Temple (Højlund & Andersen 1997 figs. 88-90. Andersen & Højlund 2003 figs. 752-756).

Fragment of object in burnt clay (fig. 1053); a straw and sand-tempered ware, dark non-oxidized core, surface darkened as if by fire; resembles a mould, but less heavily tempered with straw and sand (E157-X840).

Phase 4

Six lumps of vitrified clay, identical to the vitrified part of the crucible fragment mentioned above. The maximum dimensions of the lumps vary from 1.9 to 3.5 cm (A129-X460).

2.8 cm thick fragment of mould produced from a sand and straw-tempered ware with a black core, light brown outer surface, partially preserved on upper side, 4 x 6 cm (E137-X647).

Copper objects

Phase 1

Two triangular pieces of sheet, 2 mm thick (figs. 1054-1055) (C135-X390).

Phase 2

An almost complete dagger (fig. 1056), found in Trench A west of wall A191 (fig. 40). The dagger is 25.8 cm long with pronounced shoulders and it has two rivet holes in the tang (A215-X748). The handle, including the layer of corrosion embedded with sand, has a thickness of 9 mm; originally, it was perhaps 5 mm. The blade was flat, without a midrib, the thickness was c. 8 mm including corrosion embedded with sand and originally perhaps 3 mm. It was bent in antiquity and broke during excavation. The shape of the tang and blade is common for the 2nd millennium (Yule 2001 vol. I, p. 54-55, fig. 5.3.1, D04 class). An ivory handle (fig. 1129) was found near this and the dagger mentioned below, but does not fit either.

Dagger blade (fig. 1057), 12 cm long, 5.2 cm wide, 0.6 cm thick (A336-X860).

Chisel (fig. 1058), 8.2 cm long, cross-section 1.3 x 1.5 cm (A336-X859) (cf. Killich & Moon 2005 fig. 5.2g-j).

Folded sheet of with an elaborate decoration in raised ridges (fig. 1059) (A220-X786).

Phase 3

Fragment of fishhook, 2.3 x 1.6 cm span (fig. 1060) (E198-X888) (cf. Killich & Moon 2005 figs. 5.3g-m. Benediková ed. 2010 figs. 54-56).

Point of dagger blade, 5 cm long, 3.3 cm wide, 0.9 cm thick (fig. 1061) (A209-X698).



Fig. 1050. 1:2.

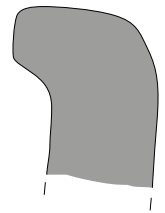
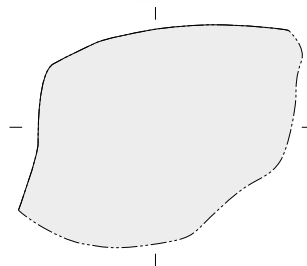


Fig. 1052. 1:2.

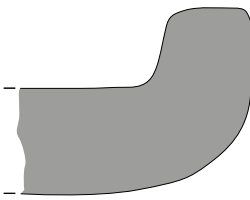


Fig. 1051. 1:2.

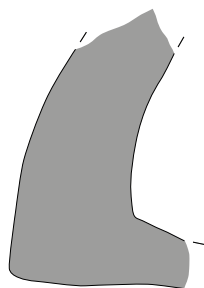


Fig. 1053. 1:2.

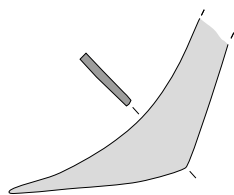


Fig. 1054. 1:2.

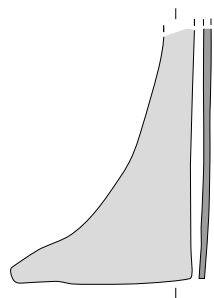


Fig. 1055. 1:2.



Fig. 1056. 1:2.



Fig. 1057. 1:2.

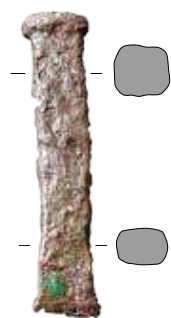


Fig. 1058. 1:2.

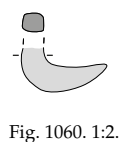
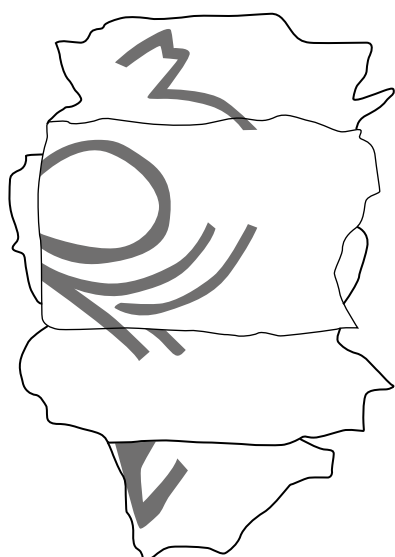


Fig. 1060. 1:2.



Fig. 1061. 1:2.



Fig. 1062. 1:2.



Fig. 1065. 1:2.



Fig. 1063. 1:2.



Fig. 1064. 1:2.

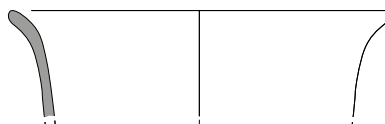


Fig. 1066. 1:2.

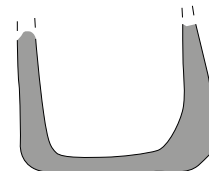


Fig. 1067. 1:2.



Fig. 1059. 1:1.



Fig. 1068. 1:2.



Fig. 1069. 1:2.

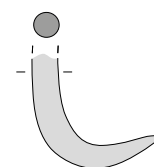


Fig. 1070. 1:2.

Phase 4

Tanged arrowhead with barbs (one broken off) (fig. 1062) (E172-X805).

Pointed pin, 8 cm long with round section, 0.6 cm thick (fig. 1063) (E165-X728).

Pointed pin, c. 6 cm long with round section, 0.5 cm thick (fig. 1064) (A172-X388).

Pointed pin, 12 cm long, 3 mm thick, broken (fig. 1065) (E130-X562).

Phase 7

Simple vessel rim with a diameter of 10 cm (fig. 1066) (E12-X49).

Square mounting (fig. 1067) (E69-X250).

Bent piece of sheet, 8 x 3 cm, 2 mm thick (fig. 1068) (E42-X152).

Undatable

Dagger tang with part of the blade preserved, bent, 6.3 x 2.8 cm, 0.6 cm thick at widest, 0.5 cm thick at mid-point (fig. 1069) (H2-X1).

Fragment of fishhook, 3.9 x 2.1 cm span, round cross section, 1.3 cm thick (fig. 1070) (A85-X818).

3.9.1. Bitumen

by Flemming Højlund

Bitumen occurred in all phases and in all layers uncovered during the 2008-2012 excavations in Tell F6, but most copiously in the two oldest phases, 1 and 2 (p. 16 and 33-34), normally as amorphous lumps, rarely carrying impressions of basketwork or as likely fragments of the waterproof coating of boats built of reed-bundles and/or planks.

Phase 1

All the Ur III layers had plenty of bitumen, especially the area immediately south of wall C28 (fig. 9) that was found to be strewn with pieces of bitumen. In the east part of the Ur III settlement, Trench J, was a stone-built pit perhaps for heating up bitumen (p. 26-29, figs. 35-36).

Impressions of possible boat construction

Fragment of bitumen with a square protuberance from which impressions of bundles of cords emerge in three directions, the other side smoothed, vegetal tempered (fig. 1071) (C89-X190.1). Probably impression of piece of wood (plank?) cut through by quadrangular hole through which cords were lashed together with three other wooden elements (adjacent planks?). Part of a sewn boat, or at least a boat with sewn elements (fig. 1072).

A similar technique is revealed by a bitumen piece found at Ra's Al-Jins in Oman (Vosmer 1996 figs. 3-4, 13, Site RJ-2 no. 422). In the case of the Ra's al-Jinn's piece, the quadrangular hole cut in the wood seems to have been closed by a wooden plug, whereas in the Failaka piece the hole was filled with bitumen. The cords have had a diameter of 3-5 mm and both Z- and S-twist occur. The thickness of the bitumen layer applied to the plank is 9 mm.

Triangular piece with impression of two bundles (of reeds?) lashed together by reeds, the third side smoothed (fig. 1073) (C89-X190.2). The minimum thickness of the bitumen layer applied to the reed bundles is 6-10 mm.

On one side impression of two bundles (of reeds?) lashed together by 5-7 mm thick reed, the other side smoothed (fig. 1074) (J-A5-X29). This piece comes from a disturbed context.

0.7-1.9 cm thick fragment of bitumen with impressions of 2-3 thick reeds lashed together by a thin reed on one side and the other side smoothed, tempered with vegetal material (J-A4-X35.1).

On one side impression of bundle of reeds tied together with thin reeds, the other side uneven (fig. 1075) (J-A4-X35.2).

A 2.5 cm thick piece of bitumen with impressions of at least three parallel, 6-8 mm thick cords on one side, the other side is evened (J-A6-X41).

The surface of one side has mostly peeled off, but there is a small impression of woven palm-leaf mat and a deep impression of a cord, probably intended to keep the mat in place (fig. 1076); the other side is smoothed (C225-X863). The cord is 3 mm thick, two threads plied together, S-twist. The thickness of the bitumen applied to the mat is 1.6 cm.

On one side impression of woven palm-leaf mat around quadrangular (5x8 mm), protruding knob (fig. 1077) (C89-X190.3). The other side is smoothed. The thickness of the bitumen layer applied to the mat is 9 mm.

Two pieces with impression of woven palm-leaf mat on one side, the other side smoothed. The thickness of the bitumen layer applied to the mat is 8 mm (fig. 1078) (C89-X190.4 and 5).

On one side impression of woven palm-leaf mat, the strips are c. 1.8 cm wide and plaited in a two-over-two-under twill pattern displaced successively by one strip (fig. 1079); the other side is broken with a preserved thickness of 1.5 cm (C13/C22-X66).

1.9 cm thick piece of bitumen with impression of woven palm-leaf or reed mat held in place by cord on one side (fig. 1080), the other side evened (C13-X77).

Possibly impression of several parallel strings (fig. 1081) (C186-X669).

Parallels to the above described impressed bitumen fragments have been found at Qala'at al-Bahrain



Fig. 1073. 1:1.



Fig. 1072. 1:10.

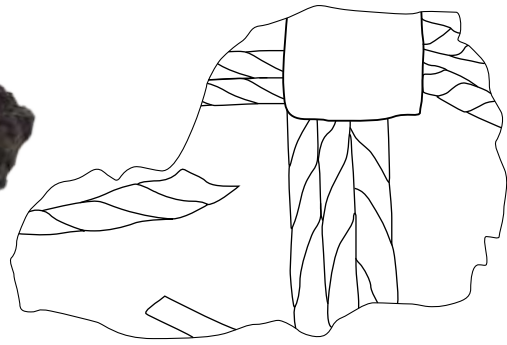


Fig. 1071. 1:1.

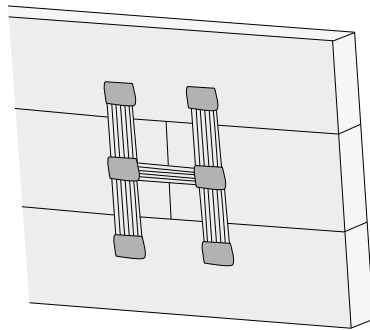


Fig. 1074. 1:1.



Fig. 1075. 1:2.



Fig. 1077. 1:2.



Fig. 1078. 1:2.



Fig. 1076. 1:2.



Fig. 1079. 1:2.



Fig. 1080. 1:2.



Fig. 1081. 1:2.



Fig. 1082. 1:2.



Fig. 1083. 1:2.



Fig. 1084. 1:2.



Fig. 1085. 1:2.

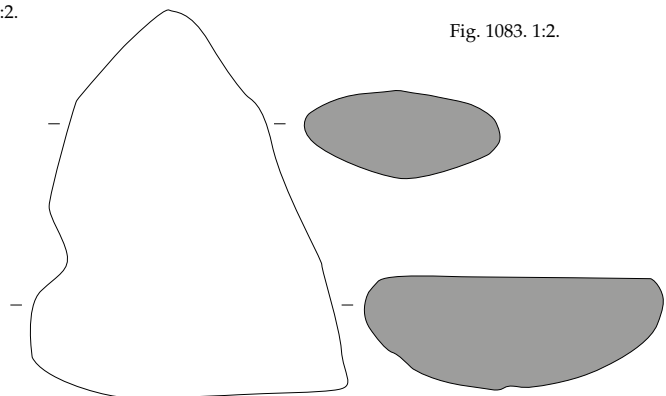


Fig. 1086. 1:3.



Fig. 1087. 1:1.



Fig. 1088. 1:2.



Fig. 1089. 1:2.



Fig. 1090. 1:2.



Fig. 1091. 1:2.



Fig. 1092. 1:2.

(Højlund & Andersen 1994 p. 409, figs. 2047), and especially at Site H3 at As-Sabiyah (Carter & Crawford 2010 p. 91-104, figs. 5.3-11, pl. 28) and Ra's Al-Jins (Site RJ-2) (Vosmer 1996 figs. 2-5, 12) where their relation to boat building is supported by the presence of barnacles (cf. also Benediková ed. 2010 fig. 197).

Net floaters?

Complete ring with a diameter of 4.7 cm, a central hole c. 1 cm in diameter, and a thickness of 1.5 cm (fig. 1082) (C124-X411). The section through the ring is rounded triangular. Similar rings in burnt clay (net sinks?) are found in this period (cf. figs. 1123-1124).

Fragment of ring with an estimated diameter of c. 5.5 cm and a hole of c. 1 cm (fig. 1083) (C187-X664).

From disturbed contexts in Trench CEE come two fragments, possibly of the same ring with a diameter of c. 9 cm and a hole with a diameter of c. 2 cm (figs. 1084-1085) (C169-X669 and C177-X732).

Other objects

Slab, the surface looks as if it has solidified/hardened after having been heated up, tempered with vegetal material (fig. 1086) (C225-X871).

Impression of a special basket sewing technique (coiling) on one side, the other side flattened, thickness 0.8 cm (fig. 1087) (C124-X413) (cf. Benediková ed. 2010 fig. 181).

Phase 2

Seven small, 0.6 to 1.5 cm thick slabs of bitumen with evened sides (fig. 1088) (A226-X794).

A 9x7 cm large and 2 cm thick piece with impressions of palm leaf matting and cord in different directions on one side and the other side flattened (A226-X794).

Phase 3

Bitumen slab, irregular five sided, 10x6 cm, c. 2.3 cm thick, two flat sides and edges at right angles to the flat sides (fig. 1089) (E188-X869).

Fragments of a c. 2 cm thick slab, the largest 7.5x8.5 cm, some with a 1.5 cm thick protruding ridge, both sides flattened/evened, without impressions (E173-X859).

Phase 4

Lump of bitumen, 10x6.5x3.5 cm, with impression of palm leaf mat on one side, a little indistinct but strips

c. 0.7 cm wide, and the other side coagulated (fig. 1090)
(E140-X626).

A 5 x 3 x 0.8 cm piece with impressions of parallel
cords on one side and flattened on the other (fig. 1091)
(E140-X626).

Impression of palm leaf mat with 6 mm wide strips,
two-over-two-under and displaced one (fig. 1092)
(E150-X663).

3.9.2. Bitumen provenience

by Thomas Van de Velde

Introduction

Several bitumen samples from the 2008-2012 excavations in Tell F6 were sent to Ghent for geochemical analysis in order to determine their origin. In the Near East, bitumen surfaces naturally at several locations and raw material from these seepages was extracted by humans for a variety of purposes. The material was especially practical as an adhesive but was also used frequently for its waterproofing properties. It has been used in architecture as a mortar or to waterproof reed mats, for the lining of pottery, repairing pottery, as a material used to make stoppers/corks, to waterproof baskets, to enhance and waterproof the hull of boats etc.

Bitumen samples from several archaeological sites in the Arabian Gulf have been sourced in the past and most came from the same seepages as the bitumen used in Mesopotamia and Iran (Van de Velde in preparation, Connan and Carter 2007, Connan *et al.* 2005, Connan *et al.* 1998, Connan and Van de Velde 2010, Connan unpubl., Van de Velde *et al.* 2015). These are to be located in northern Iraq (around Mosul), at Hit (Iraq, middle Euphrates river), the Dead Sea, the Burgan Hill in Kuwait, and a variety of seepages in western Iran (fig. 1093)

Bitumen on Bronze-Age Failaka

We can mark the 21st century BC as an important era for both Bahrain and Failaka, for it is then that the settlements on the islands start to become major players in the trade and interregional contacts that characterise the Bronze Age in the Arabian Gulf. The fate of the island of Failaka is closely intertwined with that of Bahrain as it is also here, on this strategically-located island in the upper Gulf, that we find strong manifestations of the Dilmun culture. It is especially the work of the Danish pioneering excavations (1958-1963), the following French missions (1983-1988) and the work of the Kuwaiti-Slovak missions (2004-2008) that uncovered the Bronze Age on this island (Calvet and Gachet 1990, Salles 1984, Calvet and Salles 1986, Kjærum and Højlund 2013, Højlund 1987, Benediková and Barta 2009, Barta *et al.* 2008). These teams and excavations worked respectively at the Bronze Age sites of Tell F3, Tell F6, Site G3, and Al-Khidr (fig. 1094).

At all these sites an abundance of bituminous material was found but the bitumen from Tell F3, Tell F6, and Site G3 has never been studied and is only sparsely mentioned in publications as the following review will show.

In House 26 in Tell F3 (dated c. 1850-1800 BC) the plastered surface of a small trough is said to have traces of bitumen (Kjærum and Højlund 2013 p. 81). From Site G3, a dwelling site north of Tell F6, two pottery sherds with traces of bitumen inside are mentioned (Salles 1984 fig. 28:68 & fig. 30:92).

More evidence of bitumen presence and usage is mentioned from the area of the temple in Tell F6 excavated by the French mission and dated to the end of the 3rd and beginning of the 2nd millennium BC. Several zones both inside and outside the structure yielded layers with shapeless lumps of bitumen. Unfortunately, there is no information available as to what sort of artefacts these bitumen lumps originally belonged, the easily-decaying nature of bituminous material obviously attributes to that. Beside the bitumen lumps, a “*zone de déchets de bitumen*” is mentioned (locus 389) (Calvet and Gachet 1990). The excavation report does not give any further information on this context, but it is thus possible that bitumen was worked in the immediate surroundings of the temple.

Bitumen was also attested in close context with pottery (Pic 2008: catalogue nos. 8, 43, 140, 168, 229, 241, 249, 250). This could either have been as a lining of the vessels, or a remnant of the storing/transporting/working of bituminous material.

Noteworthy is a sculptured piece made from a black bituminous stone from the temple at Tell F6 (Calvet and Pic 1986 p. 76-77). The material was originally identified as an artificial ‘stone’ and named “*mastic de bitume*” (Connan and Deschesne 1996). It was, however, later identified as a natural-occurring rock from the Sargelu formation in Iran (Connan 2012 p. 156-177) and should not be considered as actual bitumen.

It is clear from the French excavation reports from Tell F6 that bitumen was found frequently on the site. No detailed study on this material has been performed, however, making it hard to make any claims on its use or to make an assessment on quantities.

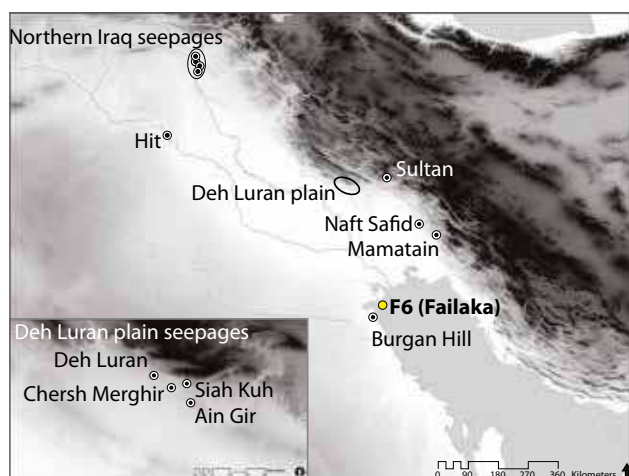


Fig. 1093. Map showing the bitumen seepages mentioned in the text. For a complete overview of all bitumen seepages in Antiquity see Connan & Van de Velde (2010) & Connan (2012).

Conversely, excavations at Al-Khidr have revealed a Dilmun-period settlement with massive amounts of bitumen used in a domestic context, and the finds from here have been properly analysed and published (Benedikova and Barta 2009). Al-Khidr was excavated by a joint Kuwaiti-Slovak team for four campaigns. The site is identified as a dwelling site similar to Tell F3. All bitumen finds from Al-Khidr have been quantified, registered and, where possible, identified. Around 200-230 kilograms of bitumen was recovered from the site, including bitumen-coated baskets and cordage, seal impressions, bottle stoppers, or just plain shapeless lumps. There are even occurrences of bitumen layers (largely in contexts with pottery or with stones in walls), between building stones as bonding materials, and thick bitumen crusts in large earthenware vessels (Belenová-Štolcová 2010). The evidence could also point towards bitumen processing on the site and the thick crusts in vessels could also be evidence of transporting.

In general, there is a lot of evidence for the use and working of bitumen at several Bronze Age sites on the island of Failaka although the material has been unevenly studied and published. The excavations at Al-Khidr show that bitumen was used for a multitude of purposes and this is supported by the excavations presented here (see above Højlund p. 210).

The origin of the bitumen from Failaka

As yet, only bitumen from the French missions at Al-Qusur, Site B6, Tell F5 and Tell F6 has been analysed



Fig. 1094. Dilmun-period sites with evidence of bitumen usage.

to obtain information on its geological origin, the latter site being the only Bronze Age site. Geochemical tests have been conducted on thirteen bitumen samples from the Bronze Age levels at Tell F6, three from the Middle Dilmun Period (ca. 1400-1300 BC) and ten from the Early Dilmun levels (ca. 2000-1700 BC). All of these samples originated from the Hit area in Iraq, with the exception of one sample which came from the Burgan Hill bitumen seepage (Kuwait) (Connan and Carter 2007) (fig. 1093). There is, however, also proof that Iranian bitumen was used at the Dilmun settlement at Saar and in several tumuli from the same period (Connan *et al.* 1998, Van de Velde in preparation). It seems that bitumen from several seepages and source areas was exported to several Dilmun settlements simultaneously, although it looks apparent that raw materials from the Hit seepages was not used in this area prior to 2000 B.C.

Bitumen samples from F6 (Kuwaiti-Danish campaigns)

Several bitumen samples from the 2008-2012 excavations in Tell F6 were selected for geochemical screening to determine their geological origin. Bitumen from several trenches and deposits were chosen in order to investigate whether or not bitumen suppliers changed through time (fig. 1095). Faint impressions are visible on several samples, but in all cases these are too small or not enough pronounced to allow any detailed identification.

The viscosity of natural bitumen does not allow the material to be properly worked, therefore a temper was added (Forbes 1964, Connan and Van de Velde 2010). These additives were not studied in detail but rather macroscopically checked. There was nothing noteworthy for most of the samples; most samples had the typical brownish-black colour, often tempered with sand. Samples 9 and 10 however are extraordi-

Sample	ID	Phase	$\delta^{13}\text{C}$	Ts/Tm	GCRN/C ₃₀	GCRN/31R	OLN/C ₃₀	Source
1	A123-X482	4	-27,05	0,26	0,14	0,41	0,04	Mixture?
2	A179-X474	4	-27,88	0,22	0,14	0,4	/	Deh Luran plain
3	A180-X493	4	-26,97	0,26	0,12	0,36	0,07	Mixture?
4	A220-X725	2	-26,54	0,36	0,13	0,4	0,1	Mixture?
5	A206-X615	3	-27,32	0,25	0,15	0,47	0,05	Mixture?
6	A209-X668	3	-27,5	0,2	0,19	0,51	/	Deh Luran plain
7	A215-X760	3	-27,49	0,29	0,19	0,49	/	Deh Luran plain
8	A218-X743	3	-27,51	0,24	0,14	0,42	/	Deh Luran plain
9	E87-X349	6		0,33	0,08	0,24	/	Deh Luran plain?
10	A140-X280	6		0,34	0,08	0,25	/	Deh Luran plain?
11	E119-X541	5	-26,77					Mixture?
12	E122-X535	6	-27,8					Deh Luran plain?
13	E67-X554	7	-27,83					Deh Luran plain?
14	J18-X79	1	-27,32					

Fig. 1095. Archaeological samples used in this research. Measured values of $\delta^{13}\text{C}$ and molecular ratios Ts/Tm (18 α (H)-22,29,30-trisnorhopane/17 α (H)-22,29,30-trisnorhopane), GCRN/C₃₀ (Gammacerane/C30 α -hopane), GCRN/31R (Gammacerane/C3122R hopane), OLN/C₃₀ (Oleanane/C30 α -hopane). NM= no measurement.

nary because of the presence of complete shells in their matrix (fig. 1096). Crushed-shell inclusions are attested quite frequently in bitumen mixtures, but these can be considered as intrusive in the initial matrix (i.e. sand) that was used to create a bitumen mixture rather than a deliberate addition. The shells present in samples 9 and 10 have been identified as members of the *Potamididae* family, a species closely associated with mud flats and mangroves. These shells are only visible on one side of the lumps, in both cases the backside is free of any inclusions, and in one case (sample 9) elongated impressions are visible. These imprints are about 2 to 5 mm wide and probably derive from reed bundles.

It is unlikely that shells were added deliberately to the bitumen mixture as that would not enhance its physical characteristics in any way, it would actually make it more porous. So either the shells in this bitumen are an accident (for example the spilling of bitumen on a shell-covered surface), or they became part of the bitumen by a physical process. There are strong indications for the usage of bitumen in naval architecture and it is not impossible to imagine shells being impressed into the bitumen coated hulls of boats when they were moored in the mangroves. Bitumen was also a product which was quite often reused and in some cases there is evidence of the stripping of bitumen from boats with the aim to store these for later re-use (Connan *et al.* 2005, Carter 2010). This could also have been the case for the bitumen samples 9 and 10 from Tell F6.

Analytical methods

Sample preparation

The idea on sourcing archaeological bitumen was tested and published for the first time in 1978 (Marchner *et al.* 1978) and further refined by Nissenbaum and Connan (Nissenbaum *et al.* 1980, Nissenbaum *et al.* 1984, Rullkotter *et al.* 1985, Rullkotter and Nissenbaum 1988, Connan 1988, Connan and Deschesne 1992, Connan *et al.* 1992). The techniques used by these and other researchers rely on those developed for the petroleum industry and have proven to be effective on archaeological samples as well. These techniques embody the identification and quantification of individual molecules from the saturated hydrocarbon fraction, and stable carbon isotope analysis ($\delta^{13}\text{C}$) on the asphaltene fraction. Bitumen consists of two more chemical fractions (resins and aromatics), which do not contain useful parameters for identifying their source.

As stated above, archaeological bitumen is generally an artificial mixture consisting of pure bitumen and a temper. Evidently, the bitumen needs to be isolated from the added ingredients prior to analysis. Archaeological samples were crushed and put in a vial for solvent extraction by dichloromethane:methanol (3:1 ratio, 1,5 ml per gram of sample). The vial was put in an ultrasonic bath and centrifuge, after which the liquid fraction was recovered. This process was repeated twice.



Fig. 1096. Bitumen sample 9 bearing shell inclusions. Note the parallel impressed lines on the backside (left photo) and the shells on the front.

Asphaltenes can be defined as the heaviest components of petroleum and are insoluble in light n-alkanes (e.g. hexane) but soluble in organic compounds (e.g. dichloromethane) (Goual 2012). Therefore the solvent was removed from the sample by evaporation after which n-hexane precipitation was used to isolate the asphaltenes fraction. Standard column chromatography was used for separation of the saturated hydrocarbon fraction (Peters *et al.* 2005a p. 200).

Stable Carbon Isotope analysis parameters

EA-IRMS (Elemental Analysis – Isotope Ratio Mass Spectrometry) was used on the asphaltenes fraction. The carbon isotopic composition of the asphaltenes fraction was determined using an elemental analyser (ANCA-SL, PDZ Europa, UK), coupled to an isotope ratio mass spectroscopy (IRMS) (20-20, SerCon, UK). The isotopic composition of natural samples (i.e. not synthetic isotopic enrichment) is reported relative to an international reference, using the so called ‘ δ ’ scale and is typically expressed in ‰.

$$\delta^{13}\text{C} = \frac{\left[\frac{^{13}\text{C}}{^{12}\text{C}} \right]_{\text{sample}} - \left[\frac{^{13}\text{C}}{^{12}\text{C}} \right]_{\text{VPDB}}}{\left[\frac{^{13}\text{C}}{^{12}\text{C}} \right]_{\text{VPDB}}}$$

For C the international reference is Vienna Pee Dee Belimnite (VPDB), which has a carbon isotopic ratio of 0.0111802 (± 0.0000028). The measured values of all samples can be found in fig. 1095.

GC-MS analysis parameters

GC-MS analysis (gas chromatography-mass spectrometry) was conducted on the saturated hydrocarbon fraction in order to determine and quantify biomarkers. These complex organic compounds are derivations

of the original organic materials from which bitumen is formed, containing source-specific information (Peters *et al.* 2005b, Peters *et al.* 2005a). A Hewlett-Packard 6890-5973 GC-MS system equipped with an Agilent Technologies HP-5MS column (30m x 0.25mm ID, 0.25 μ m) was used with Helium as a carrier gas (gas flow of 1.5 ml/min). One microliter of sample dissolved in CH₂Cl₂ was injected (in splitless mode). The oven temperature increased incrementally from 40°C to 250°C at 6°C per minute, and from 250°C to 300°C at 2°C per minute. The temperature was then held at 300°C for 30 minutes. The terpane fingerprint (m/z 191) was used to identify and quantify individual molecules present in the samples (fig. 1097). Due to the nature of the samples it is impossible to use a purely quantitative approach, therefore molecular ratios are used in fingerprinting rather than individual compounds.

For both techniques several replicate analyses of samples were conducted to cross-check consistency of the measurements. No discrepancies were identified. If a sample was measured more than once, the average of all measurements was calculated and used for further analyses and interpretation. The measured values can be found in fig. 1095.

The origin of the F6 samples

The geologic origin of the sample was investigated using the information from the terpane fingerprints and the values obtained through stable carbon isotope analysis on the asphaltenes fraction. The $\delta^{13}\text{C}$ values of all samples are too high to contain any bitumen from the Hit area. In several samples (samples 1, 3, 4 & 5) 18 α -oleanane was observed. The base for this compound is angiosperms and originates in the Triassic Period or earlier and is for this region mainly linked to oils originating from the Padbeh source-rock formation in Iran (Peters *et al.* 2005b p. 572, Connan 2012 p. 117).

Fig. 1098 shows several cross-plots combining several parameters; the molecular ratios Ts/Tm and 18 α -oleanane/C₃₀ $\alpha\beta$ -hopane, but also the values retrieved from stable carbon isotope analysis. The bitumen from Tell F6 seems to cluster nicely together with bitumen samples from the Hellenistic period site of Akkaz, which have been published in detail (Connan 2011). The exact source of these samples has not been established as their genetic parameters do not match any of the reference seepages, nor any other archaeological bitumen. It has therefore been suggested that this bitumen was either a mixture between two types of bitumen (one from the seepages on the Deh Luran

Terpane (m/z 191) fingerprint of Bitumen Sample 4

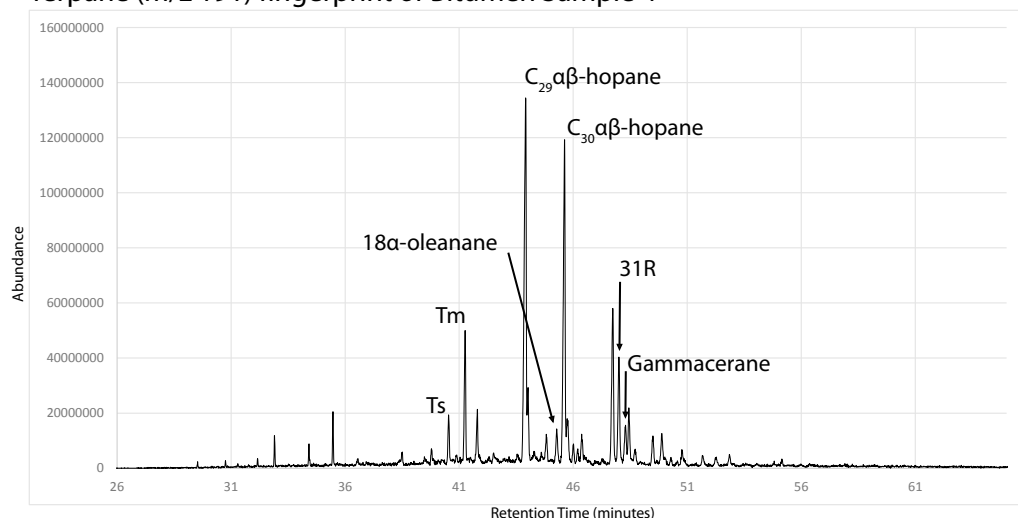


Fig. 1097. Terpane fingerprint of bitumen sample 4.

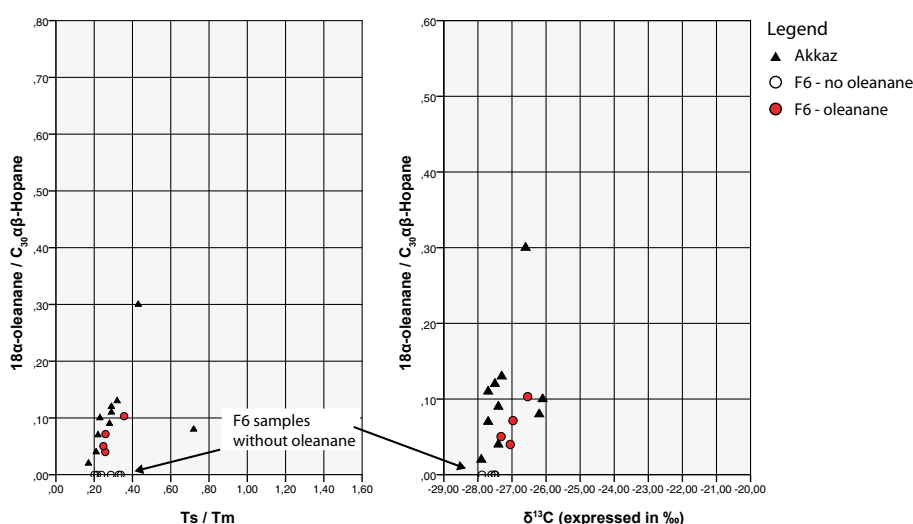


Fig. 1098. Cross-plots of Ts/Tm vs. 18α-oleanane/C30αβ-hopane (left) and δ¹³C vs. 18α-oleanane/C30αβ-hopane. The bitumen samples from Tell F6 seem to correlate with those from Akkaz (molecular ratios and δ¹³C retrieved from Connan 2011).

plain mixed with oleanane-containing bitumen) or alternatively this bitumen came from a yet-unidentified seepage (Connan 2011). We should however mention that the Tell F6 bitumen does not match those from Akkaz on all parameters, the Gammacerane/C₃₀αβ-hopane ratio²⁴ for instance is remarkably higher in the samples from Akkaz and the range in which the Akkaz bitumen δ¹³C values fall is much wider (fig. 1099). Of course, the extreme wide range of δ¹³C values in the Akkaz samples could be an indication of the presence of bitumen from multiple seepages.

If the bitumen from Akkaz is indeed a mixture of raw material from different seepages (one with a genetic pattern reminiscent of those from the Deh

Luran plain and one containing oleanane), it is striking that the 18α-oleanane/C₃₀αβ-hopane ratio is very similar in both datasets. That would indicate that the composition of the mixtures is similar, which would be quite coincidental considering the 2000 year gap between the two sites. In that respect, it seems more likely that the bitumen from both sites were extracted from an unidentified seepage. The deviating Gammacerane/C₃₀αβ-hopane, the wide δ¹³C range of the Akkaz samples, and the fact that as of yet no oleanane-containing bitumen with a δ¹³C of asphaltenes within the -28/-27‰ range has been identified in bitumen samples (Connan 2011) on the other hand conflict with this hypothesis.

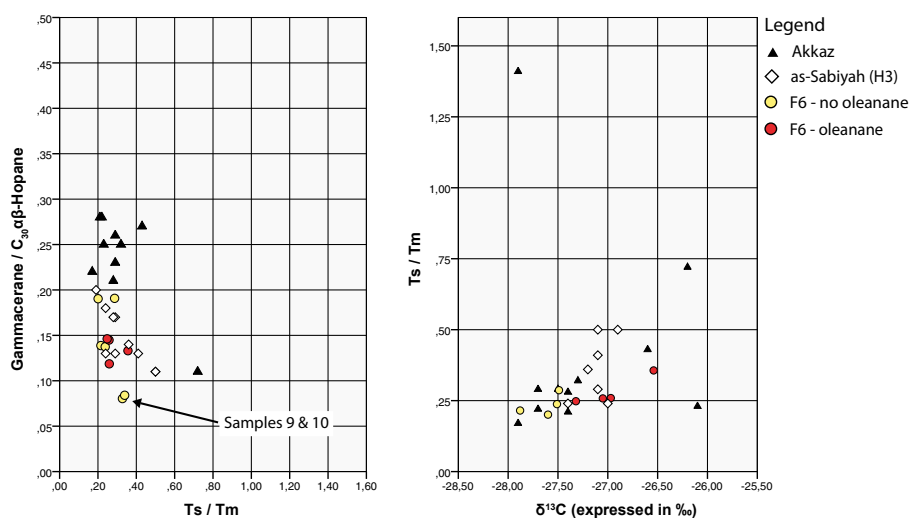


Fig. 1099. Cross-plots of Ts/Tm vs. Gammacerane/ C₃₀αβ-hopane and δ¹³C/ vs. Ts/Tm. The bitumen samples excavated at the Neolithic site of as-Sabiyah have been identified as coming from the Burgan Hill (Kuwait).

From the ten bitumen samples which have been subjected to GC-MS analyses, only 4 showed a presence of oleanane. The same genetic parameters as described above were used to identify the source of the other 6 samples. Both molecular ratios and δ¹³C match with bitumen from Susa, which lacks oleanane and has been identified as coming from one of the seepages on the Deh Luran plain.²⁵ Bitumen from the Burgan Hill shows a very similar molecular pattern (Connan 2010, Connan *et al.* 2005), but as illustrated in fig. 1099, the δ¹³C of this bitumen does not match with the values from the Tell F6 samples. Consequently, we can securely state that the Tell F6 bitumen samples without oleanane are coming from one of the seepages on the Deh Luran plain. The same type of bitumen has also been identified in archaeological bitumen samples from Bronze Age tumuli in the A'ali burial field in Bahrain (Van de Velde in prep.) and possibly also in the archaeological bitumen samples from Saar Settlement (Connan *et al.* 1998).

If the bitumen containing oleanane are mixtures, then it would make sense that one of the components is material from the seepages on the Deh Luran plain as the molecular parameters (Ts/Tm, Gammacerane/C₃₀αβ-hopane, Gammacerane/C₃₁22R hopane) between the two groups show great similarities (fig. 1099). There is, however, a distinction between the two groups based on the δ¹³C measurements, which are lower in the samples without oleanane. This, however, does not necessarily mean anything as this value of course could be influenced by the addition of bitumen from another seepage.

For three samples no molecular data was retrieved and only δ¹³C is available. The values of samples 12 & 13 are both low in number (-27,8‰ and -27,83‰) and correlate with the δ¹³C of the samples coming from the Deh Luran plain. Sample 11 on the other hand shows a relatively low δ¹³C (-26,77‰) in such a degree that it falls within the range of the samples containing oleanane and should therefore probably be placed in the same group.

Samples 9 & 10 are outliers in the dataset (fig. 1099) and do not seem to belong to either of the two groups defined above. These two samples do not exhibit any oleanane and are characterised by their low Gammacerane/C₃₀αβ-hopane. Alteration of the Gammacerane/C₃₀αβ-hopane parameter is possible because of biodegradation of the sample. Unfortunately no isotopic data is available for these samples. Either this bitumen comes from an unidentified seepage, or we are encountering a chemical alteration of the samples. It is remarkable that these samples do not only distinguish themselves from the rest of the dataset because of their molecular ratio, but also because of the high number of shell inclusions (cf. supra). It remains most likely that this bitumen shares the same origin as the other non-oleanane containing samples, i.e. one of the seepages on the Deh Luran plain.

Bitumen sample J18

One specific feature that was excavated at Tell F6 was a stone set pit (J16, figs. 36 and 1095: Sample 14) with a very fine-grained black deposit at the bottom. It was

thought that this pit was a fire pit specifically for the heating and processing of bitumen, hence the black colour of the lower deposit, and a sample was taken for geochemical screening. GC-MS on this sample revealed a very obscure chromatogram lacking all the distinctive peaks identified in bitumen. However, $\delta^{13}\text{C}$ on this sample gave a value of -27.32‰ , which correlates nicely with that of bitumen. Previous experimental work on bitumen revealed that the recycling – or more specifically, the reheating – of bitumen caused the evaporation of entire compound classes, yet that the isotopic signature on the asphaltenes remains the same independent of the duration of the reheating of the sample (Hollander and Schwartz 2000). If the fire-pit discussed here was indeed used as an installation for the working of bitumen, its contents would naturally be subject to numerous processes of (re-) heating, causing the archaeological sample to behave like it does on the chemical analyses. So quite possibly, the black deposit in the stone set pit is the silent witness of bitumen processing, but we should not take this for granted as no bitumen-specific molecules could be identified. Unfortunately, the experimental work discussed above does not include detailed chromatograms and consequently it was impossible to compare those results with that of sample J18, nor was it possible in the current research to duplicate a similar test.

Conclusion

Thirteen bitumen samples from the 2008-2012 excavations at the Bronze Age site of Tell F6 on Failaka have been investigated with the aim to determine their original seepage. GC-MS analysis was conducted on ten samples, and stable carbon isotope analysis on eleven. Using these analytical techniques, it was possible to identify two different main groups in the bitumen dataset, discriminated by either the absence or presence of oleanane – a chemical molecule only present in bitumen coming from the Padbeh source

rock formation (Iran). The group of samples containing oleanane shows close resemblance to a group of bitumen samples from the Hellenistic site of Akkaz. No specific origin for these samples was given, but it was suggested that this bitumen was either a mixture of raw materials from different sources, or coming from an unidentified seepage. The latter option being the most plausible. The other group of bitumen samples from F6, lacking oleanane, is coming from one of the bitumen seepages on the Deh Luran plain (Iran). The dataset knows two true outliers, samples 9 and 10, for which no reference data was found. It is most likely that this bitumen is also coming from the Deh Luran plain.

As Iranian bitumen has been identified in several Dilmun-period datasets from both Failaka and Bahrain, it should come as no surprise that this type of material is also found at Tell F6. Remarkable, however, is the difference in bitumen between the Tell F6 bitumen datasets from the French and the Kuwaiti-Danish missions. Most of the material from the French excavations of the temple is Iraqi in origin (Hit-area), whereas one sample was identified as coming from the Burgan Hill seepage (Kuwait).

Acknowledgements

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3.10. Other finds

By Flemming Højlund

Plaster

Numerous fragments of plaster show that this material was extensively used in the construction of buildings in Tell F6. Floors were made of plaster and/or clay. Rooms were roofed by palm trunks, sometimes coated with bitumen and covered by palm leaf stems and plaster. While walls were mostly built of stone set in clay, the faces were often coated in plaster (cf. Killich & Moon 2005 p. 199. Kjærøum & Højlund 2013 p. 11). In a few cases fragments of what may have been wall plaster was coloured red. Old buildings were demolished to make way for new construction, and the debris, i.e. stone and plaster fragments, was regularly reused in the walls and floors of new buildings where they were found during excavation. These re-used plaster fragments normally were well preserved, with sharp break edges, indicating that they do not derive from houses which have been allowed to fall into disrepair over a long time, but rather that they derive from houses that were demolished intentionally to give way to new buildings. A small selection of the material is described in the following.

Phase 3

3 pieces of 1.5-2.5 cm thick, white roof plaster (fig. 1100) with the characteristic impressions of palm wood on one side – one of them slightly concave –, and the other side uneven (A184-X634)

Two fragments of white roof plaster, 6 cm and 2.5 cm thick, with impressions of palm branch stems (A207-X630).

White roof plaster fragment (fig. 1101), 7 cm thick, with impressions of palm branch stems on one side and on the other impressions of two rounded ends of poles on which the stems were resting (A209-X633).

Large lump of white plaster with impressions of three parallel palm branch stems on one side (fig. 1102) (E172-X801).

1.4 cm thick fragments of white wall plaster with one side flattened and the other side more rough

with a protrusion filling the gap between wall stones (A187-X1923).

C. 1.5 cm thick fragment of white wall plaster with one side flattened and the other side more rough with a protrusion filling the gap between wall stones (fig. 1103) (A187-X592).

Floor plaster with small oolitic limestone chips embedded (H13-X19).

Phase 4

Fragment of white roof plaster with palm wood impression, 1 cm thick (A180-X494).

Plaster roof fragment, 4 cm thick, with impression of palm trunk covered with thin layer of bitumen (E137-X631).

Several roof plaster fragments, 2-2.5 cm thick, with one flat side and the other with possible impression of palm trunks, covered with a thin layer of bitumen (figs. 1104-1105) (E137-X645).

2 cm thick fragment of roof plaster with palm trunk impression covered with thin layer of bitumen (E153-X678). On the edge of the plaster fragment were impressions of palm branch stems perpendicular to the trunk impression (fig. 1106) (cf. Højlund & Andersen 1997 p. 68-70, figs. 294-95).

5 cm thick fragment of roof plaster with concave impression of palm trunk (fig. 1107) (E150-X699).

Fragment of white roof plaster, 5 cm thick, with concave impression of palm trunk (E200-X1047).

2.5 cm thick fragment of roof plaster with one side even and the other side with impressions of two parallel poles covered with thin layer of bitumen (E282-X1248).

Fragment of white plaster with impressions of at least five parallel palm branch stems; these impressions were later covered with a layer of plaster into which a new layer of branches were laid (fig. 1108) (E168-X815).

Plaster fragment with impressions of parallel palm branch stems (E200-X1048).

Fragment of plaster with impressions of two palm trunks covered with a thin layer of bitumen (fig. 1109) (E279-X1188).

Fragment of plaster with impressions on two sides of palm trunks covered with a thin layer of bitumen, perpendicular to each other (fig. 1110) (E279-X1192).

Impression of bundle of palm branch stems tied together with reed or string (fig. 1111) (E320-X1362).

Fragment of plaster disc, 3 cm thick, with a diameter of c. 28 cm (E150-X852).

Fragment of a round plate of plaster with incised lines, 2.8 cm thick, c. 25-30 cm in diameter (fig. 1112) (A180-X829).

Three lumps, 3 x 3 x 1.5 cm, 2 x 2 x 1 cm and 5 x 4 x 3.5 cm, of clayey plaster with a smooth, flat face painted red (figs. 1113) (A180-X549 and A180-X511). Found embedded in floor A180. Perhaps painted wall plaster from the previous Phase 3.

Grinding stones

Phase 1

Fragment of 4 cm thick grinding stone (fig. 1114), 10 x 9 cm, fine beach-rock, one side smooth, the other rough, with regular rounded edge (C195-X702) (cf. Killich & Moon 2005 p. 214-216).

Phase 2

Half of 3 cm thick, oval grinding stone in fine beach-rock, 8 x 7 cm (A224-X716).

Fragment of 3 cm thick grinding stone in beach-rock, 8 x 6 cm (A215-X694).

Fragment of 4 cm thick grinding stone, 9 x 6 cm (A215-X695).

Phase 3

Oval grinding stone (fig. 1115), 8 cm thick, one end damaged, 23 x 18 cm, in fine beach-rock; one side slightly convex and smooth, the other very convex and rough (A190-X851).

Fragment of 4.5 cm thick grinding stone in beach-rock with round edge, 12 x 11 cm (A186-X532).

Fragment of grinding stone in beach-rock (conglomerate) (fig. 1116), one side smooth, the other side rough, regular round edge (E172-X803).

Phase 4

Fragment of 4 cm thick grinding stone in fine beach-rock, 10 x 5 cm, one side slightly convex, the other rounded, regular edge (A180-X827).

Fragment of 5 cm thick grinding stone in fine beach-rock, 8 x 9 cm (A179-X559).

Fragment of 4.5 cm thick, heavy fine beach-rock (fig. 1117), 13 x 7.5 cm, one side smooth and slightly convex, the other more rough and rounded (E168-X737).

Fragment of oval, 3 cm thick, grinding stone in fine beach-rock, 9 x 5 cm (fig. 1118) (A180-X551).

Fragment of 2.5 cm thick grinding stone in fine beach-rock, one side smooth, the other more rough, 7 x 4.5 cm (E112-X503).

Phase 6

Fragment of 7 cm thick grinding stone in fine beach-rock, 15 x 14 cm, one side and the edge very smooth, the other side rough (E81-X353).

Phase 7

Fragment of 4 cm thick grinding stone in fine beach-rock, 6 x 10 cm (E74-X263).

Hammerstones

Phase 2

Flint-block with crushing marks on edge all around, 4 x 7 x 7.5 cm (E225-X1014) (cf. Killich & Moon 2005 p. 218-227).

Round quartz stone, one side very smooth, other side rough, and crushing marks all around edge, 4×7×8 cm (A225-X862).

Dark, fine-grained stone with crushing marks and one side smooth, 4×5×6 cm (A215-X749).

Phase 3

Half of rounded quartz stone with crushing marks on one end, 4×5×6 cm (E301-X1281)

Two flint nodules with crushing marks, 5×5×9 cm and 6×6×7 cm (figs. 1119-1120) (A186-X532).

Fine-grained black stone with crushing marks on two opposing ends, 3×5×6.5 cm (E198-X1026).

C. 18×9×9.5 cm ashlar with a runnel cut in the upper side, oolitic limestone, lying loose in the settlement layer (fig. 1121) (E173-X849).

Miscellaneous small finds

Shell ladle (fig. 1122). Fragment of a ladle made of a *Chicoreus Ramosus* shell (C116-X283), probably derived from the Gulf of Kutch. It occurs in the Indus area during the Mature Indus period and is not found after the decline of the major urban cities. A shell ladle represented a considerable amount of labour and was probably a highly prized object (Kenoyer 1984 p. 55-56, fig. 7, fig. 13:1). Found in a Phase 1 context.

Pottery rings. A pottery ring (fig. 1123) in buff, straw-tempered ware with a diameter of c. 8.6 cm, a central, 3 cm wide hole, a thickness of 2 cm and a rounded triangular section through the ring (C125-X319). From a Phase 1 context.

A pottery ring (fig. 1124) in straw-tempered, light greenish ware with a diameter of c. 8 cm, a 1.7 cm wide hole, a thickness of 2 cm and a triangular section (C116-X282). From a disturbed context.

Pottery rings of the same ware, size and shape have been found in Period Ia and Ib at Qala'at al-Bahrain (Højlund & Andersen 1994 figs. 1789-1790), but they do not occur in other periods in Bahrain.

Lapis lazuli. Two small fragments, 5 mm across, of lapis lazuli, both from Phase 1 contexts (figs. 1125-1126) (C7-X22 and C186-X658).

Pearls. Unpierced pearl (fig. 1127a), 6 mm in diameter, probably deriving from a grey layer with Ur III pottery above wall C28 in the west part of Trench C3 (C70-X166).

Unpierced pearl (fig. 1127b), 5.2 mm in diameter, probably from the sandy layers with Ur III pottery below the temple terrace floors C68-69 in the north part of Trench C3, found during section cleaning (C60-X141).

Carnelian weight (fig. 1128). A cubic weight of red carnelian, weighing 6.6 g ±0.1 g, found out of context while cleaning Trench A near wall A219 before beginning excavation in 2010 (A229-X803). The weight corresponds to four times a basic weight of 1.7 grams found within the Indus civilization (Bibby 1971 p. 350).

Ivory handle (fig. 1129). Handle with mushroom-shaped pommel and expanding, but damaged lower part, surface intensely polished (A220-X938). The interior was chiselled out to receive the tang of a dagger, which was fastened with bitumen and two rivets through holes drilled through the ivory. Diameter of holes 2.6 mm ± 0.1 mm, marks of rotation in both holes. Found in a Phase 2 context.

A copper dagger (fig. 1056) with at least one hole in the tang was found close by, also in a Phase 2 context, but the shape of the tang and the position of the rivet hole do not fit with the ivory handle.

Ostrich-egg shell. Small ostrich-egg shell fragments, 1-2 cm across, occurred in Trenches A and E in several phases. From Phase 3 E186-X847 (fig. 1130) and E202-X984; from Phase 4 A165-X381 (fig. 1131), A169-X382 (fig. 1132), E192-X882 (fig. 1133), E135-X601 (burnt) and E187-X878; from Phase 5 E272-X1230; and from Phase 6 E77-X315, E81-X358 (2 pieces), E83-X364 (2 pieces) and E83-X369.

From Trench C two fragments were found in disturbed contexts, C149-X573 and C177-X651.

Bone handle. The off-cut of a sheep metatarsal (fig. 1149) (E89-X373) in a Phase 6 context is evidence for the production of bone handles found in previous excavations in Tell F3 and F6 and on Al-Khidr (Benediková 2010 fig. 50:f, 108-109).

Head of snake (fig. 1134). Naturalistic modelled raised head of snake in burnt, light brownish-grey clay, traces of red colour (C139-X423). Probably a rear-fanged snake like *Malpolon monspessulanus* or *Malpolon moileensis*

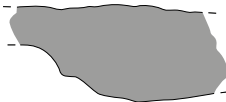


Fig. 1100. 1:2.



Fig. 1101. 1:2.



Fig. 1103. 1:2.



Fig. 1104. 1:2.

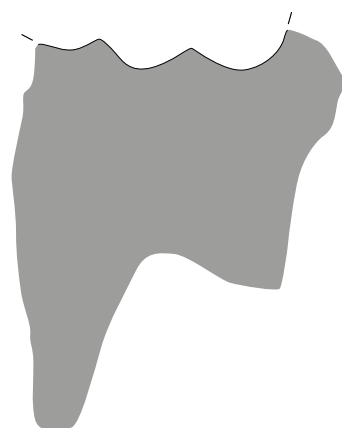


Fig. 1102. 1:2.

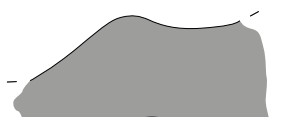


Fig. 1105. 1:2.

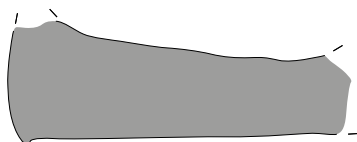


Fig. 1106. 1:2.

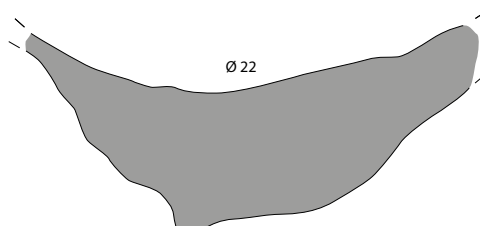
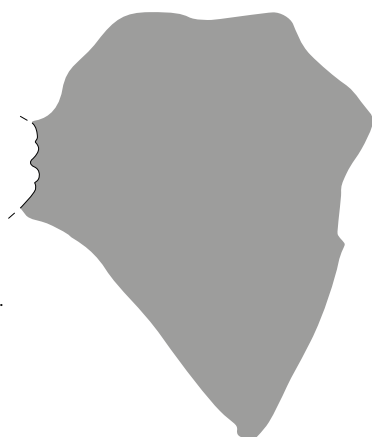


Fig. 1107. 1:2.

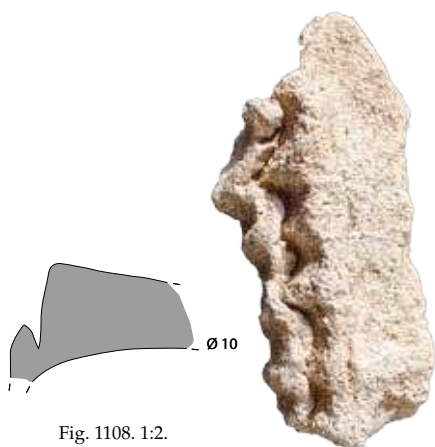


Fig. 1108. 1:2.

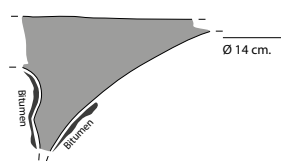


Fig. 1109. 1:2.



Fig. 1110. 1:2.



Fig. 1112. 1:2.

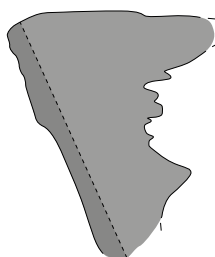


Fig. 1111. 1:2.



Fig. 1113. 1:2.



Fig. 1114. 1:3.



Fig. 1115. 1:3.



Fig. 1116. 1:3.

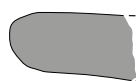


Fig. 1117. 1:3.

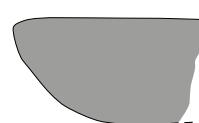


Fig. 1118. 1:3.

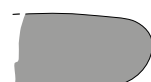


Fig. 1119. 1:2.



Fig. 1120. 1:2.



Fig. 1121. 1:4.



Fig. 1125. 1:1.



Fig. 1126. 1:1.



Fig. 1127. 1:1.



Fig. 1123. 1:2.



Fig. 1128. 1:1.



Fig. 1130. 1:1.



Fig. 1131. 1:1.



Fig. 1132. 1:1.



Fig. 1133. 1:1.



Fig. 1122. 1:2.



Fig. 1124. 1:2.



Fig. 1134. 1:1.



Fig. 1135. 1:1.



Fig. 1136. 1:2.

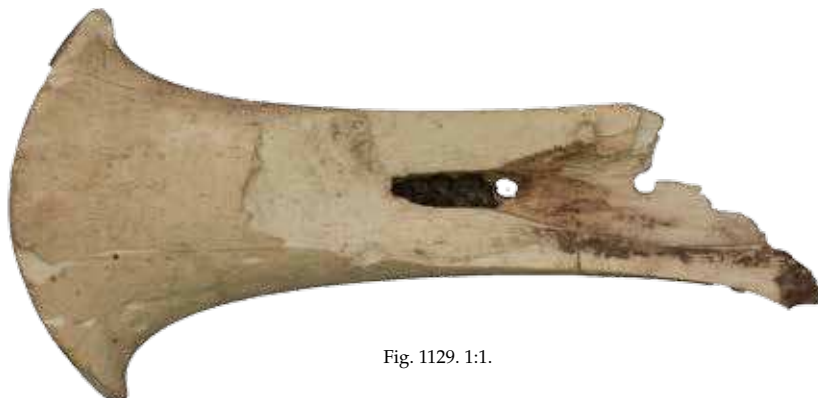


Fig. 1129. 1:1.



Fig. 1137. 1:1.



Fig. 1138. 1:1.

(identified by Flemming Nielsen, Curator at Copenhagen Zoo). From disturbed context in Trench CES.

Gold sheet (fig. 1135). Smooth sheet gold fragment (C177-X598). From disturbed context in Trench CEE. For similar material from the Barbar temple, cf. Andersen & Højlund 2003 p. 315, figs. 809-813.

Polished oval stone (fig. 1136). An oval, finely polished stone, 1.75 cm thick (E319-X1352) from Trench E, in a Phase 4 context.

Shaped calcareous stone with rivet hole (fig. 1137). Part of composite statuette? Cream coloured, from a disturbed context (C128-X346).

Female terracotta figurine (fig. 1138). Head of a Hellenistic terracotta statuette made in a mould (C196-X770) (cf. Mathiesen 1982 figs. 25-32). From disturbed context in Trench CEE.

Flint. Flint occurred only rarely in the excavations reported here, as natural nodules or natural flakes, without evidence of any human interference. Note that human worked flints were found in some number at the contemporary site of Al-Khidr (Benedikova 2010 p. 56, figs. 93-94).

Seashells. Shells found during the excavation were only partially kept and have not been studied.

3.11. Animal bones

by Lisa Yeomans

Introduction

A substantial quantity of animal bones were recovered during the 2008-2012 seasons providing evidence for the utilisation of animal resources throughout the occupation of the settlement and these were studied in the 2013 season. The faunal remains from the earlier (1958-1963) excavations at Tell F6 were not studied as the recovery method is unlikely to have been consistent and the sample is probably biased; in the future, given available time, it will be worth comparing these two samples. Overall, the analysis of the animal bones indicates that the population inhabiting the settlement was sustained by the herding of livestock as well as fishing with the occasional hunting of birds. Substantial differences in the exploitation of resources occurred between the Mesopotamian Ur III and the Dilmun Phases of occupation with the former relying heavily on sheep and goat herding whilst fishing became more frequent in the subsequent occupation (fig. 5). Additionally, there is evidence for slaughtering of livestock at the Dilmun temple probably as sacrifices and the ritual burning of the heads of these animals took place in oven structures. Fishing strategies changed over time with evidence for intensive summer exploitation of the marine environment during the Ur III Phase of settlement. During the Dilmun occupation, the coastal waters surrounding the island provided the main fishing grounds but occasionally fishing boats would target deeper waters and the pelagic fish they supported.

Methodology

Only the faunal remains from undisturbed contexts were kept for analysis and the bones were recovered using a 4mm sieve. Flotation samples were not taken during the excavations so the smallest species of fish that would probably have been exploited are unlikely to have been recovered. The mammal and bird faunal remains that are identifiable were recorded on a database providing information on the element, taxonomic grouping, taphonomy, aging information and any osteometric data for each of the bone fragments. Not all elements of the fish skeleton are recorded due to the difficulty in identifying some elements to specific families or beyond. Previous researchers working

on Arabian Gulf fish assemblages (Beech 2004) and on fish bone assemblages in general (Colley 1990) selected specific elements to be recorded further. Elements recorded were the articular, basioccipital, cleithrum, dentary, hyomandibular, maxilla, post-temporal, premaxilla, opercular, quadrate, vertebra and vomer. A few additional elements specific to certain species were recorded such as the distinctive neurocranial fragments.

Phase 1, the Mesopotamian Ur III Settlement

The animal bone from the Mesopotamian Ur III Settlement derives from Trenches C, J and I (figs. 4-5 and 9). These faunal remains are often covered in a concreted sandy deposit probably as a result of the layers they were deposited in remaining saturated for long periods of time. This resulted in the relatively poor level of preservation of the bone and often it was difficult to expose the original bone surface making recording of cutmarks problematic. Fig. 1139 shows the number of identified specimens of mammal bone, bird bone and other remains from the various phases of occupation. Contexts C210 and C218 are the backfill of a construction cut for the outer Dilmun Temple wall (cf. p. 76); however it was dug through and backfilled with Ur III strata and exclusively contains material culture of this date. The faunal remains from these two contexts have a high degree of coherence and can be confidently assigned to the Ur III phase of occupation and have, in later analysis, been included under the Ur III phase of occupation.

Overall the faunal assemblage from the Mesopotamian Ur III layers are dominated by domestic livestock. Caprines are the most common of which both sheep and goat are present with the former more frequent. Cattle were present but in very low numbers probably because of the amount of water required by these animals. No wild mammals are present in the assemblage but wild birds were occasionally hunted. These were the cormorants and egrets that would have frequented the shores of Failaka especially during the winter and would have formed a welcome variation to the diet. One bone of a bird of prey within the

Taxon	Common name	Phase 1	Temple trench	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
<i>Bos taurus</i>	Cattle	12	27	12	47	71	19	6	9
<i>Ovis aries</i>	Sheep	87	126		5	40	3	11	4
<i>Capra hircus</i>	Goat	58	33	1	29	13	1	3	1
<i>Ovis aries/Capra hircus</i>	Sheep/goat	504	243	1	47	115	33	31	61
Micromammal							1		1
Ardeidae	Egret	3			1				
Accipitridae	Hawk, kite or eagle	1							
Charadriiform		1							
Gruidae	Crane				1	1			
<i>Phalacrocorax</i> sp.	Cormorant	22	3		2	6	2	21	4
Passeriform					1				
Phasinidae	fowl				1				
Bird		3	2			2	1	1	
Turtle				5	12				
Crab		1			1	7	3	1	1
Sepiidae		11			7		1		1

Fig. 1139. Number of identified specimens (NISP) of bone from mammals, birds and other remains from various phases of occupation of Tell F6.



Fig. 1140. Tarso-metatarsus of a bird of prey in the Accipitridae Family (note the sand concretion adhering to the bone recovered from this Phase of the occupation sequence). 1:1.

Family Accipitridae (fig. 1140) was also identified and the possibility that these birds were used for hunting cannot be discounted.

Both contexts C210 and C218 (fills of the foundation trench for the Dilmun Temple) contained a faunal assemblage dominated by lower limb bones of sheep and goat. These comprise of the bones mainly below the distal row of carpals in the fore-limb and below the astragalus in the hind-limb. In total 402 identified sheep/goat bones were recorded with

the highly skewed skeletal representation shown in fig. 1141 deriving from a minimum of 13 individuals. This biased body part representation is typical of primary butchery waste although more skulls would be expected in such a deposit. The high representation of lower limb bones could also be typical of waste from leather-workers as the skins are often left attached to the lower limbs when transported to leather workshops. There is also evidence that a high proportion of the sheep and goat represented by these bones were slaughtered within a relatively narrow age range. Nearly three quarters of the phalanges are fused suggesting that the livestock were over about 10 months of age whilst the majority of the metapodials were unfused and therefore from animals below 30 months of age (fig. 1142) based on the fusion ages used by Uerpmann and Uerpmann (1994) although fusion ages given by other researchers suggest an earlier age for the fusion of the metapodials. This is a typical mortality profile for a livestock kept mainly as a source of meat with the animals killed as they approached full size with a smaller percentage kept alive longer for breeding.

Fig. 1143 shows the number of fish bones identified throughout the stratigraphic sequence. In the Ur III phase the most commonly caught fish are the Sparidae or seabream and the Serranidae or groupers. These species can be taken using a variety of methods but commonly they are caught using *gargours* (dome-

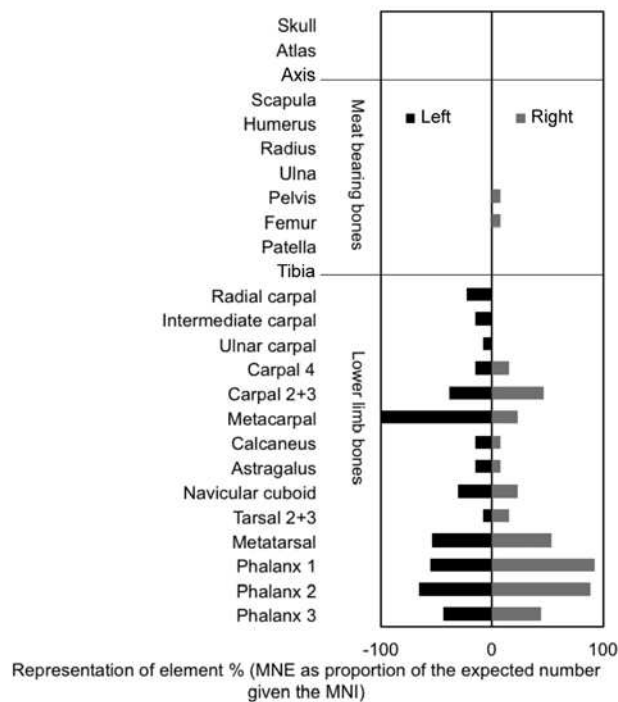


Fig. 1141. Body part representation of sheep and goat bones from the fill of a foundation trench of the Dilmun Temple wall in Trench C (contexts C210 and C218). MNE = minimum number of elements and MNI = minimum number of individuals.

Element	Fused	Fusing	Unfused epiphysis	Unfused shaft
Metacarpal	7	1	16	8
Metatarsal	3	1	10	5
Metapodial not identified to fore- or hind-limb	3	2	2	4
Phalanx 1	37	7	15	12
Phalanx 2	65	5	3	9
Percentage over 10 months	73,9			
Percentage over 30 months	28,9			

Fig. 1142. Summary of the sheep/goat fusion data from contexts C210 and C218.

shaped mess traps with a funnelled entrance) set on the sea floor in coastal or near-shore waters or hook and line fishing. Various species of Chondichthyes are also well represented; these cartilaginous fish are often caught in open waters in the case of the sharks and inshore shallow waters in the case of the saw-fish. Traditionally the flesh of these fish is sun-dried providing a preserved source of protein that could be taken on long sea voyages for instance.

Phase 2, the thick-walled buildings

The faunal remains assigned to this phase of occupation are dominated by fish (fig. 5). These include frequent cobia, barracuda and large carangids which would most likely have been caught in open waters using handlines or by trolling. Other species found in coastal waters were also present including Lethrinidae (emperors) and Sparidae. Of the Sparidae the haffara/goldlined seabream (*Rhabdosargus* sp.) were particularly common and they often found in lagoon environments. Turtle is represented by carapace fragments with one of these displaying cutmarks on the external surface (fig. 1144) suggesting that a knife was used to remove the outer shell possibly to be used as raw material.

Hearth E216 (fig. 40) is located below the oven constructed in Phase 3 (see below) and there is the possibility that this functioned as a precursor to this later feature. However minimal bone was recovered from this feature; the fill of the hearth contained several unidentifiable burnt bird bone fragments. A few cattle bones were recovered from one context in Phase 2 and several of these were burnt but there is no evidence to support the suggestion that the hearths in Phase 2 functioned in the same manner as the oven in the subsequent phase (see below).

Overall fish were the main source of food in this phase with other animals used as a source of raw material and occasionally herded livestock was consumed. Many of the fish remains are low temperature burnt suggesting their remains were discarded into cooling fires and the hearths found in this phase may have been used to prepare the fish rather than being a precursor to the sacrificial fire installations.

Phase 3, the thin-walled building

A building with thin walls comprising of several rooms with an external pavement and an oven structure were excavated from this phase (figs. 5 and 46). Analysis of the animal bone from the different rooms and areas can be used to indicate differences in function and utilisation of animals. No animal bone was recovered from the floor of Room 1. Several pits truncated this floor layer although they may relate to later activity and the construction of the "Palace". The fill (A334) of one pit (A210) contained the remains of at least six haffara/goldlined seabream the bones of which had mostly been low temperature burnt (fig. 1145). The fill (E223) of another pit (E180) contained some burnt and fragmented cattle bone and appears to be disturbed remains associated with the oven rather than relating to the use of Room 1.

Family	Species	Common name	Phase 1	Temple trench	Phase 2	Phase 3	Phase 4	Phase 5	Phase 6	Phase 7
Carcharhinidae	<i>Carcharhinus</i> sp.	Requiem shark	23	1		21		1	2	
Sphyrnidae	<i>Sphyrna</i> sp.	Hammerhead shark	13							
Pristidae		Sawfish	31	5						
Chondrichthyes		Unidentified cartilaginous fish	2							
Ariidae	<i>Arius</i> sp.	Catfish	6			1				
Mugilidae		Mullet	2							1
Belonidae		Needlefish	9							
Platycephalidae	<i>Platycephalus</i> sp.	Flathead	13	1			4			
Serranidae		Grouper	58	7	92	381	392	49	86	52
Rachycentridae	<i>Rachycentron canadum</i>	Cobia	4		21	5	12	1	2	
Carangidae	<i>Alectis</i> sp.	Indian treadfin			1	4	1			
Carangidae	<i>Carangoides</i> sp.	Trevally/jack			1	20	14	3		
Carangidae	<i>Caranx</i> sp.	Trevally/jack	4							
Carangidae	<i>Gnathanodon speciosus</i>	Toothless trevally		1		1				
Carangidae	<i>Scomberoides</i> sp.	Queenfish	1			11	59	18	10	3
Carangidae	<i>Seriola</i> sp.	Amberjack	1							
Carangidae		Unidentified jack/trevally/scad	9	1	14	39	133	26	12	7
Haemulidae		Grunt	8							
Sparidae	<i>Acanthopagrus</i> sp.	Seabream	11	1	3	5	30	2	5	1
Sparidae	<i>Argyrops spinifer</i>	King soldierbream	8		5	7	12	3		2
Sparidae	<i>Rhabdosargus</i> sp.	Haffara/goldlined seabream	6	2	51	47	9	2	2	9
Sparidae		Unidentified seabream	70	6	9	16	15	7	18	19
Lethrinidae	<i>Lethrinus</i> sp.	Emperor	13	1	76	98	207	61	40	217
Nemipteridae		Threadfin bream						2	1	
Sphyaenidae	<i>Sphyaena</i> sp.	Barracuda			25	31	51	5	10	
Scombridae	<i>Scomberomorus</i> sp.	Kingfish					5			1
Scombridae (Thunninae)		Tuna					2	5		
		Total	292	26	298	687	946	185	188	312

Fig. 1143. Number of identified specimens (NISP) of fish bone from the various phases of occupation of Tell F6 given in taxonomic order.



Fig. 1144. Turtle carapace fragment with cutmarks on external surface. 1:1.



Fig. 1145. Burnt dentary bones of haffara/goldlined seabream. 1:1.



Fig. 1146. Crab remains from Room 2 of the thin-walled building. 1:1.

Room 2 contained a series of floor layers and a small quantity of bone was recovered from the occupation deposits that accumulated on the floors. Most of the faunal remains were fish from cooking and consumption activities within the room. Bones of Serranidae and Lethrinidae were the most common but one bar-

racuda and one Carangidae bone were also present. Several fragments of crab (fig. 1146) also indicate the use of this marine resource.

To the north of Trench A, a series of layers (A206, A218 and A223) interpreted as levelled occupation debris were excavated. Faunal remains were recovered

Family	Species	Common name	Unburnt	Calcined	Carbonised	Burnt
Serranidae		Grouper	17	1	2	19
Carangidae	<i>Alectis</i> sp.	Indian threadfin		1		
Carangidae	<i>Scomberoides</i> sp.	Queenfish	2	1	1	
Carangidae		Unidentified jack/trevally/scad	2			
Sparidae	<i>Acanthopagrus</i> sp.	Seabream	1			
Sparidae	<i>Rhabdosargus</i> sp.	Haffara/goldlined seabream	4			
Sparidae		Seabream	1			
Lethrinidae	<i>Lethrinus</i> sp.	Emperor	1		1	
Sphyrinaeidae	<i>Sphyrna</i> sp.	Barracuda	2	1		
Bovidae	<i>Bos taurus</i>	Cattle	6	13	7	1
Bovidae	<i>Ovis aries</i>	Sheep			2	
Bovidae	<i>Capra hircus</i>	Goat	1	8	2	
Bovidae	<i>Ovis aries/Capra hircus</i>	Sheep/goat	4	14	7	
Phalacrocoracidae	<i>Phalacrocorax</i> sp.	Cormorant	1			

Fig. 1147. Number of identified specimens (NISP) identified within the oven contexts at different stages of burning.

from these layers in relatively large quantities. These were dominated by fairly large groupers Serranidae, some Carangidae as well as a few Lethrinidae, Sphyrinaeidae, Sparidae and some bones of *Rachycentron canadum*. Mammal bone was represented by a single fragment of an unburnt goat horncore and overall the faunal remains can be interpreted as occupation debris of the fish-based diet of the inhabitants of the thin-walled building or buildings in the immediate vicinity. Minimal bone was recovered from the pavement located around the east and south parts of the thin-walled building but the material present is typical of occupation debris from the occupation of the building. A series of floors (E308/E311 and E314) were excavated to the northeast of the pavement closer to the oven. Aside from occasional fish bone these contained a few unburnt bones of the lower limb bones of sheep/goat.

East of the pavement area was the oven (E138/E143) comprising of two parallel walls. The bone in the fills of this structure differed substantially to the faunal material recovered from the occupation debris described above. Fig. 1147 provides information on the number of bones identified within the oven and the level of burning that these bones reached ranging from low temperature heated, through carbonised to calcined indicating high temperature burning for a relatively long time. Cattle and sheep/goat bones were frequent in the oven and comprise mainly of parts of skulls. Goat horncores were found in the upper fill and these were relatively large suggesting that they may have been from males. In general, the

bones are covered in a concreted ash deposit which adhered to the bones. Fish bones are present but less heavily burnt. Large fish are particularly well represented in the fills of the oven perhaps indicating the consumption of a large fish on the occasion of the firing of the oven. The unburnt cormorant bone and the smaller fish bones would not have originated from the firing of the oven.

Large quantities of the faunal remains from Phase 3 derive from the makeup of a floor A180. This material is likely to be mixed debris from the occupation contemporary to the thin-walled building and the use of the oven. This is evident from the faunal remains with both burnt mammal bone and unburnt fish bone found within the same depositional contexts showing minimal coherence and they are therefore not particularly informative about the different utilisation of animal resources in Phase 3 spatially. A large amount of faunal remains were recovered from one context in the construction of the floor A188 containing numerous articulated portions of various fish suggesting that the material within the construction of the floor was not exclusively the disturbed remains from earlier occupation and additional waste was discarded into the floor makeup layers.

In summary the faunal remains from Phase 3 derive from two different activities. Domestic refuse from the occupation of the thin-walled building and probably other contemporary structures is heavily dominated by fish remains. The oven contains the burnt heads of domestic livestock which is possibly associated with ritual activities at the temple.

Phases 4-5

In this period the thin-walled building was demolished and the so called “Palace” constructed whilst the stratigraphic evidence suggests that the oven remained in use (figs. 5, 53 and 63). Faunal remains from the renewed excavations in the southeast corner of this large structure provide the only evidence for utilisation of animal resources in the “Palace”. Only a small area of the interior of the “Palace” floor was exposed and provided no faunal remains. Evidence for the consumption practices of those occupying the “Palace” has to rely on the occupation deposits that accumulated outside the “Palace” during this phase. As already mentioned the construction of the earliest external floor (A180) contains mixed material from the previous phase of occupation and are not discussed in detail. The occupation between the construction of the surface A180 and surface E109 (Phase 4) was characterised by faunal assemblages dominated by fish (figs. 58-59, 1148). Overall in Phase 4 the frequency of fish formed 79.9% of the NISP and in the occupation deposits which accumulated outside the “Palace” the frequency was even higher at 89.1%. The sheep and

goat remains were all elements of the skeleton and indicate the consumption of these livestock probably within the “Palace” structure. The cattle remains present are mostly isolated teeth and as these are not burnt they do not appear to be material from the oven. It seems, from the occupation deposits that accumulated outside the “Palace” in this phase, the diet of the inhabitants was dominated by fish including a number of larger fish like tuna and kingfish that would have been caught in open waters. In addition livestock and wild birds were occasionally consumed.

The stratigraphic sequence above floor E109 (Phase 5) provides a temporal comparison to the occupation sequence above floor A180. Minimal bone was incorporated into the plaster floor itself but substantial quantities of occupation debris were found above the surface (data also in fig. 1148). In these contexts 83.2% of the identified fish and mammal bone was fish remains and overall there is no significant difference in evidence for the consumption of fish and mammal when the occupation sequence is divided above and below floor E109. A number of pits and layers located closer to the oven contained additional faunal

Family	Species	Common name	Above A180	Above E109	Pits	South near oven
Carcharhinidae	<i>Carcharhinus</i> sp.	Requiem shark			1	
Platycephalidae	<i>Platycephalus</i> sp.	Flathead	4			
Serranidae		Grouper	167	71	21	7
Rachycentridae	<i>Rachycentron canadum</i>	Cobia	2		1	1
Carangidae	<i>Carangoides</i> sp.	Trevally/jack	2	1	2	
Carangidae	<i>Scomberoides</i> sp.	Queenfish	33	6	9	4
Carangidae		Unidentified jack/trevally/scad	63	17	4	2
Sparidae	<i>Acanthopagrus</i> sp.	Seabream	21	4	2	
Sparidae	<i>Argyrops spinifer</i>	King soldierbream	6	1	2	1
Sparidae	<i>Rhabdosargus</i> sp.	Haffara/goldlined seabream	2	4	3	
Sparidae		Unidentified seabream	1	8	4	1
Lethrinidae	<i>Lethrinus</i> sp.	Emperor	134	44	30	6
Nemipteridae		Threadfin beam		2		
Sphyracidae	<i>Sphyracna</i> sp.	Barracuda	29	1		2
Scombridae	<i>Scomberomorus</i> sp.	Kingfish	1		1	
Scombridae (Thunninae)		Tuna			2	2
Bovidae	<i>Bos taurus</i>	Cattle	14	5	7	2
Bovidae	<i>Ovis aries</i>	Goat	3	1		1
Bovidae	<i>Capra hircus</i>	Sheep	10	11	5	2
Bovidae	<i>Ovis aries/Capra hircus</i>	Sheep/goat	30	15	13	12
		Micromammal	1	1		
Phalacrocoracidae	<i>Phalacrocorax</i> sp.	Cormorant	3	1		
Portunidae	<i>Portunus</i> sp.	Crab	2	1	1	
Sepiidae		Cuttlefish		1		

Fig. 1148. Distribution of the identified faunal remains by subphase and context type for Phases 4-5 occupation.



Fig. 1149. Sheep metatarsal offcut from bone working. 1:1.

remains. In these contexts more cattle remains were recovered and probably are mixed occupation debris and material from the firing of the oven.

Phase 6

After a period of abandonment the “Palace” was reoccupied and additional buildings were constructed (figs. 5 and 64) and there is evidence for both Dilmun and Mesopotamian influences in the cultural evidence. The quantity of animal bone recovered from this phase of occupation is substantially lower but still provides some information about the utilisation of animal resources. The plaster floor in the northeast of the trench contained the wing bones of at least two cormorants which accounts for the high representation of cormorant bones in this phase. There is also a notable decrease in the frequency of cattle probably corresponding to the oven going out of use. In other respects the representation of the different taxonomic groups is not substantially different to that in the preceding phase. Both open water and inshore fishing still took place and sheep/goat herding occurred on the island. An offcut from bone working was found in one of the occupation debris layers (E89) that accumulated during this phase of the occupation. This sheep metatarsal (fig. 1149) documents the process of preparing bones to be used in the manufacture of bone handles.

Phase 7

Continued accumulation of occupation deposits occurred within this phase (figs. 5 and 68). The influence of only Mesopotamian culture was noted in the cultural evidence. The earliest deposit in this phase was a thick white plaster floor A35 containing pot sherds from the preceding period so the animal bone from this layer is probably also mixed. Some bone was recovered from a series of occupation layers dated to this phase and also the fill of a large clay pit (fig. 1150). There is a slightly higher representation of sheep/goat in the clay pit and overall the material seems to be redeposited occupation debris. In this phase there is a notable decrease in the fish caught in open waters,

most of the fish are likely to have been taken in inshore waters with the fish within the family Lethrinidae very common.

Changing animal exploitation patterns

The sections above provide an overview of the faunal remains from each phase of occupation highlighting individual contexts of particular interest and faunal remains of special note. To illustrate the dramatic change in the focus of the animal economy between the Mesopotamian Ur III settlement (Phase 1) and the subsequent phases of occupation, the relative proportion of the main taxonomic groups are shown in fig. 1151. There was a complete shift in the provisioning of the settlement with the main source of animal protein being herds of sheep/goat in the Mesopotamian Ur III period but afterwards fishing was key to supporting the population. Some fishing did occur in the Mesopotamian Ur III period but even the target fish were not the same with cartilaginous fish (Chondichthyes) more common. These fish are frequently dried in the sun providing a transportable source of protein and may have been targeted as a source of food to supply traders passing through the area. No Scombridae and very few of the large Carangidae were caught during Phase 1 whereas in later occupation the fishermen would have also exploited the deeper waters fishing from boats with hook and line to catch these fish.

The importance of livestock in Phases 4-5 may also be exaggerated by fig. 1151 as many of the cattle and sheep/goat remains are associated with the Temple (see below) possibly representing sacrifices made at the Temple rather than the everyday diet of the inhabitants.

The architecture in the Ur III phase consists of a probable courtyard area rather than domestic structures and this could explain the general absence of fish remains. Perhaps this location functioned as a provisioning area processing the carcasses of livestock on an industrial scale. Fish were probably consumed within the houses of the inhabitants but such structures have not yet been recovered from this phase of occupation.

Fish processing during the Dilmun occupation

The large quantity of fish remains from the Dilmun period (Phases 2-5) of the occupation sequence allows examination of the fish processing methods in more

Family	Species	Common name	Occupation above A35	Clay pit
Mugilidae		Mullet		1
Serranidae			17	34
Carangidae	<i>Scomberoides</i> sp.	Queenfish	2	1
Carangidae		Unidentified jack/trevally/scad	1	4
Sparidae	<i>Acanthopagrus</i> sp.	Seabream	1	
Sparidae	<i>Argyrops spinifer</i>	King soldierbream	1	1
Sparidae	<i>Rhabdosargus</i> sp.	Haffara/goldlined seabream	2	7
Sparidae		Unidentified seabream	8	11
Lethrinidae	<i>Lethrinus</i> sp.	Emperor	95	120
Scombridae	<i>Scomberomorus</i> sp.	Kingfish	1	
Bovidae	<i>Bos taurus</i>	Cattle	1	8
Bovidae	<i>Ovis aries</i>	Goat		1
Bovidae	<i>Capra hircus</i>	Sheep		4
Bovidae	<i>Ovis aries/Capra hircus</i>	Sheep/goat	12	49
		Micromammal	1	
Phalacrocoracidae	<i>Phalacrocorax</i> sp.	Cormorant	2	2
Portunidae	<i>Portunus</i> sp.	Crab	1	
Sepiidae		Cuttlefish		1

Fig. 1150. Number of identified specimens (NISP) from occupation above A35 and the clay pit in Phase 7.

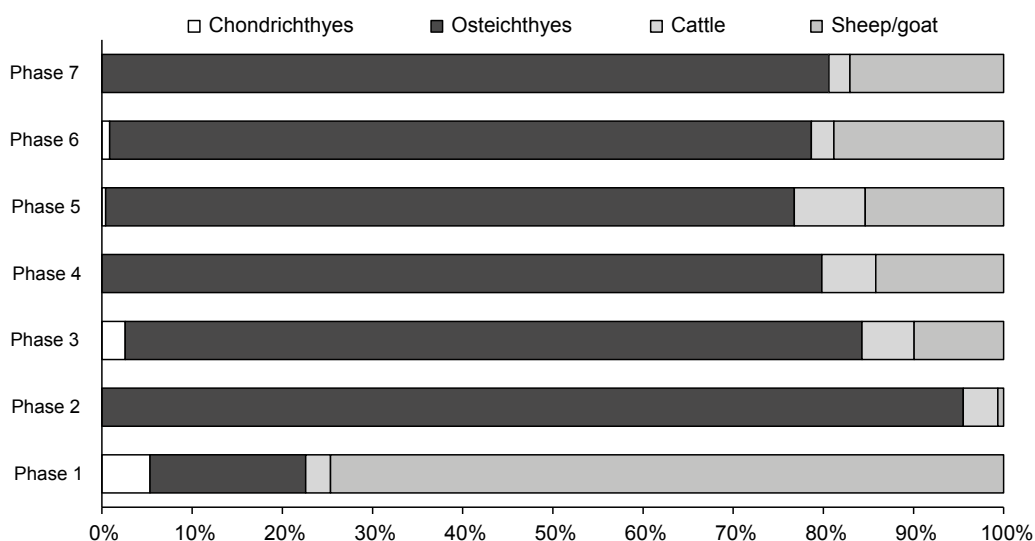


Fig. 1151. Proportional representation of the NISP of the main groups of faunal remains in the stratigraphic phases of Tell F6.

detail. Cutmarks on the bones of the large fish made during butchery are common. Fig. 1152 shows the number of Serranidae bones with cutmarks and which surface of the bone these were located on. Cutmarks on the medial (inner) sides of several skull bones suggest the use of large heavy blades during the gutting of the fish. The force used in slitting the fish from the stomach upwards towards the head resulted in the blade continuing through the head where it came

into contact with the inner side of the skull bones. Fig. 1153 shows an articulating maxilla and premaxilla of a grouper (Serranidae) with parallel cutmarks on both bones with the direction of the cut originating from the posterior side of the fish. As the heads were still attached when the fish were gutted it seems that the butchery was systematic with gutting first followed by head removal. Also most of the cuts to the first vertebra, as well as the basioccipital and second

			Percentage cut on side	
			left	right
Basioccipital	40	3	66,7	33,3
First vertebra	43	6	100	0
Abdominal vertebrae	348	1	100	0
Caudal vertebrae	198	0	-	-
			Medial	Lateral
Articular	28	0	-	-
Cleithrum	29	4	100	0
Dentary	43	2	100	0
Hyomandibular	33	0	-	-
Maxilla	26	1	100	0
Opercular	9	0	-	-
Premaxilla	44	1	100	0
Post-temporal	11	2	50	50
Quadrate	23	1	100	0

Fig. 1152. Number of Serranidae bones from layers dated to Phases 2-5 with cutmarks and their distribution.



Fig. 1153. Parallel cuts on the inside of the left premaxilla and maxilla of a grouper (Serranidae). 1:1.

vertebra, created during head removal, are directed from the left side of the fish perhaps suggesting that the stomach was turned away from a right-handed person as the head was removed.

Fishing strategies

As already mentioned there is a significant shift in consumption patterns between the Ur III and Dilmun Phases of occupation. In terms of the fish represented, one of the most notable shifts is the decrease in sharks and sawfish in the Dilmun period. Numerous species of requiem and hammerhead shark are only present in Kuwaiti waters in the summer months (Goubanov and Shleib 1980). Sawfish arrive also into these waters at the beginning of summer (Clayton and Pilcher 1983; see also Moore 2012) and it is possible that the high representation of the elasmobranchs in the faunal

record from the earliest phase of the settlement is associated with seasonal variation in the level of marine exploitation. As already mentioned, the flesh of sharks is traditionally often dried for future consumption (Costa 1988) and as such would be a suitable source of protein for trading and sea voyages. The evidence suggests that fishermen were harvesting the shark and sawfish species when they were present in Kuwaiti waters during the summer months possibly to dry their flesh.

During the Dilmun occupation there is evidence for an increase of fishing activities and in particular many of the fishermen focused on the inshore or lagoon waters. The haffara/goldlined seabream are well represented especially in Phases 2 and 3 and these fish inhabit these inshore, shallow waters. This more intensive exploitation of the inshore and lagoon waters in Phases 2 and 3 may also provide an explanation for the notable difference in size of the Lethrinidae compared to later occupation (fig. 1154).

Barracuda also form part of the assemblage from the Dilmun Phases and, whilst the juveniles inhabit inshore and lagoon waters, those found in the assemblage are from adult sized fish. The larger of the groupers and Carangidae were probably also caught by hook and line fishing for which there is the additional evidence of fish hooks. Intertidal fish traps are common around Failaka today and these, in addition to basket traps, probably formed part of the repertoire of fishing techniques used. Two families that would be expected to be present in the assemblage and for which there is no evidence is the Gerridae (silver biddies) and the Siganidae (rabbitfish). Both of these groups are very small in size and, particularly for the rabbitfish, the bones are thin and fragile. The absence of these species from the archaeological record is probably related to the taphonomic differences in the preservation and recovery of the faunal remains.

Comparison of the fish represented from the Dilmun layers at Tell F6 to other excavated Bronze Age assemblages from the island indicates that the fishing methods and catches are similar. In the Dilmun deposits also from the Temple in Tell F6 excavated by the French team during the 1980's a limited sample of fish remains were recovered (Desse and Desse-Berset 1990). A similar range of species were identified with the small assemblage dominated by Serranidae and Carangidae. At al-Khiḍr the list of species recovered (Hajnalová *et al* 2009) is also the same with the exception that no parrotfish (Scaridae) were identified at Tell F6; this may relate to differences in the local marine environment.

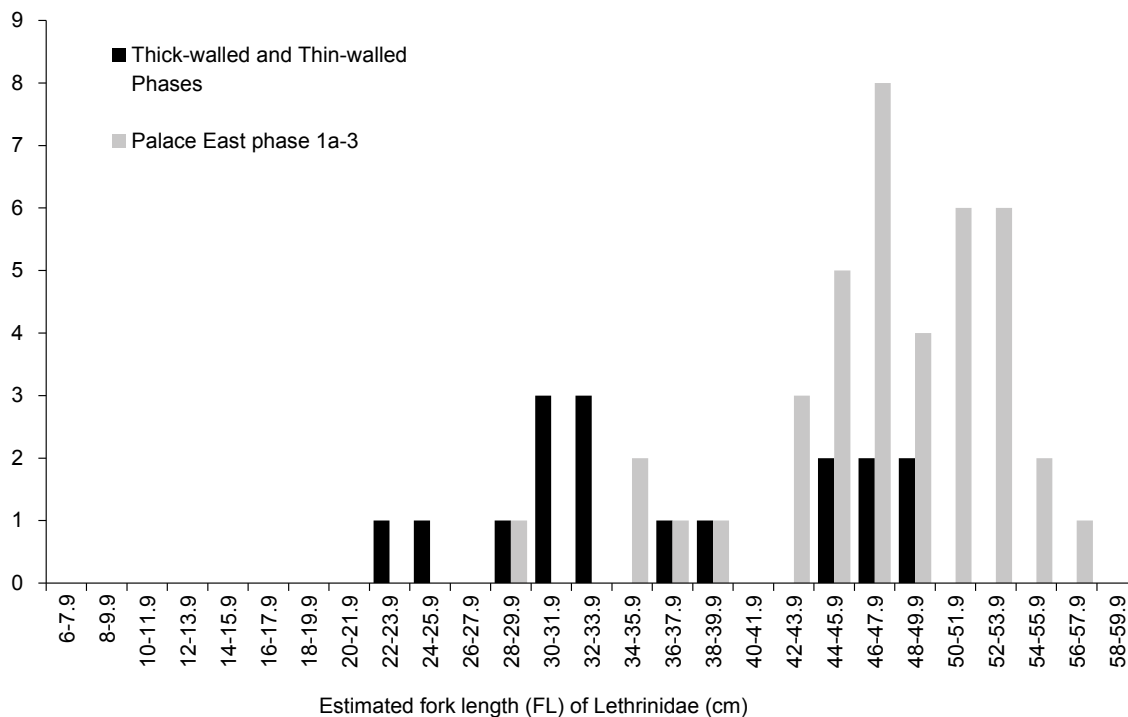


Fig. 1154. Comparison of the reconstructed fish size (FL) of Lethrinidae in Phases 2-3 with Phases 4-7.

Livestock types

Comparison of the size of the sheep and goat bones to those found at other sites allows some discussion of possible similarities to the types of livestock herded in other regions. Sheep were more common than goat and they are often more frequent at sites in the region and thereby provide the most evidence to examine livestock types. The use of the log standard animal index method allows comparison of various osteometric measurements on the same graph using the formula $LSI = \log x - \log S$ where x is the measurement of an archaeological bone and S is the same anatomical measurement taken on a standard animal. The standard animal values published by Uerpmann and Uerpmann (1994) were used in this analysis.

Fig. 1155 compares the LSI values of the sheep from Tell F6 with those published from Qala'at al-Bahrain (Uerpmann and Uerpmann 1994) and shows that overall the size range of the sheep was similar with possibly a few more sheep at the small end of the size range present in the Mesopotamian layers in Tell F6 and a few of the larger size of sheep in the Dilmun layers. Uerpmann and Uerpmann (1994) demonstrate that the sheep from Qala'at al-Bahrain are similar in size to those found at sites of a similar date in Lower Mesopotamia whilst sheep from sites further East, at Hili 8 and Maysar, were notably smaller. Meadow (1991) argued that the sheep size at Harappa was larger than the local sheep populations and their import from

Mesopotamia was suggested as one explanation (Uerpmann and Uerpmann 1994). If this was indeed the case, it also seems probable that the sheep on Failaka also resulted from this trade. Not all of the sheep in Bahrain were of this larger type however; Uerpmann and Uerpmann (1994; 2005) noted that at Saar in Bahrain, as well as the larger type of sheep, a smaller more common type of sheep similar in size to those from sites located further east along the Gulf were present and argued that these were the local type.

Faunal remains associated with the Oven E138/E143

The assemblage of burnt cattle skulls and mandibles (fig. 1156) found in the oven and the surrounding contexts are clearly reminiscent of the faunal assemblage from the Barbar Temple in Bahrain (Bangsgaard 2003) where cattle were the most numerous. The assemblage of cattle bone from the Barbar Temple was dominated by skulls and tooth fragments and even more tooth fragments were only identified as large ungulate but could only really possibly be from cattle. These teeth fragments were "very fragmented and for some part severely burnt" (Bangsgaard 2003 p. 7). Such a high percentage of cattle are very unusual of sites in this region and the skewed body-part representation and high degree of burning are an indication that the cattle bones in and around the oven structure at Tell F6 accu-

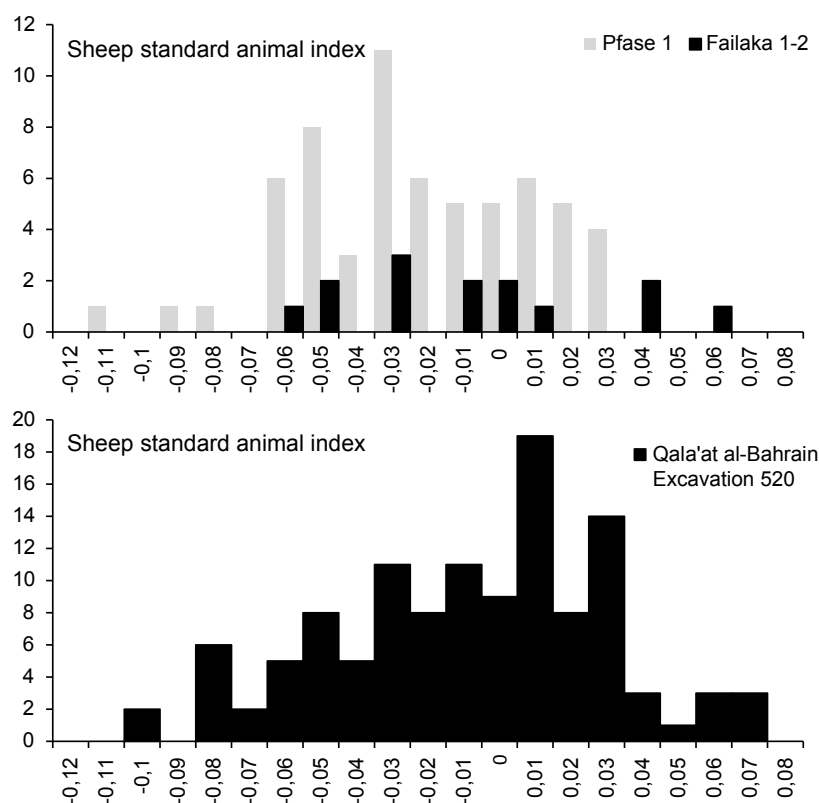


Fig. 1156. A well preserved burnt cattle tooth recovered from the oven. 1:1.

Fig. 1155. Comparison on the LSI values from Tell F6 and Qala'at al-Bahrain Excavation 520.

mulated as a result of a ritual practice. The heads of sheep and goat (more commonly goat) were also used in these offerings. Some of the larger fish remains were also burnt. At Saar burnt fish bones were recovered from areas of ash on and around one of the altars (de Moulins *et al* 1997). Many of the teeth of the cattle are not in a very good state of preservation but it is possible to get some idea of the ages of animals that were slaughtered from the different teeth. Unfortunately, none of the teeth were found in the mandibles but where the teeth were identifiable they were all permanent dentition including two lower M₃'s suggesting that fully adult animals were present.

Conclusions

The faunal remains from the 2008-2012 excavations at Tell F6 provide evidence for the changing patterns of animal utilisation between the Ur III settlement and

the Dilmun occupation which may reflect the changing use of this area of the settlement. In the Ur III Phase of occupation there is evidence for the seasonal fishing of sharks and sawfish probably to be dried. Additionally, there is evidence for the specialised processing of sheep and goats. It remains uncertain if the archaeological remains reflect an area where butchery was taking place or if this area could be related to the processing of animal skins in some manner. During the latter part of the sequence, many of the faunal remains were recovered from domestic structures and as such reflect the diet of the inhabitants. Animals were also clearly important in the ritual life of the Temple and with livestock slaughtered and the heads of the animals burnt in ovens. Cattle were the most frequently chosen to fulfil this role probably because of their high value in terms of comparable difficulty in rearing these animals in the local environment and the quantity of meat that one animal could provide.

4. Radiocarbon chronology

by Jan Heinemeier & Flemming Højlund

The layers excavated in 2008-2012 were referred to eight stratigraphic phases, Phases 1-8. Based on the pottery excavated from these layers these phases have been related to existing chronologies in Failaka, Bahrain and Mesopotamia as argued above (p. 93, fig. 5).

Whereas the chronologies of the Oman peninsula, Iran and the Indus Valley have been based on the radiocarbon method, the relative chronology of the upper Gulf (i.e. Bahrain and Kuwait) have been related through imported finds to the absolute, "Middle Chronology" of Mesopotamia, and radiocarbon dates have been too few and imprecise to be of much value (Højlund 1987 p. 157-161, fig. 707. Højlund & Andersen 1994 p. 168, fig. 395. Højlund 2007a fig. 3, p. 15).

From the onset of the 2008-2012 excavations in Tell F6 it was, however, decided to utilise radiocarbon dates in order to test and refine the Arabian Gulf Bronze Age chronology and the *Kuwait National Council for Culture, Arts and Letters* generously agreed to support this project.

Finding carbon samples for dating was not without difficulties. Very little organic or charred material was encountered in the excavation and, apart from two charred date stones, charcoal was found mostly in the shape of small spots and lumps without any clear wood structure.

All excavated layers contained plenty of bitumen that could be mistaken for charcoal, so only lumps large enough to perform a "bitumen/charcoal test" were considered – part of the lump was smeared between thumb and index finger. If it left a *smooth, fat and black* impression it was accepted as charcoal, whereas if it left a *crumbling, dry and brown* impression it was rejected as bitumen.

A total of 79 samples from the 2008-2012 excavations (fig. 1160) were processed at the AMS ^{14}C Dating Centre of Aarhus University. Of these, seven were excluded from further analysis (AAR-15412, 15413, 15414, 15421, 16145, 16147 and 16161) because they gave excessively high ^{14}C ages (c. 11.000-43.000 BP) probably due to contamination with bitumen. A further six were excluded because they were considered too old in relation to the context in which they were found, possibly because

of wood being long dead before use or excessively high built-in age ("old wood effect", inner tree rings being dated) in the charcoal (c. 2100-2700 BC) (AAR-13776, 16144, 16153, 16154, 16156, 16163 and 18321). One was excluded because it was considered too young, probably a modern intrusion (close to present-day atmospheric ^{14}C level) (AAR-15416). One sample did not have sufficient material for dating (AAR-16155).

Sixty-six of the 79 radiocarbon-dated samples came from the 2008-2012 excavations in Tell F6 that are the subject of this volume. The Phase 1 samples derive from Trench C below the Temple and the samples from Phases 2-7 derive from Trenches A and E, the area between the "Palace" and the Temple.

The remaining 13 samples were found in 2012 in Trenches BC and BD in Tell F3, immediately west of Trench AI, which was excavated in 1958-1963 (Kjærøum & Højlund 2013 fig. 5). Excavations resumed in Tell F3 in 2012, continued in 2013, 2014 and 2015 (Højlund & Abu-Laban 2013. Højlund & Hilton 2014, 2015 and 2016) and are planned to continue for two more years.

The 13 measurements from Tell F3 are brought into this section because they derive from a period that was not represented in the 2008-2012 excavations in Tell F6, namely Failaka Period 3A. The samples were recovered from above the floors of a couple of rooms in the western continuation of House 1 (Kjærøum & Højlund 2013 fig. 7). House 1 is part of a row of houses (1-11, 16-17, and 22) all dating to Period 3A, and this is the date given to eight of the 13 samples included here (AAR18307-18313 and 18320). The remaining five samples (AAR18314-18318) were originally thought to predate Period 3A but they seem to derive from a disturbed area and are therefore discarded from the analysis.

Forty-nine of the 79 measurements are ordered according to the stratigraphically defined Phases 1-7 plus Period 3A (figs. 1157-1158). Within each phase an attempt has been made to order the individual samples in a stratigraphic sequence. The samples from Phases 4-7 derive from relatively thin layers corresponding to horizontal floors or layers of refuse

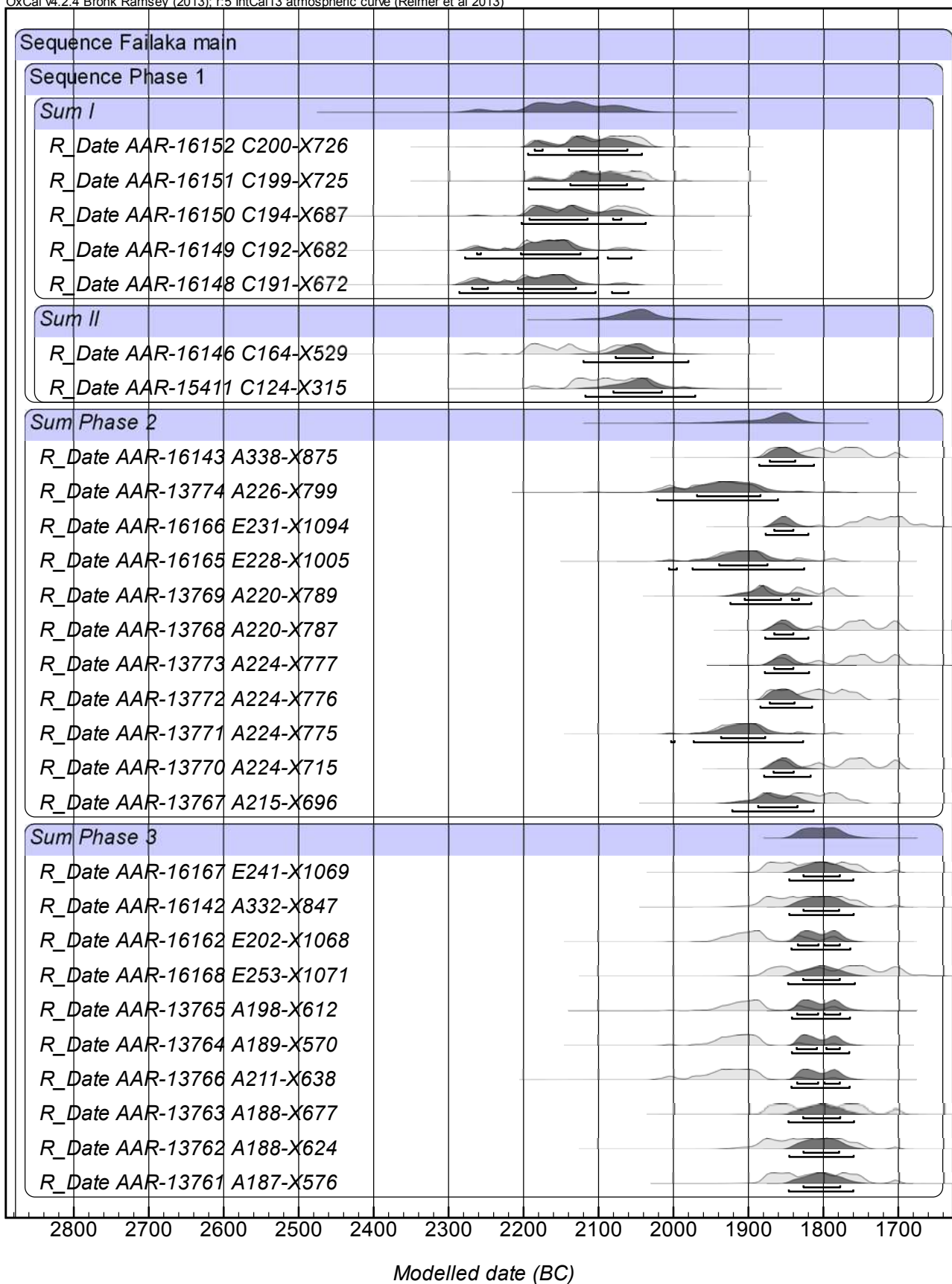


Fig. 1157. Main sequence prior (unmodeled) calibrated age probability distributions shown in light shades and posterior (modeled) distribution in dark shades. Series continued in fig. 1158.

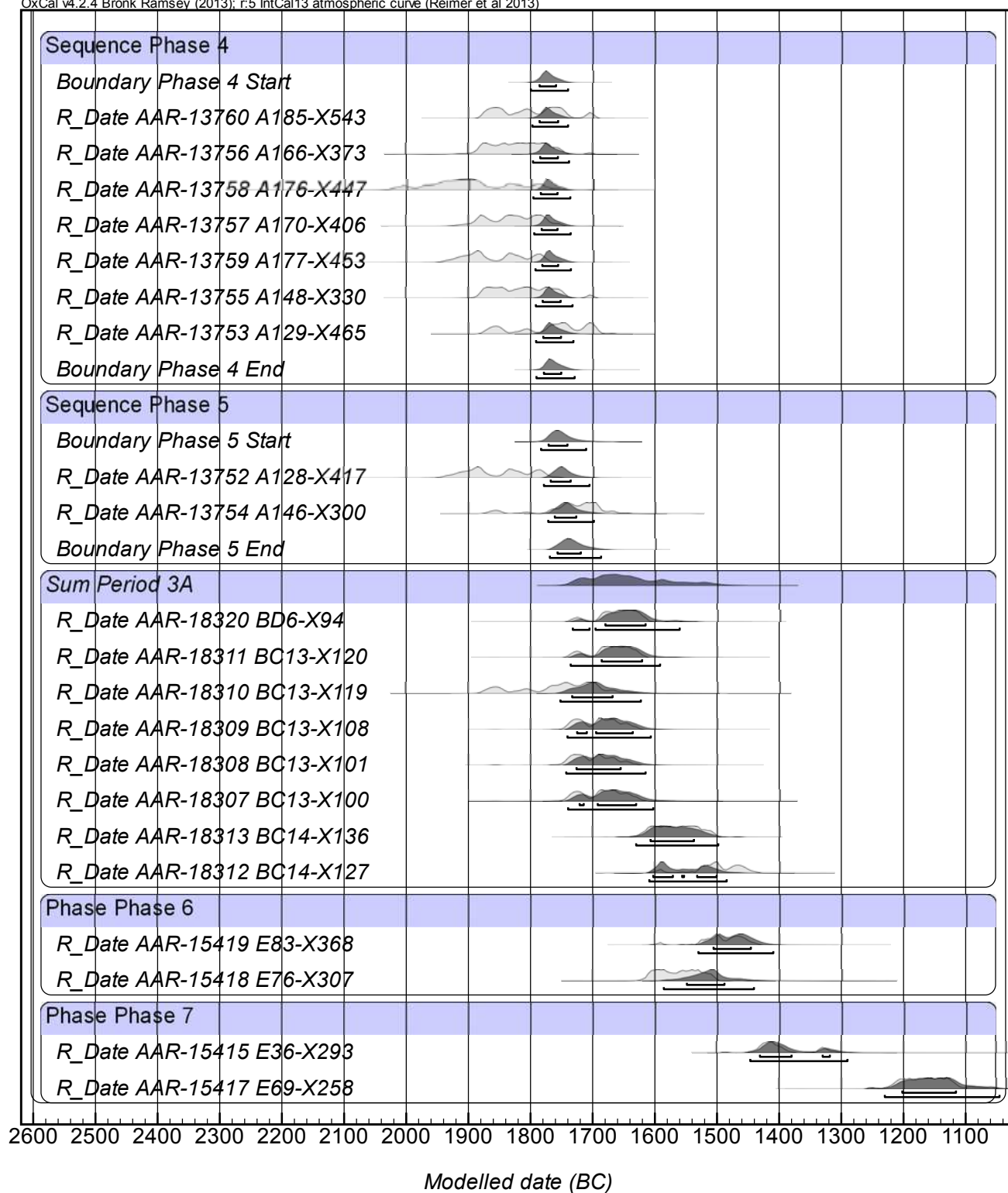


Fig. 1158. Main sequence prior (unmodeled) calibrated age probability distributions shown in light shades and posterior (modeled) distribution in dark shades. Series continued from fig. 1157.

deposited on these floors (cf. figs. 41, 52, 83, and 87), so that the stratigraphic sequence of these samples is securely determined.

The samples from Phases 2-3 are likewise ordered according to their height above sea level but layers from these phases are more disturbed by demolition and levelling of older architecture.

The samples from Phase 1 have been ordered in two groups; one consisting of two samples is stratigraphically contemporaneous with a house/courtyard (C28 cf. p. 15), the other derives from layers below this house/courtyard and consists of five samples.

No stratigraphic ordering within the series has been attempted for the Period 3A samples, but the sam-

ples of this period are assumed to fall chronologically between Phase 5 and 6 (fig. 5).

In addition, a small number of samples are treated separately from the preceding sequence, i.e. six samples from two ovens, two samples from the Temple and one sample from the Buttress Wall of the Temple (fig. 1159).

The method of Bayesian statistics implemented in the OxCal program (v. 4.2, Bronk Ramsey 2009a) was used to calibrate the ^{14}C dates with the latest calibration curve, IntCal13 (Reimer *et al.* 2013). The probability distributions of the unmodeled (prior) calibrated radiocarbon dates are shown in grey shades in the graphs (figs. 1157-1159). These dates have been further

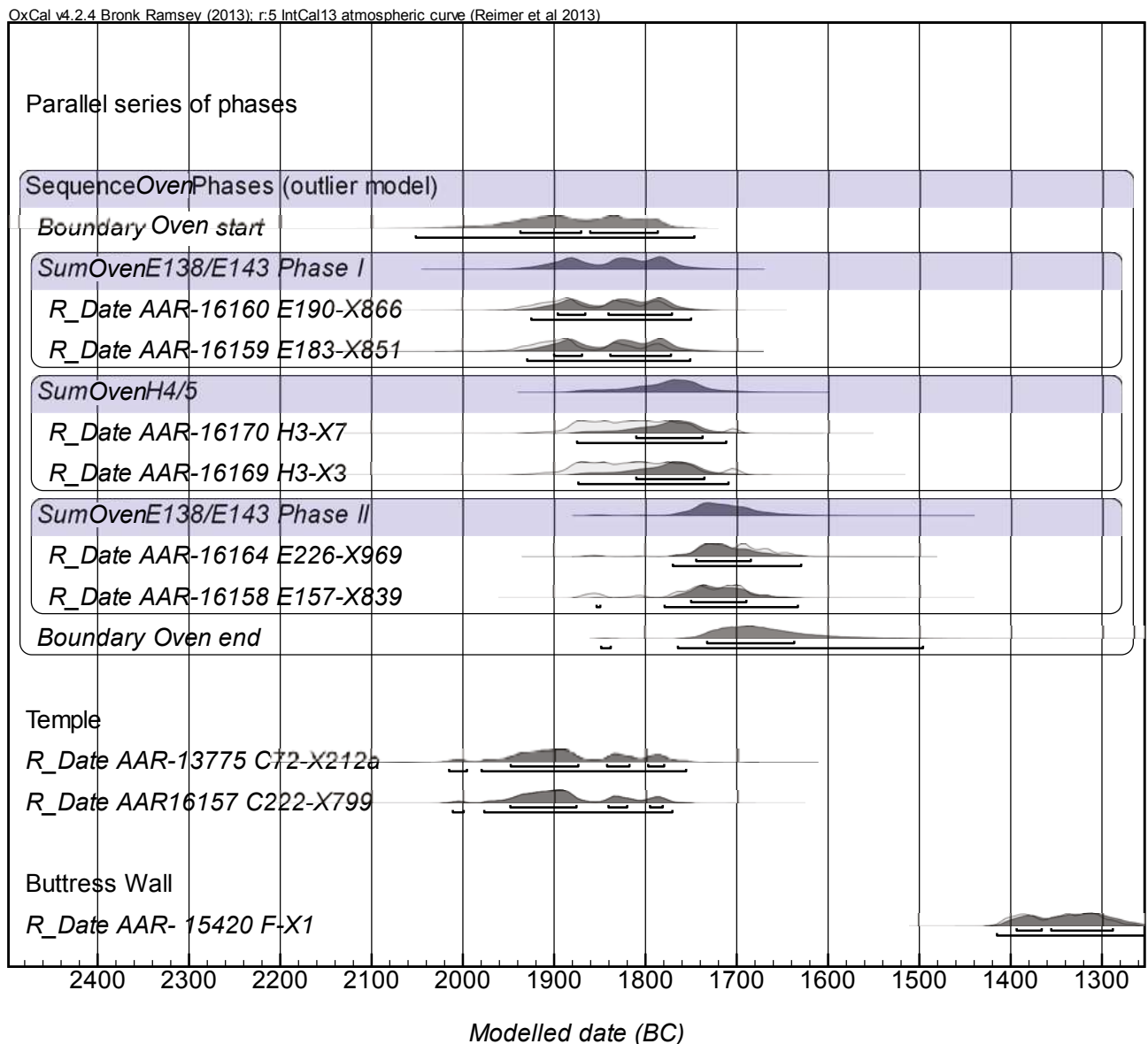


Fig. 1159. Separate Oven/Temple series prior (unmodeled) calibrated age probability distributions shown in light shades and posterior (modeled) distribution in dark shades.

analysed using a Bayesian model resulting in more precise dates shown in black shades. In order to achieve this precision certain constraining assumptions have been incorporated into the model. First of all the stratigraphic ordering of the samples has been used as constraints on the calibrated dates and this was done in two different ways. Within Phases 1-3 and Period 3A where the stratigraphic ordering within the groups is less consistent, the samples are treated as a "phase" in a statistical sense, i.e. all dates within a group are assumed not necessarily to be stratigraphically ordered and their probability distributions have been summed to give an impression of the distribution in time of these samples. In the plots the group descriptions start with the word "Phase" or "Sum". However, in Phases 4 and 5, where the stratigraphic evidence is more consistent, the samples within each group are assumed to be stratigraphical (i.e. in strict time order) and the groups are designated "Sequence" in the plots. For Phases 6-7, which only includes two samples each, no ordering has been assumed (designated as "phase") and no summing performed. All the phases and sequences of figs. 1157-1158 are constrained to be in sequence, nested within a main sequence ("Sequence Failaka main").

The Bayesian model further makes use of the fact that in general charcoal dates will always be older than their context as they relate to the growth of wood in living trees. There must always be some passage of time between this growth, through the death or cutting of the tree to its use in a building or as firewood. The subsequent conversion to charcoal may also be sometime later than this, though in this particular instance it is the use of the wood that we are primarily interested in (Bronk Ramsey 2009b).

In order to assess the reliability of the model the individual index of agreement (A: Bronk Ramsey 1995 p. 429) for each sample has been calculated. In general, the A values are fine, i.e. >60% (standard limit for good agreement), but four samples are significantly below 60%, either due to excessively high in-built age or younger intrusions and consequently the A value for the model as a whole (54%) is slightly below the critical 60% value.

Several charcoal samples from Phase 1 (the Ur III period) were rejected from the analysis because, based on their radiocarbon dates, they were deemed to be too old (c. 22-2400 BC) in relation to the context. They may have been residual in the sediment, i.e. the charcoal may already have been on the site for a long time when it was occupied in Phase 1.

The first group of dates from Phase 1 has a sum distribution of 2142 ± 56 cal BC and the second group

has a sum distribution of 2047 ± 33 cal BC. These dates partly overlap with the Mesopotamian Ur III period according to the Middle Chronology (2112-2004 BC) and deviate markedly from the New Low Chronology (2018-1911 BC) (Gasche *et al.* 1998).

The dates from Phase 2 have a sum distribution of 1872 ± 38 cal BC, and there seems thus to be a clear gap between Phase 1 and 2, i.e. a substantial period of abandonment after the termination of the Mesopotamian settlement and until the island is settled by Dilmunites, at least judging from the 2008-2012 excavations.

Thirty radiocarbon dates derive from Phases 2-5 and their measurements are very similar suggesting that these phases were short and close to each other in time. This corroborates the conclusions drawn from the study of their contents of pottery and stamp seals which showed no or very small typological changes (cf. p. 95 and 133). Phase 2 has a sum of 1872 ± 38 cal BC and Phase 3 of 1803 ± 23 cal BC. Phase 4 has a boundary start of 1771 ± 15 cal BC and a boundary end of 1762 ± 16 cal BC. Phase 5 has a boundary start of 1751 ± 19 cal BC and a boundary end of 1732 ± 21 cal BC.

The eight dates from Period 3A sum to 1641 ± 62 cal BC. Their distribution consists of two rather different subgroups, two dates later than 1600 cal BC and six dates earlier than 1600 cal BC. This suggests a somewhat later dating of Period 3A than hitherto thought (Højlund 2007a fig. 3, note 8).

The two Phase 6 samples (Period 3B) have mean dates of 1473 ± 32 and 1511 ± 36 cal BC partly overlapping with the previously suggested dating of this period (Højlund 2007a fig. 3).

The ongoing excavations in Tell F3 will hopefully provide a series of new dates for Periods 3A and 3B that will help clarify their start and end dates.

The two samples from Phase 7 (Period 4A) have mean dates of 1382 ± 47 cal BC and 1149 ± 47 cal BC. The pottery assemblage of Phase 7 seems rather homogeneous and devoid of any diagnostic Period 4B types. More radiocarbon dates for Periods 4A and 4B are needed in order to clarify their relative dating.

The separate series of samples (fig. 1159) that can only be partly stratigraphically related to the main series discussed above, have been analysed in a similar way, however, the stratigraphic constraints are fewer. Two phases can be distinguished in the walls of oven E138/E143. Convincing stratigraphic and architectural evidence places the construction of the first oven in Phase 3 (cf. above p. 39). The two radiocarbon dates of charcoal from this oven phase are summed to 1835 ± 49 BC and that is entirely compatible with the

AAR	Sample Reference	Material	Context	C14 age	d13C (dual-inlet)	Calibrated age
Phase 7						
15416	OA895-E37, X286	Charcoal	Tell F6, Trench E, layer E37	-302 ± 23	-26.69 ± 0.05	68.2% probability 1956.3AD (68.2%) 1956.5AD 95.4% probability 1956.18AD (95.4%) 1956.62AD
15417	OA895-E69, X258	Charcoal	Tell F6, Trench E, layer E69	2952 ± 24 (ext)	-23.52 ± 0.05	68.2% probability 1252BC (5.8%) 1242BC 1213BC (62.4%) 1128BC 95.4% probability 1264BC (89.3%) 1111BC 1103BC (4.5%) 1074BC 1066BC (1.5%) 1056BC
15415	OA895-E36, X293	Charcoal	Tell F6, Trench E, layer E36	3131 ± 24 (ext)	-25.87 ± 0.05	68.2% probability- 1434BC (68.2%) 1394BC 95.4% probability 1490BC (1.3%) 1481BC 1455BC (87.3%) 1374BC 1341BC (6.8%) 1318BC
Phase 6						
15418	OA895-E76, X307	Charcoal	Tell F6, Trench E, layer E7	3280 ± 23 (ext)	-23.5 ± 0.05	68.2% probability- 1608BC (35.1%) 1570BC 1561BC (11.7%) 1547BC 1541BC (21.3%) 1518BC 95.4% probability 1616BC (95.4%) 1501BC
15419	OA895-E83, X368	Charcoal	Tell F6, Trench E, layer E83	3228 ± 25 (ext)	-22.27 ± 0.05	68.2% probability- 1519BC (37.2%) 1490BC 1481BC (31.0%) 1455BC 95.4% probability 1605BC (2.9%) 1587BC 1536BC (92.5%) 1431BC
Period 3A						
18307	OA896- BC13, X100	Charcoal	Tell F3, Trench BC, layer BC13	3389 ± 26 (ext)	-26.01 ± 0.05	68.2% probability 1737BC (25.8%) 1711BC 1695BC (32.3%) 1662BC 1652BC (10.1%) 1640BC 95.4% probability 1746BC (95.4%) 1621BC
18308	OA896- BC13, X101	Charcoal	Tell F3, Trench BC, layer BC13	3403 ± 25 (ext)	-25.26 ± 0.05	68.2% probability 1743BC (68.2%) 1682BC 95.4% probability 1756BC (95.4%) 1626BC
18309	OA896- BC13, X108	Charcoal	Tell F3, Trench BC, layer BC13	3394 ± 25 (ext)	-10.13 ± 0.05	68.2% probability- 1739BC (33.7%) 1707BC 1698BC (31.3%) 1665BC 1648BC (3.1%) 1644BC 95.4% probability 1747BC (95.4%) 1626BC
18310	OA896- BC13, X119	Charcoal	Tell F3, Trench BC, layer BC13	3437 ± 35 (ext)	-24.12 ± 0.09	68.2% probability- 1867BC (9.3%) 1848BC 1774BC (58.9%) 1689BC 95.4% probability 1880BC (94.4%) 1664BC 1650BC (1.0%) 1643BC
18311	OA896- BC13, X120	Charcoal	Tell F3, Trench BC, layer BC13	3374 ± 25 (ext)	-10.11 ± 0.05	68.2% probability- 1727BC (3.4%) 1723BC 1691BC (64.8%) 1629BC 95.4% probability 1741BC (95.4%) 1613BC
18320	OA896- BD6, X94	Charcoal	Tell F3, Trench BD, layer BD6	3362 ± 27 (ext Small sample: 0.399 mg C)	-29.27 ± 0.54 (ext Small sample: 0.399 mg C)	68.2% probability- 1689BC (68.2%) 1621BC 95.4% probability 1741BC (89.4%) 1606BC 1577BC (6.0%) 1537BC
18312	OA896- BC14, X127	Charcoal	Tell F3, Trench BC, layer BC14	3234 ± 25 (ext)	-22.28 ± 0.05	68.2% probability- 1524BC (43.1%) 1491BC 1480BC (25.1%) 1456BC 95.4% probability 1606BC (6.3%) 1575BC 1536BC (89.1%) 1434BC
18313	OA896- BC14, X136	Charcoal	Tell F3, Trench BC, layer BC14	3288 ± 30 (ext)	-25.3 ± 0.12	68.2% probability- 1608BC (35.5%) 1569BC 1563BC (32.7%) 1525BC 95.4% probability 1659BC (0.4%) 1655BC 1637BC (95.0%) 1496BC

Fig. 1160. List of 79 radiocarbon samples from the 2008-2012 excavations processed at the AMS 14C Dating Centre of Aarhus University.

18321	OA896- BD9, X118	Charcoal	Tell F3, Trench BD, layer BD9	4115 ± 29 (ext Small sample: 0.439 mg C)	-30.01 ± 0.58 (ext Small sample: 0.439 mg C)	68.2% probability- 2854BC (20.2%) 2813BC 2746BC (8.9%) 2726BC 2697BC (39.1%) 2621BC 95.4% probability 2866BC (24.6%) 2804BC 2775BC (0.5%) 2770BC 2762BC (70.3%) 2577BC
18314	OA896- BC16, X126	Charcoal	Tell F3, Trench BC, layer BC16	3272 ± 25 (ext)	-22.42 ± 0.05	68.2% probability- 1606BC (30.1%) 1574BC 1558BC (5.1%) 1551BC 1538BC (33.1%) 1505BC 95.4% probability 1619BC (95.4%) 1494BC
18315	OA896- BC16, X153	Charcoal	Tell F3, Trench BC, layer BC16	3450 ± 25 (ext)	-23.46 ± 0.05	68.2% probability- 1870BC (16.3%) 1846BC 1809BC (1.9%) 1805BC 1775BC (36.3%) 1735BC 1714BC (13.7%) 1694BC 95.4% probability 1878BC (21.5%) 1840BC 1827BC (11.5%) 1795BC 1783BC (62.4%) 1690BC
18316	OA896- BC16, X154	Charcoal	Tell F3, Trench BC, layer BC16	3472 ± 25 (ext)	-24.95 ± 0.05	68.2% probability- 1876BC (25.7%) 1842BC 1820BC (15.4%) 1797BC 1781BC (27.0%) 1746BC 95.4% probability 1883BC (93.0%) 1738BC 1708BC (2.4%) 1698BC
18317	OA896- BC16, X156	Charcoal	Tell F3, Trench BC, layer BC16	3549 ± 27 (ext)	-23.44 ± 0.05	68.2% probability- 1941BC (59.2%) 1878BC 1840BC (6.5%) 1829BC 1791BC (2.5%) 1786BC 95.4% probability 1972BC (68.6%) 1864BC 1850BC (26.8%) 1773BC
18318	OA896- BC16, X157	Charcoal	Tell F3, Trench BC, layer BC16	3394 ± 25 (ext)	-23.62 ± 0.05	68.2% probability- 1739BC (33.7%) 1707BC 1698BC (31.3%) 1665BC 1648BC (3.1%) 1644BC 95.4% probability 1747BC (95.4%) 1626BC
Phase 5						
13754	OA895- A146, X300	Charcoal	Tell F6, Trench A, layer A146	3429 ± 26	-10.09 ± 0.05	68.2% probability- 1756BC (68.2%) 1689BC 95.4% probability 1875BC (9.0%) 1842BC 1818BC (3.0%) 1798BC 1780BC (82.4%) 1663BC 1651BC (1.0%) 1641BC
13752	OA895- A128, X417	Charcoal	Tell F6, Trench A, layer A128	3532 ± 27	-9.86 ± 0.05	68.2% probability- 1917BC (33.1%) 1874BC 1844BC (20.5%) 1816BC 1799BC (14.6%) 1779BC 95.4% probability 1941BC (95.4%) 1770BC
Phase 4						
13753	OA895- A129, X465	Charred date stone	Tell F6, Trench A, layer A129	3449 ± 26	-22.71 ± 0.05	68.2% probability- 1870BC (15.6%) 1846BC 1810BC (2.5%) 1805BC 1776BC (35.3%) 1734BC 1715BC (14.8%) 1694BC 95.4% probability 1878BC (21.0%) 1839BC 1828BC (11.9%) 1793BC 1784BC (62.5%) 1689BC
13755	OA895- A148, X330	Charcoal	Tell F6, Trench A, layer A148	3484 ± 26	-10.19 ± 0.05	68.2% probability- 1877BC (27.8%) 1842BC 1822BC (20.5%) 1796BC 1782BC (19.9%) 1756BC 95.4% probability 1886BC (95.4%) 1741BC
13759	OA895- A177, X453	Charcoal	Tell F6, Trench A, layer A177	3532 ± 26	-10.25 ± 0.05	68.2% probability- 1916BC (33.2%) 1874BC 1844BC (20.4%) 1817BC 1799BC (14.6%) 1779BC 95.4% probability 1941BC (95.4%) 1771BC

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13757	OA895- A170, X406	Charred date stone	Tell F6, Trench A, layer A170	3512 ± 24	-23.36 ± 0.05	68.2% probability- 1886BC (13.2%) 1870BC 1846BC (30.2%) 1811BC 1804BC (24.8%) 1776BC 95.4% probability 1907BC (95.4%) 1753BC
13758	OA895- A176, X447	Charcoal	Tell F6, Trench A, layer A176	3558 ± 47	-10.7 ± 0.05	68.2% probability- 1972BC (52.1%) 1876BC 1843BC (9.5%) 1819BC 1798BC (6.6%) 1780BC 95.4% probability 2026BC (95.4%) 1759BC
13756	OA895- A166, X373	Charcoal	Tell F6, Trench A, layer A166	3496 ± 25	-10.61 ± 0.05	68.2% probability- 1881BC (12.6%) 1862BC 1852BC (55.6%) 1772BC 95.4% probability 1890BC (95.4%) 1747BC
13760	OA895- A185, X543	Charcoal	Tell F6, Trench A, layer A185	3470 ± 24	-10.11 ± 0.05	68.2% probability- 1875BC (25.5%) 1843BC 1819BC (13.9%) 1798BC 1780BC (28.8%) 1745BC 95.4% probability 1882BC (92.6%) 1738BC 1709BC (2.8%) 1698BC
Phase 3						
13761	OA895- A187, X576	Charcoal	Tell F6, Trench A, layer A187	3483 ± 26	-10.76 ± 0.05	68.2% probability- 1877BC (27.5%) 1842BC 1822BC (20.1%) 1796BC 1782BC (20.6%) 1755BC 95.4% probability 1886BC (95.4%) 1741BC
13762	OA895- A188, X624	Charcoal	Tell F6, Trench A, layer A188	3500 ± 32	-9.66 ± 0.05	68.2% probability- 1883BC (13.5%) 1862BC 1851BC (54.7%) 1772BC 95.4% probability 1915BC (95.4%) 1742BC
13763	OA895- A188, X677	Charcoal	Tell F6, Trench A, layer A188	3474 ± 29	-10.54 ± 0.05	68.2% probability- 1877BC (25.5%) 1842BC 1822BC (17.3%) 1796BC 1782BC (25.5%) 1747BC 95.4% probability 1885BC (91.6%) 1737BC 1712BC (3.8%) 1695BC
13766	OA895- A211, X638	Charcoal	Tell F6, Trench A, layer A211	3569 ± 32	-11.43 ± 0.05	68.2% probability- 1964BC (68.2%) 1882BC 95.4% probability 2024BC (85.4%) 1873BC 1845BC (6.0%) 1815BC 1800BC (3.9%) 1778BC
13764	OA895- A189, X570	Charcoal	Tell F6, Trench A, layer A189	3560 ± 26	-10.11 ± 0.05	68.2% probability- 1946BC (68.2%) 1883BC 95.4% probability 2012BC (1.9%) 2000BC 1977BC (81.4%) 1874BC 1844BC (7.5%) 1816BC 1800BC (4.6%) 1779BC
13765	OA895- A198, X612	Charcoal	Tell F6, Trench A, layer A198	3547 ± 25	-10.12 ± 0.05	68.2% probability- 1938BC (58.6%) 1878BC 1840BC (7.0%) 1829BC 1791BC (2.7%) 1786BC 95.4% probability 1956BC (67.4%) 1864BC 1850BC (28.0%) 1773BC
16168	OA895- E253, X1071	Charcoal	Tell F6, Trench E, layer E253	3459 ± 40 (ext)	-23.34 ± 0.05	68.2% probability- 1877BC (20.2%) 1842BC 1822BC (12.7%) 1797BC 1781BC (27.6%) 1737BC 1711BC (7.7%) 1695BC 95.4% probability 1891BC (95.4%) 1682BC
16162	OA895- E202, X1068	Charcoal	Tell F6, Trench E, layer E202	3544 ± 27 (ext)	-10.13 ± 0.05	68.2% probability- 1937BC (51.6%) 1877BC 1841BC (10.1%) 1825BC 1795BC (6.5%) 1783BC 95.4% probability 1956BC (95.4%) 1771BC
16142	OA895- A332, X847	Charcoal	Tell F6, Trench A, layer A332	3497 ± 30 (ext)	-10.45 ± 0.05	68.2% probability- 1881BC (12.9%) 1862BC 1851BC (55.3%) 1772BC 95.4% probability 1906BC (95.4%) 1741BC

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16167	OA895- E241, X1069	Charcoal	Tell F6, Trench E, layer E241	3487 ± 27 (ext)	-10.91 ± 0.05	68.2% probability- 1878BC (27.4%) 1840BC 1827BC (25.6%) 1792BC 1785BC (15.2%) 1760BC 95.4% probability 1889BC (95.4%) 1741BC
Phase 2						
13767	OA895- A215, X696	Charcoal	Tell F6, Trench A, layer A215	3512 ± 28	-10.17 ± 0.05	68.2% probability- 1887BC (13.5%) 1869BC 1847BC (54.7%) 1775BC 95.4% probability 1913BC (95.4%) 1751BC
13770	OA895- A224, X715	Charcoal	Tell F6, Trench A, layer A224	3457 ± 24	-10.28 ± 0.28	68.2% probability- 1872BC (21.5%) 1845BC 1812BC (6.1%) 1803BC 1777BC (36.6%) 1739BC 1706BC (4.0%) 1699BC 95.4% probability 1880BC (25.7%) 1838BC 1831BC (58.0%) 1730BC 1720BC (11.7%) 1692BC
13771	OA895- A224, X775	Charcoal	Tell F6, Trench A, layer A224	3562 ± 26	-10.68 ± 0.05	68.2% probability- 1945BC (68.2%) 1884BC 95.4% probability 2013BC (2.3%) 1999BC 1978BC (82.9%) 1875BC 1844BC (6.4%) 1818BC 1799BC (3.8%) 1780BC
13772	OA895- A224, X776	Charcoal	Tell F6, Trench A, layer A224	3485 ± 20	-9.79 ± 0.05	68.2% probability- 1877BC (28.5%) 1842BC 1823BC (22.1%) 1796BC 1783BC (17.6%) 1758BC 95.4% probability 1881BC (95.4%) 1749BC
13773	OA895- A224, X777	Charcoal	Tell F6, Trench A, layer A224	3450 ± 24	-12.85 ± 0.05	68.2% probability- 1869BC (16.3%) 1847BC 1775BC (37.9%) 1736BC 1713BC (13.9%) 1694BC 95.4% probability 1877BC (21.4%) 1841BC 1826BC (10.9%) 1796BC 1782BC (63.2%) 1690BC
13768	OA895- A220, X787	Charcoal	Tell F6, Trench A, layer A220	3452 ± 22	-11.12 ± 0.05	68.2% probability 1870BC (17.4%) 1846BC 1809BC (1.9%) 1805BC 1775BC (38.0%) 1737BC 1711BC (10.9%) 1695BC 95.4% probability 1878BC (22.6%) 1840BC 1827BC (11.3%) 1795BC 1783BC (61.5%) 1691BC
13769	OA895- A220, X789	Charcoal	Tell F6, Trench A, layer A220	3527 ± 20	-9.8 ± 0.05	68.2% probability- 1899BC (23.7%) 1874BC 1844BC (25.9%) 1815BC 1800BC (18.6%) 1778BC 95.4% probability 1926BC (95.4%) 1771BC
16161	OA895- E197, X884	Charcoal	Tell F6, Trench E, layer E197	39125 ± 354 (ext)	-27.37 ± 0.05	68.2% probability- 41715BC (68.2%) 41015BC 95.4% probability 42118BC (95.4%) 40776BC
16165	OA895- E228, X1005	Charcoal	Tell F6, Trench E, layer E228	3560 ± 30 (ext)	-9.91 ± 0.05	68.2% probability- 1952BC (68.2%) 1880BC 95.4% probability 2016BC (3.4%) 1997BC 1980BC (75.5%) 1869BC 1847BC (16.5%) 1775BC
16163	OA895- E216, X950	Charcoal	Tell F6, Trench E, layer E216	3685 ± 28 (ext)	-10.73 ± 0.05	68.2% probability- 2134BC (44.3%) 2080BC 2061BC (23.9%) 2030BC 95.4% probability 2194BC (2.5%) 2178BC 2144BC (86.4%) 2009BC 2003BC (6.5%) 1977BC
16166	OA895- E231, X1094	Charcoal	Tell F6, Trench E, layer E231	3429 ± 29 (ext)	-9.57 ± 0.05	68.2% probability- 1768BC (68.2%) 1688BC 95.4% probability 1876BC (10.3%) 1842BC 1821BC (4.3%) 1797BC 1781BC (78.9%) 1661BC 1653BC (2.0%) 1638BC

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13774	OA895- A226, X799	Charcoal	Tell F6, Trench A, layer A226	3587 ± 34 (Small sample: 0.329 mg C)	-24.36 ± 0.05 (Small sample: 0.329 mg C)	68.2% probability- 2008BC (3.2%) 2003BC 1976BC (65.0%) 1894BC 95.4% probability 2034BC (92.8%) 1877BC 1841BC (1.7%) 1825BC 1794BC (0.9%) 1784BC
16143	OA895- A338, X875	Charcoal	Tell F6, Trench A, layer A338	3474 ± 28 (ext)	-18.82 ± 0.05	68.2% probability- 1877BC (25.7%) 1842BC 1822BC (16.8%) 1797BC 1782BC (25.7%) 1747BC 95.4% probability 1885BC (92.0%) 1737BC 1711BC (3.4%) 1696BC
Phase 1						
15411	OA895- C124, X315	Charcoal	Tell F6, Trench C, layer C124	3695 ± 25 (ext)	-26.01 ± 0.05	68.2% probability- 2134BC (45.1%) 2076BC 2064BC (23.1%) 2036BC 95.4% probability 2195BC (3.9%) 2176BC 2145BC (89.4%) 2019BC 1995BC (2.1%) 1981BC
15412	OA895-C124, X316	Charcoal	Tell F6, Trench C, layer C124	4550 ± 25 (ext) Small sample: 0.782 mg C	-17.74 ± 0.05	68.2% probability- 3363BC (30.0%) 3333BC 3213BC (21.0%) 3188BC 3155BC (17.2%) 3132BC 95.4% probability 3368BC (35.3%) 3316BC 3293BC (0.5%) 3289BC 3274BC (1.0%) 3267BC 3238BC (58.7%) 3106BC
15413	OA895-C124, X395	Charcoal	Tell F6, Trench C, layer C124	11017 ± 37 (ext) Small sample: 0.723 mg C	-24.72 ± 0.05	68.2% probability- 11100BC (3.6%) 11084BC 11019BC (64.6%) 10815BC 95.4% probability 11134BC (95.4%) 10762BC
15414	OA895-C126, X346	Charcoal	Tell F6, Trench C, layer C126	29271 ± 125 (ext)	-27.49 ± 0.05	68.2% probability- 32504BC (15.4%) 32336BC 32202BC (52.8%) 31658BC 95.4% probability 32589BC (95.4%) 31478BC
16144	OA895- C153, X525	Charcoal	Tell F6, Trench C, layer C153	3817 ± 29 (ext)	-17.47 ± 0.05	68.2% probability- 2296BC (68.2%) 2202BC 95.4% probability 2431BC (0.6%) 2424BC 2403BC (2.7%) 2381BC 2349BC (85.9%) 2193BC 2177BC (6.3%) 2143BC
16145	OA895- C153, X596	Charcoal	Tell F6, Trench C, layer C153	4426 ± 29 (ext)	-13.44 ± 0.05	68.2% probability- 3264BC (7.0%) 3245BC 3101BC (56.2%) 3011BC 2978BC (3.6%) 2965BC 2949BC (1.4%) 2944BC 95.4% probability 3323BC (7.1%) 3272BC 3266BC (9.2%) 3235BC 3173BC (1.3%) 3161BC 3117BC (77.9%) 2924BC
16146	OA895- C164, X529	Charcoal	Tell F6, Trench C, layer C164	3732 ± 29 (ext)	-24.35 ± 0.05	68.2% probability- 2198BC (24.8%) 2166BC 2151BC (15.6%) 2129BC 2087BC (27.8%) 2048BC 95.4% probability 2264BC (0.4%) 2261BC 2206BC (95.0%) 2033BC
16147	OA895- C180, X624	Charcoal	Tell F6, Trench C, layer C180	34045 ± 255 (ext)	-27.71 ± 0.05	68.2% probability- 37311BC (68.2%) 36692BC 95.4% probability 38071BC (95.4%) 36501BC
16148	OA895- C191, X672	Charcoal	Tell F6, Trench C, layer C191	3770 ± 30 (ext)	-10.37 ± 0.05	68.2% probability- 2274BC (11.9%) 2256BC 2208BC (56.3%) 2140BC 95.4% probability 2290BC (89.0%) 2131BC 2086BC (6.4%) 2050BC
16149	OA895- C192, X682	Charcoal	Tell F6, Trench C, layer C192	3760 ± 28 (ext)	-10.17 ± 0.05	68.2% probability- 2272BC (6.6%) 2258BC 2207BC (61.6%) 2137BC 95.4% probability 2286BC (13.4%) 2240BC 2235BC (70.6%) 2126BC 2090BC (11.5%) 2044BC

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16150	OA895- C194, X687	Charcoal	Tell F6, Trench C, layer C194	3729 ± 28 (ext)	-11.02 ± 0.05	68.2% probability- 2197BC (21.8%) 2169BC 2148BC (14.9%) 2128BC 2088BC (31.5%) 2046BC 95.4% probability 2203BC (95.4%) 2035BC
16151	OA895- C199, X725	Charcoal	Tell F6, Trench C, layer C199	3702 ± 28 (ext)	-25.63 ± 0.05	68.2% probability- 2138BC (18.5%) 2113BC 2101BC (49.7%) 2037BC 95.4% probability 2198BC (8.9%) 2166BC 2151BC (85.2%) 2022BC 1993BC (1.3%) 1983BC
16152	OA895- C200, X726	Charcoal	Tell F6, Trench C, layer C200	3710 ± 27 (ext)	-10.54 ± 0.05	68.2% probability- 2141BC (18.8%) 2115BC 2100BC (49.4%) 2038BC 95.4% probability 2199BC (12.8%) 2164BC 2152BC (82.6%) 2029BC
16153	OA895- C207, X744	Charcoal	Tell F6, Trench C, layer C207	3853 ± 35 (ext)	-25.8 ± 0.5 (ext)	68.2% probability- 2452BC (2.5%) 2445BC 2438BC (6.7%) 2420BC 2405BC (11.6%) 2378BC 2350BC (36.2%) 2280BC 2250BC (8.2%) 2230BC 2220BC (3.0%) 2212BC 95.4% probability 2461BC (77.2%) 2270BC 2260BC (18.2%) 2206BC
16154	OA895- C215, X786	Charcoal	Tell F6, Trench C, layer C215	3844 ± 35 (ext)	-13.9 ± 0.6 (ext)	68.2% probability- 2429BC (1.4%) 2425BC 2401BC (7.5%) 2382BC 2348BC (37.5%) 2274BC 2256BC (21.8%) 2208BC 95.4% probability 2459BC (95.4%) 2203BC
Kiln H4/5						
16169	OA895- H3, X3	Charcoal	Tell F6, Trench H, layer H3	3487 ± 40 (ext)	-10.56 ± 0.05	68.2% probability- 1878BC (24.0%) 1840BC 1828BC (44.2%) 1756BC 95.4% probability 1916BC (90.8%) 1732BC 1718BC (4.6%) 1693BC
16170	OA895- H3, X7	Charcoal		3490 ± 35 (ext)	-10.4 ± 0.05	68.2% probability- 1879BC (26.4%) 1838BC 1831BC (41.8%) 1767BC 95.4% probability 1907BC (93.1%) 1737BC 1711BC (2.3%) 1696BC
Temple						
16155	OA895- C222, X797	Charcoal	Tell F6, Trench C, layer C222			
16156	OA895- C222, X798	Charcoal	Tell F6, Trench C, layer C222	3987 ± 30 (ext) Small sample: 0.673 mg C	-13.59 ± 0.05	68.2% probability- 2566BC (41.6%) 2525BC 2496BC (26.6%) 2472BC 95.4% probability 2575BC (95.4%) 2465BC
16157	OA895- C222, X799	Charcoal	Tell F6, Trench C, layer C222	3551 ± 35 (ext)	-27.2 ± 0.6 (ext)	68.2% probability- 1949BC (52.2%) 1877BC 1842BC (9.5%) 1822BC 1796BC (6.5%) 1782BC 95.4% probability 2013BC (2.3%) 1999BC 1979BC (93.1%) 1771BC
13775	OA895- C72, X212a	Charcoal	Tell F6, Trench C, layer C72	3549 ± 40	-25.87 ± 0.05	68.2% probability- 1948BC (48.1%) 1875BC 1843BC (11.7%) 1819BC 1798BC (8.4%) 1780BC 95.4% probability 2015BC (3.0%) 1997BC 1980BC (92.4%) 1756BC
13776	OA895- C72, X212b	Charcoal	Tell F6, Trench C, layer C72	3704 ± 24 (Small sample: 0.391 mg C)	-10.49 ± 0.05	68.2% probability- 2137BC (16.4%) 2118BC 2097BC (51.8%) 2040BC 95.4% probability 2196BC (7.6%) 2169BC 2146BC (87.8%) 2027BC

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Kiln E138/E143						
16158	OA895- E157, X839	Charcoal	Tell F6, Trench E, layer E157, ash layer in kiln E138/E143	3431 ± 29 (ext)	-9.6 ± 0.05	68.2% probability- 1769BC (68.2%) 1689BC 95.4% probability 1877BC (11.3%) 1841BC 1824BC (5.1%) 1796BC 1782BC (77.7%) 1662BC 1651BC (1.3%) 1641BC
16164	OA895- E226, X969	Charcoal	Tell F6, Trench E, layer E226, ash layer in kiln E138/E143	3410 ± 27 (ext)	-10.4 ± 0.05	68.2% probability- 1747BC (68.2%) 1683BC 95.4% probability 1862BC (1.2%) 1853BC 1771BC (94.2%) 1629BC
16159	OA895- E183, X851	Charcoal	Tell F6, Trench E, layer E183, clay and ash layer in kiln E138/E143	3540 ± 28 (ext)	-9.74 ± 0.05	68.2% probability- 1927BC (43.8%) 1877BC 1842BC (14.4%) 1821BC 1797BC (10.0%) 1781BC 95.4% probability 1952BC (95.4%) 1771BC
16160	OA895- E190, X866	Charcoal	Tell F6, Trench E, layer E190, ash layer in kiln E138/E143	3533 ± 30 (ext)	-9.86 ± 0.05	68.2% probability- 1921BC (33.7%) 1873BC 1844BC (19.7%) 1815BC 1801BC (14.8%) 1778BC 95.4% probability 1947BC (95.4%) 1760BC
Buttress Wall						
15420	OA895-F, X1	Charcoal	Tell F6, Trench F	3082 ± 25 (ext)	-26.07 ± 0.05	68.2% probability- 1408BC (37.6%) 1371BC 1346BC (30.6%) 1316BC 95.4% probability 1421BC (94.6%) 1292BC 1278BC (0.8%) 1272BC
15421	OA895-F, X2	Charcoal	Tell F6, Trench F	42775 ± 383 (ext)	-27.84 ± 0.05	68.2% probability- 44265BC (68.2%) 43571BC 95.4% probability 44677BC (95.4%) 43247BC

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sum of 1803 ±23 BC given to Phase 3. The sum of two dates from the second oven phase to 1716 ±37 BC supports the notion that the oven continued into Phase 5.

The sum of two radiocarbon dates from another oven, H4/5, to 1779 ±39 BC suggests that this oven may have been used in between the first and the second phase of oven E138/E143, a possibility that is not contradicted by the model results.

A heavy plaster floor (H13) relates the low terrace northwest of the Temple (H17) to the pavement (E203) fronting the thin-walled building and thereby relates the Temple to Phase 3 (cf. p. 72-74). However, from the thin activity layers identified in Section 1 (fig. 11)

immediately to the southeast of the outer temple wall and related to the construction of the Temple two radiocarbon measurements of rather wide calibrated age range have mean dates of 1882 ±65 and 1886 ±60 BC that would fit better with a dating to Phase 2. More dates from the construction levels of the Temple are evidently needed to decide whether these two dates were made on old wood or whether the construction of the Temple began already in Phase 2.

The Buttress Wall, which is a late addition to the Temple, has one radiocarbon measurement with a calibrated age of 1402 - 1302 cal BC (at 68.2% probability) referring it to Phase 7 (Period 4A) (cf. p. 91-92).

5. Summary and conclusions

by Flemming Højlund

For millennia the island of Failaka was a small inhospitable rocky outcrop hardly sheltering anything but seabirds. However, this landmass stilled the waters of the Gulf inviting loam, sand and clay brought down the great Mesopotamian rivers to collect around it. With the lowering of the sea level during the 3rd millennium BC, human settlement on the island became possible.

The first archaeological excavations on Failaka in 1958-1963 revealed that the island was inhabited in the early 2nd millennium BC by Dilmunites from Bahrain who used it to support their trade with Mesopotamia (fig. 2). The excavations carried out in 2008-2012 and reported in this volume prove that the potential of Failaka for facilitating long distance trade was already realized several hundred years earlier by the Mesopotamian Ur III state.

Phase 1. The Mesopotamian Ur III settlement

The oldest evidence of settlement so far identified on Failaka was found in the southeast part of Tell F6 (figs. 3-6). Over an area of at least 22 x 40 m, the sterile beach sand was covered by a 55-65 cm thick layer of sand with occupation remains, potsherds, animal bones, oyster shells, bitumen fragments, cylinder seals, etc. In the middle of this area, the corner of a walled enclosure was uncovered (figs. 9-10). The lack of any internal walls or any substantial floor perhaps point towards a courtyard rather than a house. To the north, the remains of this settlement were removed by the later construction of a Dilmun Temple.

With the exception of a few sherds imported from Bahrain, the Oman peninsula and the Indus Valley region, all the pottery found here were in a pure Mesopotamian tradition with respect to production method, ware, shape, as well as decoration and finish (figs. 131-257). This was undoubtedly a settlement of people from Mesopotamia and is as such the southernmost known Mesopotamian settlement during the Ur III period. The dating based on the finds is supported by radiocarbon dates.

Amongst the finds recovered from the excavations are six cuneiform inscribed cylinder seals adhering

strictly to the Mesopotamian Ur III tradition (figs. 752-757). These seals, together with other similar seals previously found on Failaka, indicate that there was a Mesopotamian administrative presence on the island of Failaka. All these cylinder seals depict, as shown by Pittman (see above p. 157), a standard presentation scene of a worshipper lead by a goddess towards an enthroned goddess indicating that all the seal-owning officials were of a lower middle rank in the administrative hierarchy (Mayr 1997). One seal was owned by a scribe (*dub-sar*) who states that his father was an official/foreman (*nubanda*).

In the Ur III period the Mesopotamian commercial activities in the Gulf were controlled by a ministry of foreign trade which commanded a fleet of “big boats” (*má-gal-gal*), also called “Makkan boats” (*má-Mí-ganki*), because the main destination of these ships were the Makkan coast, i.e. the present day Oman peninsula (Steinkeller 2013 p. 417-418). The Ur III fleet of “big boats” was based in the port of Gu’abba (Steinkeller 2013 fig. 3. Laursen & Steinkeller forth.). From Gu’abba (still not located) to Failaka there was a distance of around 250 km, and a settlement with a sheltered and accessible anchorage here would undoubtedly be a valuable waypoint on the 1000 km journey to the Emirate coast. It could have supplied fresh water and harboured boats in the case of bad weather, unsuitable winds or when repairs were necessary. We may thus conclude that the settlement on Failaka was in all likelihood established by the Ur III state’s bureau of foreign trade with a view to supporting and perhaps controlling the trade towards the southeast.

Some of the finds from the Ur III settlement on Failaka, such as the many beads of carnelian, agate, chalcedony, rock crystal, and lapis lazuli (figs. 832-892), the calcite and soft-stone vessels (figs. 770-775), a shell ladle of Harappan type (fig. 1122), Harappan and Umm an-Nar potsherds (figs. 267-273) point towards international connections. Numerous fragments of bitumen probably derive from the water-proof coating of reed-bundle or plank boats (figs. 1071-1081), and a stone-lined pit may have been used for heating up bitumen (figs. 35-36). These finds all indicate that the repair of boats took place on this site and that this may have been an important part of the rationale behind

establishing a settlement on Failaka. Yeomans' study of the animal bones (see above p. 228) reveals that there was a heavy reliance on sheep and goat herding in this period and less of a focus on fishing compared to later periods. There was, however, evidence for intensive summer exploitation of the marine environment, e.g. types of fish traditionally sundried as food for long sea voyages, as well as numerous oyster shells and two genuine pearls (figs. 1127).

If the pottery strainers (figs. 244-256) were used for making cheese as seems very likely (cf. Gouin 1991 p. 46-48, fig. 6), animal husbandry of some importance must be presumed. This hypothesis is supported by Yeomans' study (see above p. 229) which documents a systematic slaughter profile of sheep/goat and although the mortality profile suggests that the herds were kept mainly for meat production it does not preclude the use of the herds for milk which was a commonly utilised secondary product.

There is no evidence for copper working in the shape of crucibles or moulds.

The little evidence of relations to Dilmun/Bahrain tallies with the fact that Dilmun hardly occurs in the written evidence of Gulf trading in the Ur III period. Bahrain must certainly have been a stopover on the route to Magan, but Dilmun seems to have played a minor role in this trading system.

The collapse of the Ur III state which took place by the end of the 21st century BC brought an end to the "big boats" and the long-distance trade to the Makkan coast and therefore also to the Failaka settlement. More than a hundred years seems to have passed before the island was again settled, this time from the south, the Dilmun centre of Bahrain.

Phases 2-5. The Dilmun settlement in Periods 1-2

The Dilmun settlement established itself on the Failaka coast in two different areas with a distance of 400 m separating them (fig. 3). Tell F3 housed domestic architecture and was presumably residential quarters for a number of families. Tell F6 held monumental structures housing a large public institution (fig. 4). The spatial separation of the two areas undoubtedly stresses the underlying social distance.

Three architectural phases are documented in Trenches A, E and H in Tell F6 excavated in 2008-2012 (figs. 5-6). Two thick-walled buildings were constructed on virgin soil (Phase 2, figs. 40 and 42-43) but they were subsequently torn down and replaced by a complex of smaller rooms bounded by thinner

walls (Phase 3, figs. 46 and 39). These structures relate to the first terrace northwest of a nearby Temple and therefore probably to the construction of this Temple. These small rooms were shortly after razed to the ground and replaced by the construction of the "Palace" (Phases 4-5, figs. 53-54).

The pace of construction, demolition and rebuilding in Tell F6 is closely paralleled in the centre of Qala'at al-Bahrain in Period IIb-c where a series of monumental buildings were erected, torn down and replaced by new monuments (Højlund & Andersen 1997 p. 40-41. Højlund 2007b).

At Qala'at al-Bahrain the evidence suggests an increase in the scale of the monumental buildings. The same can be seen in Tell F6 where the development from 1) thick-walled buildings in uncut beach-rock stones to 2) a limestone ashlar temple with attached thin-walled buildings to 3) a replacement of the thin-walled buildings with a large production and storage facility – the "Palace" – suggests a great increase of investment in resources and manpower (fig. 6).

The speed of construction, the radical demolition followed by rebuilding and the increase in investment in public buildings in Tell F6 stand in stark contrast to the situation in Tell F3. Here the houses in the north part of the tell (Kjærums & Højlund 2013 fig. 7: 26-29) were occupied throughout the same period (fig. 5) with reparations and some alterations but without any major rebuilding.

In general, the development of the Dilmun civilization shows an exponential growth with respect to investments in public architecture that accelerated in the first centuries of the 2nd millennium BC (Højlund 2007b). The colonization of Failaka was part of this dynamic expansion, and the fact that Failaka was colonized by Dilmun in the early 2nd millennium, after being a Mesopotamian trade post in the late 3rd millennium, is clear testimony that control of trade through the Gulf had moved from Mesopotamia to Dilmun.

The similarity in the pottery and the stamp seals from Phases 2-5 is notable (cf. above p. 95 and 133) and therefore they must represent a short time span in the Dilmun culture. The radiocarbon dates suggest that Phase 2, i.e. the establishment of the Dilmun settlement on Failaka, began c. 1900 BC and that Phase 5 ends in the late 18th century BC (cf. p. 243).

Phase 2. The thick-walled buildings in Periods 1-2

The size of the two walls, A191 and A219, reveals a considerable investment of work and resources and

suggest some kind of public buildings (figs. 40-43). This notion is supported by the number of hearths encountered between the two buildings (figs. 40 and 44), and especially their size which surpass that of similar hearths in the contemporary Saar Village on Bahrain (Killick & Moon 2005 p. 156-157). The cohesion of this complex is underlined by the fact that walls and several of the hearths were oriented along the same northeast-southwest axis.

The lack of any workshop remains in and around the hearths perhaps suggests that they were related to food preparation and their size indicates that they may have been for larger scale production. Finds of pottery utensils that can be used for baking bread or cakes (*tannur's*, and rimmed plates, cf. p. figs. 289-293) support such an identification of the functional character of the area.

The rimmed plate is probably an imitation of a Mesopotamian shape that appears in the late Isin-Larsa period (Armstrong & Gasche 2014 pl. 1). It is interesting that a Mesopotamian utensil apparently related to bread baking is adopted at the same time on Failaka and Bahrain (Barbar Temple III, Højlund & Andersen 2003 figs. 579-583). It could be testimony to increased import of grain from Mesopotamia and consumption of bread according to Mesopotamian recipes.

That bread baking was part of what public authority was expected to provide to the community perhaps in return for construction work may be seen in the rows of bread baking ovens adjacent to public buildings, *vide* the city wall around Qala'at al-Bahrain (and in the centre of Qala'at al-Bahrain) and the oval terrace wall around the Barbar temple (Højlund & Andersen 1994 p. 470. 1997 p. 40. Andersen & Højlund 2003 p. 119, 227, figs. 142-144, 453-456, plan 3: 36-37).

Phase 3. The thin-walled building in Periods 1-2

At some point the thick-walled buildings were demolished down to the level of their lowest course and replaced by a building with at least three small rooms with fine plaster floors (figs. 39 and 46). The walls were only about 25 cm wide and built of uncut beach rock set in clay or plaster. To the east of the rooms was an 8.8 m long and almost 1 m wide pavement that continued south of the building with a corner towards the southeast.

The architecture of these buildings is, in general, similar in scale to the domestic houses in Tell F3 dated to Periods 2 and 3A, with respect to wall thickness and perhaps also to room size (Kjærøum & Højlund

2013). The first plaster floor in Room 1 was, however, especially solid (fig. 46) and a pavement like the one along the thin-walled building (E203) has not been encountered in Tell F3 or in any contemporary settlement in Bahrain (Qala'at al-Bahrain or Saar). The investment of labour and materials in providing a firm surface both inside and outside the house suggests a certain quality and status.

As argued above, the construction of the thick-walled buildings indicates a considerable investment of resources, and the same can be said about its demolition. It is unlikely that the thick-walled buildings would have been dismantled down to its first course just for constructing the thin-walled building. Such small-scale constructions could have avoided the area of the previous monumental buildings or made use of their existing walls. The thin-walled building should, in all likelihood, be considered part of a much larger project, namely the Temple. The stones of the thick-walled building would have been excellent building material for the substantial foundations of the Temple.

That the construction of the Temple is contemporary with that of the thin-walled building is plausible for several reasons. Firstly, in contrast to the thick-walled phase, the walls, floors and pavement of the thin-walled building contained chips of oolitic limestone stemming from the cutting of the temple ashlar. This shows conclusively that the Temple was at least in preparation when the thin-walled building was being constructed. Secondly, the floor built up against the northwest terrace of the Temple meets with the pavement east of the thin-walled building in such a way that contemporaneity is implied.

However, two radiocarbon dates from the construction levels of the Temple suggest that it may have been constructed earlier and possibly contemporary with Phase 2 (cf. p. 250). These dates may, however, have been obtained on old wood.

The floor outside the first terrace northwest of the Temple included fragments of an earlier plaster floor constructed above a layer of fine oolitic limestone chips, indicating some earlier temple activities. It could also be argued that the lack of oolitic limestone chips in the walls of Phase 2 may be due to the limited exposure of architecture from this phase. On the whole, the possibility that the Temple was built earlier than Phase 3 cannot be excluded.

The oven E138/E143 positioned between the Temple and the thin-walled building (figs. 46, 49-50) contained burnt animal bones, especially the horned heads of probably male goats (cf. Yeomans above p. 232), and it is suggested that it was used for purification of the

Temple by disposing of sacrificial remains such as the heads of animals. The position of a temple-ritual related oven close to the thin-walled building and the later "Palace" seems to suggest that these buildings were related to the Temple with respect to function and meaning.

The size, structure, contents and location of the oven E138/E143 is very similar to an oven excavated behind the Barbar Temple (Andersen & Højlund 2003 p. 140, 327, figs. 182-183, section 31).

Phases 4-5. The "Palace" in Periods 1-2

At some point the thin-walled building was demolished down to its lowest course and replaced by a c. 20x20 m large building known as the "Palace". A major part of this structure was excavated between 1960 and 1963 (Kjærum & Højlund 2013), but its south-east corner was included in Trench A that was opened in 2008 (fig. 53).

Inside this corner was a plaster floor at a level of c. 1.59 m a.s.l. and laid above a 50 cm thick layer of clay (fig. 55). Outside the "Palace" a similar thick layer of clay covered the demolished remains of the earlier thick- and thin-walled buildings of Phases 2-3, and this surface at a level of c. 1.80-1.60 m a.s.l. could be followed for at least 10 m to the east (fig. 57). Upon the floor was a c. 50 cm thick sequence of thin layers of new floors with occupational waste such as pottery, stamp seals, bones and ashes lenses sandwiched between.

A few meters from the "Palace" was a large hearth (A164, fig. 60) which was later replaced by a new hearth (A148, fig. 61) in the same location. There were no finds in or around the hearths that indicated any kind of craft production. Instead the layers around the hearths contained a lot of ash, charcoal, potsherds and animal bones, especially fish bones including articulating skeletons (figs. 58-59). Perhaps they were used for communal food preparation. No finds were recovered inside the "Palace" from the first phase of occupation (cf. Kjærum & Højlund 2013 p. 107).

The original interpretation of this large building as a palace with administrative functions (Glob 1968 p. 132. Kjærum 1986 p. 79) rested on the size of the building (c. 425 m²), which is considerable compared to the small houses in Tell F3 (c. 12-50 m²), and on its general layout especially the long corridor leading up to an inner gate giving access to a large pillared hall.

The finds in the building were, however, not of the exceptional quality one would expect from a palace.

The installations in the building point towards more menial functions such as manufacture and storage. The pillared hall has several features suggesting manufacture and storage of fluids (date juice?) including a waterproof cistern below the floor that could have held 1300 litres (Kjærum & Højlund 2013 figs. 149-150), and a channel with a grid leading out to an external cistern with a volume of 500 litres (*op.cit.* figs. 151-152). Besides these features three cylindrical constructions, possibly ovens (*op.cit.* figs. 139, 142-143 and 154), a *pithos* by the doorway to room 3 (Højlund 1987 figs. 111-112 and 629) and a great number of storage vessels in several of the adjoining small rooms were identified (Højlund 1987 figs. 625-27 and 630-32). The "Palace" was used in several phases with major rebuilding taking place in Period 4A (phase II). However, the fact that the general structure of the building remained unaltered with very few changes from phase I to II, suggests it had a stable function throughout.

The dense distribution of large storage pots south of the "Palace" (Højlund 1987 fig. 621, p. 146-48) also indicates that this was a production area, perhaps used for large-scale food preparation. Additionally, the small buildings uncovered to the west of the "Palace" by the American excavation (Kjærum & Højlund 2013 fig. 131 and p. 153) may perhaps best be interpreted as related to production and storage (cf. Calvet in Callot 2005 p. 46).

The excavators of the Temple lying 15 m east of the "Palace" (fig. 4) argued that the Temple had its back to the NW wind and its entrance towards the SE, *i.e.* towards the shore.²⁶ It follows then, that the "Palace" was situated *behind* the Temple, a location indicating lower status compatible with functions that are more menial.

Though only part of the "Palace" has so far been excavated and though the building is heavily disturbed by stone plundering it may be tentatively interpreted as a production and storage facility related to the nearby Temple. How much of this production was intended for consumption by the worshippers and tenders of the Temple and how much was intended to be transferred back to the centre of Dilmun or exported to the north (cf. the parallel to one of the most common storage vessels in the "Palace" found at Nippur, Failaka type 57C, McCown & Haines 1967 pl. 99: 1) is a matter of speculation.

The oven, E138/E143, that was built in Phase 3 seems still to have been in use in Phase 5 (fig. 63). A second oven of similar type, H4/5, was located immediately to the southwest of the temple terrace and may have been in use in Phase 4 (fig. 53 and 98) (cf. p. 74).

Phases 2-5 in general

The pottery from Phases 2-5 (figs. 274-534) belong almost exclusively to the Barbar tradition, i.e. there are relatively few imports from Mesopotamia (datable to the Isin-Larsa/Old Babylonian period) and the wider Indus Valley. Most of the pottery are from storage vessels or cooking jars, and finer types of pottery related to eating and drinking on a domestic scale are scarce. A small pot in red, fine A or H ware (fig. 314) and a uniquely ridged vessel (fig. 422) should be noted. Both clearly belong in another sphere, perhaps related to temple rituals, as may an exquisitely shaped ivory dagger handle (fig. 1129).

Stamp seals were plentiful in the 2008-2012 excavations. Most of them were found in secure stratigraphic contexts (Abu-Laban above p. 133), and this has significantly improved our understanding of the different stamp seal styles originally distinguished by Kjærøum (1980).

There is no doubt that seals of Arabian Gulf type are the oldest in this series, dating to Qala'at al-Bahrain Period IIa (fig. 5), i.e. earlier than the Dilmun occupation on Failaka (Kjærøum 1994, Laursen 2010). They were followed by seals of Style IA, which dominate in Bahrain in Qala'at al-Bahrain Period IIb-c (Kjærøum 1994). It is equally clear that the final style of the Dilmun seals, Style III, should be dated to somewhere in the late Old Babylonian-Early Kassite period (Kjærøum 1994 p. 348. Denton 1997).

It is the dating of Style IB and Style II seals that has been clarified by the excavations in Tell F6 presented here (Abu-Laban above p. 133). Though Kjærøum advocated for a substantial overlap in time between the different stamp seal styles, he nevertheless proposed that there was a gradual chronological development from Style IA to Style IB to Style II and that this development largely took place after the decline of the Dilmun civilization (Kjærøum 1994 p. 348, fig. 1758. Højlund 2007a p. 13-14). This theory has now to be revised. According to the 2008-2012 excavations, it is clear that stamp seals of Styles IA, IB and II coexisted in Phases 2-5 (fig. 5) and in almost equal percentages in all four phases (Abu-Laban above fig. 674). Therefore, Style IB and Style II cannot anymore be proposed as late developments belonging to a Post IIC Period (cf. Kjærøum 1994 fig. 1758), but must be understood as local Failaka styles existing side by side to Style IA and dating to the floruit of Dilmun civilization. The three co-existing styles may be interpreted as products from three different workshops or, considering the small size of the Failaka population, perhaps rather three different "hands" or different generations in the same shop.

Style II seems to be influenced by the point drill style of Mesopotamian cylinder seals (Kjærøum 1980 p. 46) which appeared towards the end of the reign of Hammurapi and even more during the reign of his son Samsuiluna (Al-Gailani Werr 1986 p. 463). The technique of drilling was suitable for modifying hard stones used for cylinder seals, but quite superfluous for working in the soft steatite. However, Style II imitates the point drill style, though sometimes the effect was achieved using traditional cutting instruments, rather than the point drill technique (Kjærøum pers. comm. 2009).

The size of the Dilmun seals decreased from Style I to II/III (cf. Abu-Laban above fig. 670), a development which is paralleled in the decrease of the size of cylinder seals in the Old Babylonian period (Al-Gailani Werr 1986 p. 463).

The existence of a workshop of seal-cutters on Failaka in this period may be another expression of the radical expansion of the institutional sphere in Dilmun at this time (Højlund 2007b). The vigorous investments in public buildings that took place in Bahrain as well as in Failaka must have been prompted by an expansion and differentiation of public administrative functions. Some of these functions, such as the authority of issuing seals, seem to have been delegated from Bahrain to Failaka. The two extraordinary large seals (figs. 691 and 697) may be an indication that an institutional hierarchization had developed in the public administration.

A few fragments of crucibles and moulds show that copper working was taking place on Failaka in Phases 2-5, though exactly where is uncertain (figs. 1050-1053). The size of the moulds is extraordinary and only paralleled in the centre of Qala'at al-Bahrain and at the Barbar temple (Højlund & Andersen 1997 figs. 87-90. Andersen & Højlund 2003 figs. 752-756) and therefore probably related to the public sphere.

Stone vessels belonging to both the Umm an-Nar and the Wadi Suq repertoire were found (figs. 776-786), as well as a few carnelian beads (figs. 893-903). The faunal remains are dominated by fish which must have been the main source of food; in addition domestic livestock (sheep, goat and cattle) were occasionally consumed (see Yeomans above p. 228).

Occupation of Tell F3 and F6 seem to have ceased at the same time, i.e. after Period 2B (fig. 5). The same picture is seen at Qala'at al-Bahrain where the last occupation uncovered in both Excavation 519 and Excavation 520 dates to Period IIC and is followed by complete abandonment. The radiocarbon dates from Tell F6 suggest that this happened sometime during

the late 18th century BC, possibly around 1732 ±21 cal BC., cf. p. 243.

Since the dynamic exponential growth of the Dilmun civilization – especially its public buildings manifesting the power of the king – must be explained by Dilmun's monopoly over the trade to the Mesopotamian cities, it seems likely that the abandonment of these same buildings was a result of a collapse of this trade and the related stately level, and that this led to a resettlement of the entire population.

It is hardly coincidental that the collapse of public institutions and the widespread scarcity of finds in Bahrain after Period IIc occur contemporaneously with a breakdown of the Babylonian states in South Mesopotamia. Around 1740 BC (according to the Middle Chronology) the southernmost Babylonian cities appear to have been abandoned and some 20 years later, the same happens to cities in Middle Babylonia (Gasche 1989). The disappearance of the lucrative South Mesopotamian market is probably responsible for the recession in the Gulf.

Failaka in Period 3A

At Failaka, the abandonment was followed by a break in the occupation that lasted until the beginning of Period 3A (fig. 5). This break was long enough for the local Barbar pottery tradition to change distinctly. A new ware type (D) appeared (Højlund 1987 p. 119-121, 129), and a marked reliance on import of Mesopotamian pottery, especially drinking and serving vessels, began (Højlund 1987 fig. 460, 1989b p. 10). Radiocarbon dates suggest that Period 3A should be placed between c. 1700 and 1500 BC, but there are still too few datings to be sure of the beginning and end of Period 3A.

Period 3A is marked by a row of more than nine houses in Tell F3. In comparison to earlier Dilmun houses dating to the height of the Dilmun civilization, e.g. Saar Village or Qala'at al-Bahrain, the Period 3A houses on Failaka are smaller, manifesting less investment of resources and less wealth (Kjærøum & Højlund 2013 figs. 7 and 19, p. 17 and 23).

There was a complete lack of sherds diagnostic of Period 3A outside and to the east of the "Palace" in the 2008-2012 excavations. During the 1960-63 excavations inside and to the north and south of the "Palace" some Period 3A diagnostic sherds were found, but a small number compared to the large amounts of Period 3B and 4A pottery. This indicates that the use of the building was sparse in Period 3A but it seems that the stone cist below the floor and the cistern to

the north of the building were still used (Højlund 1987 p. 139-140).

It is of special interest that the excavations carried out in Period 3A layers in Tell F3 in 2012-2014 (which will be published elsewhere) have failed to uncover any seals (Højlund & Abu-Laban 2013. Højlund & Hilton 2014 and 2015). However, if seals were issued from a royal office as has been proposed (Højlund 2015) it makes sense that they disappear after the collapse of royal authority.

At around the same time in Bahrain, a new phase of occupation by the West Wall at Qala'at al-Bahrain referred to Period "IIF" (Højlund 1987 p. 154, figs. 665-682) is documented. The "IIF" pottery is typologically related to that of Period IIc, but with several new shapes, e.g. a new version of the lip-shaped hole-mouth rim Type B19 which is diagnostic of Period IIc and bowls/plates with rims influenced by Mesopotamian shapes of the late Isin-Larsa and early Old Babylonian periods (Højlund 2007a p. 11-13). Several plates and bowls have transverse painted strokes on the upper rim resembling Kaftari-related pottery from Phases 15 and 18 at Tol-e Spid (Petrie *et al.* 2005 figs. 8 and 10). No pottery similar to that of Period "IIF" has been found in Tells F3 or F6. Again, it is noteworthy that no seals were found with Period "IIF" by the West Wall. Neither were any seals found in the late, collective graves from Karanah 1 published by Velde and dating to the same post IIc-Period (1998 p. 255).

The so-called "Enigmatic Pottery" known from Bahrain and probably dating to the late Old Babylonian period has not been identified on Failaka (Denton 1997. 1999 p. 135, 145. Velde 1998 p. 250-255. Højlund 2007a p. 13-14). In a presentation of a number of Style III seals found in Bahrain, Denton argued that some of them were found with "Enigmatic pottery" (Denton 1997 p. 181-184). This is certainly a possibility, but the finds are still doubtful, deriving from collective burials with very slight documentation of the individual interments.

It is remarkable how much of the archaeological material found in the Gulf and dating to the Post IIc-Period is affected by Mesopotamian influence. Perhaps this is an indication that the former Dilmun polity had lost its independence. These emulations of Mesopotamian style occur after a long period (IIa-c) during which a unique Dilmun identity was created and may be explained as a loss of cultural self-confidence brought about by economic recession and the consequential fragmentation of the Dilmun kingdom after Period IIc. The Dilmunites may have sought, in this situation, to accommodate themselves to the

Babylonians and in this process adopt elements of Mesopotamian identity. The imported Mesopotamian pottery and the local Mesopotamian influenced pottery could be seen in the context of new eating and drinking habits.

The introduction of collective burials in Bahrain (Velde 1998) in this period could also be explained as an emulation of Mesopotamian culture, perhaps encouraged or stimulated by economic necessity. The investment in burial constructions was reduced considerably if a burial mound could be used, not just by one person, but by a larger family or by several generations.

Although the abandonment of large institutions and a general dearth of finds suggest problems in Dilmun, the continuation of Barbar pottery production indicates a certain degree of social and cultural continuity. It should also be noted that the increase in luxury pottery – wheel-made and/or painted – that can be followed through Periods IIa-c (*Qala'at* 1 fig. 393, p. 175-176) continues into Period "IIF" (Højlund 2007b fig. 14. Cf. Carter 2005 p. 256-259). There must have been pockets of settlements where life continued with some trading but on a lower social level than previously.

Phase 6. The "Palace" in Period 3B

The "Palace" was reoccupied in Period 3B and several rooms contained storage vessels dating to this period. To the east two new buildings with thick walls on solid foundations – apparently of some importance – were erected (figs. 64-65). The Period 3B deposit was limited to a few horizontal floor and occupation layers. Several narrow channels were cut in the floors, perhaps to transport water e.g. to prepare clay as mortar.

The potsherds from this phase were rather small and their edges worn in comparison to other phases (figs. 535-605). This perhaps indicates, that for some reason, accumulation of refuse was slower in this area compared to the preceding and subsequent phases.

The lack of precise parallels in Mesopotamia to this corpus of pottery that seems nevertheless to follow a Mesopotamian tradition is not surprising. The South Alluvial Plain was deurbanized from around 1720 BC (Middle Chronology) leading to a complete disappearance of material culture in the archaeological record that lasted for several hundred years well into the Kassite period (Gasche *et al.* 1998 p. 30).

In this "dark age" South Mesopotamia was the centre of a little known polity called the *Sealand Dynasty* (Dalley 2009) that was conquered with some diffi-

culty by the Kassite kings around 1465 BC. Nothing is known about the pottery development on the South Alluvial Plain during this particular political phase, but it is likely that we see a reflection of a specific South Babylonian pottery tradition in Eastern Arabia in Failaka Period 3B and Qala'at al-Bahrain Period IIIa (Højlund 1987 p. 160; 1989b p. 10).

Two radiocarbon dates of Period 3B lie around 1500 BC (cf. above p. 243), but more are needed to clarify the beginning and end of Period 3B.

One stamp seal possibly of Style II was found in layers belonging to this phase (fig. 746), but in all likelihood intruded from lower layers as it is typologically indistinguishable from seals firmly embedded in Phases 2-5. It is, however, possible that seals of Style III should be referred to Period 3B (cf. Højlund 1989b p. 11. 2007a p. 14), a possibility that was strengthened after the find of a Style III stamp seal in a period 3B context in Tell F3 in 2015 (Højlund & Hilton 2016).

A recent study of stone vessels found on Failaka during the 1958-1963 excavations (Hilton 2014) has shown that a certain style of soft stone vessels, the *Failaka Figurative Style*, has identical iconographical details to those found in Style III seals and that these similarities are so close and convincing that it must be concluded that both seals and stone vessels were produced within the same culture (Hilton 2014 p. 105-107). The seals of Style III clearly derive from a Dilmun tradition, but at the same time, this seal style as well as the Failaka Figurative Style of soft stone vessels is heavily influenced by Mesopotamian glyptic art to the extent that it suggests a break with the old cultural tradition.

It is therefore significant that a fragment of a Failaka Figurative Style bowl was found in a Phase 6 context, the oldest securely dated example of the style on Failaka.

No beads were found in this phase. The population continued to be sustained by herding of livestock (sheep/goat) and fishing, both open water and inshore (see Yeomans above p. 234).

Phase 7. The "Palace" in Period 4A

The "Palace" continued to be occupied in Period 4A according to the 1960-63 excavations (Højlund 1987 p. 143). During the new excavations a wall (A104) was uncovered parallel to the northeast outer wall of the "Palace", probably a second building separated from the "Palace" by a 1.5 m wide alley (fig. 68).

Sometime during this period, a large pit was dug between the "Palace" and the Temple (fig. 89), probably

for mixing and tempering clay to be used as mortar in a large building, perhaps for restoration of the “Palace” or for building the Buttress wall around the Temple, both projects took place in this period.

A 60 cm thick sequence of horizontal floor and occupation layers were deposited in this period, with remains of ash, bones, potsherds, etc. The pottery is all in the Mesopotamian tradition, there is no further Barbar pottery, and the majority derives from storage vessels, with a few bowls and beakers, all dating to Failaka Period 4A (figs. 606-667).

Two radiocarbon dates of Period 4A have mean dates in the 14th and 12th century BC, but more dates are needed to clarify the beginning and end of Period 4A and its relationship with Failaka period 4B (fig. 5)

A number of faience cylinder seals in Common Mitanni style and cylinder seals in chalcedony (Pittman above p. 158, figs. 759-764) belong to this period. Two stamp seals in Style IB must be redeposited from earlier occupation (figs. 706 and 723). A single stamp seal in Style III (fig. 743) may also be intrusive.

A few stone vessels were recovered, in the Wadi Suq and Plain Failaka Styles, besides two stone plates, one of which cuneiform inscribed (figs. 790-798). Beads are mostly in carnelian, glass and faience (figs. 904-917).

More sheep/goat was consumed with less fish compared to the previous periods. The fish caught in open waters decreased in relation to inshore fish (Yeomans above p. 234).

Phase 8. The Post-“Palace” period

No pottery later than Period 4A was found in the upper layers of Trenches A, E and H, e.g. none of the types diagnostic of Period 4B found abundantly in houses 23 and 30 in Tell F3 (Højlund 1987 p. 127-128). Neither were any of the types diagnostic of Period IIIc at Qala’at al-Bahrain (c. 1000 BC) found (Højlund & Andersen 1994 p. 185-197).

A certain amount of deflation of the surface of Tell F6 must, however, be envisaged, since no pottery or other cultural material could be related to the large walls uncovered in 1960-1963 in the uppermost level of Tell F6 (Kjærums & Højlund 2013 p. 108). These walls evidently post-date the Period 4A phase of the “Palace” (fig. 6), and they may relate to a number of ashlar in a fine beach-rock found in Trench A and in the area of the “Palace” (figs. 73-77). An undated layer of chips from cutting fine beach-rock ashlar found south of the Temple (cf. p. 60, fig. 78) may belong to the construction of this building.

A plundering of the “Palace” area for stone is evidenced by numerous stone plundering pits dug deep into the tell (figs. 81-84). The layout of the plundering pits in Trench A (fig. 72) indicates that this plundering was part of a synchronous operation with a high level of organization as the individual pits seem to respect each other. Some of the fine beach-rock ashlar were found in these pits, indicating that the plundering post-dates this architectural phase. The pits contained pottery from all the occupation phases present in Trenches A, E and H, but notably no Hellenistic sherds, indicating that the plundering operation predated the Hellenistic period.

The removal of the oolitic limestone ashlar from the Dilmun Temple probably also took place before the Hellenistic period. If the Temple had been robbed of its ashlar in the Hellenistic period (cf. Calvet 1989 p. 6. 1987 p. 21. 1991 p. 140. Callot, Gachet-Bizollon & Salles in Callot 2005 p. 67), one would expect more than sporadic occurrence of oolitic limestone in the fortifications and houses of the Hellenistic town of Tell F5 lying only 50 m south of Tell F6 (Kjærums & Højlund 2013 p. 158-159. Højlund 2012). When the Hellenistic temple and the surrounding town in Tell F5 were built, the Dilmun Temple had long since been plundered and the fine blocks transported away. The Hellenistic constructions, therefore, had to be made of a coarse local stone that could be quarried from the beach (Massih in prep.). Only the Temple in Tell F5 was built of the fine white oolitic limestone, which was, most probably, obtained directly from a quarry.

The Dilmun Temple

The Temple in Tell F6 was demolished to its foundation. Originally, it was built of finely dressed ashlar, of which a few fragments were found still embedded in the floor plaster (Calvet & Pic 1990 plan 1: 382) and could be identified on the site as oolitic limestone (fig. 91). Close to the Temple a few oolitic ashlar were found buried in the tell (figs. 119-122).

Ashlar of the same oolitic limestone were used also in the later phases (IIb and III) of the Barbar temples, and this is one of many close parallels to these temples in Bahrain (Højlund 2012. Kjærums & Højlund 2013).

Between the Temple and the shore, a c. 30 m wide area was covered by chips of oolitic limestone that must have been produced when dressing the ashlar for the Temple.

The relation between the Dilmun Temple, the oolitic chips and the loose oolitic ashlar is almost certain because no oolitic ashlar have been found in any

other Bronze Age buildings on Failaka other than in the Dilmun Temple. Not a single piece of oolitic limestone has been found in the walls of the nearby "Palace" or the domestic quarters in Tell F3 lying 200 m to the SW. They were all built of beach rock (Kjærsum & Højlund 2013).²⁷ Even in the foundations for the Temple walls there were no oolitic limestone at all, only a coarse, gravelly beach rock, probably local, and some rough sandstone also of beach rock character but less coarse. Oolitic chips and fragments first occurred in the plaster layer on top of the foundation and in the foundation for the temple floor flagstones. They also occurred in the walls and floors of the thin-walled house, the pavement, the oven and the floor extending to the temple terrace belonging to Phase 3 (fig. 6).

There is no outcrop of oolitic limestone on Failaka so the stone would have been brought to the island by ship or raft. At present, the shore is 250 m south of Tell F6, but it may have been closer in the 2nd millennium BC. The nearest source of oolitic limestone is said to have been on the mainland south of Kuwait city (Jeppesen 1989 p. 13), but no observations from this site have been published, and its exact location is unknown. In Bahrain, the island of Jiddah has extensive occurrences of oolitic limestone, and many (undatable) traces of quarrying have been observed there (Højlund 2013 p. 174-176, figs. 1-6). The typical way of extracting an ashlar at Jiddah was to cut a vertical narrow channel around the ashlar and then to prise it out.

Oolitic limestone is very soft and easily damaged, and it is therefore unlikely that the ashlars would have received more than coarse shaping at the quarry. Instead they would have been shipped as roughly shaped blocks to be cut into their precise form at the building site. On the standing walls of the Barbar temple, traces of cutting show that the façade received a final trimming after being positioned in the wall (Andersen & Højlund 2003, figs. 88-220 passim, p. 327-328).

Each of these production stages must have left a characteristic variety of chips that may eventually be defined through typological classification. A basic description of the dimensions of the chips is presented in fig. 115.

Oolitic limestone seems to have had a special significance in Dilmun as seen by its use exclusively in temple contexts perhaps because of its pure white colour. It is probably not a coincidence that this highly valued stone was used to repair the temple front (cf. above p. 88, fig. 123), though it does not seem to have been a freshly quarried stone. The heavily eroded surface of the block indicates that it could have been lost in the tidal zone only to be salvaged from there later.

Temple phases. Outside the southeast wall of the Temple was a series of thin layers of clay, plaster or sand, with some bitumen and oolitic limestone chips, probably formed during the construction of the Temple as they consist of materials used in construction (see above p. 79-81, figs. 11 and 16).

The central Temple building constructed in white oolitic limestone ashlar was enclosed by a terrace with a thick floor of clayey plaster that has been uncovered both to northwest and southeast of the Temple (figs. 94 and 99). A channel leading from the centre of the building through the north wall and extending for a number of meters further north was related to this first phase of the Temple (fig. 105). This phase was contemporary with the thin-walled building of Phase 3 (see above p. 73-74).

At some point, the terrace south of the temple was modified and a stone floor was laid (fig. 99). In this floor a new channel was constructed and cut in limestone ashlars (fig. 104). This channel originally connected with a channel excavated by the French Mission along the southwest and the northwest side of the Temple, and its direction of flow would have been northwards. The construction of this second terrace bordered by a channel may well have been contemporary with the erection of the "Palace" and the floor outside the "Palace" in Phase 4 (see above p. 79).

It seems that the first channel leading from the inside of the building to its rear was replaced by a new channel extending from the entrance of the Temple around the building to its rear. If the function of the channels was to evacuate water from cleaning altars after bloody sacrifices as has been argued in relation to the Barbar Temples (see above p. 79), then one may suggest that the main altar was moved from inside the building to the front of the building.

It should be noted that the first drain leading out of the Temple is located slightly off the centre of the main room (fig. 91:4), as if making room for something in the middle of the back wall, like a cult figure, to which the altar was associated. Right here an oolitic block was found *in situ* on the inside of the wall as a further indication that there was originally an important construction in this place (fig. 91:3).

Parallel to the southeast side of the Temple and at a distance of c. 12 m from the Temple a four meter wide clay platform was raised (figs. 99 and 110) which may have carried a wall enclosing the Temple, possibly continuing as a wall encountered northwest of the Temple (fig. 53: 314).

Due to a lengthy exposure to the elements, the southwest temple façade was severely eroded (fig. 125) and this erosion damage was subsequently repaired

(figs. 123-124) at a time, which cannot be specified (see above p. 88).

Later a 4 m wide wall was constructed in buttress-like sections around the ashlar temple building (figs. 105 and 128). This Buttress Wall was built of uncut beach-rock stones laid in clay, and carbon embedded in the plaster which covered its faces has been radiocarbon dated to c. 13-1400 BC, placing it in Period 4A (cf. p. 250). The function of this notable enlargement is unknown, perhaps it was to create a

high platform and the coarse building materials may argue against a continued religious purpose of the building.

Only few finds can be attached to the Temple, a bead on the temple terrace stone floor C68 (figs. 897), a stamp seal on the clay platform C48 (figs. 704), and perhaps a few unusual finds, a gold bead (fig. 991) and a fragment of gold sheet (fig. 1135) found in the disturbed layers in Trench C/J that might originate from the Temple.

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Five years of excavations in Tell F6 have significantly improved our understanding of the history of the site. The evidence documents that Failaka was first inhabited by a community from Mesopotamia, who about 2100 BC established a settlement at the site of Tell F6. The settlement served to support the long-distance-trading “big boats” sent out by the Ur III state from the port of Gu’abba in South Mesopotamia to the coast of Magan in present day United Arab Emirates.

Some time after the collapse of the Ur III state Failaka was colonized by Dilmunites from Bahrain who chose the site of Tell F6 for their public buildings, whereas the nearby site of Tell F3 was reserved for mundane domestic quarters.

A sequence of architectural phases has been identified in the two great buildings, the “Palace” and the Temple and in structures in between, and the dating of the different phases has been narrowed down by means of radiocarbon dating.

The rate of construction, demolition and rebuilding suggest an exponential growth of investment of resources and man-power in public architecture in Tell F6 through the first centuries of the 2nd millennium BC, a development that has previously been ascertained in Bahrain. The colonization of Failaka was part of this dynamic expansion of the Dilmun civilisation, and the fact that Failaka was colonized by Dilmun in the early 2nd millennium, after being a Mesopotamian trade post in the late 3rd millennium, is clear testimony that control of trade through the Gulf had moved from Mesopotamia to Dilmun.

While previous studies of Dilmun stamp seals have been marred by the scarcity of precise stratigraphical information, the 2008-2012 excavations produced 75 stamp seals of which 63 were found under controlled conditions. Based on this new and improved dataset it was demonstrated that stamp seal Styles IB and II were local Failaka styles that were coterminous with the late Style IA and dating to the Dilmun floruit around 1800 BC. When the Dilmun civilisation collapsed in the late 18th century, Dilmun seals disappeared and public buildings in Tell F6 were abandoned, thus confirming the strong connection between royal authority and the issue of stamp seals in Dilmun.

While settlement in Tell F3 continued in the 17th century after the collapse of the Dilmun state level, the public buildings in Tell F6 were not re-inhabited till around 1600 BC. Dilmun material culture continued into this period, heavily influenced from Mesopotamia. There are indications that the *Failaka Figurative Style* of stone vessels as well as stamp seal Style III belong in this period and demonstrate that Dilmun had recovered in some as of yet poorly understood relationship with the Mesopotamian Sealand Dynasty.

A hundred years or so later, the public buildings in Tell F6 were refurbished and extended, and the temple was rebuilt possibly with a non-religious purpose. The material culture now belongs to a Mesopotamian Kassite tradition and probably relates to a situation where Dilmun (i.e. both Failaka and Bahrain) had become a Babylonian province.

Notes

1. When the foundation wall of the Temple was excavated in the 1980s its stones were still embedded in clay mortar and its vertical faces were coated with clay (Calvet & Pic 1990 p. 104). In 2008, twenty years later, all clay in the exposed parts of the foundation had disappeared, whereas the unexposed parts were still covered with clay (cf. fig. 100).
2. In the houses of Tell F3, it was not unusual for a wall to be built on a plaster floor (Kjærøum & Højlund 2013 p. 25).
3. Note that the knob foot continues in Failaka Period 4A (Højlund 1987 p. 124, fig. 572) and is found again around 1000 BC in Qala'at al-Bahrain Period IIIc (Højlund & Andersen 1994 fig. 798), parallel to its occurrence in late 2nd millennium BC Mesopotamia (Armstrong & Gasche 2014 pl. 90).
4. I am grateful to Henk Mienis (Tel Aviv University) for the assistance in identifying the shell species (2015).
5. On similar scarabs from Tell el-Ajul see Keel 1997 p. 104-525, Nos. 43, 111, 119, 351, 378, 398, 480, 523, 565, 622, 837, 894, 922, 979, 1103, 1109, 1110. For other Canaanite sites see Keel 1997, 2010a, 2010b, 2013; see also Eggler and Keel 2006. For a similar shape of the side and the legs see in particular Keel 1997 p. 238-239, No. 398.
6. Besides the scaraboid seal published by Kjærøum (1983 p. 128-129, No. 316) and the one discussed by Abu-Laban in this volume (p. 154) there are a few pieces from Oman on display in the National Museum in Muscat (personal communication by Hélène David-Cuny; I am very grateful to her for providing this information).
7. From the previous 1958-1963 excavations on Failaka c. 170 beads were recovered from Tell F3 and c. 350 from Tell F6 (Andersson in prep.).
8. Elsewhere, such discolouration of beads is ascribed to a process arising "... from the environment that they were preserved in, rather than from any specific treatments (e.g. heating)" (Lombard 1999: 71) or a complete etching of a bead surface (Højlund 2007 p. 76). The assemblage at Failaka contains both perfectly red transparent carnelian beads along with discoloured examples, therefore the explanation of purely environmental causes is found unlikely by this author. In a number of cases beads from Failaka are only partially discoloured, which does not point to a deliberate etching of the bead surface, unless the beads represent failed attempts at a complete etching. No beads with decorative white etched patterns, which are traditionally called etched carnelian beads, commonly ascribed to the Indus region, have been found at Failaka (cf. De Waele and Haerinck 2006; Possehl 1996 p. 152-158).
9. Law cites several other sources in the central and eastern part of Iran, as well as Afghanistan and Yemen (Law 2011 p. 282). In addition, Moorey includes Egypt and Turkey (Moorey 1994 p. 97).
10. In addition, a characteristic bead form commonly associated with the Indus region, long bicones made of carnelian that may have been produced only for a restricted period of time from c. 2450-1900 BC. (Lankton 2003 p. 35; Possehl 1996 p. 159-160), is only represented by a fragmented example in the 1958-1963 assemblage.
11. Moorey notes that chrysoprase has not been reliably registered in Mesopotamia before the Seleucid period (Moorey 1994 p. 96). The complicated issue of identifying "greenstones" is discussed by Moorey (1994 p. 83).
12. However, two beads made of porphyry are reported from burial contexts at Kish (Mackay 1925 p. 188).
13. Several of the mines have no firm evidence for ancient exploitation (Moorey 1994 p. 101).
14. No faunal evidence suggests the presence of the ostrich at Failaka (pers. comm. Lisa Yeomans 2014).
15. Ostrich eggshell is also reported from 3rd and 2nd millennium contexts in Saudi Arabia, Sharjah, Ras al-Khaimah and Abu Dhabi (UAE) (Potts 2001 p. 186-187). Cluzan also notes that fragments of ostrich eggshell is found at many sites and that the near total absence of ostrich eggshell beads outside Africa is difficult to reconcile (Cluzan 2008 p. 326).
16. Thanks are owed to Henk K. Mienis (Tel Aviv University), who identified the *Cypraea* species and the *Conus* genus of the shell bead assemblage.
17. A reason for this could be that the cowrie shells were not recognized as worked shells, but rather as naturally broken shells. However, one worked cowrie was reported from the Hellenistic fortress at Tell F5 (Cataliotti-Valdina 1990 p. 79, fig. 1).
18. At the Saar Settlement bone examples of similar rings are also mentioned (Killick and Moon 2005 p. 172, fig. 5.5c).
19. At early 2nd millennium Usiyeh, located at the Euphrates 700 km from the Gulf, a great quantity of shell rings were found indicating "...a significant trade in the middle Euphrates region through the Gulf and southern Mesopotamia, in this period." (Oguchi 1992 p. 61).
20. Thanks are owed to Helle Strehle (Conservator, M.Sc.) who identified the material.
21. This parallel is attributed to level IIa i.e. 1st millennium (Hellenistic) activity at the Temple, represented by stone looting pits disturbing the older occupation levels (Calvet and Pic 1986: 16). The parallel may therefore be older than this date.
22. A gold foil fragment was also recorded by the Johns Hopkins University Expedition in Trench FH-9 just west of the "Palace" at Tell F6 (Howard-Carter 1984).
23. Though no sieving took place during the French excavations (pers. comm. S.A.H. Shehab 2012).
24. It should, however, be mentioned that this parameter is considered to be less reliable than the others.

25. This is true for a part of the Susa dataset, some bitumen samples of this dataset do contain oleanane, and several have also been identified as coming from Sultan (Connan 2012 p. 127).
26. Compare the clearly defined front and rear of the Barbar Temple (Andersen & Højlund 2003 p. 327).
27. In the 1980s, twenty years after being uncovered in 1958-1963, the architecture of Tells F3 and F6 were in a sad state of preservation. A restoration made in 1985 was rather rough, so that the present position of most walls is inaccurate and the building material not always original. Especially in the restoration of the "Palace" in Tell F6 some rather fine ashlar, some in oolitic limestone, were included which were not originally used in this building.

Bibliography

- AAE: Arabian archeology & epigraphy.
- Abu-Laban, A. (in prep.): *The Dilmun stamp seals from the Danish excavations in Failaka 1958-63 and 2008-2012*. Ph.D. dissertation at the University of Copenhagen.
- Al-Gailani Werr, L. 1986: The Sippar Workshop of Seal Engraving. *AJA* 90:4, p. 461-463.
- Al-Sindi, K.M. 1999: *Dilmun Seals*. Bahrain.
- Andersen, H.H. & Højlund, F. 2003: *The Barbar Temples*, vols. 1-2. Aarhus. *JASP* 48.
- Andersson, A. (in prep.): *The beads from the Danish excavations at Tells F3 and F6 on Failaka, Kuwait in 1958-1963*.
- André-Salvini, B. 1999: Cylindrical Seal with Inscription... In: Lombard (ed.) 1999 p. 91.
- Armstrong, J.A. & Gasche, H. 2014: *Mesopotamian pottery. A guide to the Babylonian tradition in the second millennium B.C.* Mesopotamian History and Environment Series II Memoirs VI. Ghent & Chicago.
- Asher-Greve, J. and Westenholz, J.G. 2013: *Goddesses in Context. On divine powers, roles, relationships and gender in Mesopotamian textual and visual sources*. Orbis Biblicus et Orientalis 259, Fribourg.
- Aston, B.G., Harrel, J.A. and Shaw, I. 2000: "Stone." In: Nicholson, P.T. and Shaw, I. (ed.), *Ancient Egyptian Materials and Technology*. Cambridge University Press p. 5-77.
- Ball, V. and Holland, S.T.H. 1898: *A manual of the Geology of India: Economic geology, Vol. 3, Part 1, Corundum*. Calcutta, Government of India.
- Bangsgaard, P. 2003: Appendix 1: Animal Bones from the Barbar Temple. In: Andersen and Højlund 2003 vol. 2, p. 7-16.
- Barta, P., Benediková, L., Hajnalová, M., Mikliková, Z., Belanova, T. & Shehab, A.H. 2008: Al-Khidr on Failaka Island: Preliminary Results of the fieldworks at a Dilmun culture settlement in Kuwait. *Tüba-ar* XI, p. 121-134.
- Barthélemy de Saizieu, B. 2003: *Les Parures de Mehrgarh: Perles et Pendentifs du Néolithique Préceramique à la Période pré-Indus: Fouilles 1974-985*. Paris, Editions Recherche sur les Civilisations.
- Beech, M.J. 2004: *In the Land of the Ichthyophagi: Modelling Fish Exploitation in the Arabian Gulf and Gulf of Oman from the Late 5th Millennium BC to the Late Islamic Period*. Oxford: British Archaeological Report: International Series 1217.
- Beck, H.C. 1928: The Classification and Nomenclature of Beads and Pendants. In: *Archaeologia. The Society of Antiquaries of London*, Vol. 77 p. 1-76.
- Belenová-Štolcová, T. 2010: *Al-Khidr, Failaka Island, State of Kuwait 2004-2008. Primary scientific report on the field research: Use of Bitumen*. Unpublished Report.
- Benediková, L. (ed.) 2010: *Al-Khidr 2004-2009, Primary Scientific Report on the Activities of the Kuwait-Slovak Archaeological Mission*. National Council for Culture, Arts & Letters, Kuwait City. Archeologicky ústav Slovenska akadémie vied, Nitra.
- Benediková, L. & Barta, P. 2009: A Bronze Age settlement at al-Khidr, Failakah Island, Kuwait. *PSAS* 39, p. 43-56.
- Ben-Tor, D. 2007: *Scarabs, Chronology, and Interconnections. Egypt and Palestine in the Second Intermediate Period*. Orbis Biblicus et Orientalis, Series Archaeologica 27, Fribourg/Göttingen.
- Beyer, D. 1986: Les sceaux. In: Calvet, Y. & Salles J.-F. (eds.), *Failaka. Fouilles Francaises 1984-1985*. Travaux de la Maison de l'Orient 18. Maison de l'Orient. Lyon p. 89-104.
- Bibby, T.G. 1969: *Looking for Dilmun*. Alfred A. Knopf. New York.
- Bibby, T.G. 1971: "...according to the standard of Dilmun". *Kuml* 1970 p. 345-353.
- Bibby, T.G. 1973: *Preliminary Survey in East Arabia 1968*. *JASP* vol. 12, Copenhagen.
- Bishop, A.C., Woolley, A.R. and Hamilton, W. R. 2005: *Guide to Minerals, Rocks and Fossils*. Firefly Ltd.
- Boehmer, R.M., Pedde, F. & Salje, B. 1995: *Uruk. Die Gräber. Ausgrabungen in Uruk-Warka, Endberichte* 10. Verlag P. von Zabern.
- Bronk Ramsey, C. 1995: Radiocarbon Calibration and Analysis of Stratigraphy: The OxCal Program. *Radiocarbon* 37(2) p. 425-30.
- Bronk Ramsey, C. 2009a: Bayesian analysis of radiocarbon dates. *Radiocarbon* 51(1) p. 337-60.
- Bronk Ramsey C. 2009b: Dealing with outliers and offsets in radiocarbon dating. *Radiocarbon* 51(3) p. 1023-45.
- Brunet, O. 2009: Bronze and Iron Age carnelian bead production in the UAE and Armenia: New Perspectives. *PSAS* 39 p. 57-68.
- Buchanan, B. 1981: *Early Near Eastern Seals in the Yale Babylonian Collection*. New Haven and London, Yale University Press.
- Callot, O. (et al.) 2005: *L'île de Failaka. Archéologie du Kuwait*. Somogy éditions d'art.
- Calvet, Y. 1987: Le pays de Dilmoun au 2^e millénaire: découvertes récentes. In: Fahd, T. (ed), *L'Arabie Préislamique et son environnement historique et culturel*. Actes du Colloque de Strasbourg 24-27 juin 1987.
- Calvet, Y. 1989: Failaka and the northern part of Dilmun. *PSAS* 19.
- Calvet, Y. 1991: Un bâtiment de l'Age du Bronze à Failaka, Koweït. In: Schippmann, K., Herling, A. & Salles, J.-F. (eds.): *Golf Archäologie. Mesopotamien, Iran, Kuwait, Bahrain, Vereinigte Arabische Emirate und Oman*. Internationale Archäologie 6.
- Calvet, Y. 2002: The Dilmun culture at its border with Mesopotamia. In: Cleuziou, S., Tosi, M. & Zarins, J. (eds.), *Essays on the late prehistory of the Arabian peninsula, Serie Orientale Roma, XCIII*, Istituto italiano per l'Africa e l'Oriente, Rome, p. 257-271.
- Calvet, Y. 2008: Le temple-tour de Failaka dans son contexte proche-oriental. In: *Calvet & Pic 2008* p. 15-19.
- Calvet, Y. & Pic, M. 1986: Un nouveau bâtiment de l'Age du Bronze sur le Tell F6. In: *Calvet & Salles 1986*, p. 13-87.
- Calvet, Y. & Salles, J.-F. (eds.) 1986: *Failaka. Fouilles Francaises 1984-1985*. Travaux de la Maison de l'Orient 12. Lyon.
- Calvet, Y. & Gachet, J. (eds.) 1990: *Failaka. Fouilles Francaises 1986-1988*. Travaux de la Maison de l'Orient 18. Maison de l'Orient. Lyon.
- Calvet, Y. & Pic, M. 1990: Un temple-tour de l'âge du Bronze à Failaka. In: *Calvet & Gachet 1990*, p. 103-122.
- Calvet, Y. & Pic, M. 2008: *Failaka Fouilles Francaises 1984-1988*.

- Matériel céramique du temple-tour et épigraphie. Travaux de la Maison de l'Orient et de la Méditerranée* 48, Lyon.
- Carter, R. 2005: Pottery vessels: typological analysis. In: Killick & Moon 2005 p. 235-278.
- Carter, R. 2010: Boat-Related Finds. In: Carter, R. & Crawford, H. (eds.) *Maritime Interactions in the Arabian Neolithic: Evidence from H3, As-Sabiyah, an Ubaid-related site in Kuwait*. Leiden: Brill.
- Carter, R. & Crawford, H. (eds.) 2010: *Maritime Interactions in the Arabian Neolithic. Evidence from H3, As-Sabiyah, an Ubaid-related site in Kuwait*. Leiden.
- Casanova, M. 2003: Alabaster and Calcite Vessels. In: Andersen & Højlund 2003 p. 283-288.
- Cataliotti-Valdina, J. 1990: Les coquillages de Failaka. In: Calvet & Gaschet 1990 p. 71-84.
- Ciarla, R. 1979: The Manufacture of Alabaster Vessels at Shahr-I Sokhta and Mundigak in the 3rd Mill. BC. A Problem of Cultural Identity. In: Gnoli, G. & Rossi, A.V. (eds.): *Iranica*. Instituto Universitario Orientale Seminario di Studi Asiatic. Series Minor. X, p. 319-335.
- Clayton, D. and Pilcher, C. 1983: *Kuwait's Natural History: An Introduction*. Ahmadi: Kuwait Oil Company.
- Cleuziou, S., Méry, S. & Vogt, B. (eds.) 2011: *Les sépultures de l'âge du Bronze*. Protohistoire de l'oasis d'al-Aïn. Travaux de la Mission archéologique française à Abou Dhabi (Emirats arabes unis). BAR International Series 2227.
- Cluzan, S. 2008: Cyprus: An International Nexus of Art and Trade. In: Aruz, J., Benzel, K. & Evans, J.M. (eds.), *Beyond Babylon. Trade, Art and Diplomacy in the Second Millennium*. The Metropolitan Museum of Art, New York. Yale University Press, New Haven and London p. 311-386.
- Colley, S.M. 1990: The Analysis and Interpretation of Archaeological Fish Remains. *Archaeological Method and Theory* 2, p. 207-253.
- Collon, D. 1982: *Catalogue of the Western Asiatic Seals in the British Museum. Cylinder Seals II. Akkadian, Post Akkadian Ur III periods*. British Museum Publications Ltd.
- Collon, D. 1986: *Catalogue of the Western Asiatic Seals in the British Museum. Cylinder Seals III. Isin-Larsa and Old Babylonian Periods*. London.
- Collon, D. 1993: *First Impressions. Cylinder Seals in the Ancient Near East*. British Museum Press.
- Connan, J. 1988: Quelques secrets des bitumes archéologiques de Mésopotamie révélés par les analyses de géochimie organique pétrolière. *Bulletin du Centre de recherches Elf Exploration Production* 12.
- Connan, J. 2010: Appendix II: Technical Report on the Comparative Compositional and Isotopic Analysis of the H3 Bitumen. In: Carter, R. & Crawford, H. (eds.), *Maritime Interactions in the Arabian Neolithic: Evidence from H3, as-Sabiyah, an Ubaid-Related Site in Kuwait*. Leiden: Brill.
- Connan, J. 2011: The bituminous mixtures of Akkaz. In: Gachet-Bizollon, J. (eds.) *Le Tell d'Akkaz au Koweït, Tell Akkaz in Kuwait*. Lyon.
- Connan, J. 2012: *Le bitume dans l'Antiquité*. Arles, Editions Errances.
- Connan, J. (unpubl.): *The bitumen of ed-Dur*.
- Connan, J. & Carter, R. 2007: A geochemical study of bituminous mixtures from Failaka and Umm an-Namel (Kuwait), from the Early Dilmun to the Early Islamic period. *AAE* 18, p. 139-181.
- Connan, J., Carter, R., Crawford, H., Tobey, M., Charrie-Duhaut, A., Jarvie, D., Albrecht, P. & Norman, K. 2005: A comparative geochemical study of bituminous boat remains from H3, As-Sabiyah (Kuwait), and RJ-2, Ra's al-Jinz (Oman). *AAE* 16, p. 21-66.
- Connan, J. & Deschesne, O. 1992: Origine et altération de quelques bitumes archéologiques de Tell es-Sawwan (5500-500 avant J.-C.). *Mesopotamia* 27.
- Connan, J. & Deschesne, O. 1996: *Le bitume à Suse. Collection du Musée du Louvre*. Paris et Pau, coédition Elf Aquitaine-Réunion des Musées nationaux.
- Connan, J., Lombard, P., Killick, R., Højlund, F., Salles, J. F. & Khalaf, A. 1998: The archaeological bitumens of Bahrain from the Early Dilmun period (c. 2200 BC) to the sixteenth century AD: a problem of sources and trade. *AAE* 9, p. 141-181.
- Connan, J., Nissenbaum, A. & Dessort, D. 1992: Molecular archaeology: Export of Dead Sea asphalt to Canaan and Egypt in the Chalcolithic-Early Bronze Age (4th-3rd millennium BC). *Geochimica et Cosmochimica Acta* 56, p. 2743-2759.
- Connan, J. & Van de Velde, T. 2010: An overview of bitumen trade in the Near East from the Neolithic (c. 8000 BC) to the early Islamic period. *AAE* 21, p. 1-19.
- Costa, P.M. 1988: Fishing Stations of the Coast of Oman: A Theme of Ethno-Archaeological Research. *PSAS* 18, p. 3-13.
- Crawford, H. 1999: "The house on the edge of the quay": The Kingdom's Warehouse. In: Lombard (ed.) 1999, p. 86-100.
- Crawford, H. 2001: *Early Dilmun seals from Saar. Art and commerce in Bronze Age Bahrain*. London-Bahrain Archaeological Expedition, Institute of Archaeology, University College London.
- Crawford, H., Killick, R. and Moon, J. 1997: *The Dilmun Temple at Saar. Bahrain and its Archaeological Inheritance*. London-Bahrain Archaeological Expedition: Saar Excavation Reports 1. Kegan Paul International. London and New York.
- DA 2003 = *Dossiers d'Archeologie* n°287 oct. 2003: Jiroft, Fabuleuse Découverte en Iran.
- Dalley, S. 2009: *Babylonian Tablets from the First Sealand Dynasty in the Schøyen Collection*. Cornell University Studies in Assyriology and Sumerology vol. 9. Maryland.
- David, H. 1996: Styles and Evolution: Soft stone vessels during the Bronze Age in the Oman Peninsula. *PSAS* 26, p. 31-46.
- David, H. 2011: Les Vases en Chlorite. In: Cleuziou et al. 2011 p. 184-201.
- David-Cuny, H. and Azpeitia, J. 2012: *Failaka Seals Catalogue vol. 1: Al-Khidr*. Ed. S.A.H. Shehab. National Council for Culture, Arts and Letters, State of Kuwait.
- De Cardi, B. 1988: The Grave-goods from Shimal Tomb 6 in Ras al-Khaimah, U.A.E. In: Potts, D.T. (ed.): *Araby the Blest. Studies in Arabian Archaeology*. Museum Tusculanum Press, p. 45-72.
- De Meyer, L. (ed.) 1984: *Tell ed-Der IV*. Progress Reports (Second Series). Louvain.
- De Waele, A. and Haerinck, E. 2006: Etched Carnelian Beads from the Northeast and Southeast Arabia. *AAE* vol. 17:1, p. 31-40.
- Denton, B. E. 1997: Style III seals from Bahrain. *AAE* 8, p. 174-189.
- de Moulins, D., Moon, J., Irving, B. and Glover, E. 1997: Food Remains. In: Crawford, H., Killick, R. and Moon, J. (eds.) 1997: *The Dilmun Temple at Saar and its Archaeological Inheritance*. London: London-Bahrain Archaeological Expedition, p. 79-82.

- Desse, J. and Desse-Berset, N. 1990: La Faune: Les mammifères et les poissons. In: *Calvet & Gachet 1990*, p. 51-70.
- During Caspers, E.C.L. 1970: A note on the carved stone vases and Incised Grayware. In: de Cardi, B. (ed) *Excavations at Bampur*. American Museum of Natural History, p. 319-325.
- During Caspers, E.C.L. 1983: Triangular stamp seals from the Arabian Gulf and their Indus Valley connection. *Instituto Universitario Orientale, Annali* 43 p. 661-670.
- Eggler, J. & Keel, O. 2006: *Corpus der Siegel-Amulette aus Jordanien. Vom Neolithikum bis zur Perserzeit*. Orbis Biblicus et Orientalis, Series Archaeologica 25, Fribourg/Göttingen.
- Feingold R. 2014: *Engraved on Stone: Mesopotamian Cylinder Seals and Seal Inscriptions in the Old Babylonian Period*. Gorgias Studies in the Ancient Near East 7.
- Forbes, R. J. 1964: *Studies in Ancient Technology*, vol. I (2nd edition). Leiden, E.J. Brill.
- Frifelt, K. 1991: *The Island of Umm an-Nar vol. 1. Third Millennium Graves*. JASP 26:1.
- Gasche, H. 1989: *La Babylonie au 17^e siècle avant notre ère: Approche archéologique, problèmes et perspectives*. Mesopotamian History and Environment. Series II, Memoirs I. Ghent.
- Gasche, H., J.A. Armstrong, S.W. Cole & V.G. Gurzadyan 1998: *Dating the Fall of Babylon. A Reappraisal of Second-Millennium Chronology*. Mesopotamian History and Environment Series II, Memoirs IV.
- Glassner, J. J. 2008: Textes cunéiformes. In: Calvet Y. and Pic, M. 2008: *Failaka Fouilles Françaises 1984-1988*. Travaux de la Maison de L'Orient et de la Méditerranée 48, p. 171-205.
- Glob, P.V. 1968: *Al-Bahrain. De danske ekspeditioner til oldtidens Dilmun*. Gyldendal.
- Golani, A. 2014: Cowrie Shells and their Imitations as Ornamental Amulets in Egypt and the Near East. In: Golani, A. and Wygnanska, Z. (eds.), *Beyond Ornamentation. Jewelry as an Aspect of Material Culture in the Ancient Near East*. Polish Archaeology in the Mediterranean, vol. XXIII:2 p. 71-83.
- Gorelick, L. and Gwinnett, A.J. 1983: Ancient Egyptian Stone Drilling. *Expedition* 25:3, p. 40-47.
- Goual, L. 2012: Petroleum Asphaltenes. In: El-Sayed Abdul-Raouf, M. (ed.) *Crude Oil Emulsion- Composition Stability and Characterization*. Rijeka.
- Goubanov, E.P. and Shleib, N.A. (eds.) 1980: *Sharks of the Arabian Gulf*. Ministry of Public Works, Agricultural Department, Fisheries Divisions, Kuwait.
- Gouin, Ph. 1991: Rapes, jarres et faisselles. La production et l'exportation des produits laitiers dans l'Indus du 3^e millénaire. *Paléorient* vol. 16:2, p. 37-54.
- Guba, I. 2007: *The Geological Wonders of Oman*. Muscat, Sultan Qaboos University.
- Gwinnett, A. J. and Gorelick, L. 1981: Beadmaking in Iran in the Early Bronze Age. *Expedition* 24:1, p. 10-23.
- Gwinnett, A. J. and Gorelick, L. 1987: Experimental Evidence for the Use of a Diamond Drill in Sri Lanka ca. A.D. 700-1000. *Archaeomaterials* 1, p. 149-152.
- Hajnalová M., Miklíková, Z. and Belanová-Štolcová, T. 2009: Environmental Research at al-Khidr, Failakah Island, Kuwait. *PSAS* 39, p. 197-202.
- Hakemi, A. 1997: *Shahdad, Archaeological Excavations of the Bronze Age Center in Iran*. ISMEO.
- Hannestad, L. 1983: *The Hellenistic Pottery from Failaka vol. 2:1. Ikaros. The Hellenistic Settlements*. JASP 16:2, Højbjerg.
- Harrell, J. 2012: Gemstones. *UCLA Encyclopedia of Egyptology*.
- Wendrich, W., Dieleman, J., Frood, E. and Baines, J. University of California, Los Angeles, UCLA Press, p. 1-23. <http://digital2.library.ucla.edu/viewItem.do?ark=21198/z0002CZX11>.
- Heimpel, W., Gorelick, L. and Gwinnett, A.J. 1988: Philological and Archaeological Evidence for the Use of Emery in the Bronze Age Near East. *Journal of Cuneiform Studies* 40:2, p. 195-210.
- Herrmann, G. 1968: Lapis Lazuli: The Early Phases of Its Trade. *Iraq* vol. 30:1 p. 21-57.
- Hilton, A., Højlund, F., David-Cuny, H. & Eidem, J. 2012: Altar plates from second millennium BC Failaka, Kuwait. *Bibliotheca Orientalis* 69, p. 411-420.
- Hilton, A. 2014: *The Stone Vessels*. Failaka/Dilmun. The Second Millennium Settlements, vol. 4. JASP 17:4. Aarhus University Press and the National Council for Culture, Arts and Letters, Kuwait.
- Hilton, A. 2013: *Qarn al-Harf. Bead Report. An analysis of beads excavated at the tombs of Qarn al-Harf, Ras al-Khaimah, U.A.E.* Unpublished Report Archived at the National Museum of Ras al-Khaimah.
- Hollander, D. & Schwartz, M. 2000: Annealing, distilling, reheating and recycling : bitumen processing in the Ancient Near East. *Paleorient* 26, p. 83-91.
- Hornung, E. & Staehelin, E. 1976: *Skarabäen und andere Siegel-amulette aus Basler Sammlungen*. Mainz.
- Howard-Carter, T. 1984: *The Johns Hopkins University Expedition to Failaka Island 1973-1974*. Unpublished report on file in the Kuwait National Museum.
- Häser, J. 1991: Soft-stone vessels from Shimal and Dhayah, Ras al-Khaimah. In: Schippmann, K., Herling, A. & Salles, J.-F. (eds.): *Golf-Archäologie, Mesopotamien, Iran, Kuwait, Bahrain, Vereinigte Arabische Emirate und Oman*. Internationale Archäologie 6. Göttingen & Lyon p. 221-232.
- Højlund, F. 1987: *The Bronze Age Pottery*. (Failaka/Dilmun. The Second Millennium Settlements 2). JASP 17:2.
- Højlund, F. 1989a: The Formation of the Dilmun State and the Amorite Tribes. *PSAS* 19, p. 45-59.
- Højlund, F. 1989b: Dilmun and the Sealand. *Northern Akkad Project Reports* 2, p. 9-14.
- Højlund, F. 2007a: *The Burial Mounds of Bahrain. Social complexity in Early Dilmun*. JASP 58.
- Højlund, F. 2007b: Transformations of the Dilmun state, c. 2050-1600 BC. Paper presented at the *Twenty Years of Bahrain Archaeology (1986-2006) Conference* in Manama, Bahrain, 9-13. December 2007.
- Højlund, F. 2008: *The Danish Archaeological Expedition to Kuwait 1958-1963. A glimpse into the archives of Moesgård Museum*. Moesgård Museum & The Kuwaiti National Council for Culture, Arts and Letters (Arabic version 2009).
- Højlund, F. 2010: *The Kuwaiti-Danish Mission to Failaka 2009*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. 2012: The Dilmun temple on Failaka, Kuwait. *AAE* 23, p. 165-173.
- Højlund, F. 2013: The stone quarry on Jiddah Island, Bahrain. *AAE* 24, p. 174-176.
- Højlund, F. 2016: Foreword. In: David-Cuny, H. & Neyme, D.: *Failaka Seals Catalogue vol. 2: Tell F6, "The Palace"*. Ed. by S.A.H. Shehab. National Council for Culture, Arts and Letters, State of Kuwait, p. 5-11.
- Højlund, F. & Andersen, H.H. 1994: *Qala'at al-Bahrain vol. 1. The Northern City Wall and the Islamic Fortress*. JASP 30:1.

- Højlund, F. & Andersen, H. H. 1997: *Qala'at al-Bahrain. The Central Monumental Buildings* vol. 2. JASP 30:2. Højbjerg.
- Højlund, F., Bangsgaard, P., Hansen, J., Haue, N., Kjærø, P. & Lund, D.D. 2005: New excavations at the Barbar Temple, Bahrain. *Arabian Archaeology and Epigraphy* 16 p. 1-24.
- Højlund, F. and Krants Larsen, L. 2009: *The Kuwaiti-Danish Mission to Failaka 2008*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Abu-Laban, A. 2011: *The Kuwaiti-Danish Mission to Failaka 2010*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Abu-Laban, A. 2012: *The Kuwaiti-Danish Mission to Failaka 2011*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Abu-Laban, A. 2013: *The Kuwaiti-Danish Mission to Failaka 2012*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Hilton, A. 2014: *The Kuwaiti-Danish Mission to Failaka 2013*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Hilton, A. 2015: *The Kuwaiti-Danish Mission to Failaka 2014*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Højlund, F. & Hilton, A. 2016: *The Kuwaiti-Danish Mission to Failaka 2015*. Report to the Kuwait National Council for Culture, Arts and Letters.
- Ibrahim, M. 1982: *Excavations of the Arab Expedition at Sar el-Jisr, Bahrain*. State of Bahrain, Ministry of Information.
- Inizan, M. L., Jazim, M. and Merimier, F. 1992: L'artisanat de la cornaline au Yémen: premières données. *Techniques et cultures* 20, p. 155-174.
- Invernizzi, A., Bergamini, G., Gabutti, A. & Valtz, E. 2002-03: La ceramica di Tell Yelkhi. *Mesopotamia* 37-38.
- Jasim, S. A. 2012: *The Necropolis of Jebel al-Buhais: Prehistoric Discoveries in the Emirate of Sharjah, United Arab Emirates*. The Department of Culture & Information.
- JASP = Jutland Archaeological Society Publications. Moesgaard Museum. Aarhus/Højbjerg.
- Jeppesen, K. 1989: *The Sacred Enclosure in the Early Hellenistic Period*. (Failaka/Ikaros. The Hellenistic Settlements 3). JASP 16:3.
- Keel, O. 1995: *Corpus der Stempelsiegel-Amulette aus Palästina/Israel. Von den Anfängen bis zur Perserzeit. Einleitung*. Orbis Biblicus et Orientalis, Series Archaeologica 10, Fribourg/Göttingen.
- Keel, O. 1997: *Corpus der Stempelsiegel-Amulette aus Palästina/Israel. Von den Anfängen bis zur Perserzeit. Katalog Band I: Von Tell Abu Farağ bis 'Atlit*. Orbis Biblicus et Orientalis, Series Archaeologica 13, Fribourg/Göttingen.
- Keel, O. 2010a: *Corpus der Stempelsiegel-Amulette aus Palästina/Israel. Von den Anfängen bis zur Perserzeit. Katalog Band II: Von Bahan bis Tel Eton*. Orbis Biblicus et Orientalis, Series Archaeologica 29, Fribourg/Göttingen.
- Keel, O. 2010b: *Corpus der Stempelsiegel-Amulette aus Palästina/Israel. Von den Anfängen bis zur Perserzeit. Katalog Band III: Von Tell Far'a Nord bis Tell el-Fir*. Orbis Biblicus et Orientalis, Series Archaeologica 31, Fribourg/Göttingen.
- Keel, O. 2013: *Corpus der Stempelsiegel-Amulette aus Palästina/Israel. Von den Anfängen bis zur Perserzeit. Katalog Band IV: Von Tel Gamma bis Chirbet Husche*. Orbis Biblicus et Orientalis, Series Archaeologica 33, Fribourg/Göttingen.
- Kenoyer, J.M. 1984: Shell working industries of the Indus civilization: A summary. *Paléorient* 10/1, p. 49-63.
- Kenoyer, J. M. 2003: Stone Beads and Pendant Making Techniques. In: *A Bead Timeline: Vol. 1 Prehistory to 1200 CE*. J. W. Lankton. Washington, DC, The Bead Museum, p. 14-19.
- Kenoyer, J.M. 2005: Bead Technologies at Harappa, 3300-1900 BC: A Comparative Summary. In: *Jarrige, C. & Lefèvre, V. (eds), South Asian Archaeology*. Paris, Editions Recherche sur les Civilisations p. 157-170.
- Kenoyer, J.M. 2010: Master of Animals and Animal Masters in the Iconography of the Indus Tradition. In: *Counts, D.B. & Arnold, B. (eds.) The Master of Animals in Old World Iconography*, Budapest p. 37-58.
- Kenoyer, J. M. and Vidale, M. 1992: A new look at stone drills of the Indus Valley Tradition. In: *Vandiver, P., Druzick, J.R., Wheeler, G.S. and Freestone, I., Materials Issues in Art and Archaeology, III*. Pittsburgh, Materials Research Society 267, p. 495-518.
- Kenoyer, J. M., Vidale, M. and Bhan, K.K. 1991: Contemporary Stone Bead Making in Khambhat India: patterns of craft specialization and organization of production as reflected in the archaeological record. *World Archaeology* 23:1, p. 44-63.
- Kenoyer, J.M. 2001: Bead Technologies at Harappa, 3300-1900 BC: A Comparative Summary. In: *Jarrige, C. & Lefèvre, V. (eds.), South Asian Archaeology*, Paris, p. 157-170.
- Killick, R. & Moon, J. 2005: *The Early Dilmun Settlement at Saar*. London-Bahrain Archaeological Expedition. Saar Excavation Report 3. Archaeology International Ltd, Ludlow, UK.
- Kjærø, P. 1980: Seals of "Dilmun-Type" from Failaka, Kuwait. *PSAS* 10, p. 45-53.
- Kjærø, P. 1983: *The Stamp and Cylinder Seals. Plates and Catalogue Descriptions*. Failaka/Dilmun. The Second Millennium Settlements vol. 1. JASP 17:1. Moesgård, Aarhus.
- Kjærø, P. 1986: Architecture and Settlement Patterns in 2nd Mill. Failaka. *PSAS* 16, p. 77-88.
- Kjærø, P. 1994: Stamp seals, seal-impressions and seal blanks. In: *Højlund & Andersen 1994*, p. 319-350.
- Kjærø, P. 1997: Cylinder seal impressions. In: *Højlund & Andersen 1997* p. 160-164.
- Kjærø, P. 1999: The hidden art of Dilmun: the Stamp-seals. In: *Lombard (ed.) 1999* p. 115-120.
- Kjærø, P. 2003: Stamp Seals and Seal Impressions. In: *Andersen & Højlund 2003*, p. 289-306.
- Kjærø, P. & Højlund, F. 2013: *The Bronze Age Architecture*. (Failaka/Dilmun. The Second Millennium Settlements 3). JASP 17:3.
- Kohl, P.L. 2001: Reflections on the Production of Chlorite at Tepe Yahya: 25 years later. In: *Lamberg-Karlovsky & Potts 2001*, p. 209-230.
- Lamberg-Karlovsky, C.C. 1970: *Excavations at Tepe Yahya, Iran, 1967-1969*. Massachusetts: Cambridge.
- Lamberg-Karlovsky, C.C. 1988: The Intercultural Style Carved Vessels. *Iranica Antiqua* 23 p. 45-95.
- Lamberg-Karlovsky, C.C. & Potts, D.T. 2001: *Excavations at Tepe Yahya, Iran, 1967-1975. The Third Millennium*. American School of Prehistoric Research Bulletin 45. Peabody Museum of Archaeology and Ethnology. Harvard University, Cambridge, Massachusetts.
- Lankton, J.W. 2003: Chapter 4. In: *The Bead Timeline, Vol. 1: Prehistory to 1200 CE* p. 38-43.
- Laursen, S.T. 2010: The westward transmission of Indus Valley sealing technology: origin and development of the 'Gulf Type' seal and other administrative technologies in Early Dilmun, c. 2100-2000 BC. *AAE* 21 p. 96-134.
- Laursen, S.T. & Steinkeller, P. forth.: *Trade, Cultural, and His-*

- torical Contacts between Babylonia and the Gulf Region during the Third and Early Second Millennium: Archaeology and Texts.
- Law, R. W. 2011: *Inter-Regional Interaction and Urbanism in the Ancient Indus Valley: A Geologic Provenience Study of Harappa's Rock and Mineral Assemblage*, Occasional Paper 11. Kyoto, Research Institute for Humanity and Nature.
- Lombard, P. (ed.) 1999: *Bahrain. The Civilisation of the Two Seas. From Dilmun to Tylos*. Paris.
- Lombard, P. 1999: Early Dilmun Burial Offerings. In: Lombard (ed.) 1999.
- Ludvik, G., Kenoyer, J.M., Pieniążek, M. and Aylward, W. 2015: New perspectives on Stone Bead Technology at Bronze Age Troy. *Anatolian Studies Journal* 65 p. 1-18.
- Mackay, E.J.H. 1925: *Report on the Excavations of the "A" Cemetery at Kish, Mesopotamia*. Field Museum-Oxford University joint expedition. Field Museum of Natural History. Chicago.
- Madjidzadeh, Y. 2003: La découverte de Jiroft. In: *Dossiers d'Archeologie* n°287 oct. 2003 p. 18-63.
- Marschner, R.-F., Duffy, L. J. & Wright, H. 1978: Asphalts from Ancient town sites in Southwestern Iran. *Paleorient* 4, p. 97-112.
- Marshall, J. 1931: *Mohenjo-Daro and the Indus Civilization Vol. III*. London.
- Massih, J. A. in prep.: *Observations on the coastal and natural rock formation*. French mission to Failaka, Ms dated 2013.
- Mathiesen, H.E. 1982: *The Terracotta Figurines*. Ikaros-The Hellenistic Settlements 1. JASP.
- Mayr, R.H. 1997: *The Seal Impressions of Ur III Umma*. PhD dissertation, Leiden.
- McCown, D.E. & Haines, R.C. 1967: *Nippur 1: Temple of Enlil, Scribal Quarter, and Soundings*. OIP 78.
- Meadow, R.H. 1991: Faunal Remains and Urbanism at Harappa. In: Meadow, R.H. (ed.), 1991 *Harappa Excavations 1986-1990*. Monographs in World Archaeology 3. Madison, p. 89-106.
- Miroschedji, P. de 1973: Vases et objets en stéatite susiens du musée du Louvre. *Cahiers de la Délégation Archéologique Française en Iran* 3, p. 9-80
- Moon, J. 2000: The Dilmunite Settlement at Saar. In: *Traces of Paradise. The Archaeology of Bahrain 2500 BC – 300 AD*.
- Moore, A.B.M. 2012: Elasmobranchs of the Persian (Arabian) Gulf: Ecology, Human Aspects and Research Priorities for their Improved Management. *Review in Fish Biology and Fisheries* 22, p. 35-61.
- Moorey, P. R. S. 1985: *Materials and Manufacture in Ancient Mesopotamia: The evidence of archaeology and art*. Oxford, BAR International Series.
- Moorey, P. R. S. 1994: *Ancient Mesopotamian Materials and Industries*. Oxford, Clarendon Press.
- Mughal, M.R. 1983: *The Dilmun Burial Complex at Sar*. The 1980-82 Excavations in Bahrain. Ministry of Information. Bahrain.
- Nissenbaum, A., Aizenshtat, Z. & Goldberg, M. 1980: The floating blocks of the Dead Sea. In: Douglas, A. G. & Maxwell, J. R. (eds.) *Advances in Organic Geochemistry*. London: Pergamon Press.
- Nissenbaum, A., Serban, A., Amiran, R. & Ilan, O. 1984: Dead Sea Asphalt from the Excavations in Tel Arad and Small Tell Malhata. *Paleorient* 10, p. 157-161.
- Oguchi, K. 1992: Shells and Shell Objects from Area A of 'Usiyeh. *Al-Rafidan*, vol. XIII, p. 61-81.
- Oppenheim, A.L. 1954: The Seafaring Merchants of Ur. *JAOS* 74 p. 6-17.
- Peters, K. E., Walters, C. C. & Moldowan, J. M. 2005a: *The Biomarker Guide Volume 1: Biomarkers and Isotopes in the Environment and Human History*, New York, Cambridge University Press.
- Peters, K. E., Walters, C. C. & Moldowan, J. M. 2005b: *The Biomarker Guide: Volume 2, Biomarkers and Isotopes in Petroleum Systems and Earth History*, New York, Cambridge University Press.
- Petrie, C.A., Chaverdi, A.A. & Seyedin, M. 2005: From Anshan to Dilmun and Magan: The spatial and temporal distribution of Kaftari and Kaftari-related ceramic vessels. *Iran* 43, p. 199-236.
- Peyronel, L. 2001-2003: Commercianti Dilmuniti nella Susiana? Evidenze archeologiche ed epigrafiche sulle relazioni culturali tra Golfo Persico e Iran Sud-Occidentale agli inizi del II Mill. A.C. *Scienze dell'Antichità* 11 p. 95-118.
- Pic, M. 2008: La céramique du tell F6, fouilles francaises de 1984 à 1988. In: *Calvet & Pic 2008*, p. 33-170.
- Piperno, M. 1973: Micro-Drilling at Shahr-i-Sokhta; the making and use of the lithic drill-heads. *South Asian Archaeology 1971*. N. Hammond. London, Duckworth p. 119-130.
- Pittman, H. 2003: La Culture du Halil Roud. In: *Dossiers d'Archeologie* n° 287 oct. 2003 p. 78-87.
- Possehl, G.L. 1996: Meluhha. In: Reade, J. (ed.), *The Indian Ocean in Antiquity*. Kegan Paul: British Museum p. 133-208.
- Postgate, N. and Moon, J.A. 1982: Excavations at Abu Salabikh 1981. *Iraq* vol. 44:2, p. 103-136.
- Potts, D.T. 1986: The booty of Magan. *Oriens Antiquus* XXV p. 271-285.
- Potts, D.T. 2001: Ostrich Distribution and Exploitation in the Arabian Peninsula. *Antiquity* vol. 75, p. 182-190.
- Potts, D.T. 2010: Cylinder seals and their use in the Arabian Peninsula. *AAE* 21, p. 20-40.
- Potts, T.F. 1989: Foreign Stone Vessel of the Late Third Millennium B.C. from Southern Mesopotamia: Their Origins and Mechanisms of Exchange. *Iraq* 51, p. 123-164.
- Potts, T.F. 1994: *Mesopotamia and the East. An Archaeological and Historical Study of Foreign Relations ca. 3400-2000 BC*. Oxford: Oxford University Committee for Archaeology.
- Prabhakar, V. N., Bisht, R.S., Law, R.W. and Kenoyer, J.M. 2012: Stone Drill Bits from Dholavira: A multifacet analysis. *Puratattva* 37, p. 8-25.
- PSAS: Proceedings of the Seminar for Arabian Studies.
- Reade, J. & Searight, A. 2001: Arabian softstone vessels from Iraq in the British Museum. *AAE* 12, p. 156-172.
- Reimer, P.J. et al. 2013: IntCal13 and Marine13 radiocarbon age calibration curves 0-50,000 years cal BP. *Radiocarbon* 55(4) p. 1869-87.
- Rullkotter, J. & Nissenbaum, A. 1988: Dead Sea asphalt in Egyptian mummies: molecular evidence. *Naturwissenschaften* 75, p. 618-621.
- Rullkotter, J., Spiro, B. & Nissenbaum, A. 1985: Biological marker characteristics of oils and asphalts from carbonate source rocks in a rapidly subsiding Graben, Dead-Sea, Israel. *Geochimica Et Cosmochimica Acta*, 49, p. 1357-1370.
- Salje, B. 1990: *Der 'Common Style' der Mitanni-Glyptik und die Glyptik der Levant und Zyperns in der späten Bronzezeit*. Mainz: Baghdader Forschungen 11.
- Salles, J. F. 1984: Failaka Fouilles Françaises 1983. *Travaux de la Maison de l'Orient*. Lyon: Maison de l'Orient.
- Sperveslage, G. 2014: *Ägypten und Arabien. Ein Beitrag zu den interkulturellen Beziehungen Altägyptens*. PhD dissertation Berlin, Freie Universität (to be published as *Alter Orient und Altes Testament* 420).
- Steinkeller, P. 2013: *Trade Routes and Commercial Networks in the Persian Gulf during the Third Millennium BC*. Paper

- presented at the Third International Biennial Conference of the Persian Gulf. Tehran p. 413-431.
- Stocks, D. A. 2003: *Experiments in Egyptian Archaeology: Stone-working technology in Ancient Egypt*. Routledge, London.
- Vallat, F. 2003: La Ziggurat. In: *Dossiers d'Archeologie* (n°287 oct. 2003) p. 92-95.
- Uerpmann, M. and Uerpmann, H.-P. 1994: Animal bone finds from Excavation 520 at Qala'at al-Bahrain. In: *Højlund & Andersen 1994* p. 417-444.
- Uerpmann, M. and Uerpmann, H.-P. 1999: The Animal Economy of Ancient Dilmun in Light of Faunal Remains from Saar and Qala'at al-Bahrain. *ISIMU Revista Sobre Orientale Próximo y Egipto en la antigüedad* 2 p. 635-46.
- Uerpmann, M. and Uerpmann, H.-P. 2005: Animal Bone Finds and their Relevance to the Ecology and Economy of Saar. In: *Killick & Moon 2005* p. 293-308.
- Van de Velde, T. in prep.: Appendix 2: Analysis of Bitumen from the Royal Mounds. In: Laursen, S.T, *The Rise of Kingship and the Early Dilmun State in Bahrain*. Moesgaard: Jutland Archaeological Society.
- Van de Velde, T., De Vrieze, M., Surmont, P., Bodé, S. & Drechsler, P. 2015: A geochemical study on the bitumen from Dosariyah (Saudi-Arabia): tracking Neolithic-period bitumen in the Persian Gulf. *Journal of Archaeological Science* p. 248-256.
- Vats, M.S. 1940: *Excavations at Harappa vol. 2*. Delhi.
- Velde, C. 1998: The Dilmun Cemetery at Karanah 1 and the change of burial customs in late City II. In: Phillips, C.S., Potts, D.T. & Searight, S.: *Arabia and Its Neighbours: Essays on Prehistorical and Historical Developments*. Abiel II. Brepolis, p. 245-261.
- Velde, C. 2003: Wadi Suq and Late Bronze Age in the Oman Peninsula. In: Potts, D., Naboodah, H.A., Hellyer, P. (eds.): *Proceedings of the First International Conference on the Archaeology of the U.A.E*. Trident press, p. 102-113.
- Vogt, B. and Franke-Vogt, U. (eds.) 1987: *Shimal 1985/1986. Excavations of the German Archaeological Mission in Ras Al-Khaimah, U.A.E. A Preliminary Report*. Berliner Beiträge zum Vorderen Orient: Band 8. Dietrich Reimer Verlag. Berlin.
- Vogt, B. 1996: Bronze Age Maritime Trade in the Indian Ocean: Harappan Traits on the Oman Peninsula. In: Reade, J. (ed.), *The Indian Ocean in Antiquity*. Kegan Paul: British Museum p. 107-132.
- Vosmer, T. 1996: Watercraft and Navigation in the Indian Ocean: An Evolutionary Perspective. In: M. Tosi: *The Prehistory of Asia and Oceania*, vol. 16 of the XIII International Congress of Prehistoric and Protohistoric Sciences, Forli, Italia 8/14 September 1996, p. 223-242.
- Weisgerber, G. 1980: "... und Kupfer in Oman". Das Oman-Projekt des Deutschen Bergbau-Museums. *Der Anschnitt* 32 p. 62-110.
- Woolley, C.L. 1934: *The Royal Cemetery*. Ur Excavations 2. Oxford.
- Yule, P. 2001: *Die Gräberfelder in Samad al-Shan (Sultanat Oman). Materialien zu einer Kulturgeschichte*. Deutsches Archäologisches Institut, Orient-Abteilung, Orient-Archäologie vol. 4. Rahden.
- Zarins, J. 1978: Typological Studies in Saudi Arabian Archaeology, Steatite Vessels in the Riyadh Museum. *ATLAL Journal of Saudi Arabian Archaeology* 2, p. 65-93.



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